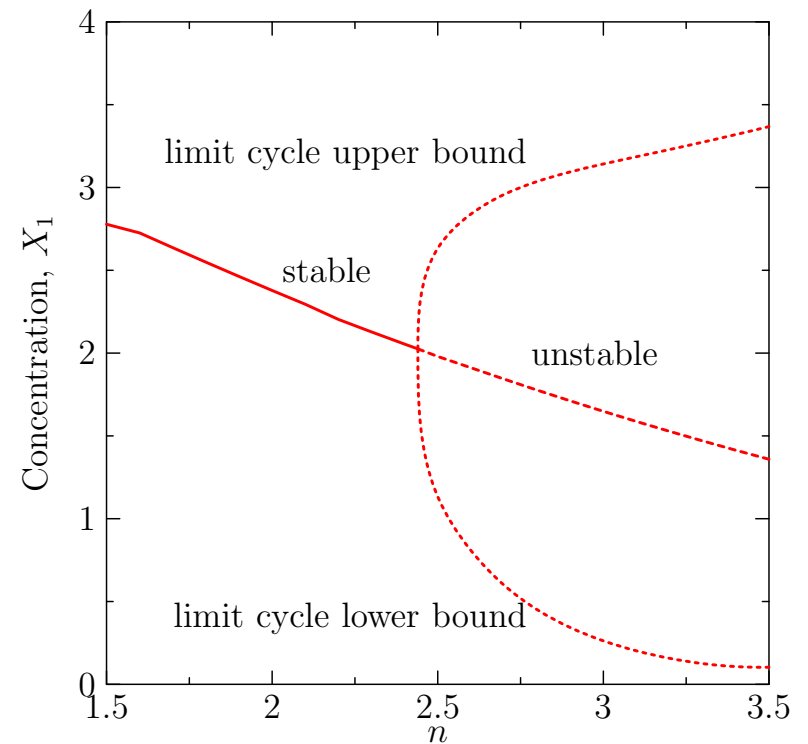
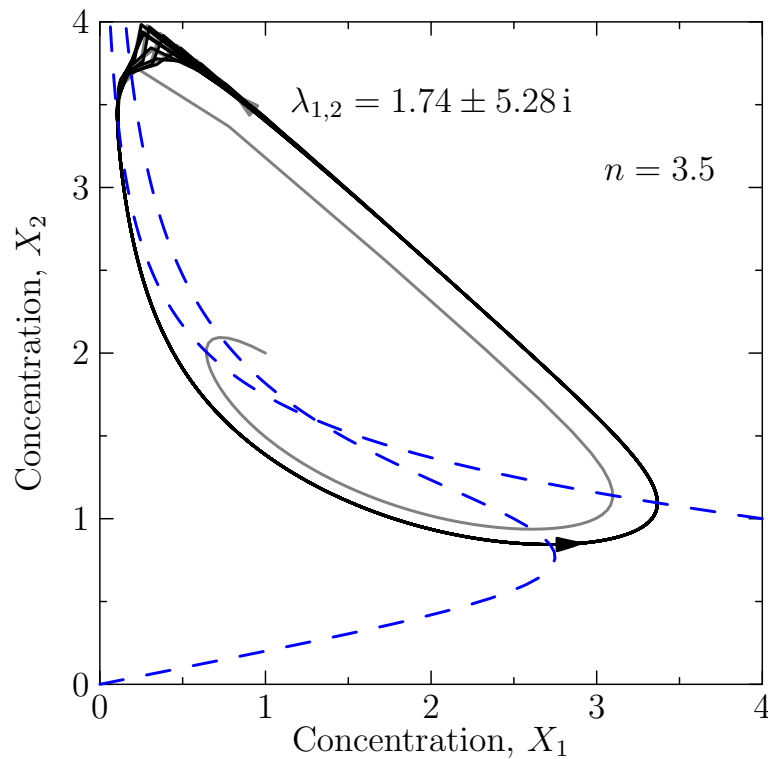


Computational Biology

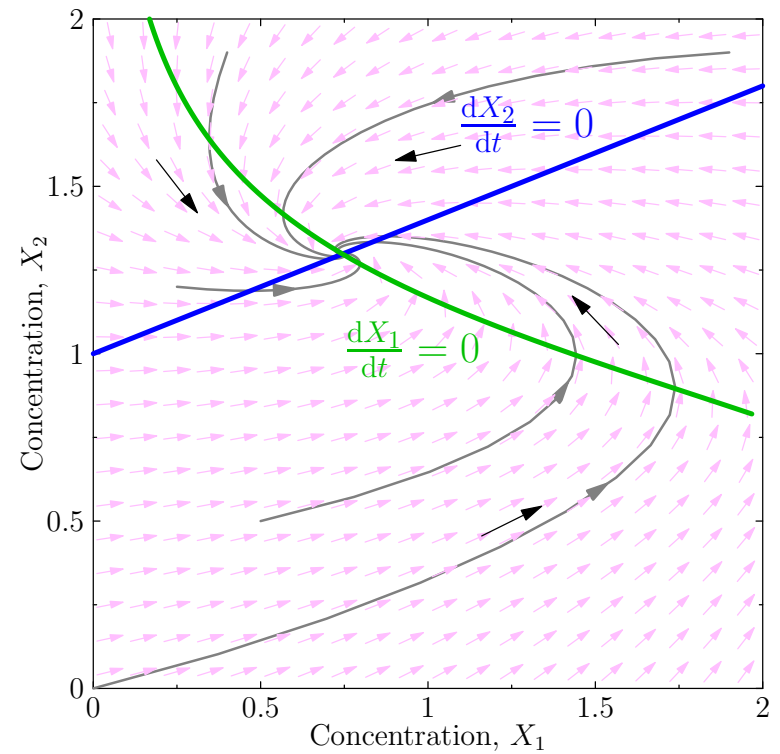
Lecture 4: *Dynamical Systems*



ODEs, nullclines, fixed points, stability analysis, bifurcations

Outline

1. **Understanding ODEs**
2. Stability
3. Bifurcations



Dynamical Systems

- In the last lecture we saw that we could model the dynamics of a system of molecules $\mathbf{X} = (X_i | i = 1, 2, \dots, n)$ in terms of differential equations

$$\frac{dX_i}{dt} = f_i(\mathbf{X})$$

- These model the mean behaviour of a set of chemical equations
- Provide the number of molecules is sufficiently large this model is accurate
- In this lecture we look at mathematical tools for understanding sets of first order differential equations

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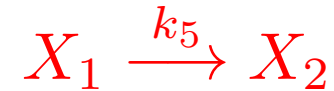
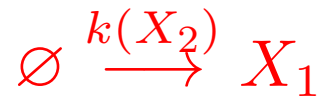
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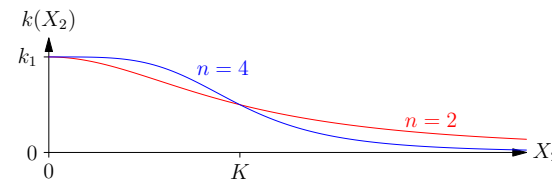
Two Species

- Consider the simple system



- Where X_2 represses the expression of X_1

$$k(X_2) = \frac{k_1}{1 + (X_2/K)^n}$$



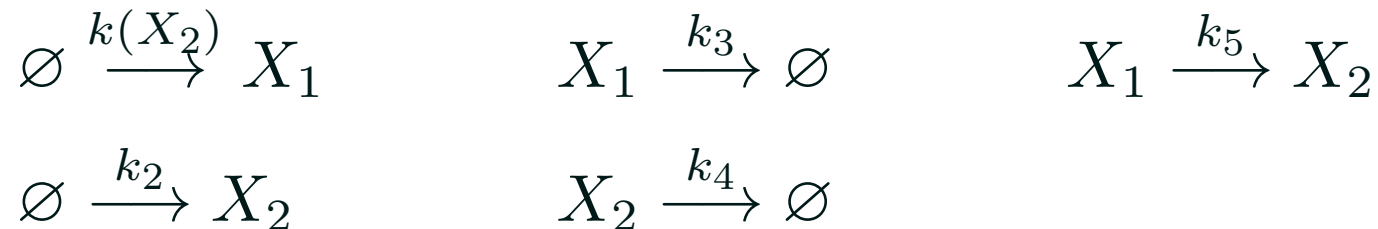
- Leading to the coupled ODEs

$$\frac{dX_1(t)}{dt} = f_1(X_1, X_2) = \frac{k_1}{1 + (X_2(t)/K)^n} - k_3 X_1(t) - k_5 X_1(t)$$

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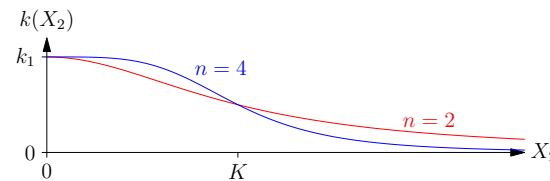
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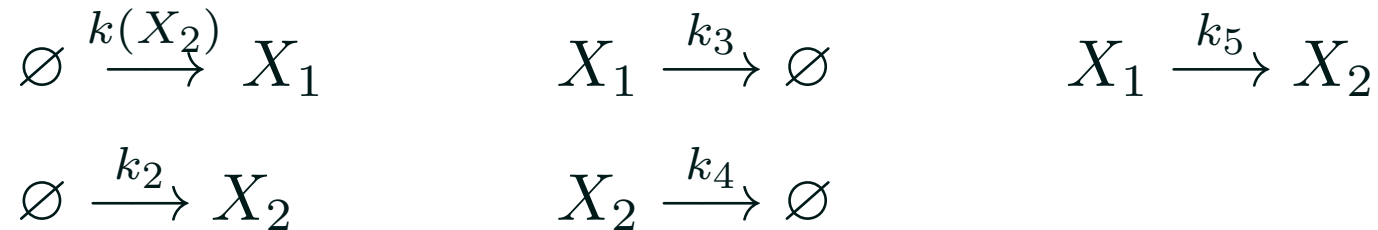


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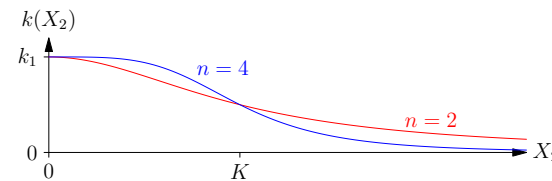
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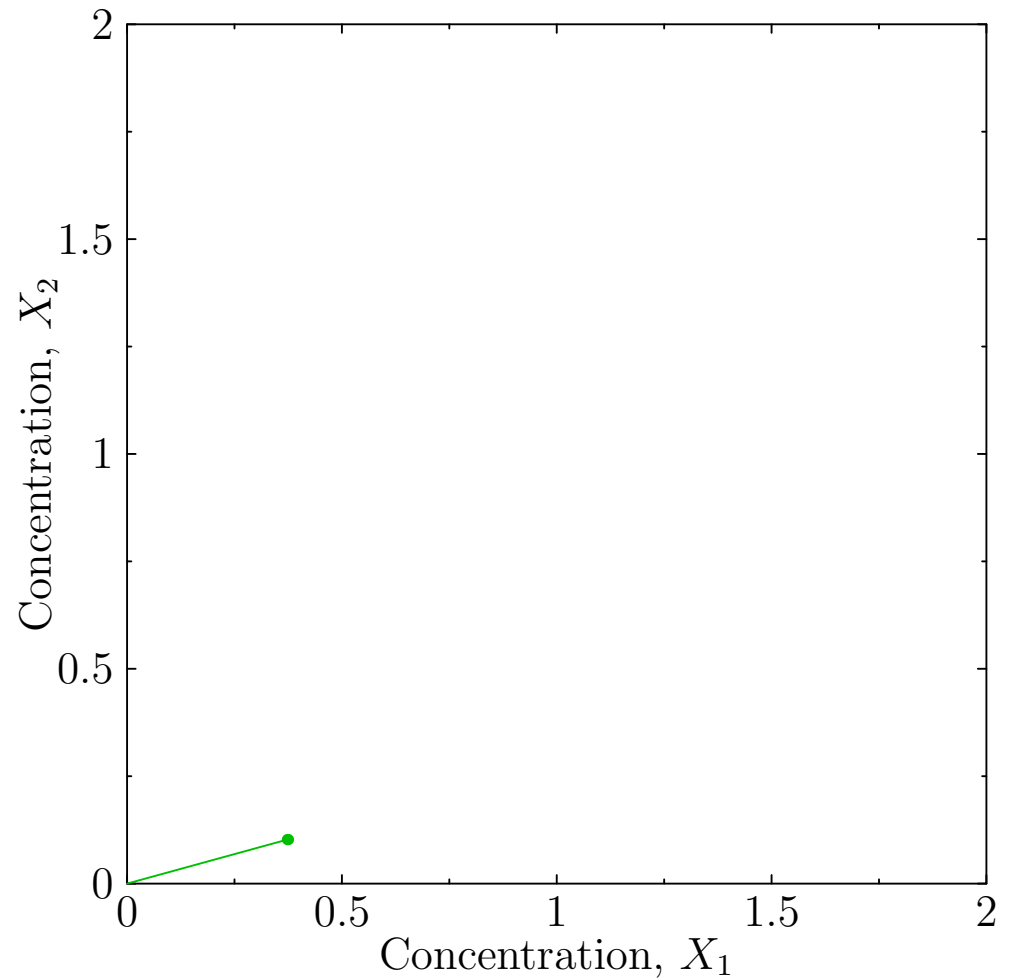
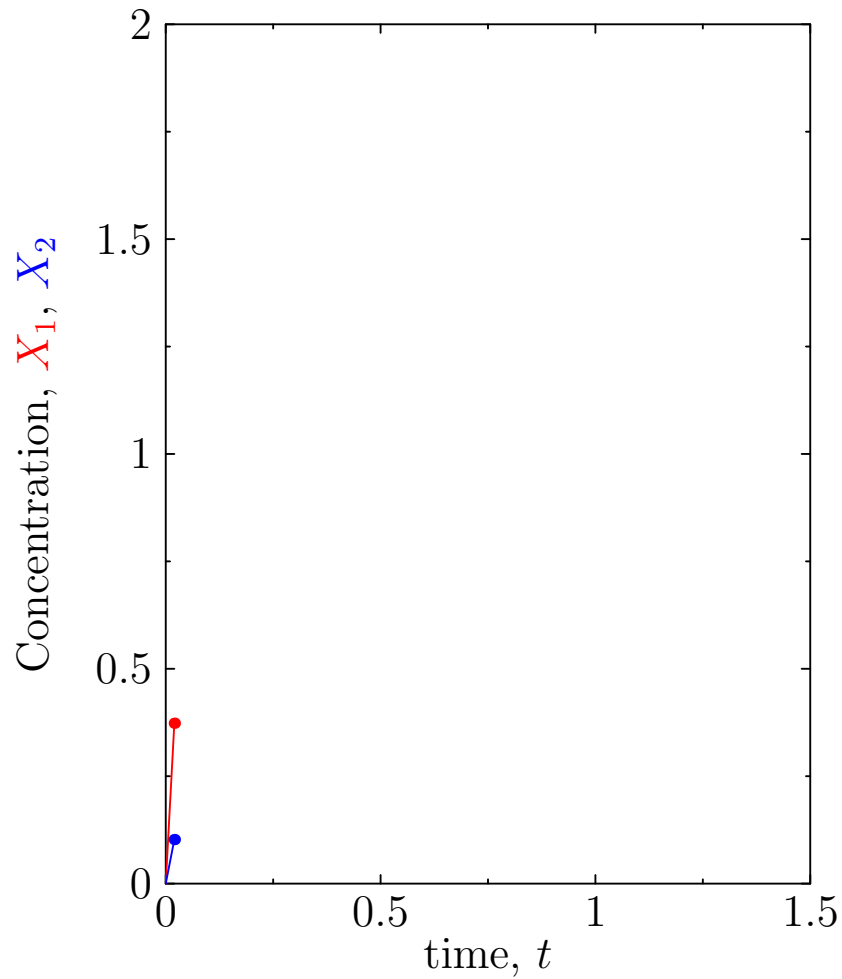
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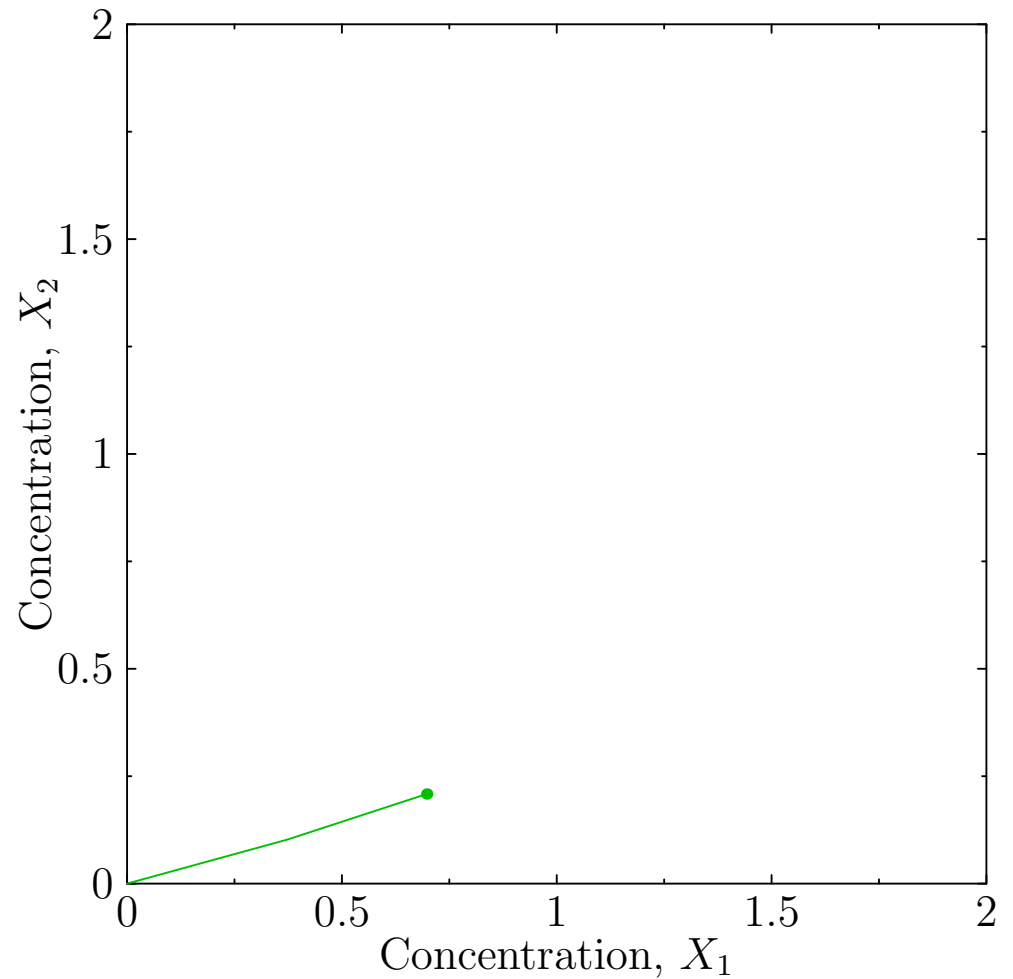
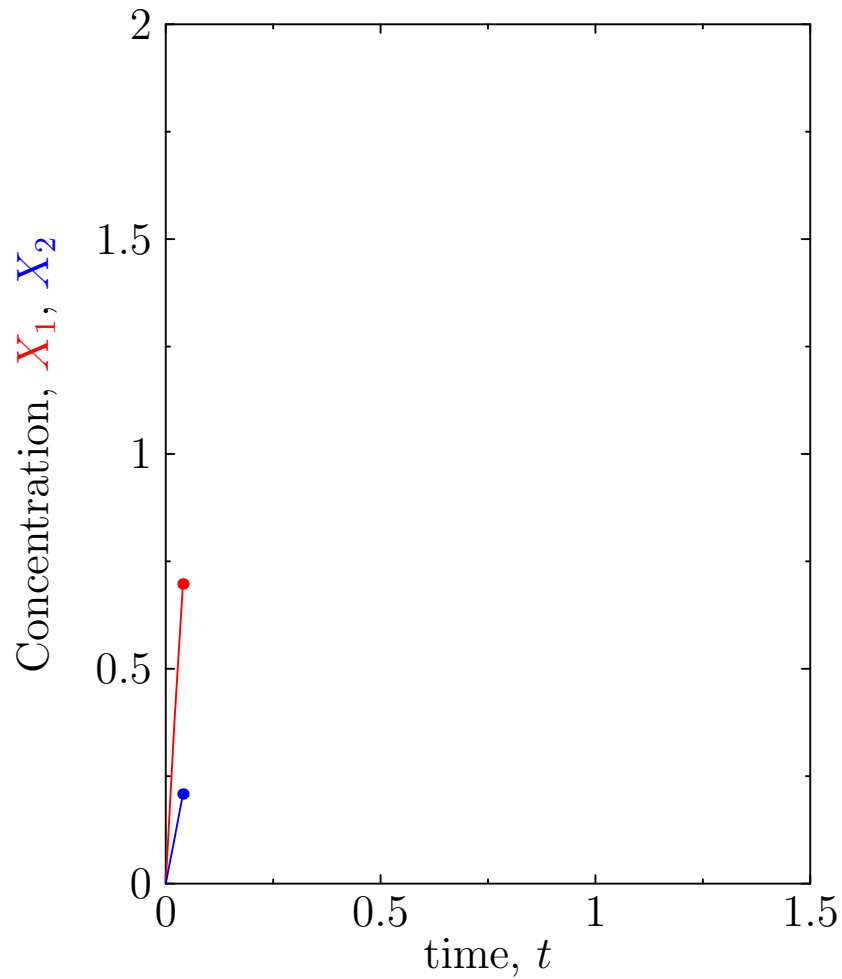
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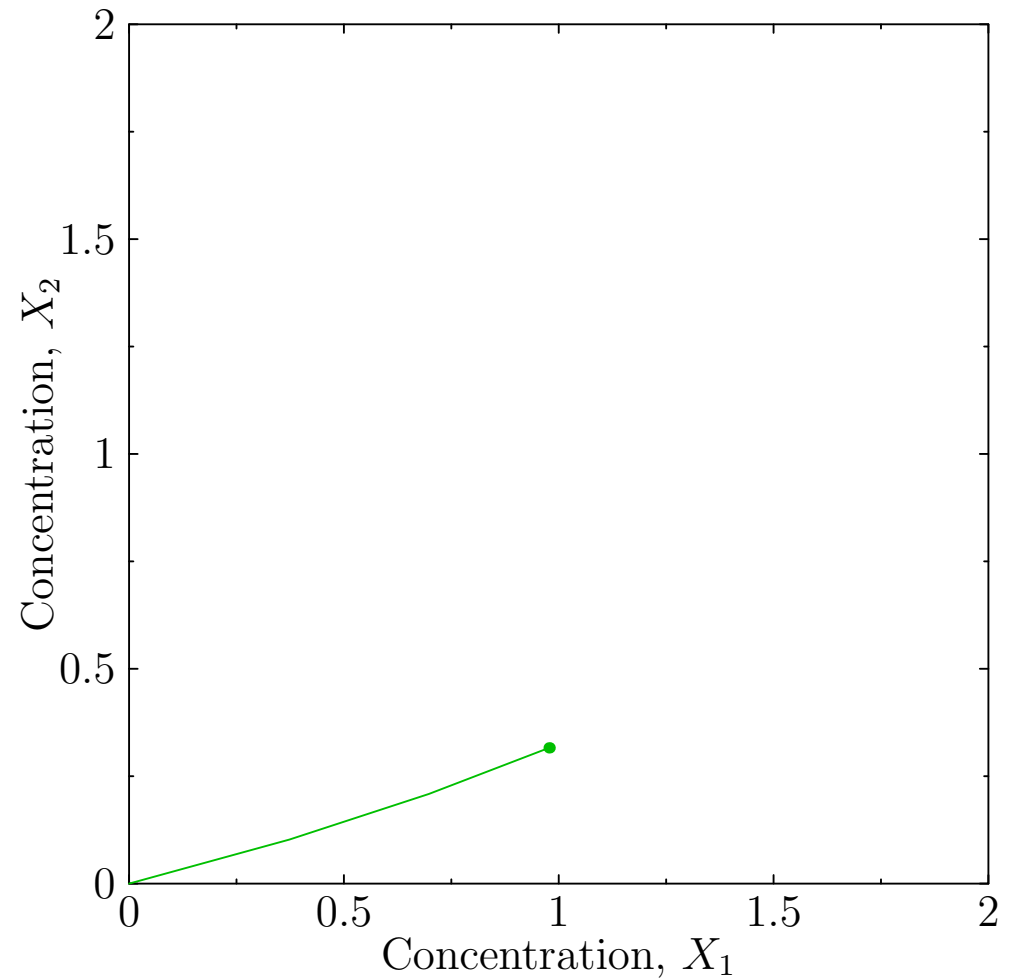
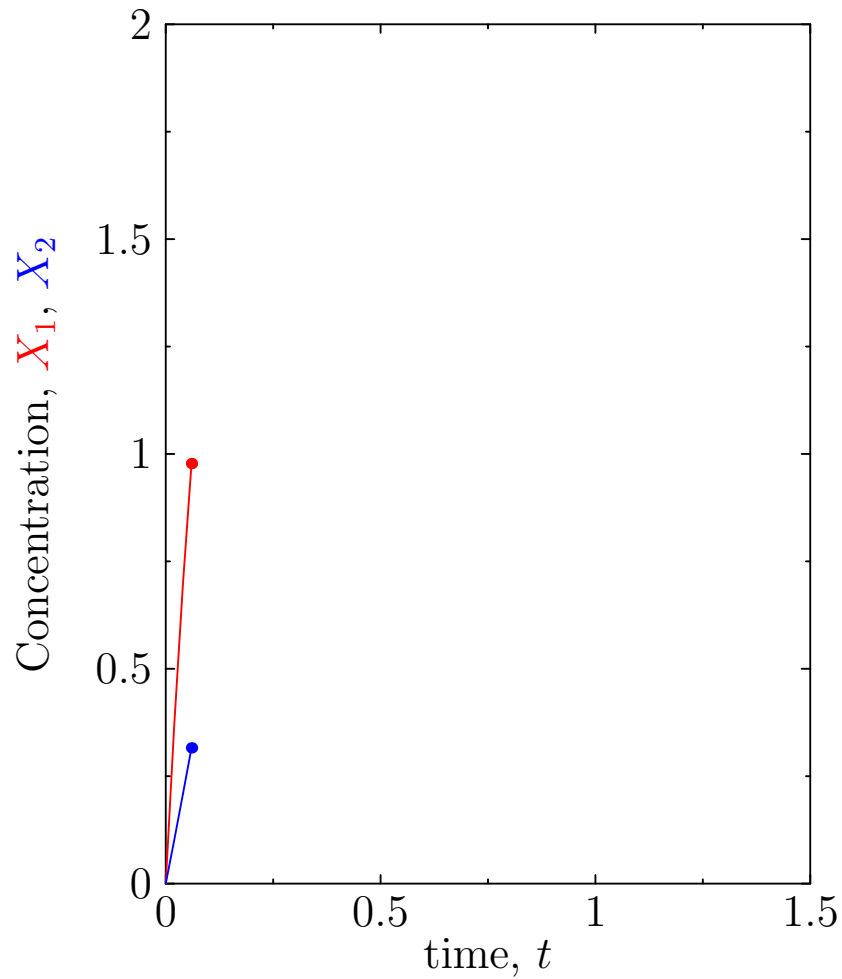
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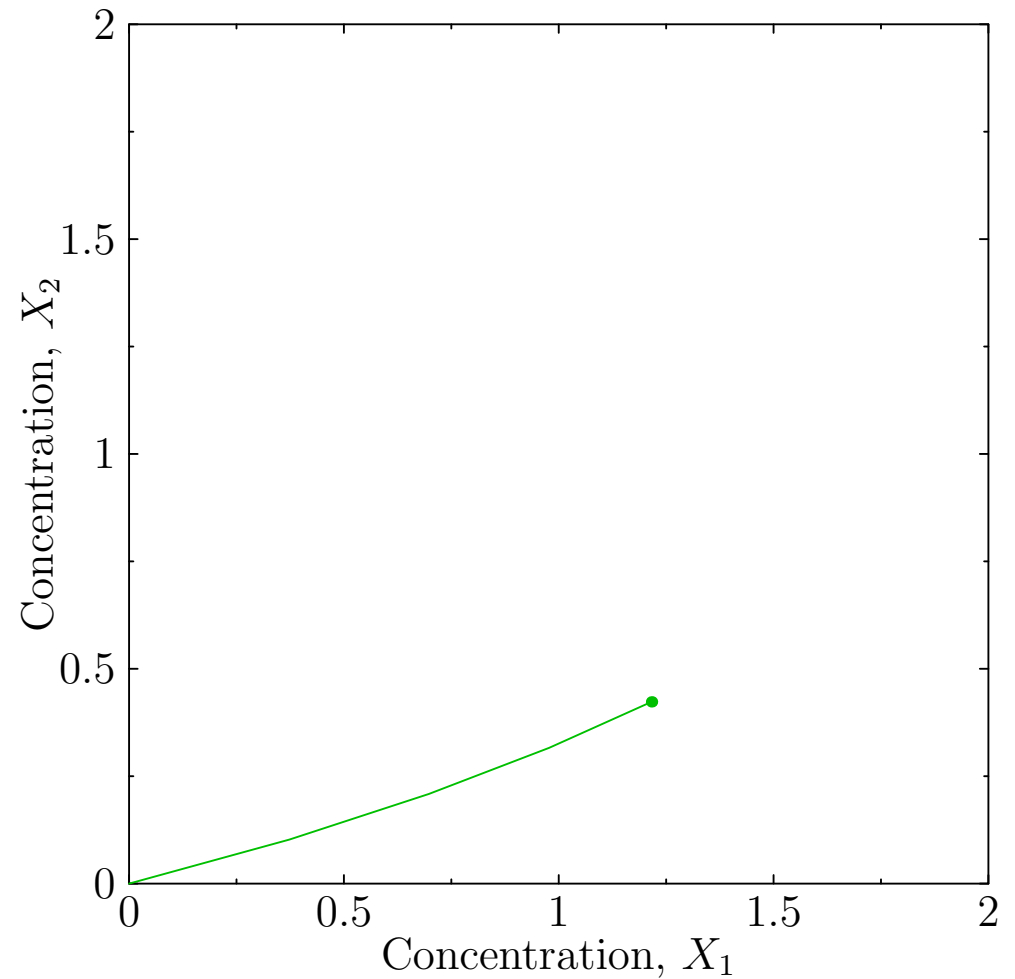
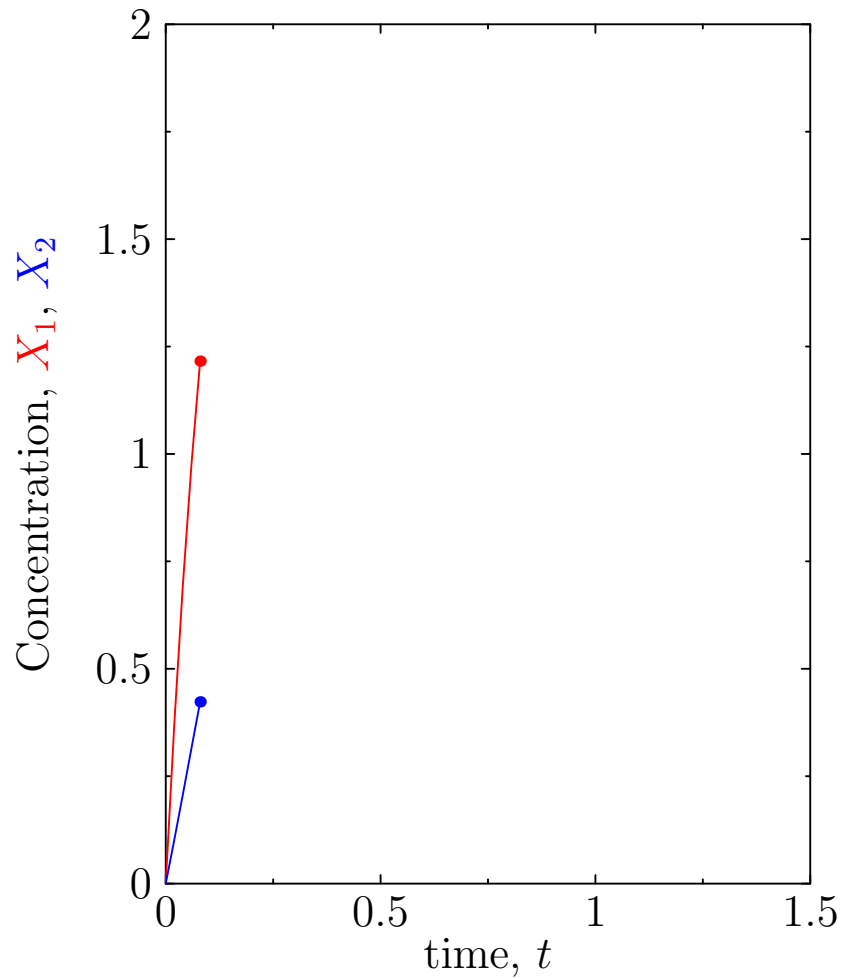
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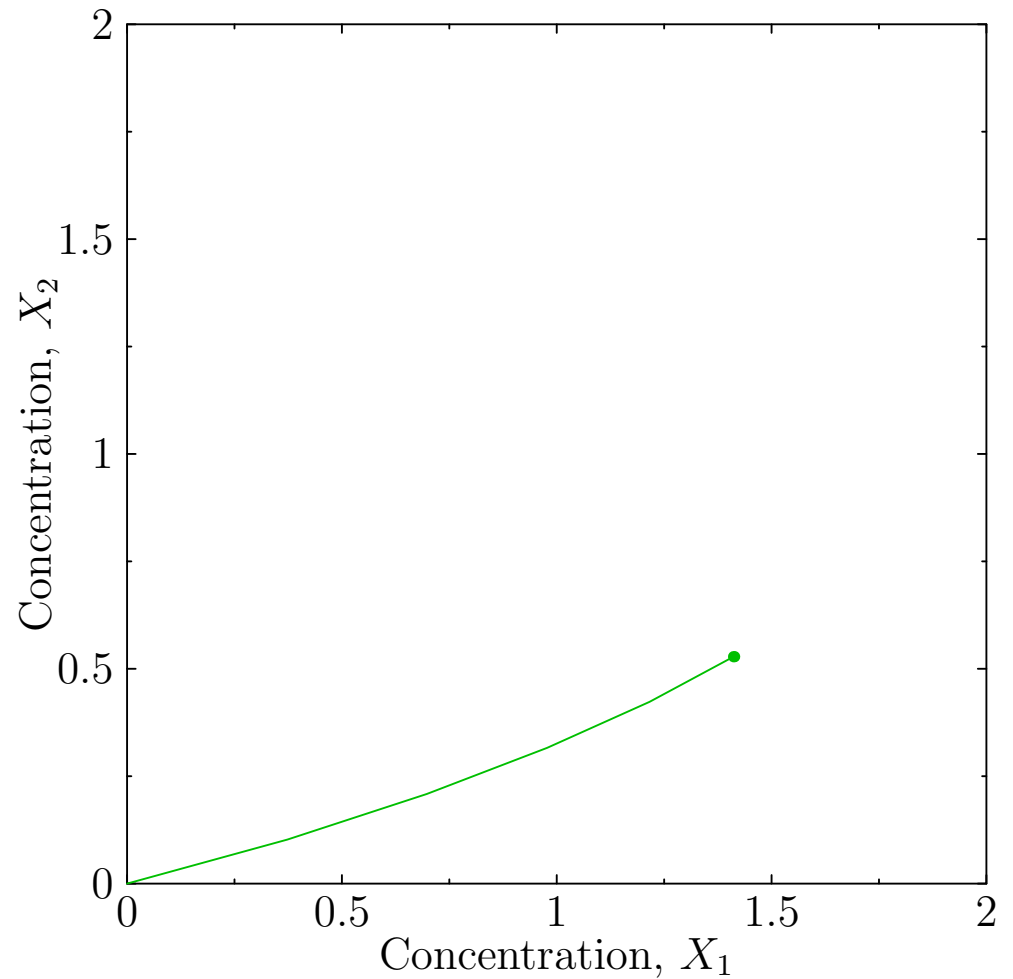
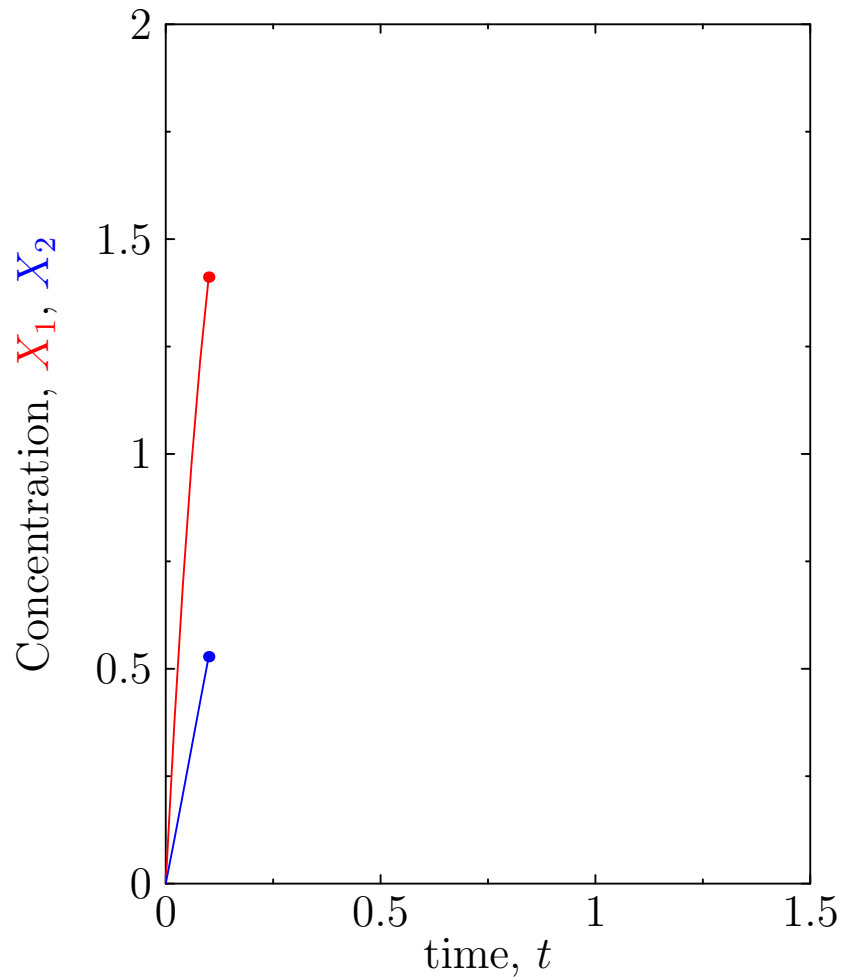
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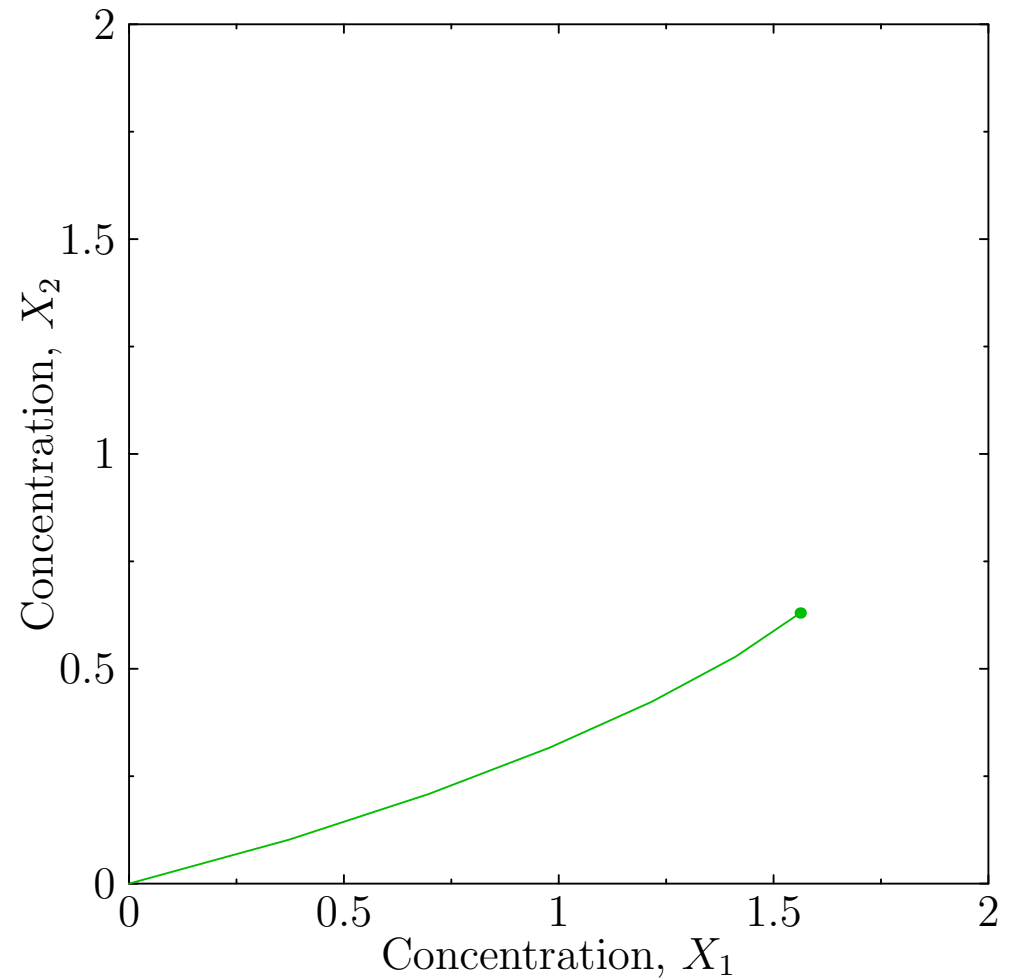
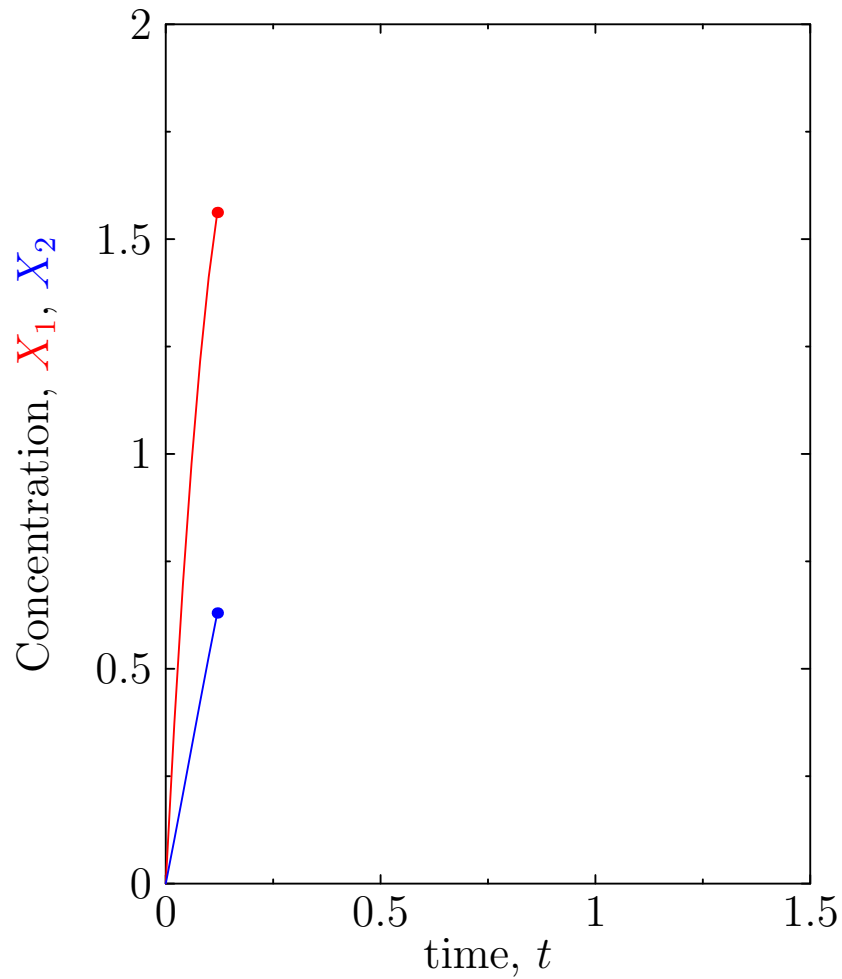
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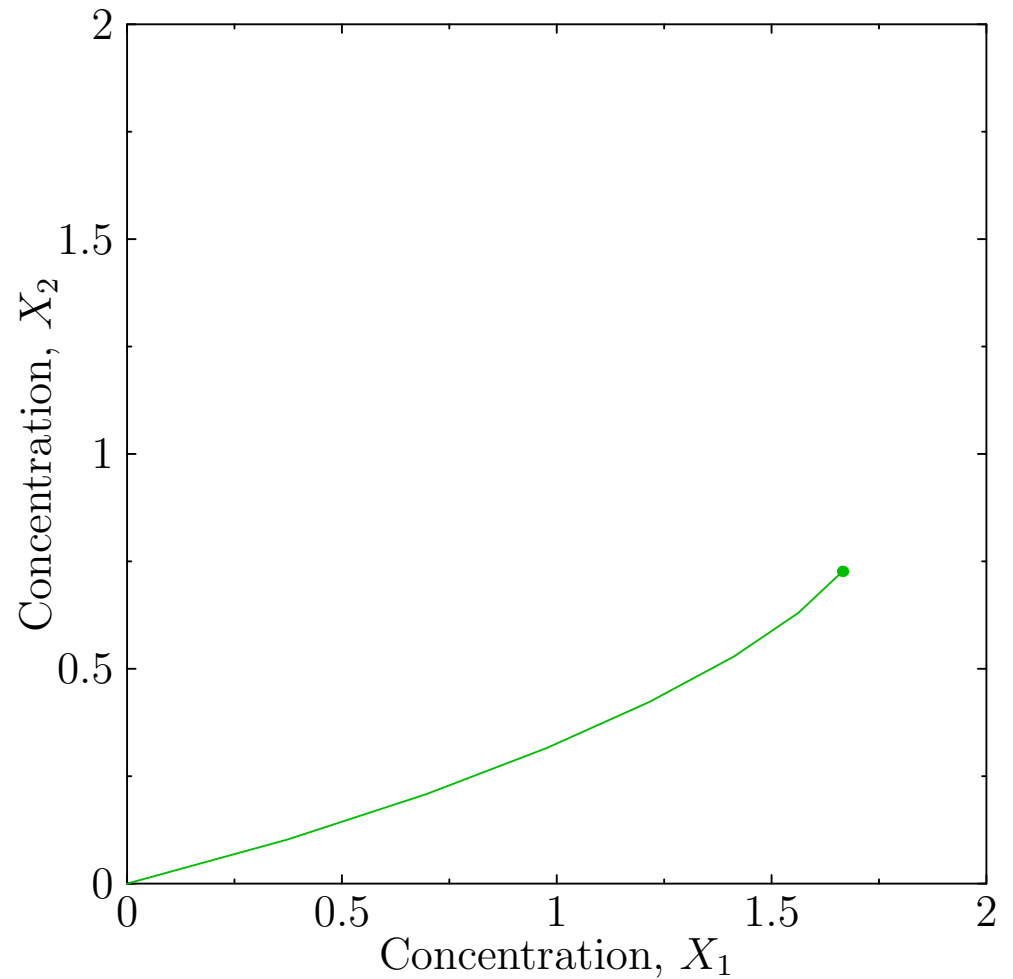
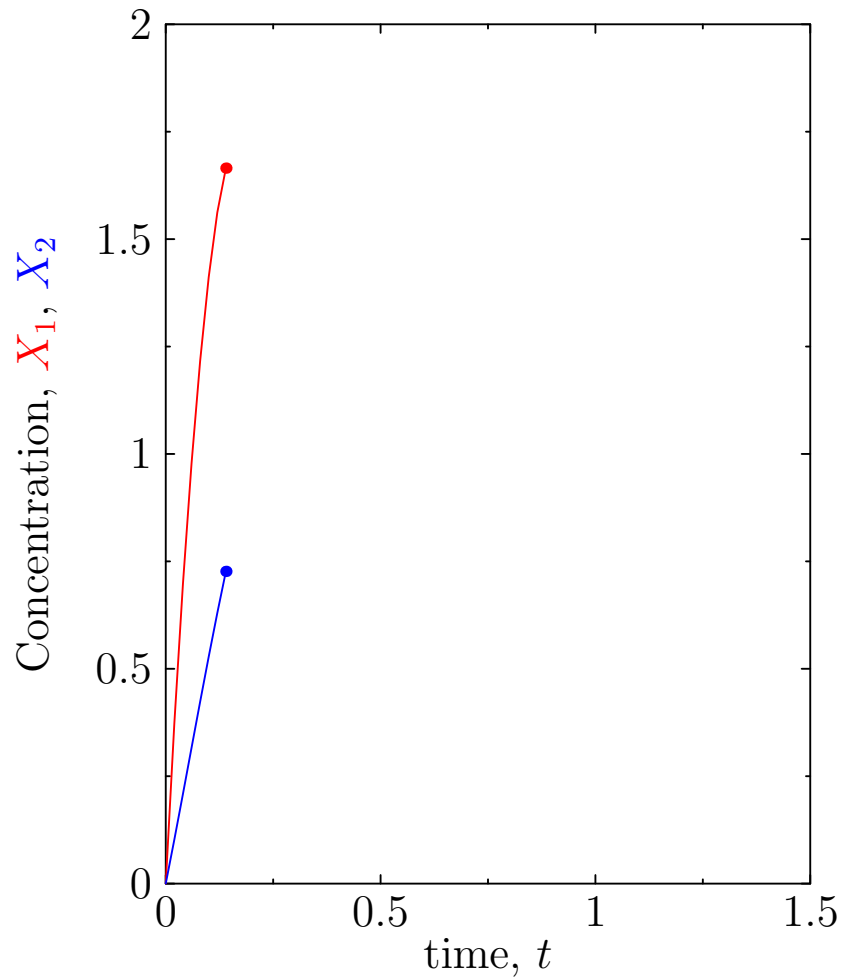
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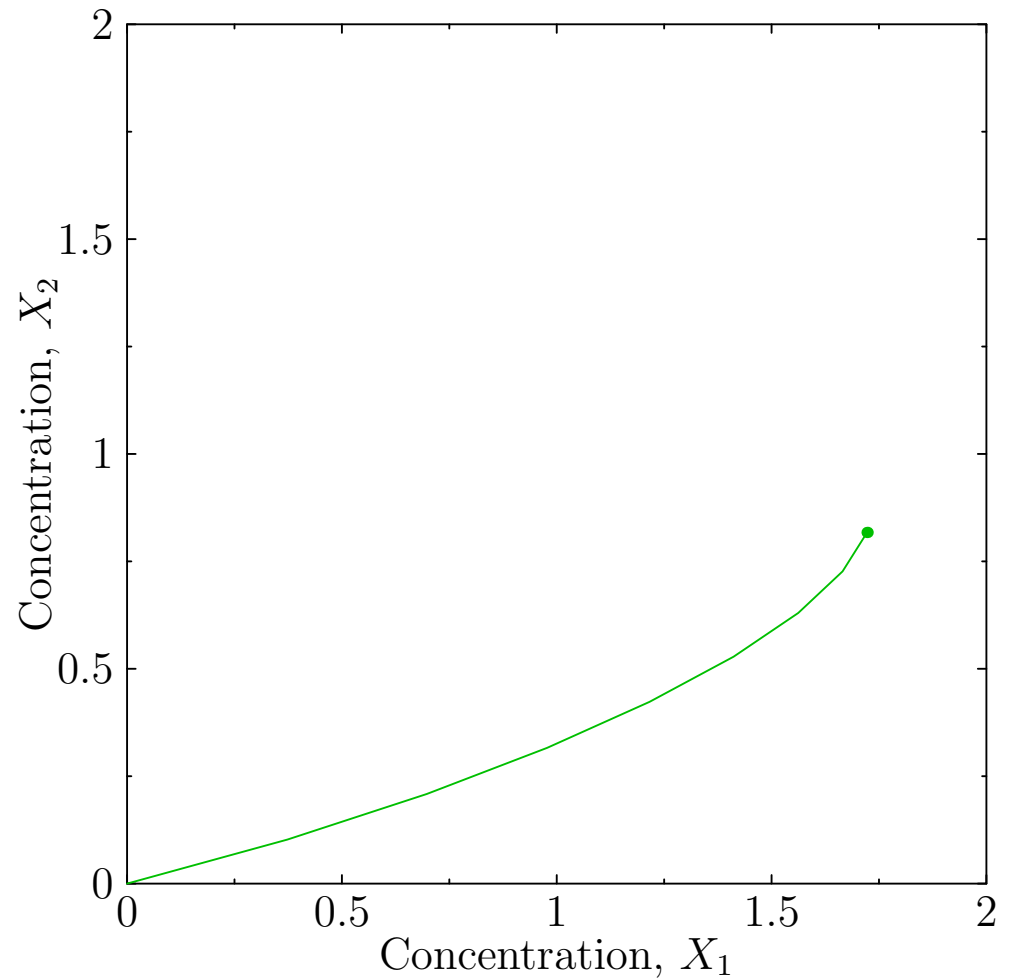
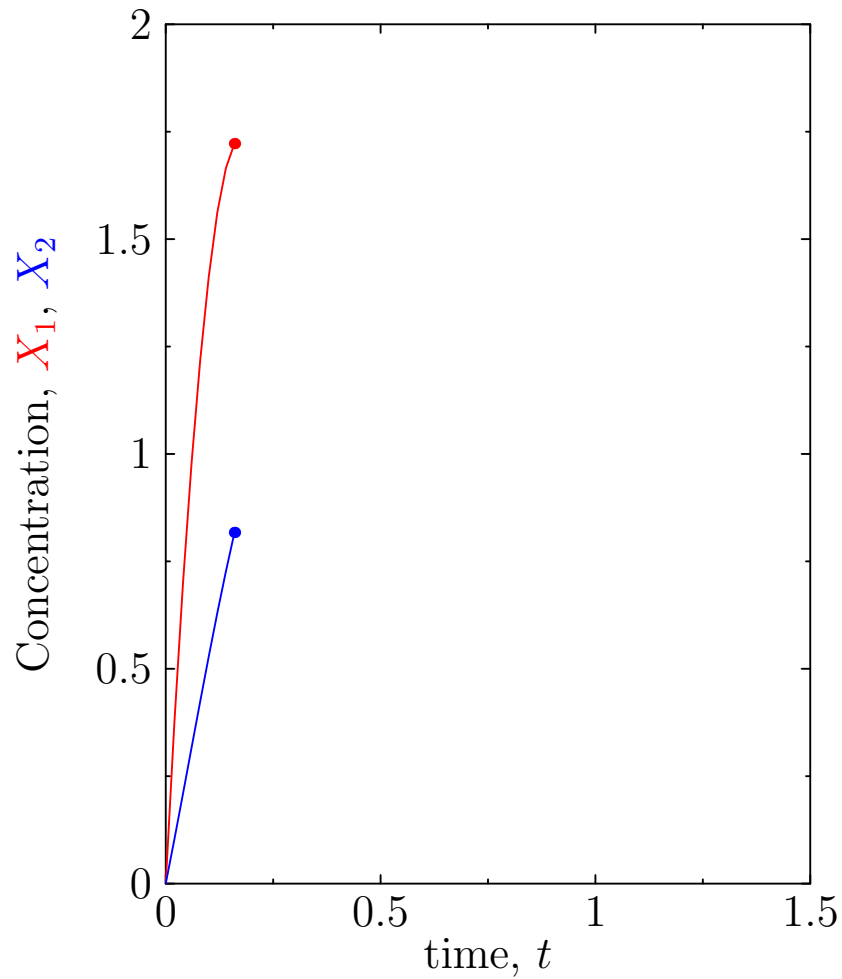
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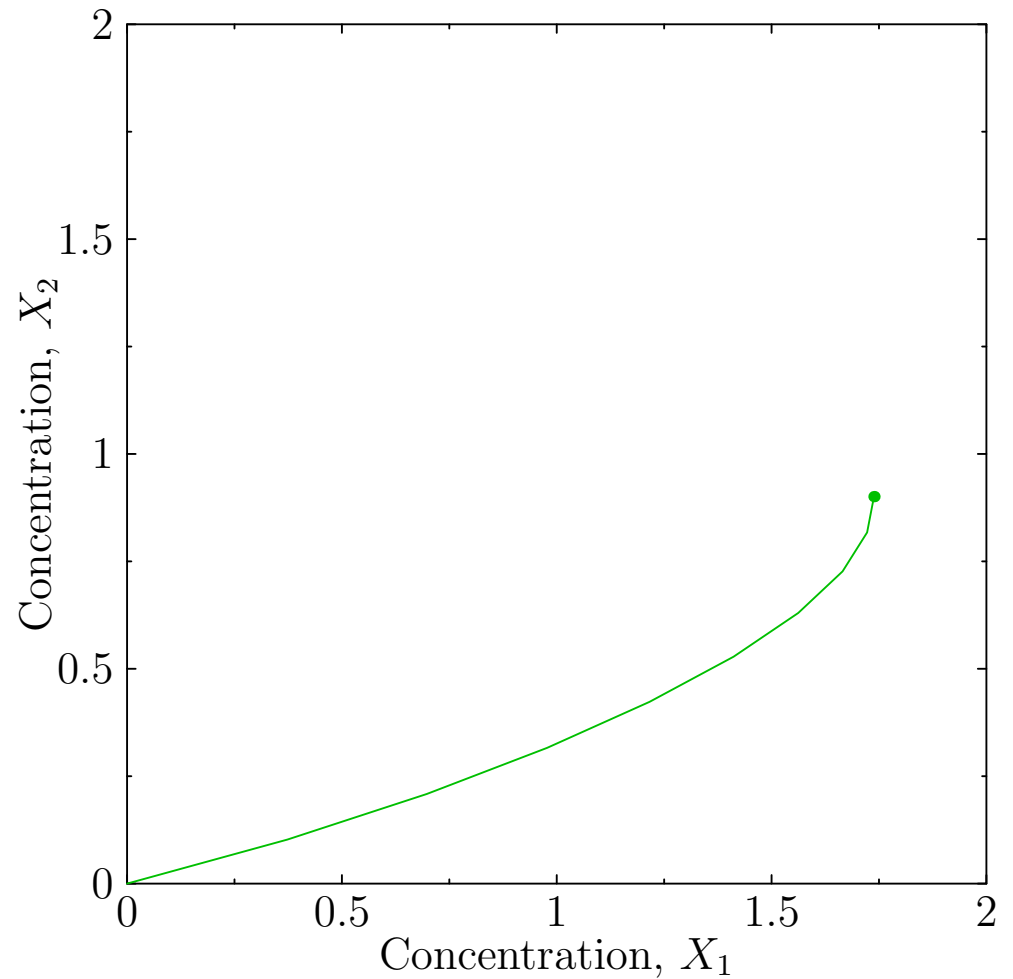
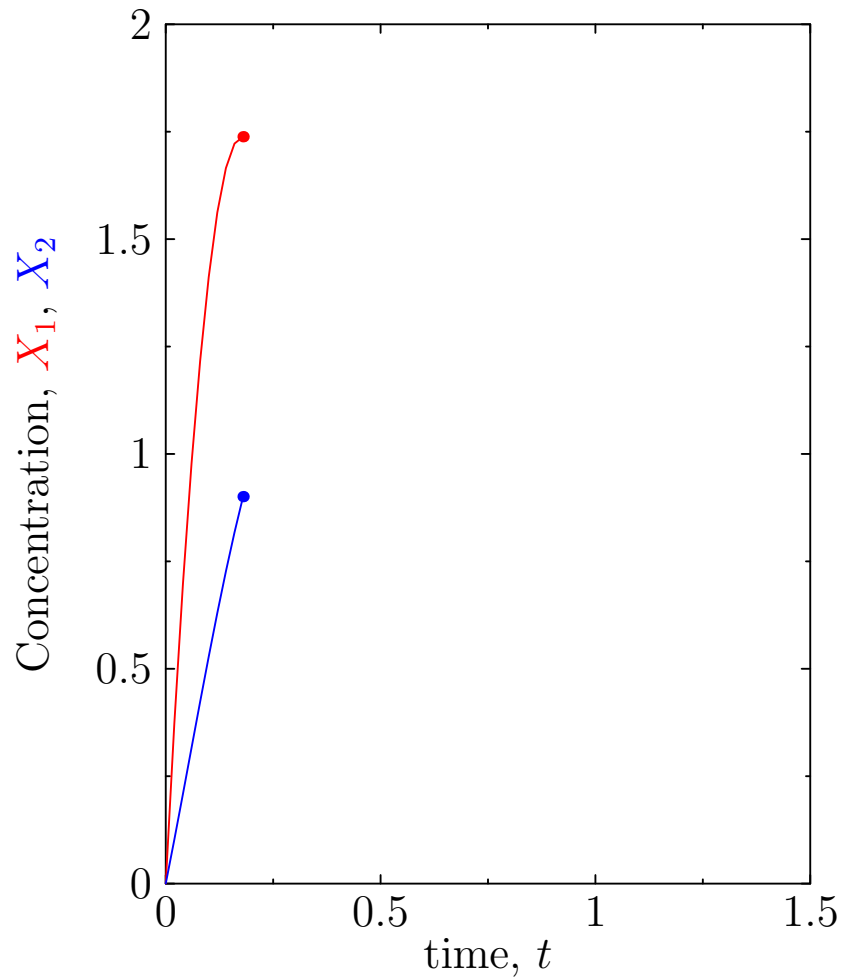
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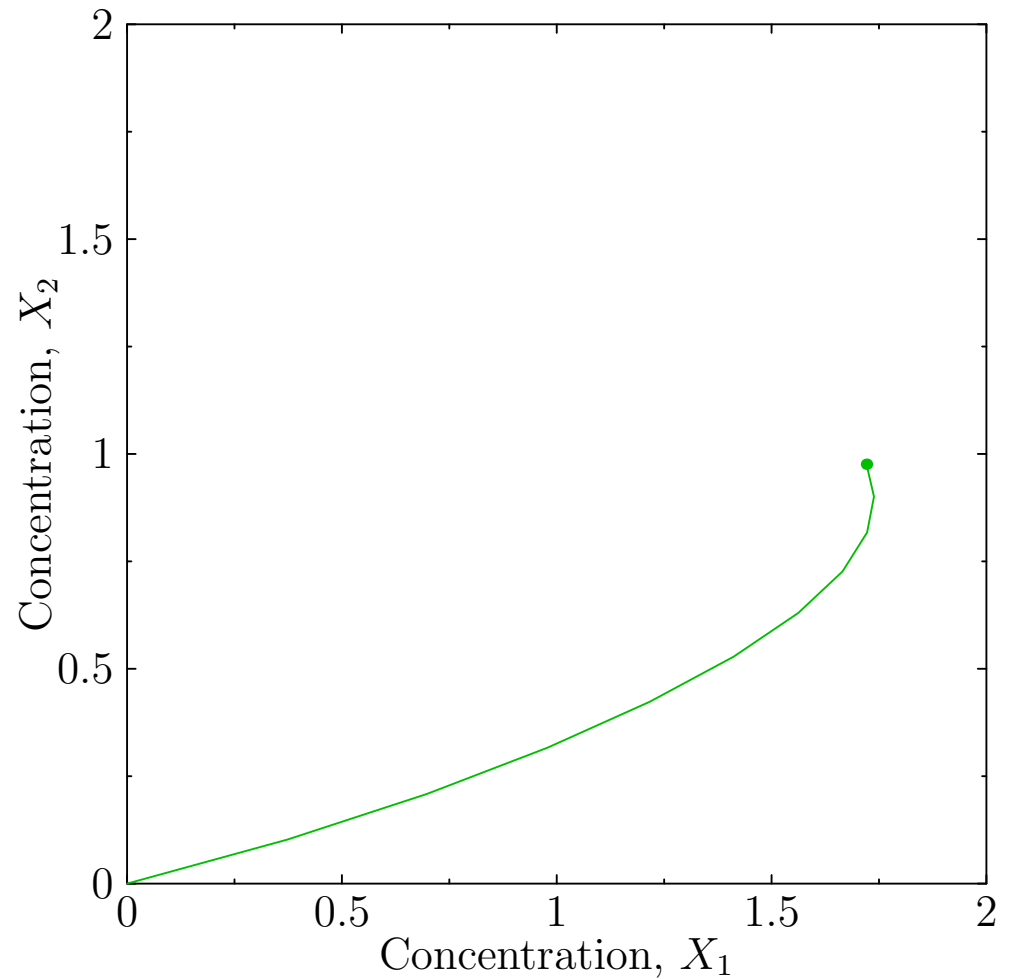
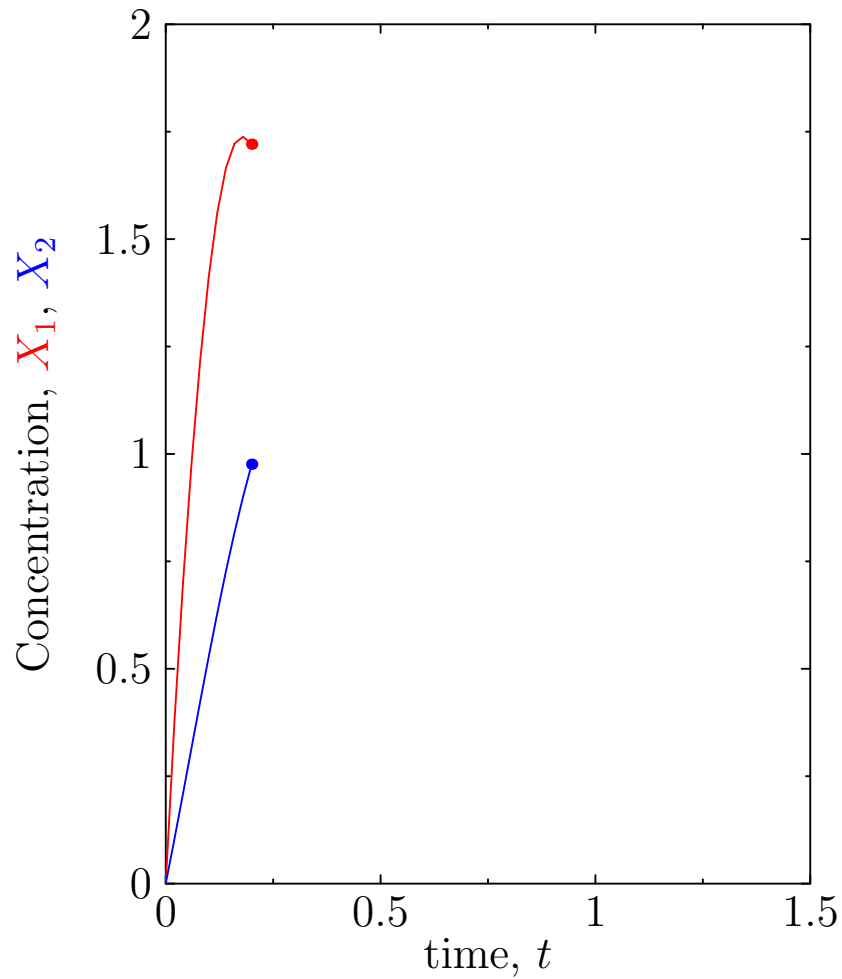
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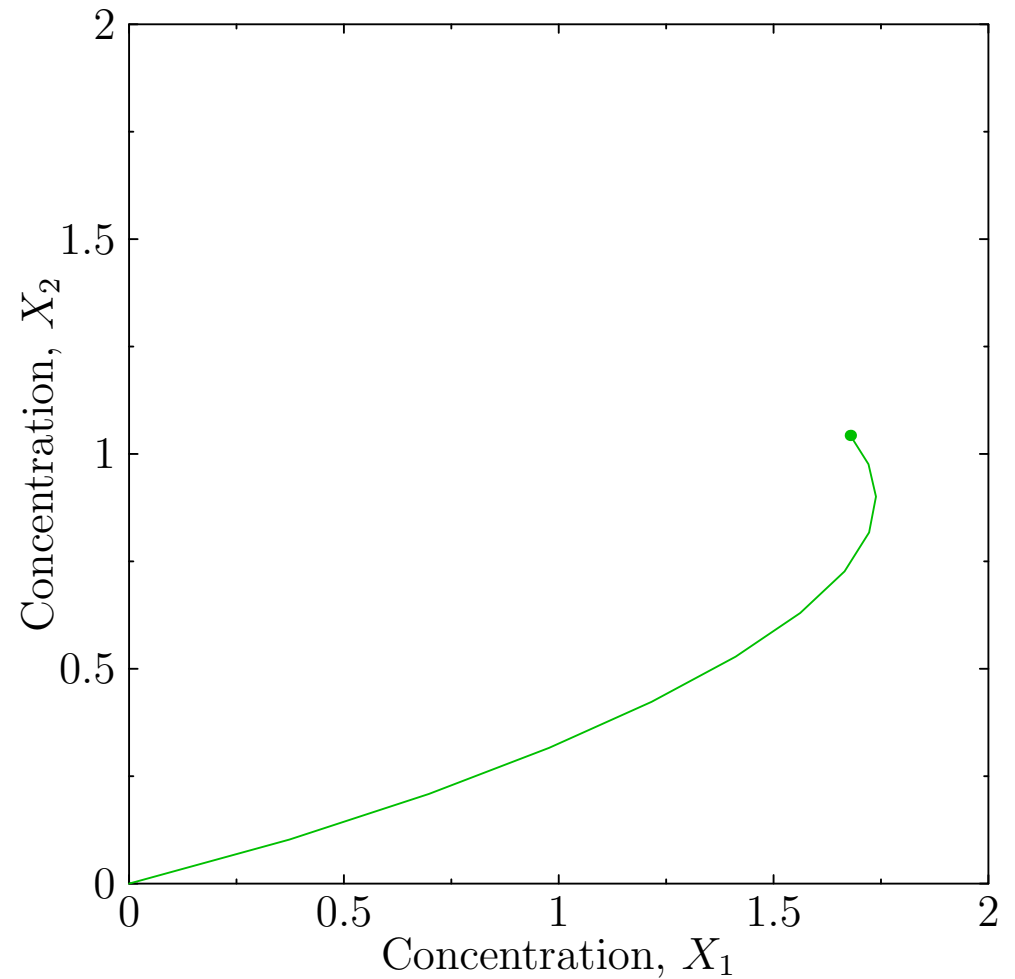
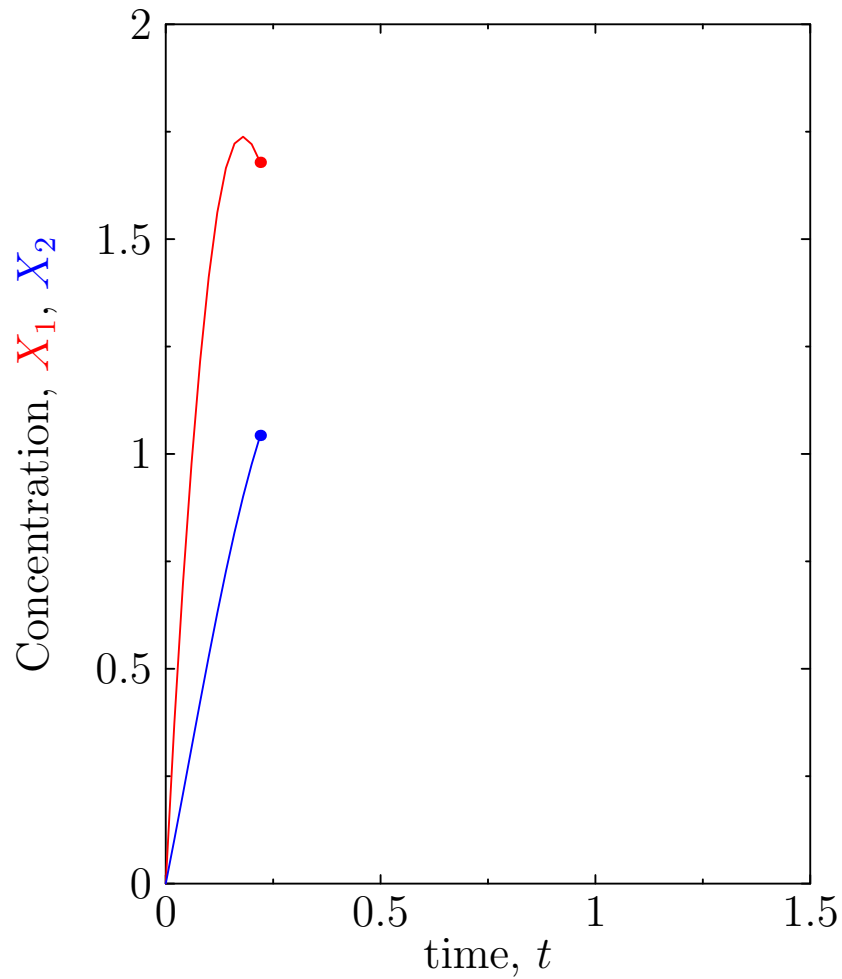
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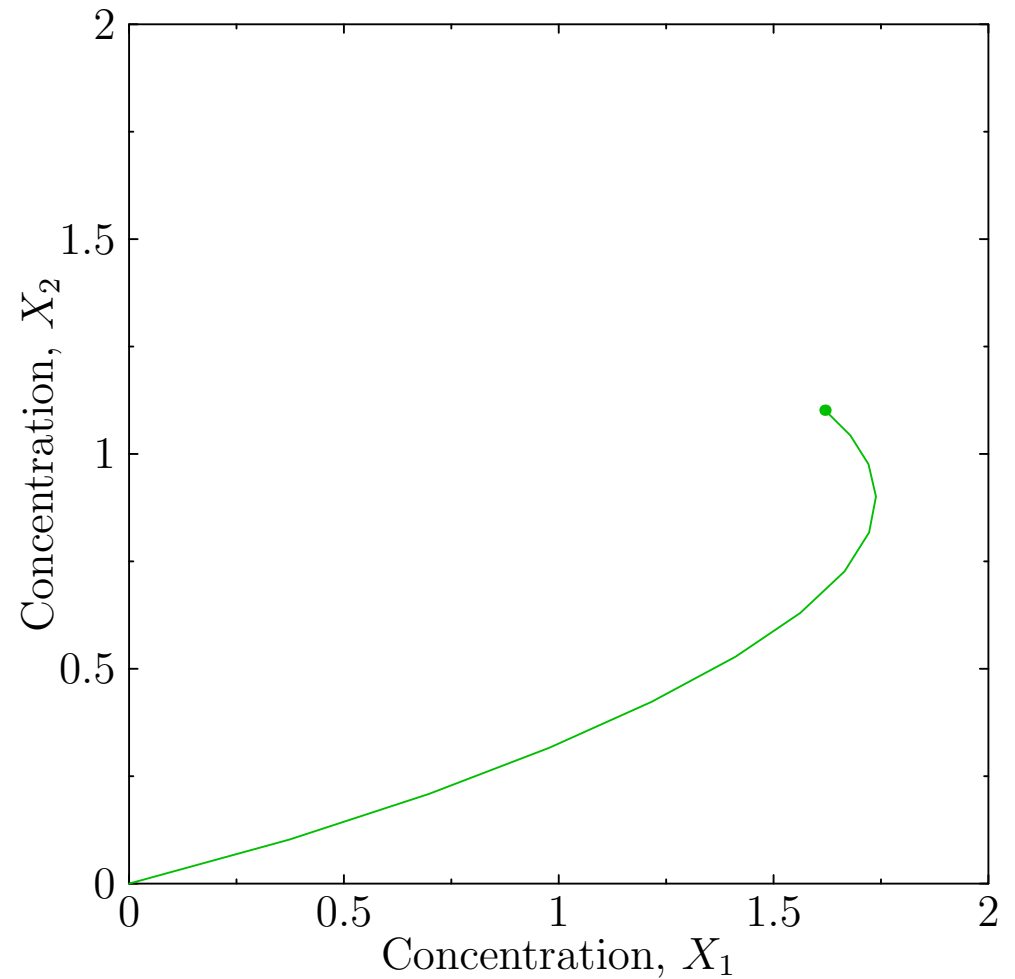
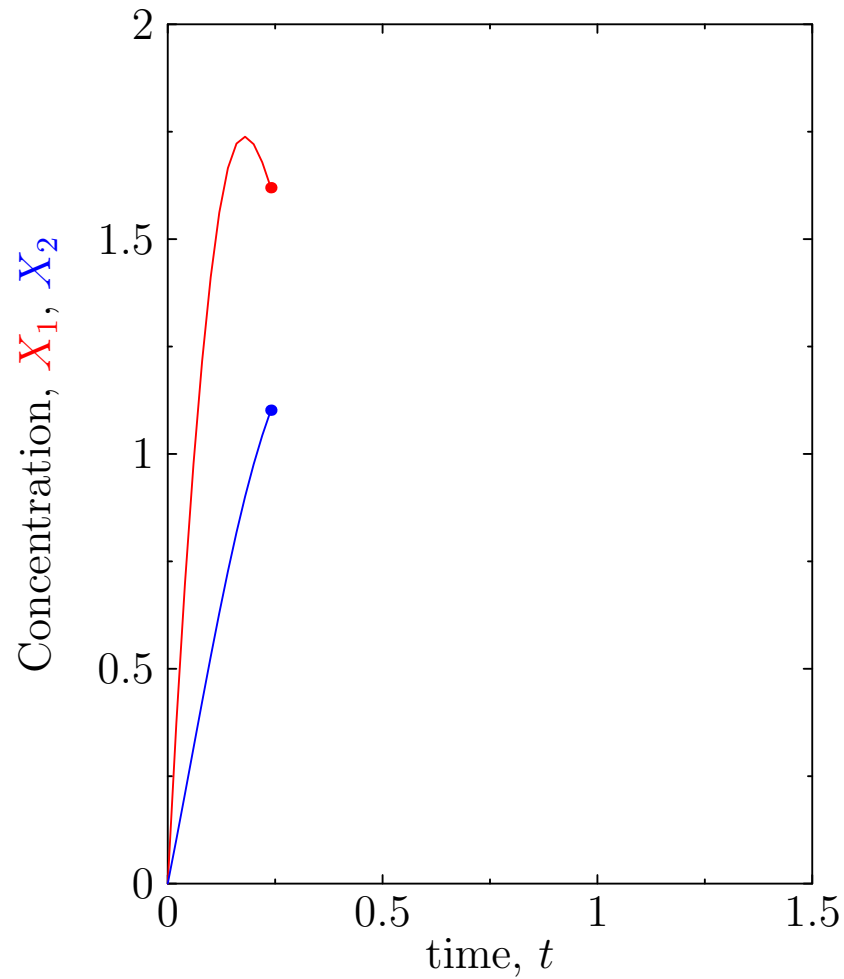
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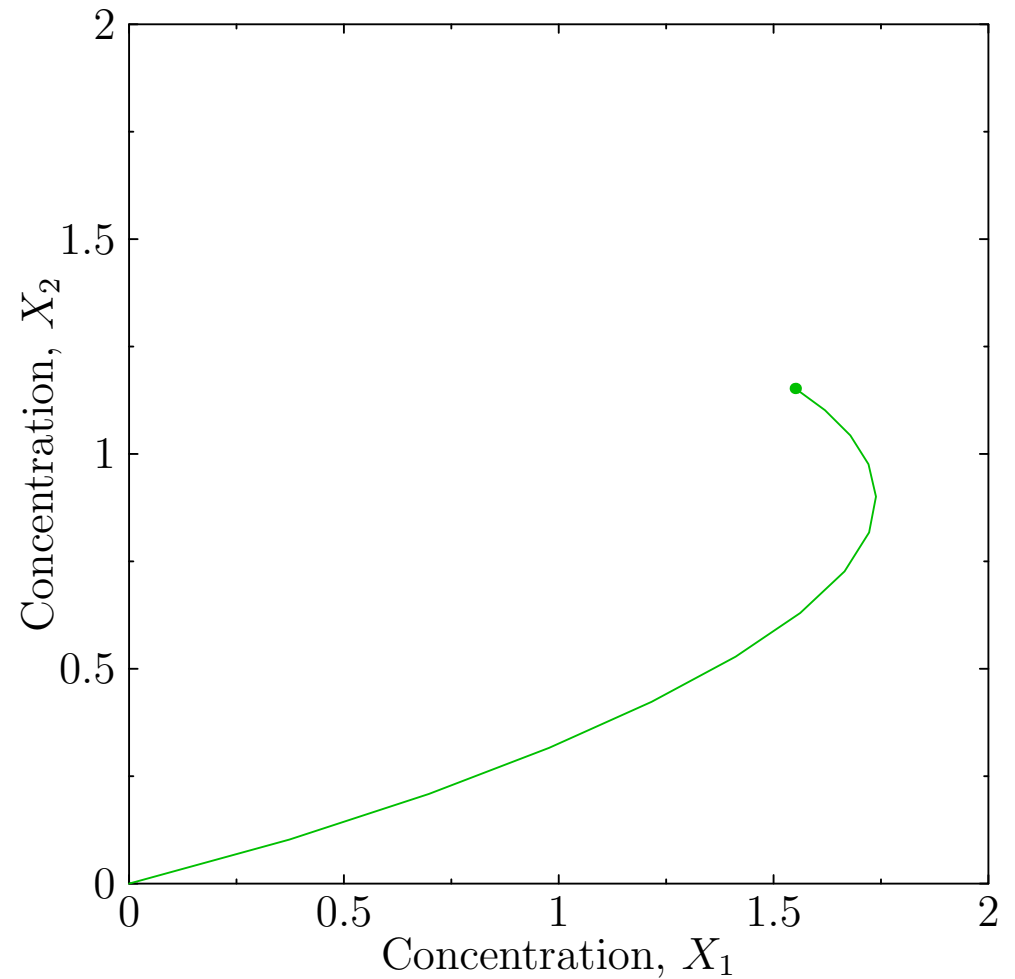
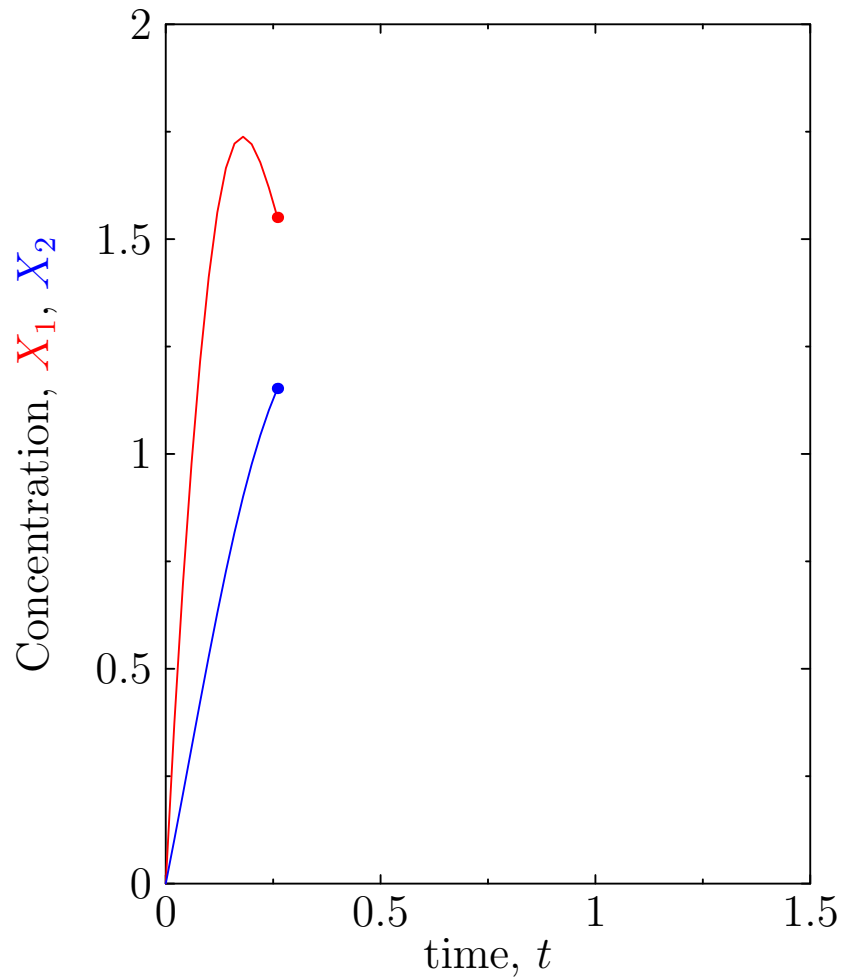
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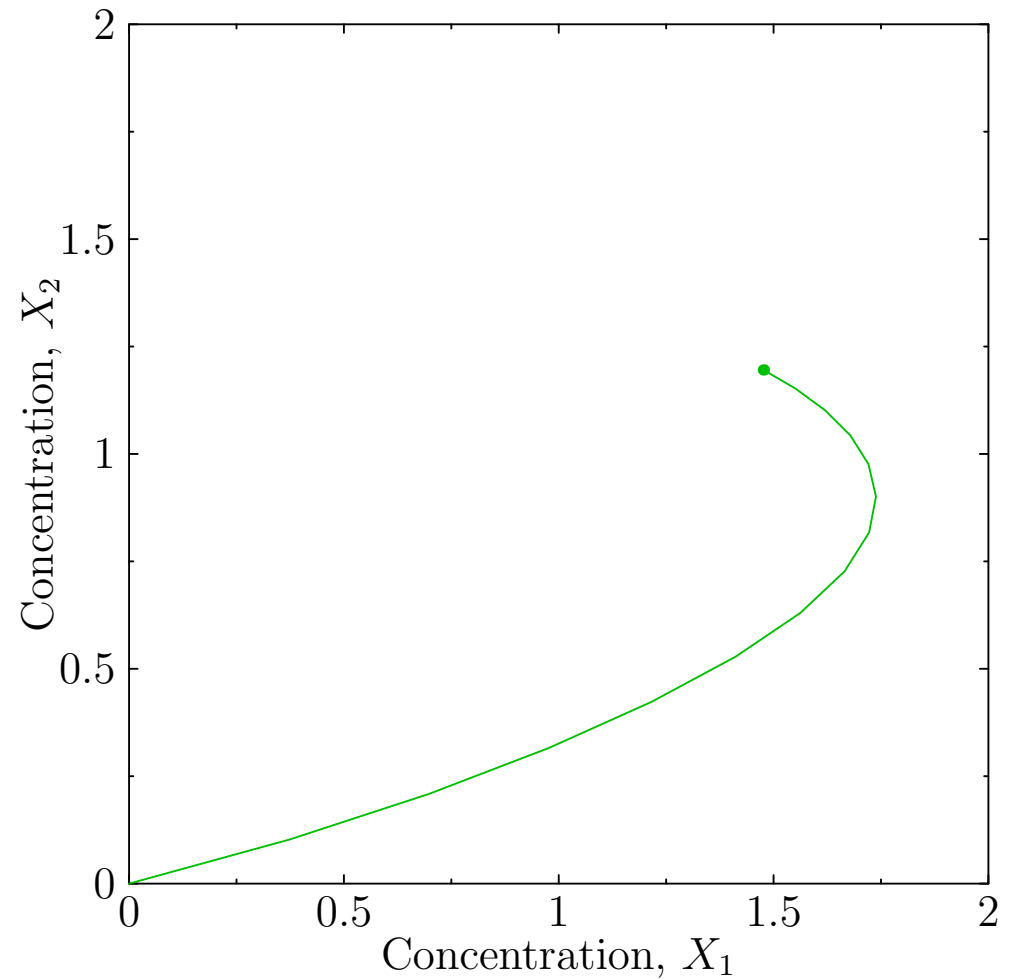
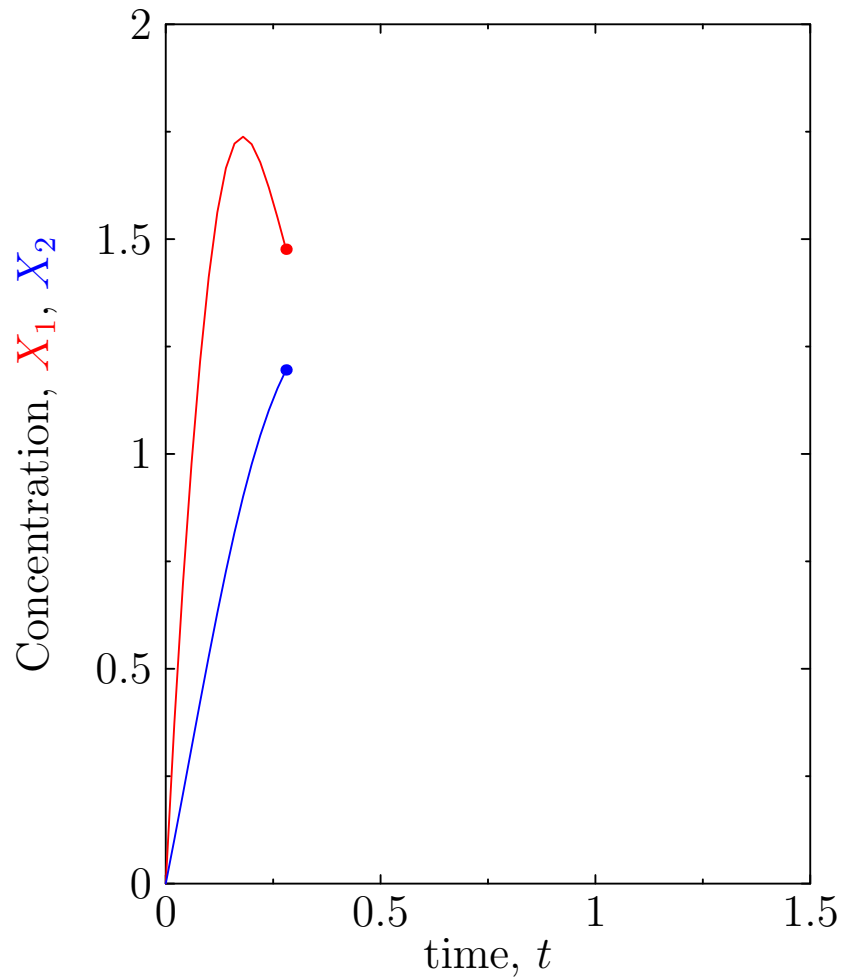
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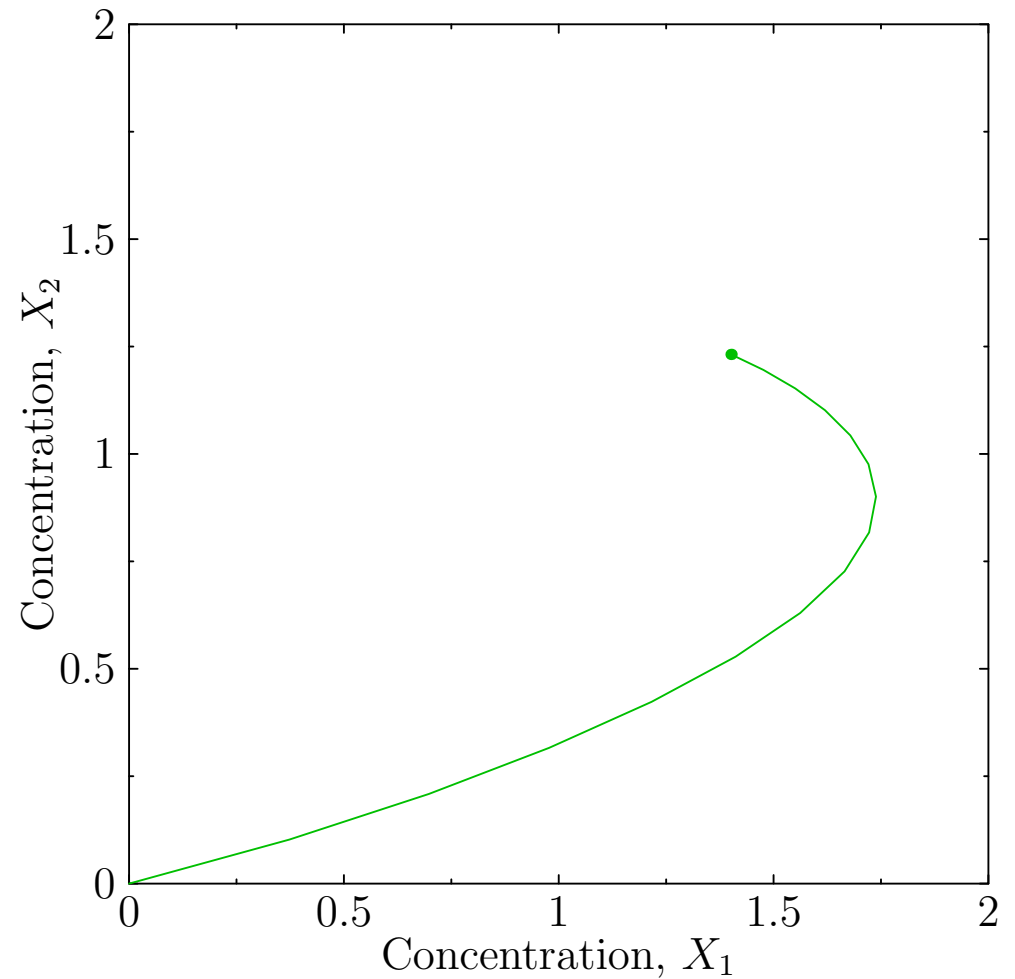
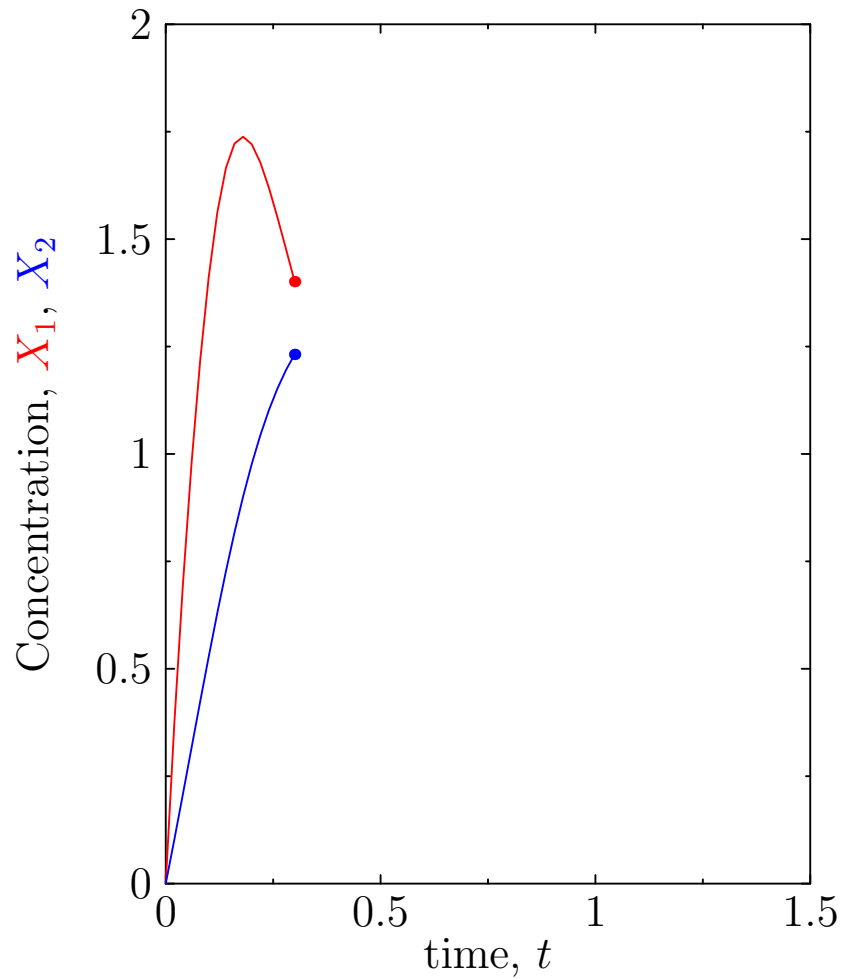
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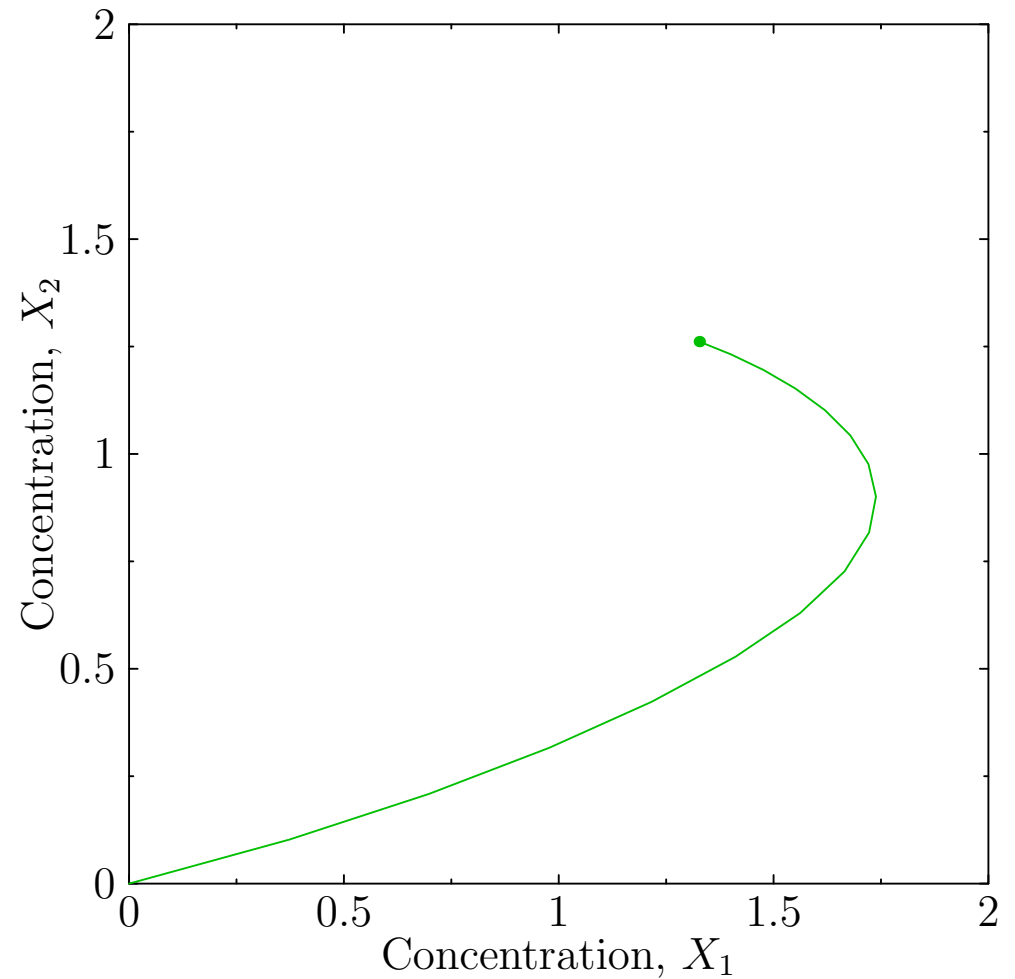
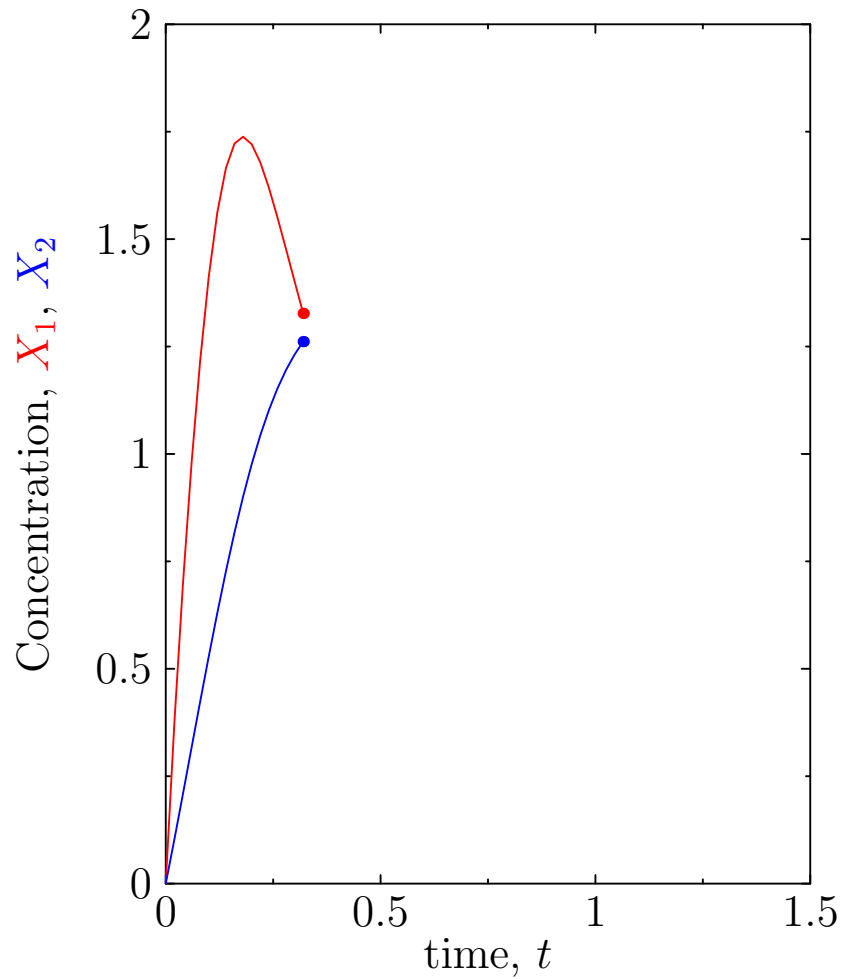
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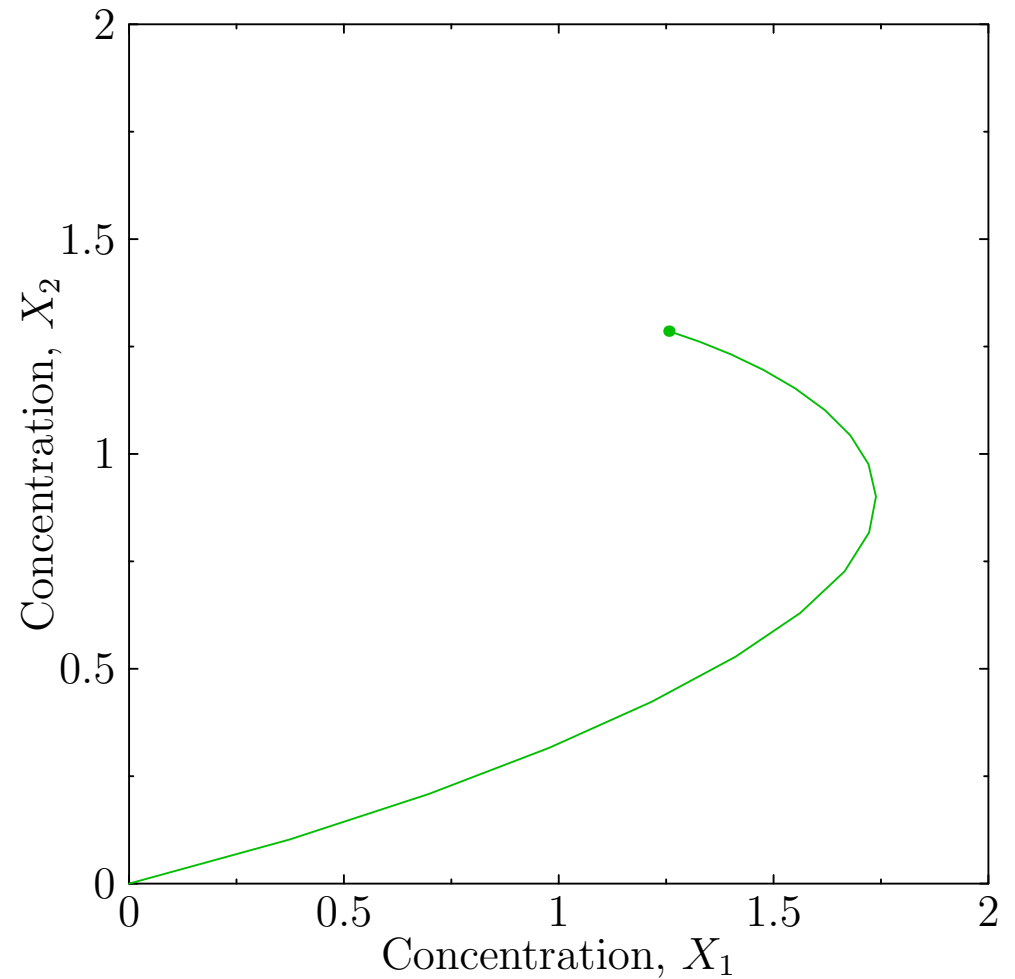
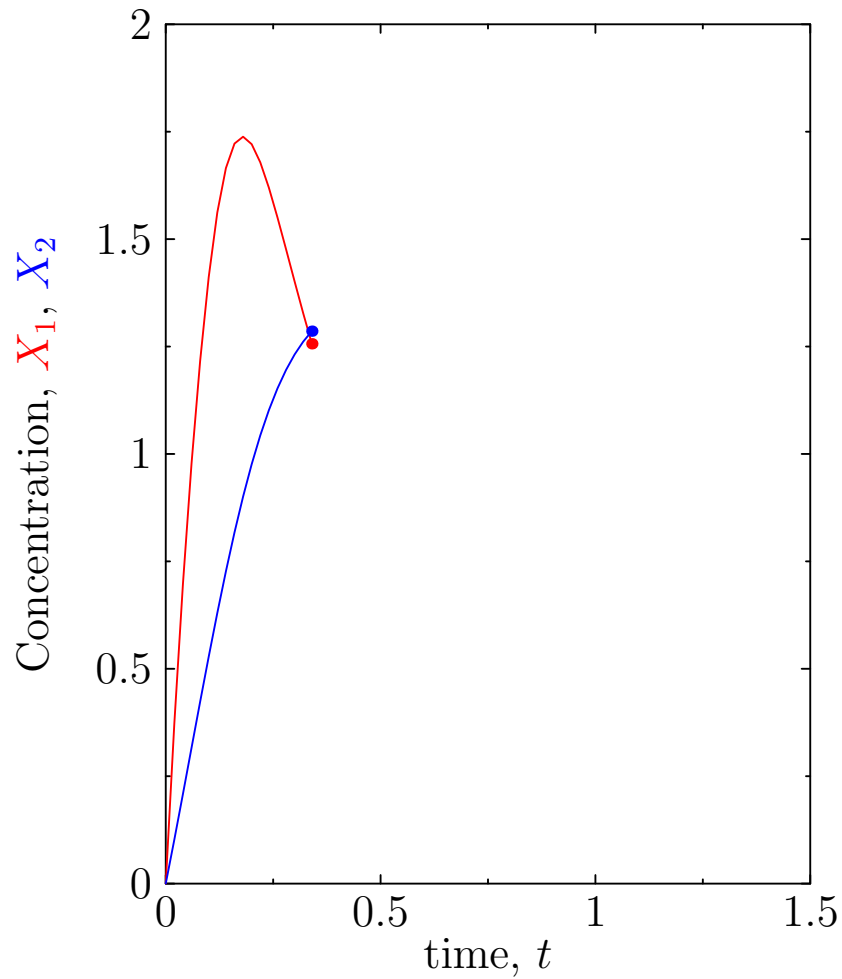
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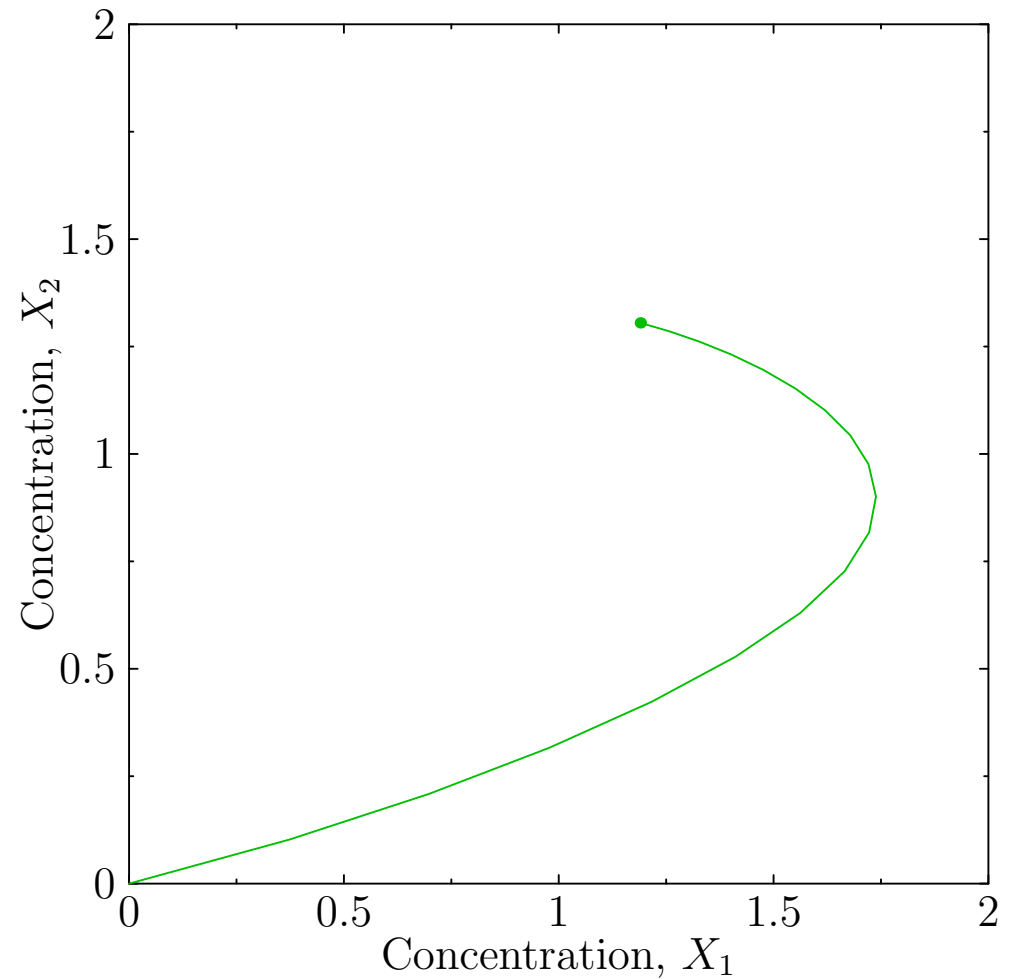
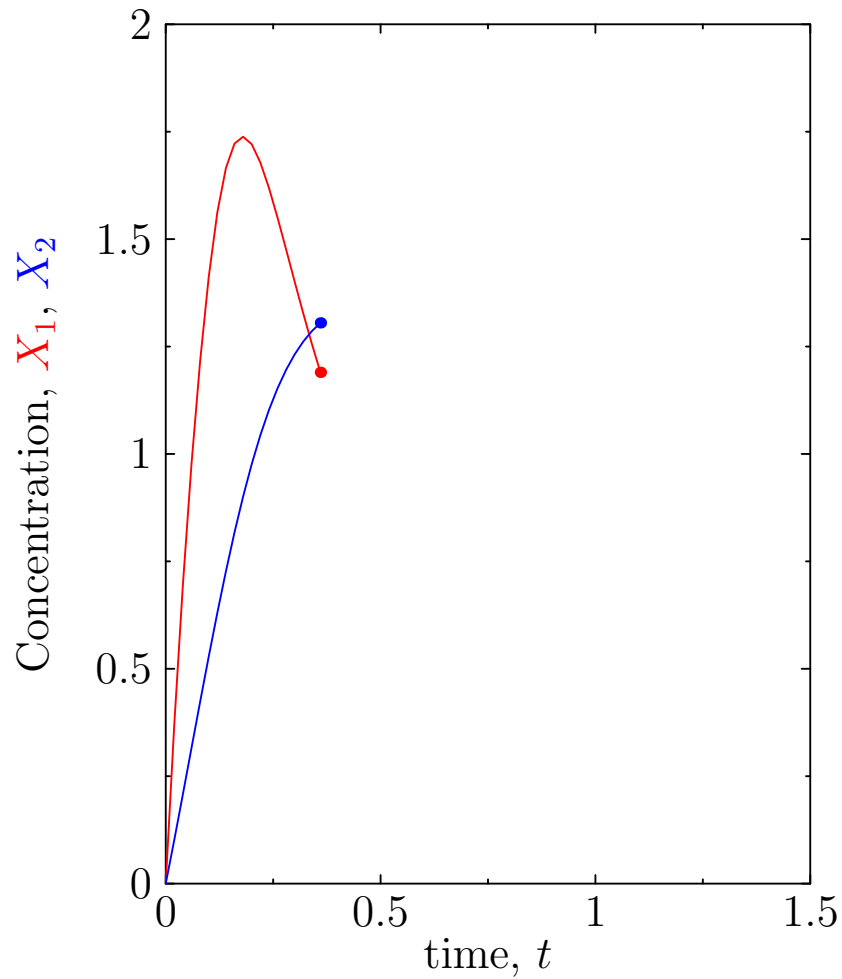
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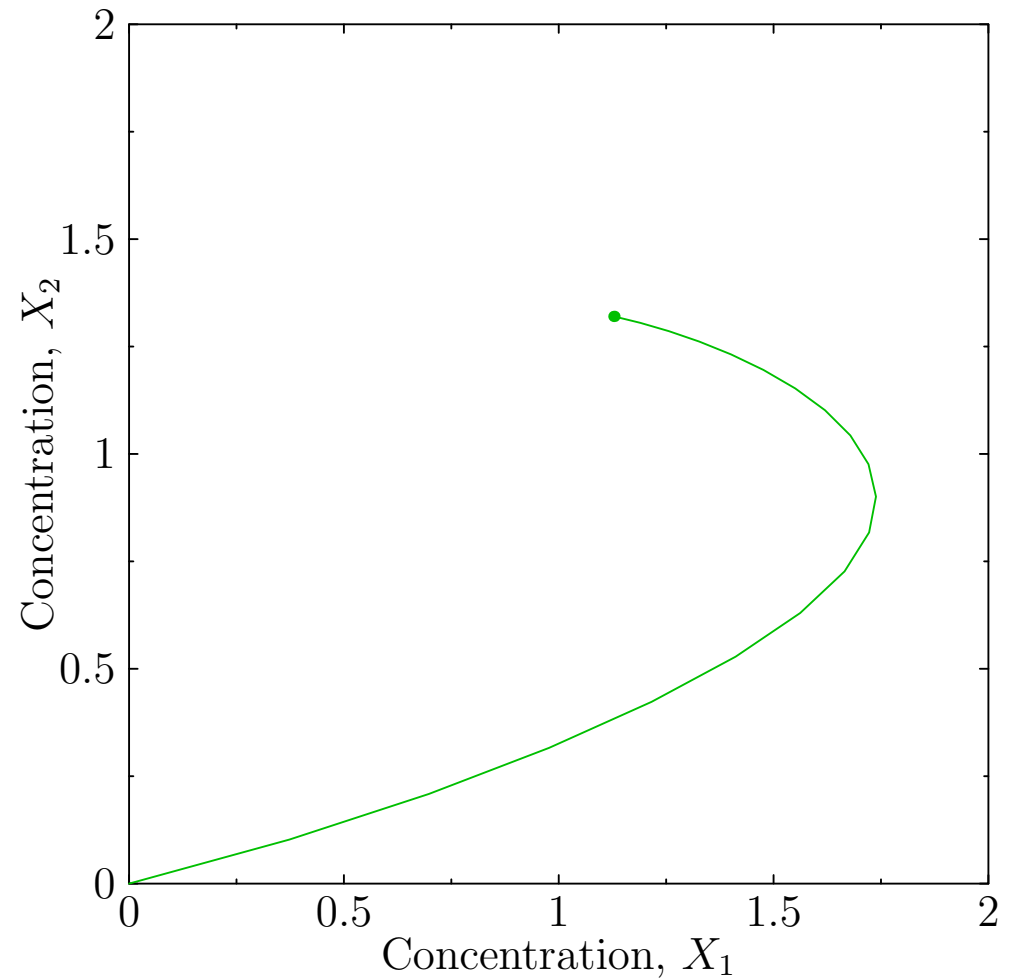
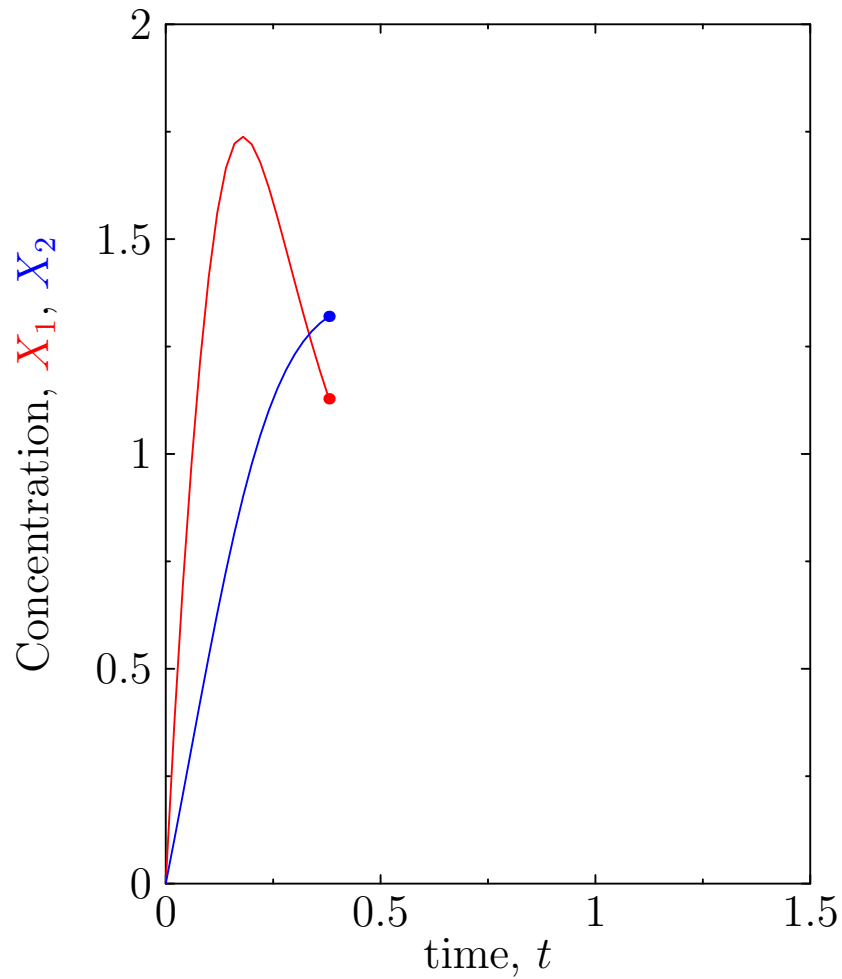
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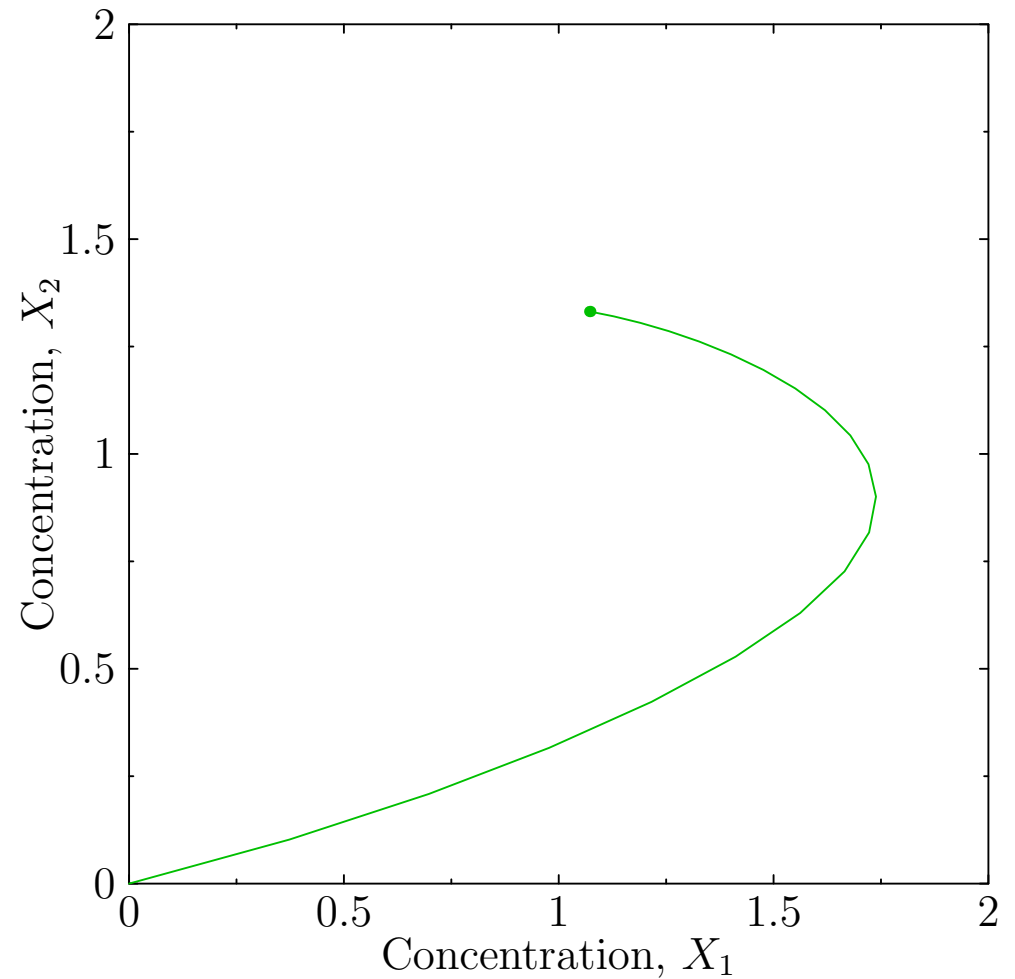
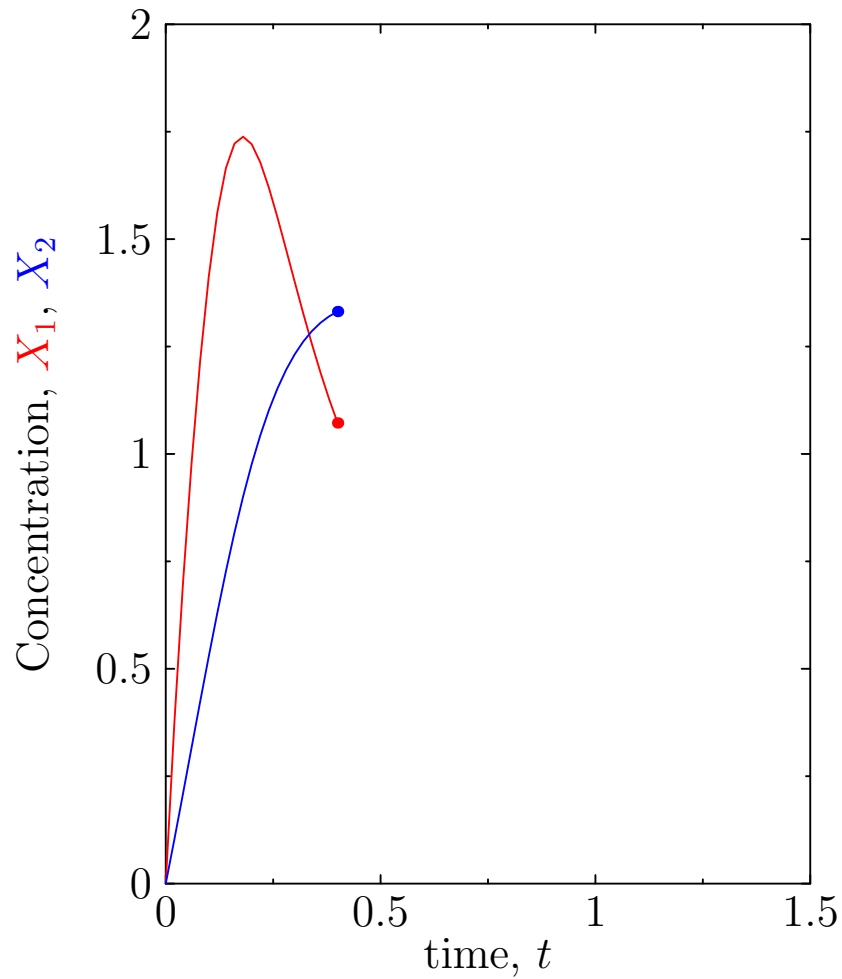
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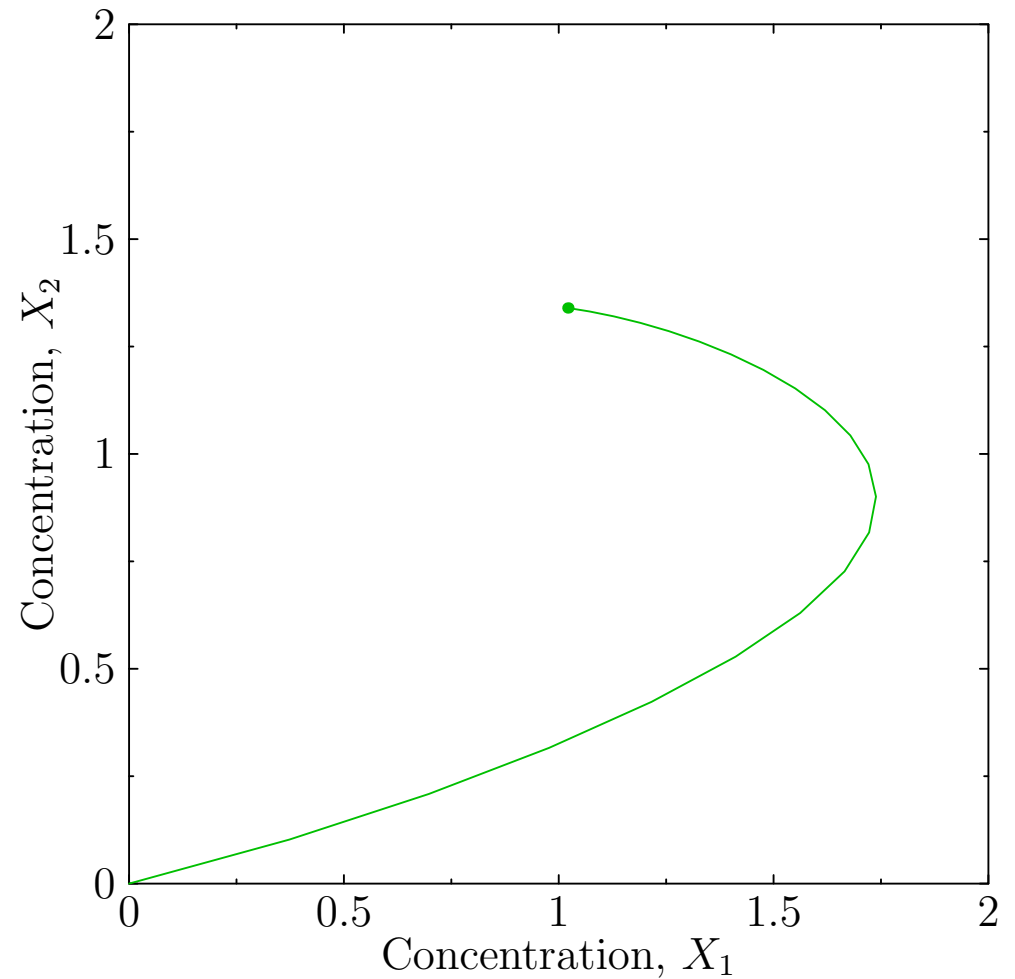
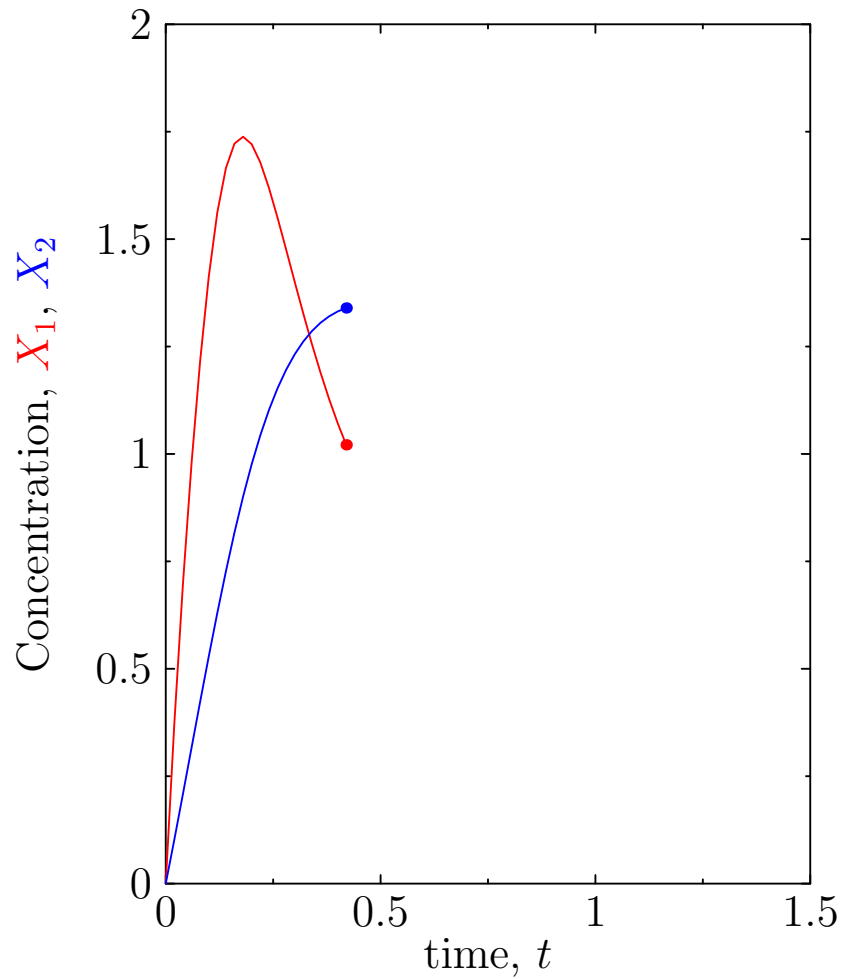
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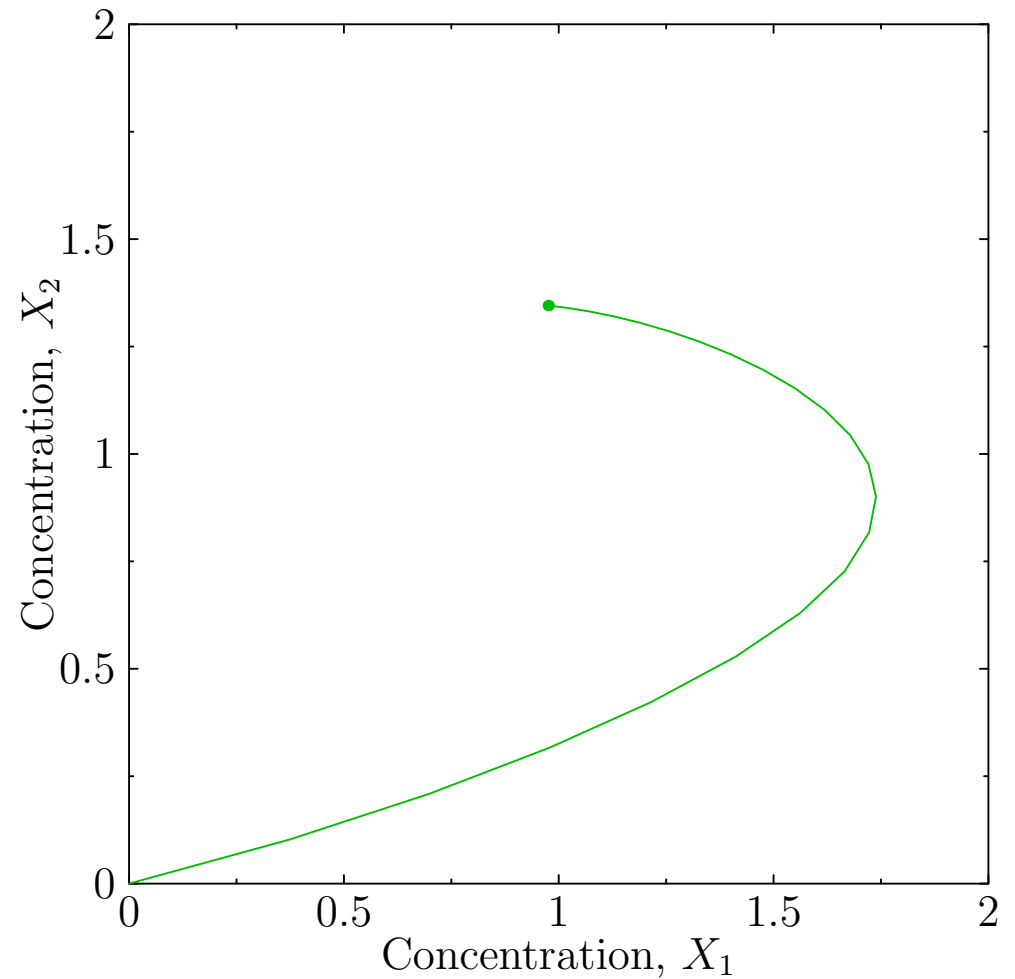
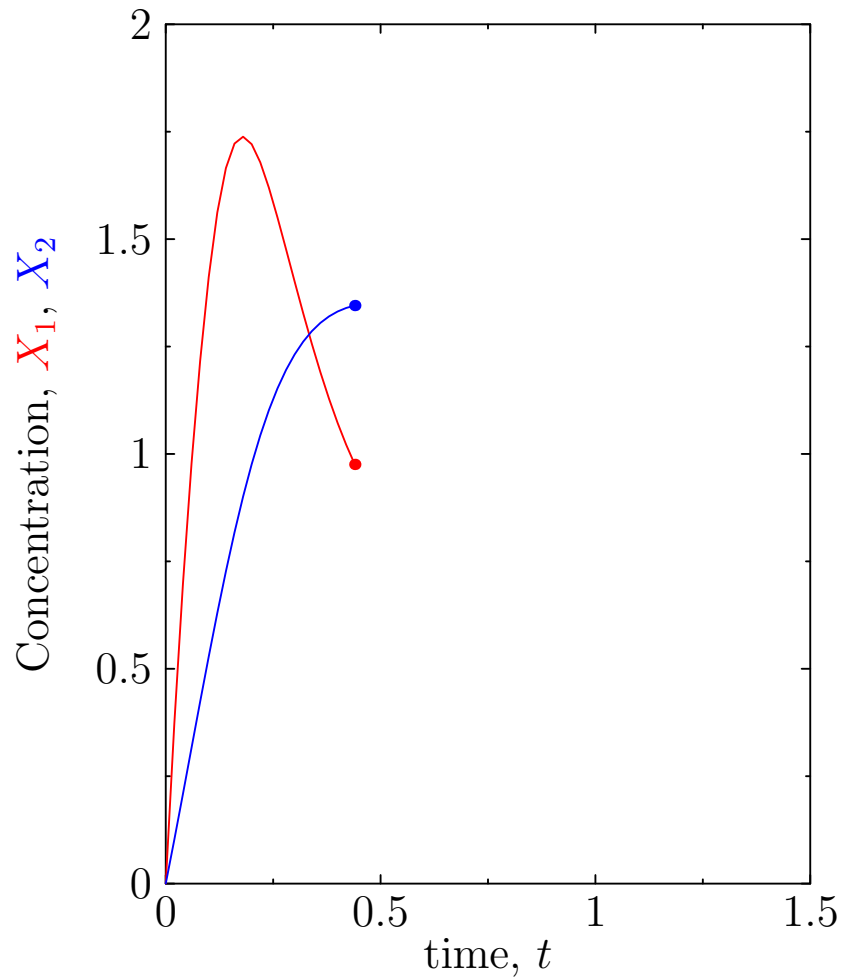
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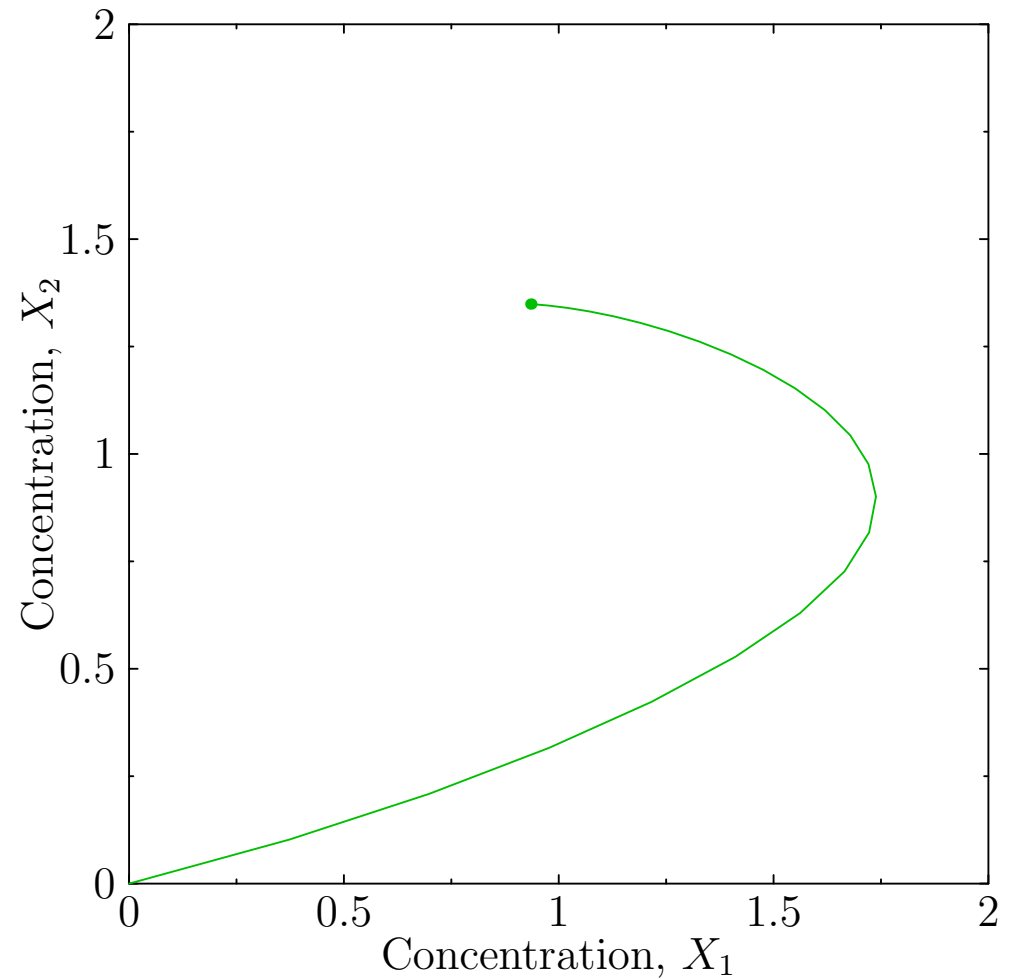
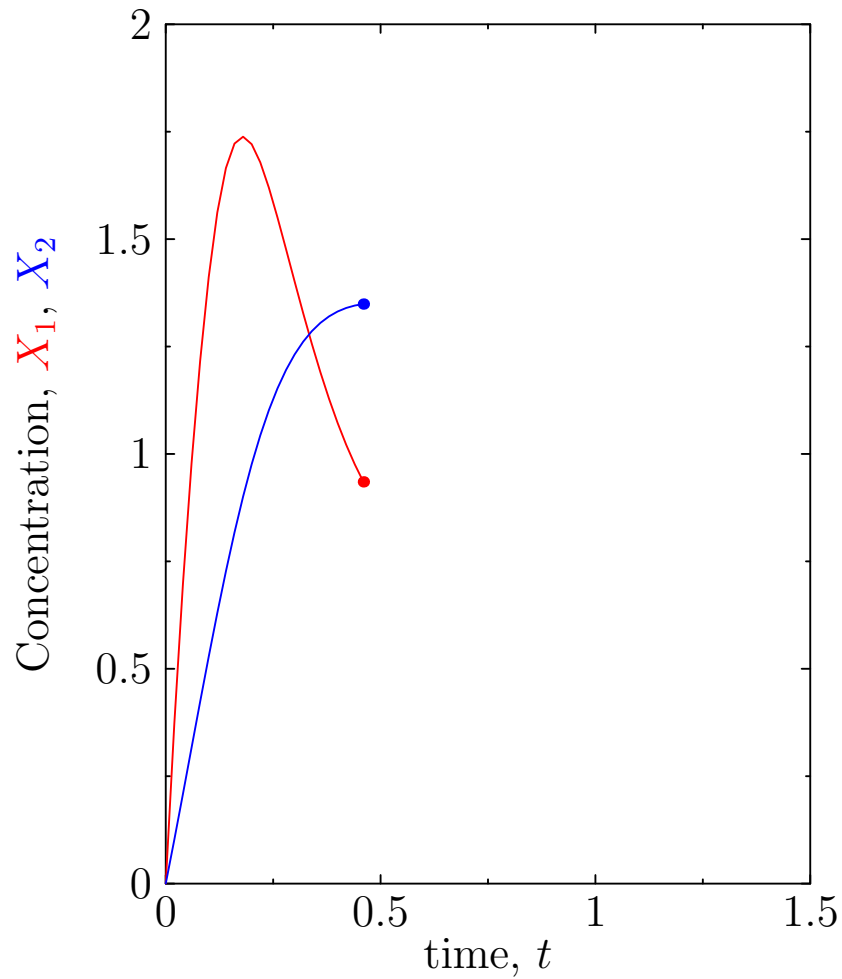
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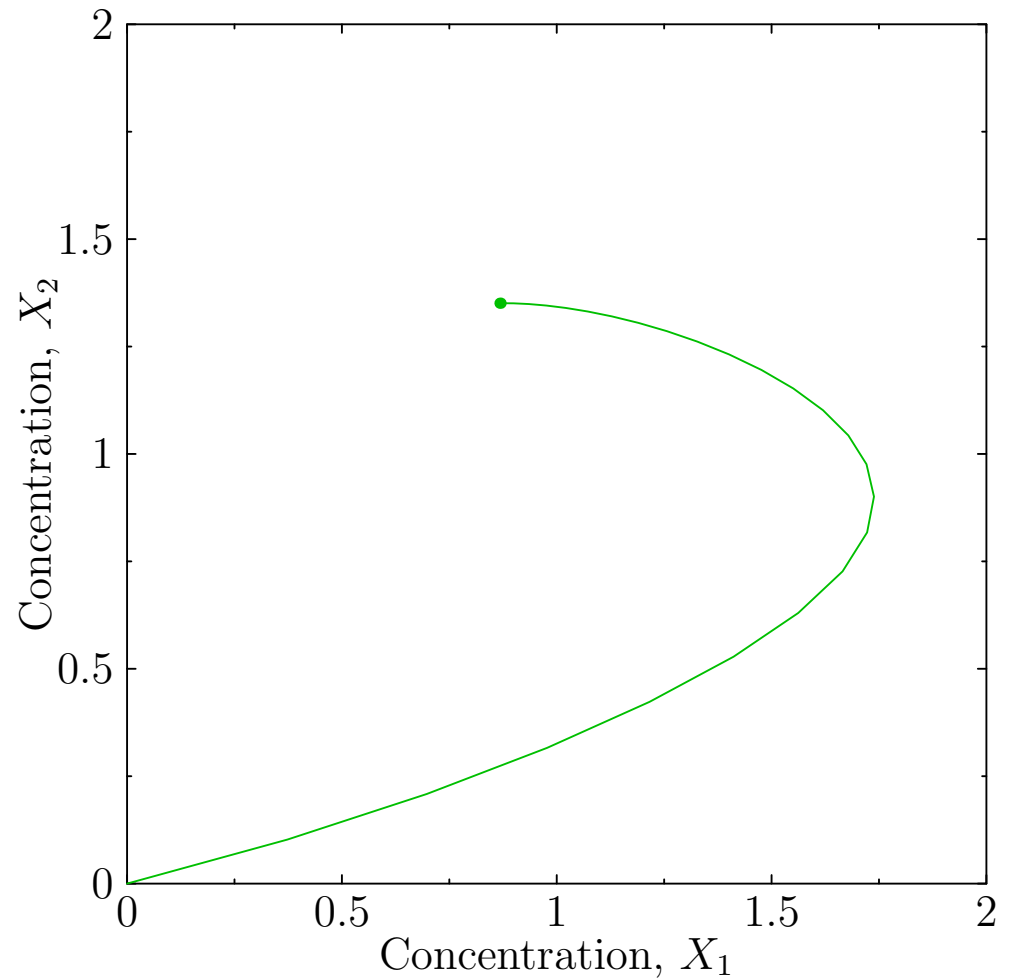
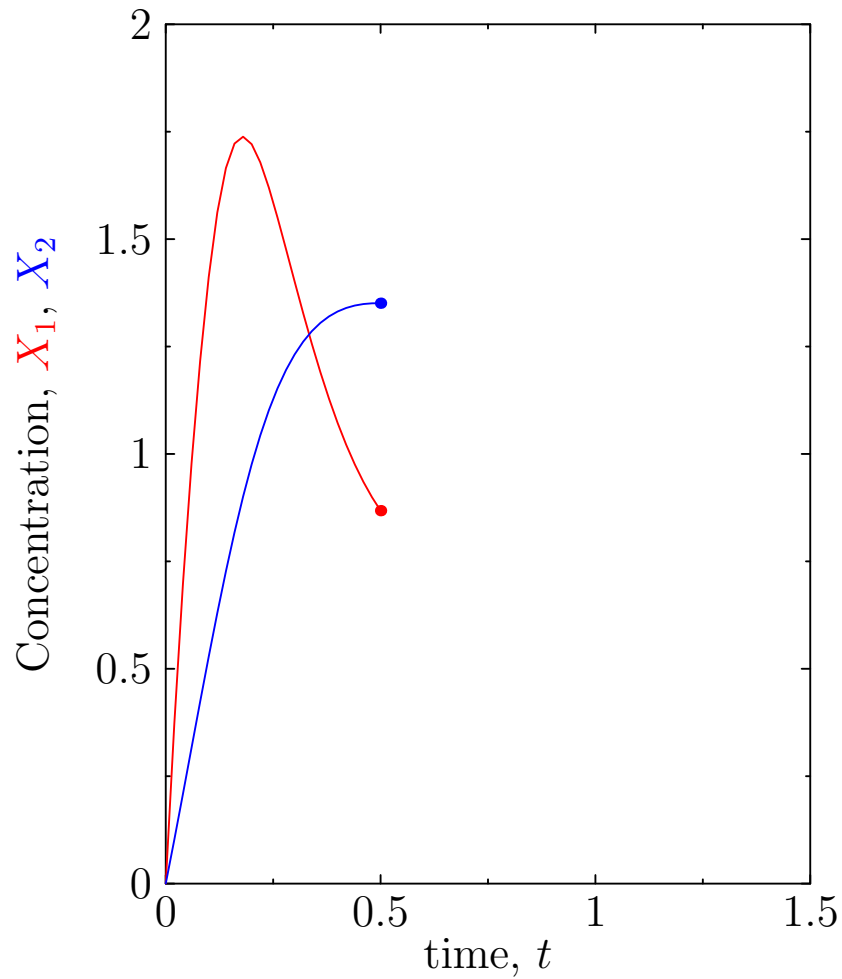
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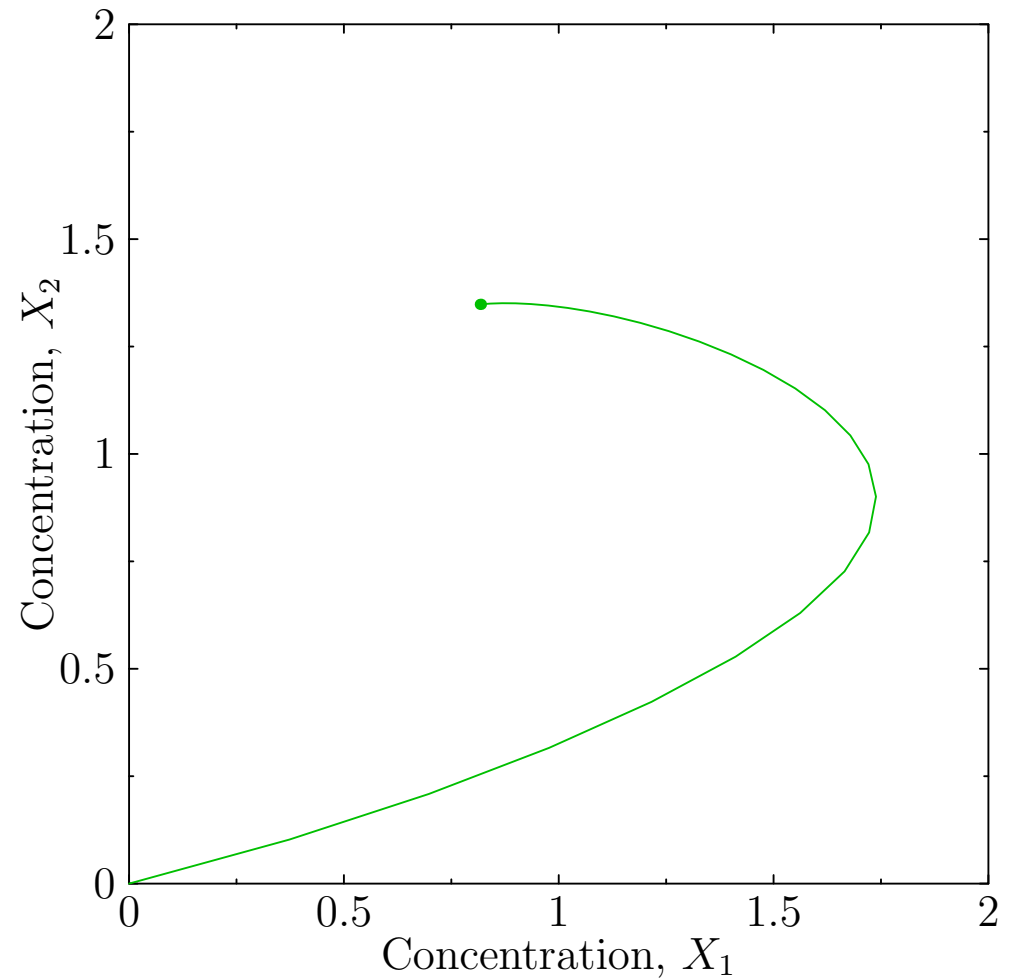
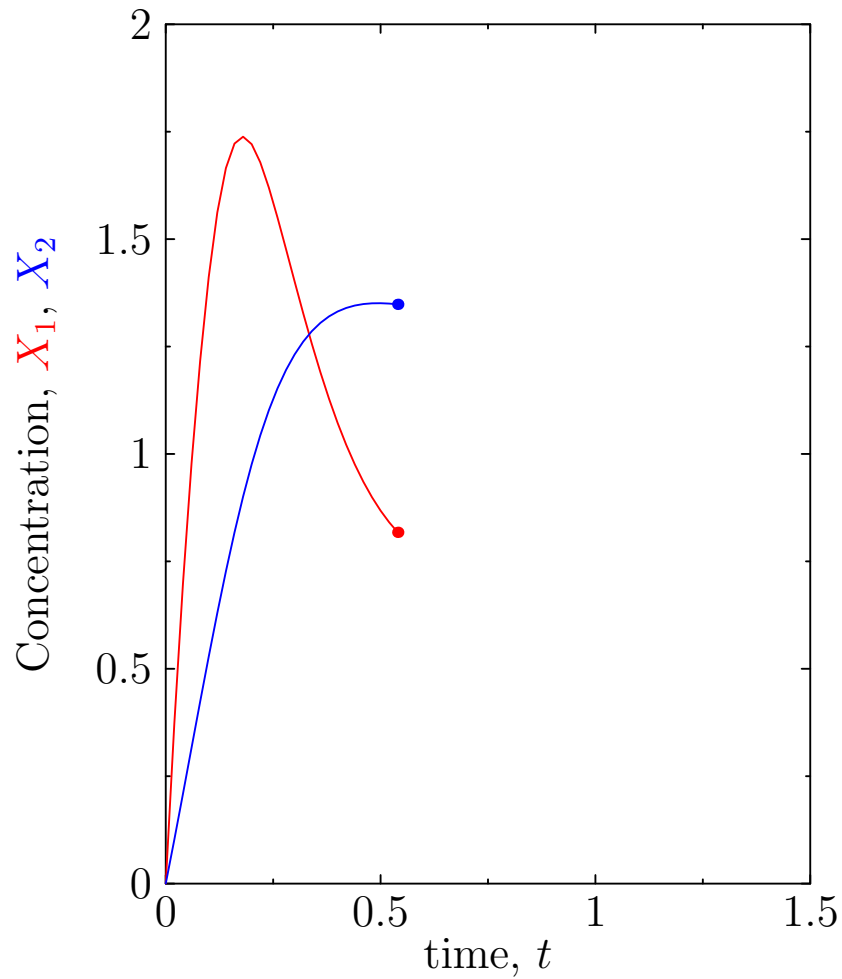
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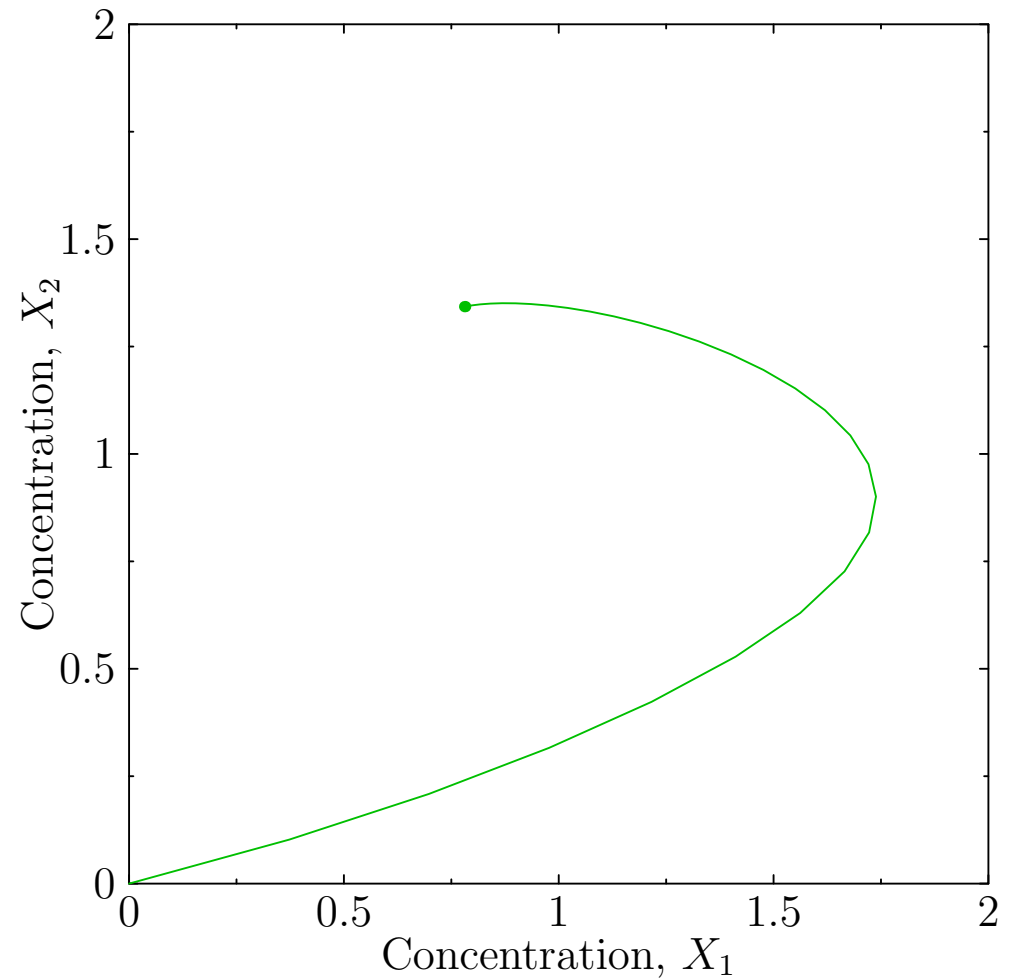
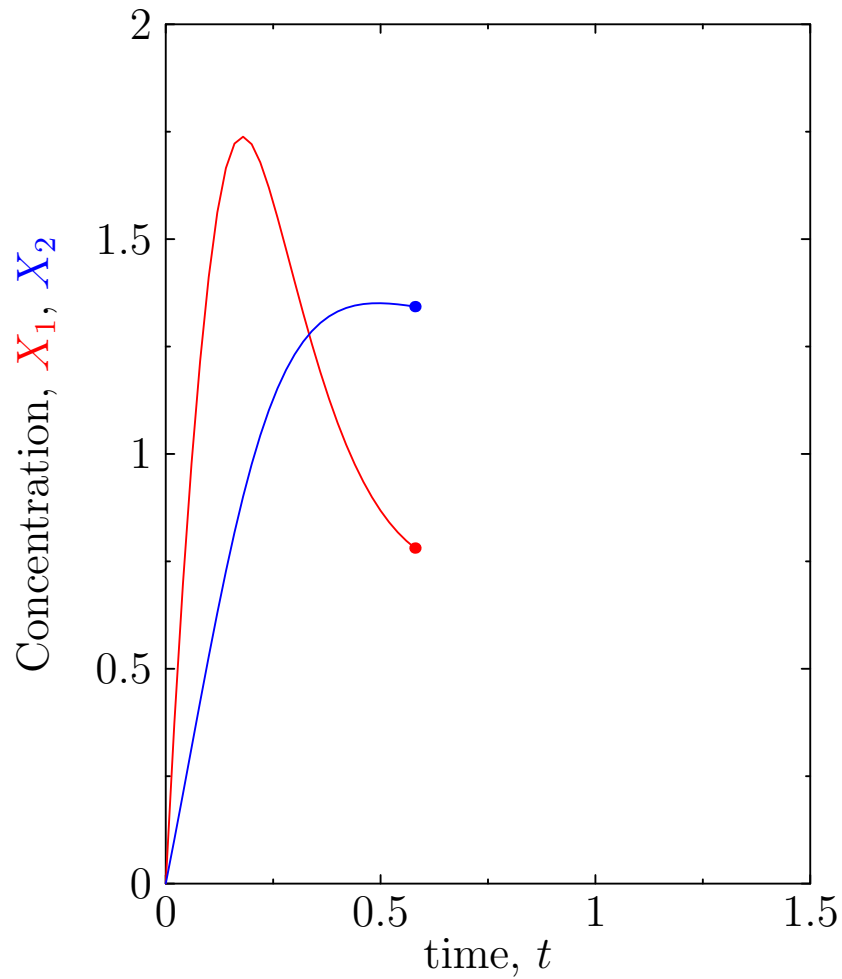
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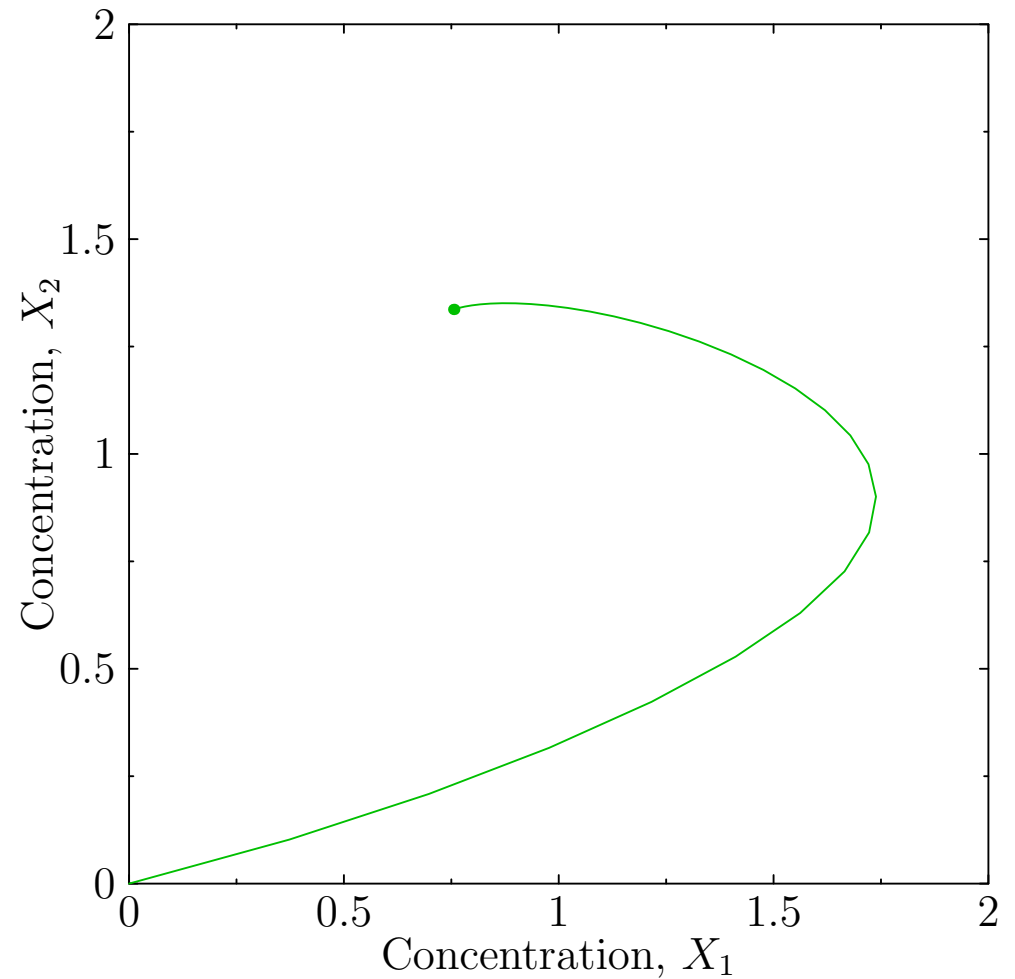
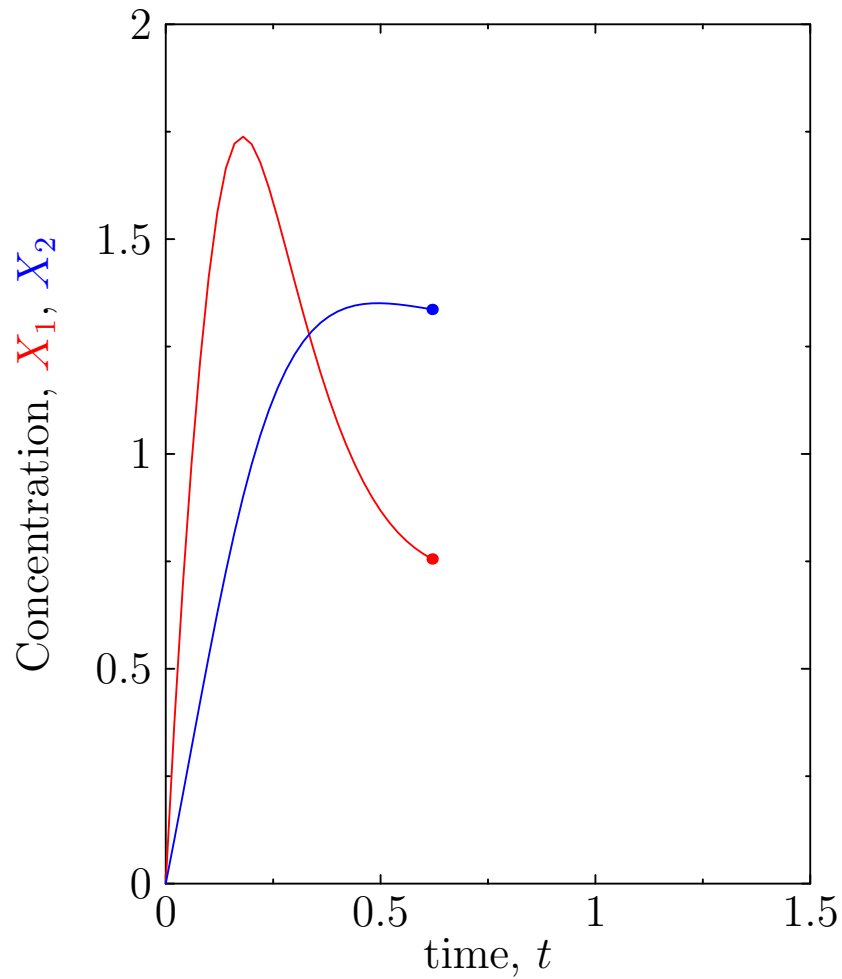
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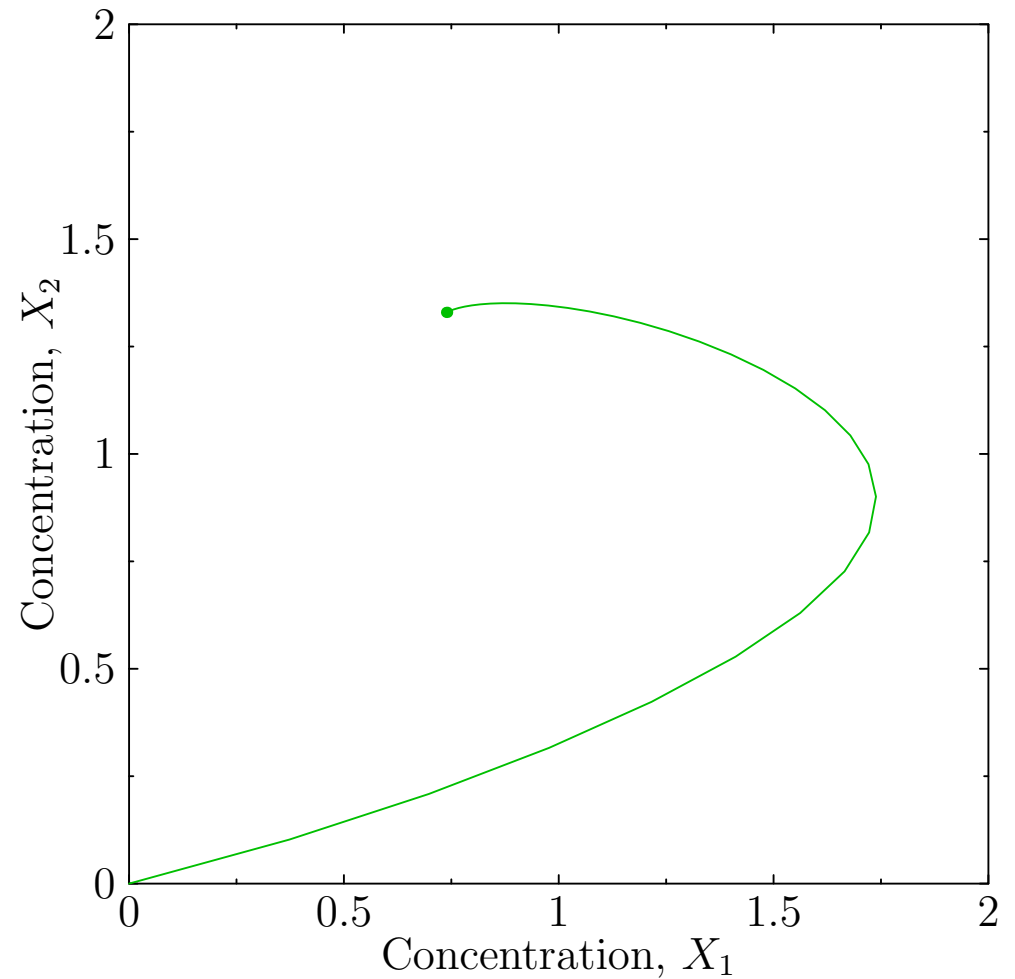
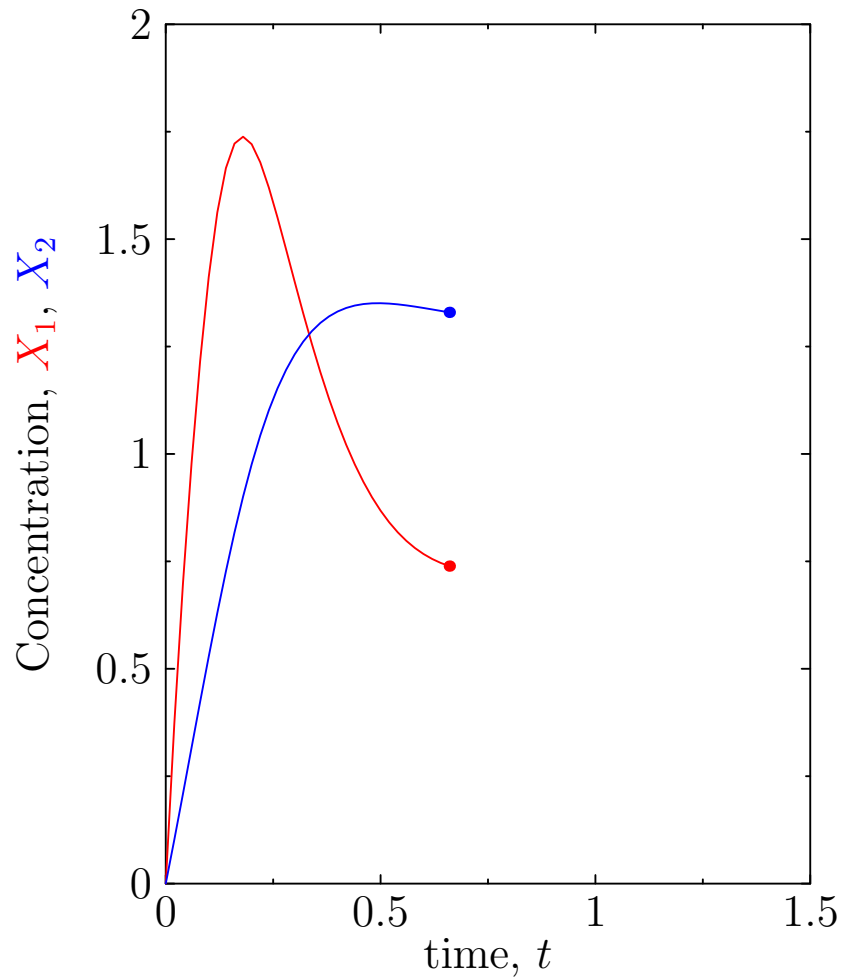
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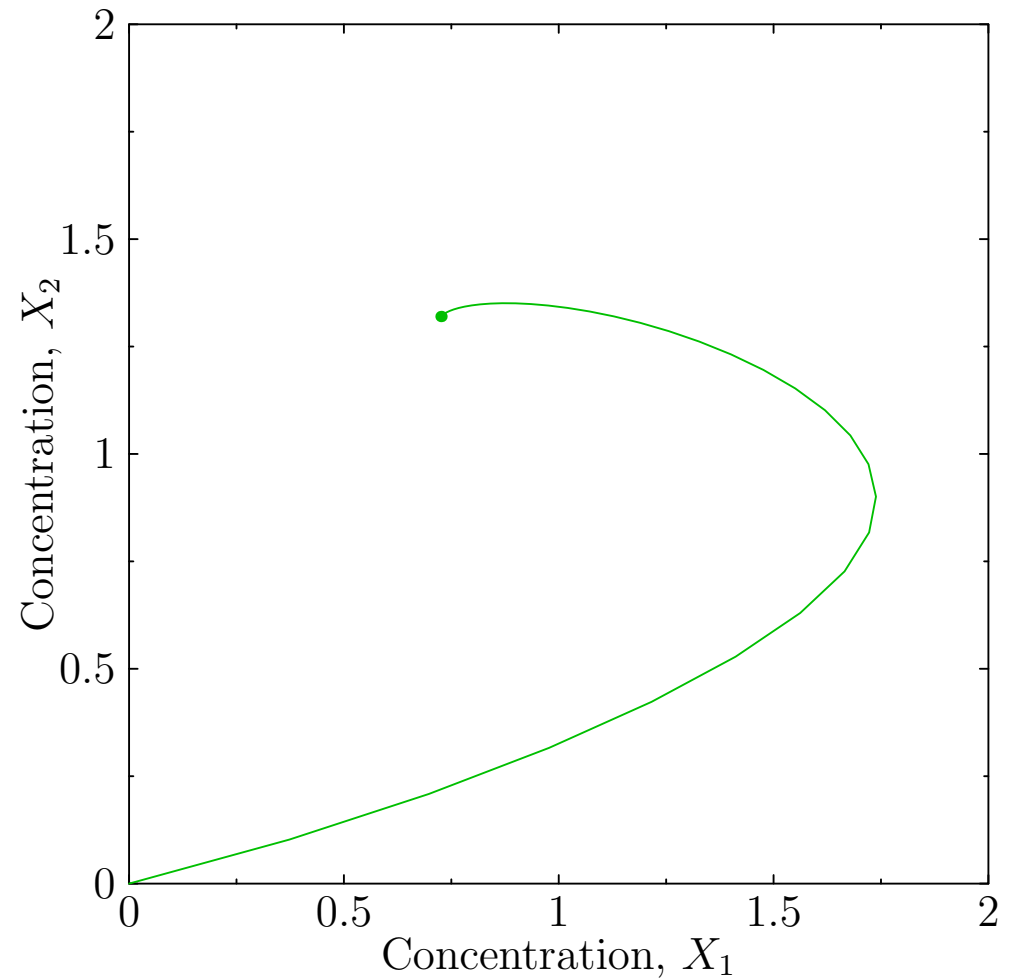
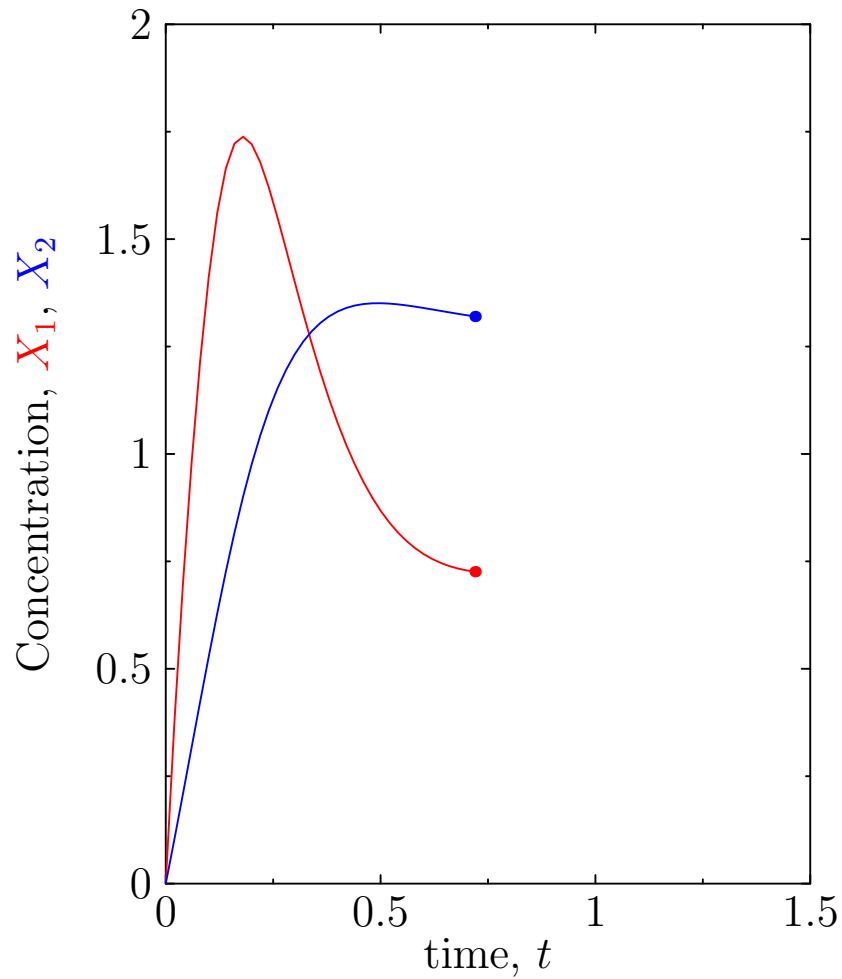
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Solving the Dynamics



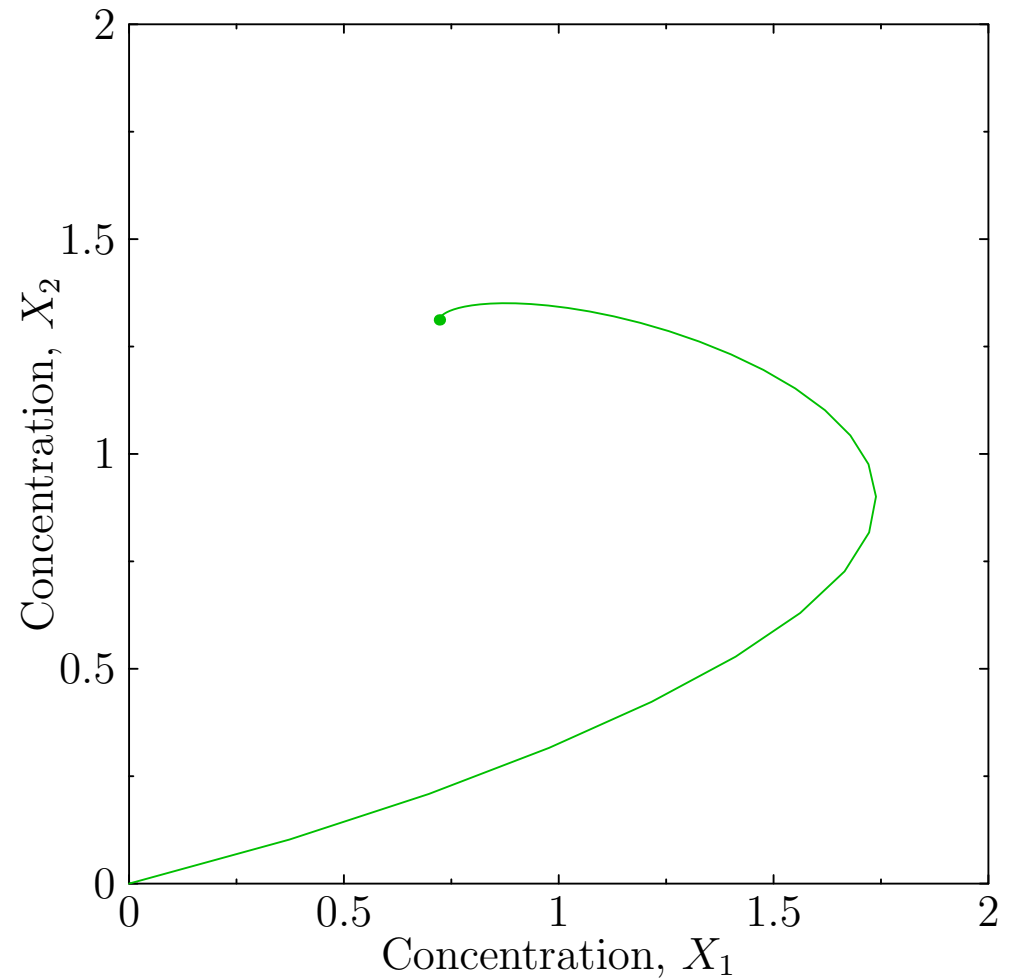
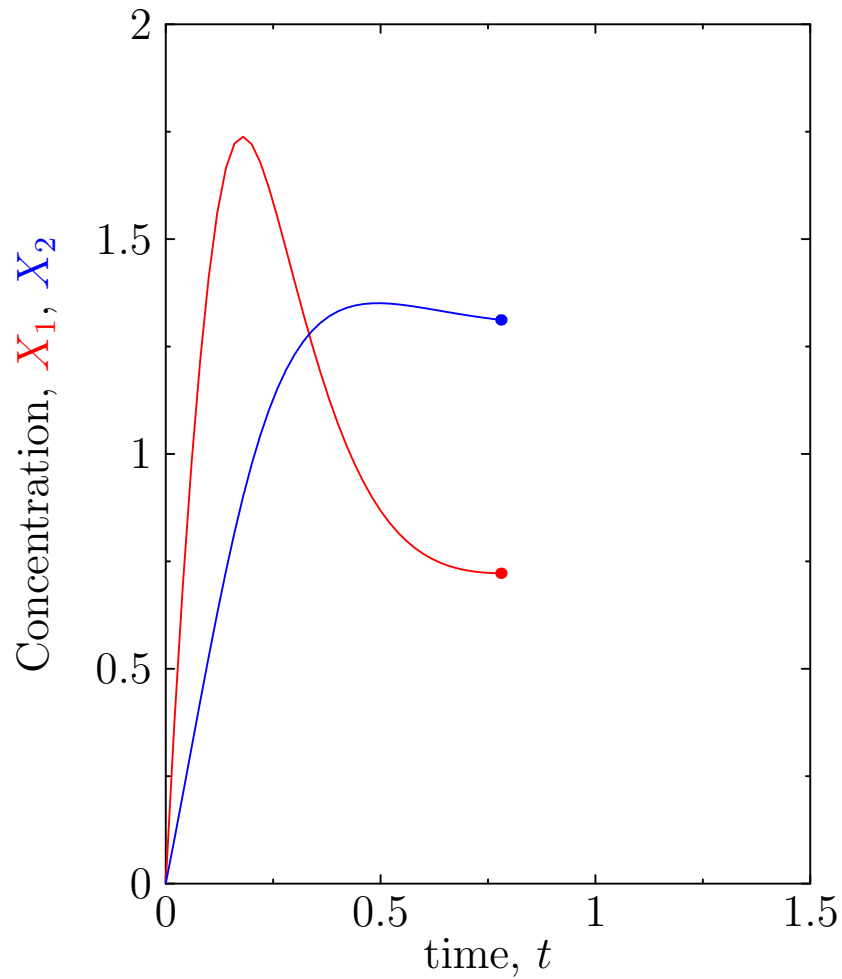
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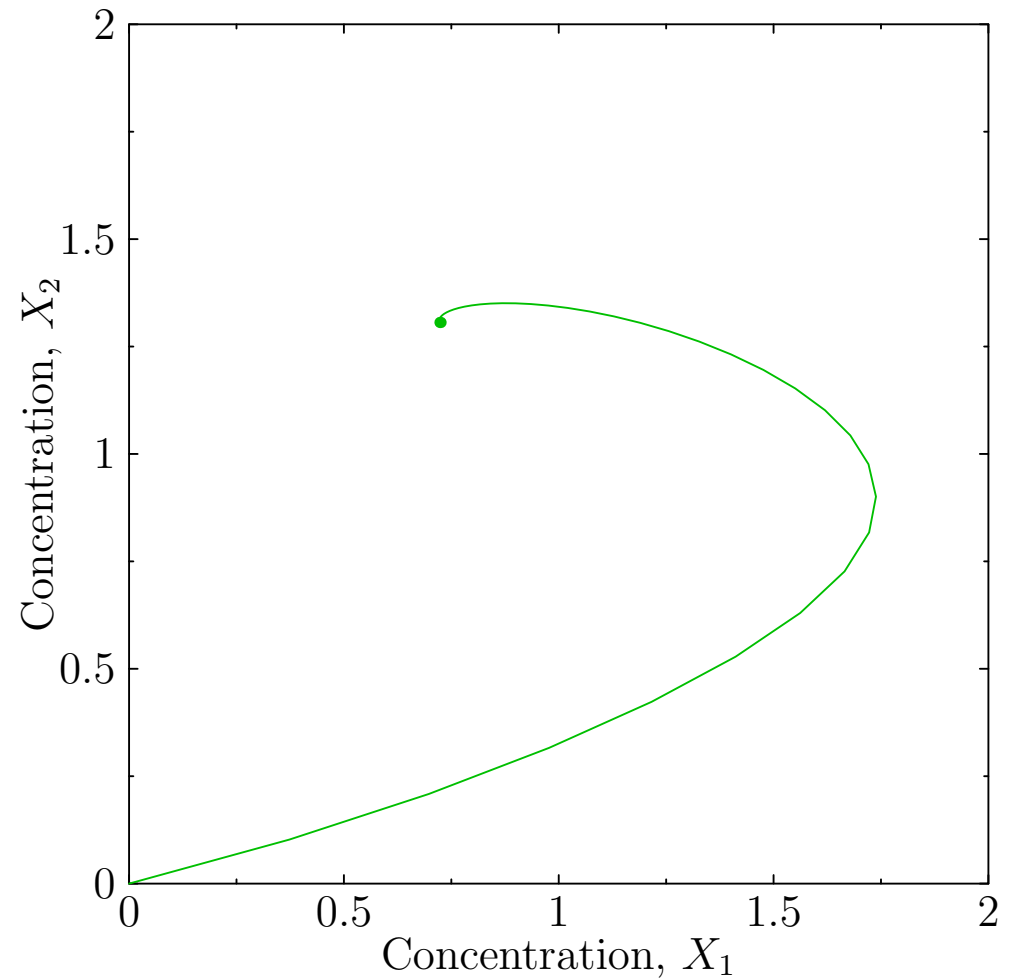
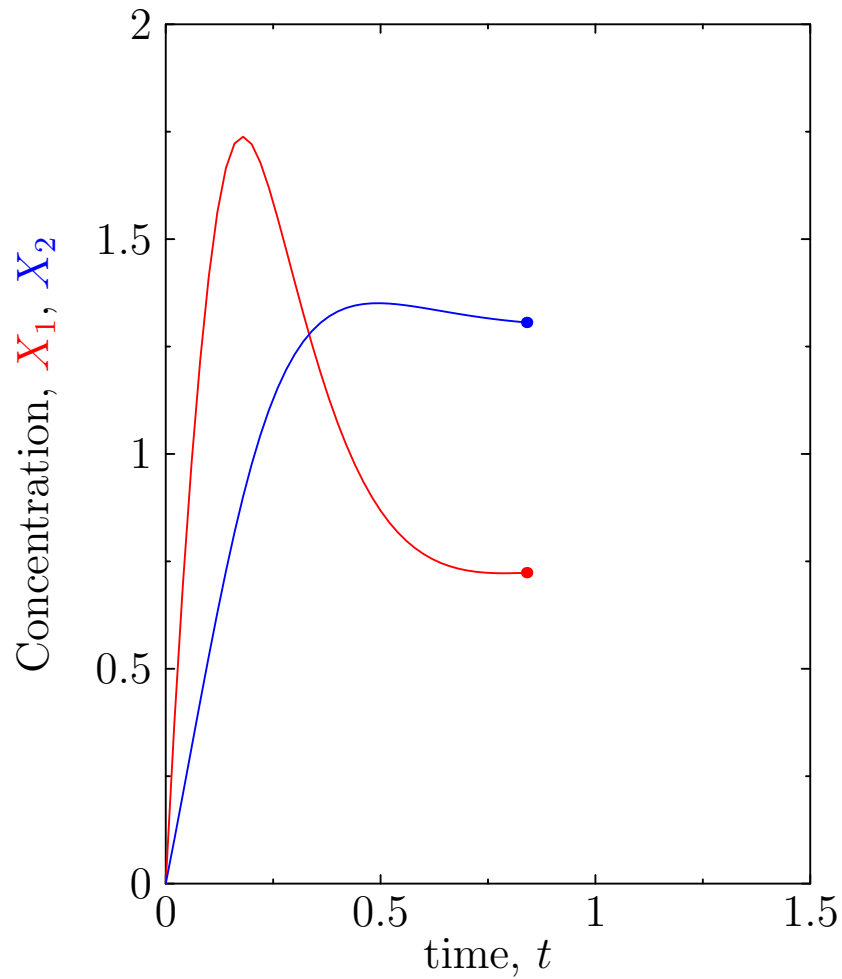
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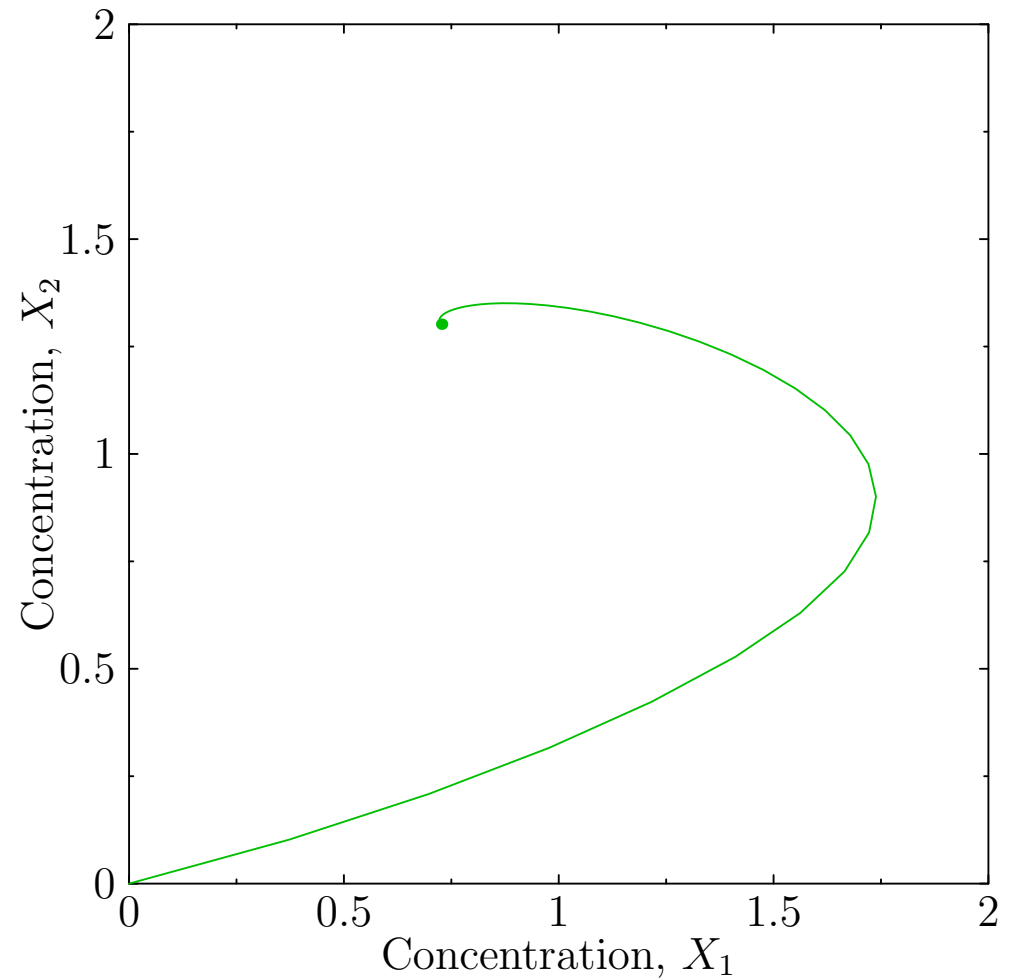
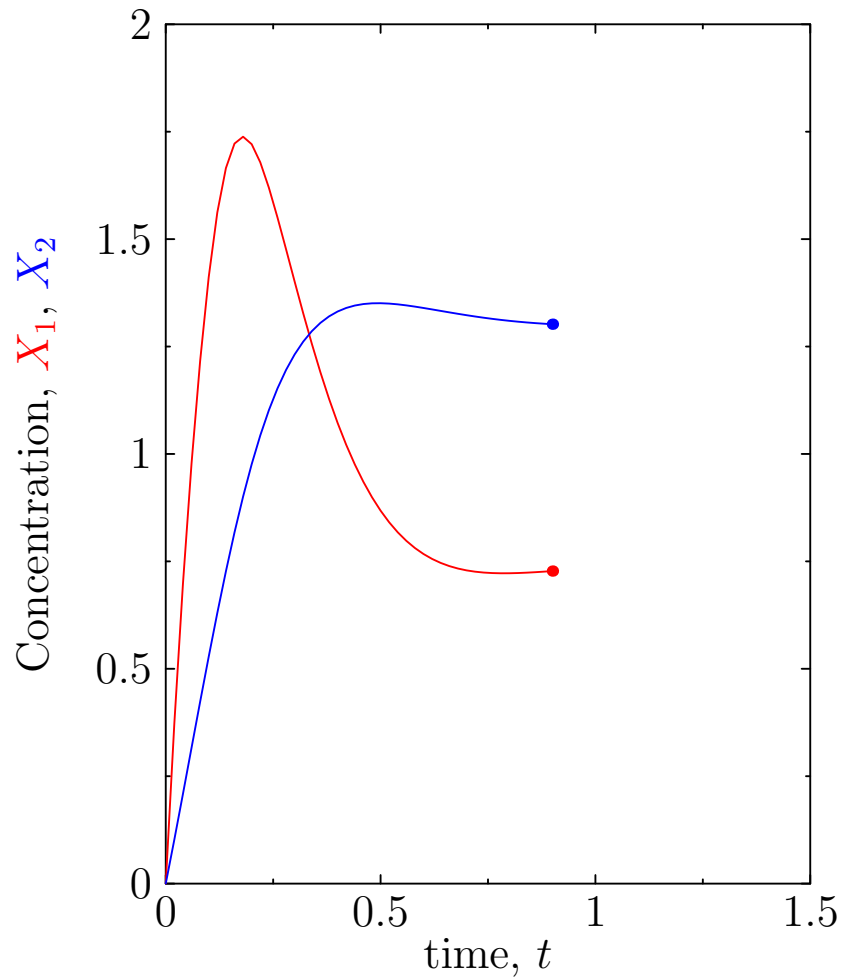
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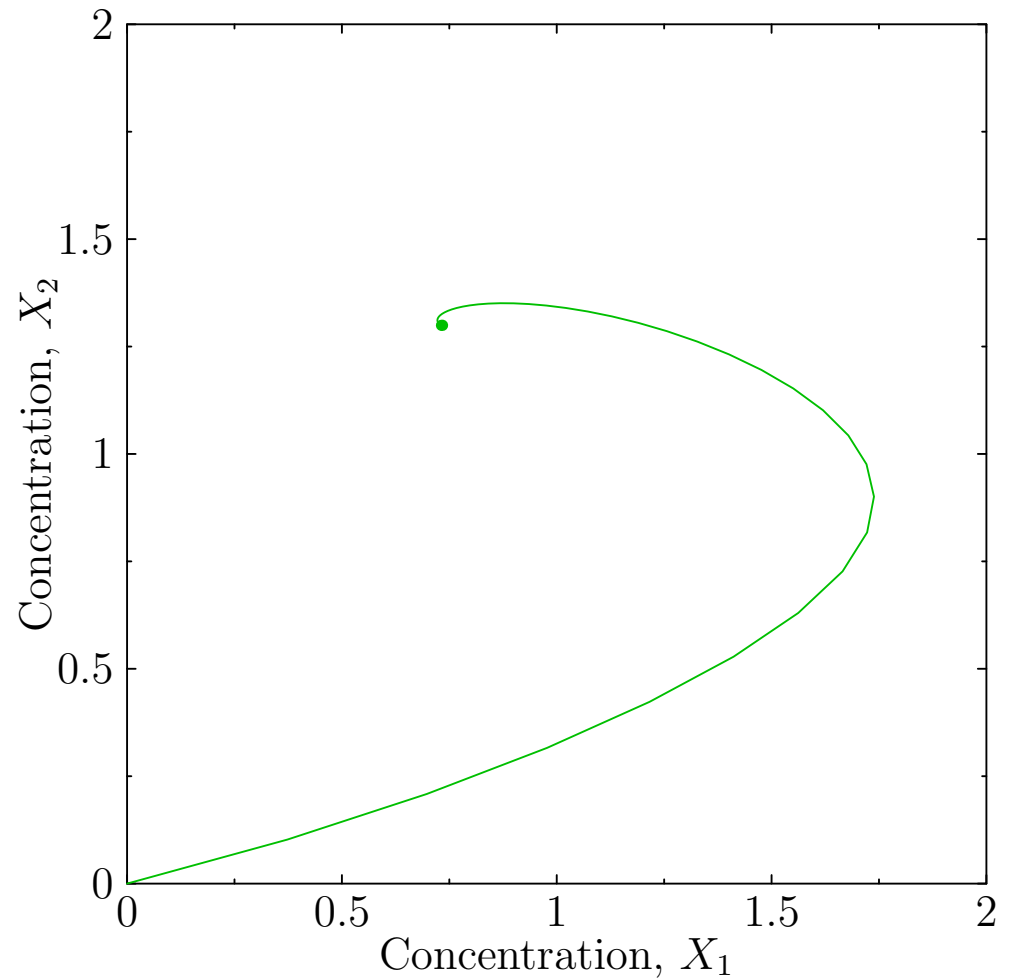
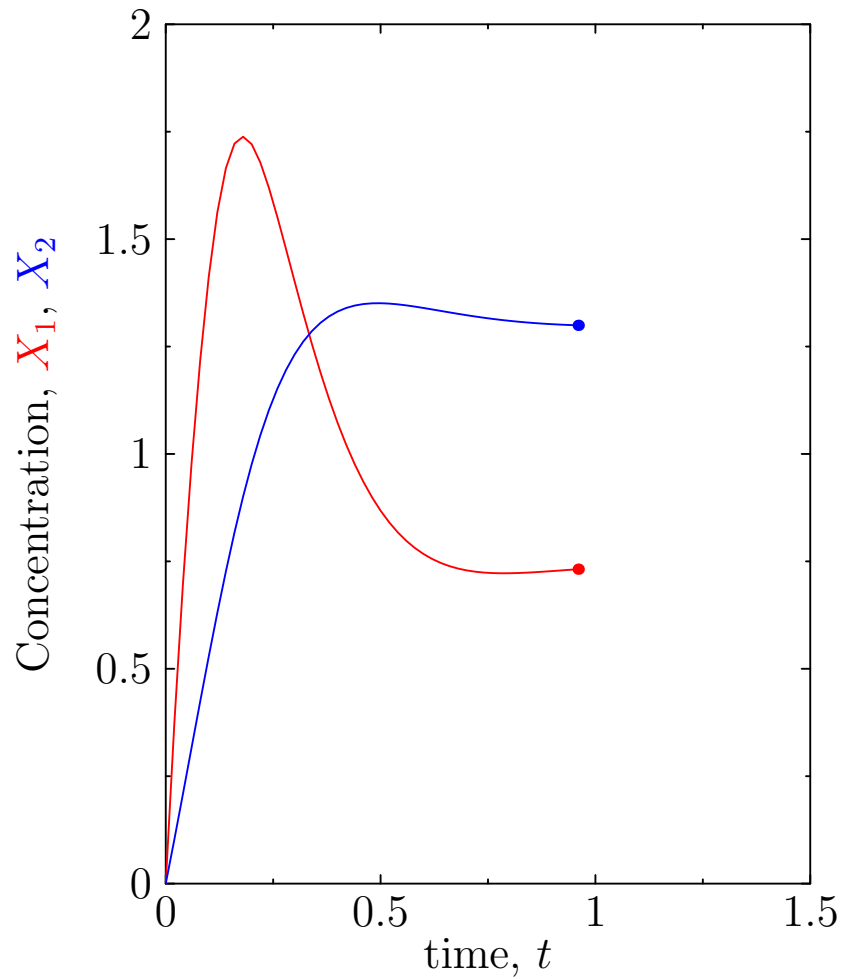
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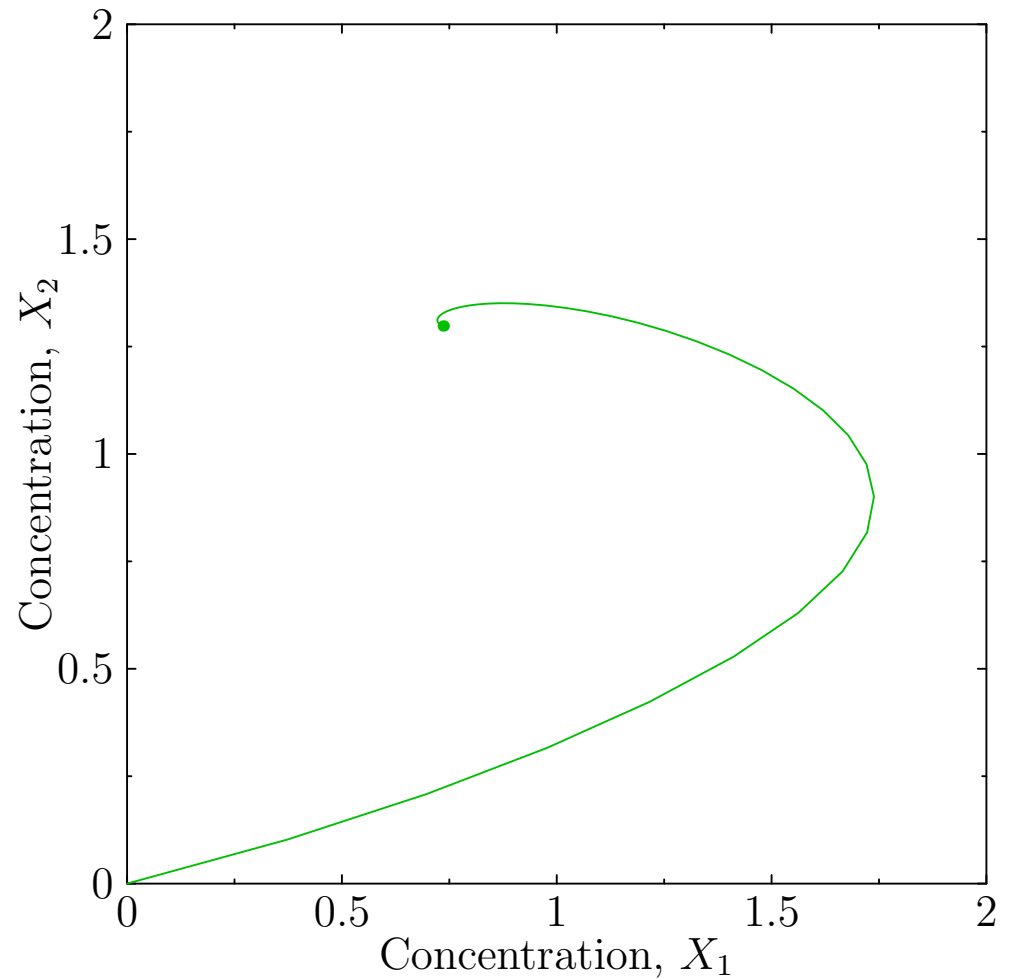
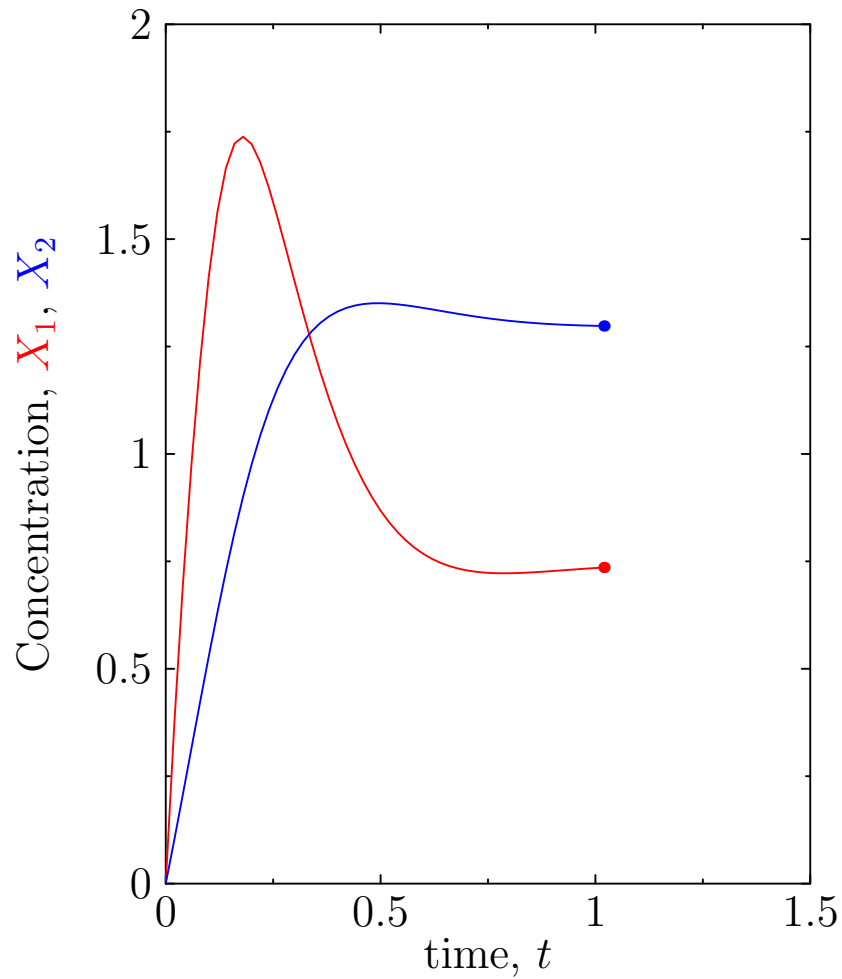
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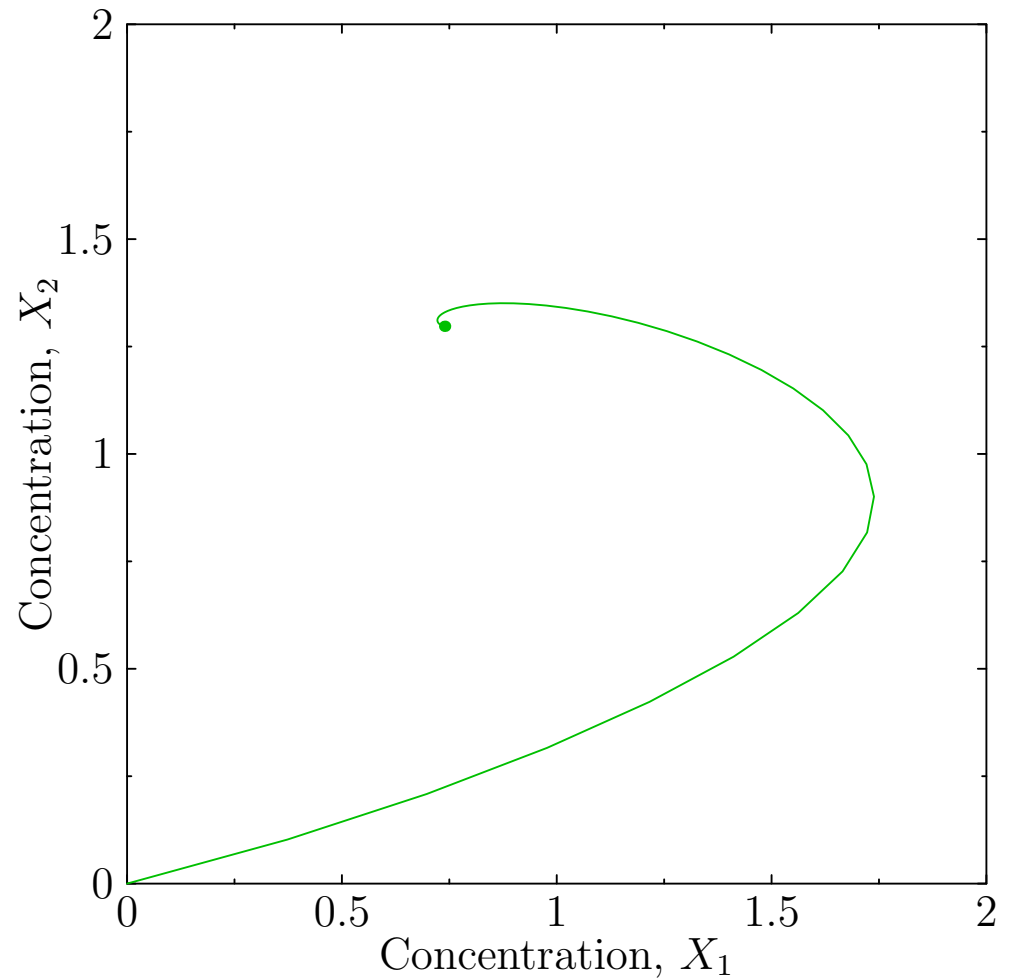
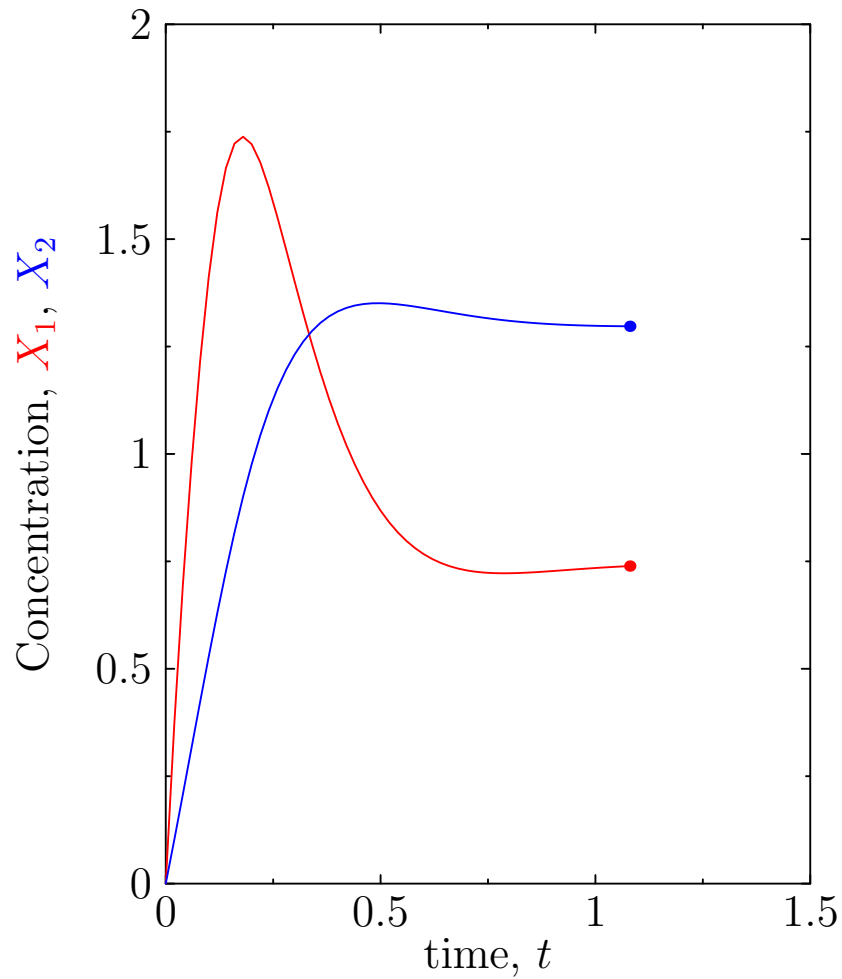
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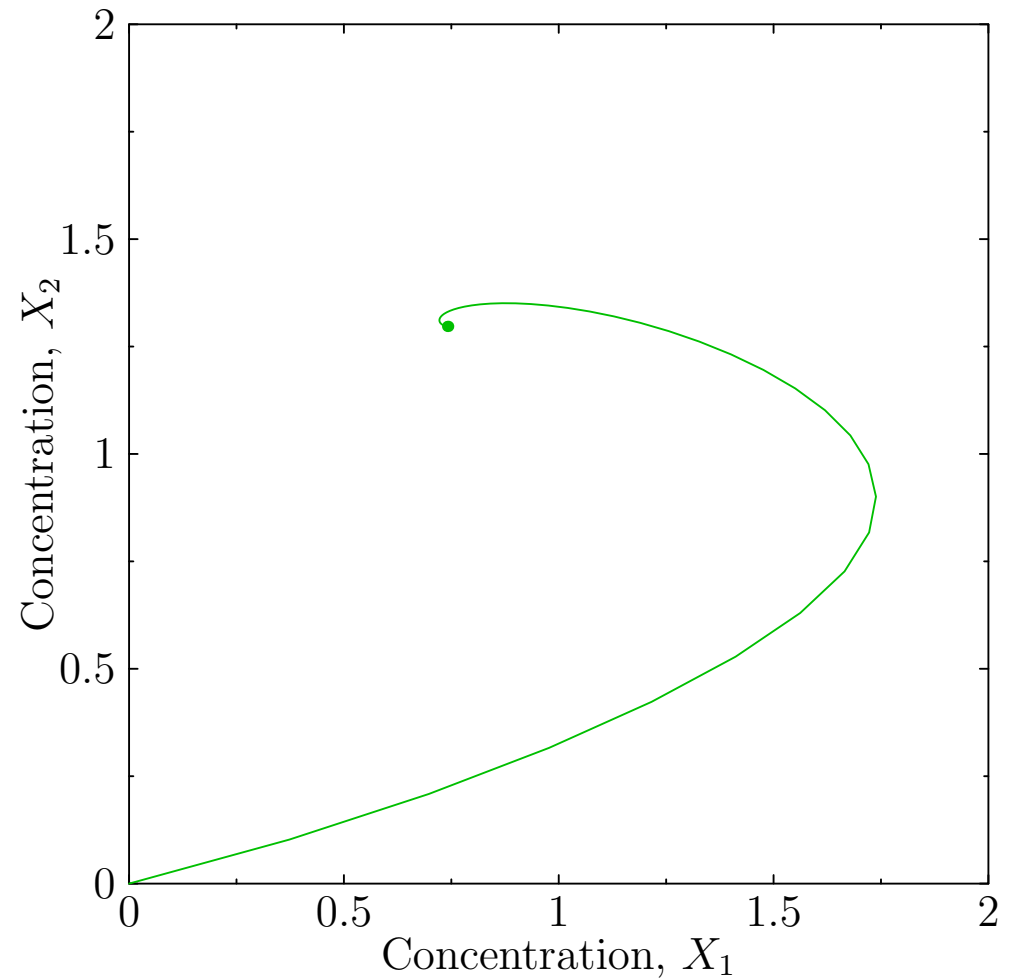
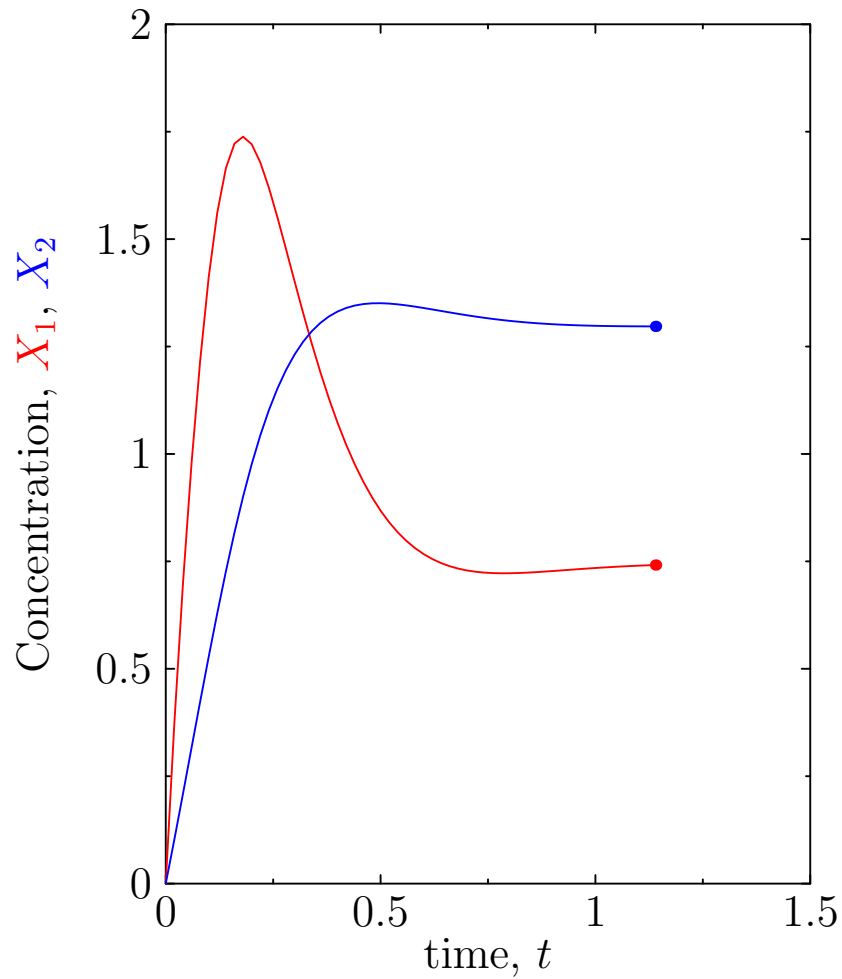
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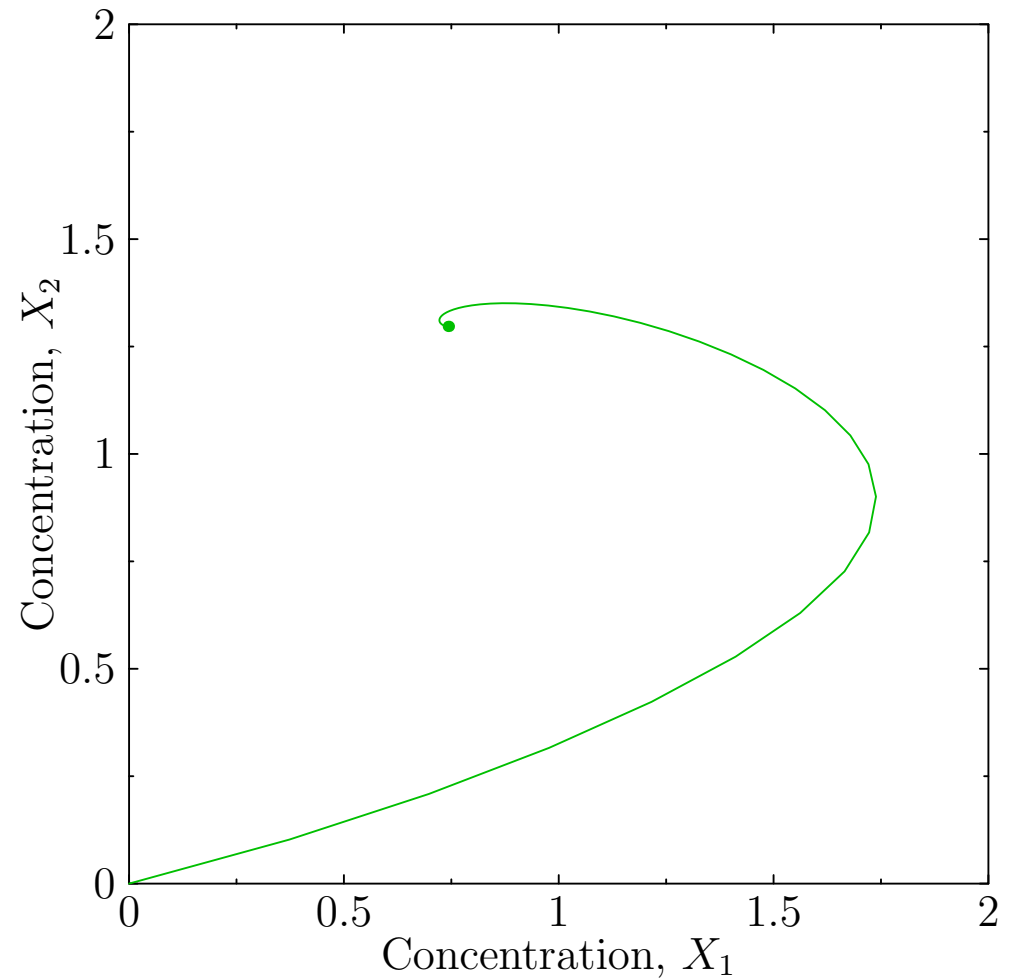
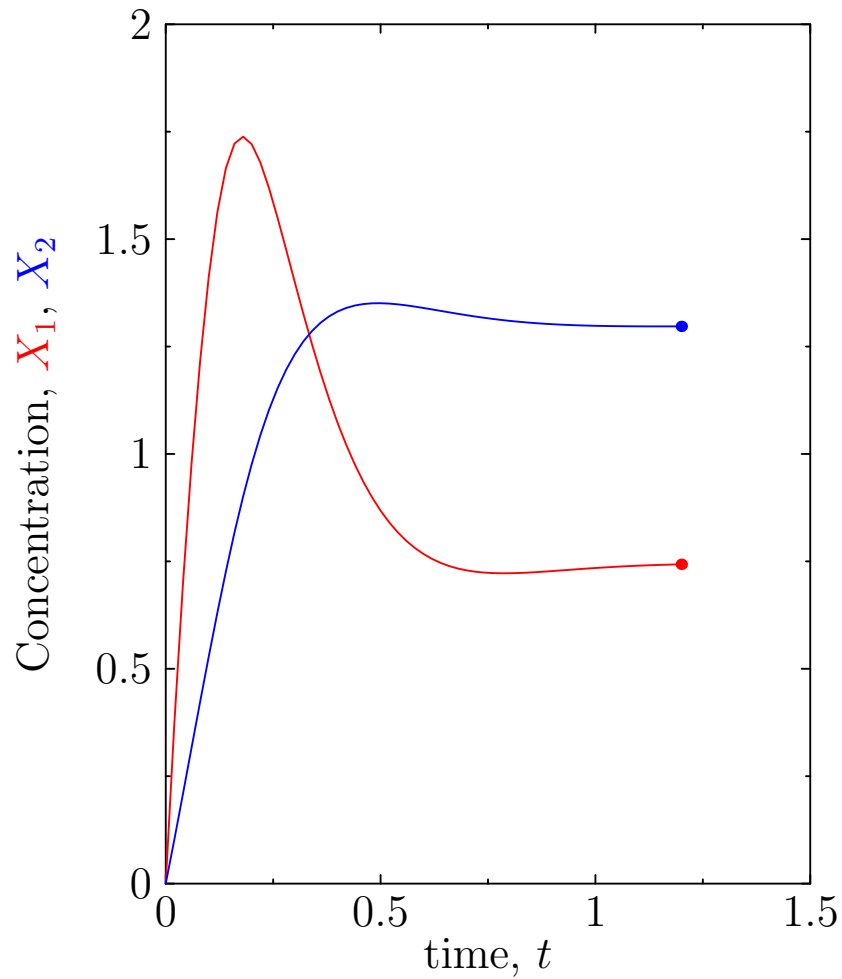
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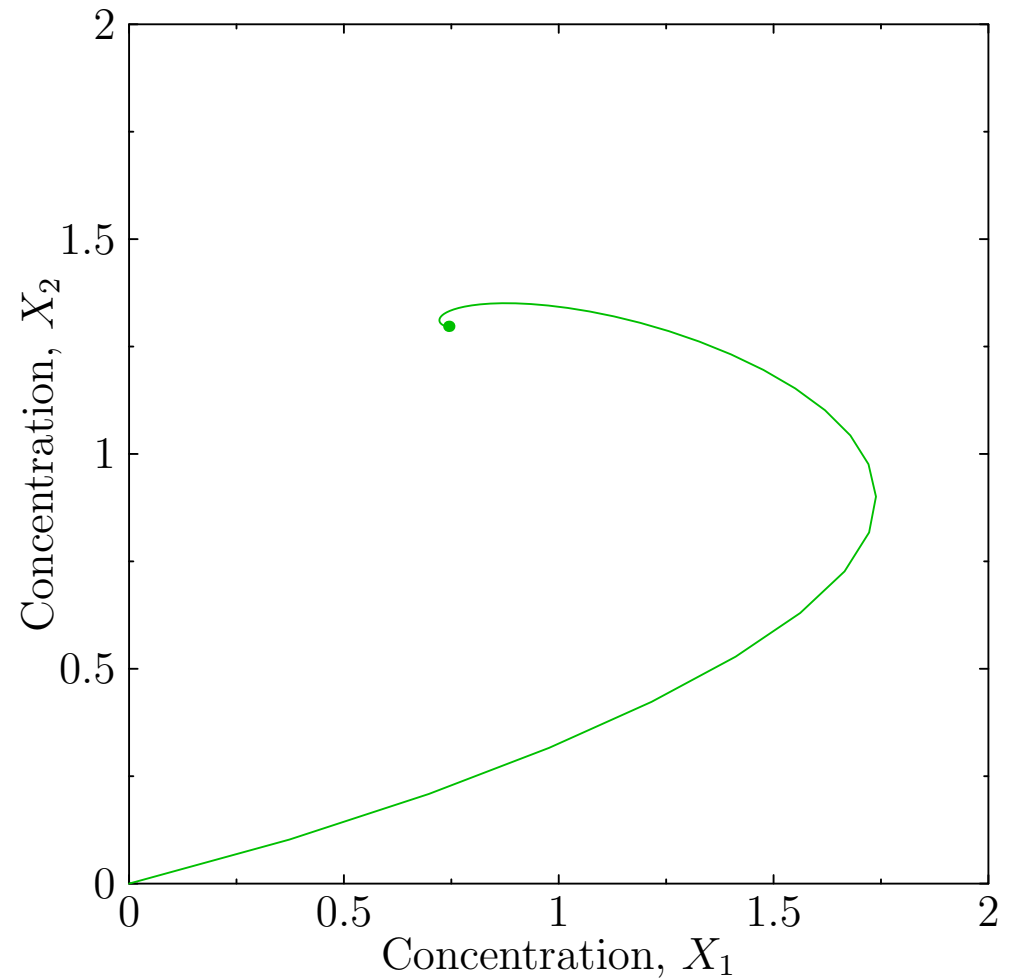
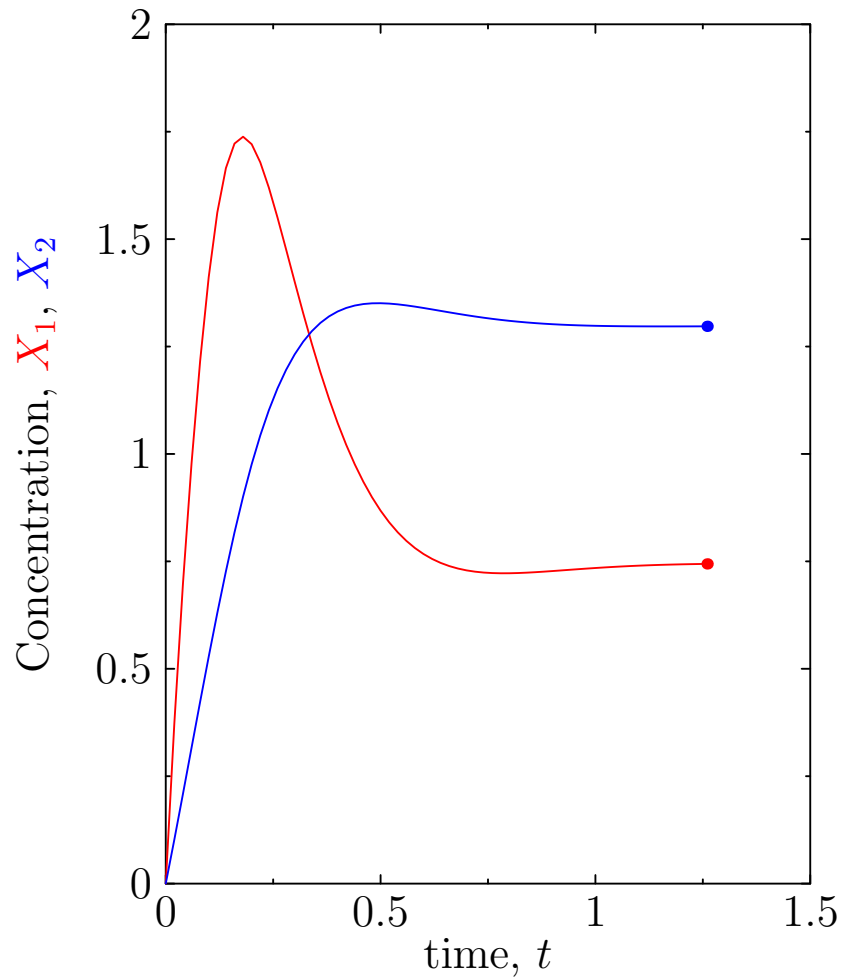
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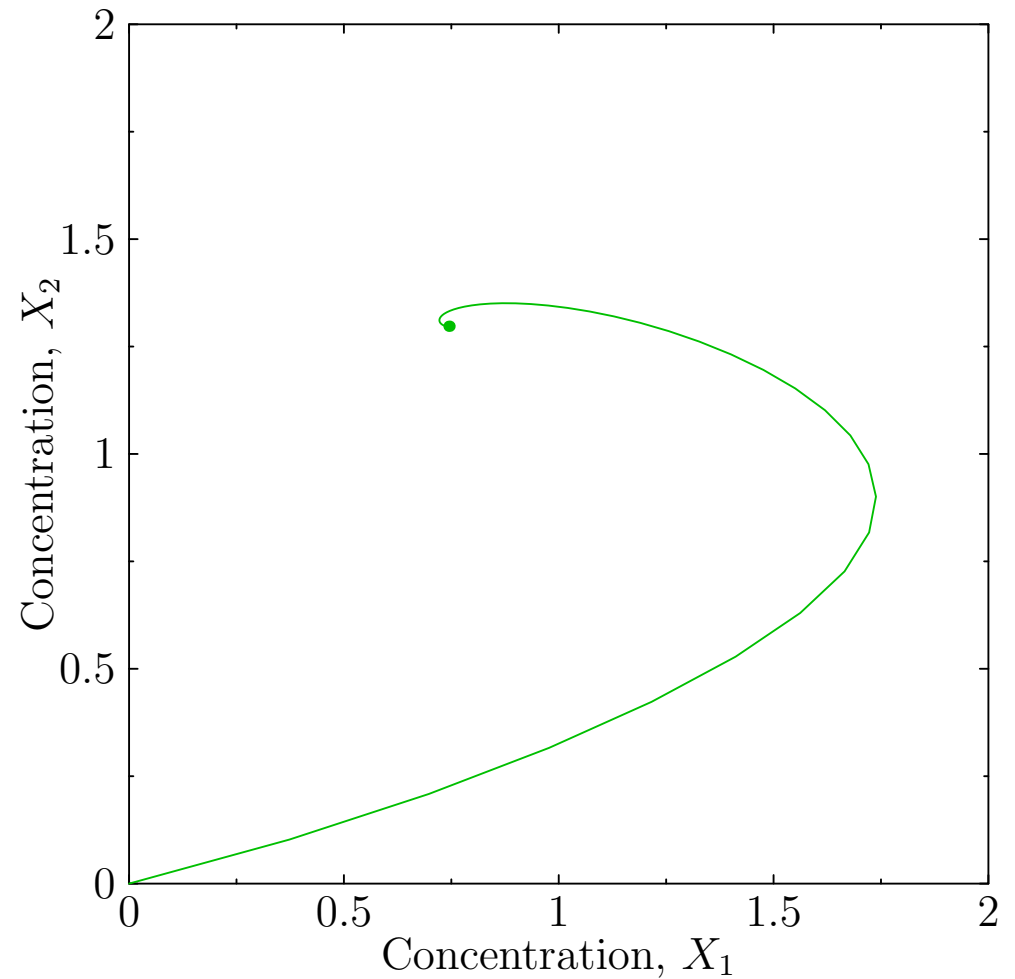
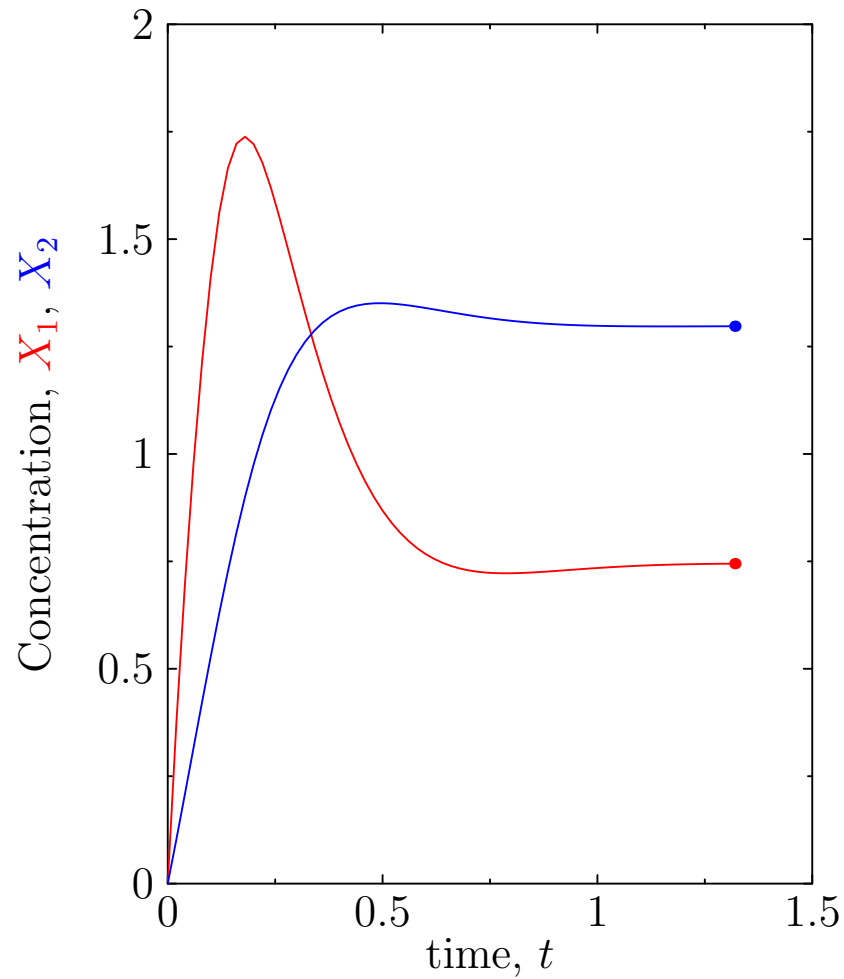
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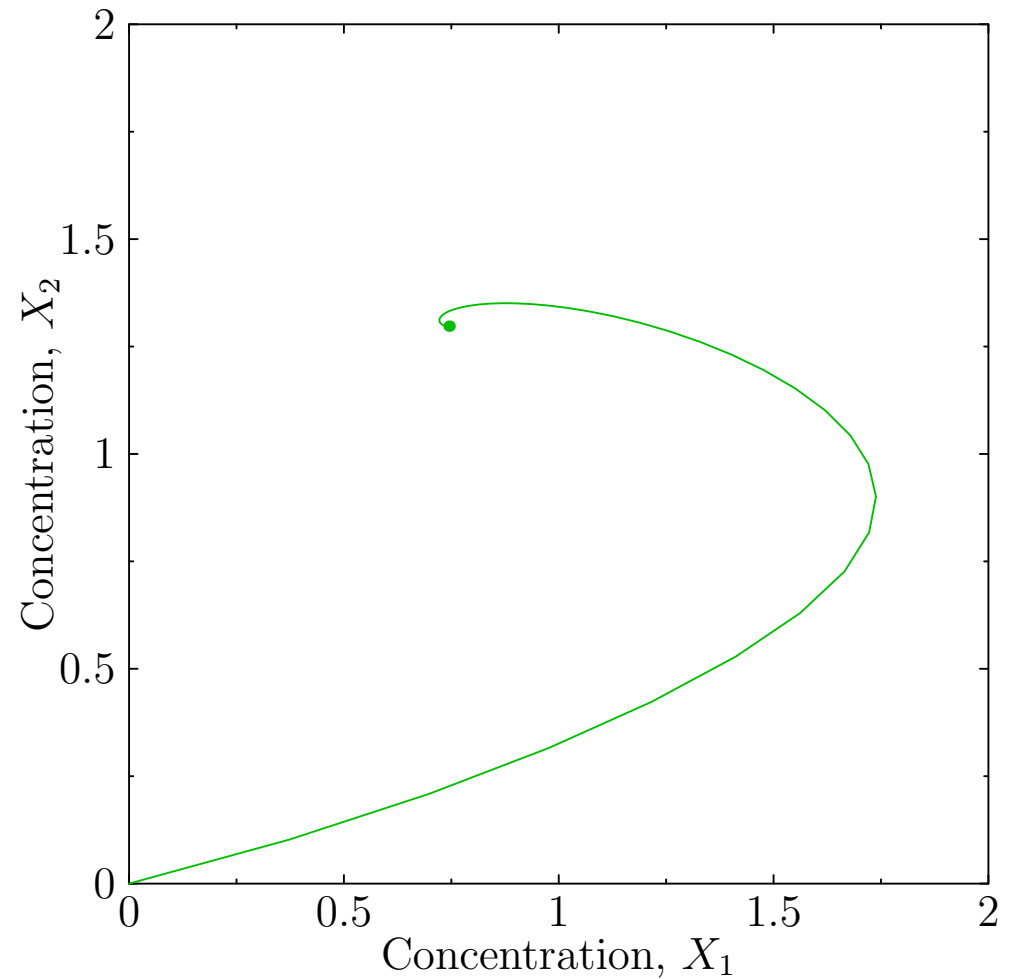
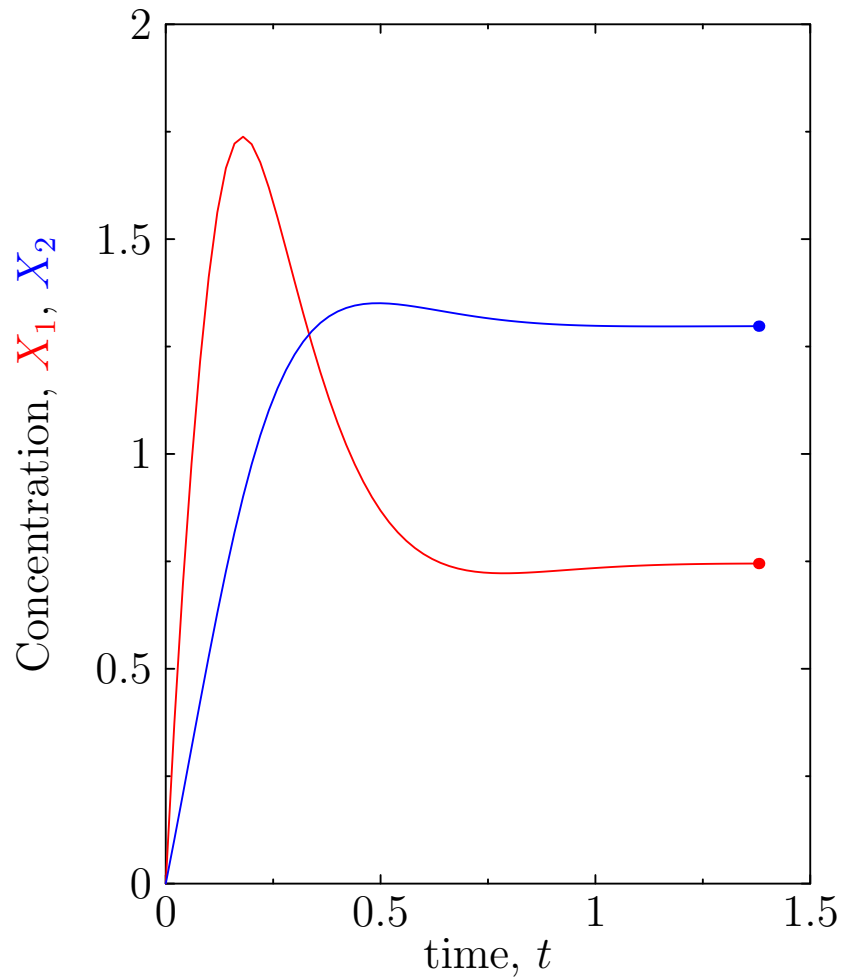
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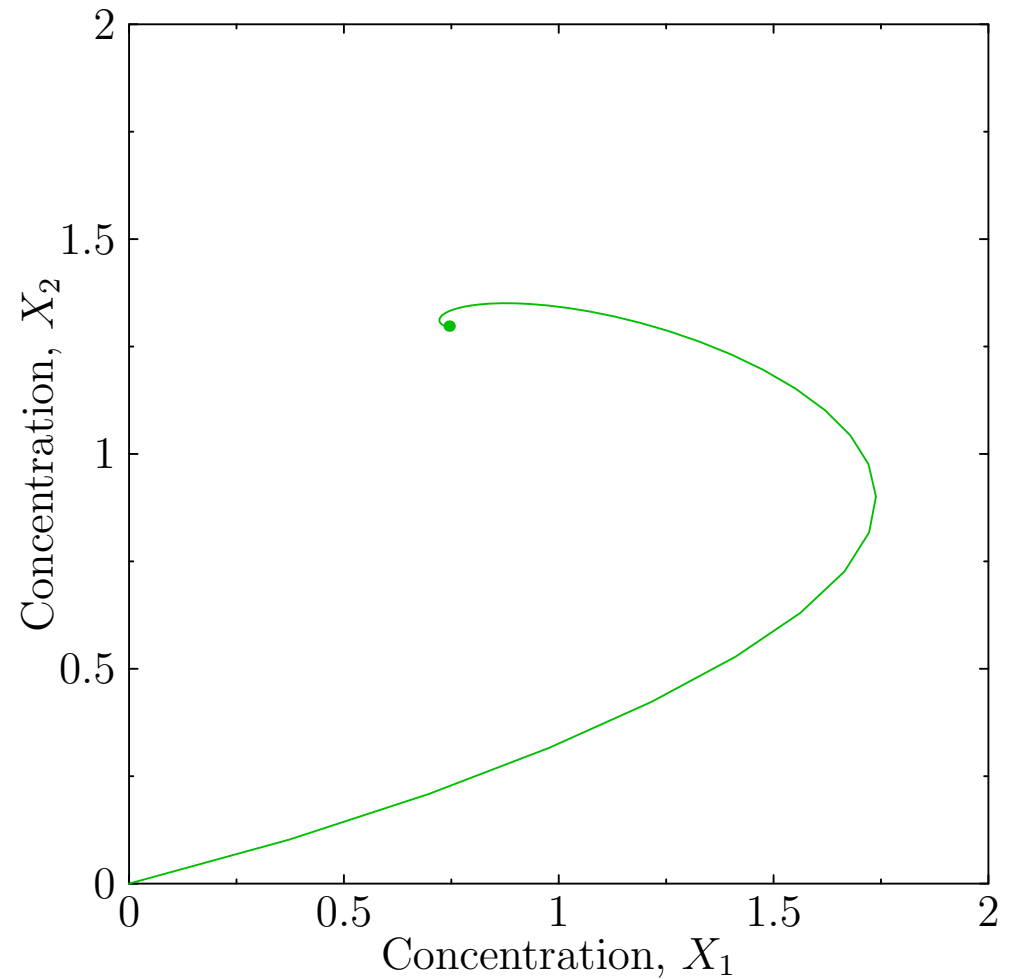
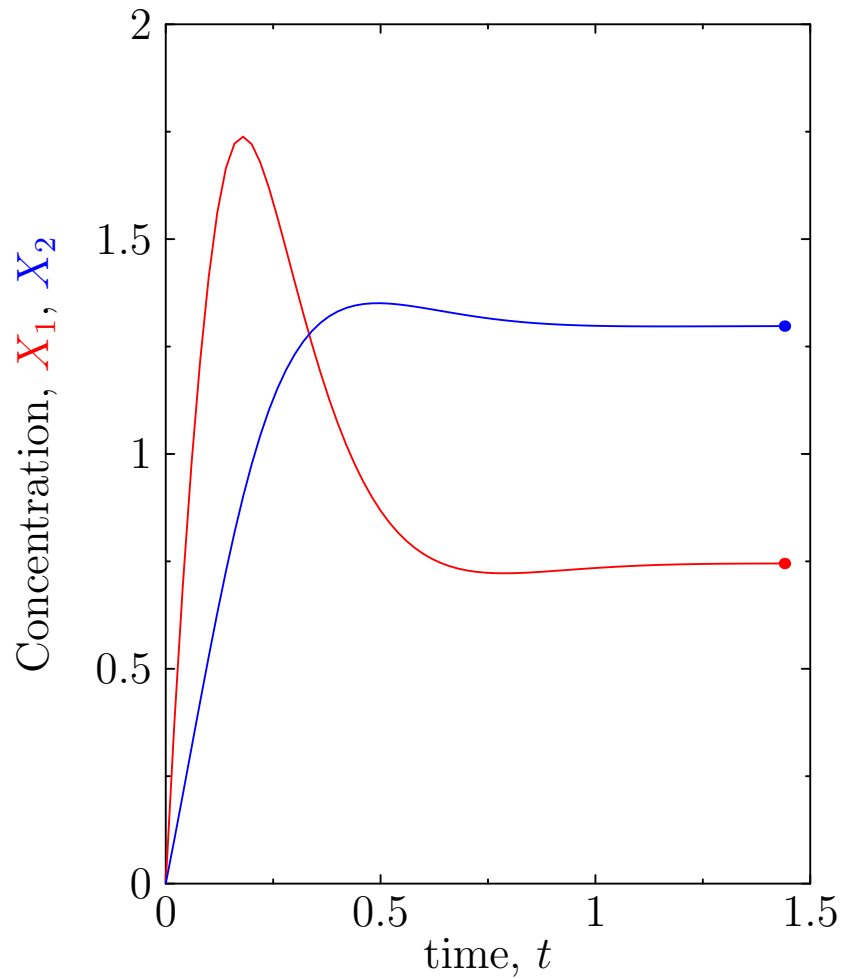
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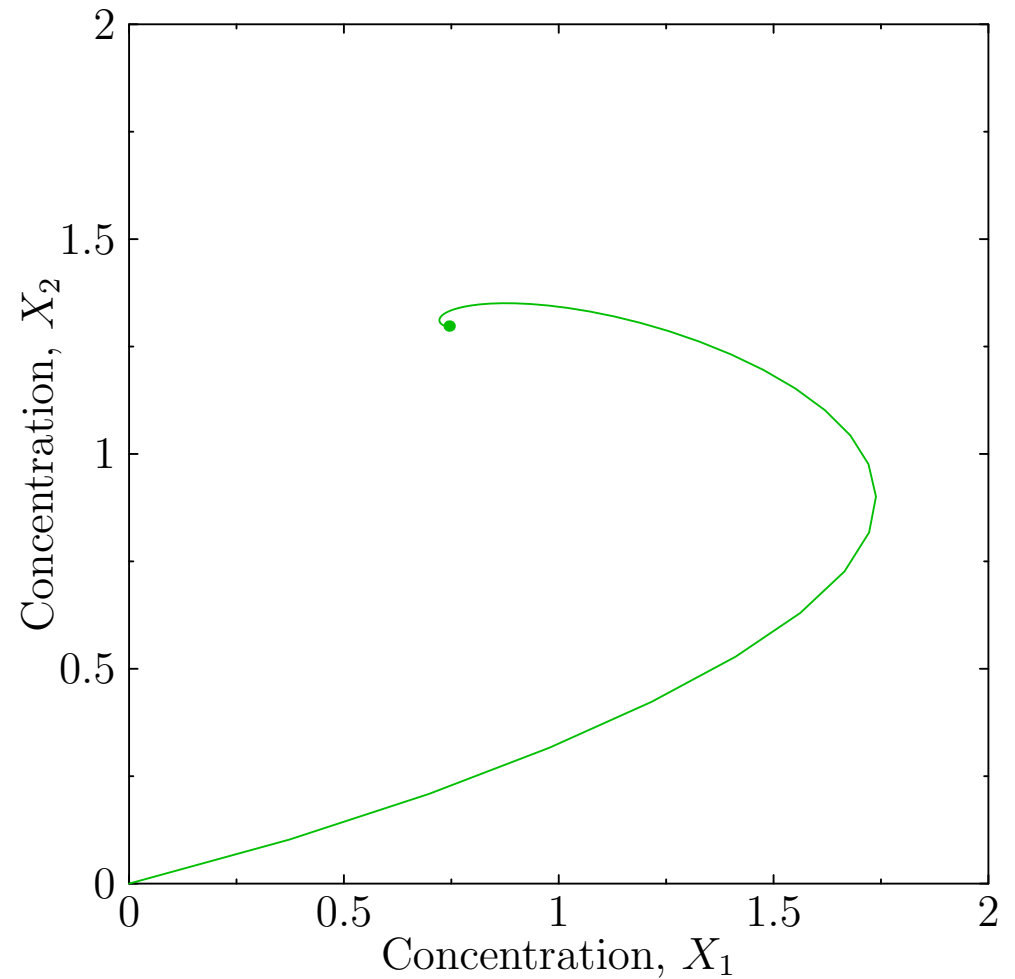
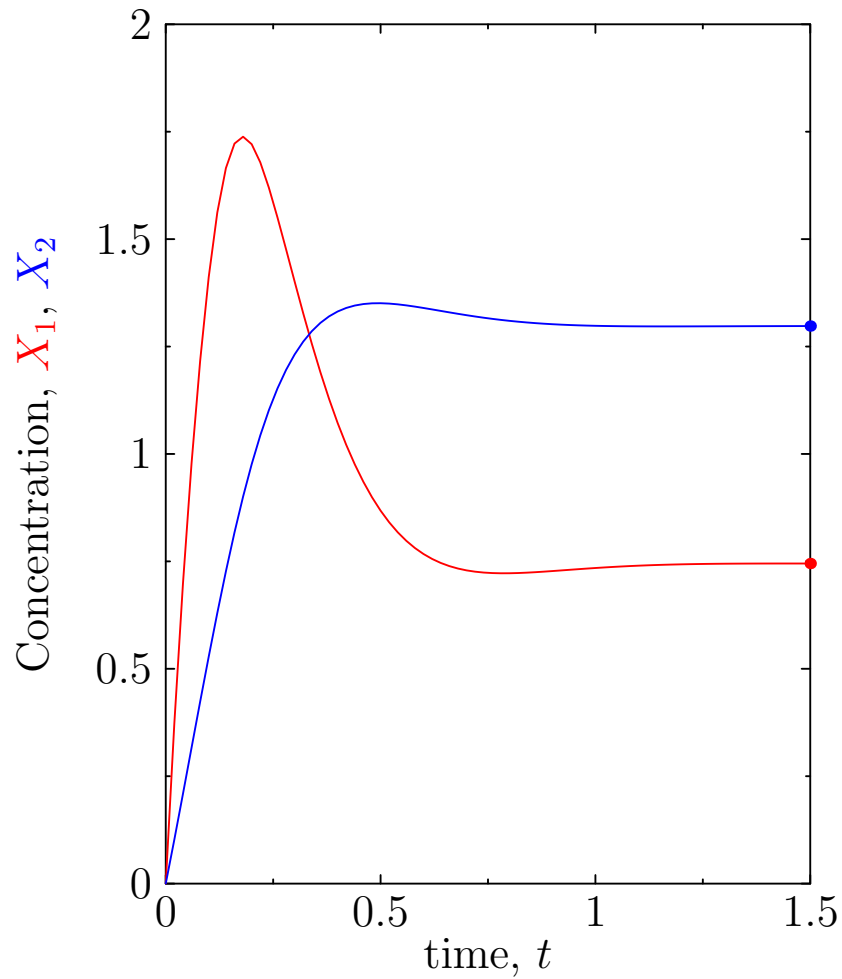
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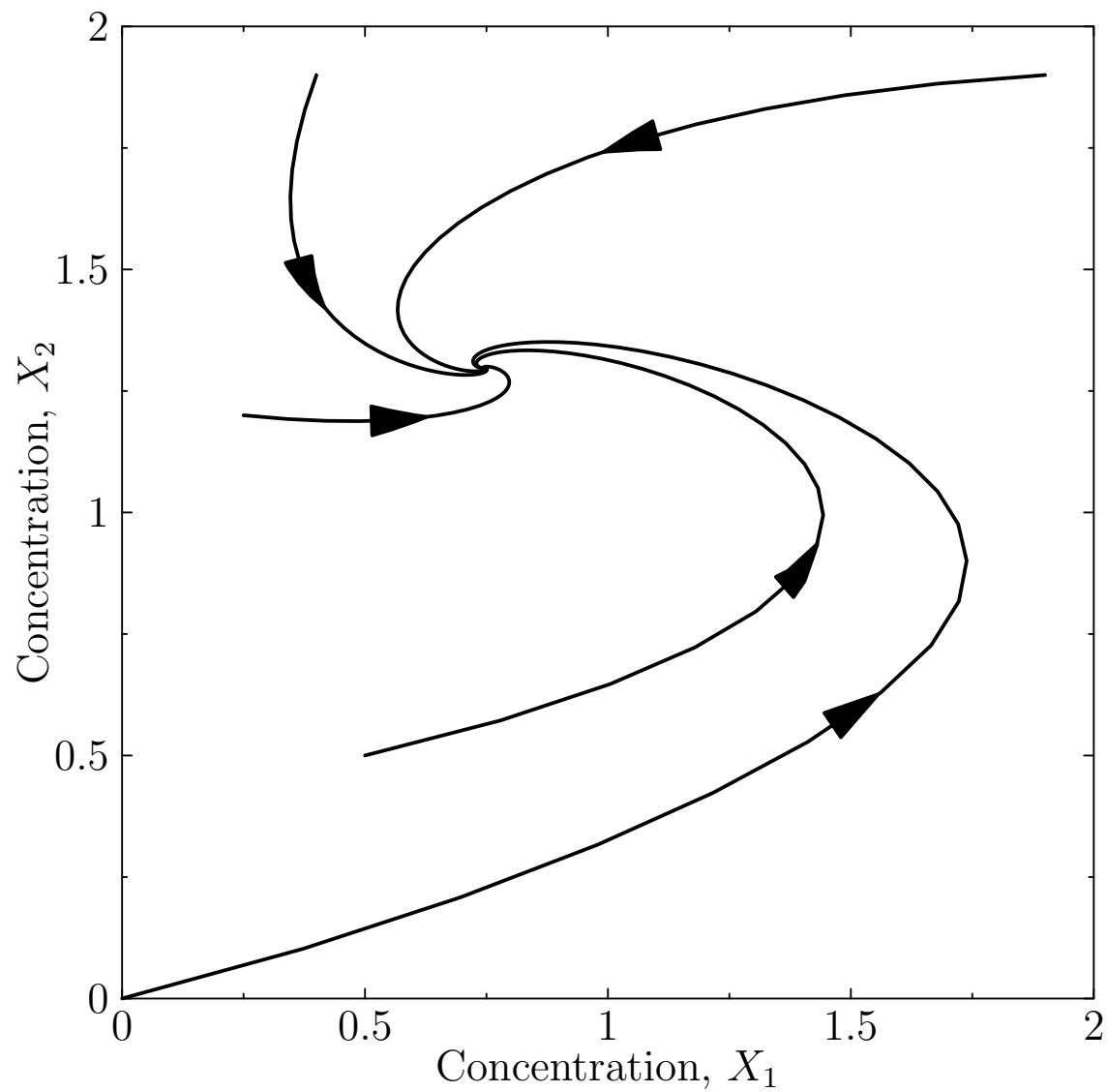
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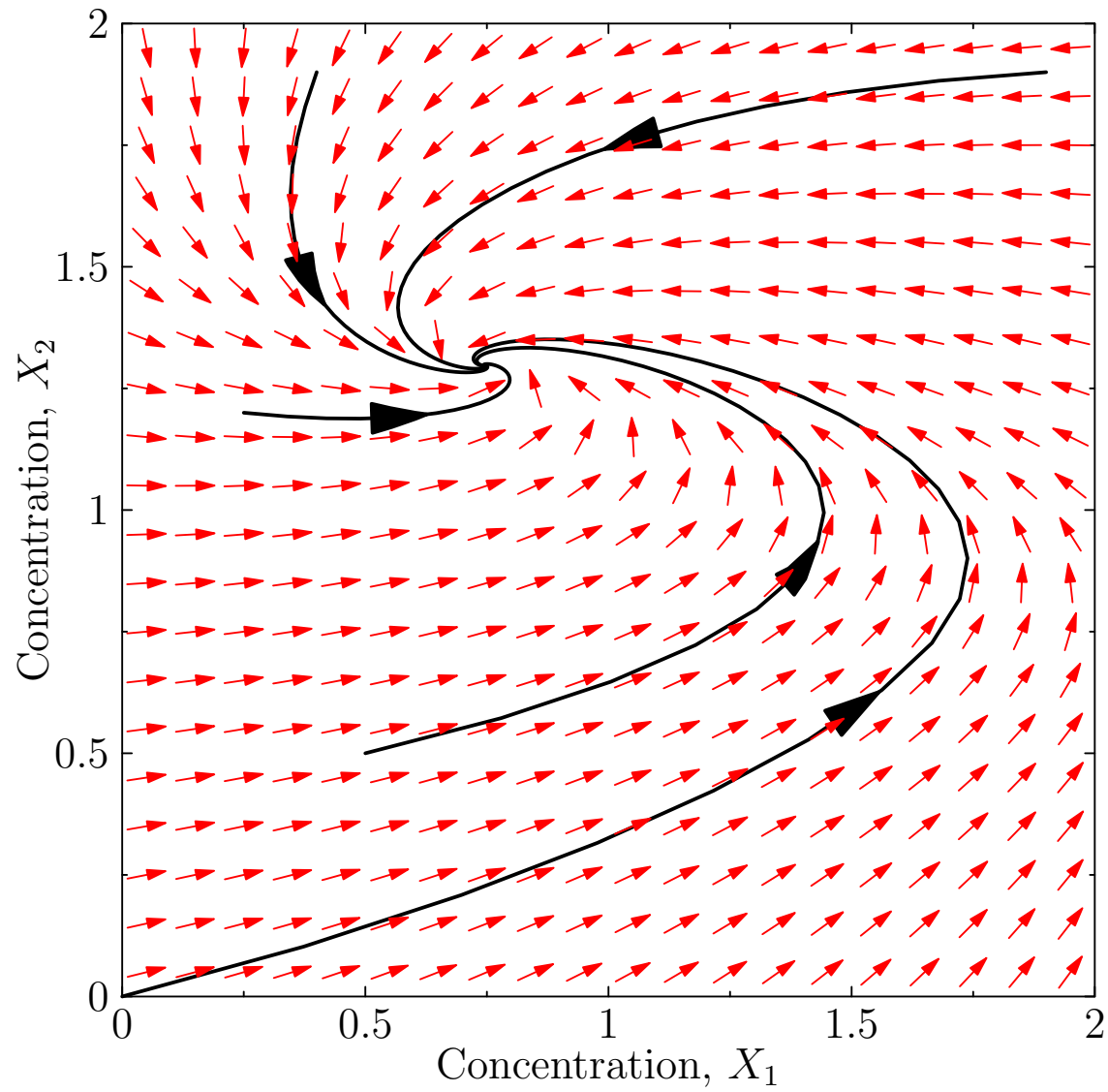


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Trajectories



Trajectories



Nullclines

- To get a better understanding of the dynamics we look at the **nullclines** defined by $dX_i/dt = 0$
- In our system

$$\frac{dX_1}{dt} = \frac{k_1}{1 + (X_2/K)^n} - k_3 X_1 - k_5 X_1 = 0 \Rightarrow X_1 = \frac{k_1}{(k_3 + k_5)(1 + (X_2/K)^n)}$$

$$\frac{dX_2}{dt} = k_2 + k_5 X_1 - k_4 X_2 = 0 \Rightarrow X_2 = \frac{k_2 + k_5 X_1}{k_4}$$

- The intersection of the nullclines defines the stationary points

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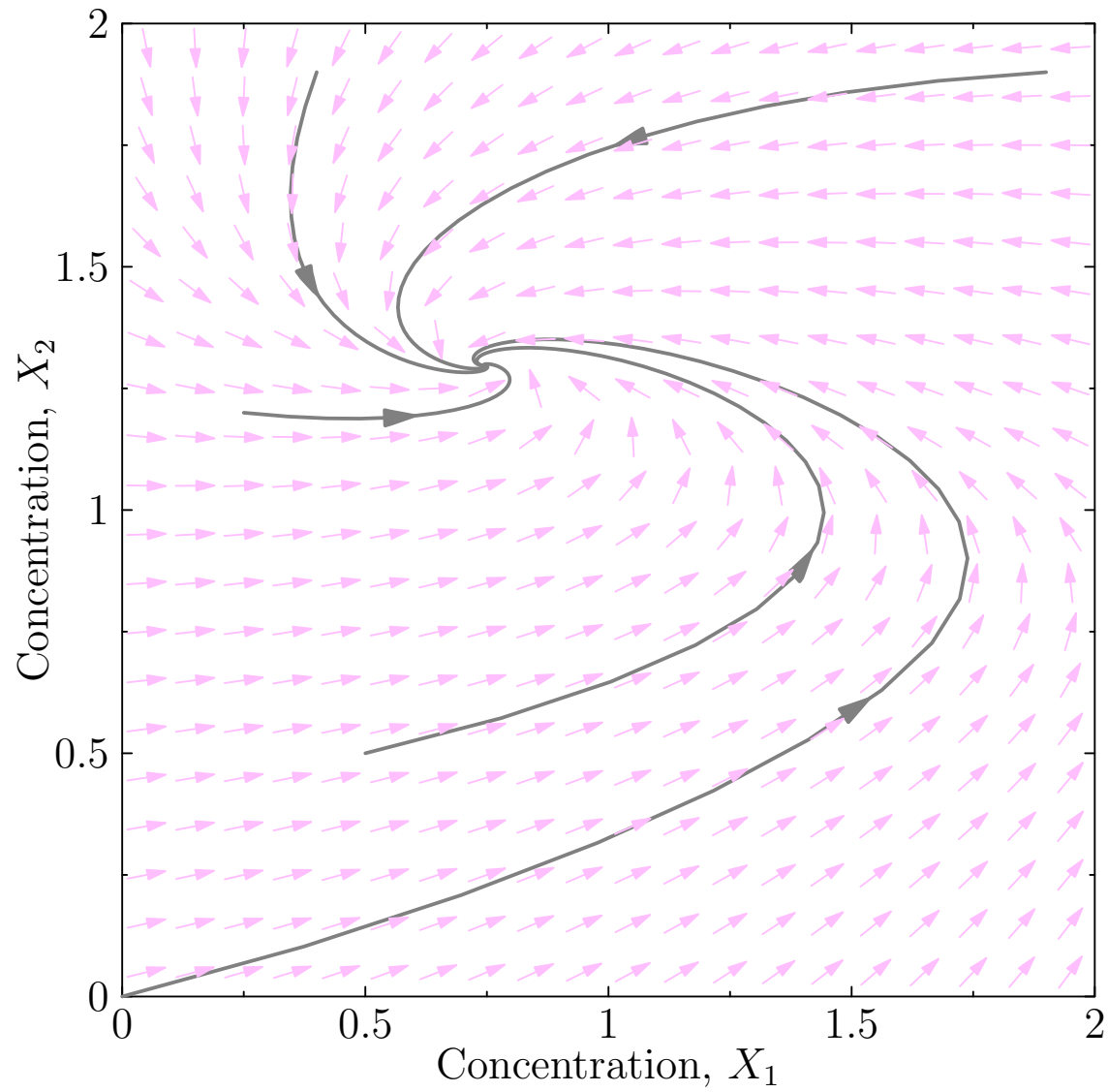
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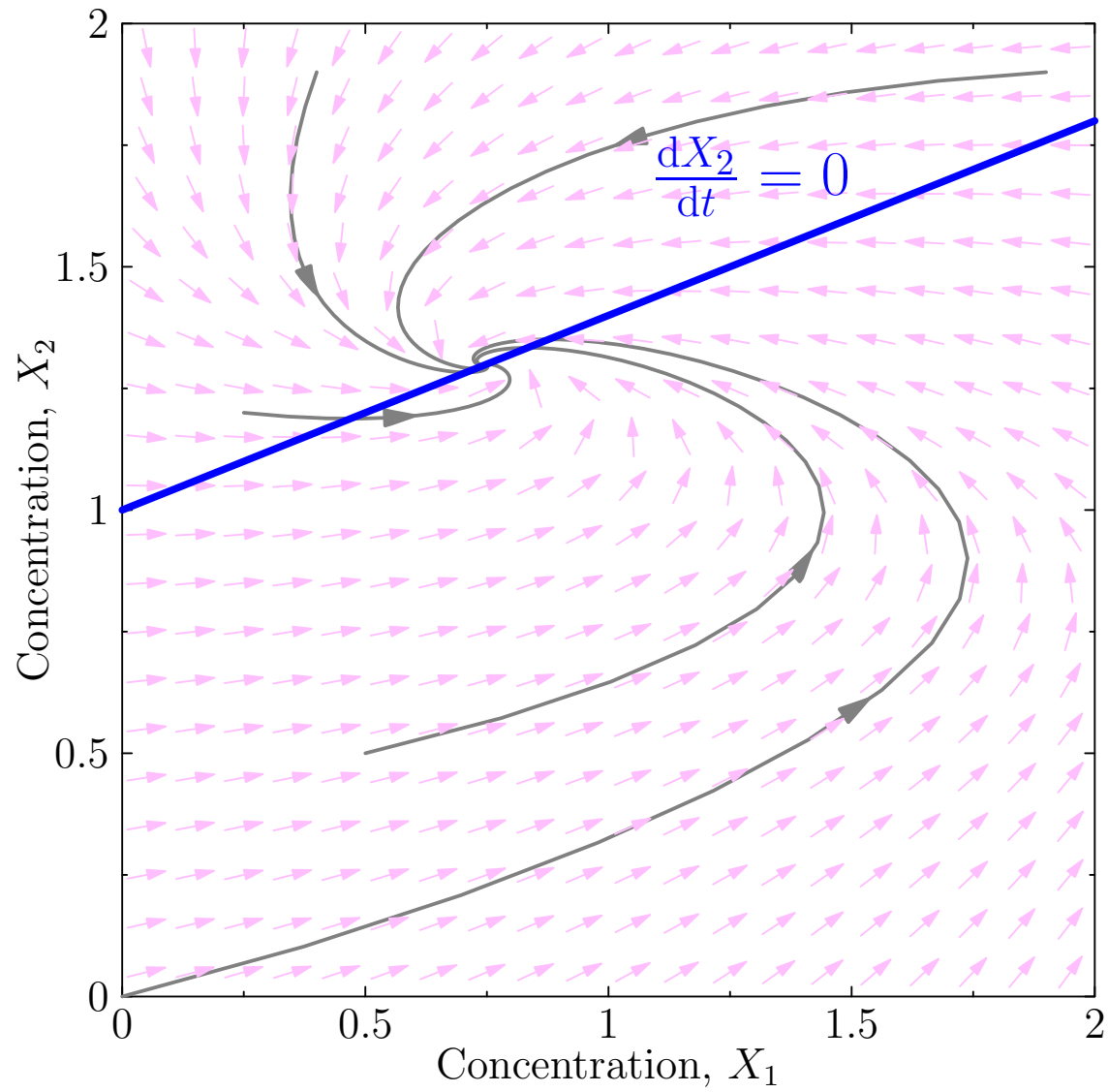
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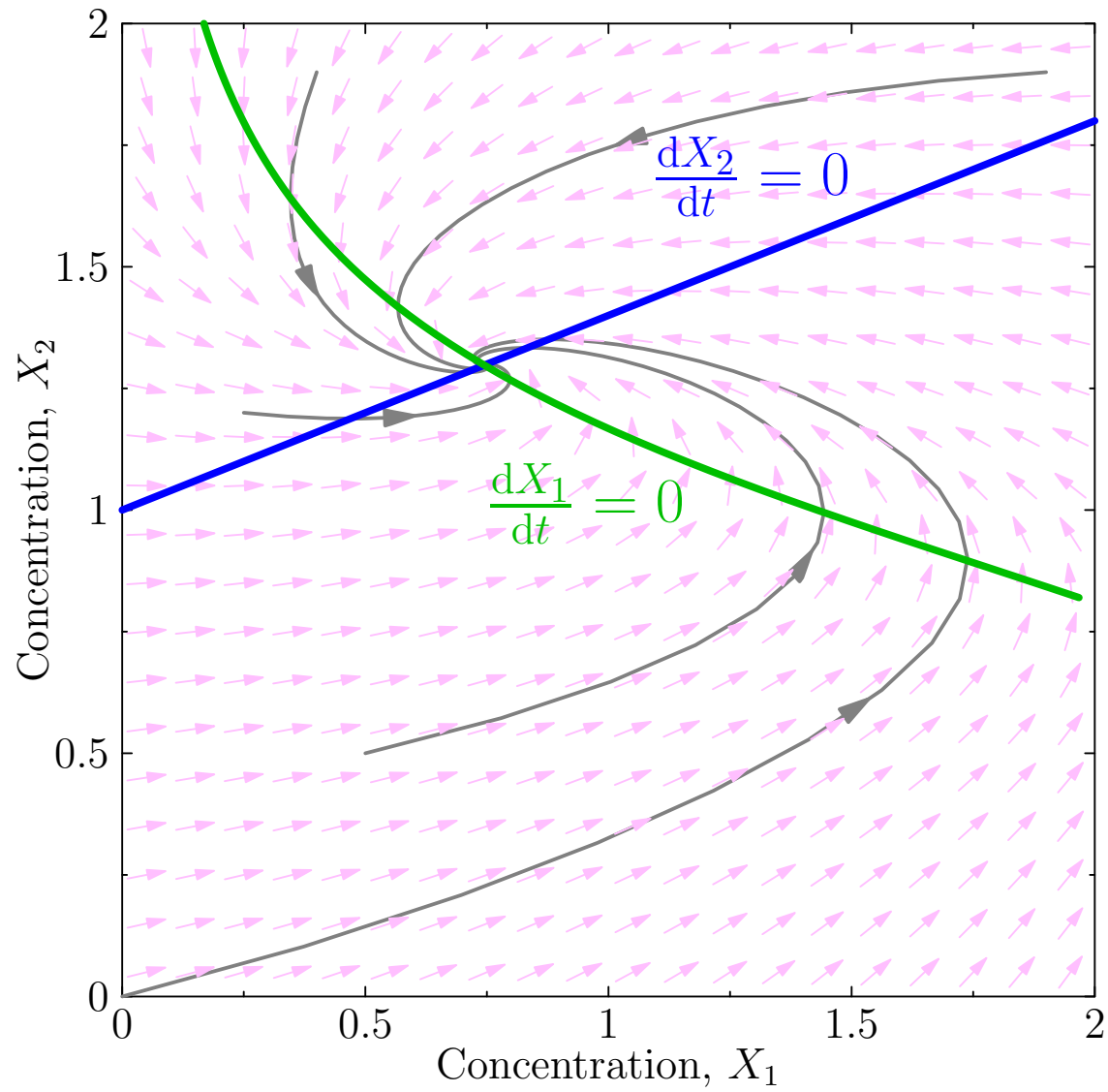
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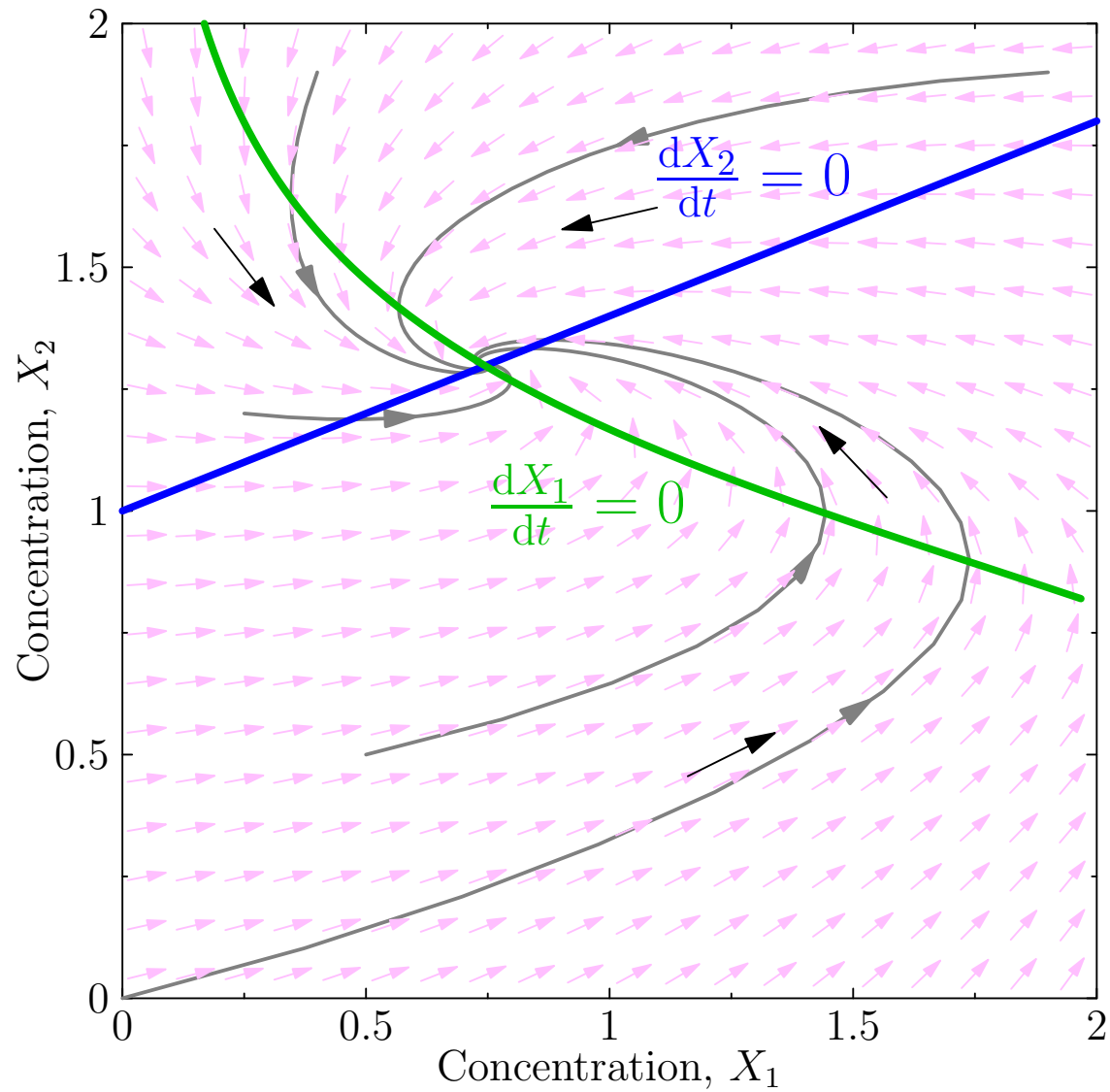
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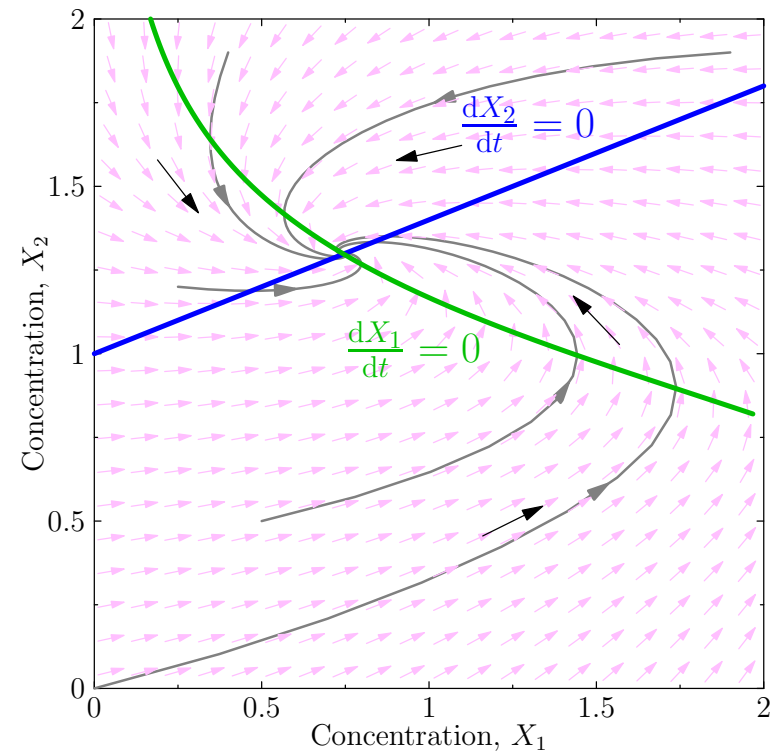


Nullclines



Outline

1. Understanding ODEs
2. **Stability**
3. Bifurcations



Bistability

- Let's consider a slightly different system where X_1 and X_2 inhibits each others' expression

$$\frac{dX_1(t)}{dt} = f_1(\mathbf{X}) = \frac{k_1}{1 + (X_2(t)/K_2)^{n_1}} - k_3 X_1(t)$$

$$\frac{dX_2(t)}{dt} = f_2(\mathbf{X}) = \frac{k_2}{1 + (X_1(t)/K_1)^{n_2}} - k_4 X_2(t)$$

- If we consider the symmetric case $k_1 = k_2 = 20$, $K_1 = K_2 = 1$, $k_3 = k_4 = 5$, $n_1 = n_2 = 4$ we end up with three fixed points

Bistability

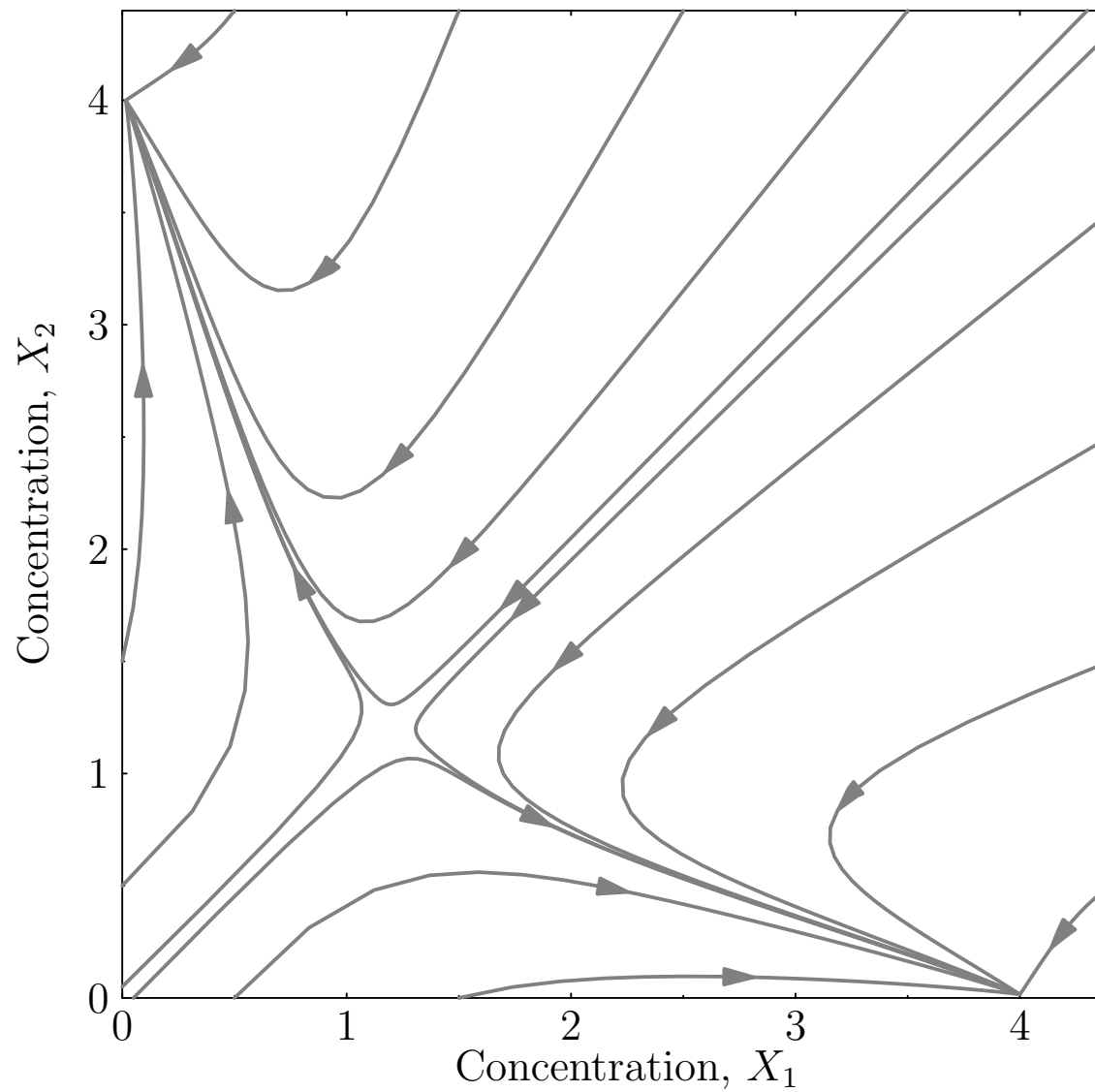
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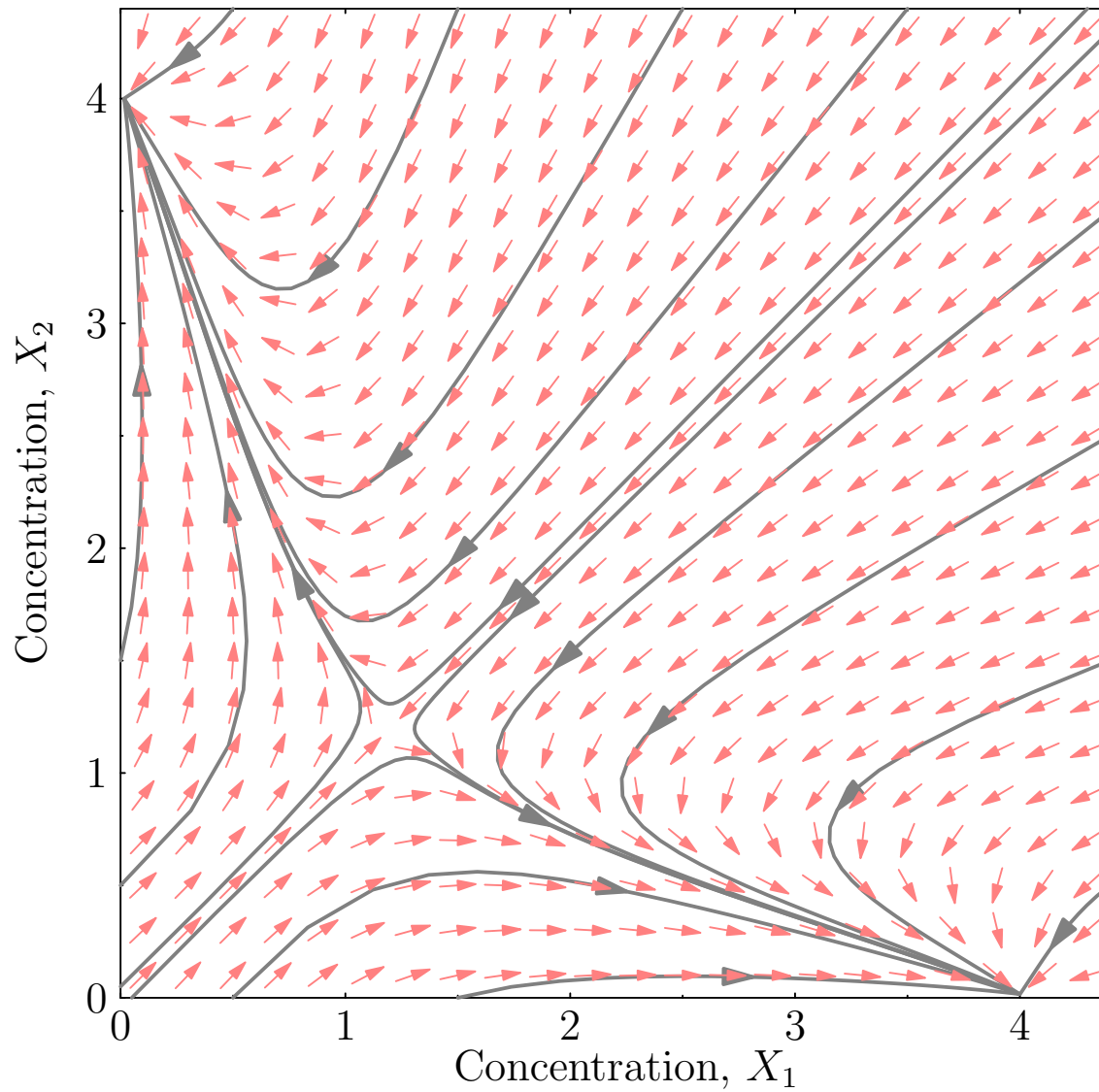
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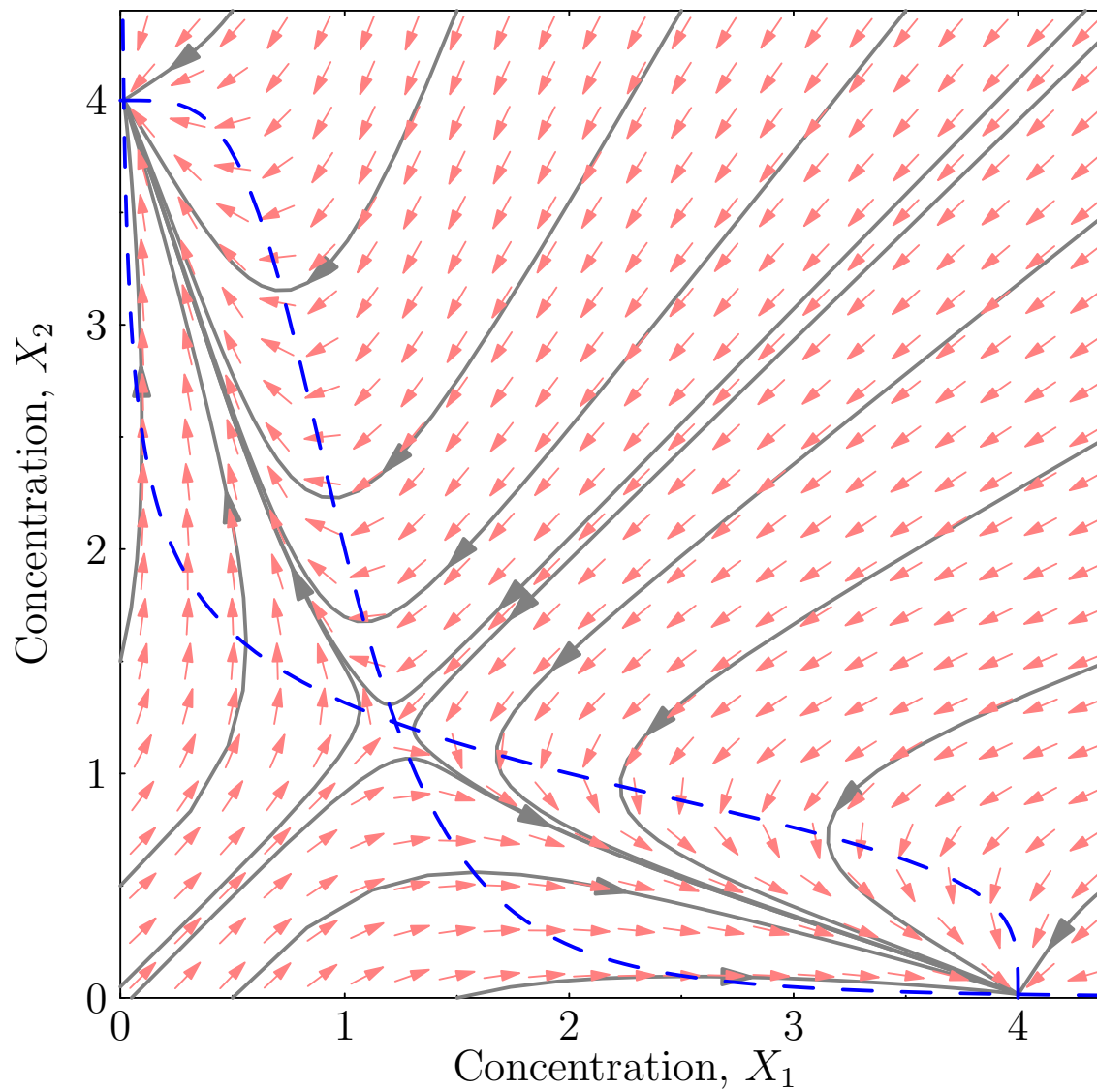
Nullclines and Bistability



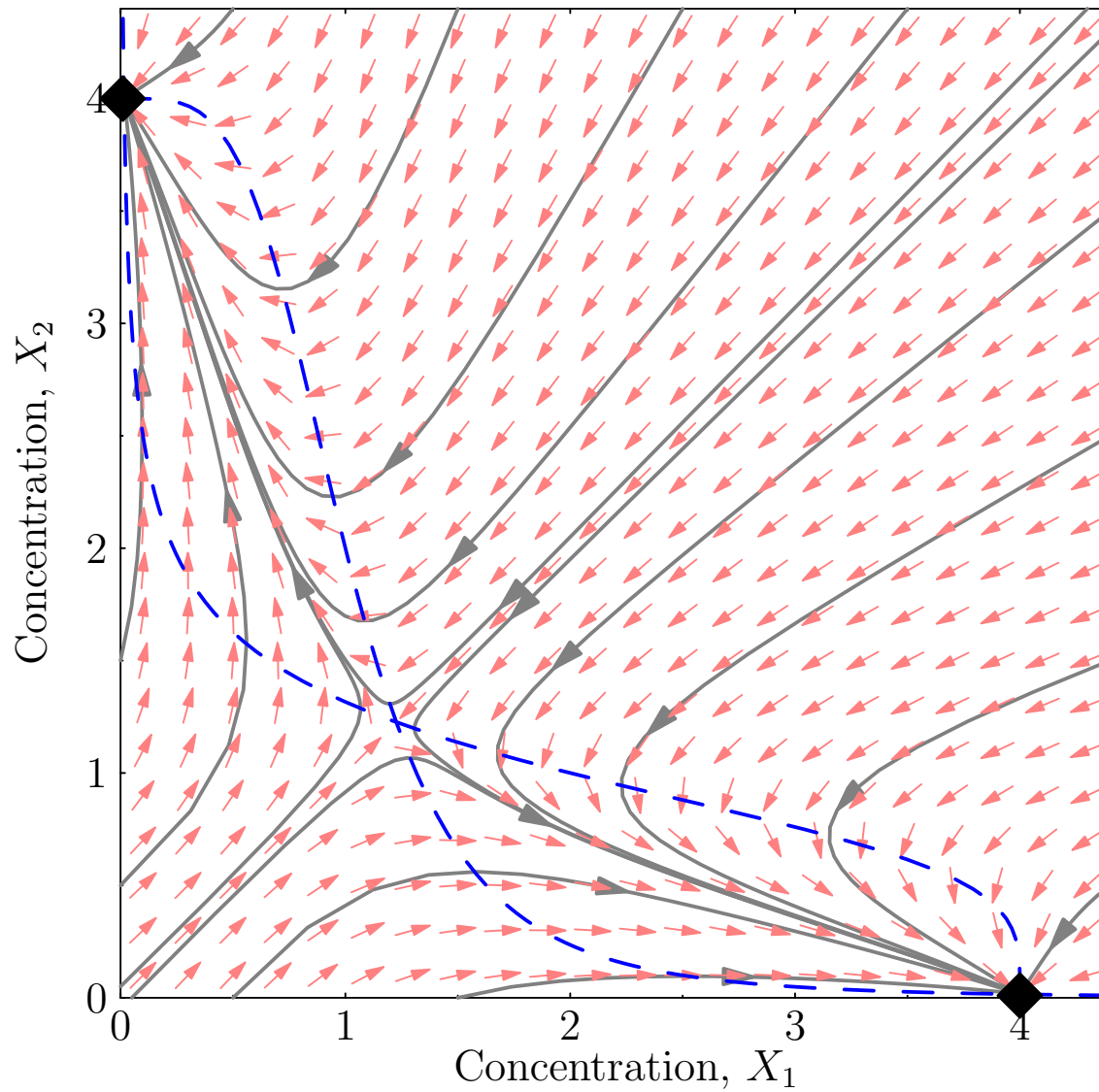
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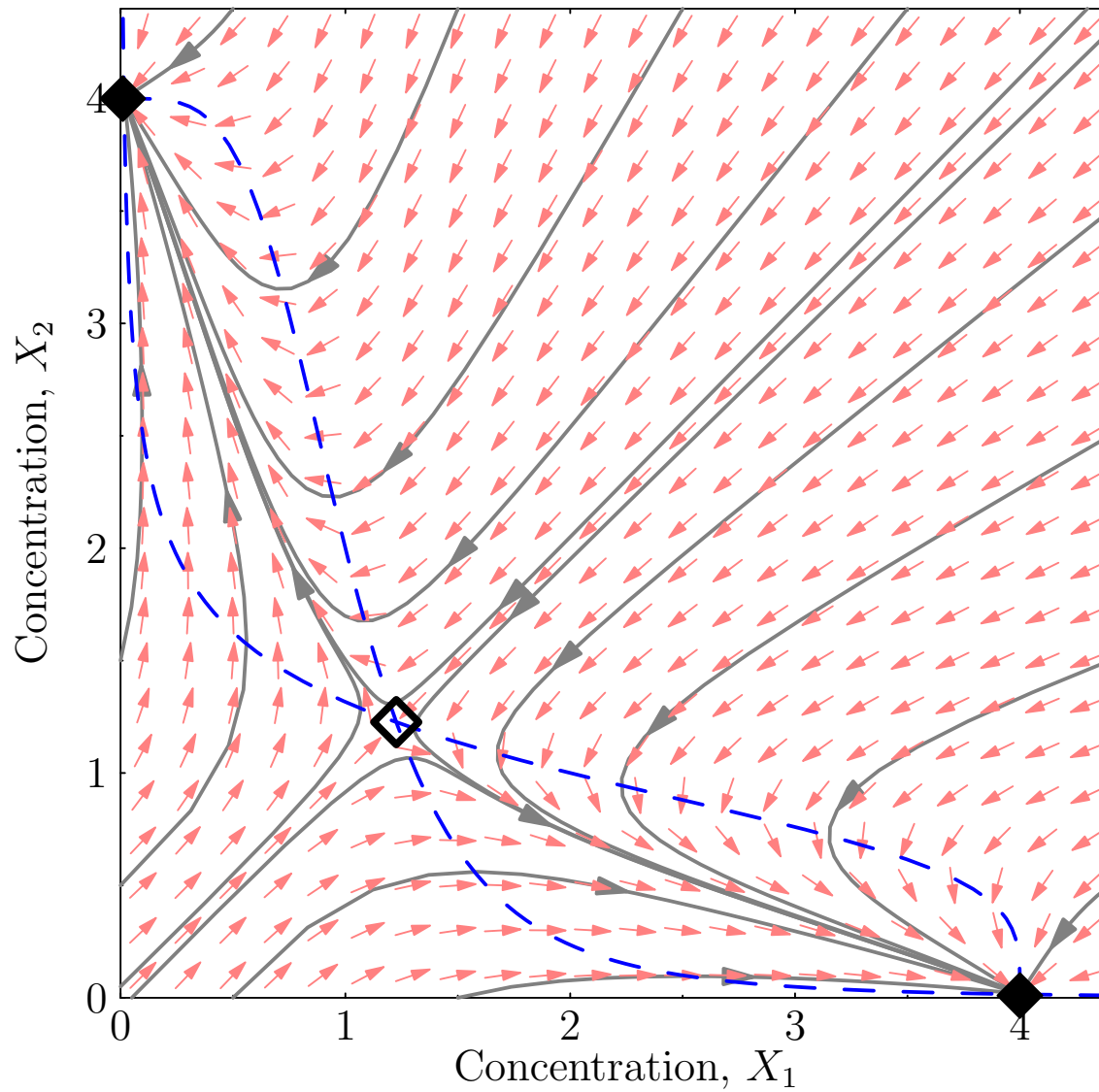
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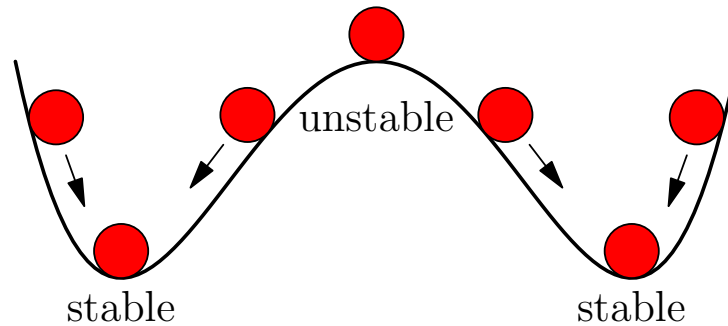


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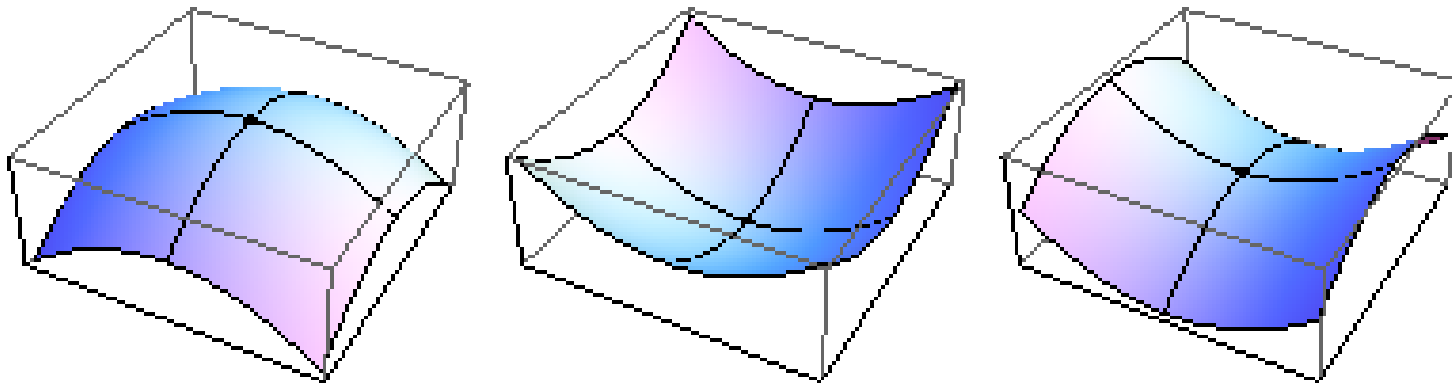


Stability

- Points can be stable or unstable

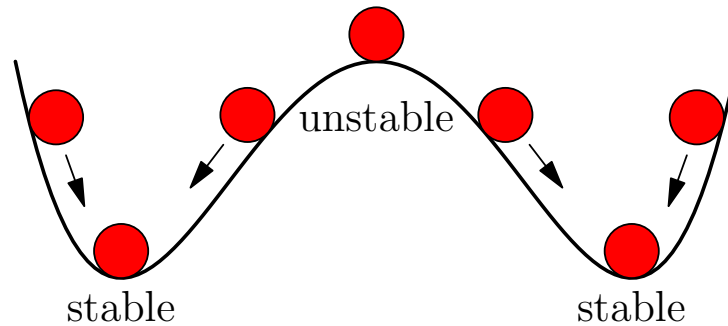


- Above 1D we can also have saddle-points

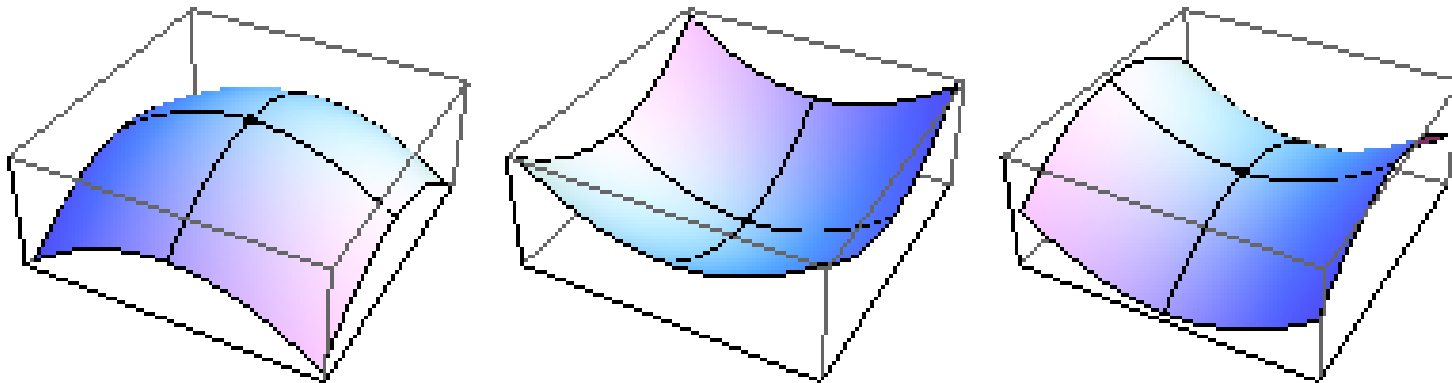


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Expand Around Fixed-Point

- To study stability at a fixed-point $\mathbf{X}^*$ we look at what happens when we make a small perturbation ϵ

$$\frac{dX_i}{dt} = f_i(\mathbf{X}^* + \epsilon) \approx f_i(\mathbf{X}^*) + \epsilon^T \nabla f_i(\mathbf{X}^*) = \epsilon^T \nabla f_i(\mathbf{X}^*)$$

since $f_i(\mathbf{X}^*) = 0$ (we are at a fixed-point)

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$$\epsilon^T \nabla f_i(\mathbf{X}^*) = \epsilon_1 \frac{\partial f_i(\mathbf{X}^*)}{\partial X_1} + \epsilon_2 \frac{\partial f_i(\mathbf{X}^*)}{\partial X_2} + \dots + \epsilon_n \frac{\partial f_i(\mathbf{X}^*)}{\partial X_n}$$

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Jacobian

- In vector form

$$\frac{d\mathbf{X}}{dt} = \mathbf{f}(\mathbf{X}^* + \boldsymbol{\epsilon}) = \mathbf{J} \boldsymbol{\epsilon} + O(\epsilon^2)$$

- where $\mathbf{J}$ is the **Jacobian** matrix with elements

$$J_{ij} = \left. \frac{\partial f_i(\mathbf{X})}{\partial X_j} \right|_{\mathbf{X}=\mathbf{X}^*}$$

- Note that this is just saying

$$\frac{dX_i}{dt} \approx \epsilon_1 \frac{\partial f_i(\mathbf{X}^*)}{\partial X_1} + \epsilon_2 \frac{\partial f_i(\mathbf{X}^*)}{\partial X_2} + \cdots + \epsilon_n \frac{\partial f_i(\mathbf{X}^*)}{\partial X_n}$$

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Linear Approximation

- Let $\epsilon(t) = \mathbf{X}(t) - \mathbf{X}^*$ then (since $d\epsilon(t) = d\mathbf{X}(t)$)

$$\frac{d\epsilon(t)}{dt} = \frac{d\mathbf{X}(t)}{dt} \approx \mathbf{J} \epsilon(t)$$

- To solve this we note that we can write most square matrices $\mathbf{J}$ as

$$\mathbf{J} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^{-1}$$

- Where $\mathbf{\Lambda}$ is a diagonal matrix whose elements $\Lambda_{ii} = \lambda_i$ are eigenvalues of $\mathbf{J}$ and the columns of $\mathbf{V}$ are the eigenvectors

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Eigensystem of a General Matrix

- For a general (non-symmetric) matrix $\mathbf{M}$ the vector $\mathbf{v}_i$ is an (right-handed) eigenvector if

$$\mathbf{M} \mathbf{v}_i = \lambda_i \mathbf{v}_i$$

- There are also left-handed eigenvectors $\mathbf{u}_i$ with the same eigenvalues, λ_i , such that

$$\mathbf{u}_i^T \mathbf{M} = \lambda_i \mathbf{u}_i^T$$

- In general $\mathbf{v}_i \neq \mathbf{u}_i$ although if $\lambda_i \neq \lambda_j$ then $\mathbf{u}_i^T \mathbf{v}_j = 0$
- The eigenvalues and eigen-vectors are either real or exist in complex conjugate pairs $\lambda_j = \lambda_i^*$ and $\mathbf{v}_j = \mathbf{v}_i^*$

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Eigensystem of a General Matrix

- For a general (non-symmetric) matrix $\mathbf{M}$ the vector $\mathbf{v}_i$ is an (right-handed) eigenvector if

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Matrix Decomposition

- If we write

$$\mathbf{V} = (\mathbf{v}_1 \quad \mathbf{v}_2 \quad \cdots \quad \mathbf{v}_n) \quad \mathbf{\Lambda} = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}$$

- Then $\mathbf{M} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^{-1} = \mathbf{V} \mathbf{\Lambda} \mathbf{U} = \mathbf{U}^{-1} \mathbf{\Lambda} \mathbf{U}$

- where $\mathbf{U} = \mathbf{V}^{-1}$ has rows equal to $\mathbf{u}_i^T$

- Multiplying $\mathbf{M}$ on the right by $\mathbf{V}$ we find

$$\mathbf{M} \mathbf{V} = (\mathbf{V} \mathbf{\Lambda} \mathbf{V}^{-1}) \mathbf{V} = \mathbf{V} \mathbf{\Lambda} \quad \Rightarrow \quad \mathbf{M} \mathbf{v}_i = \lambda_i \mathbf{v}_i$$

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Linearised Solution

- Multiply the linear approximation by $\mathbf{V}^{-1}$

$$\frac{d\boldsymbol{\epsilon}(t)}{dt} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{-1}\boldsymbol{\epsilon}(t) \quad \Rightarrow \quad \frac{d\mathbf{V}^{-1}\boldsymbol{\epsilon}(t)}{dt} = \boldsymbol{\Lambda}\mathbf{V}^{-1}\boldsymbol{\epsilon}(t)$$

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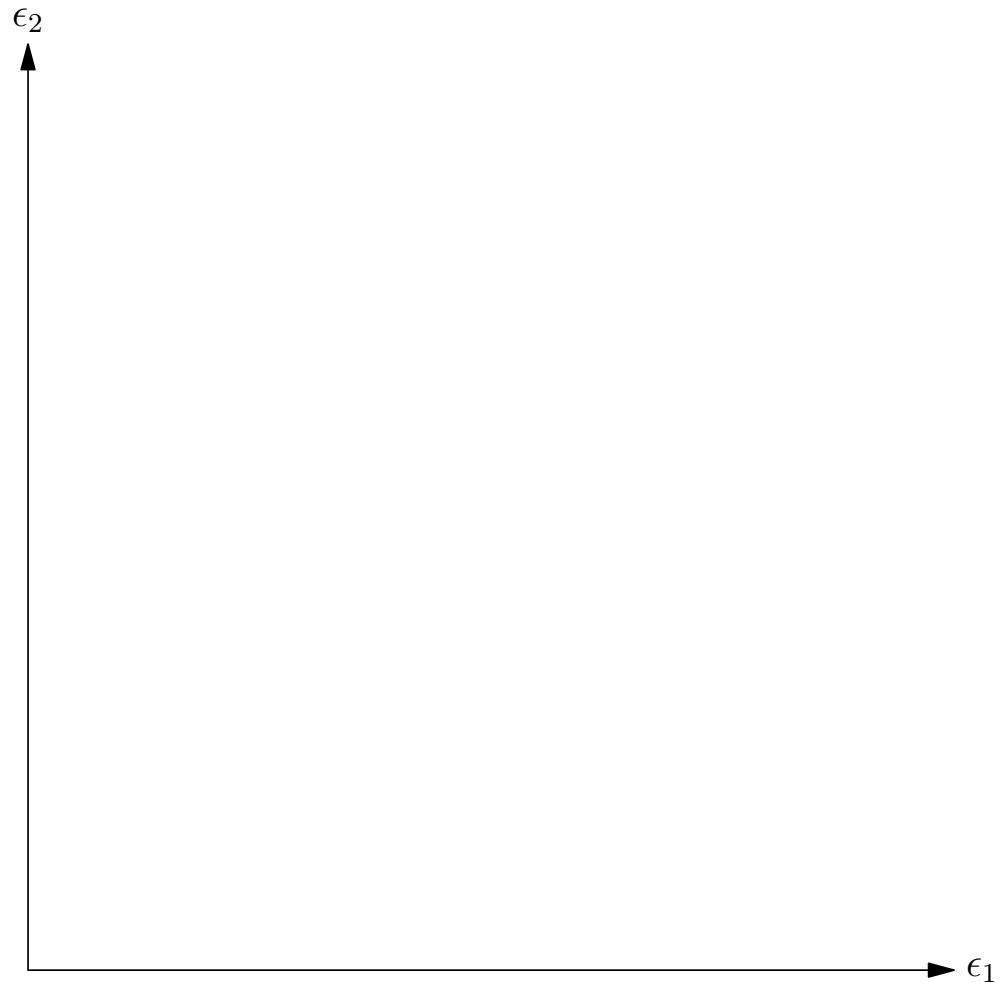
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Example

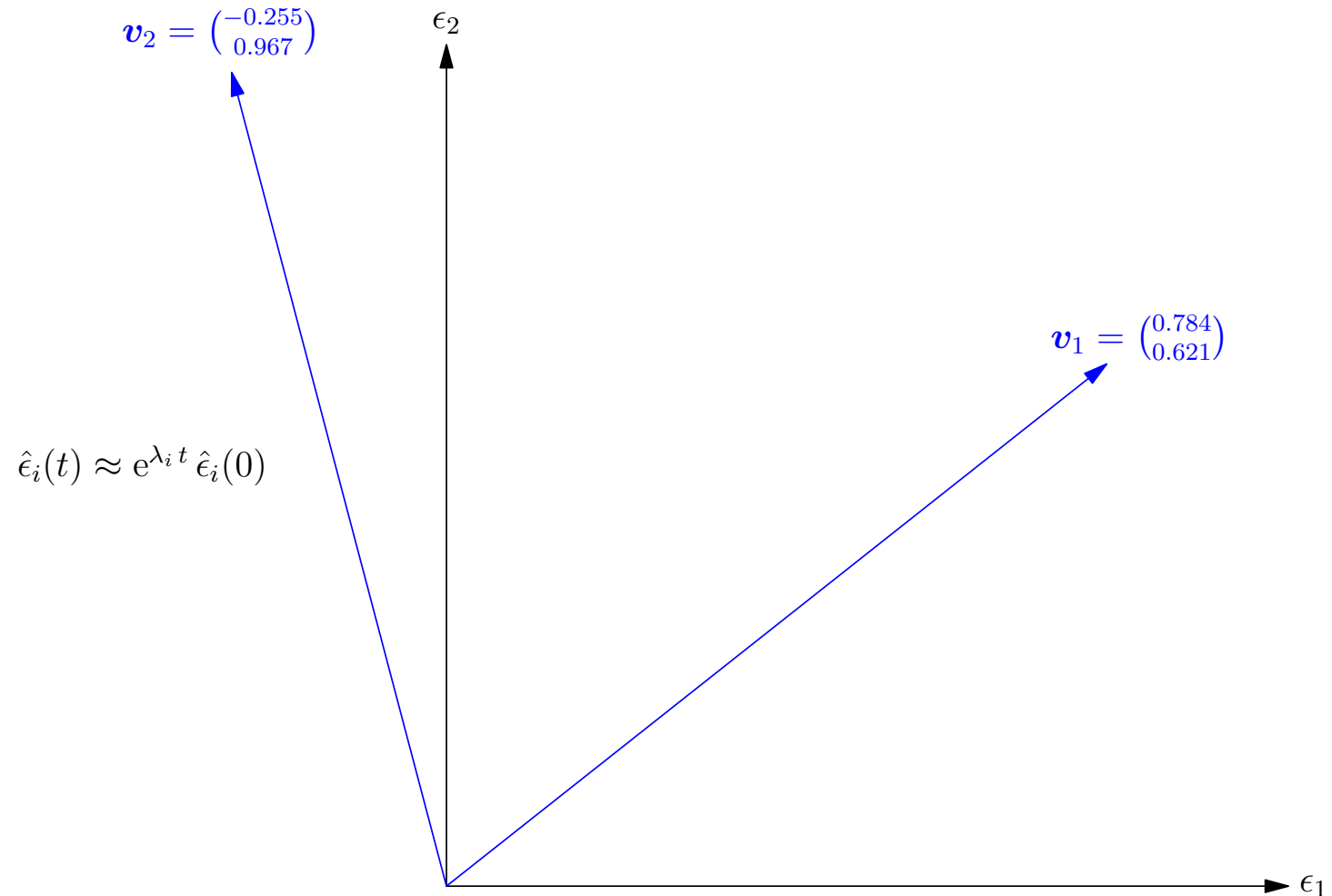
$$\mathbf{J} = \begin{pmatrix} 2 & 1 \\ 3 & -1 \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.784 & -0.255 \\ 0.621 & 0.967 \end{pmatrix} \begin{pmatrix} 2.791 & 0 \\ 0 & -1.791 \end{pmatrix} \begin{pmatrix} 1.055 & 0.278 \\ -0.677 & 0.856 \end{pmatrix}$$

$$\hat{\epsilon}_i(t) \approx e^{\lambda_i t} \hat{\epsilon}_i(0)$$



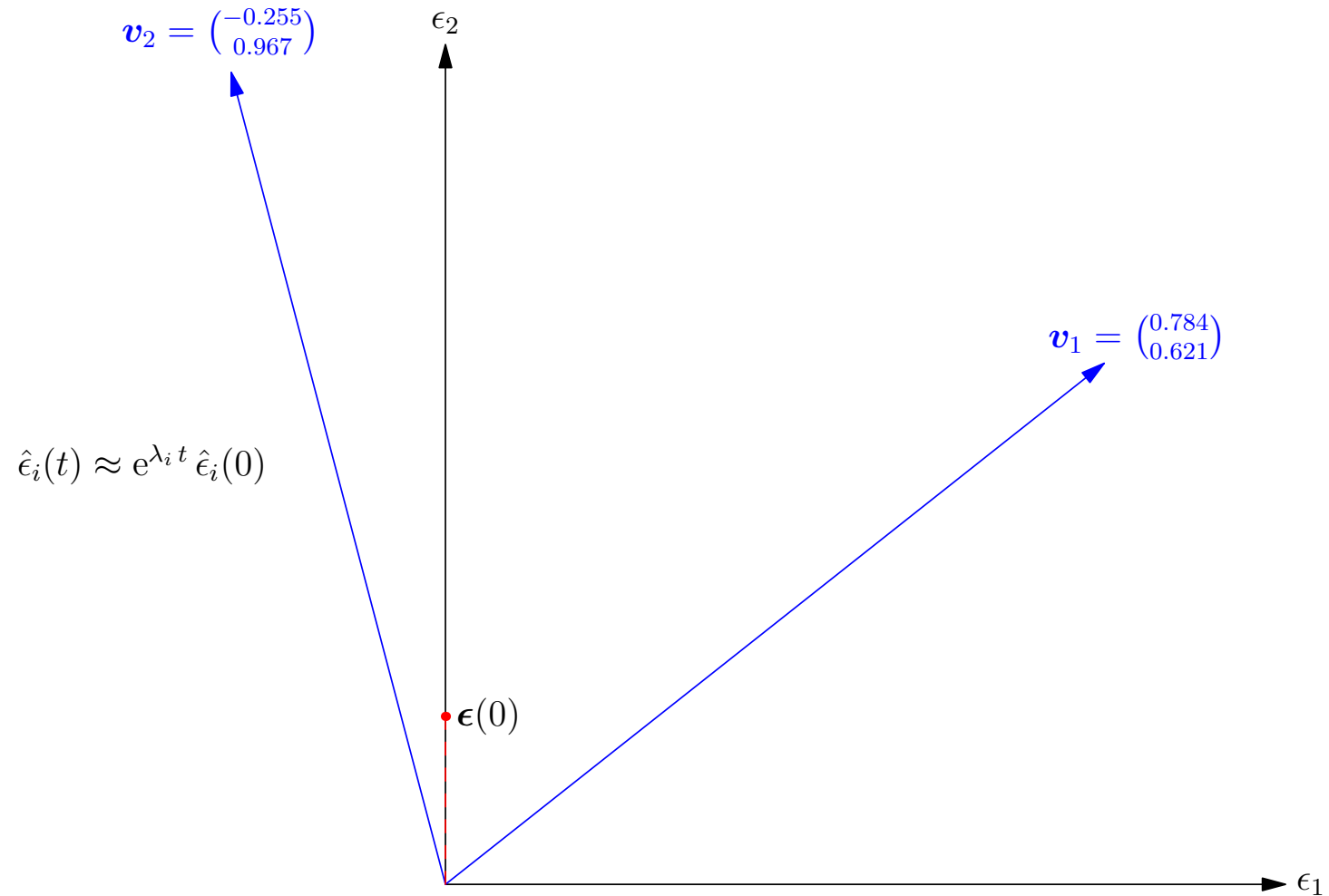
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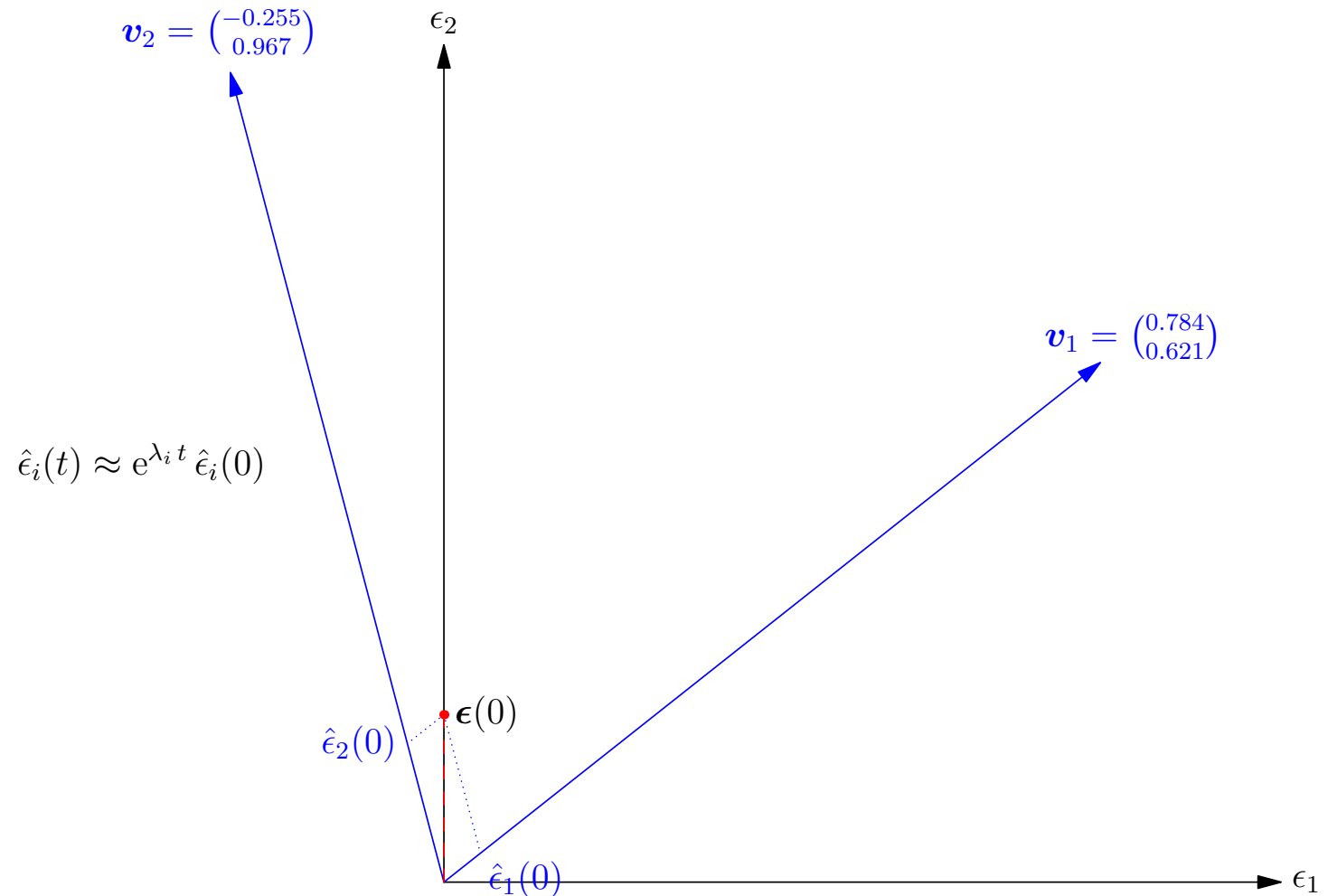
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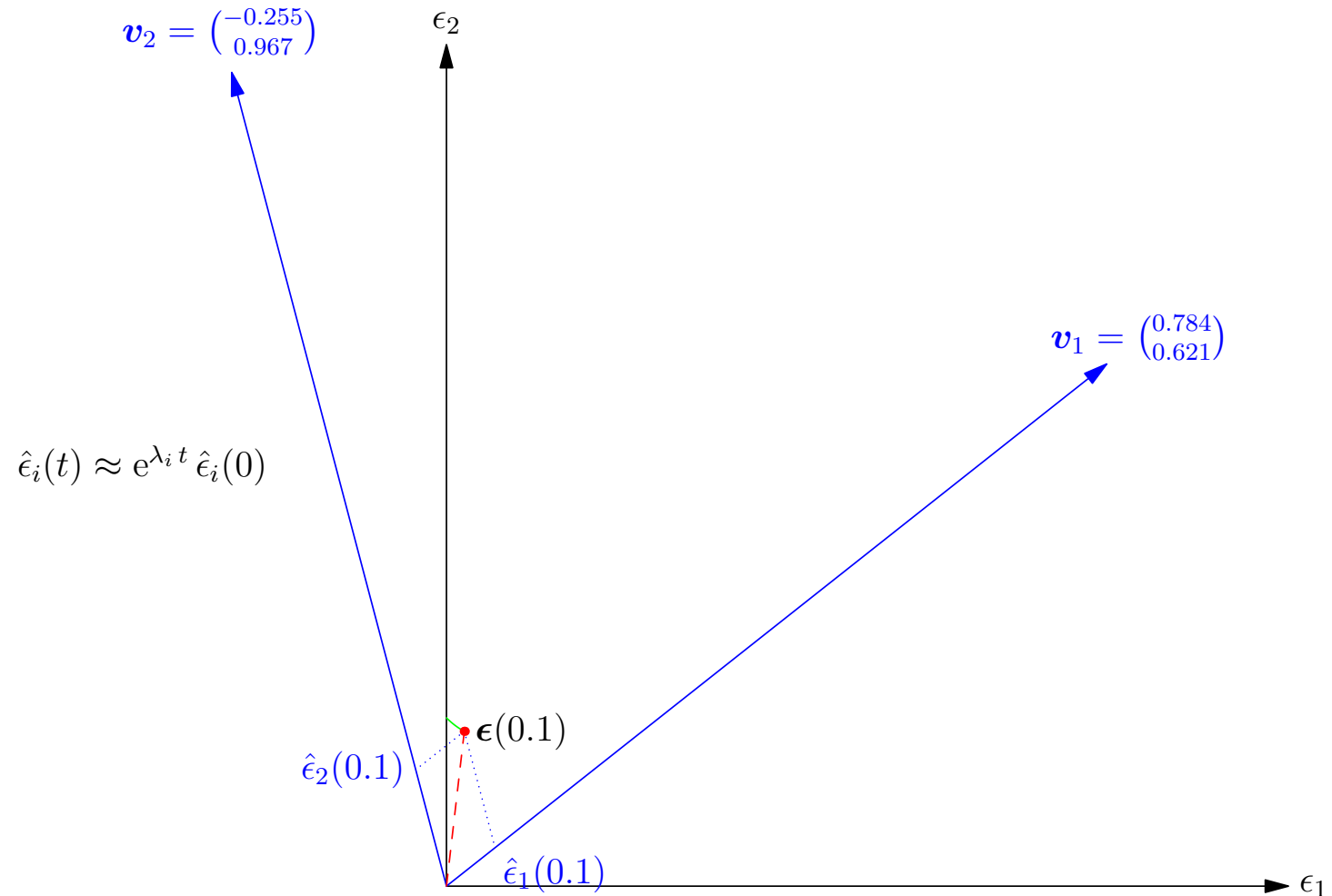
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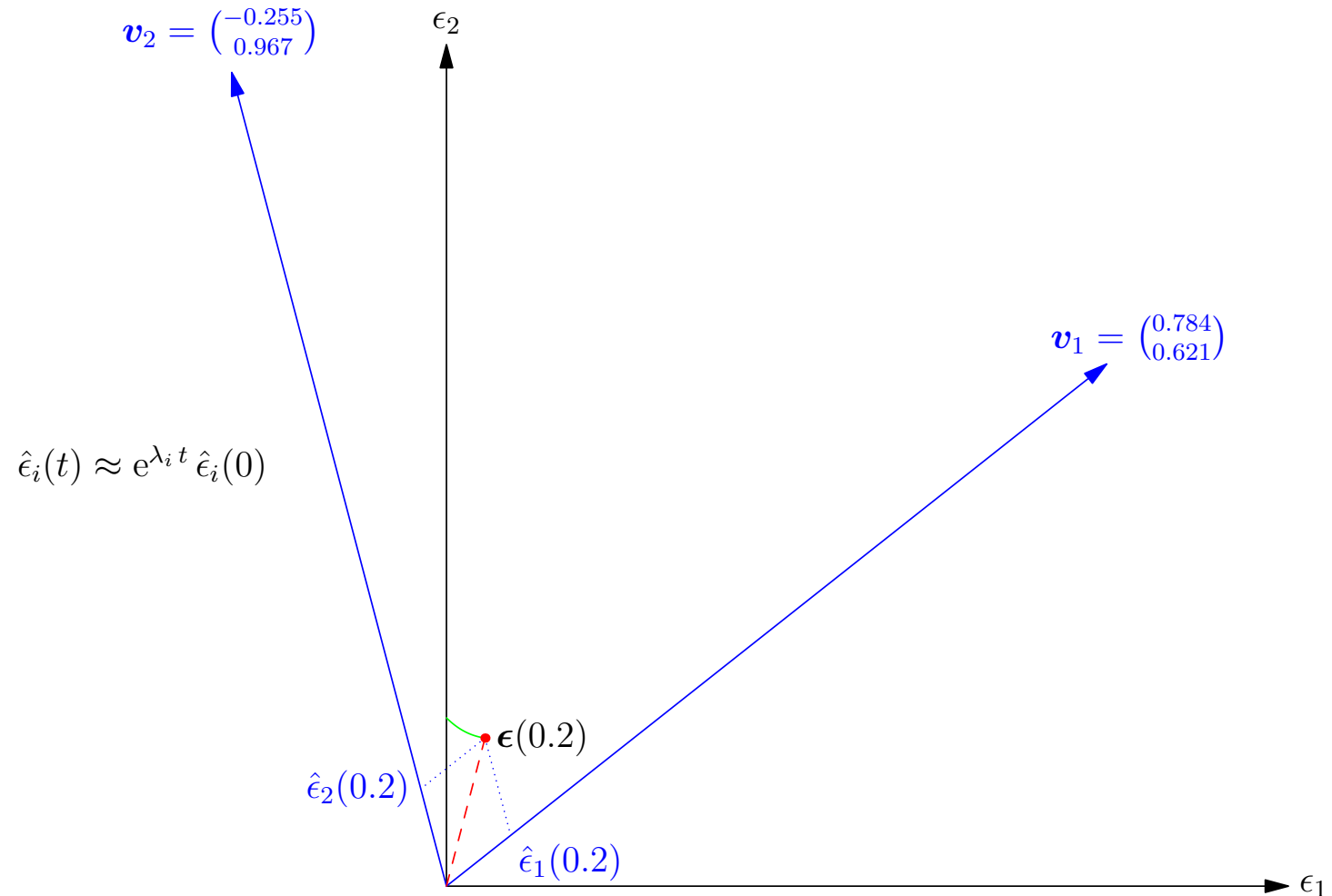
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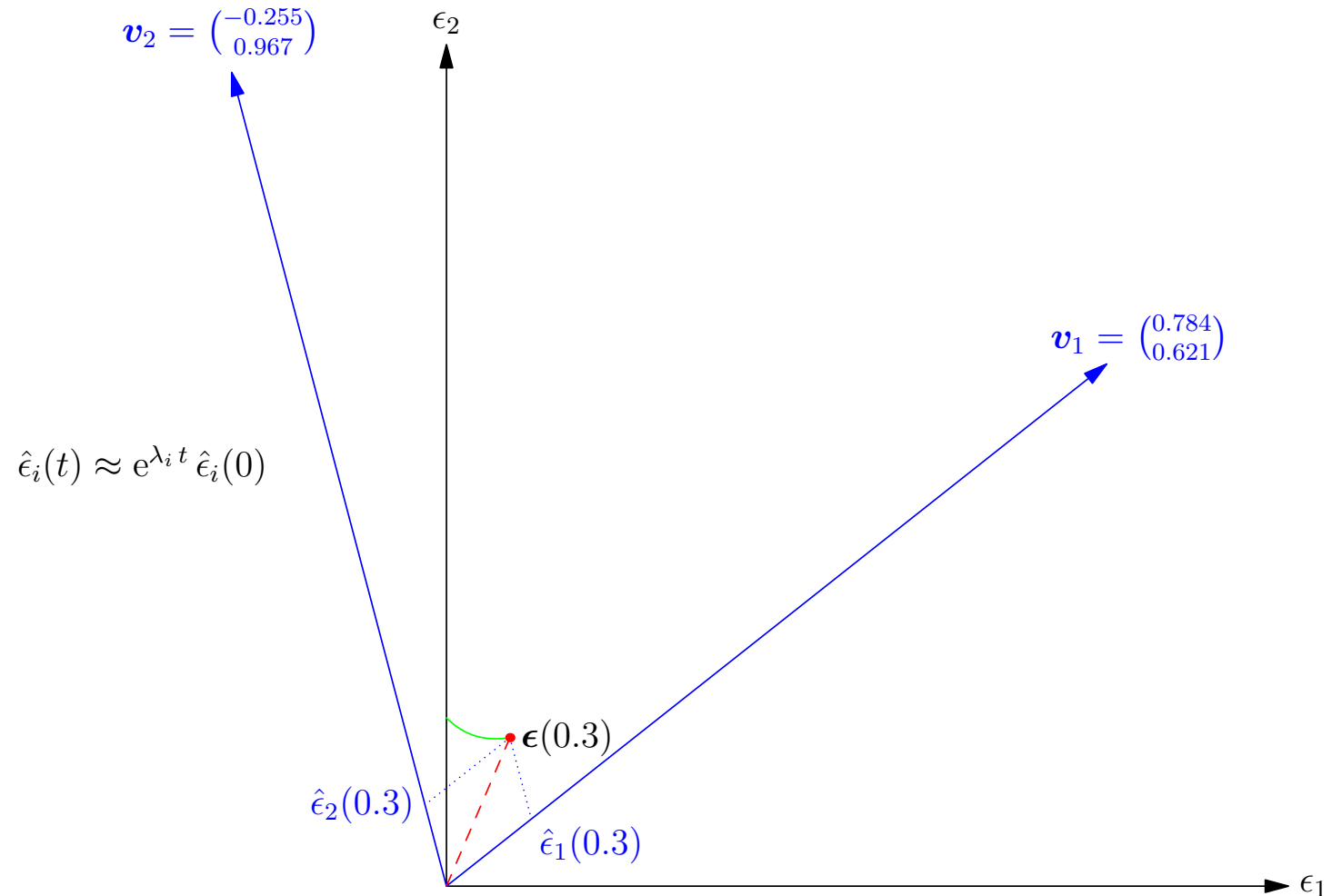
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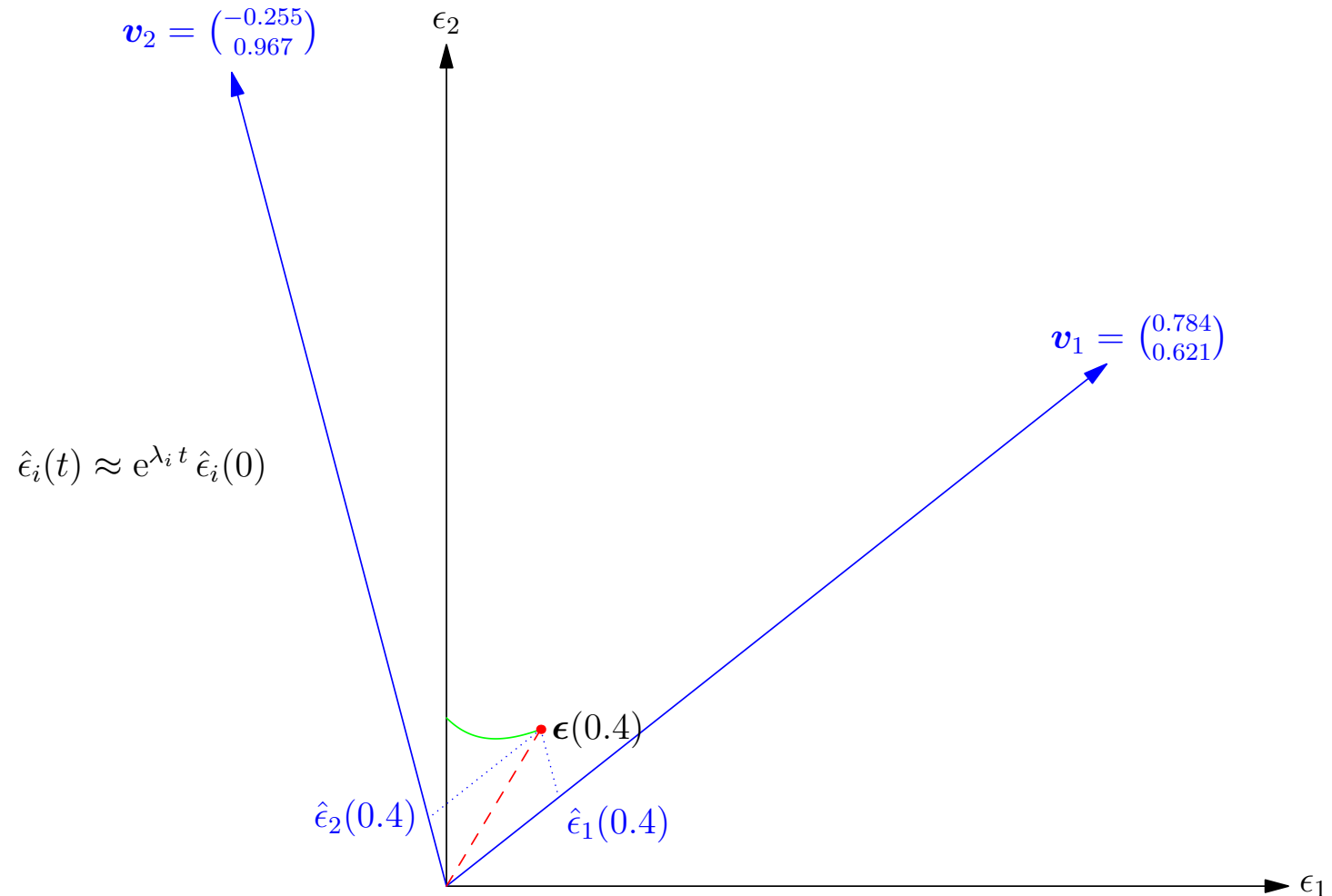
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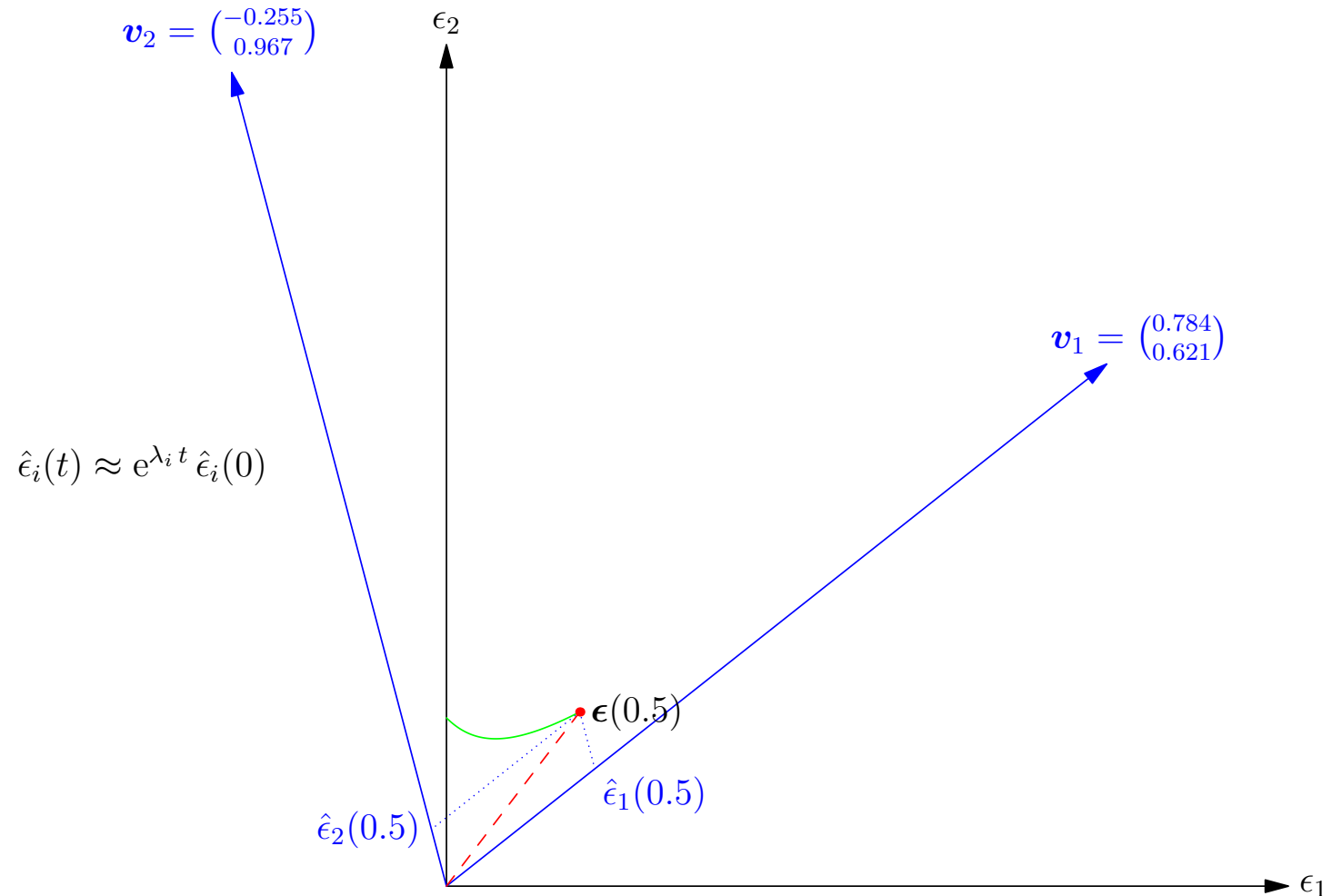
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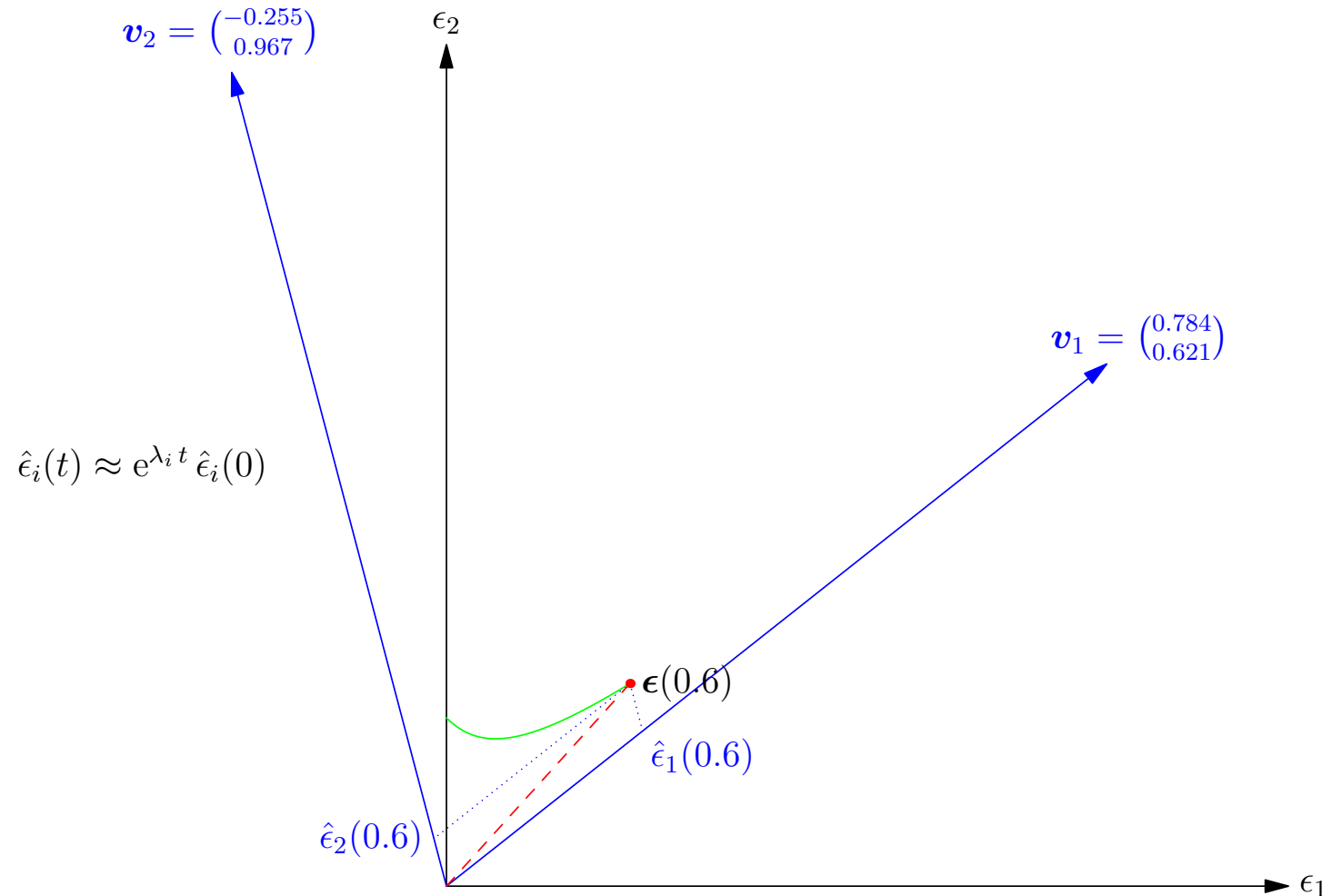
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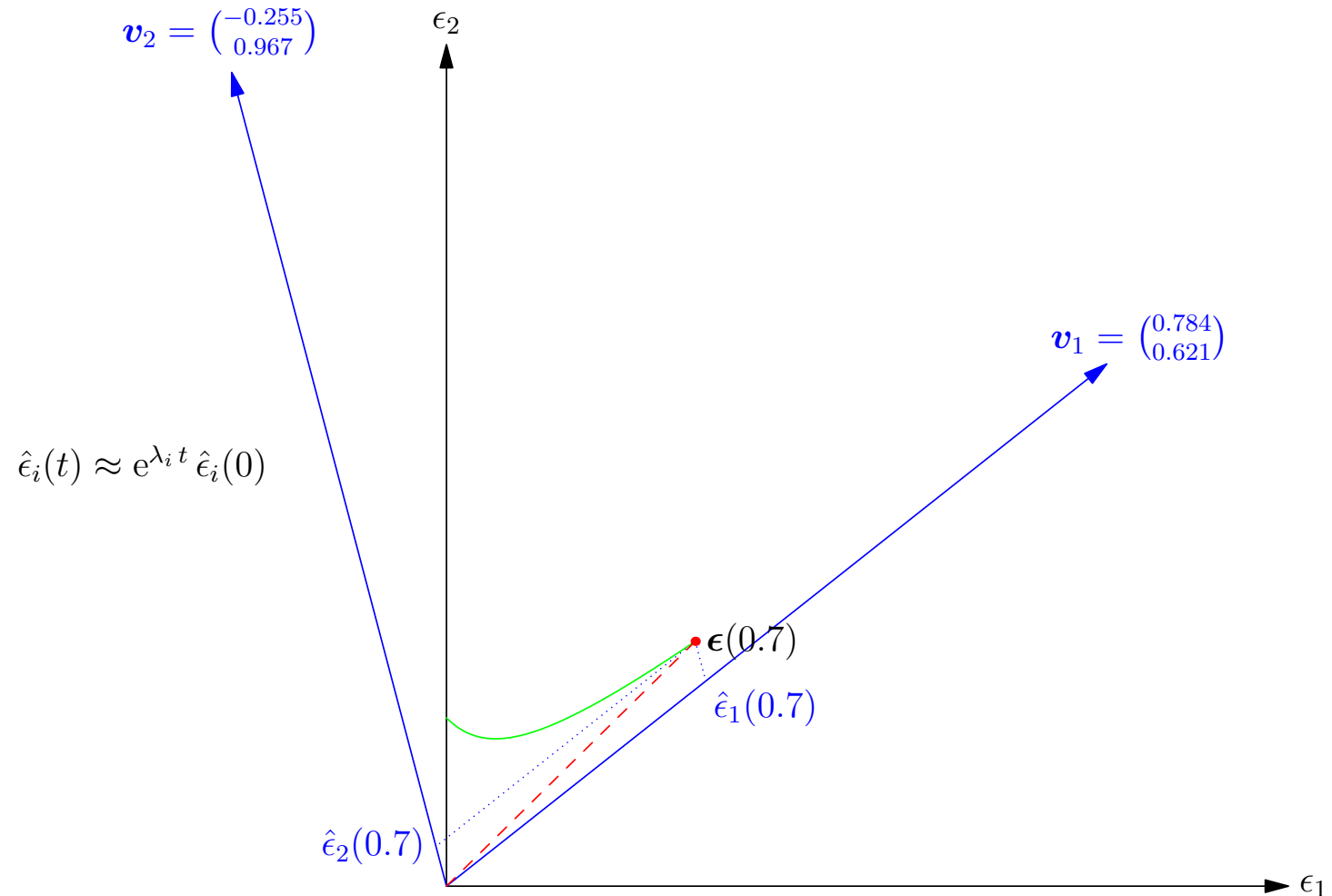
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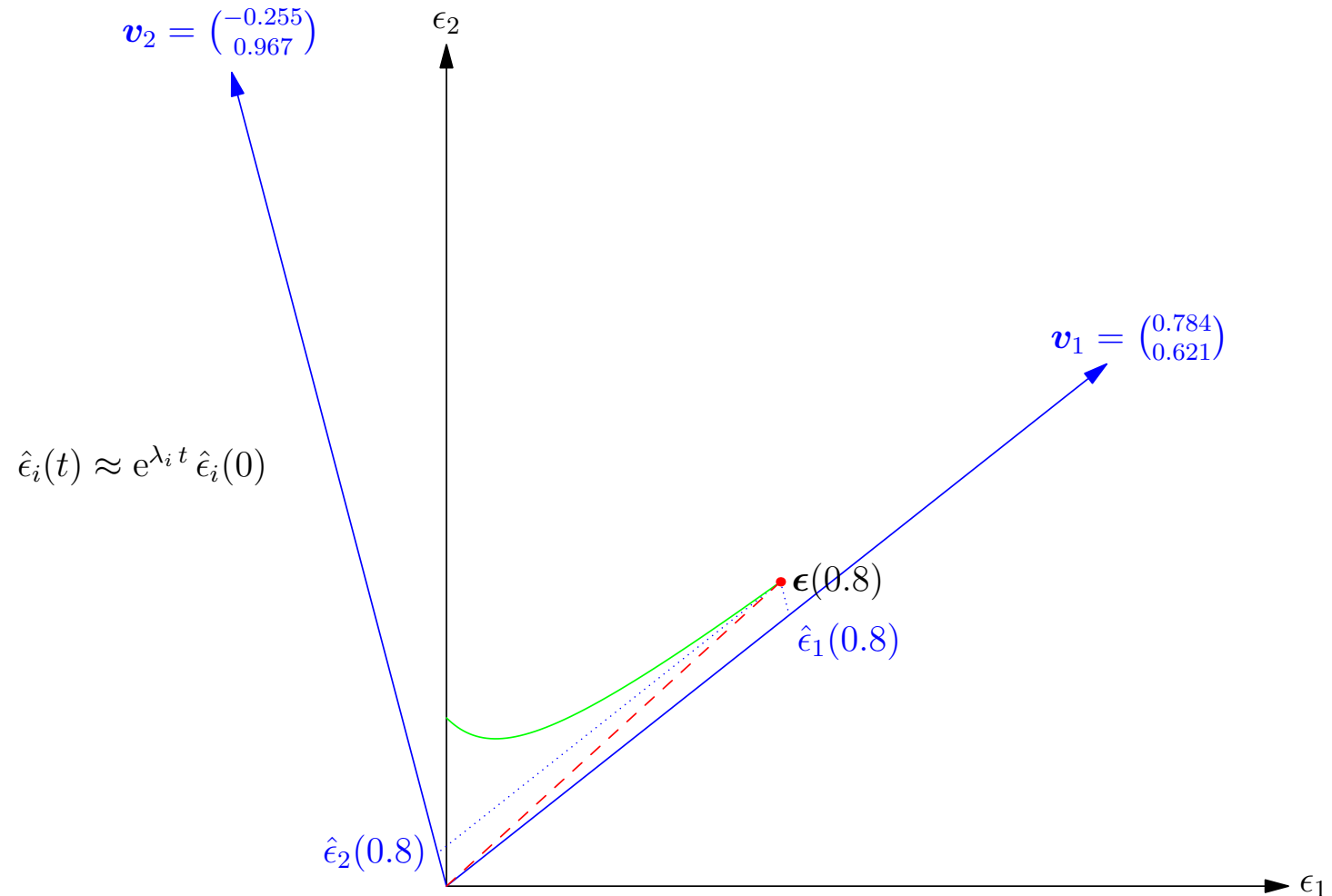
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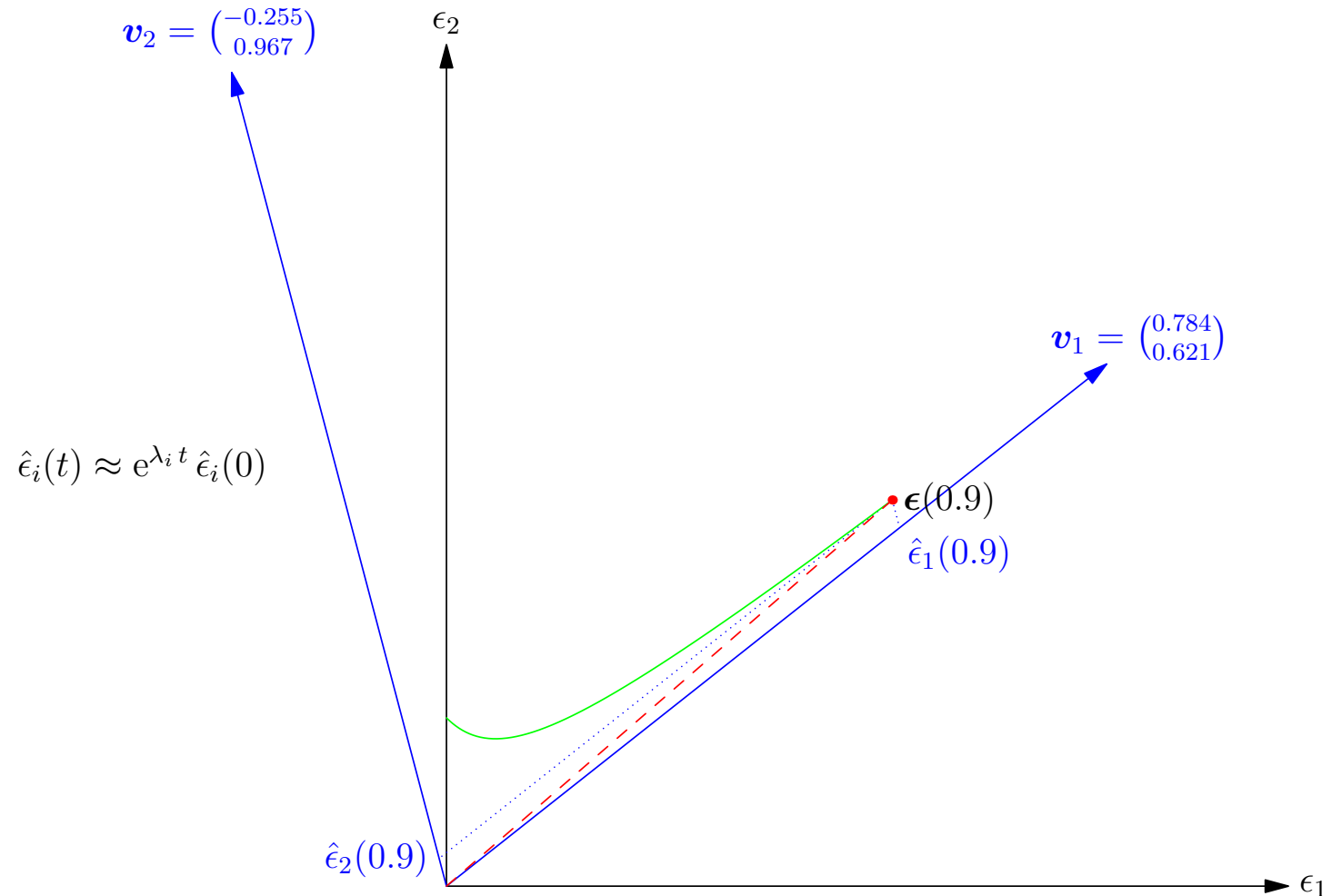
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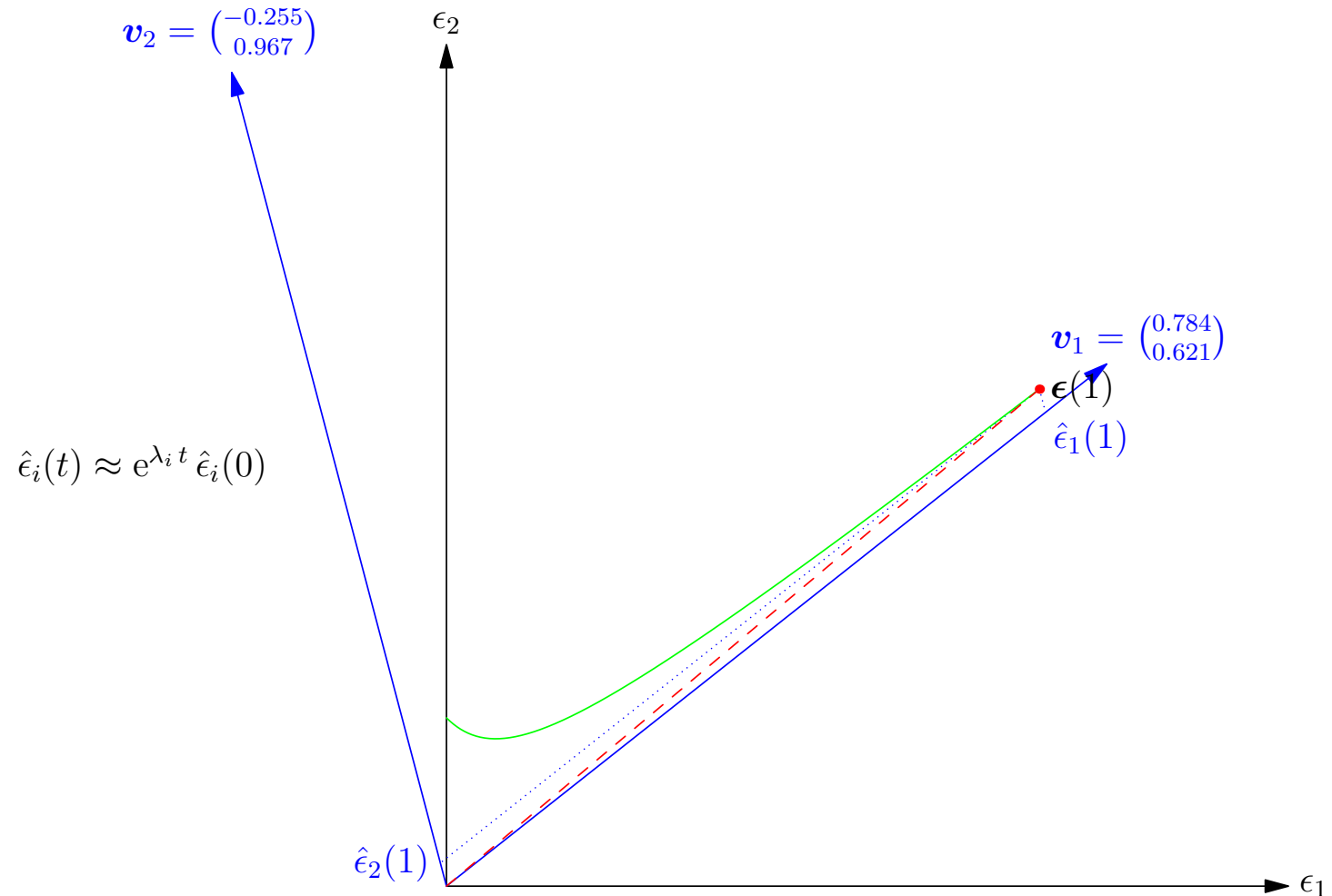
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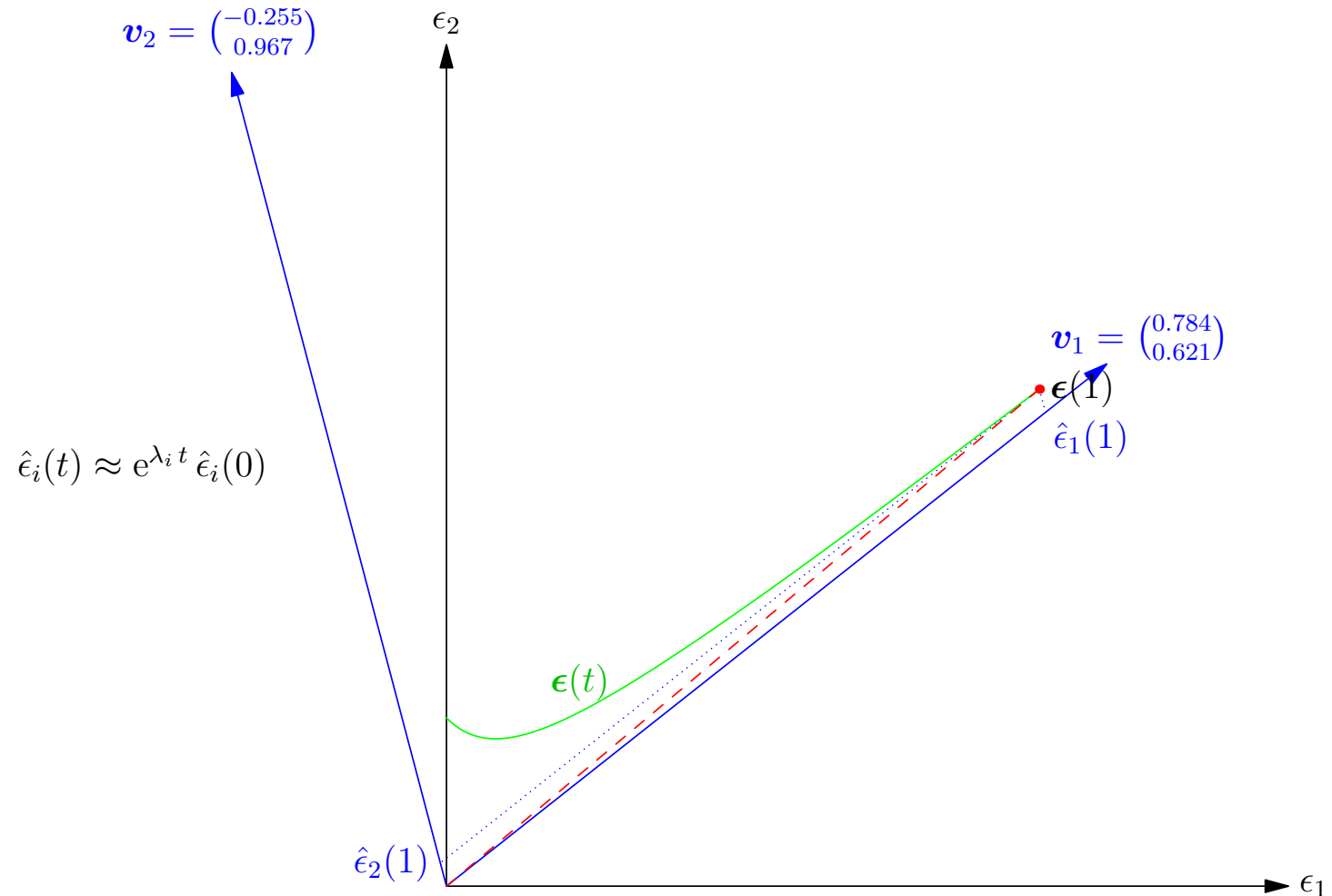
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Stability Analysis

- If $\Re(\lambda_i) < 0$ for all i the fixed point is stable (all sufficiently small perturbations shrink)
- If there exists an i such that $\Re(\lambda_i) > 0$ then the fixed point is unstable
- If $\Re(\lambda_i) < 0$ and $\Re(\lambda_j) > 0$ then we are at an unstable saddle point
- Note that the Jacobian is not symmetric which means the eigenvalues can occur in complex conjugate pairs (i.e. $\lambda_i^* = \lambda_j$)
- In this case the trajectories will either spiral in to the fixed point or spiral away from the fixed point

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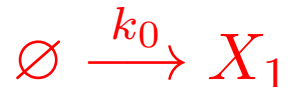
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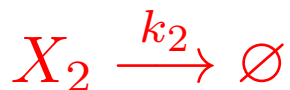
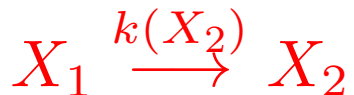
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Oscillator

- Consider a chain with a positive feedback



$$k(X_2) = k_1 (1 + (X_2/K)^n)$$



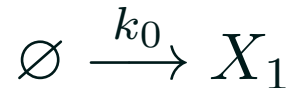
$$\frac{dX_1(t)}{dt} = k_0 - k_1 \left(1 + \left(\frac{X_2(t)}{K} \right)^n \right) X_1(t)$$

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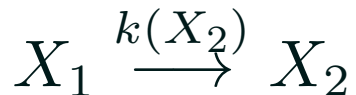
- Consider $k_0 = 8$, $k_1 = 1$, $K = 1$, $k_2 = 5$ and $n = 2$
- Large values of X_2 will result in most of X_1 being turned into X_2 , then X_2 will decay rapidly allowing X_1 to increase

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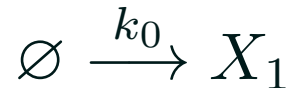
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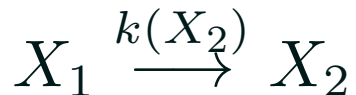
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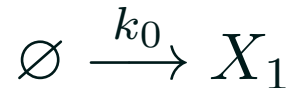
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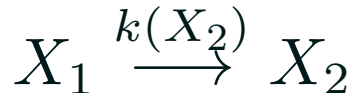
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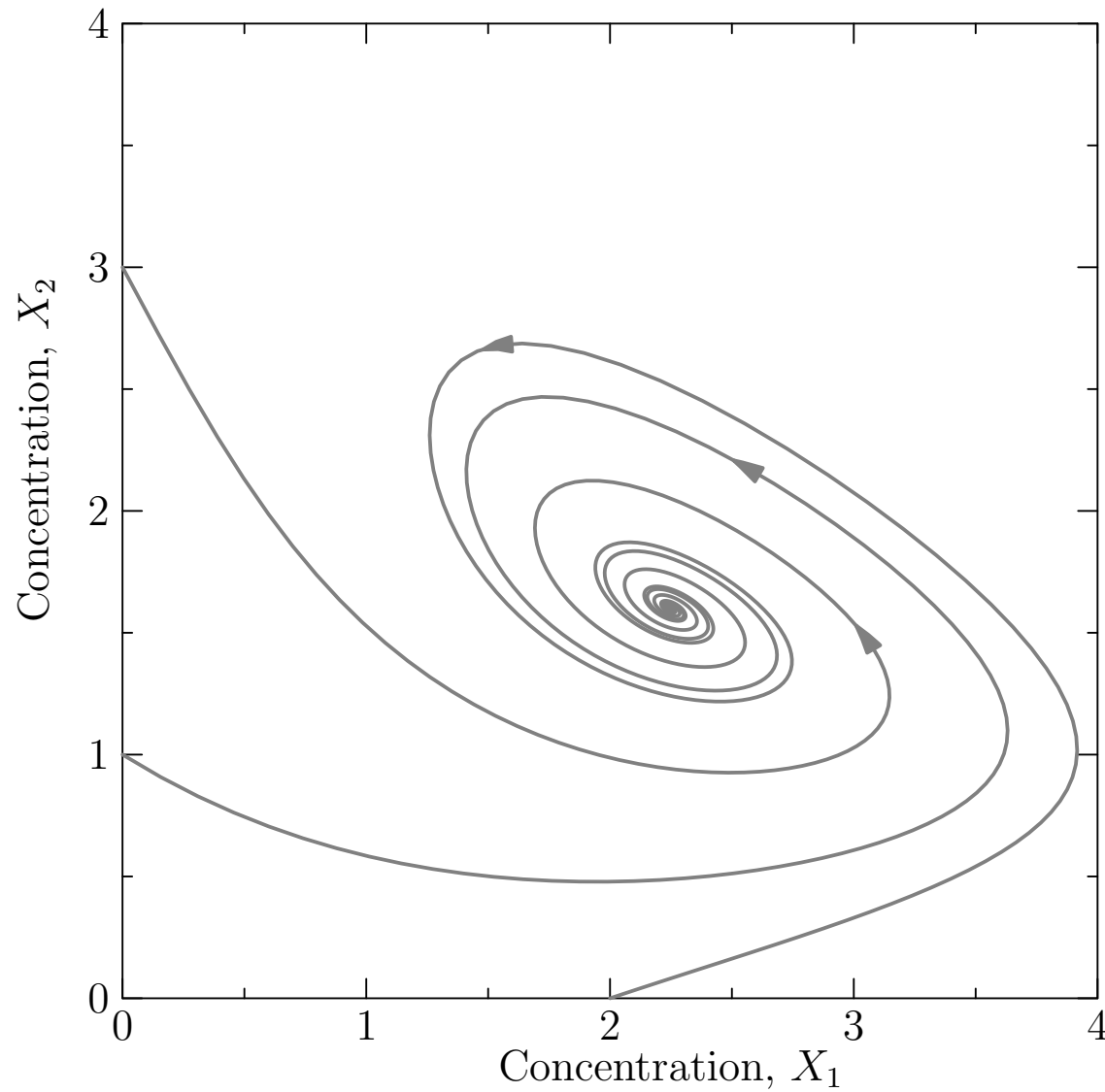


$$\frac{dX_1(t)}{dt} = k_0 - k_1 \left(1 + \left(\frac{X_2(t)}{K} \right)^n \right) X_1(t)$$

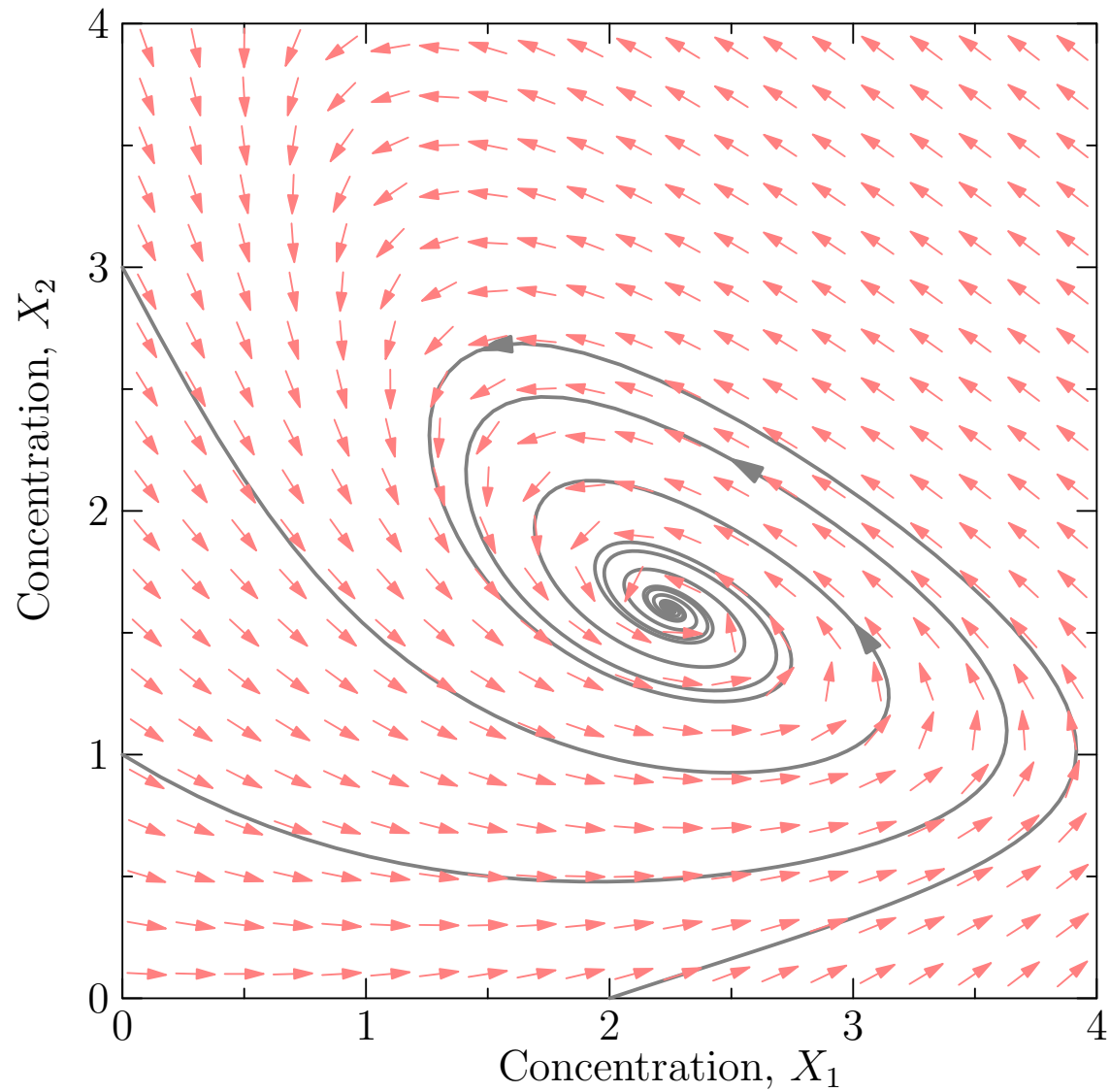
$$\frac{dX_2(t)}{dt} = k_1 \left(1 + \left(\frac{X_2(t)}{K} \right)^n \right) X_1(t) - k_2 X_2(t)$$

- Consider $k_0 = 8$, $k_1 = 1$, $K = 1$, $k_2 = 5$ and $n = 2$
- Large values of X_2 will result in most of X_1 being turned into X_2 , then X_2 will decay rapidly allowing X_1 to increase

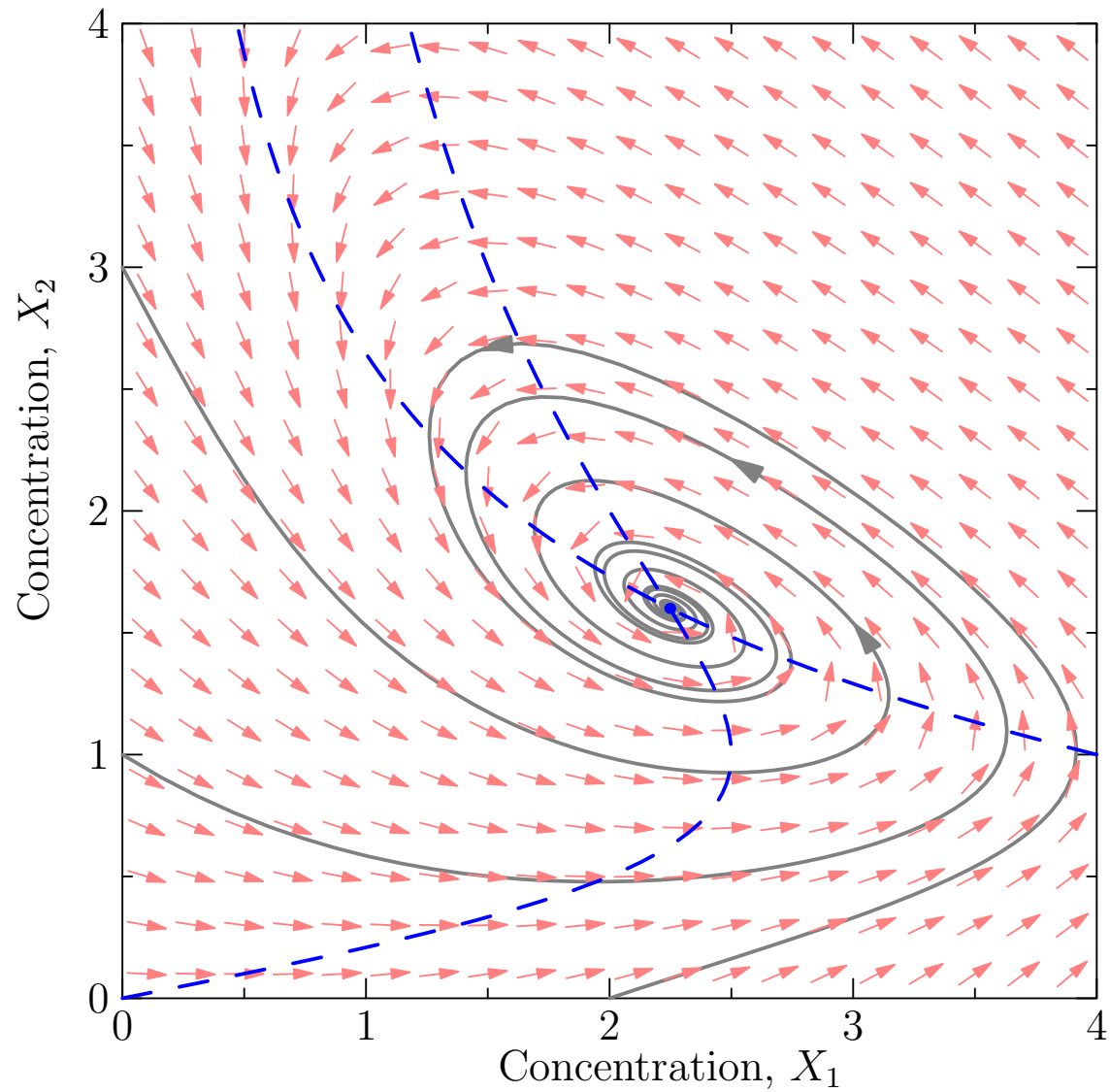
Nullclines and Bistability



Nullclines and Bistability



Nullclines and Bistability



Fixed Point Equations

- Our dynamical equations are

$$\begin{aligned}\frac{dX_1(t)}{dt} &= f_1(\mathbf{X}(t)) = k_0 - k_1 \left(1 + \left(\frac{X_2(t)}{K} \right)^n \right) X_1(t) \\ \frac{dX_2(t)}{dt} &= f_2(\mathbf{X}(t)) = k_1 \left(1 + \left(\frac{X_2(t)}{K} \right)^n \right) X_1(t) - k_2 X_2(t)\end{aligned}$$

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- The fixed point is at $\mathbf{X}^* = \left(\frac{k_0}{k_1(1+(k_0/(K k_2))^n)}, \frac{k_0}{k_2} \right) = \left(\frac{200}{89}, \frac{8}{5} \right)$
- The Jacobian $J_{ij} = \partial f_i(\mathbf{X}(t))/\partial X_j$ is

$$\mathbf{J} = \begin{pmatrix} -k_1 \left(1 + \left(\frac{X_2^*}{K} \right)^n \right) & -\frac{n k_1}{K} \left(\frac{X_2^*}{K} \right)^{n-1} X_1^* \\ k_1 \left(1 + \left(\frac{X_2^*}{K} \right)^n \right) & \frac{n k_1}{K} \left(\frac{X_2^*}{K} \right)^{n-1} X_1^* - k_2 \end{pmatrix} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix}$$

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Eigen Decomposition

- Plugging this into Matlab (using $\text{eig}(\mathbf{J})$)

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V} \begin{pmatrix} -0.68 + 4.16i & 0 \\ 0 & -0.68 - 4.16i \end{pmatrix} \mathbf{V}^{-1}$$

- where

$$\mathbf{V} = \begin{pmatrix} 0.82 & 0.82 \\ -0.33 - 0.47i & -0.33 + 0.47i \end{pmatrix}$$

- Or $T = \text{tr } \mathbf{J} = J_{11} + J_{22}$ and $D = \det(\mathbf{J}) = J_{11} J_{22} - J_{12} J_{21}$
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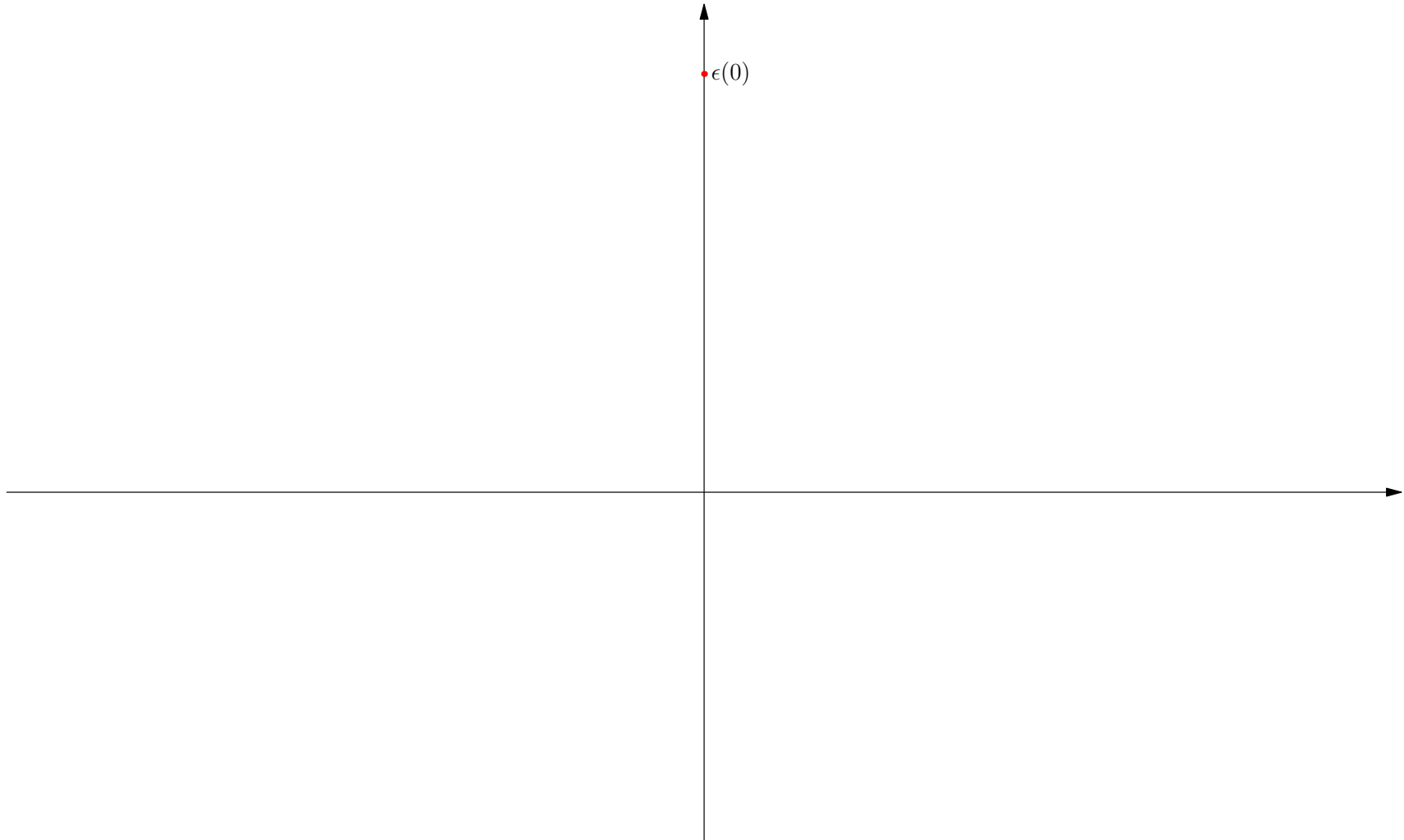
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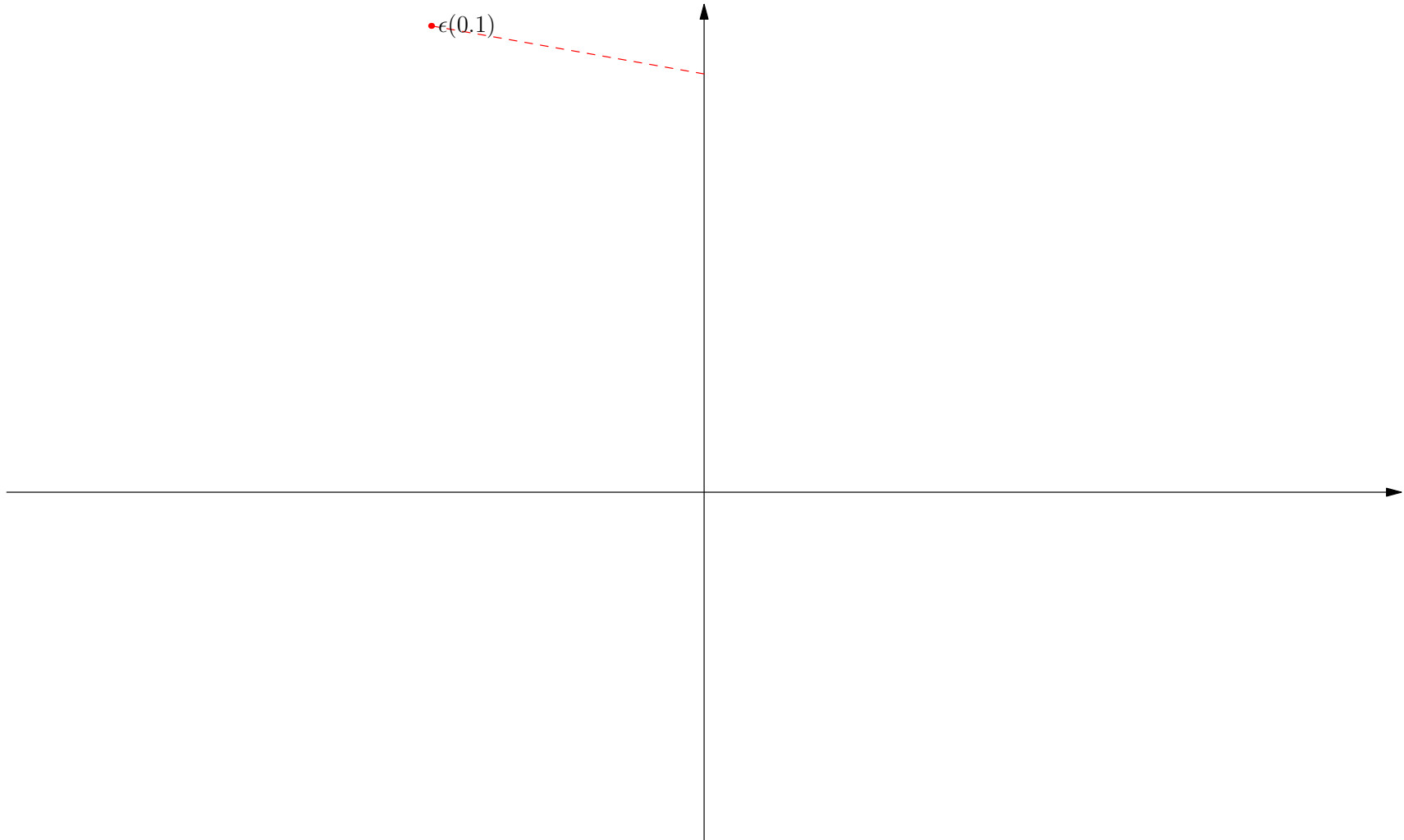
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473i & -0.327 + 0.473i \end{pmatrix} \begin{pmatrix} -0.685 + 4.163i & 0 \\ 0 & -0.685 - 4.163i \end{pmatrix} \begin{pmatrix} 0.611 + 0.422i & 1.056i \\ 0.611 - 0.422i & -1.056i \end{pmatrix}$$



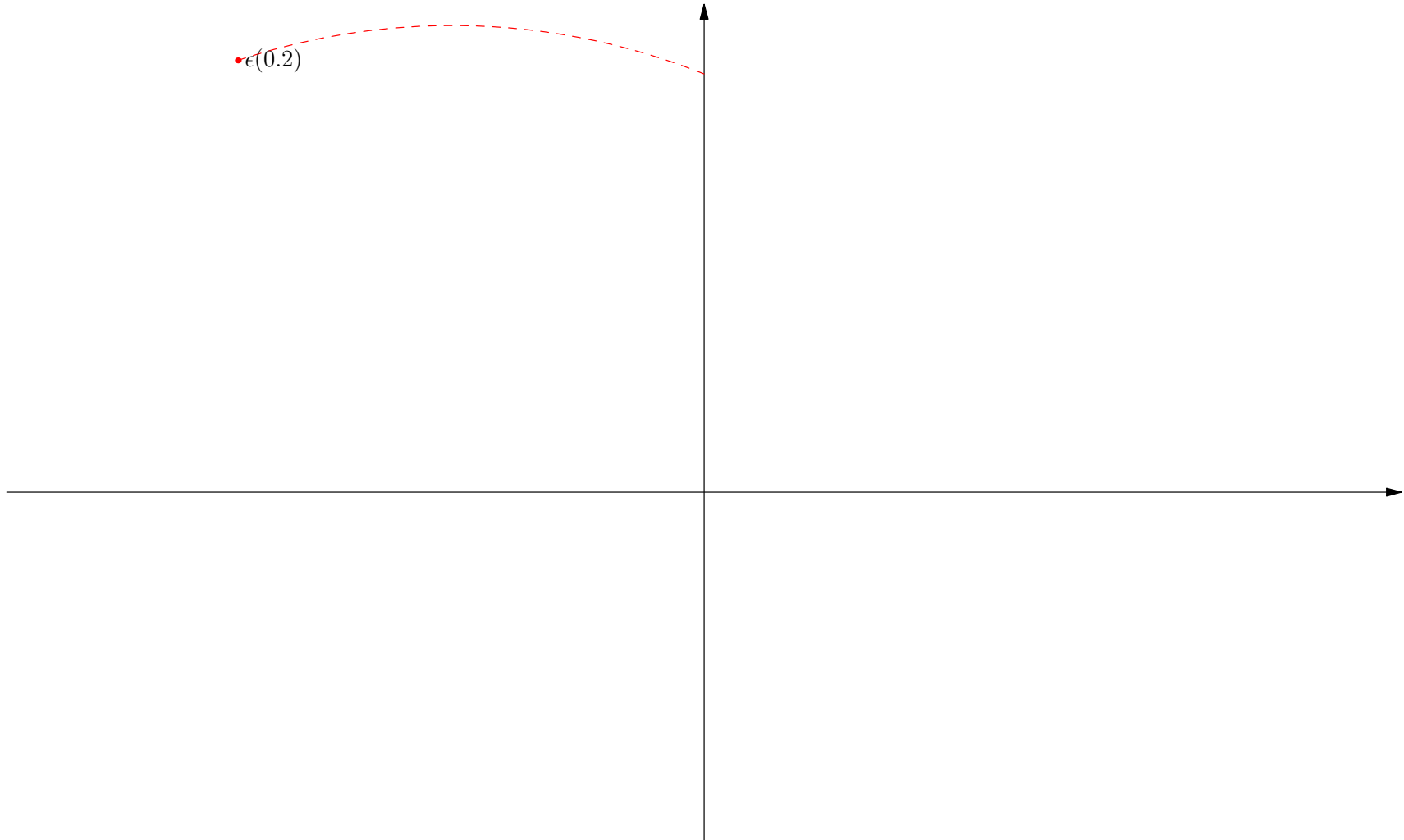
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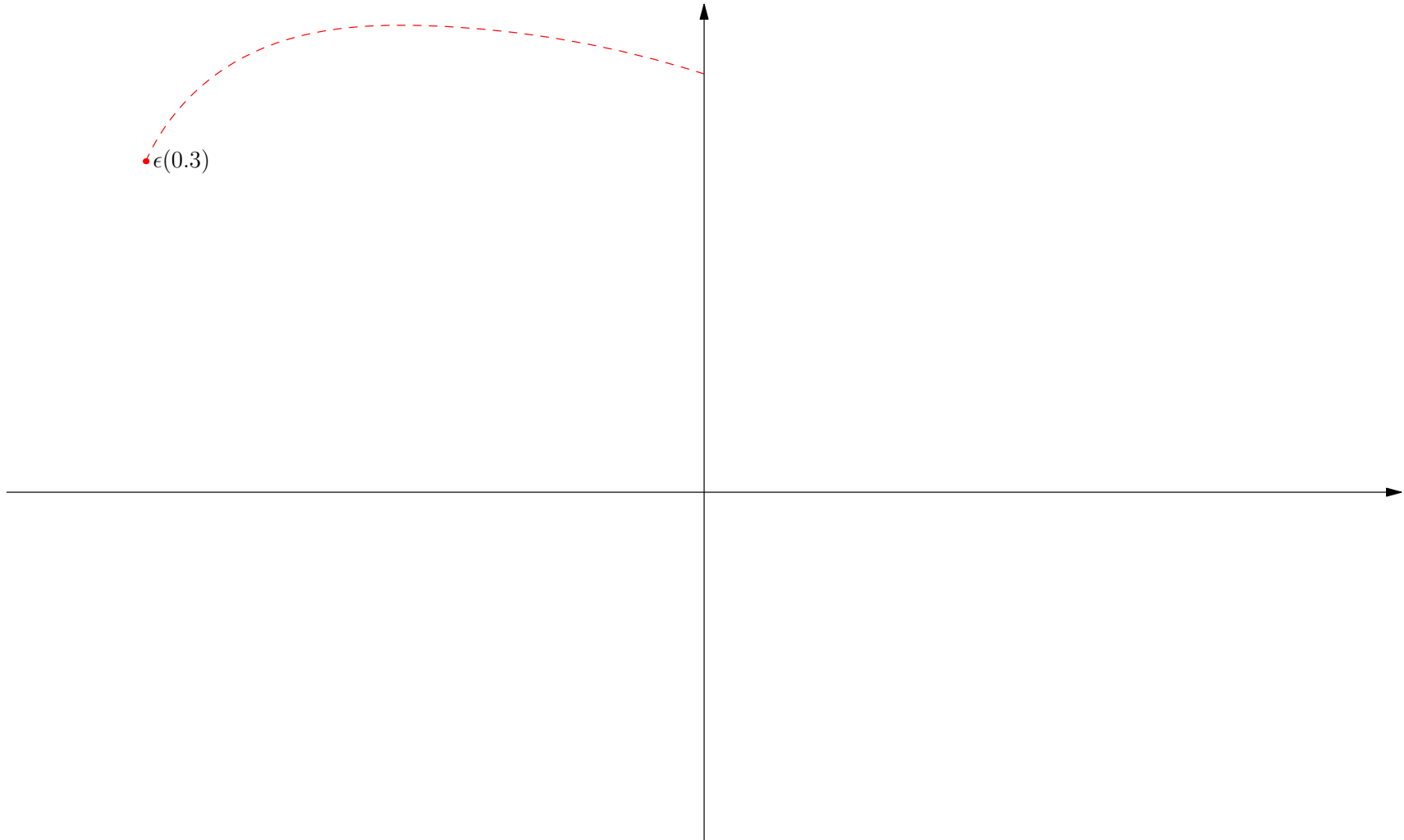
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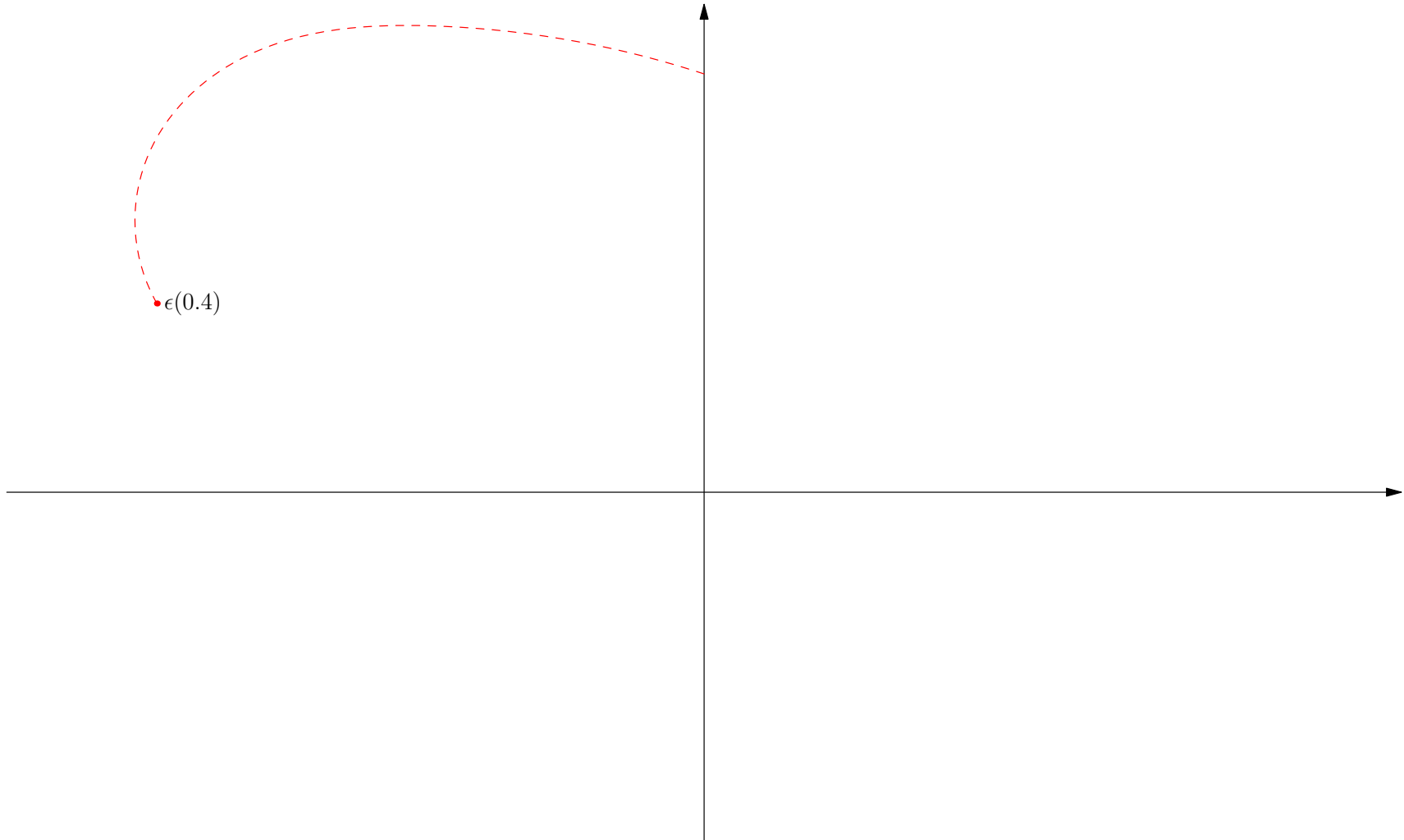
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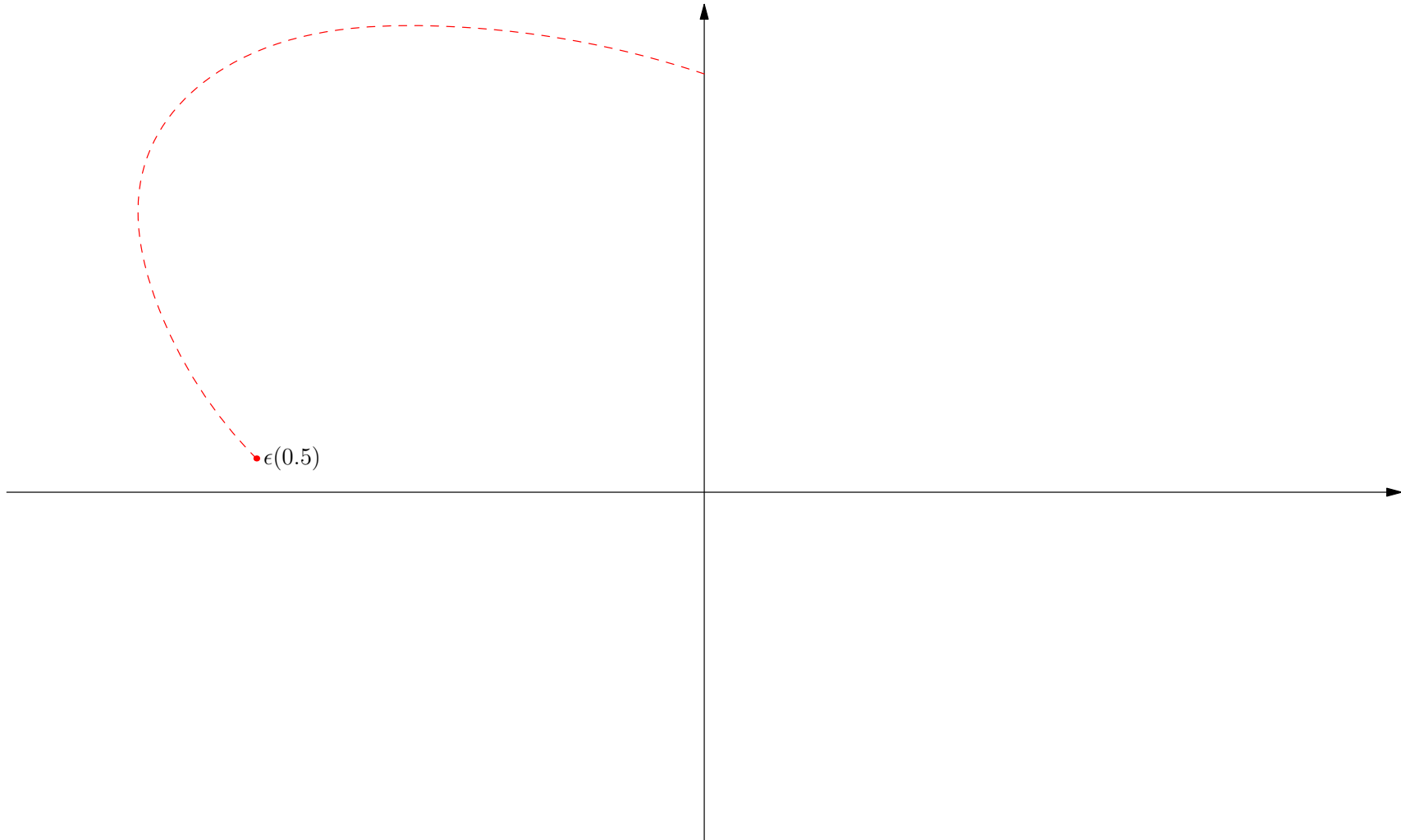
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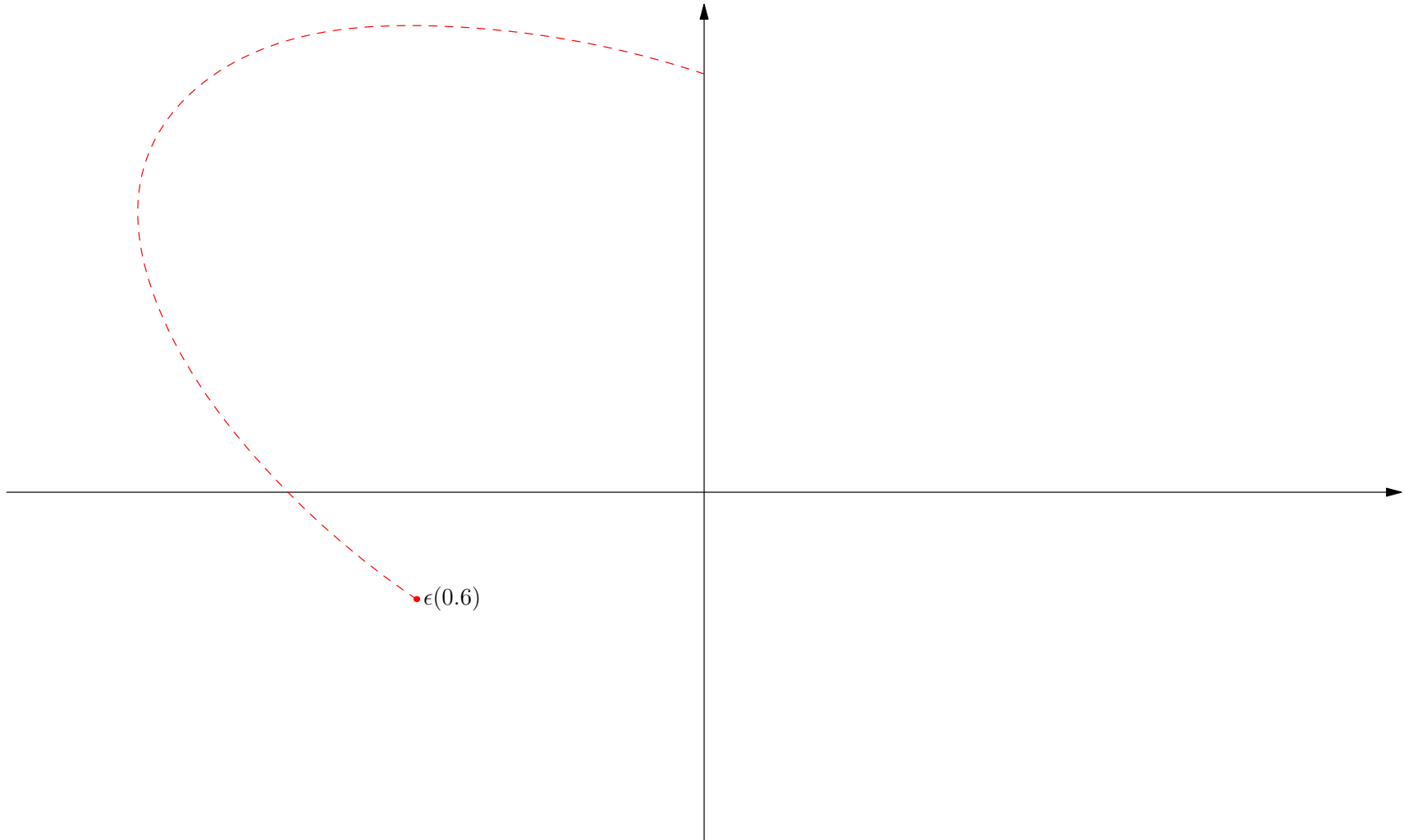
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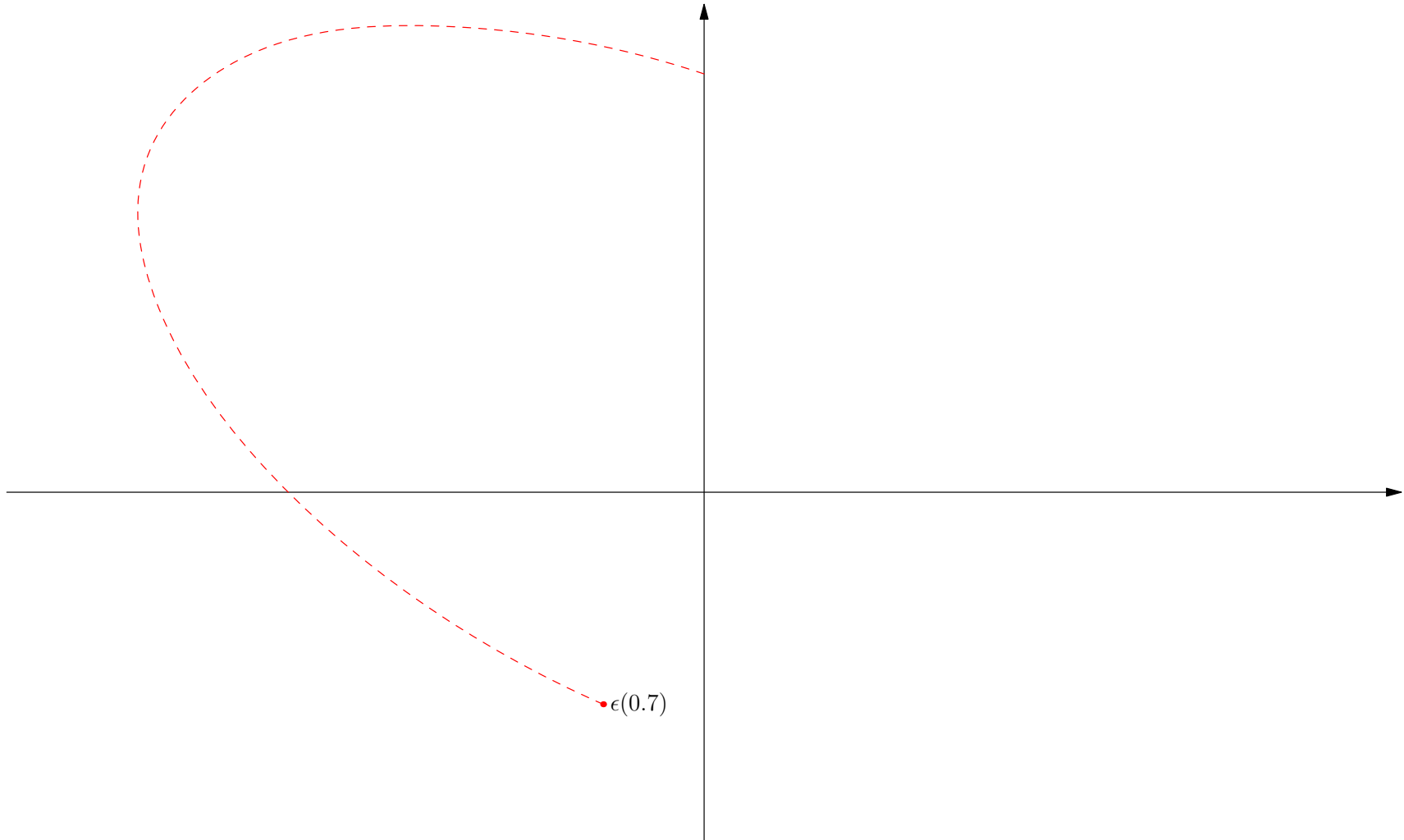
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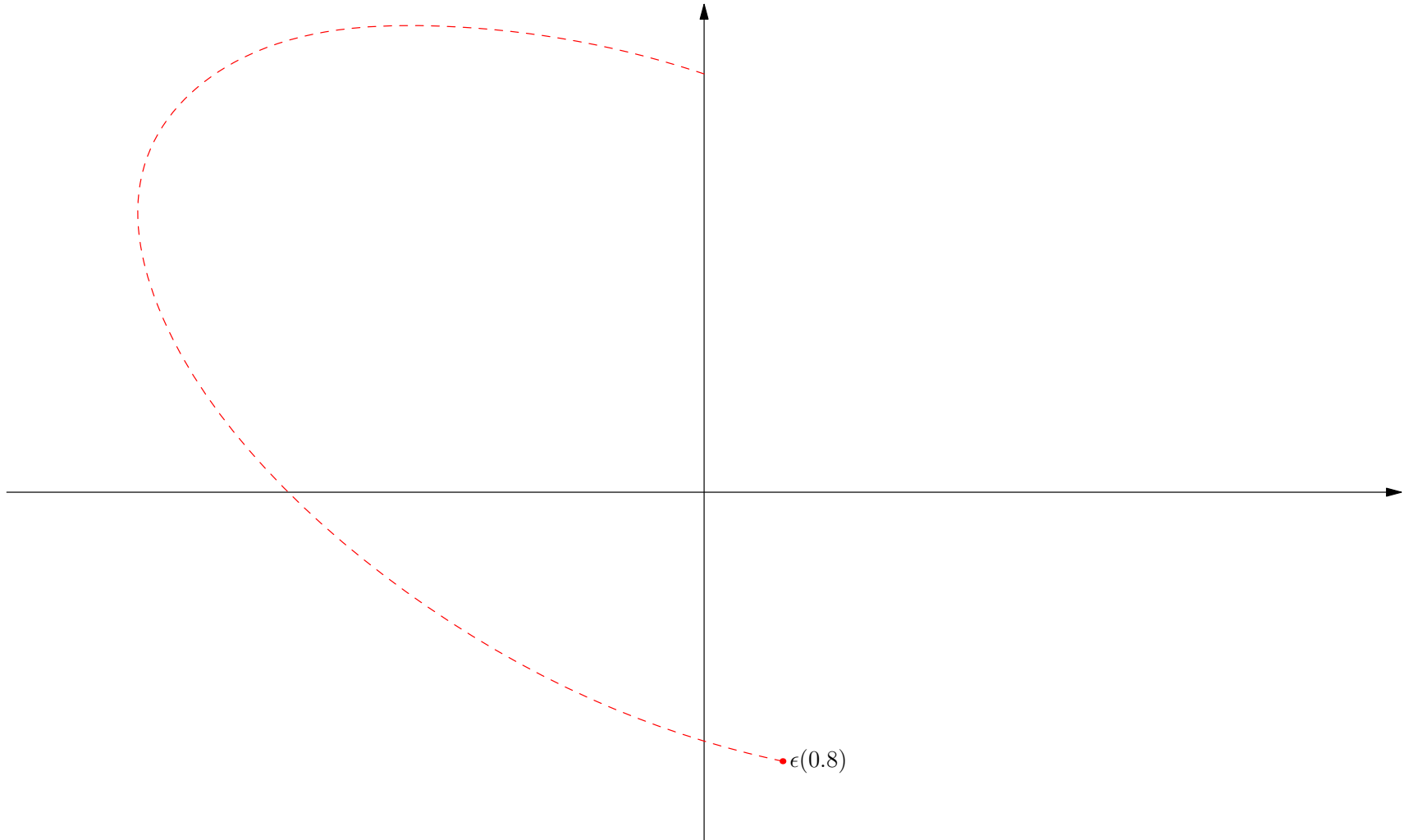
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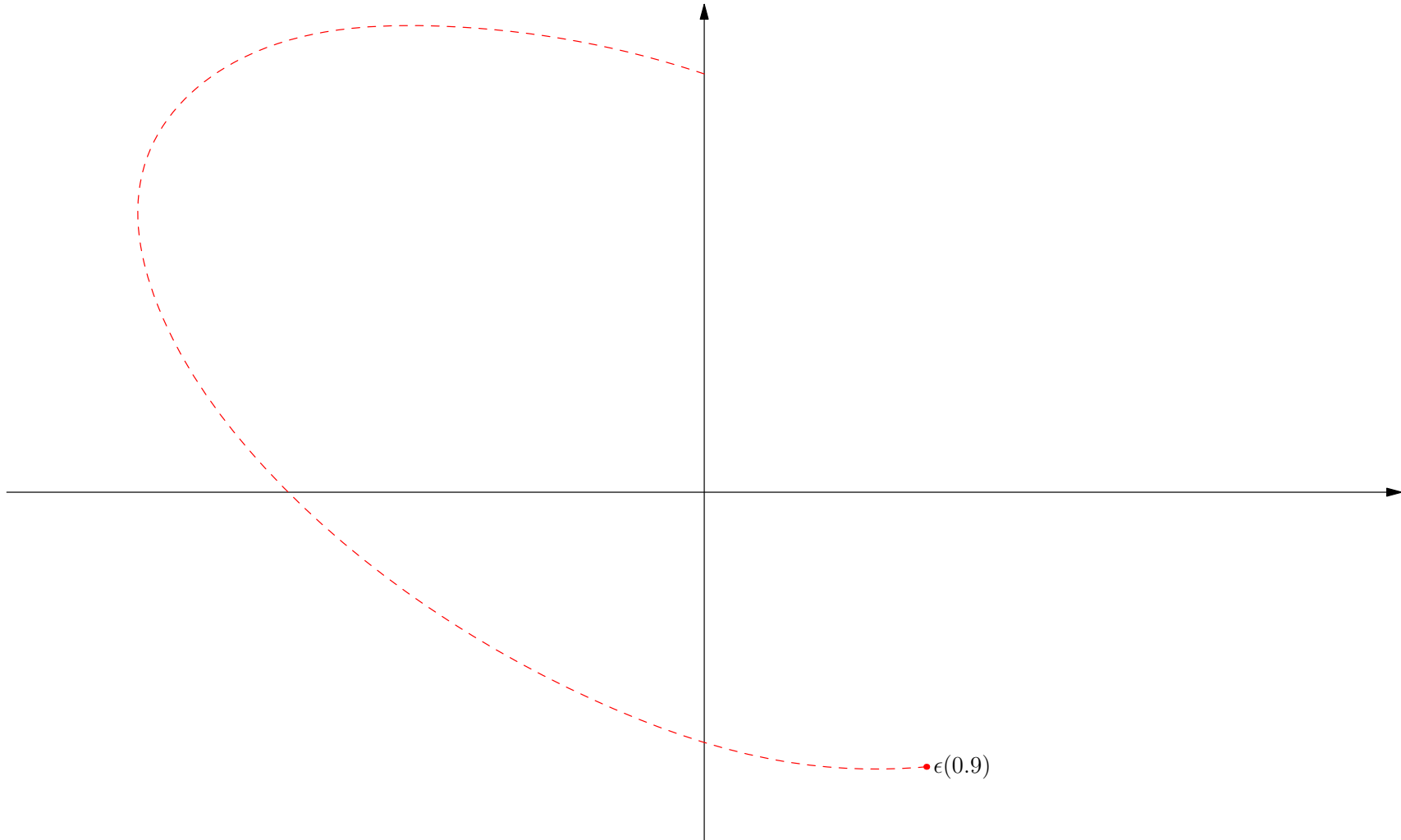
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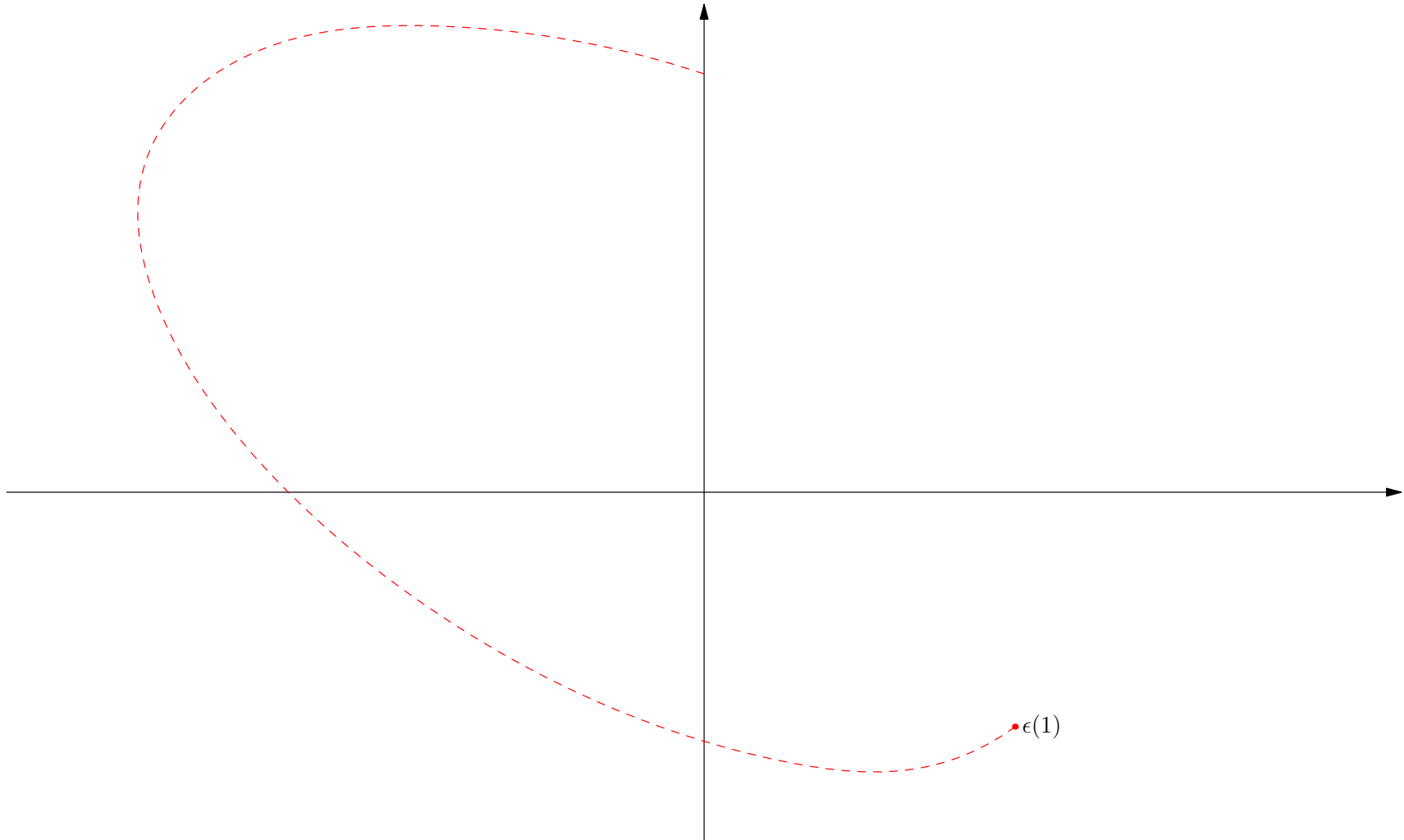
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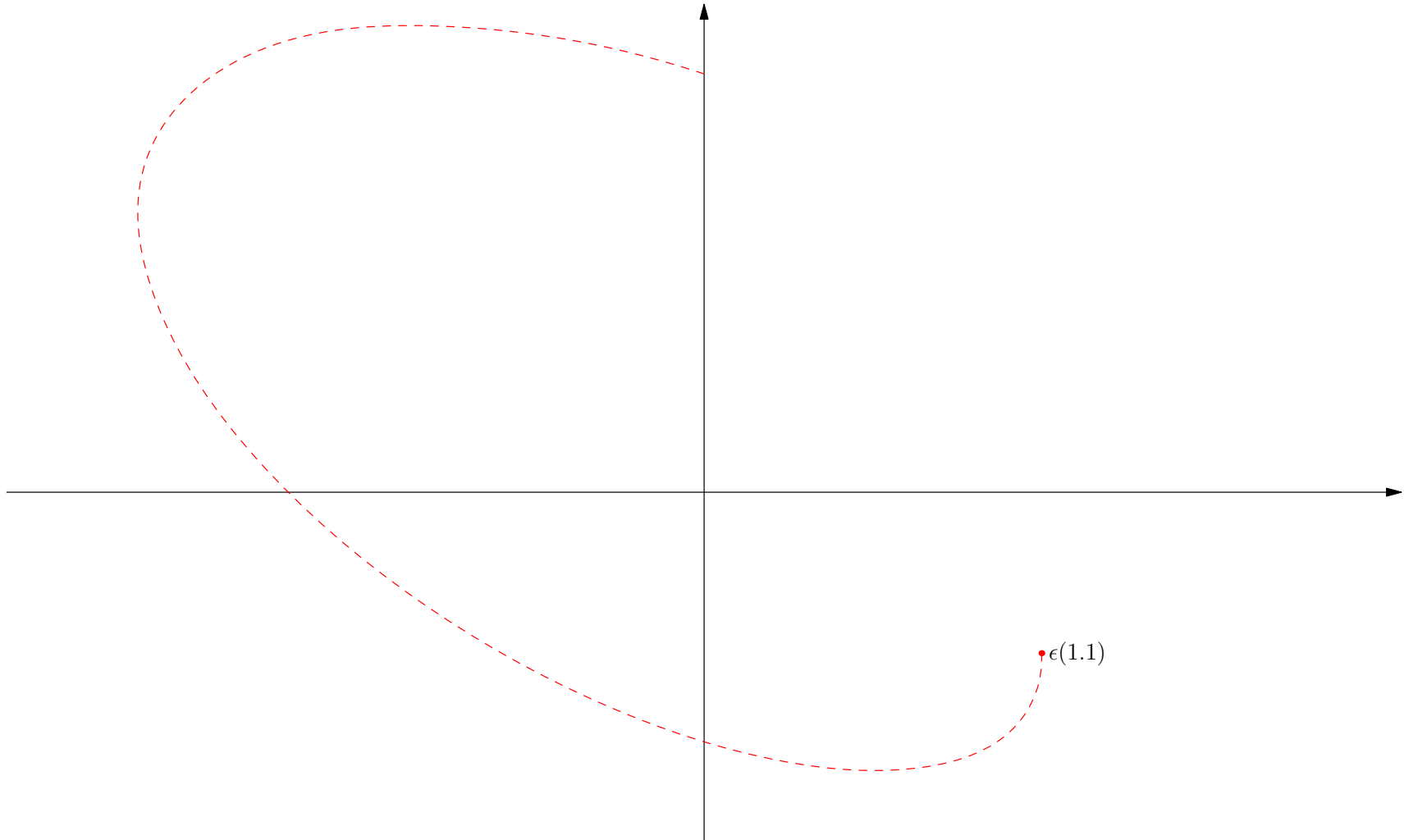
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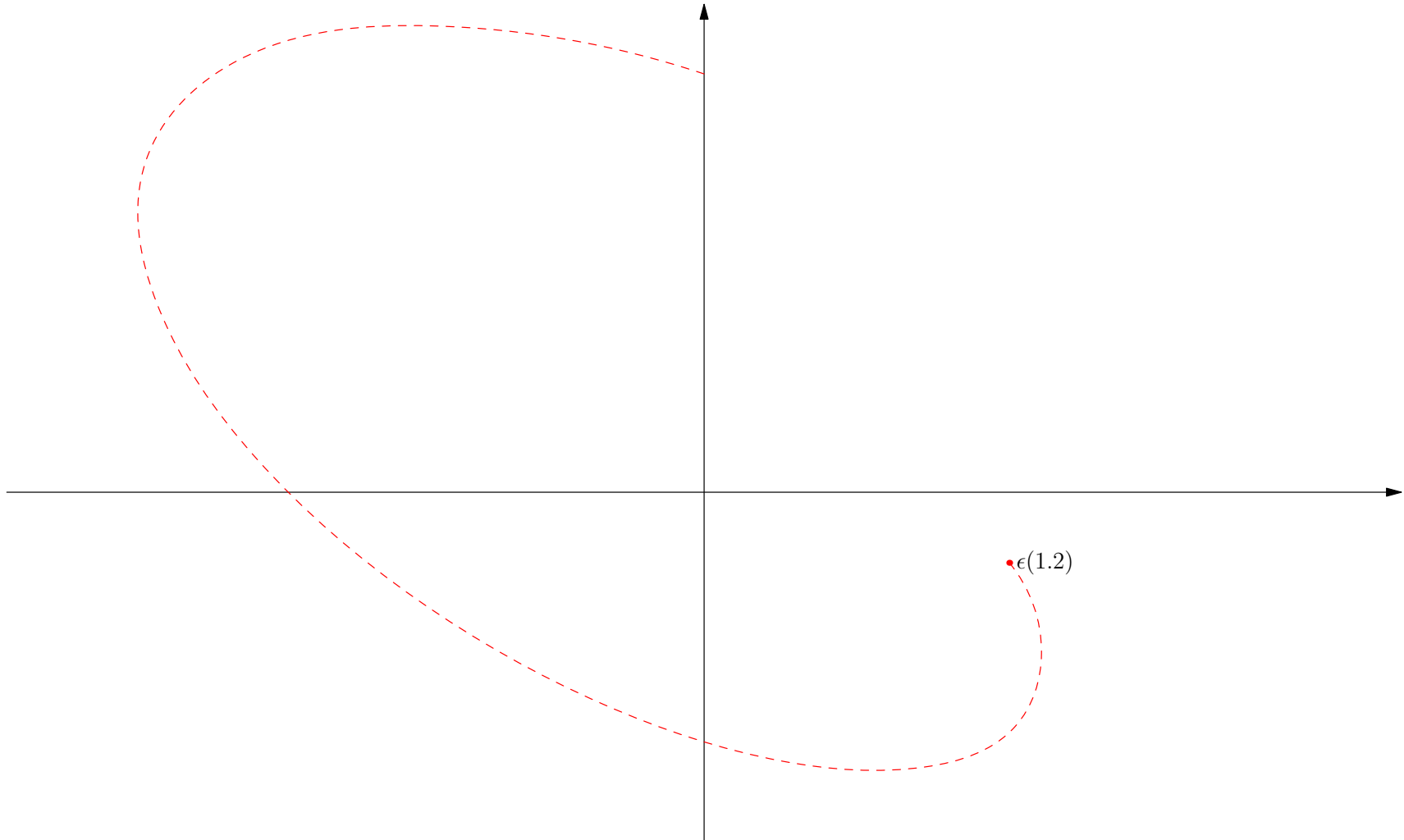
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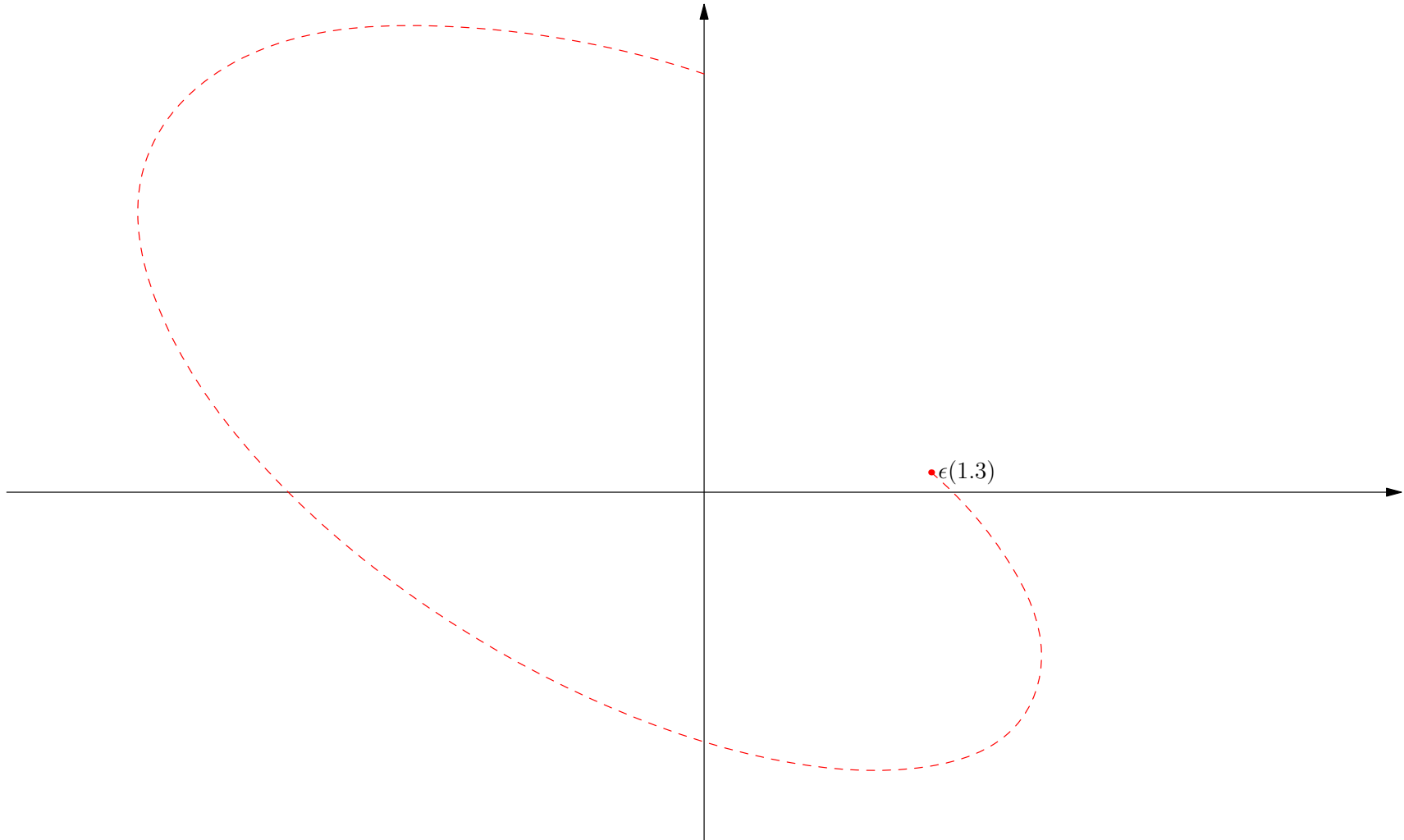
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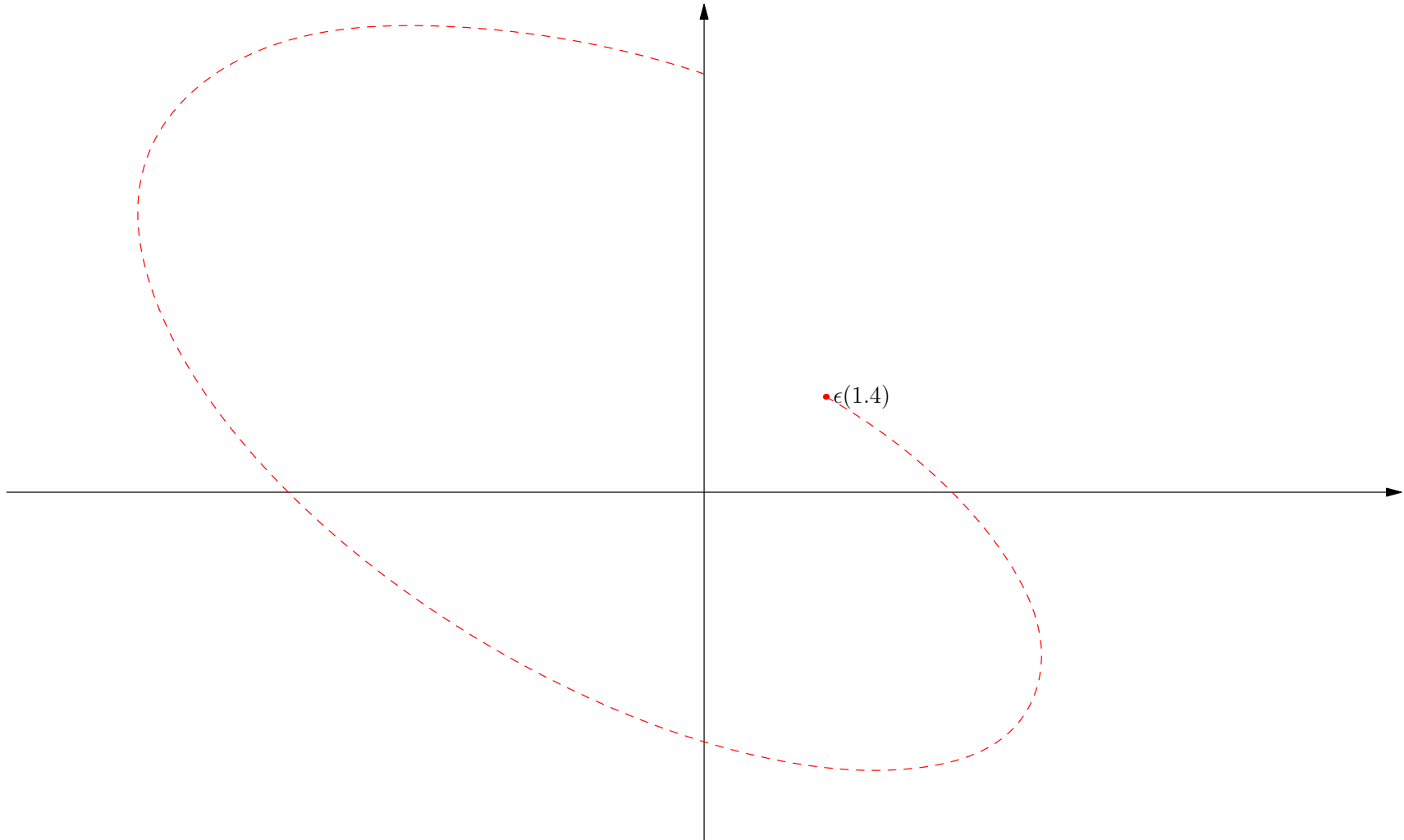
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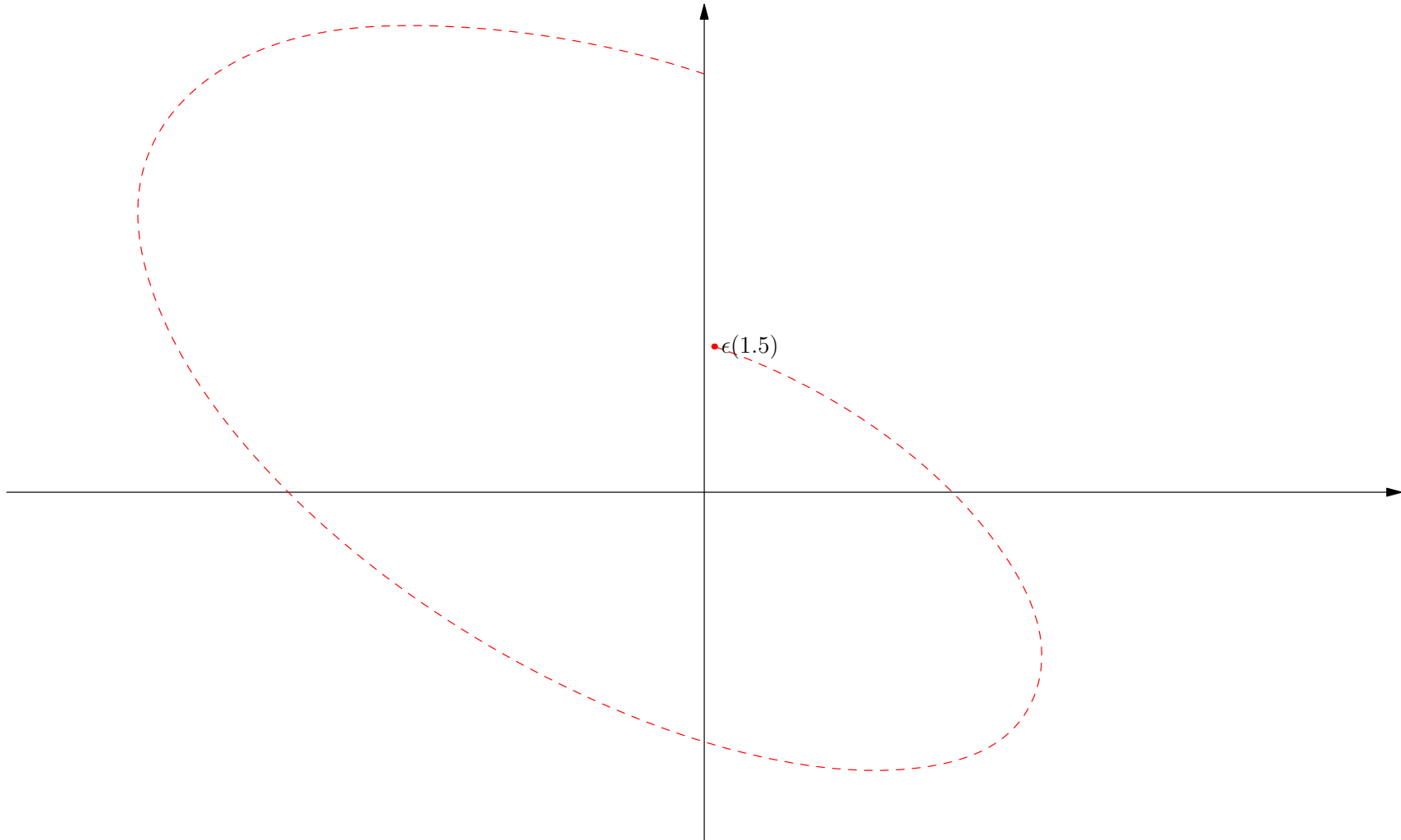
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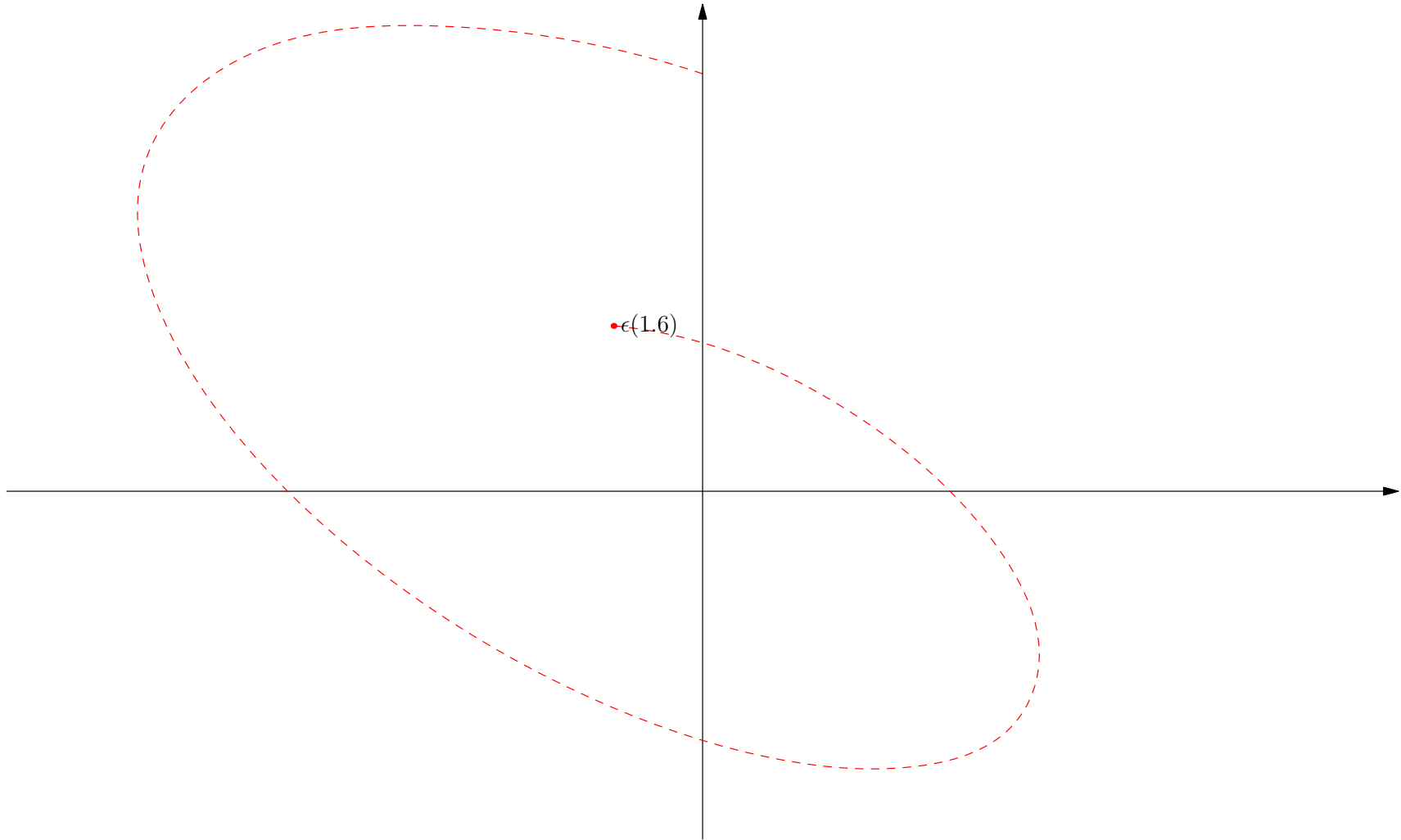
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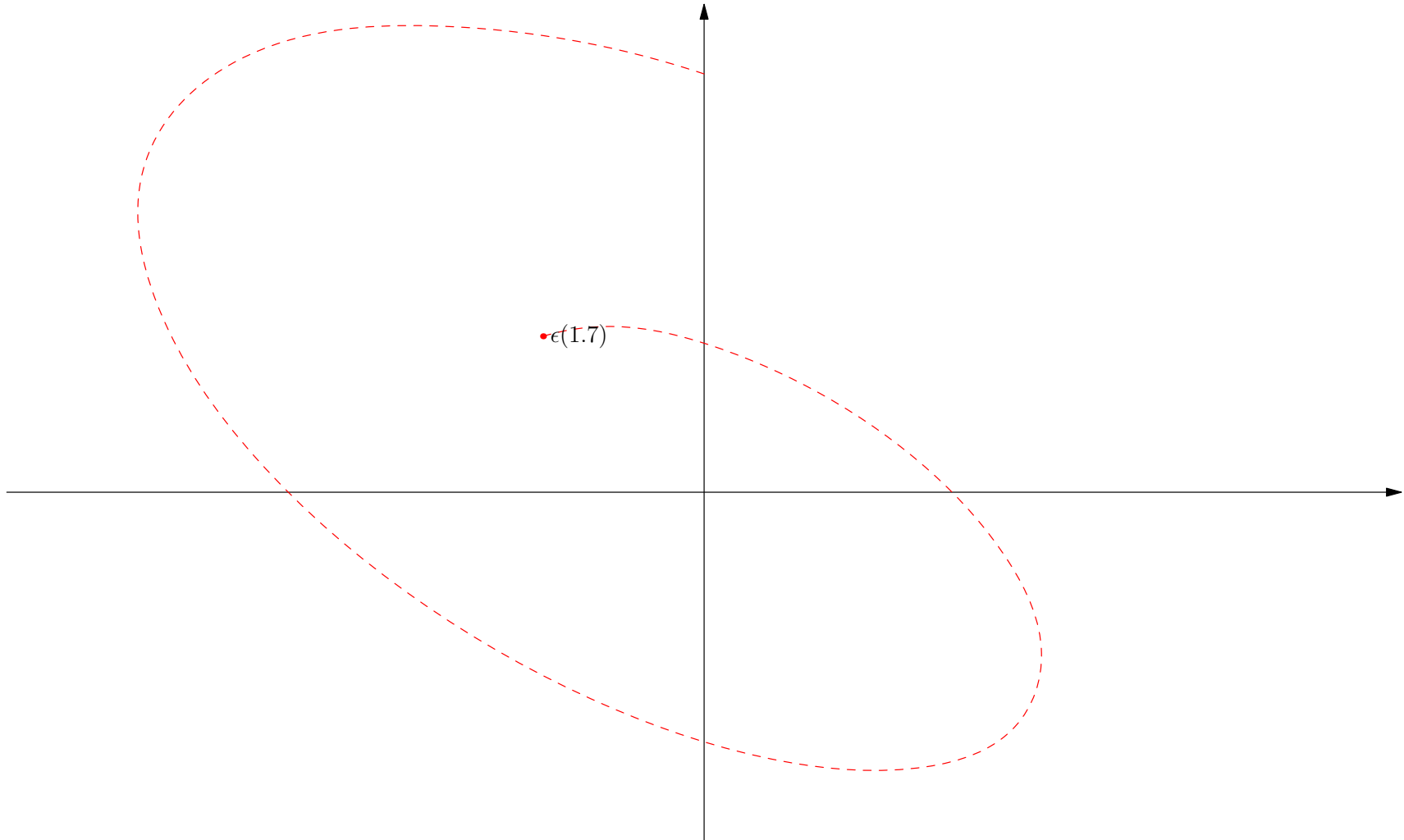
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Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473i & -0.327 + 0.473i \end{pmatrix} \begin{pmatrix} -0.685 + 4.163i & 0 \\ 0 & -0.685 - 4.163i \end{pmatrix} \begin{pmatrix} 0.611 + 0.422i & 1.056i \\ 0.611 - 0.422i & -1.056i \end{pmatrix}$$



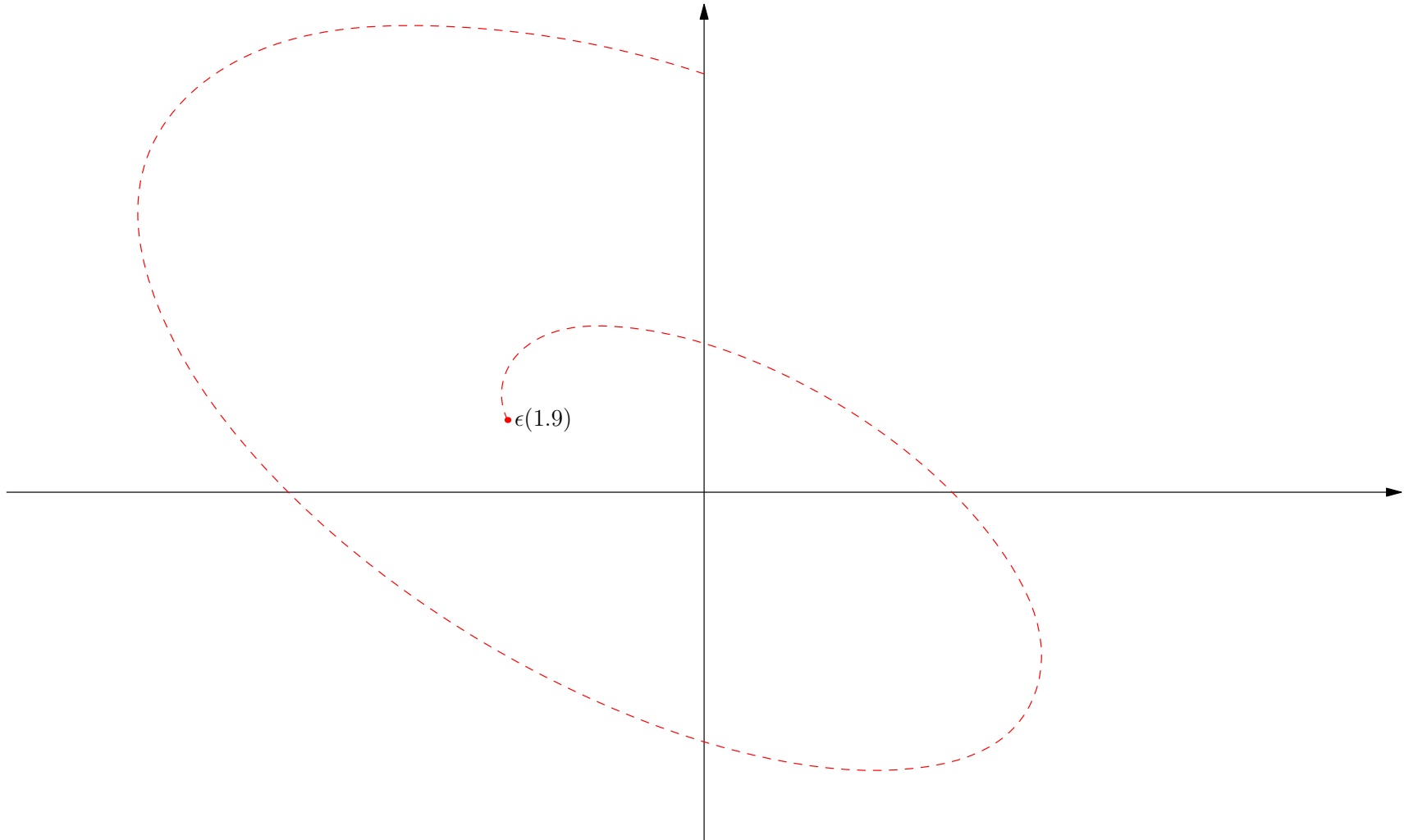
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473i & -0.327 + 0.473i \end{pmatrix} \begin{pmatrix} -0.685 + 4.163i & 0 \\ 0 & -0.685 - 4.163i \end{pmatrix} \begin{pmatrix} 0.611 + 0.422i & 1.056i \\ 0.611 - 0.422i & -1.056i \end{pmatrix}$$



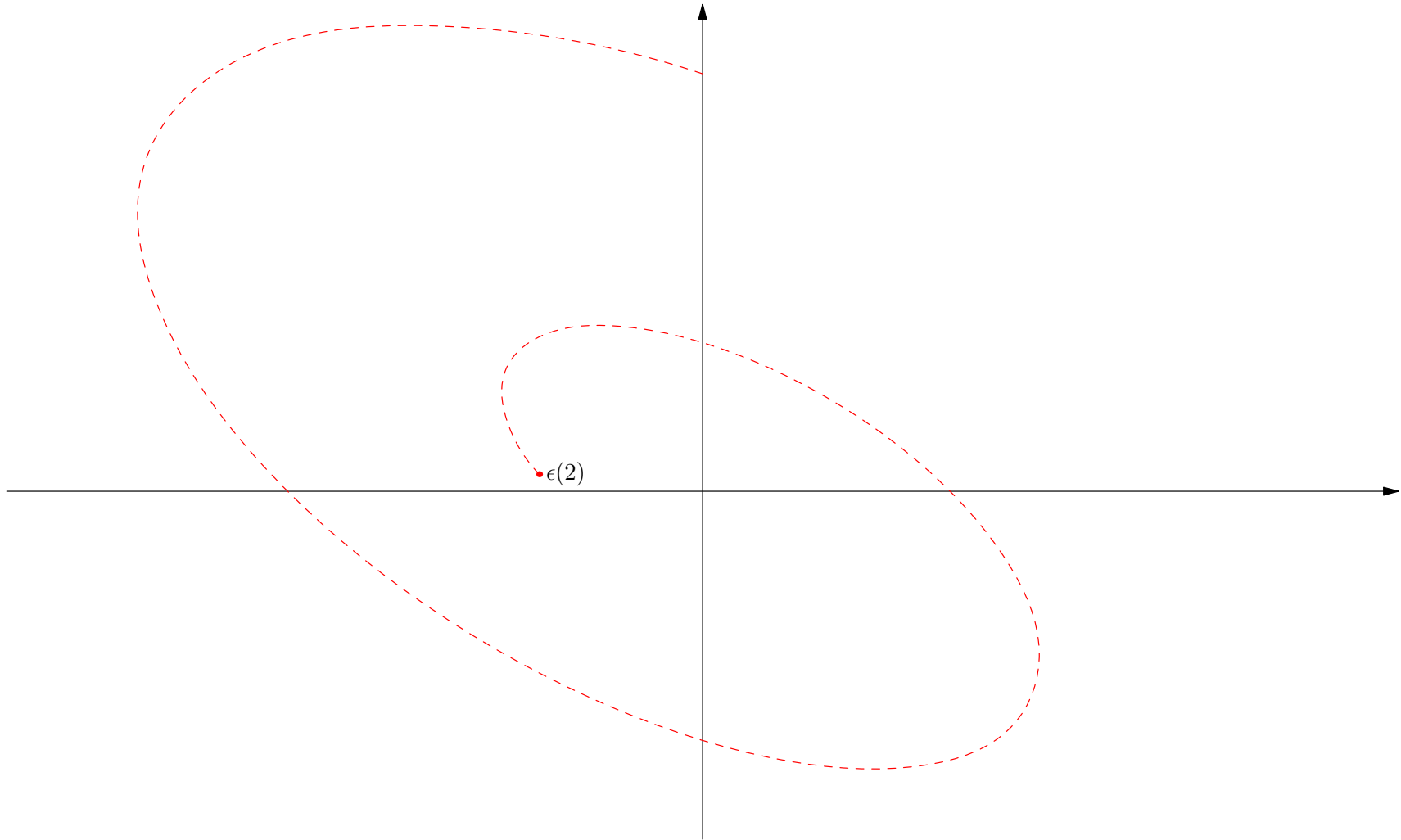
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473i & -0.327 + 0.473i \end{pmatrix} \begin{pmatrix} -0.685 + 4.163i & 0 \\ 0 & -0.685 - 4.163i \end{pmatrix} \begin{pmatrix} 0.611 + 0.422i & 1.056i \\ 0.611 - 0.422i & -1.056i \end{pmatrix}$$



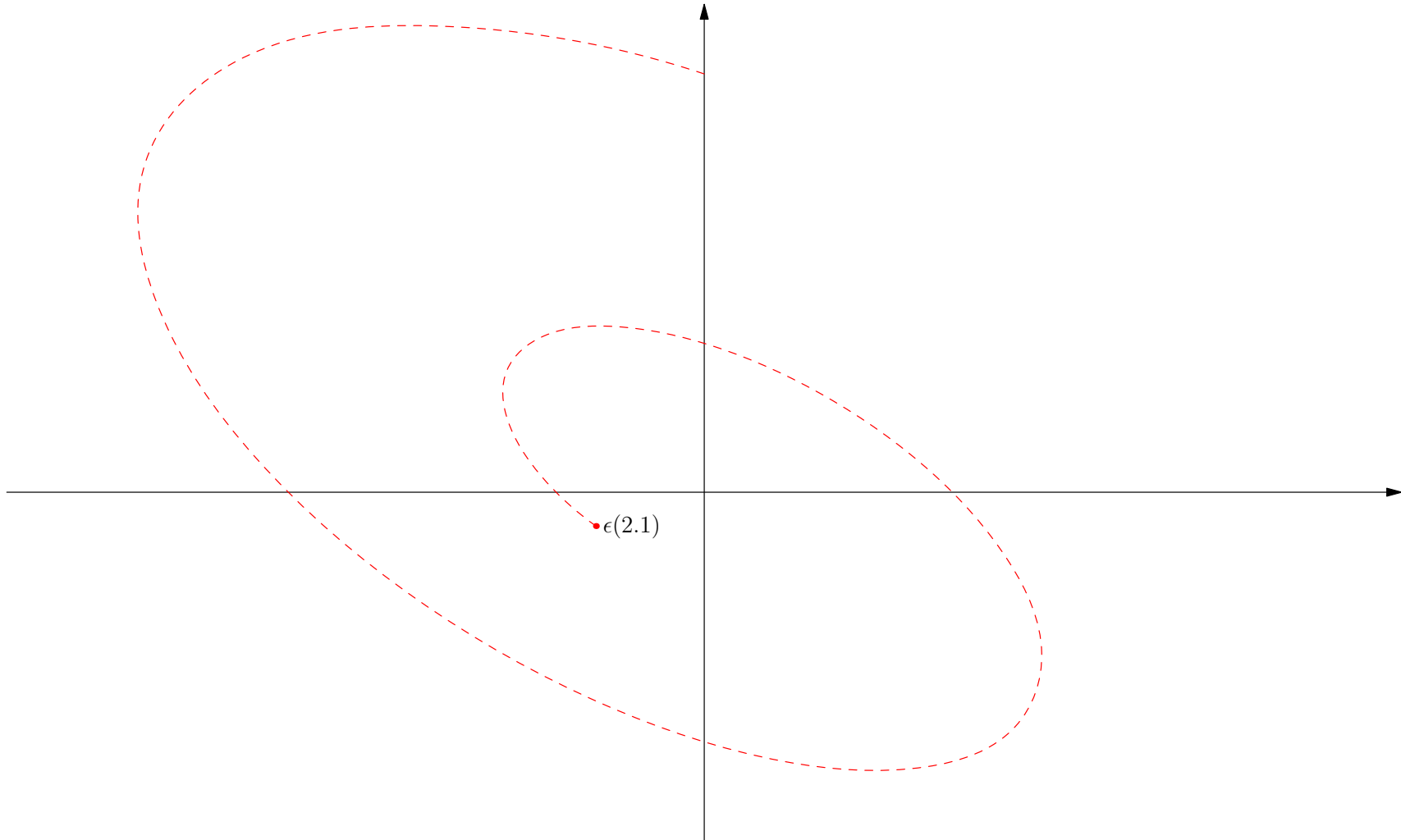
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473i & -0.327 + 0.473i \end{pmatrix} \begin{pmatrix} -0.685 + 4.163i & 0 \\ 0 & -0.685 - 4.163i \end{pmatrix} \begin{pmatrix} 0.611 + 0.422i & 1.056i \\ 0.611 - 0.422i & -1.056i \end{pmatrix}$$



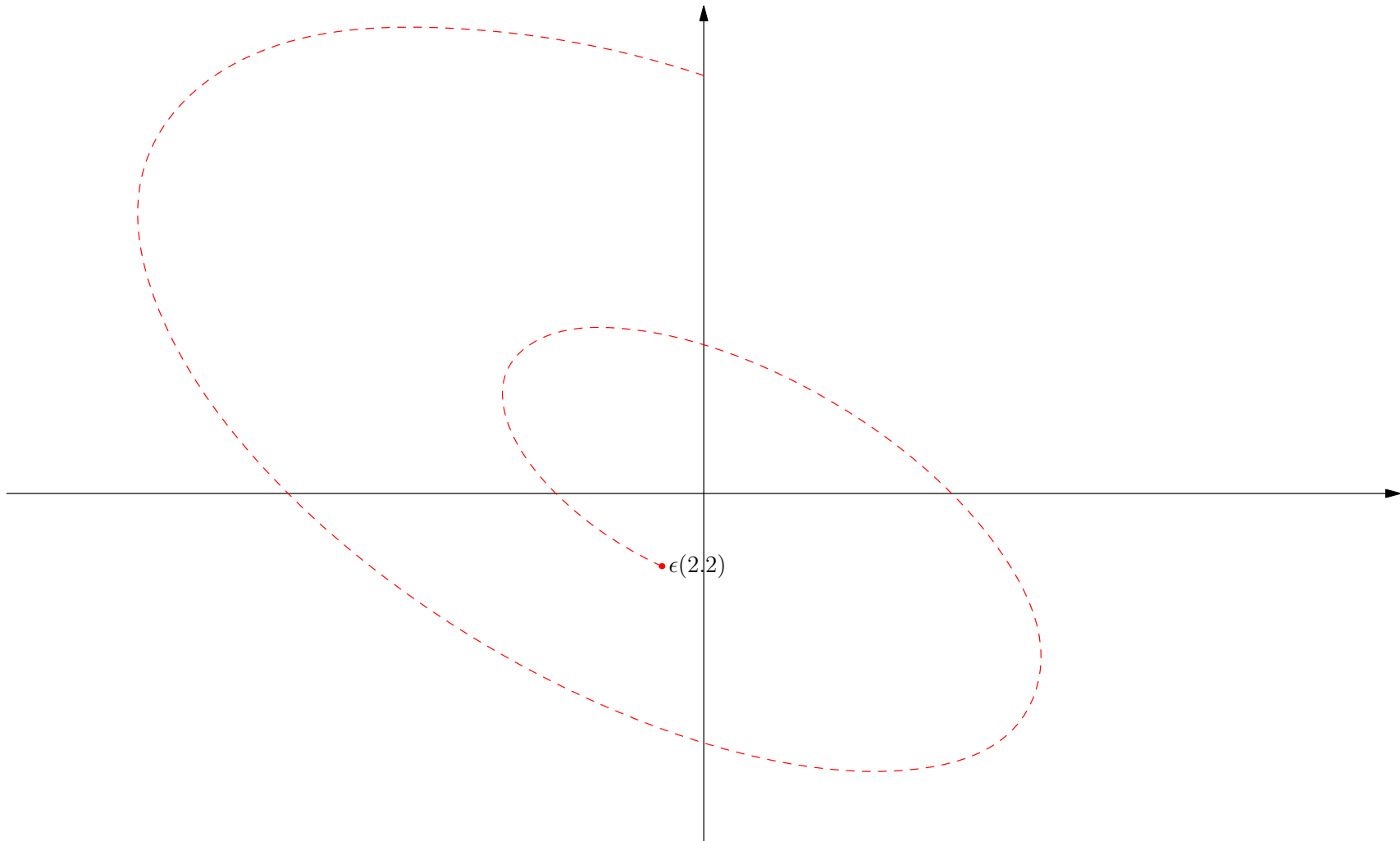
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473i & -0.327 + 0.473i \end{pmatrix} \begin{pmatrix} -0.685 + 4.163i & 0 \\ 0 & -0.685 - 4.163i \end{pmatrix} \begin{pmatrix} 0.611 + 0.422i & 1.056i \\ 0.611 - 0.422i & -1.056i \end{pmatrix}$$



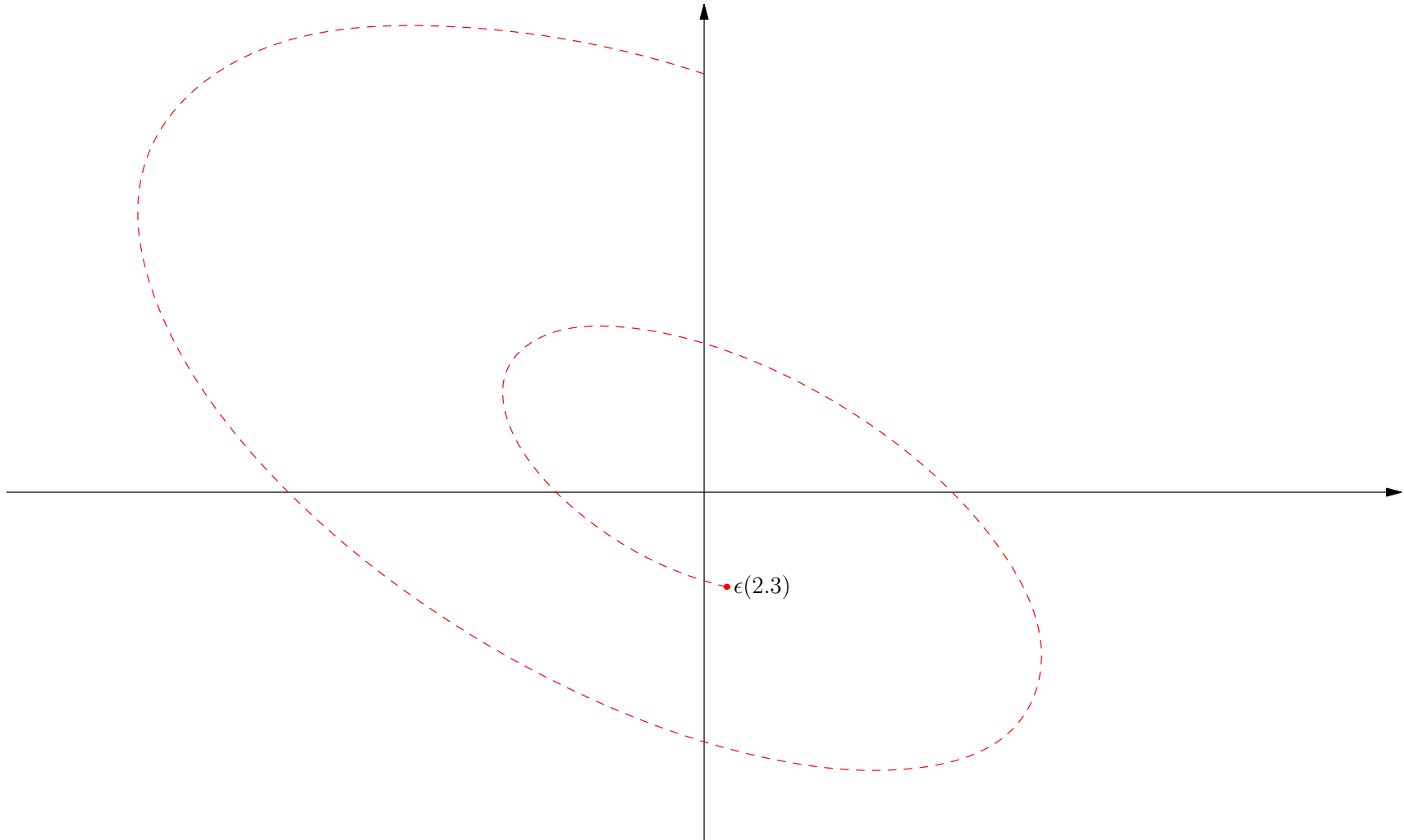
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473i & -0.327 + 0.473i \end{pmatrix} \begin{pmatrix} -0.685 + 4.163i & 0 \\ 0 & -0.685 - 4.163i \end{pmatrix} \begin{pmatrix} 0.611 + 0.422i & 1.056i \\ 0.611 - 0.422i & -1.056i \end{pmatrix}$$



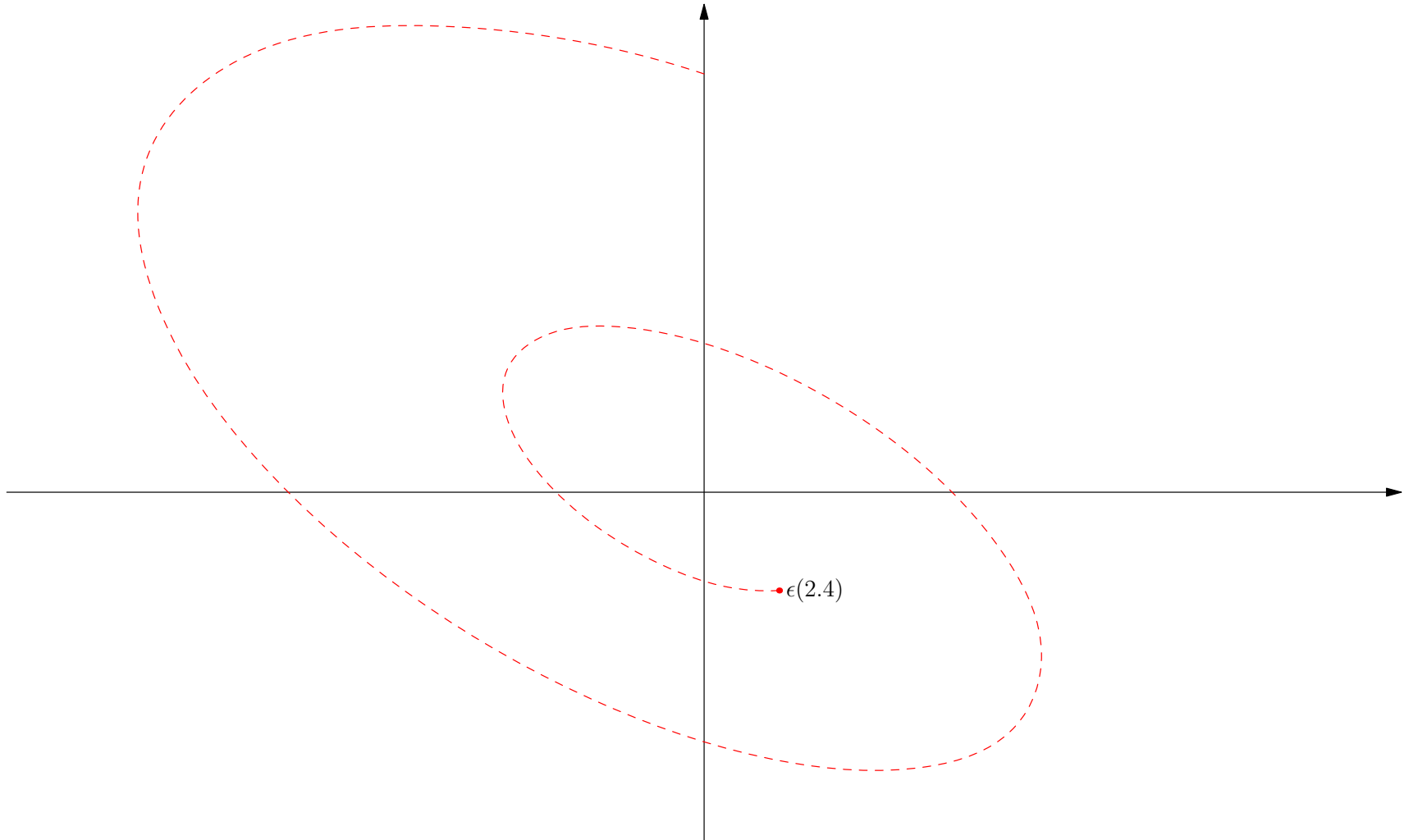
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473i & -0.327 + 0.473i \end{pmatrix} \begin{pmatrix} -0.685 + 4.163i & 0 \\ 0 & -0.685 - 4.163i \end{pmatrix} \begin{pmatrix} 0.611 + 0.422i & 1.056i \\ 0.611 - 0.422i & -1.056i \end{pmatrix}$$



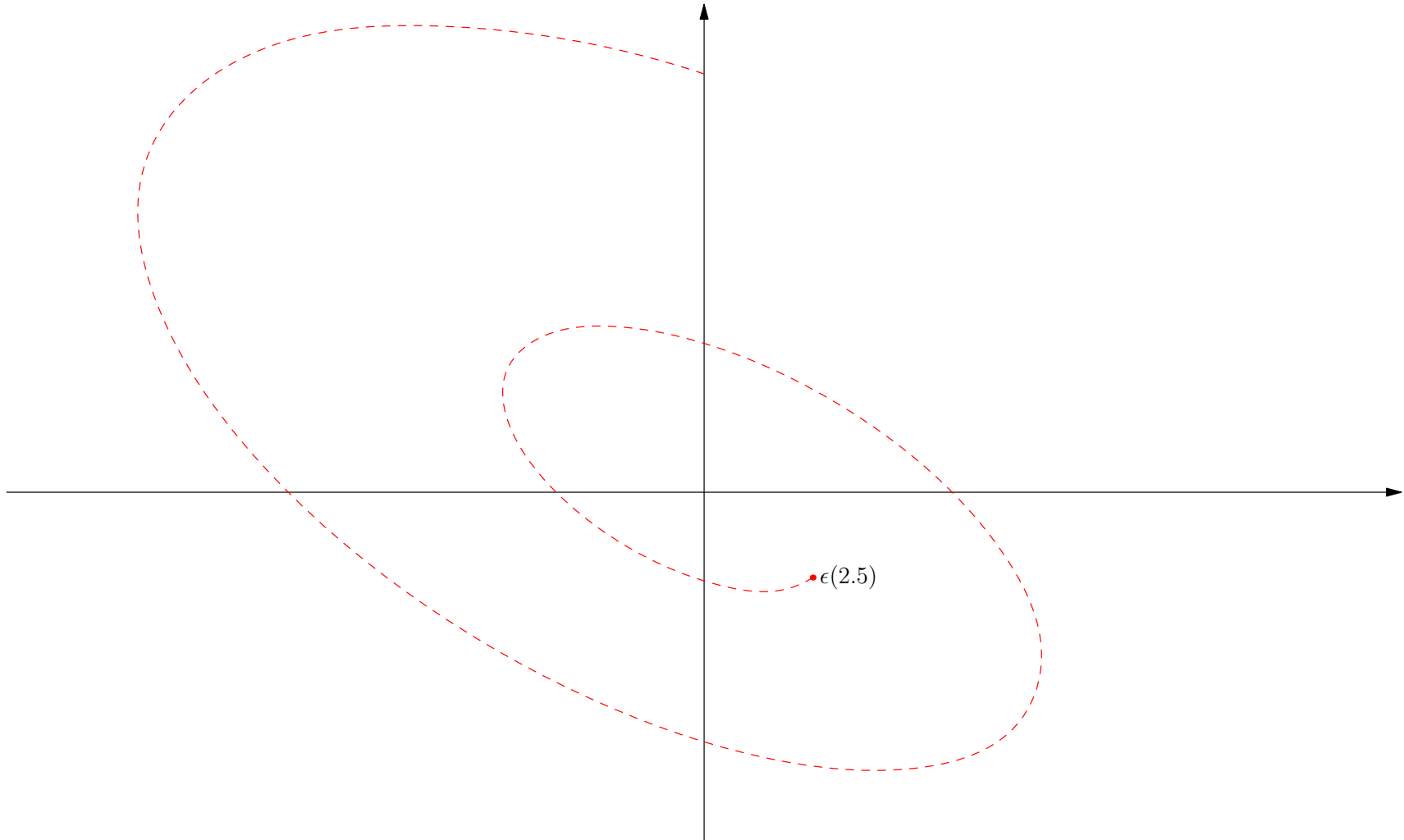
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473i & -0.327 + 0.473i \end{pmatrix} \begin{pmatrix} -0.685 + 4.163i & 0 \\ 0 & -0.685 - 4.163i \end{pmatrix} \begin{pmatrix} 0.611 + 0.422i & 1.056i \\ 0.611 - 0.422i & -1.056i \end{pmatrix}$$



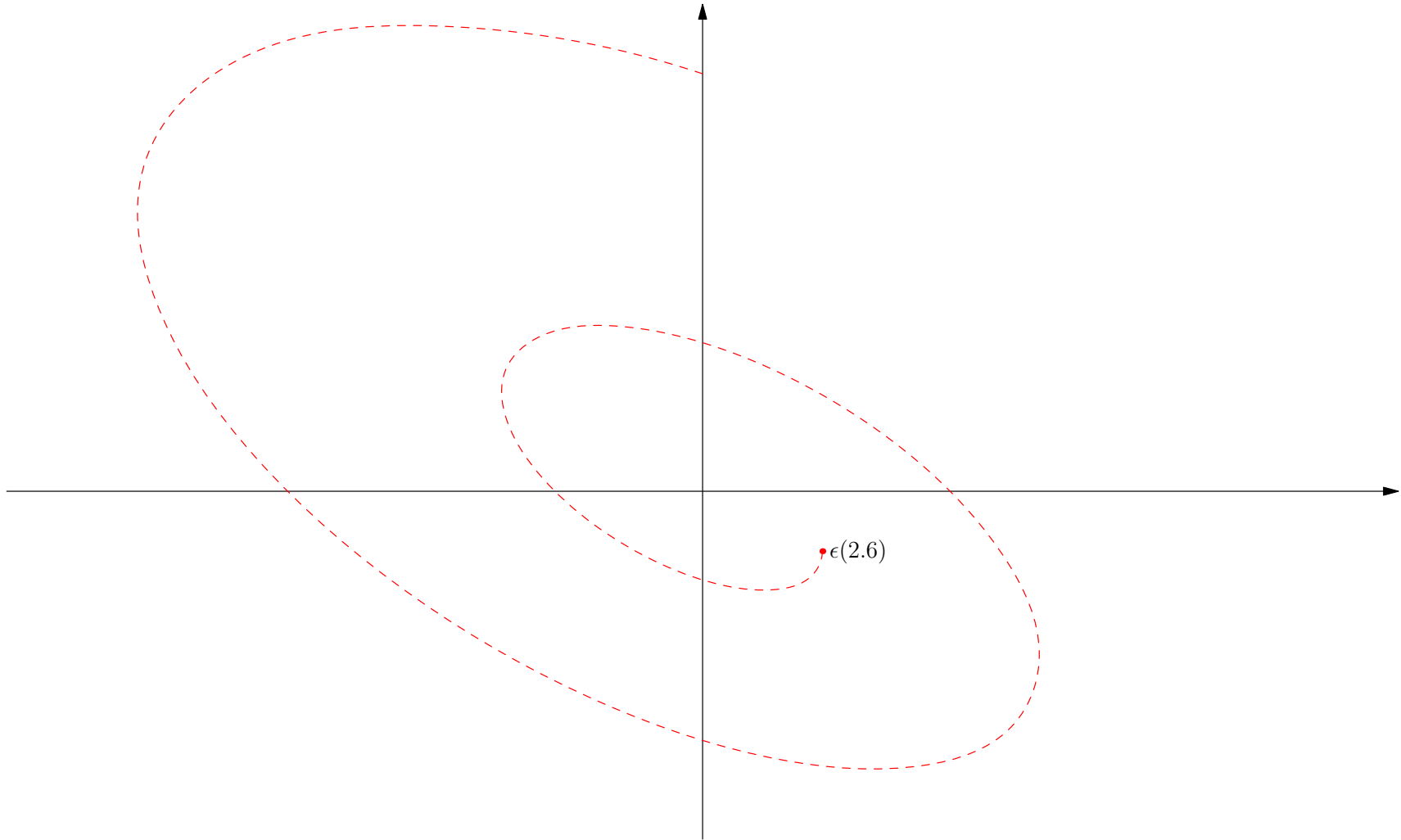
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473i & -0.327 + 0.473i \end{pmatrix} \begin{pmatrix} -0.685 + 4.163i & 0 \\ 0 & -0.685 - 4.163i \end{pmatrix} \begin{pmatrix} 0.611 + 0.422i & 1.056i \\ 0.611 - 0.422i & -1.056i \end{pmatrix}$$



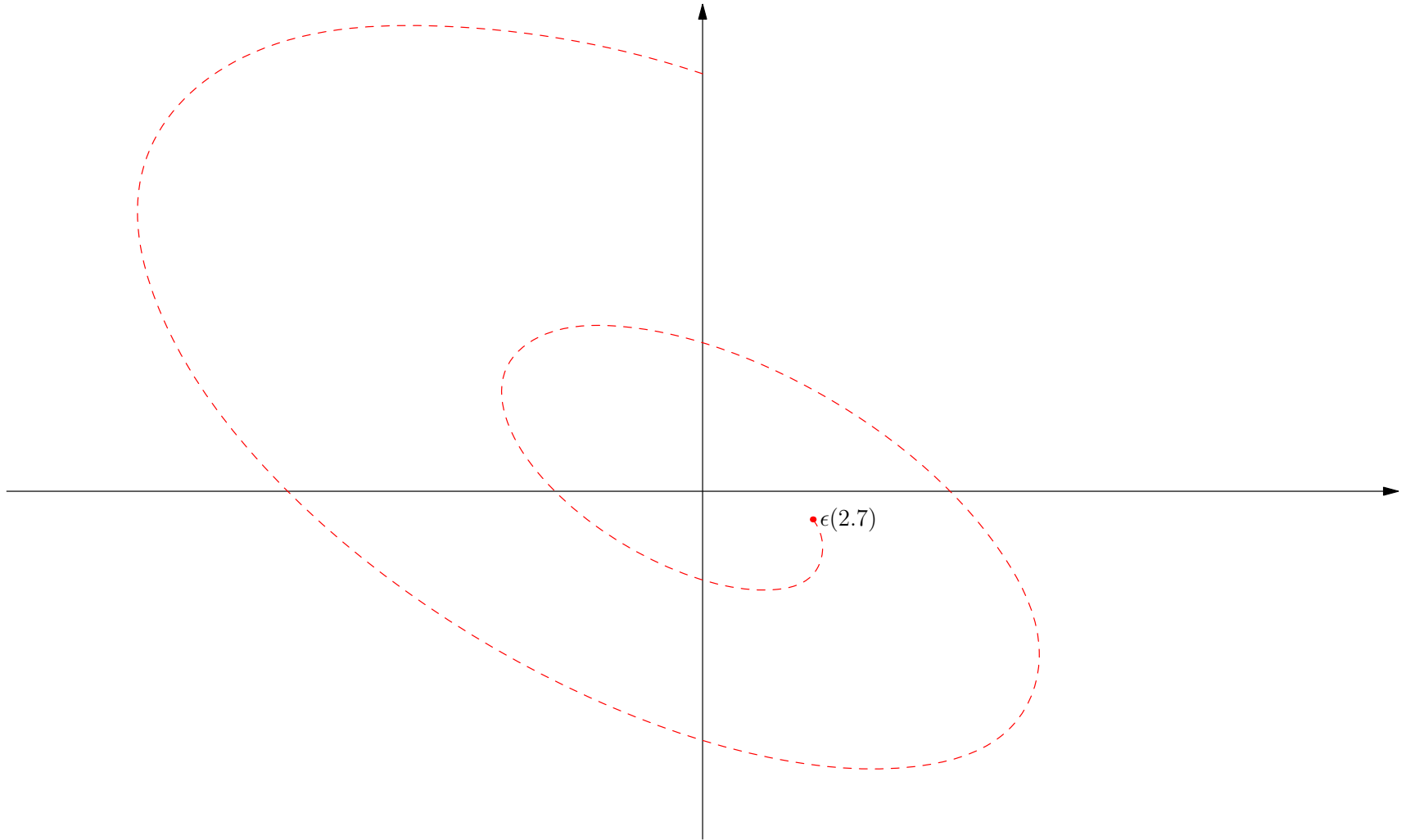
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473i & -0.327 + 0.473i \end{pmatrix} \begin{pmatrix} -0.685 + 4.163i & 0 \\ 0 & -0.685 - 4.163i \end{pmatrix} \begin{pmatrix} 0.611 + 0.422i & 1.056i \\ 0.611 - 0.422i & -1.056i \end{pmatrix}$$



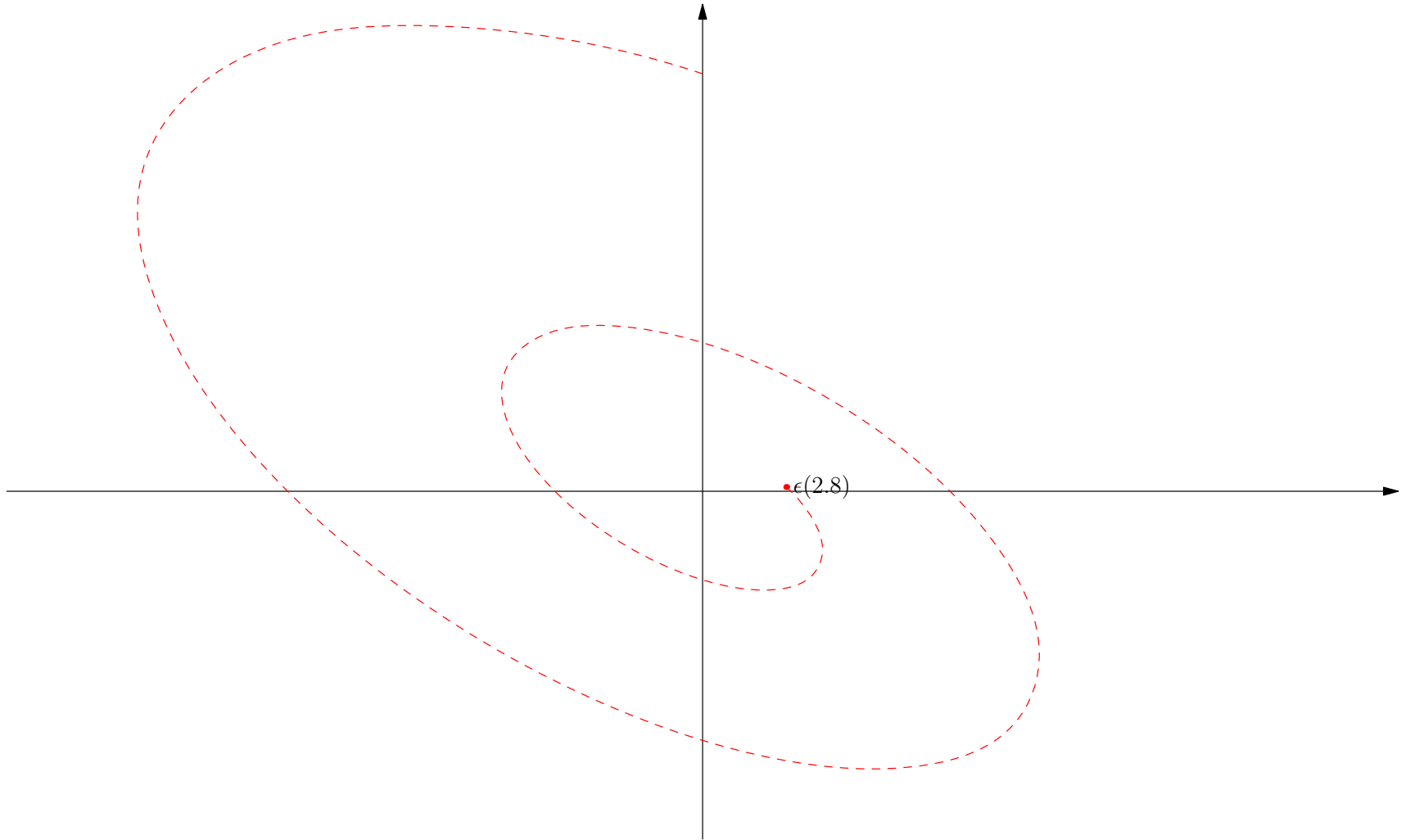
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473i & -0.327 + 0.473i \end{pmatrix} \begin{pmatrix} -0.685 + 4.163i & 0 \\ 0 & -0.685 - 4.163i \end{pmatrix} \begin{pmatrix} 0.611 + 0.422i & 1.056i \\ 0.611 - 0.422i & -1.056i \end{pmatrix}$$



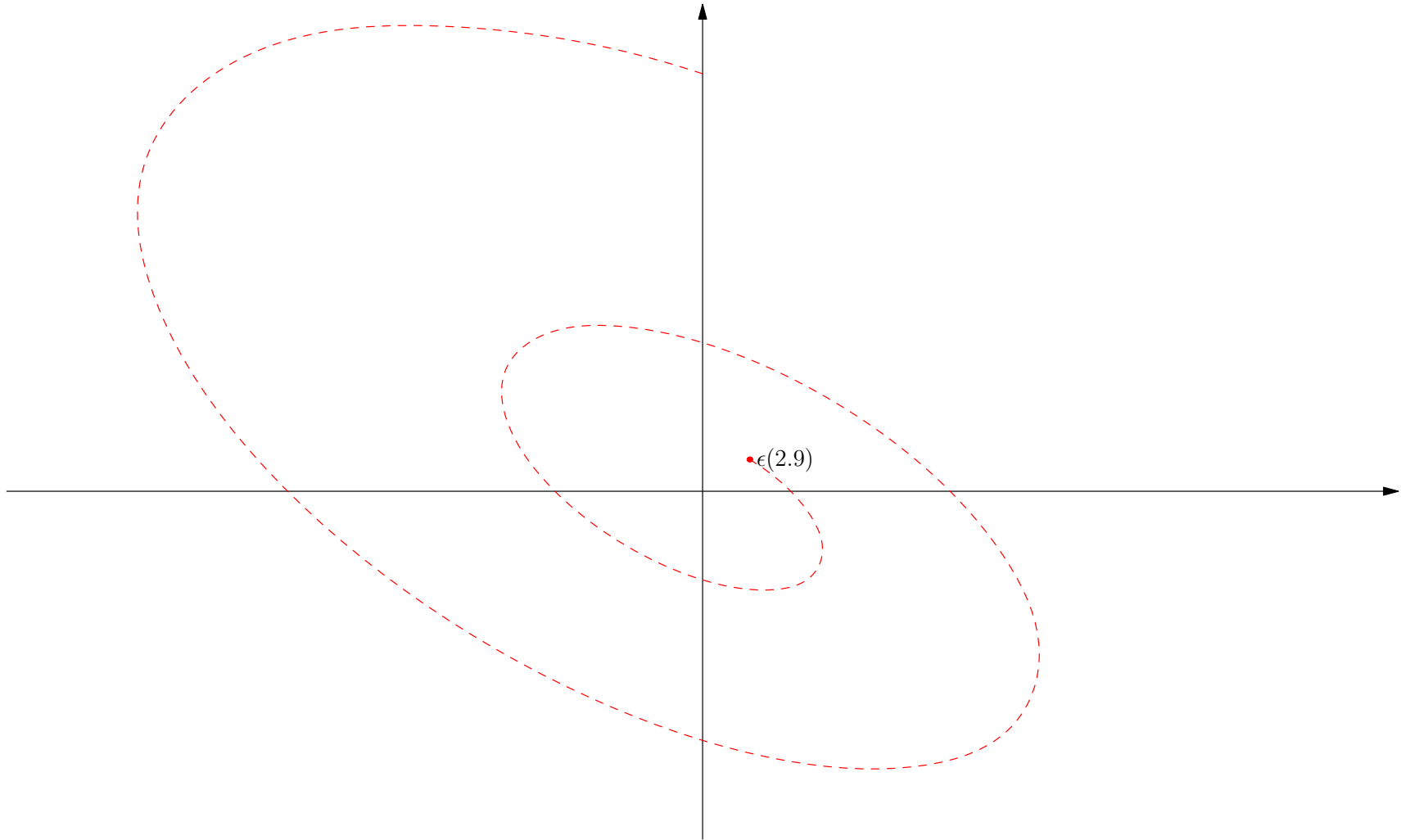
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473i & -0.327 + 0.473i \end{pmatrix} \begin{pmatrix} -0.685 + 4.163i & 0 \\ 0 & -0.685 - 4.163i \end{pmatrix} \begin{pmatrix} 0.611 + 0.422i & 1.056i \\ 0.611 - 0.422i & -1.056i \end{pmatrix}$$



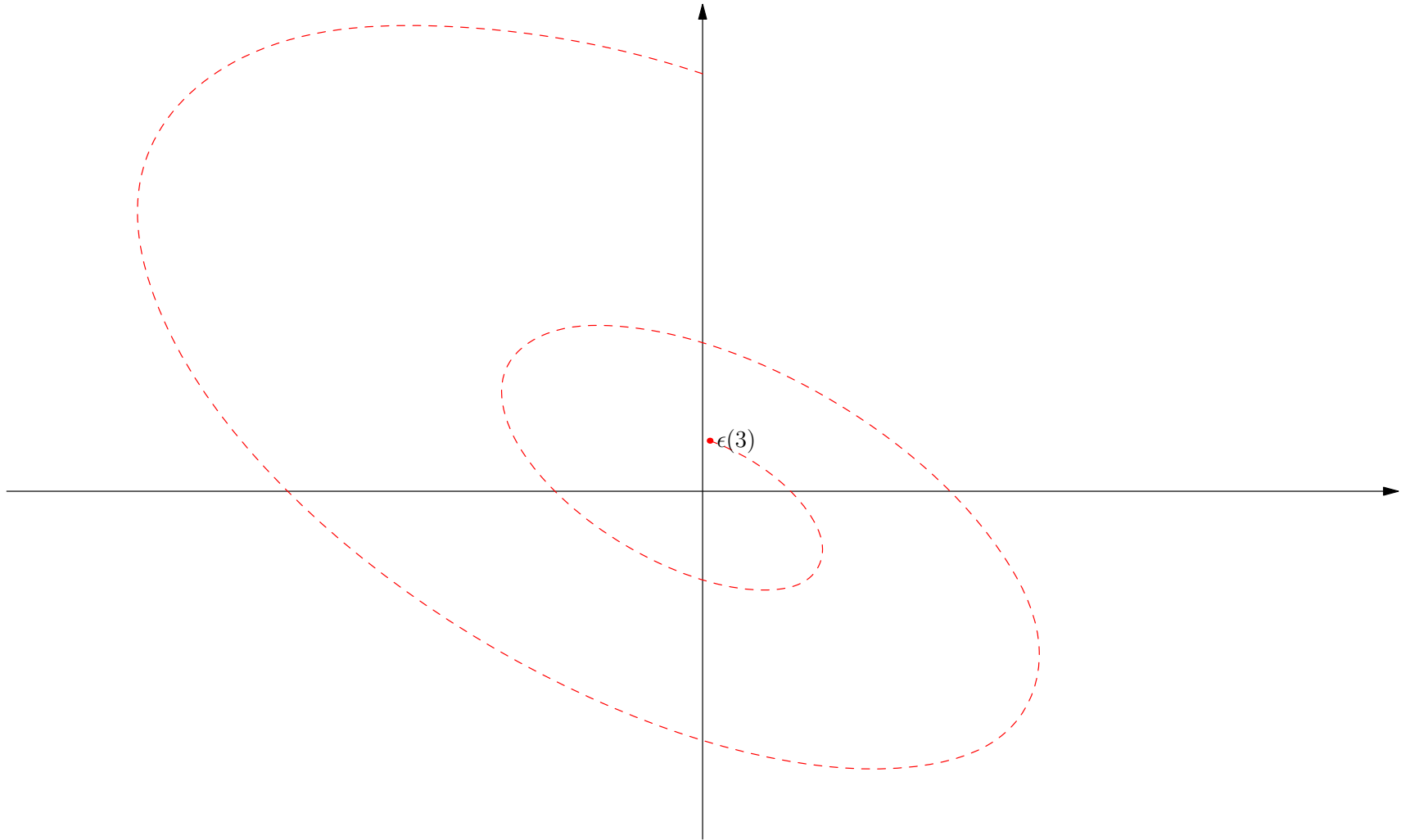
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473i & -0.327 + 0.473i \end{pmatrix} \begin{pmatrix} -0.685 + 4.163i & 0 \\ 0 & -0.685 - 4.163i \end{pmatrix} \begin{pmatrix} 0.611 + 0.422i & 1.056i \\ 0.611 - 0.422i & -1.056i \end{pmatrix}$$



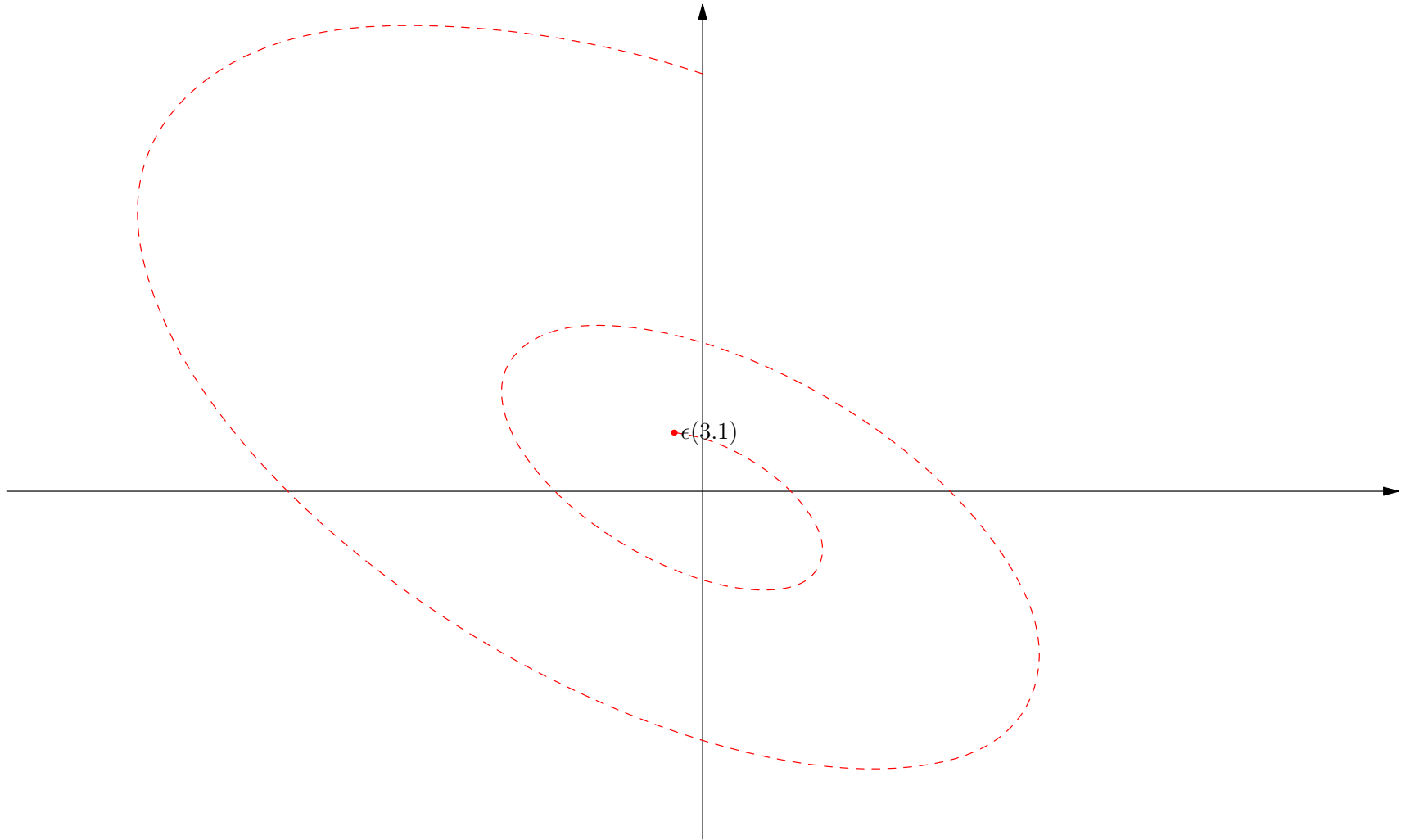
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473i & -0.327 + 0.473i \end{pmatrix} \begin{pmatrix} -0.685 + 4.163i & 0 \\ 0 & -0.685 - 4.163i \end{pmatrix} \begin{pmatrix} 0.611 + 0.422i & 1.056i \\ 0.611 - 0.422i & -1.056i \end{pmatrix}$$



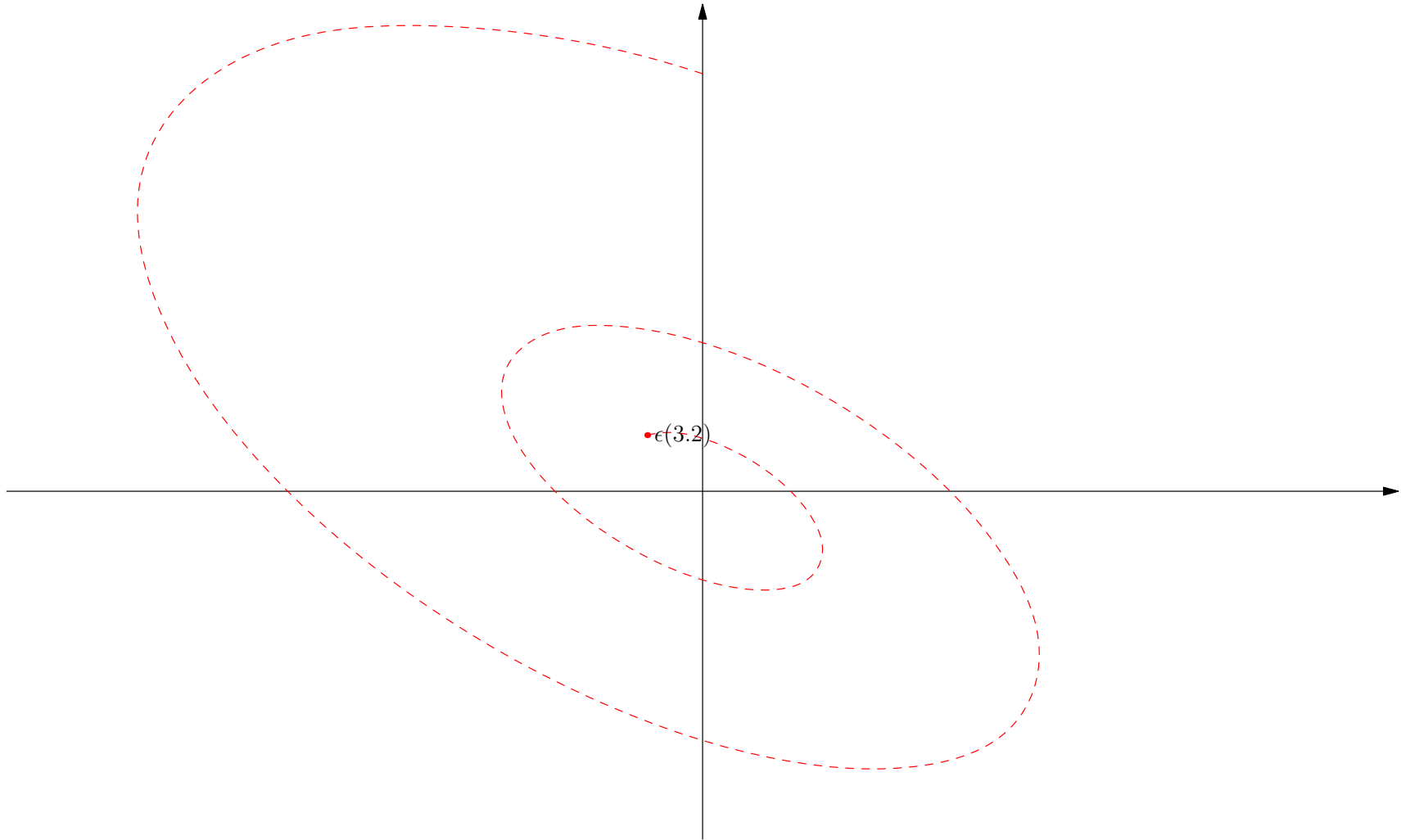
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473i & -0.327 + 0.473i \end{pmatrix} \begin{pmatrix} -0.685 + 4.163i & 0 \\ 0 & -0.685 - 4.163i \end{pmatrix} \begin{pmatrix} 0.611 + 0.422i & 1.056i \\ 0.611 - 0.422i & -1.056i \end{pmatrix}$$



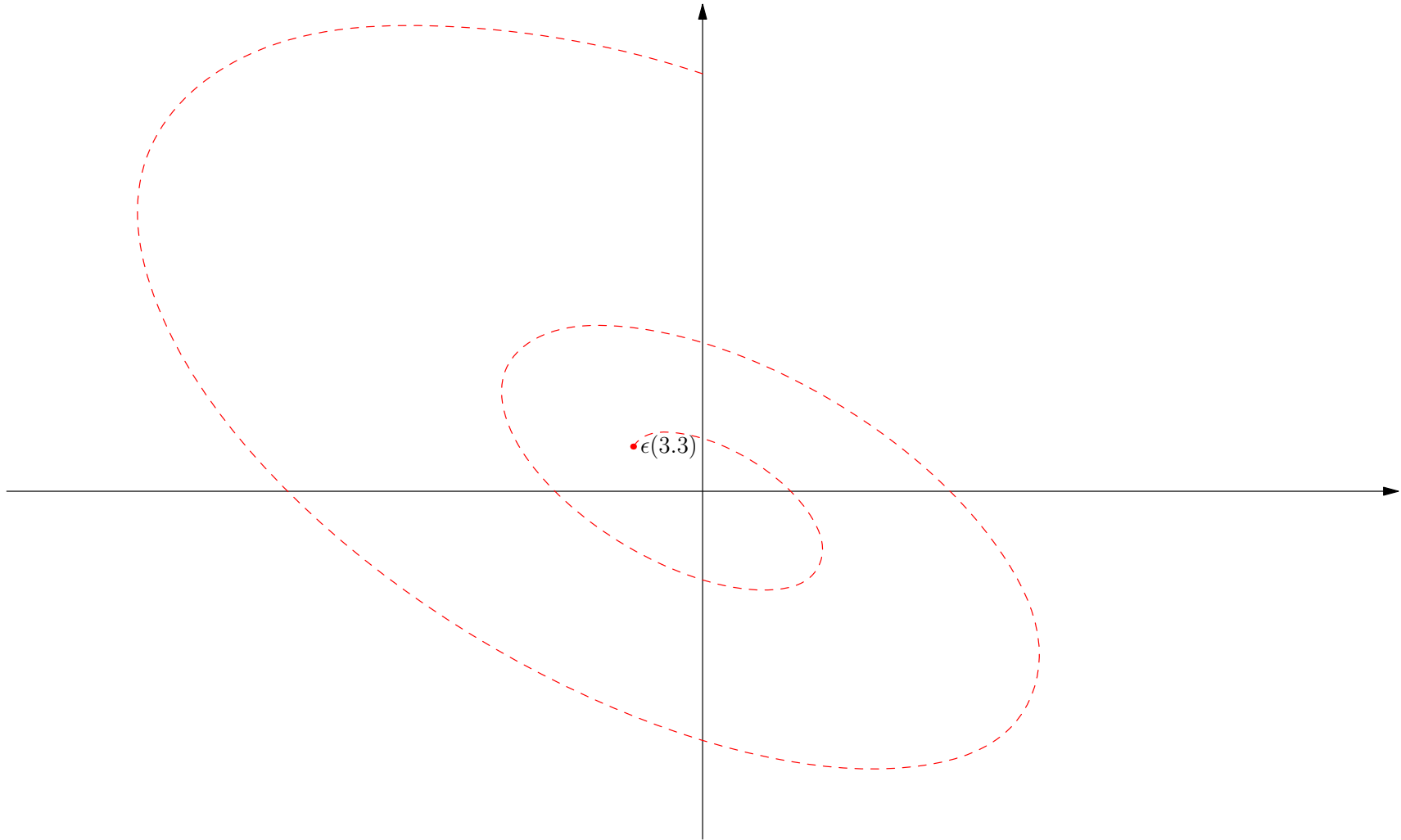
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473i & -0.327 + 0.473i \end{pmatrix} \begin{pmatrix} -0.685 + 4.163i & 0 \\ 0 & -0.685 - 4.163i \end{pmatrix} \begin{pmatrix} 0.611 + 0.422i & 1.056i \\ 0.611 - 0.422i & -1.056i \end{pmatrix}$$



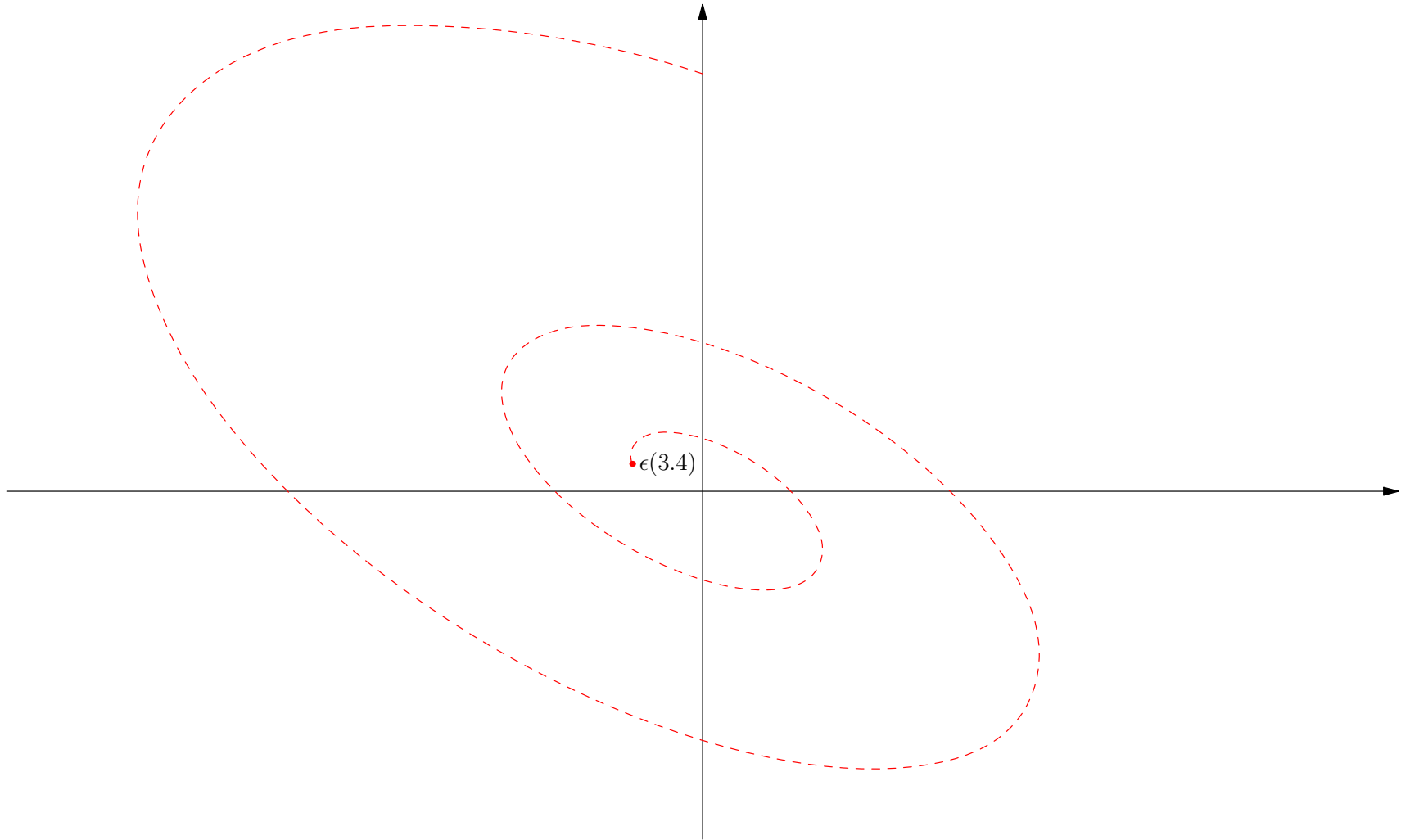
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473i & -0.327 + 0.473i \end{pmatrix} \begin{pmatrix} -0.685 + 4.163i & 0 \\ 0 & -0.685 - 4.163i \end{pmatrix} \begin{pmatrix} 0.611 + 0.422i & 1.056i \\ 0.611 - 0.422i & -1.056i \end{pmatrix}$$



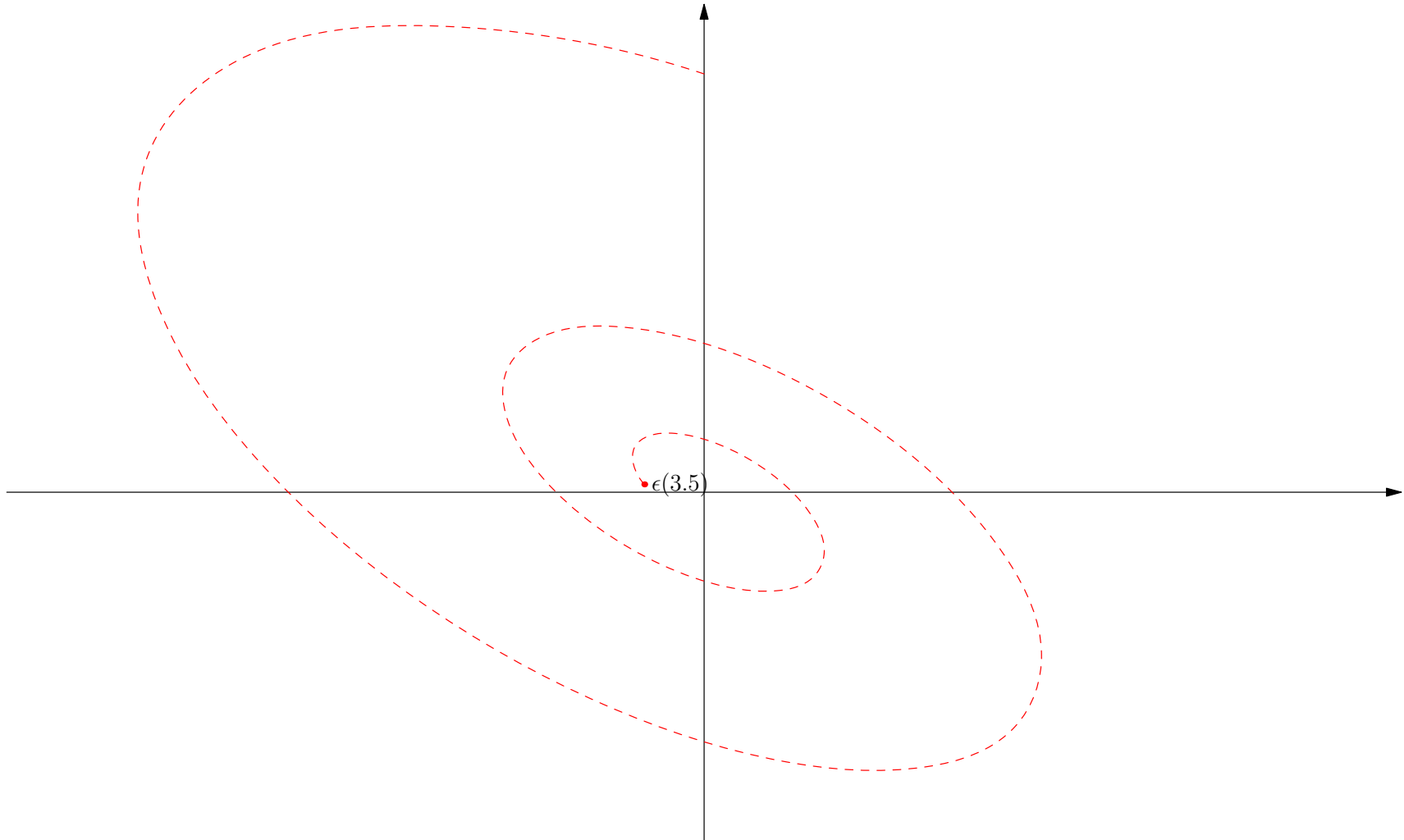
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473i & -0.327 + 0.473i \end{pmatrix} \begin{pmatrix} -0.685 + 4.163i & 0 \\ 0 & -0.685 - 4.163i \end{pmatrix} \begin{pmatrix} 0.611 + 0.422i & 1.056i \\ 0.611 - 0.422i & -1.056i \end{pmatrix}$$



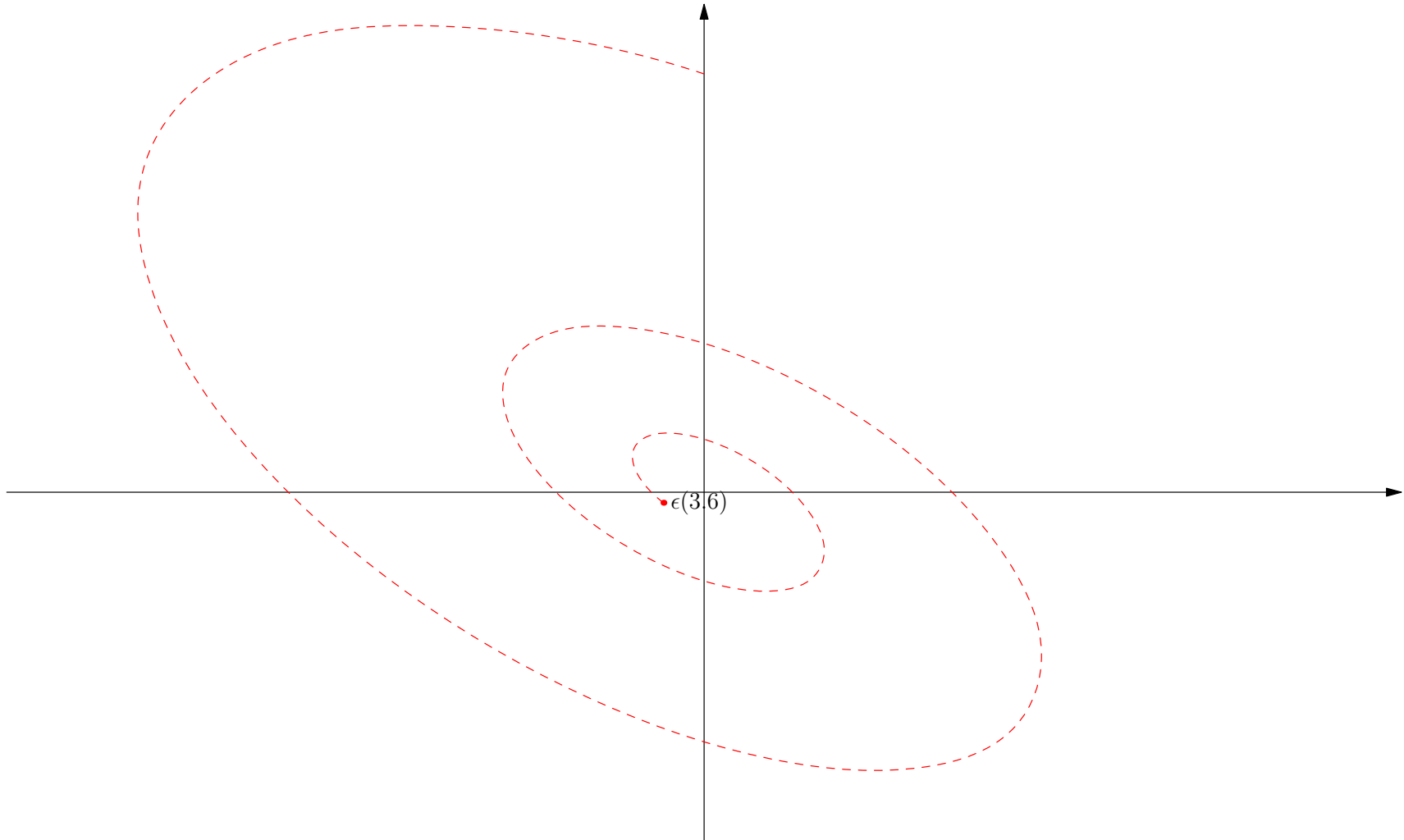
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473i & -0.327 + 0.473i \end{pmatrix} \begin{pmatrix} -0.685 + 4.163i & 0 \\ 0 & -0.685 - 4.163i \end{pmatrix} \begin{pmatrix} 0.611 + 0.422i & 1.056i \\ 0.611 - 0.422i & -1.056i \end{pmatrix}$$



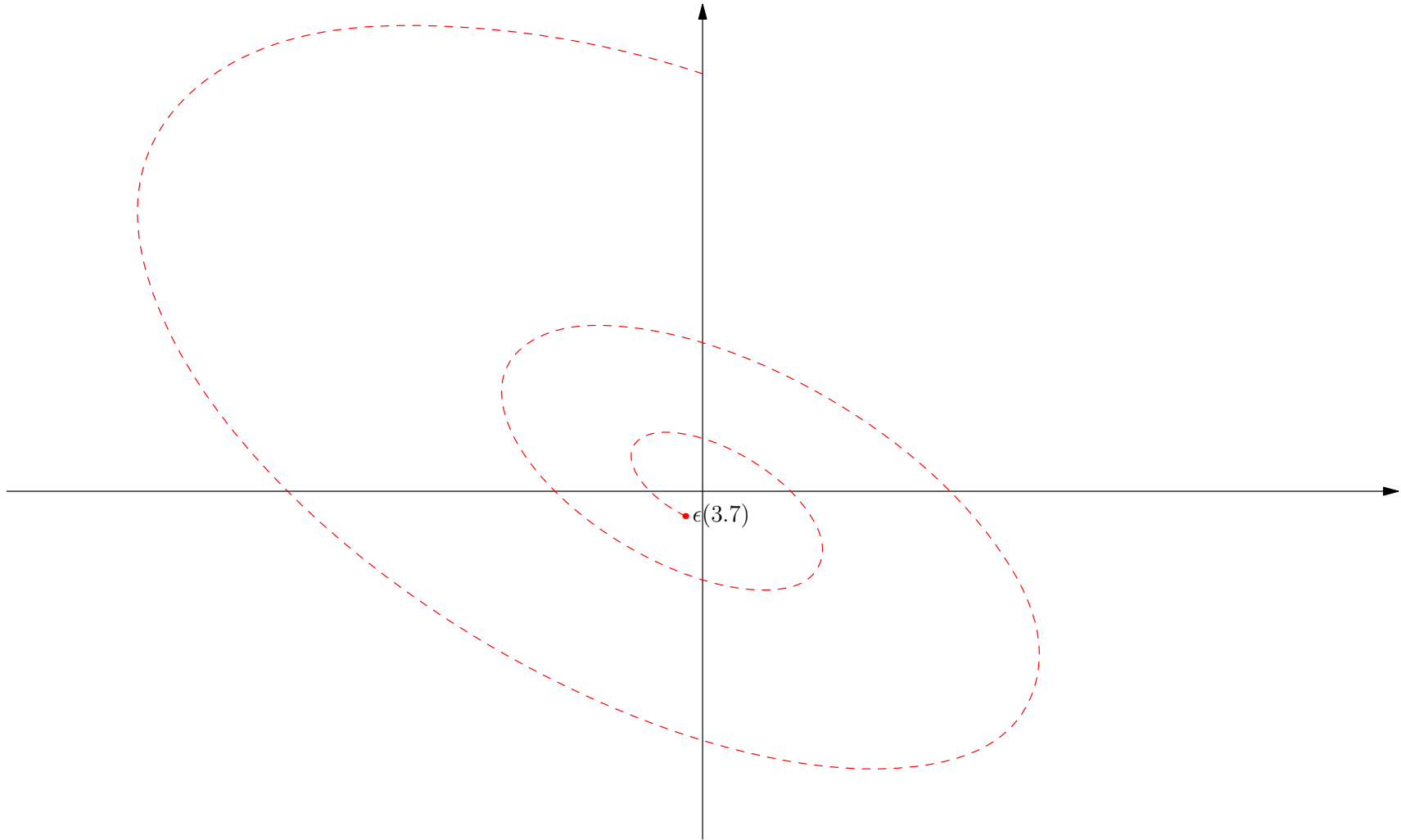
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473i & -0.327 + 0.473i \end{pmatrix} \begin{pmatrix} -0.685 + 4.163i & 0 \\ 0 & -0.685 - 4.163i \end{pmatrix} \begin{pmatrix} 0.611 + 0.422i & 1.056i \\ 0.611 - 0.422i & -1.056i \end{pmatrix}$$



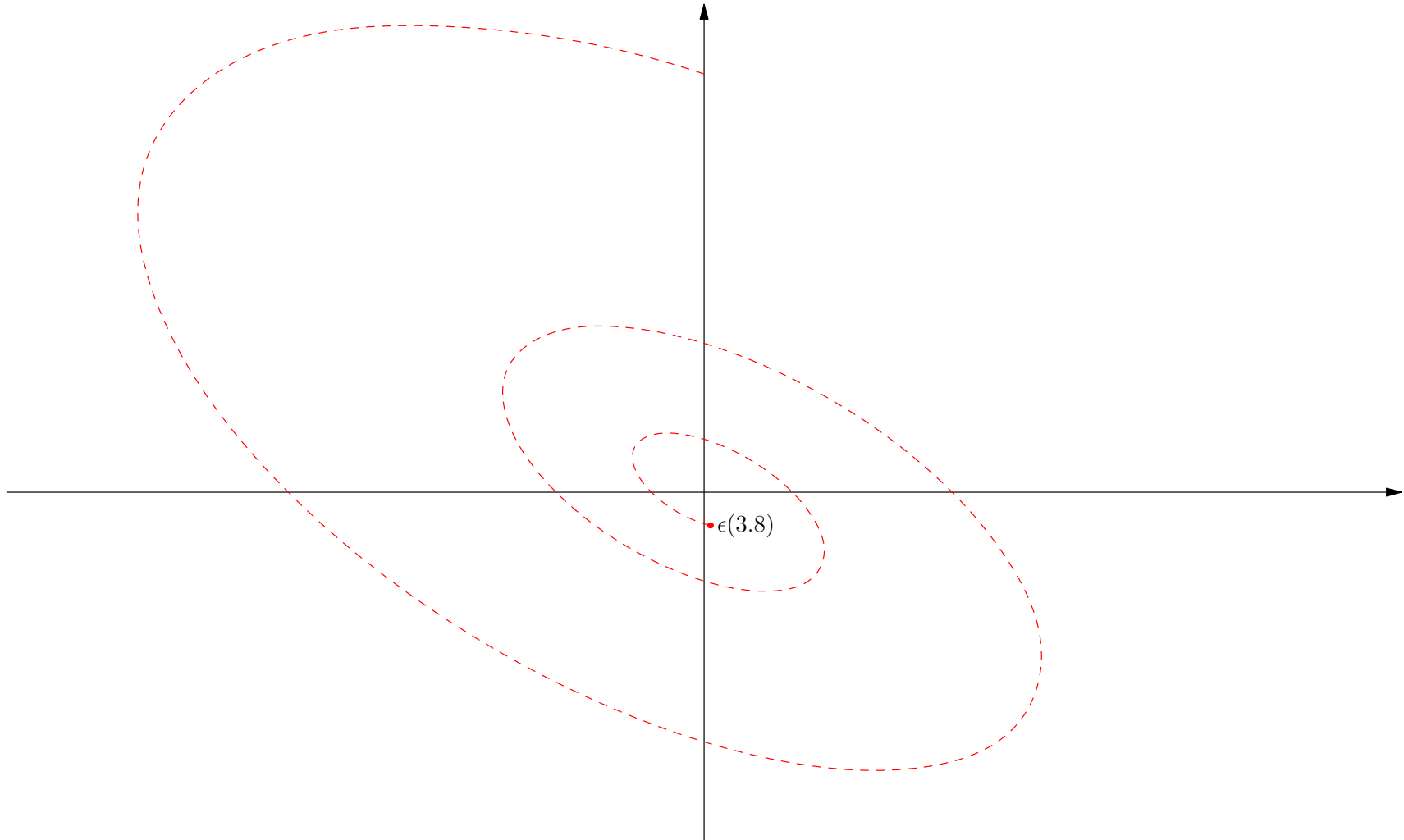
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473i & -0.327 + 0.473i \end{pmatrix} \begin{pmatrix} -0.685 + 4.163i & 0 \\ 0 & -0.685 - 4.163i \end{pmatrix} \begin{pmatrix} 0.611 + 0.422i & 1.056i \\ 0.611 - 0.422i & -1.056i \end{pmatrix}$$



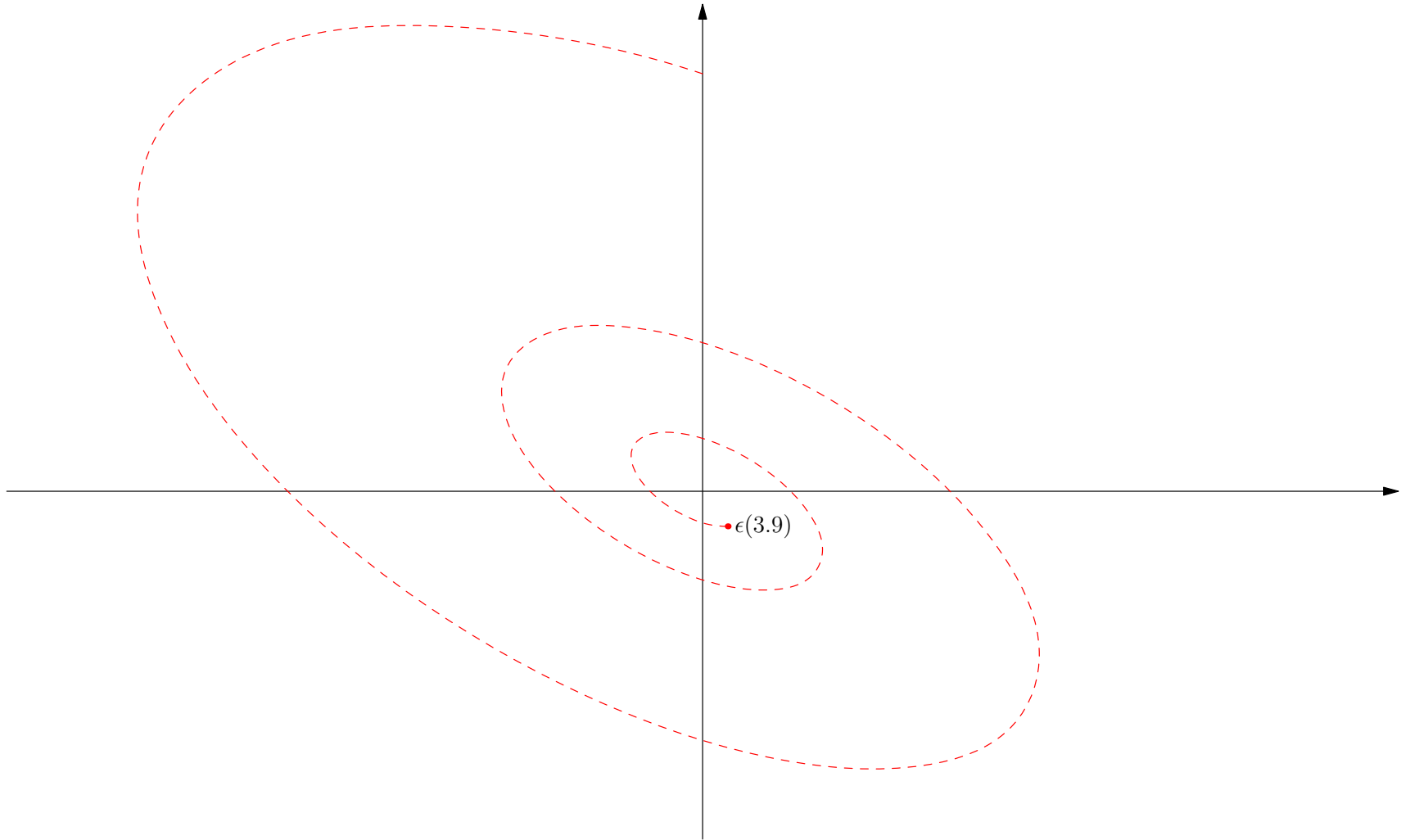
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473i & -0.327 + 0.473i \end{pmatrix} \begin{pmatrix} -0.685 + 4.163i & 0 \\ 0 & -0.685 - 4.163i \end{pmatrix} \begin{pmatrix} 0.611 + 0.422i & 1.056i \\ 0.611 - 0.422i & -1.056i \end{pmatrix}$$



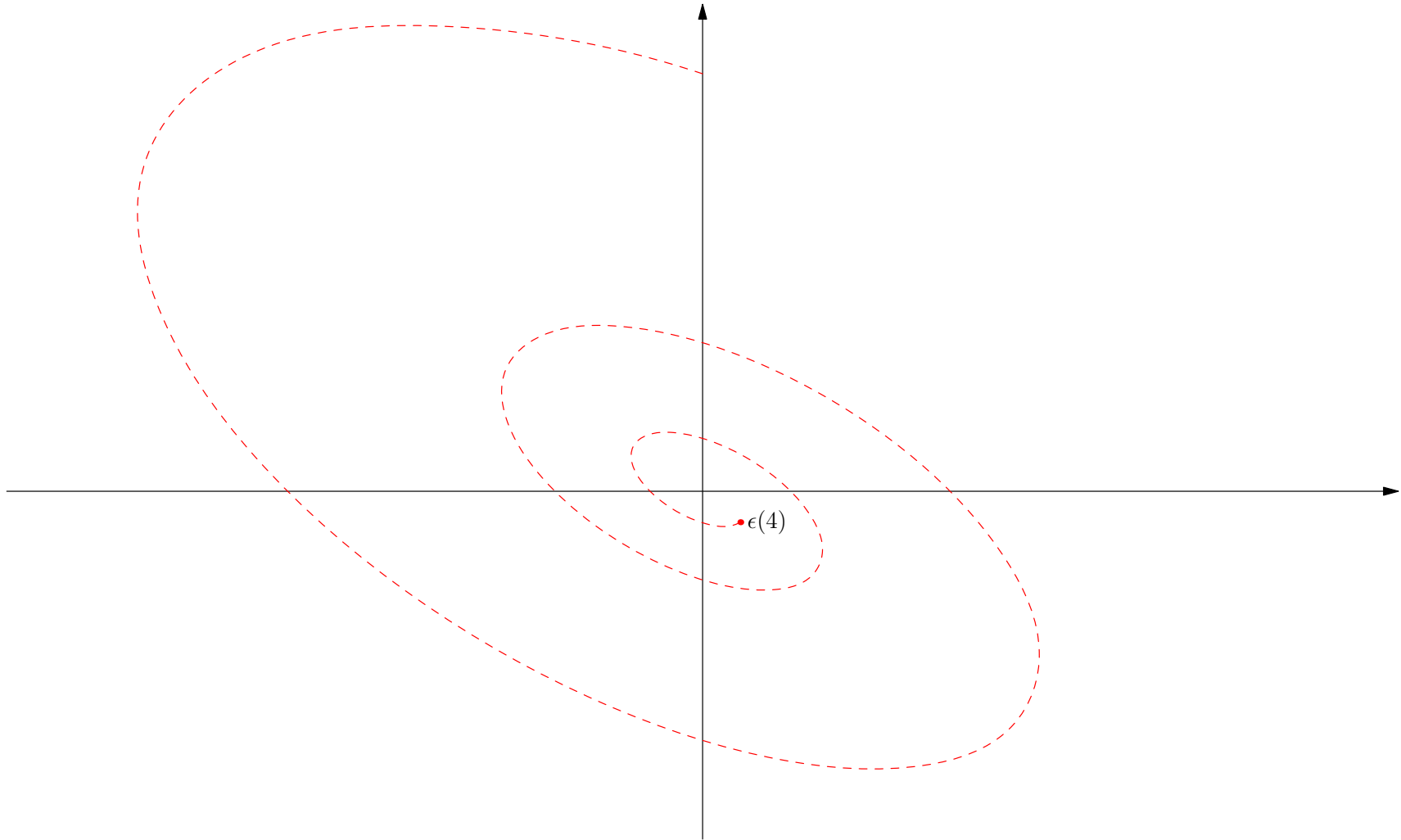
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473i & -0.327 + 0.473i \end{pmatrix} \begin{pmatrix} -0.685 + 4.163i & 0 \\ 0 & -0.685 - 4.163i \end{pmatrix} \begin{pmatrix} 0.611 + 0.422i & 1.056i \\ 0.611 - 0.422i & -1.056i \end{pmatrix}$$



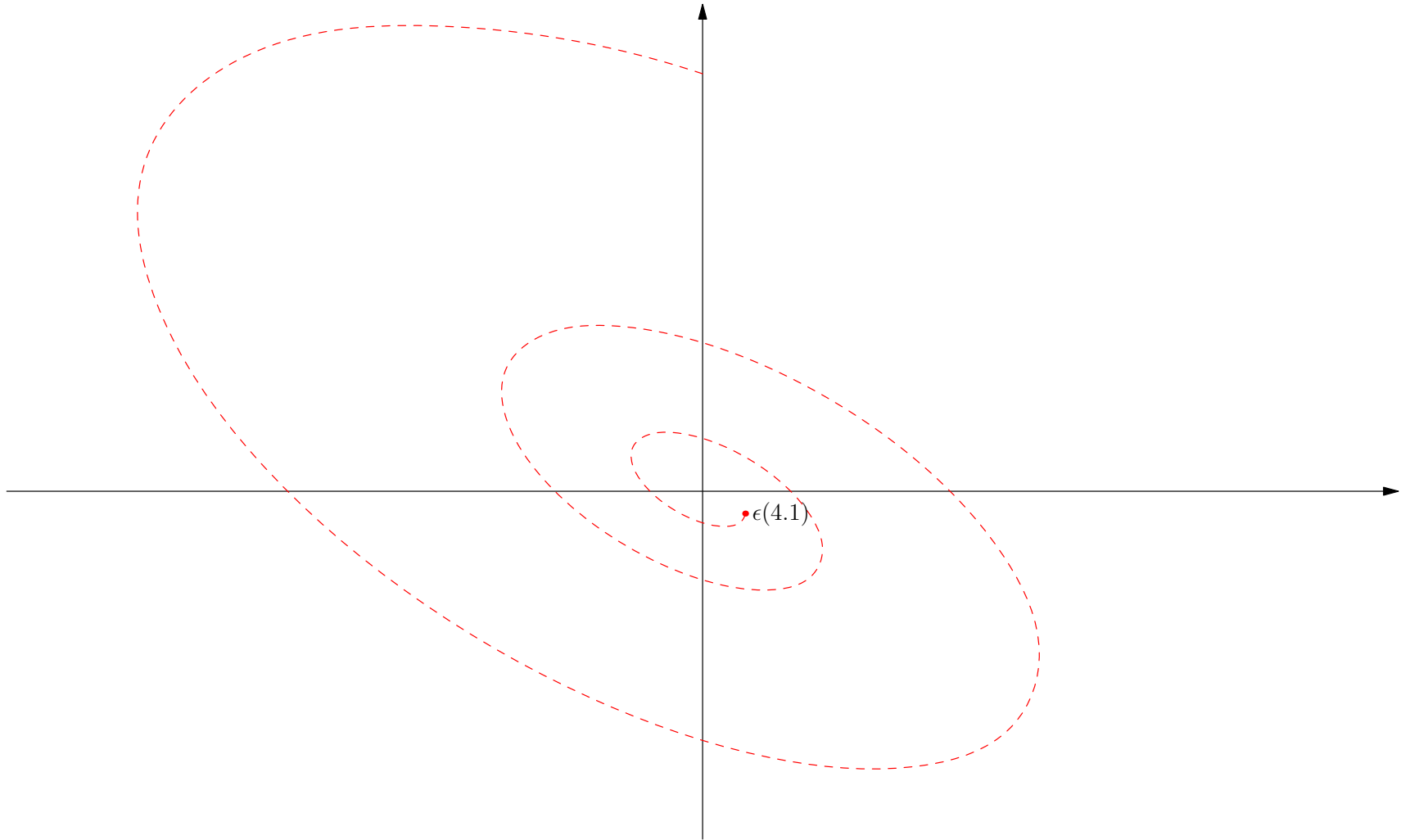
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473i & -0.327 + 0.473i \end{pmatrix} \begin{pmatrix} -0.685 + 4.163i & 0 \\ 0 & -0.685 - 4.163i \end{pmatrix} \begin{pmatrix} 0.611 + 0.422i & 1.056i \\ 0.611 - 0.422i & -1.056i \end{pmatrix}$$



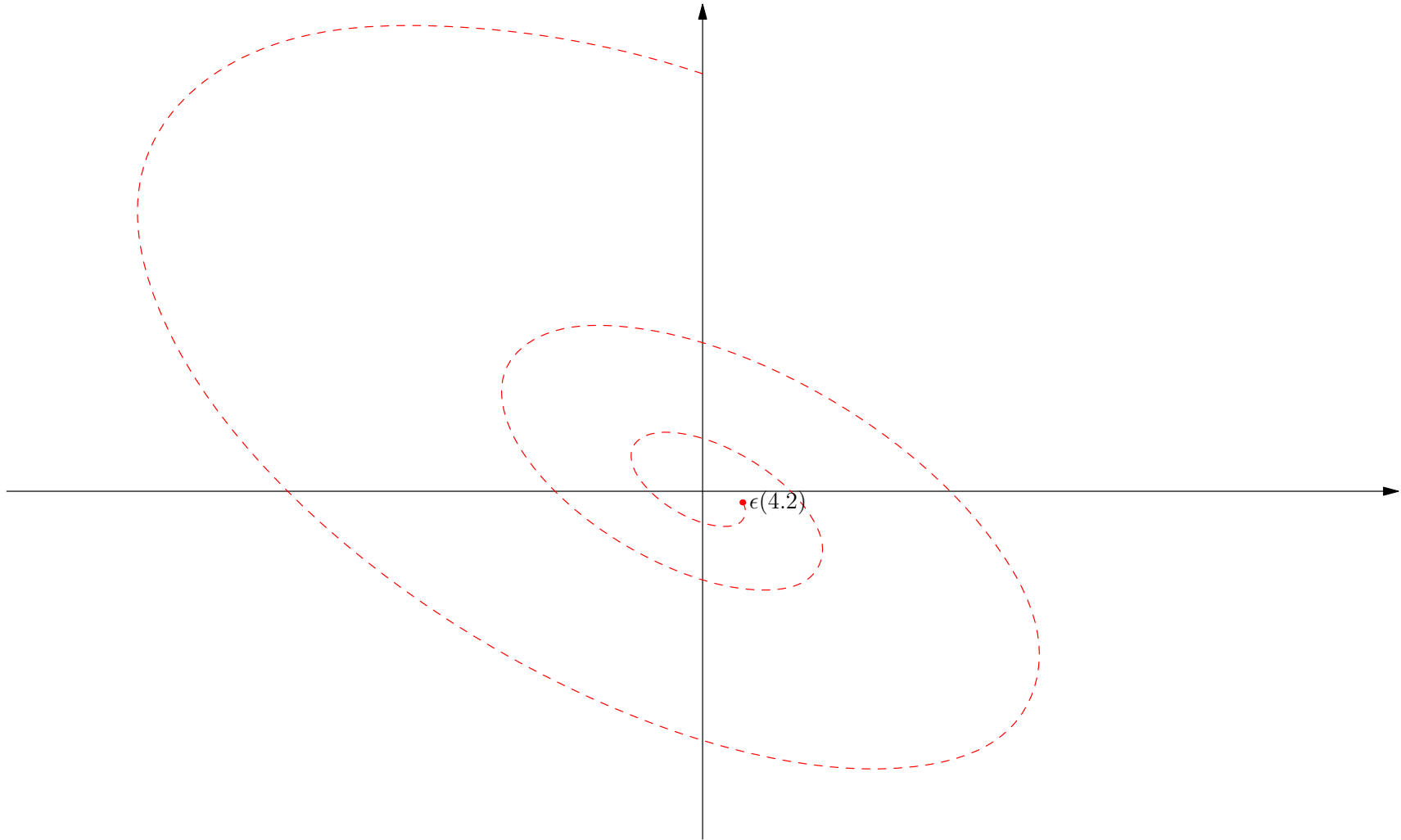
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473i & -0.327 + 0.473i \end{pmatrix} \begin{pmatrix} -0.685 + 4.163i & 0 \\ 0 & -0.685 - 4.163i \end{pmatrix} \begin{pmatrix} 0.611 + 0.422i & 1.056i \\ 0.611 - 0.422i & -1.056i \end{pmatrix}$$



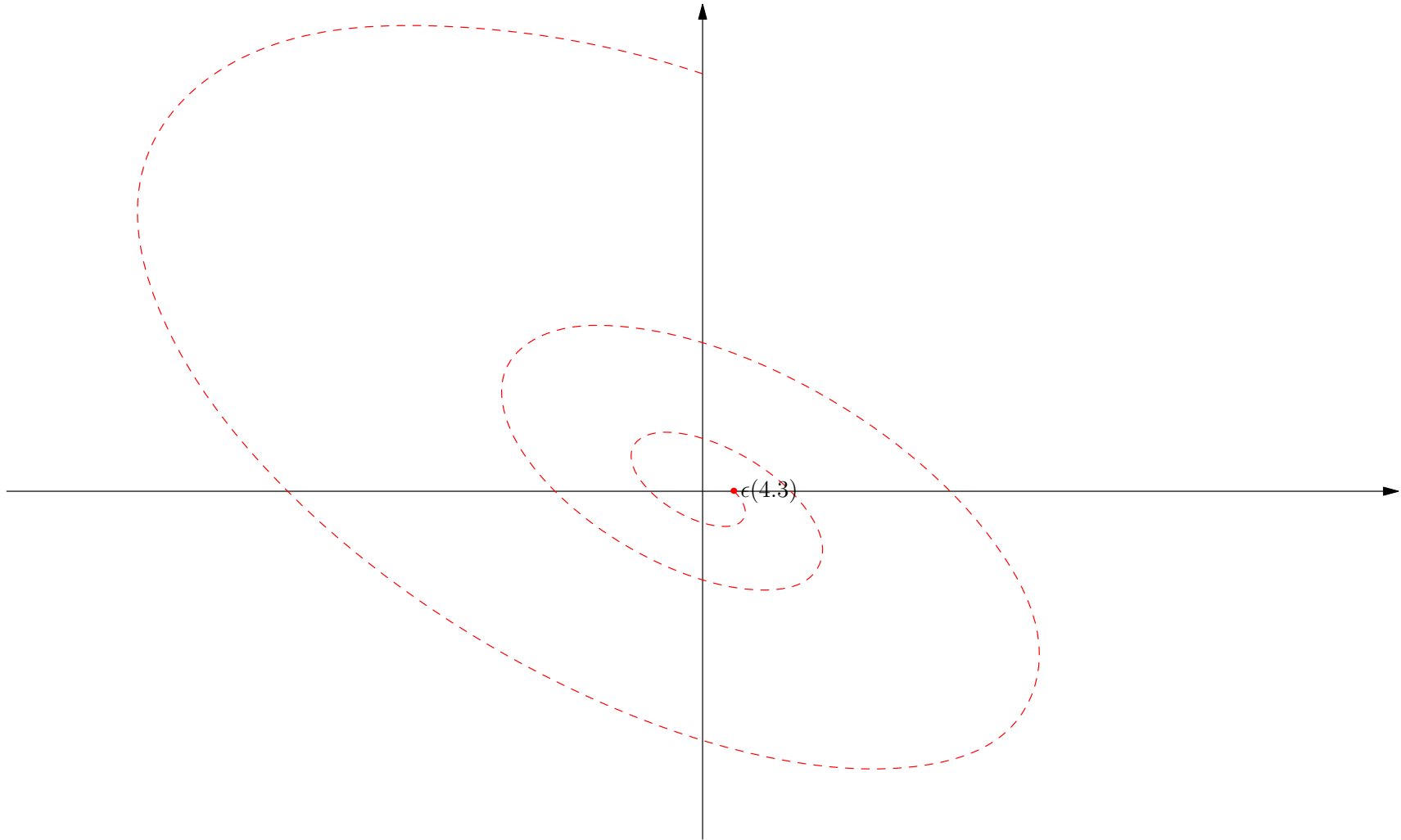
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473i & -0.327 + 0.473i \end{pmatrix} \begin{pmatrix} -0.685 + 4.163i & 0 \\ 0 & -0.685 - 4.163i \end{pmatrix} \begin{pmatrix} 0.611 + 0.422i & 1.056i \\ 0.611 - 0.422i & -1.056i \end{pmatrix}$$



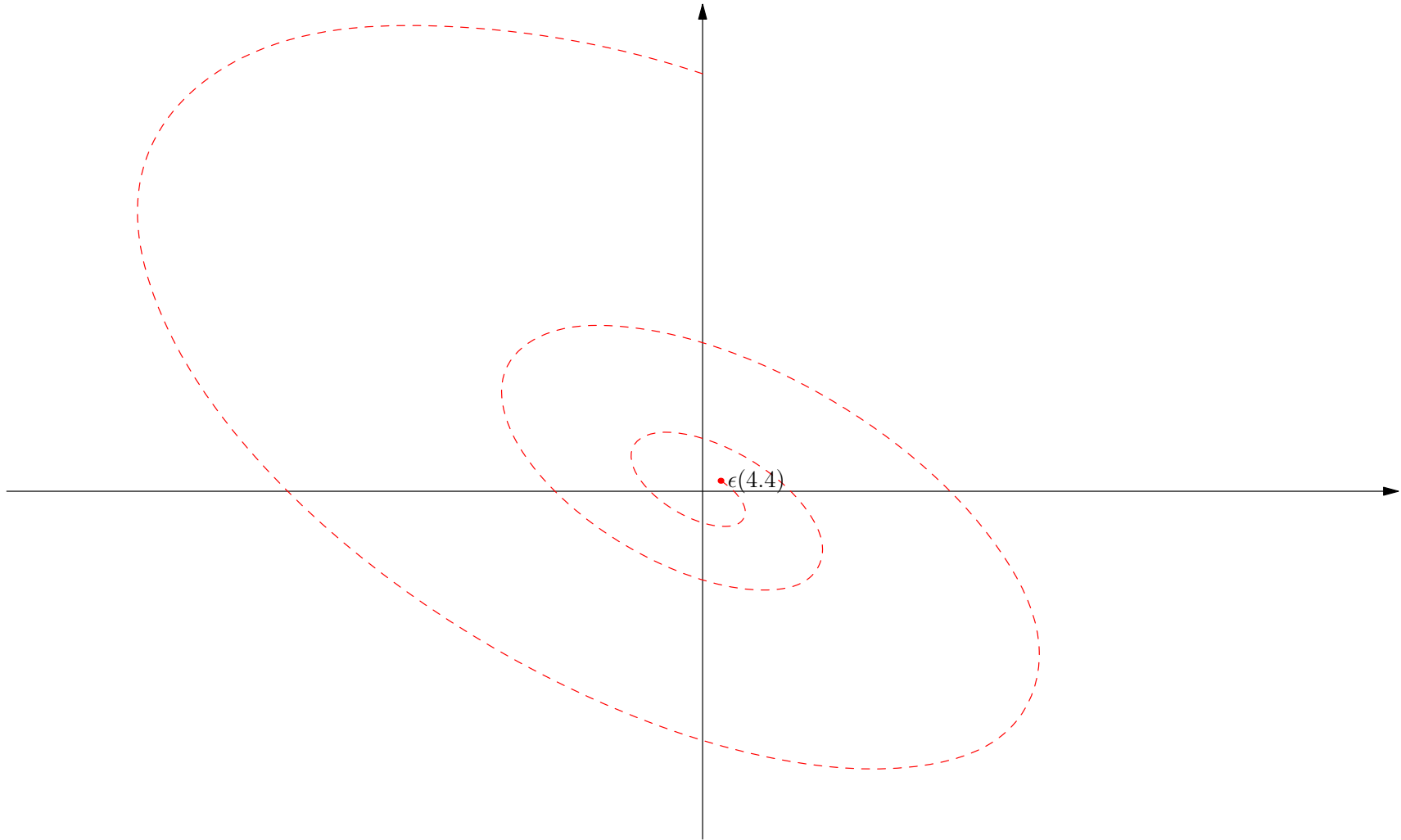
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473i & -0.327 + 0.473i \end{pmatrix} \begin{pmatrix} -0.685 + 4.163i & 0 \\ 0 & -0.685 - 4.163i \end{pmatrix} \begin{pmatrix} 0.611 + 0.422i & 1.056i \\ 0.611 - 0.422i & -1.056i \end{pmatrix}$$



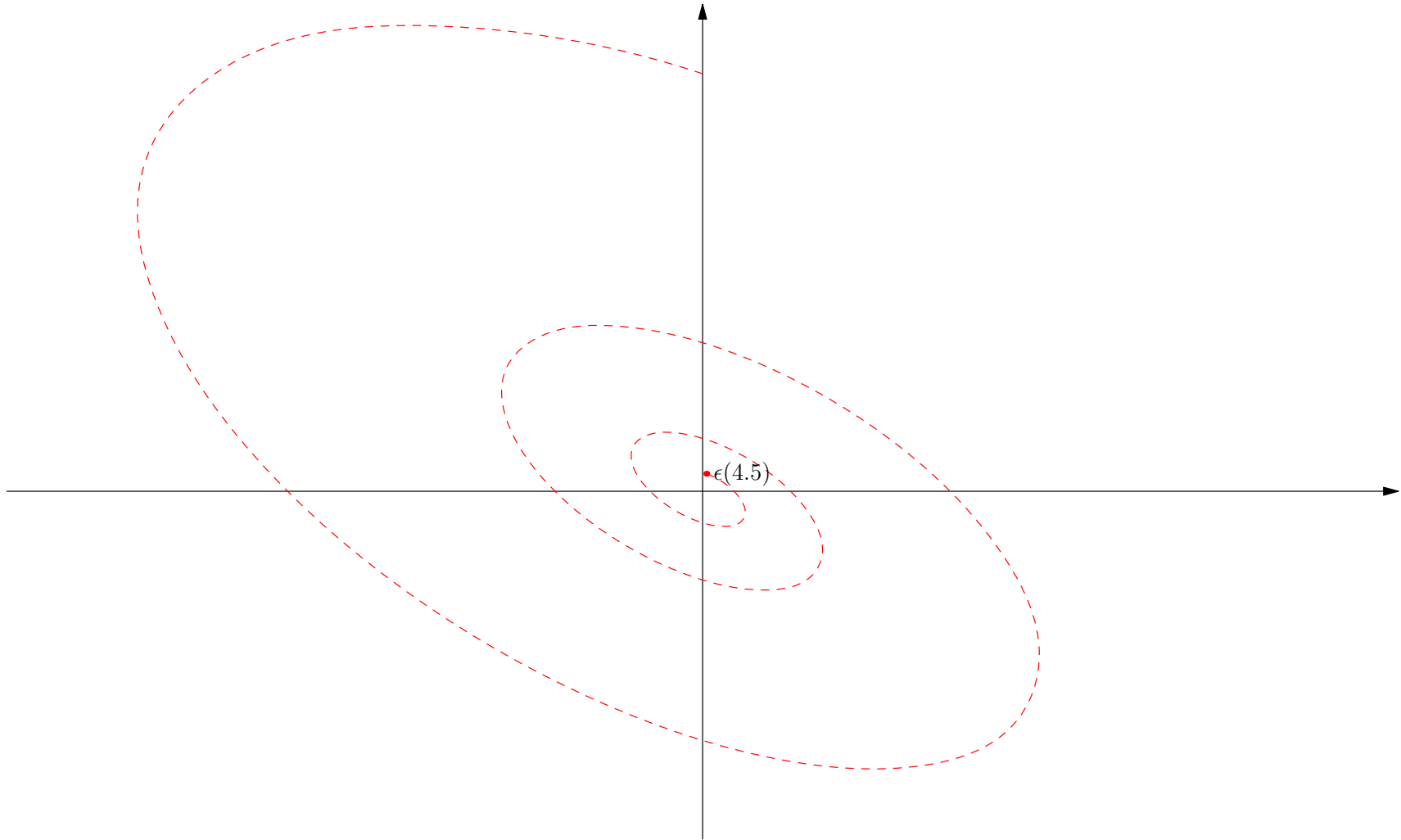
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473i & -0.327 + 0.473i \end{pmatrix} \begin{pmatrix} -0.685 + 4.163i & 0 \\ 0 & -0.685 - 4.163i \end{pmatrix} \begin{pmatrix} 0.611 + 0.422i & 1.056i \\ 0.611 - 0.422i & -1.056i \end{pmatrix}$$



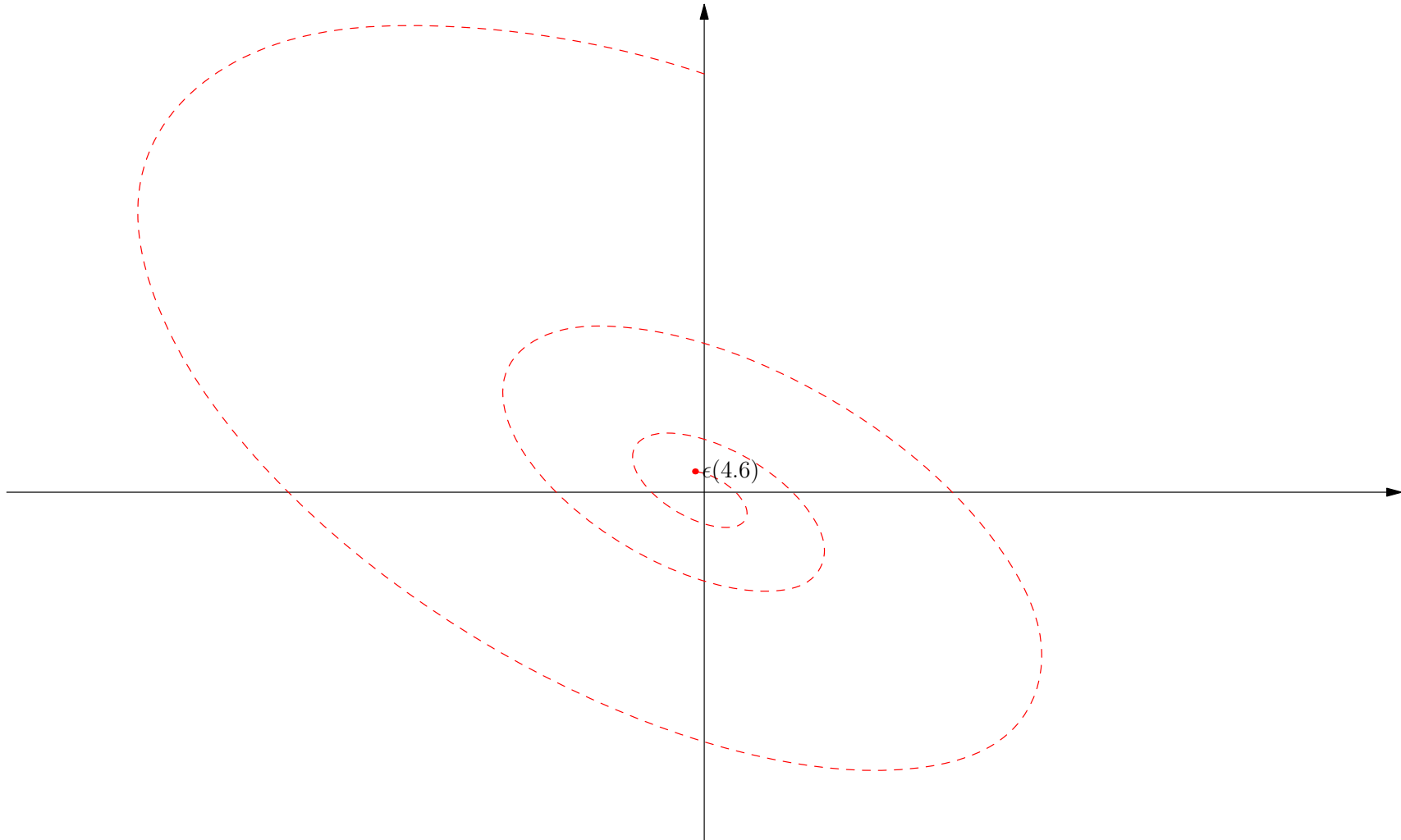
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473i & -0.327 + 0.473i \end{pmatrix} \begin{pmatrix} -0.685 + 4.163i & 0 \\ 0 & -0.685 - 4.163i \end{pmatrix} \begin{pmatrix} 0.611 + 0.422i & 1.056i \\ 0.611 - 0.422i & -1.056i \end{pmatrix}$$



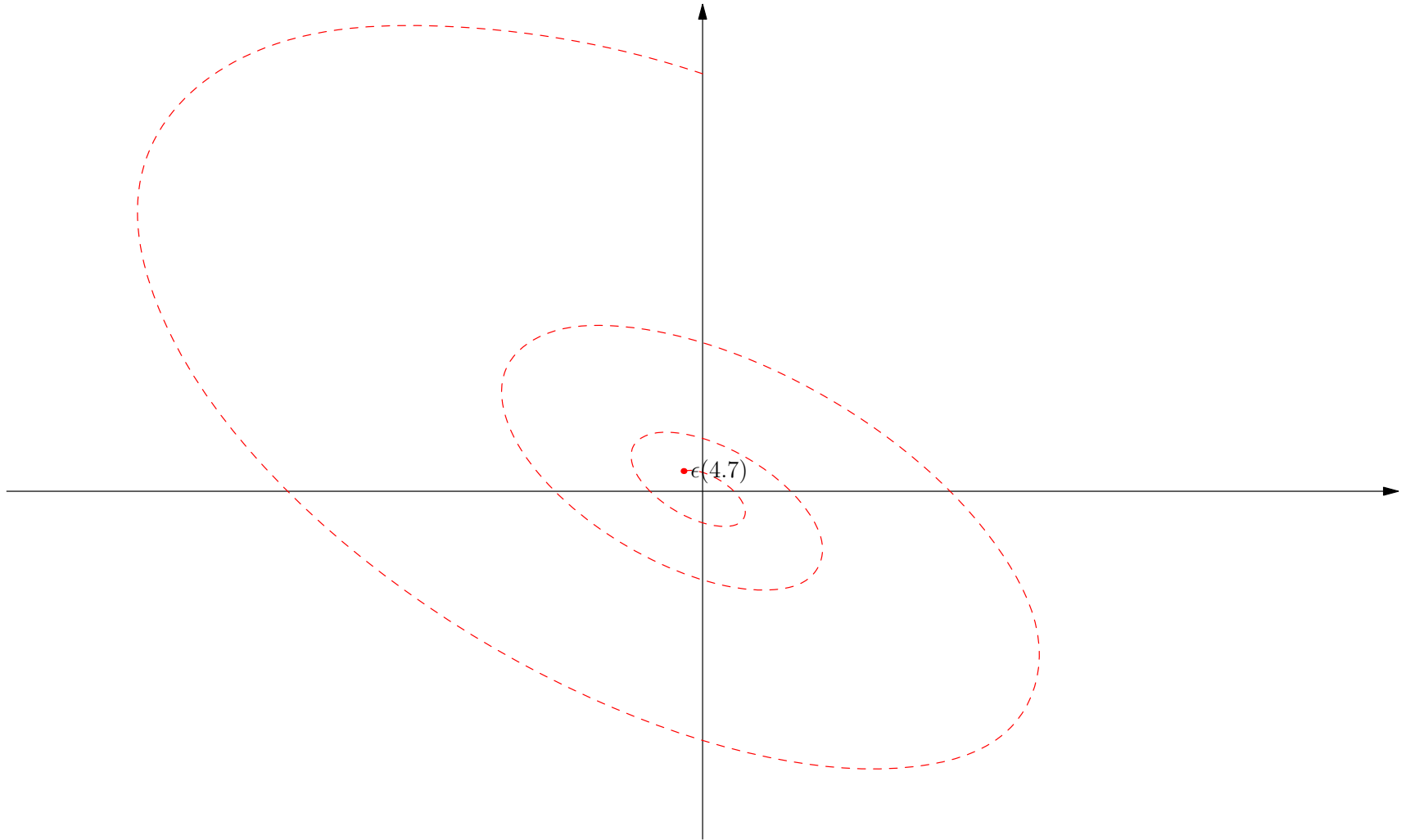
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473i & -0.327 + 0.473i \end{pmatrix} \begin{pmatrix} -0.685 + 4.163i & 0 \\ 0 & -0.685 - 4.163i \end{pmatrix} \begin{pmatrix} 0.611 + 0.422i & 1.056i \\ 0.611 - 0.422i & -1.056i \end{pmatrix}$$



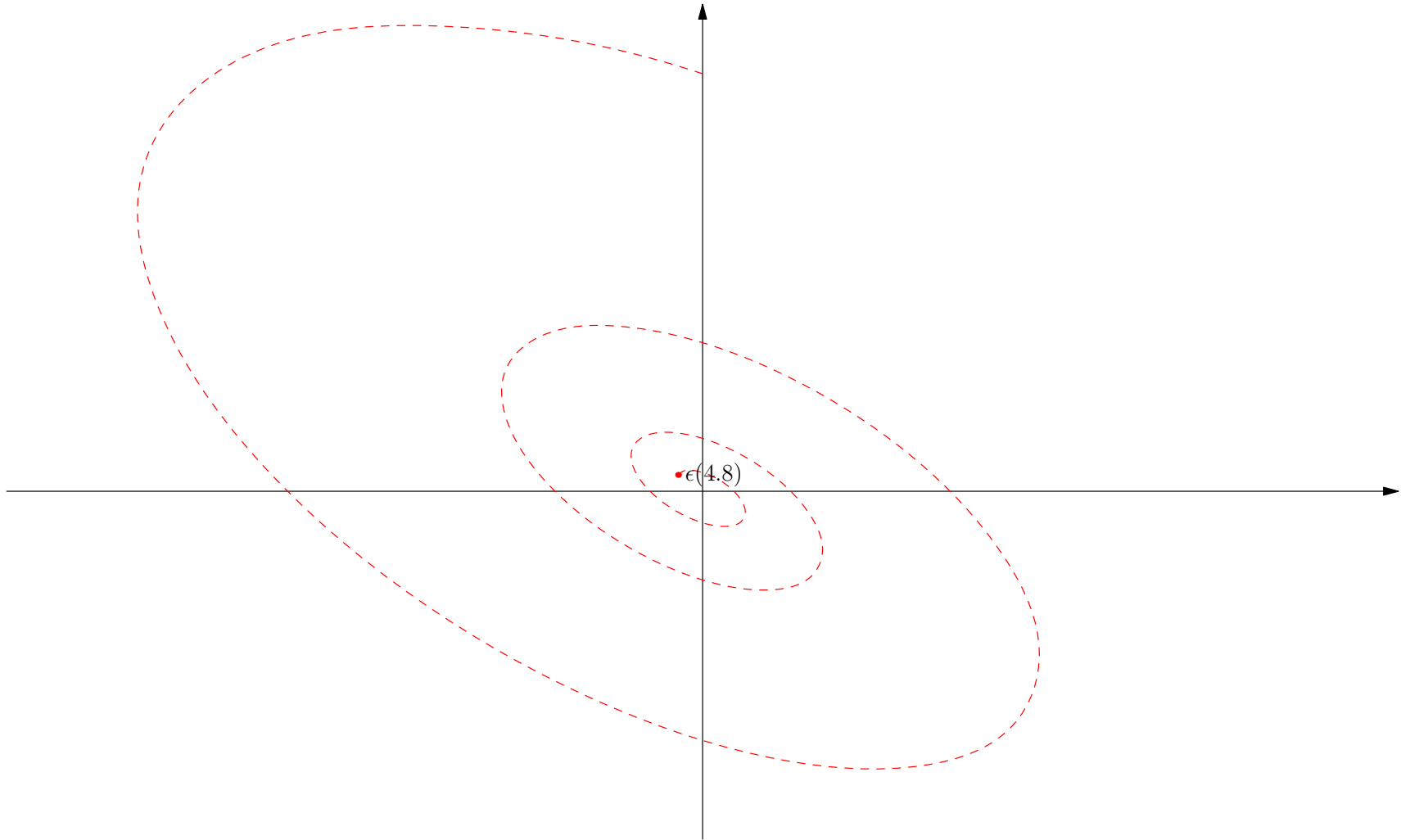
Dynamics

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{pmatrix} 0.818 & 0.818 \\ -0.327 - 0.473i & -0.327 + 0.473i \end{pmatrix} \begin{pmatrix} -0.685 + 4.163i & 0 \\ 0 & -0.685 - 4.163i \end{pmatrix} \begin{pmatrix} 0.611 + 0.422i & 1.056i \\ 0.611 - 0.422i & -1.056i \end{pmatrix}$$



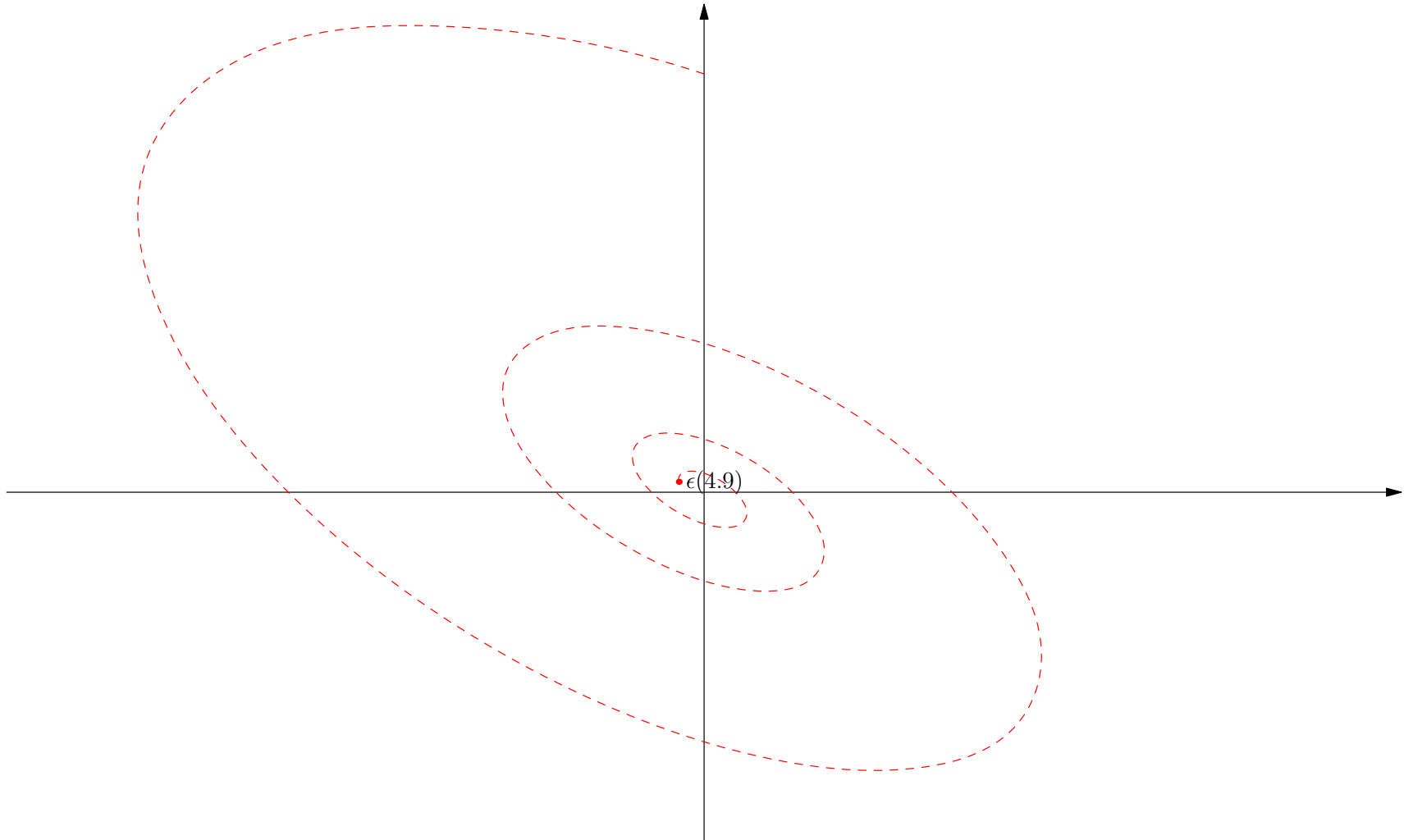
Dynamics

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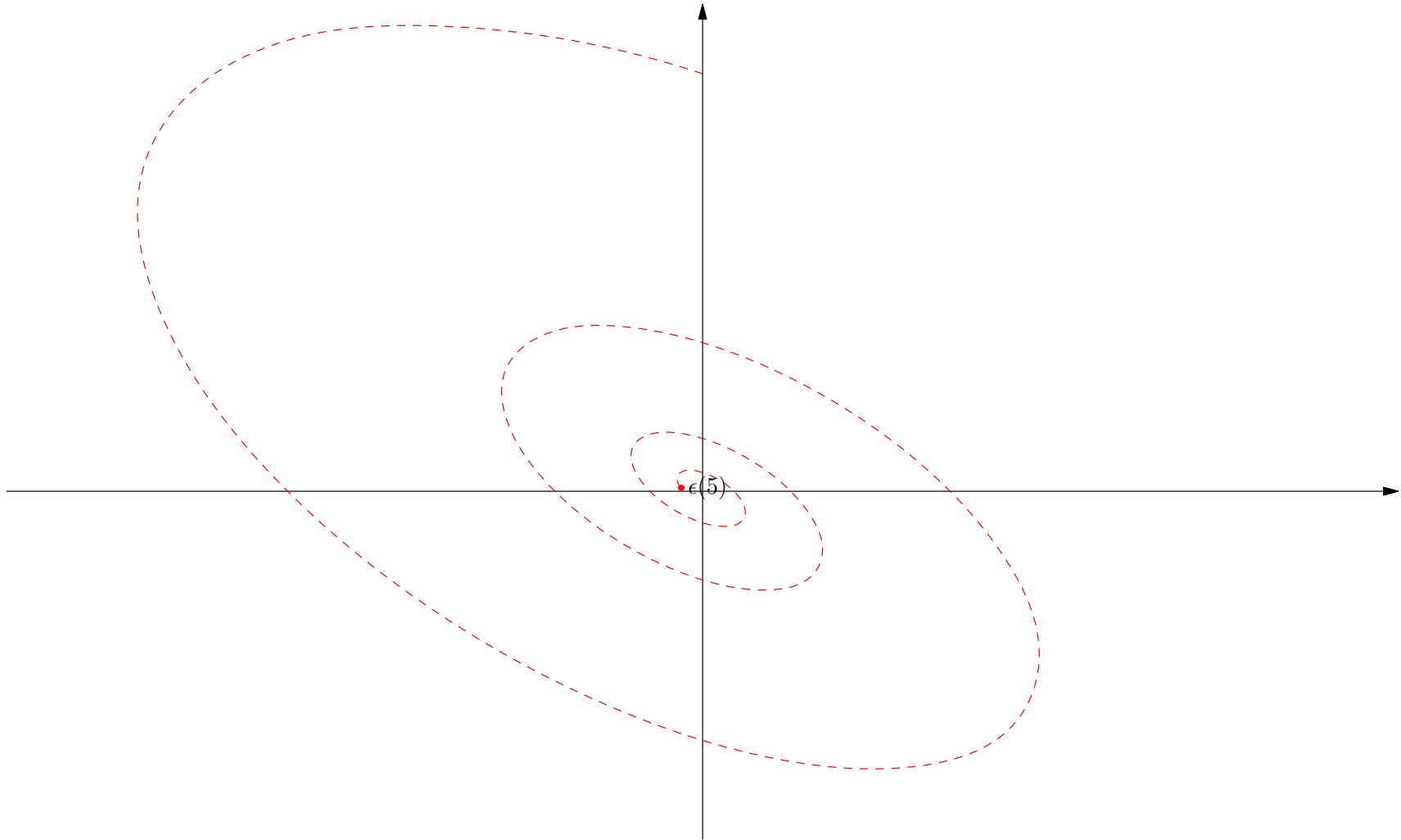
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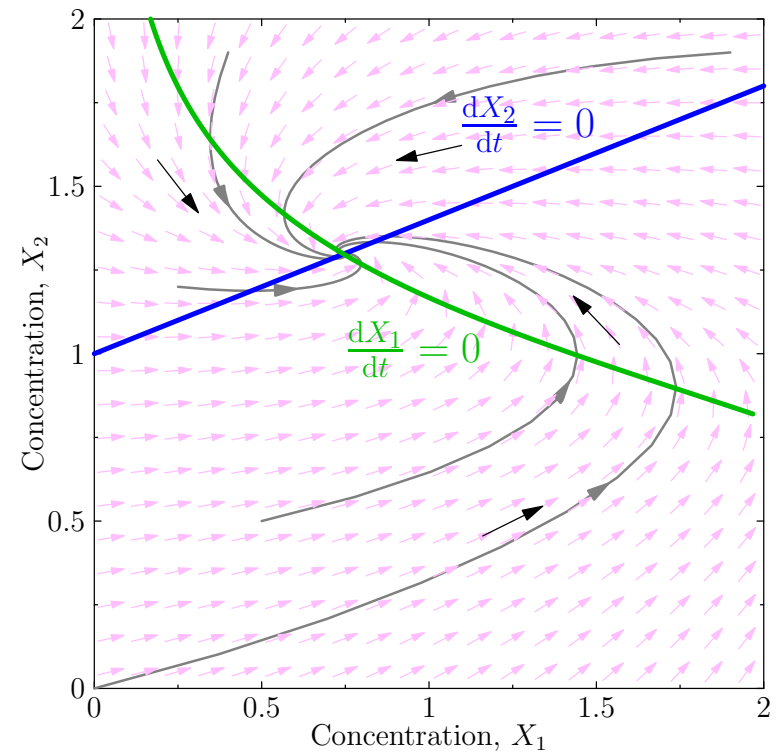
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Outline

1. Understanding ODEs
2. Stability
3. **Bifurcations**



Back to Bistability

- Recall that we could create a bistable system through mutual inhibition

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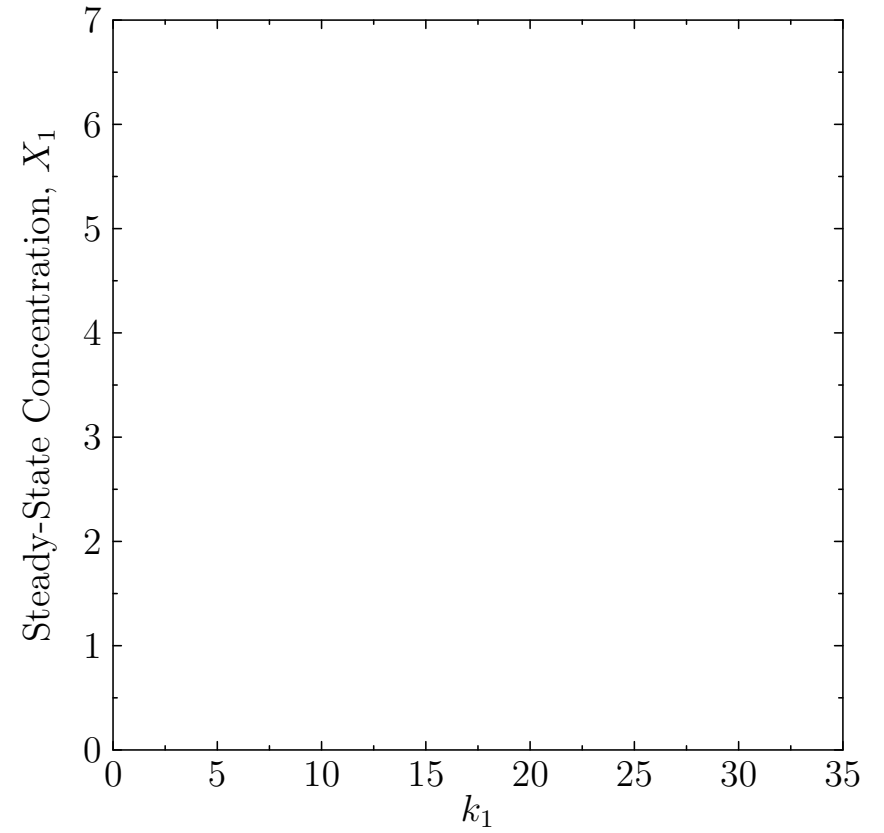
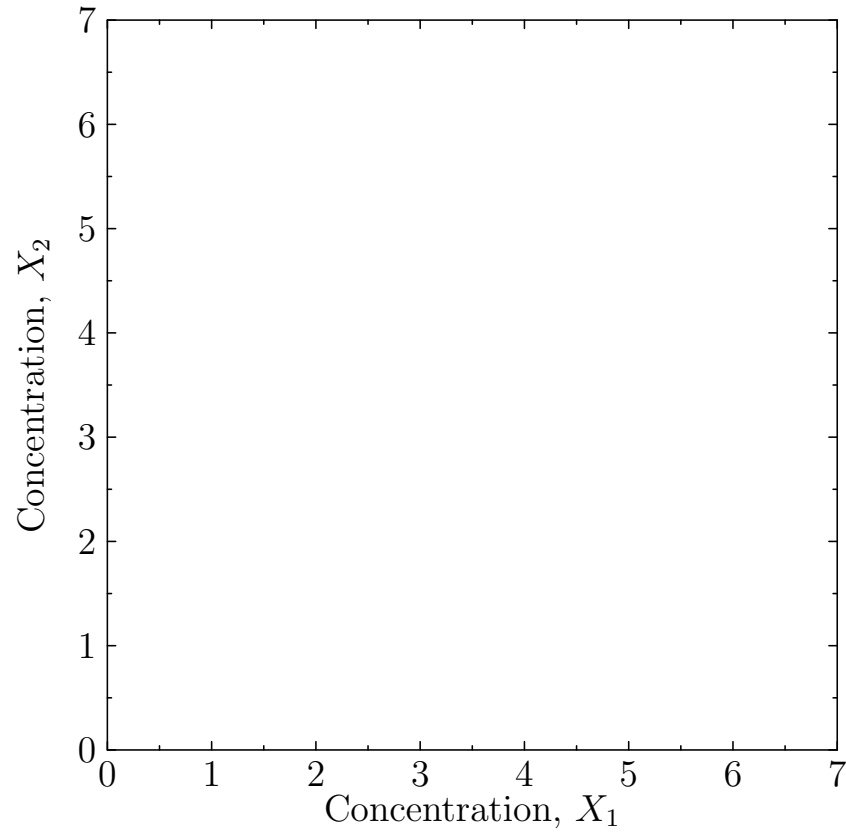
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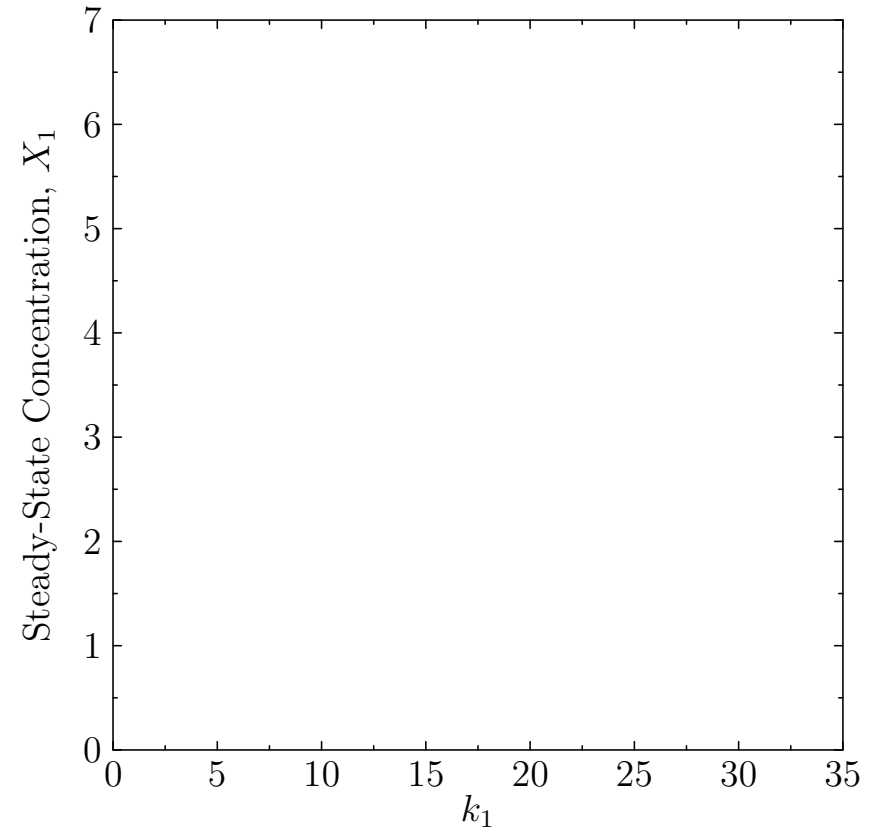
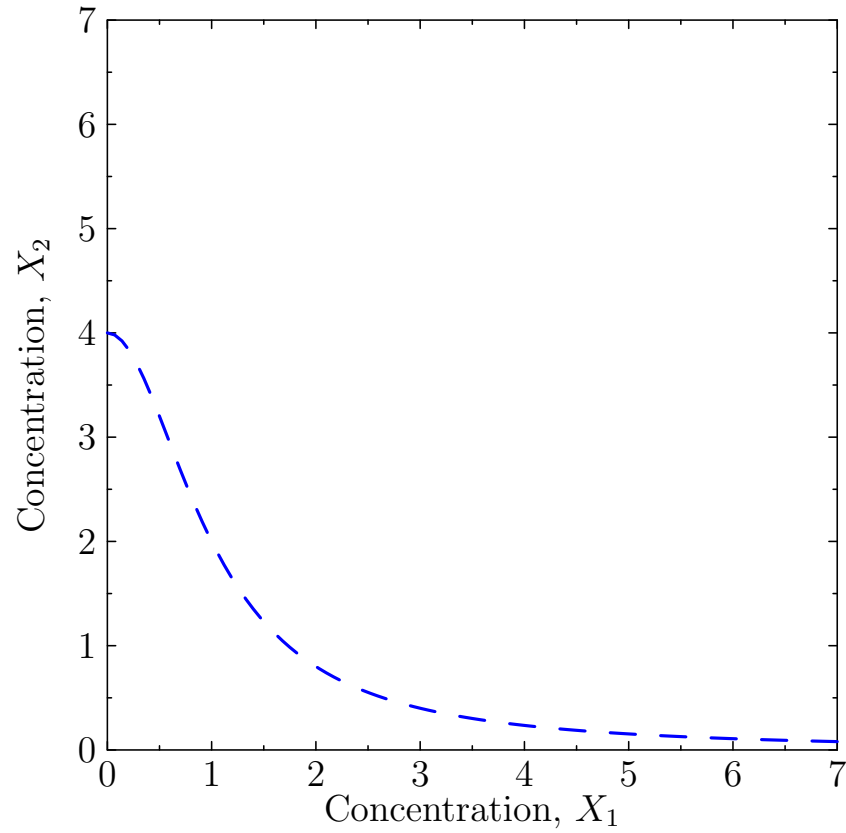
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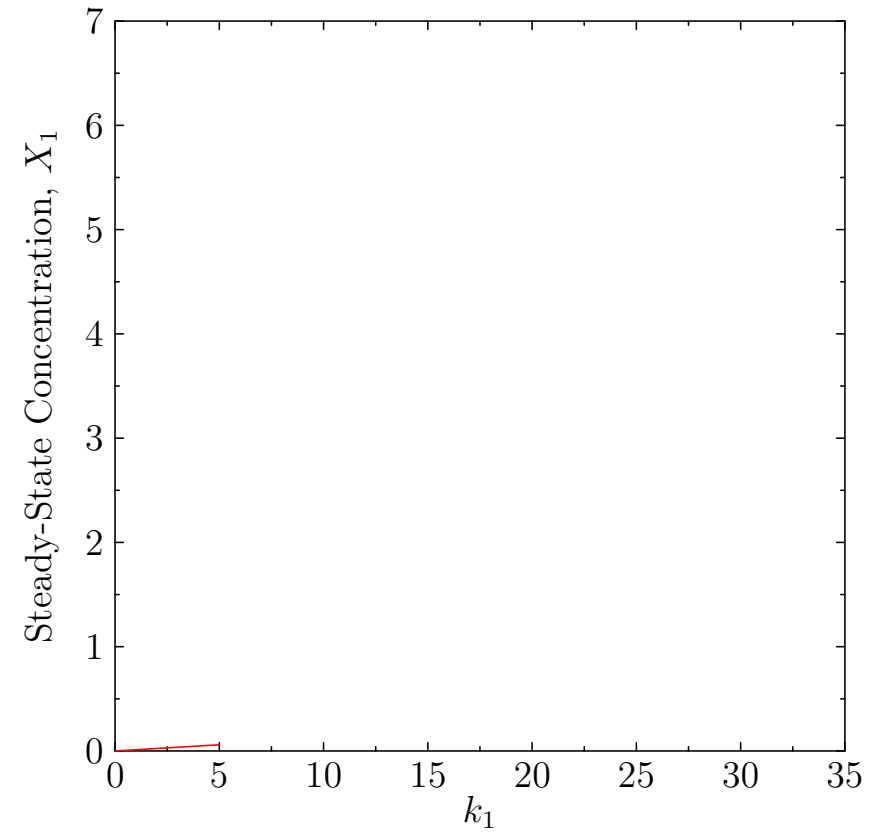
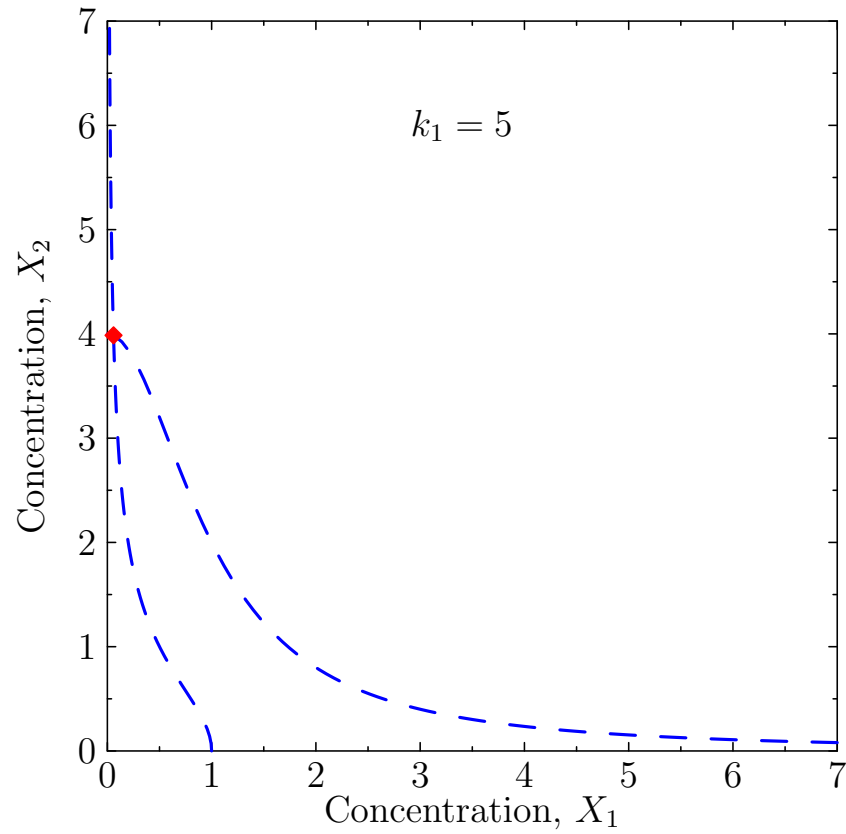
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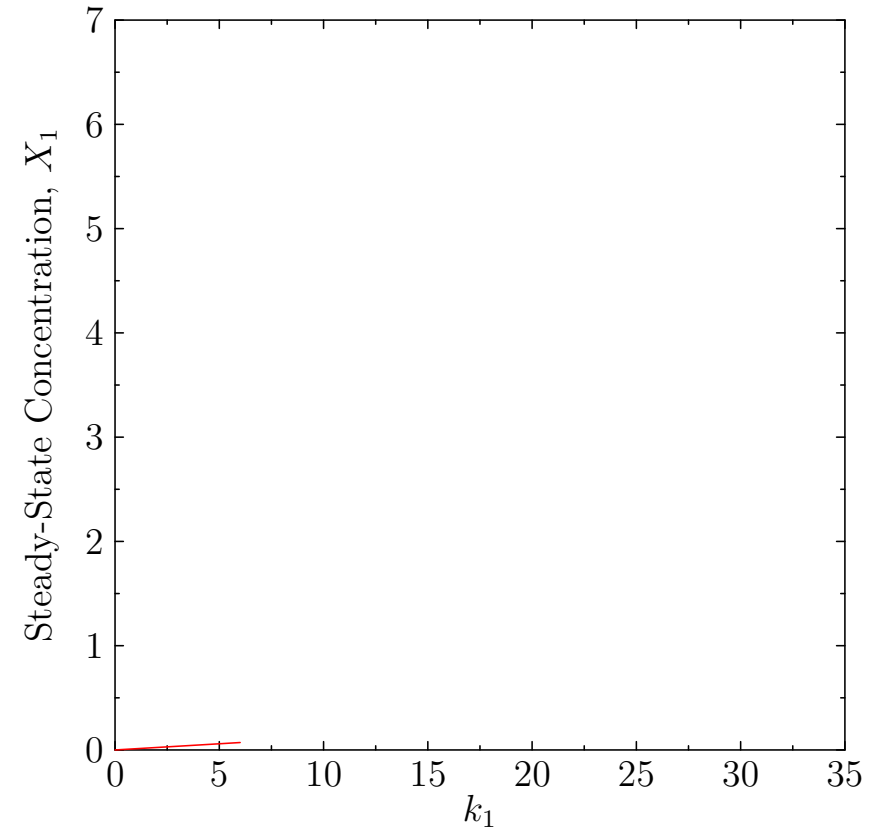
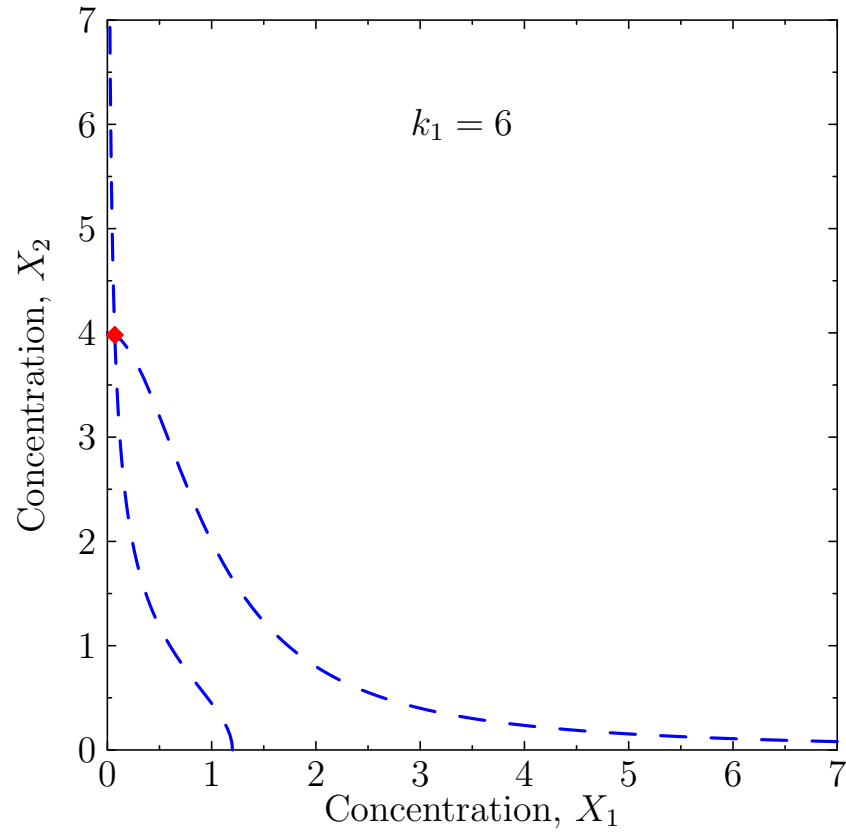
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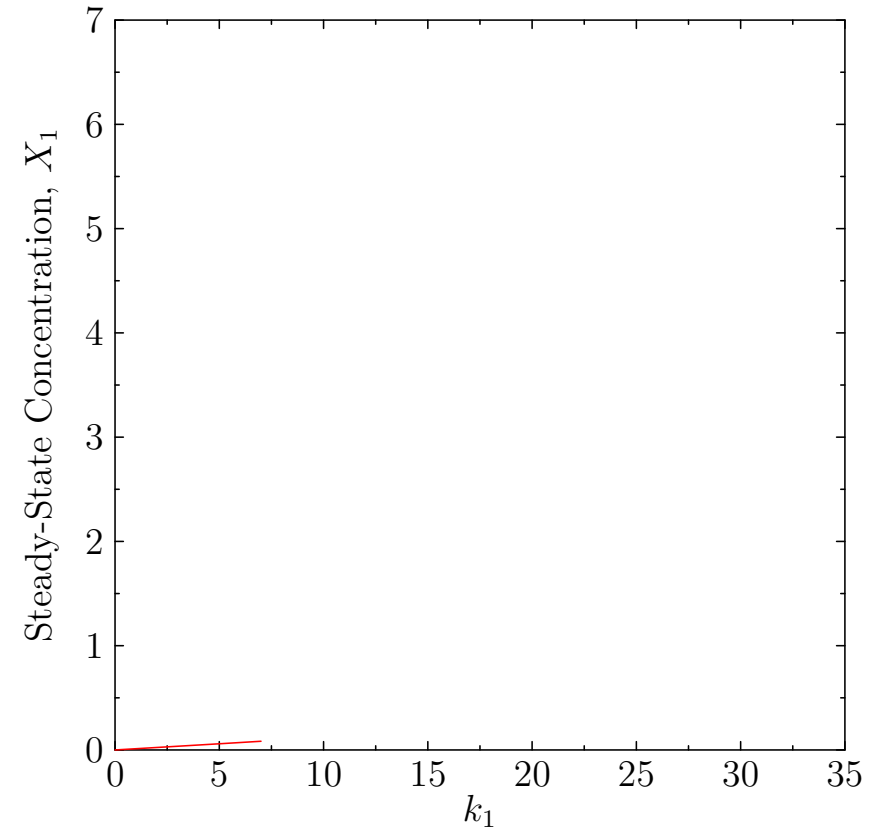
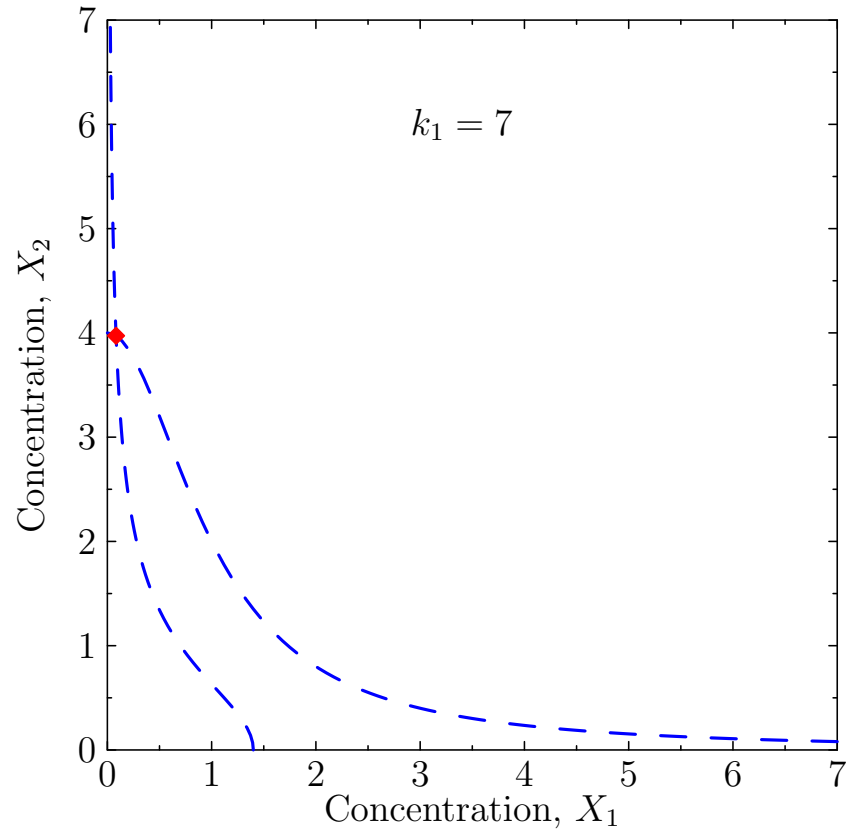
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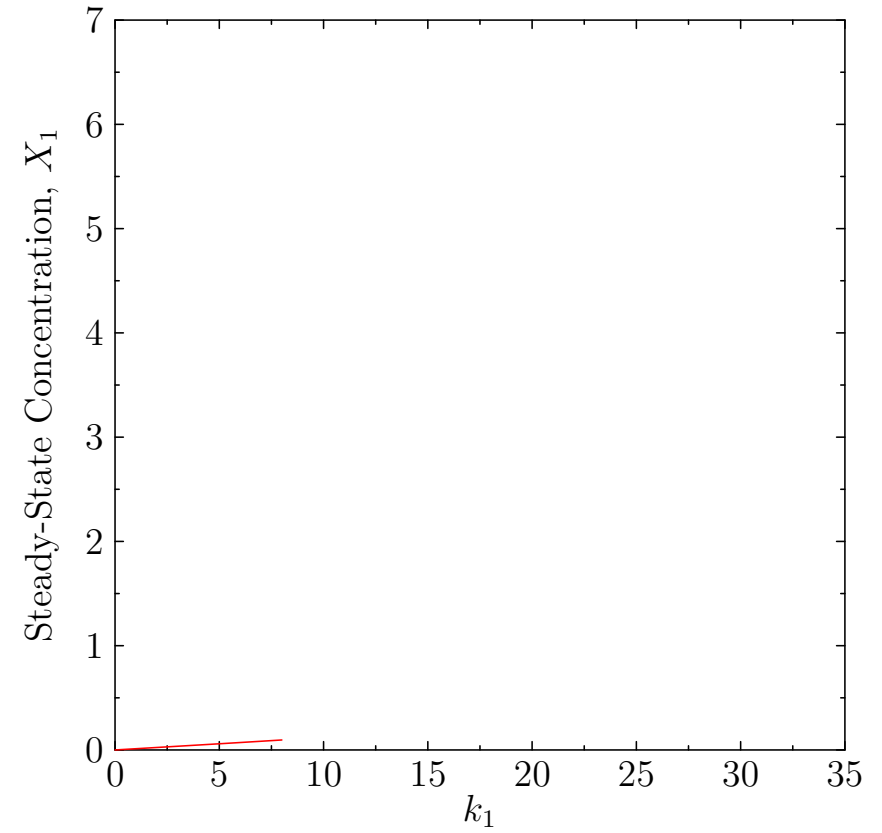
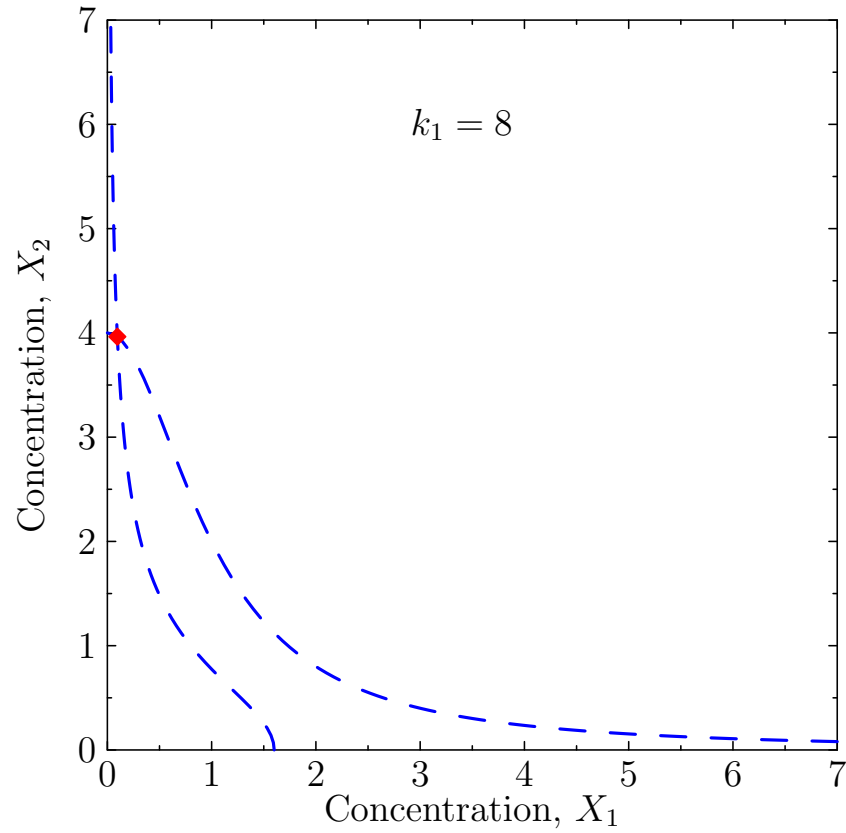
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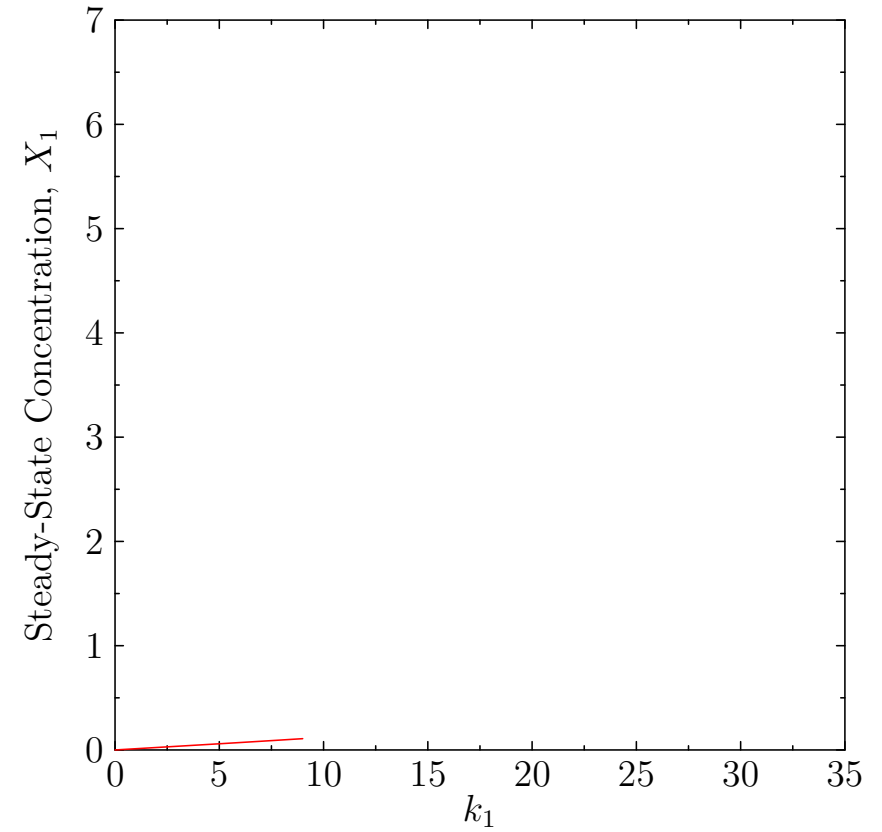
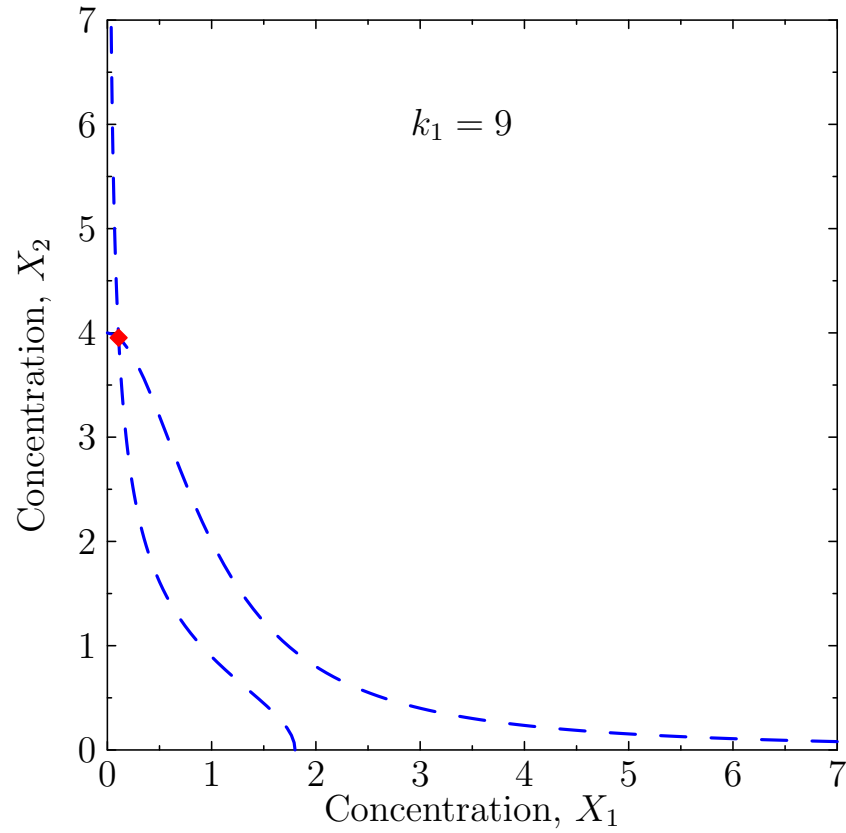
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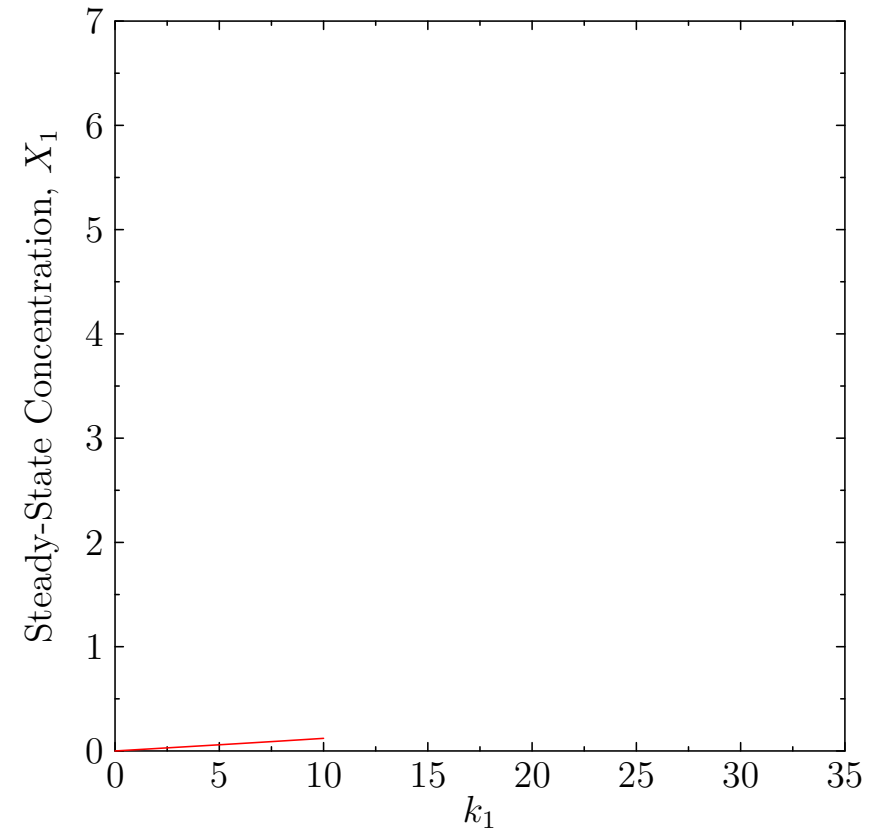
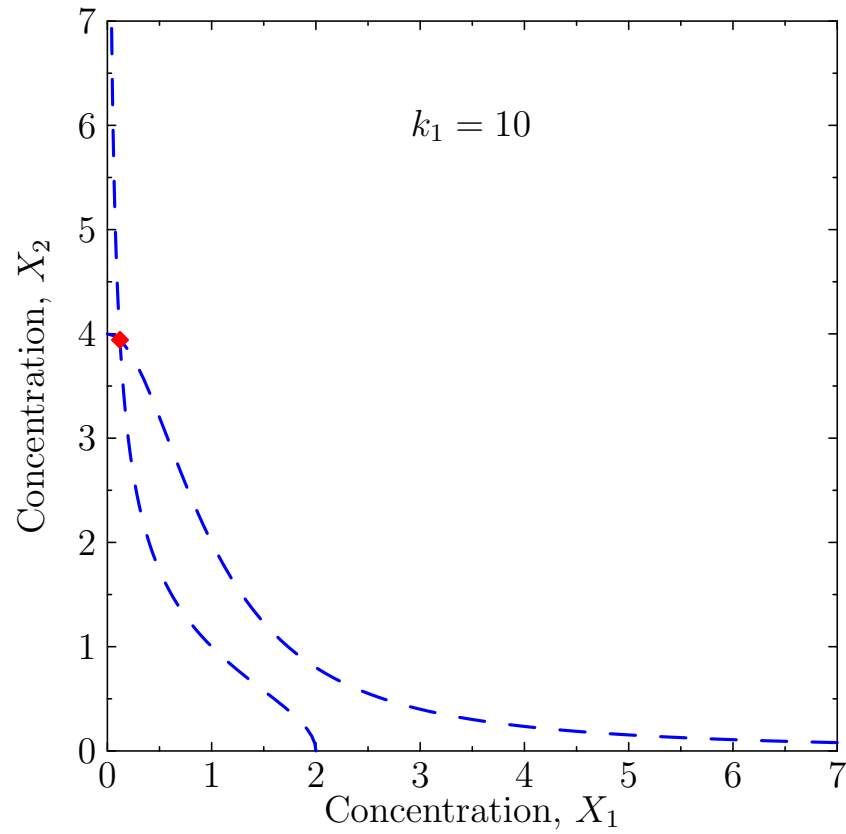
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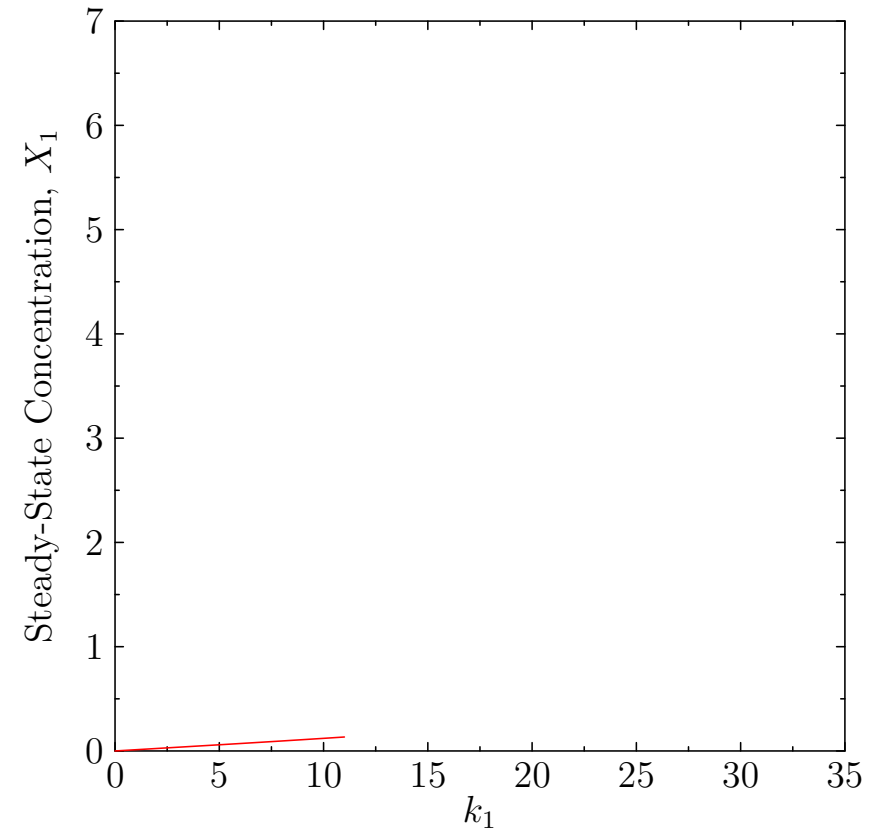
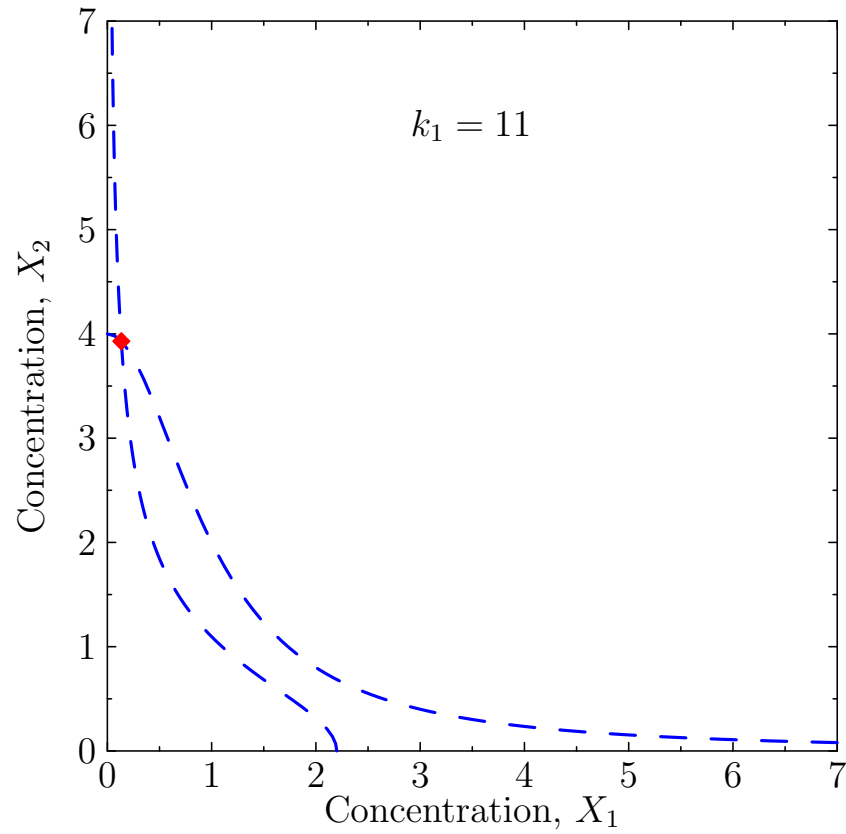
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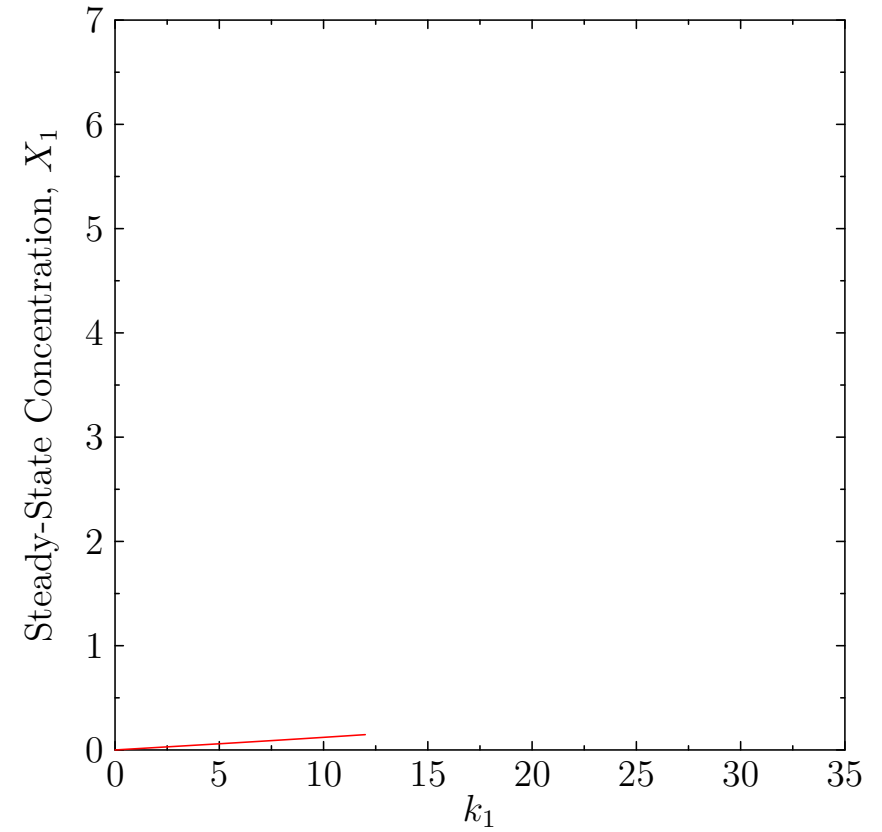
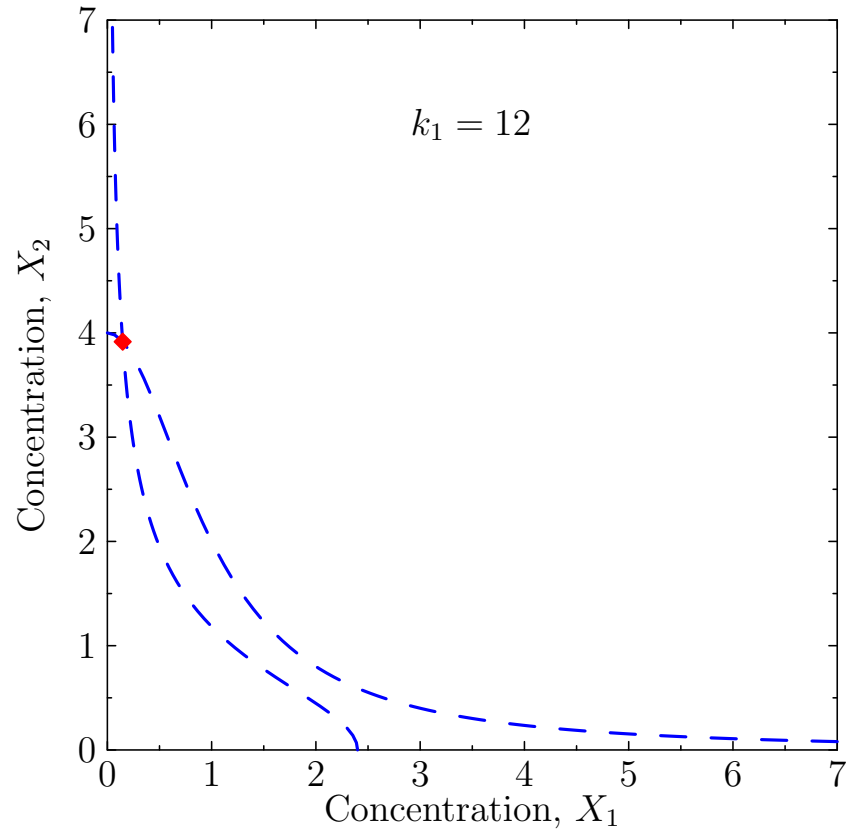
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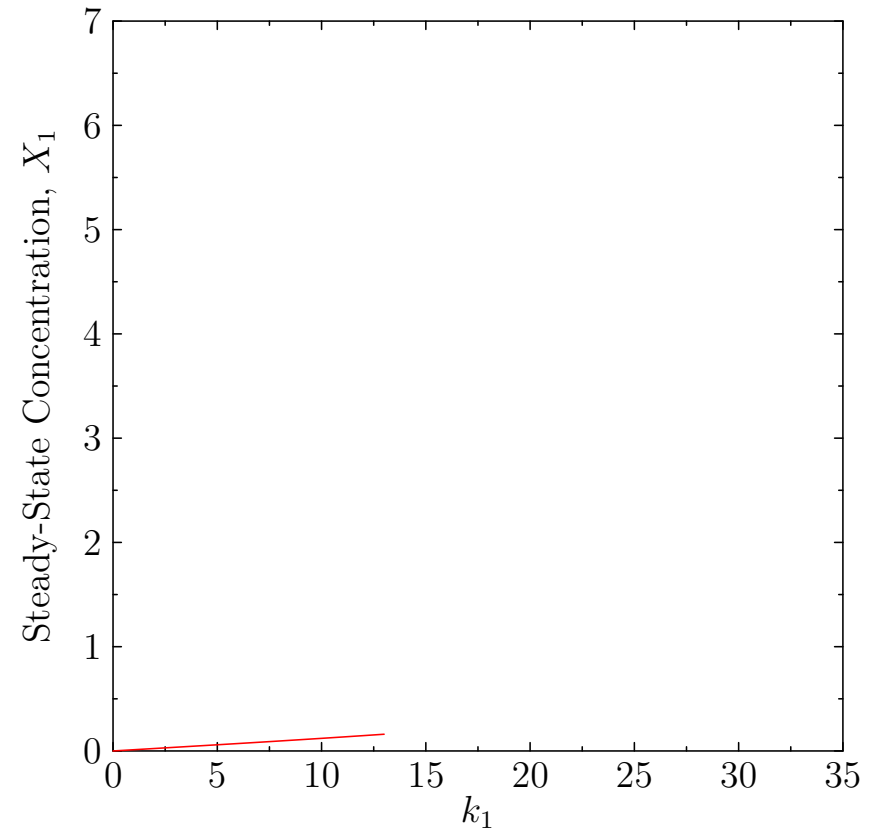
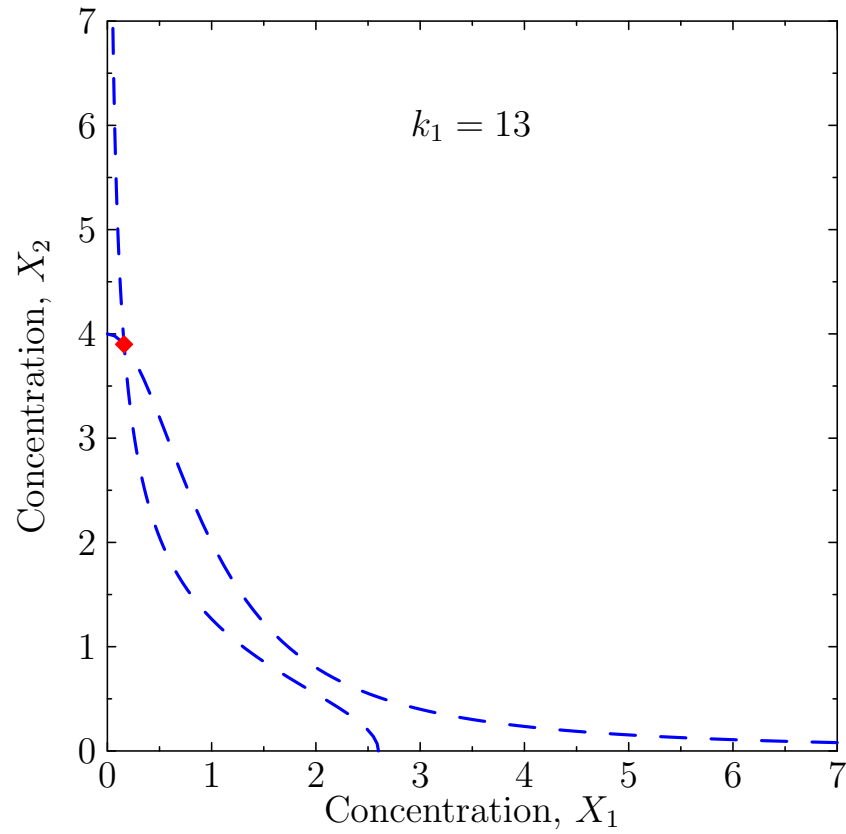
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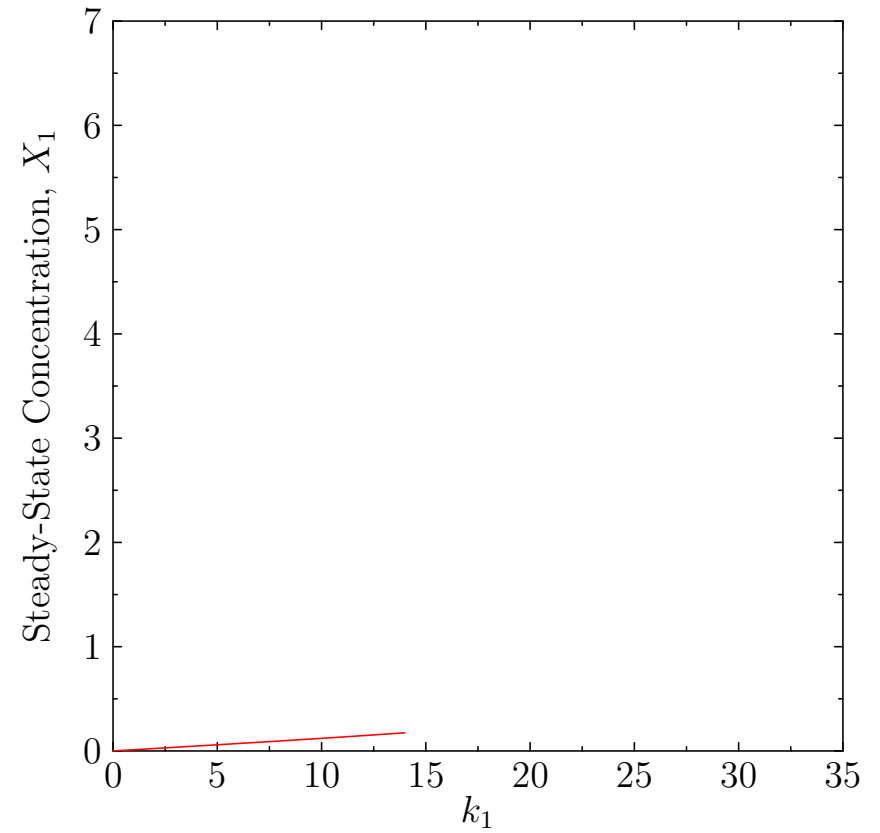
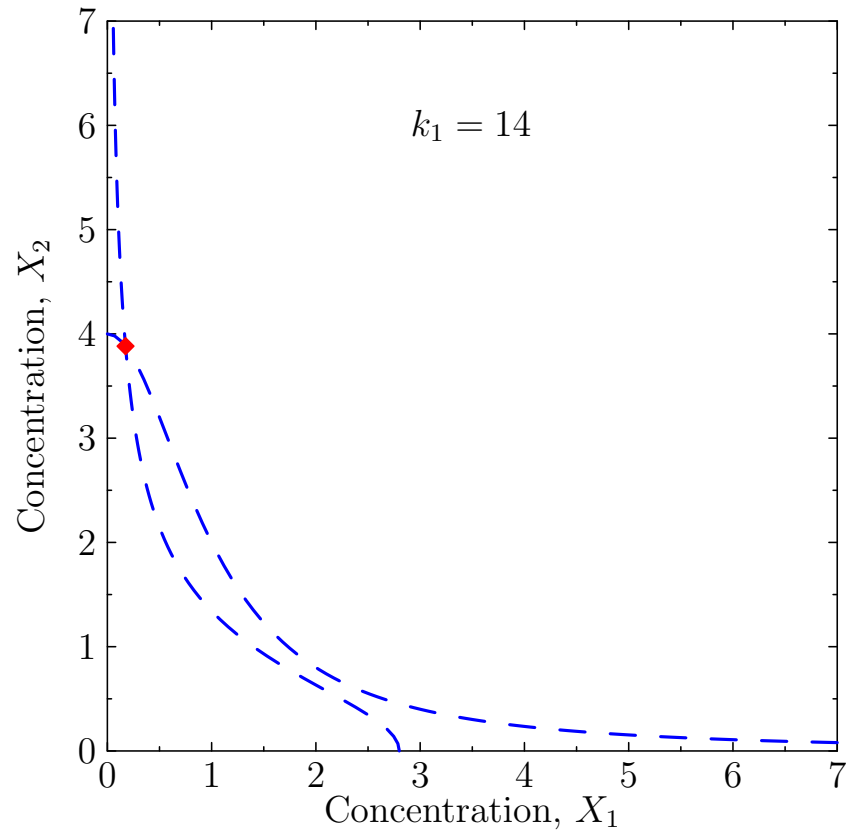
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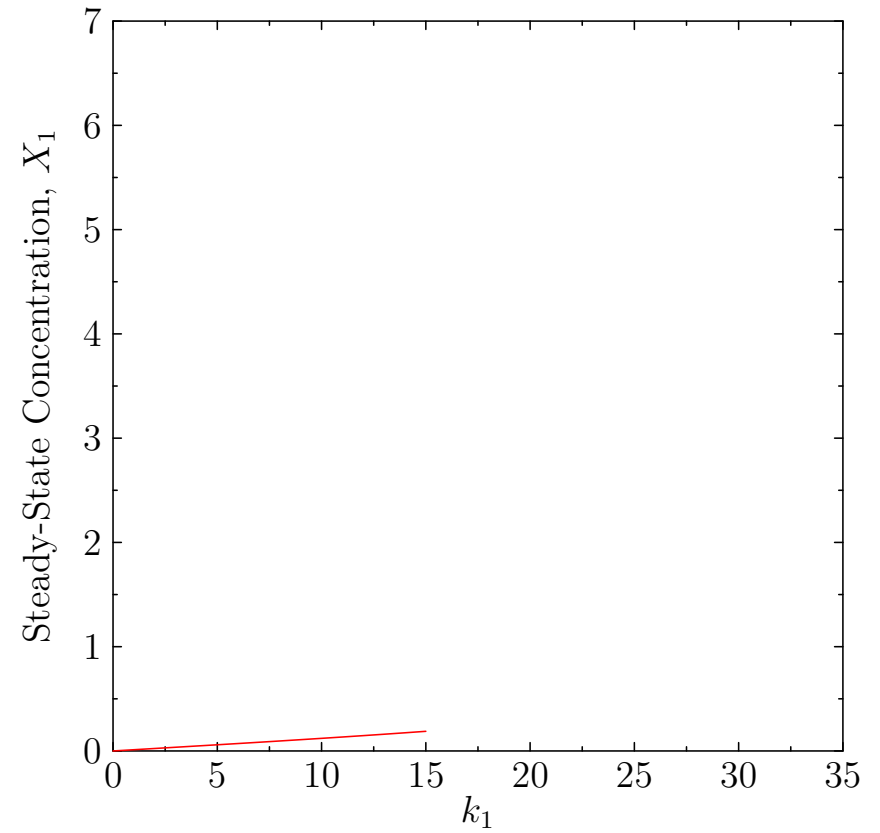
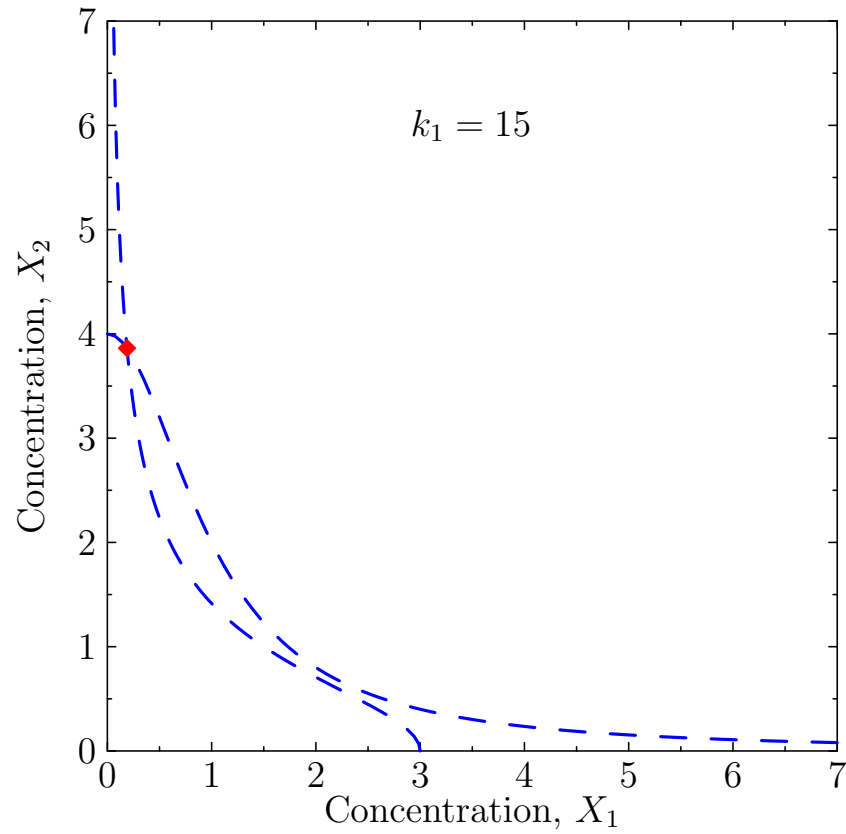
Varying k_1



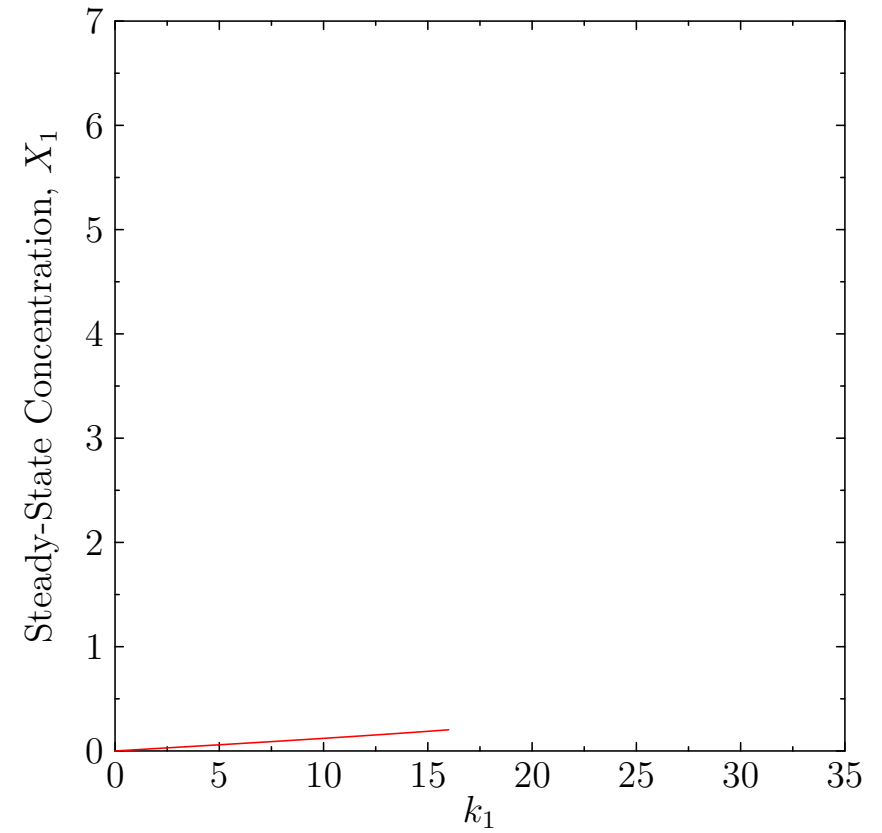
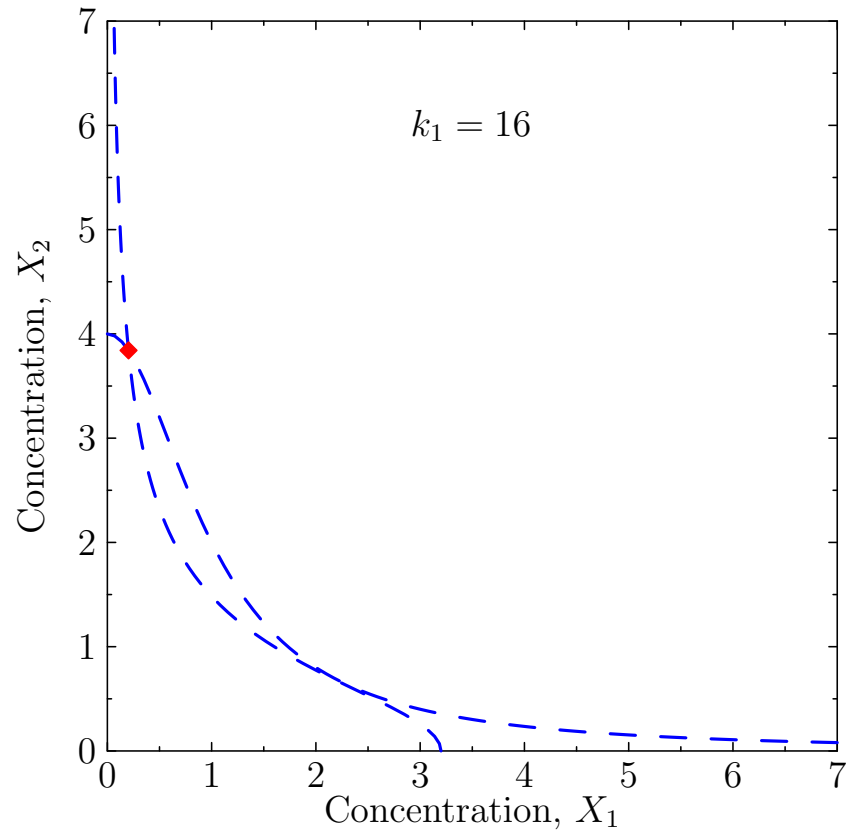
Varying k_1



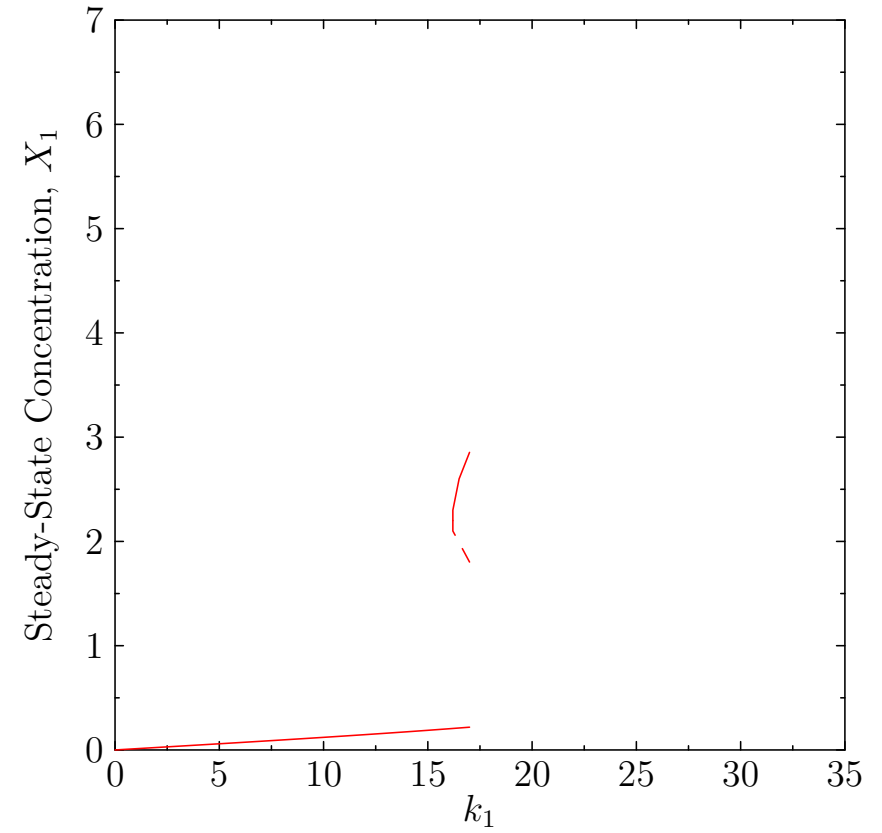
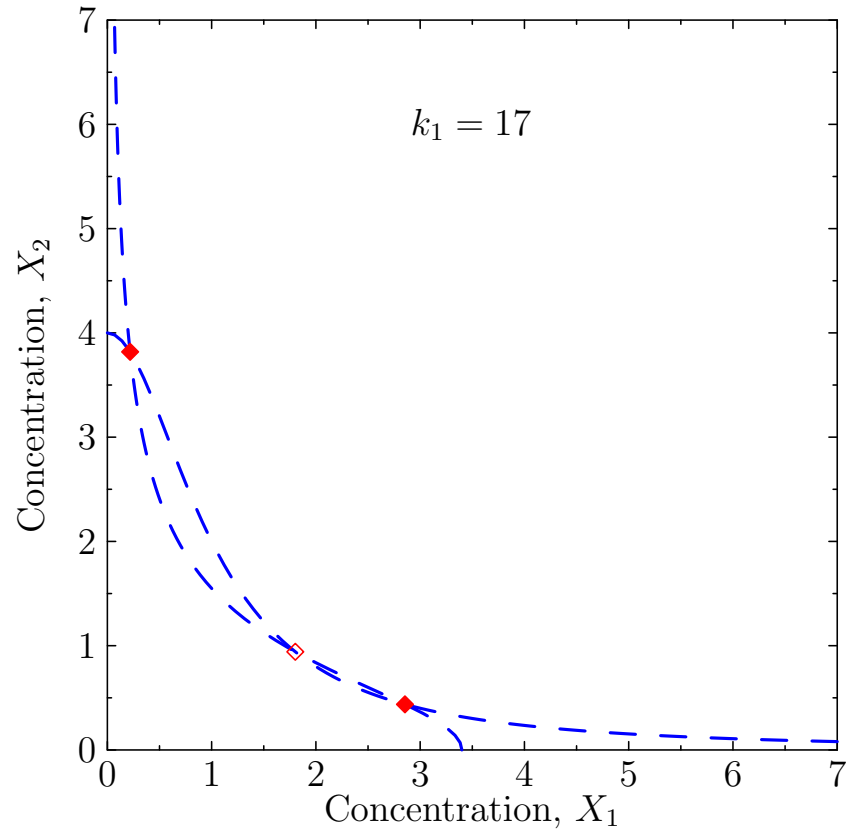
Varying k_1



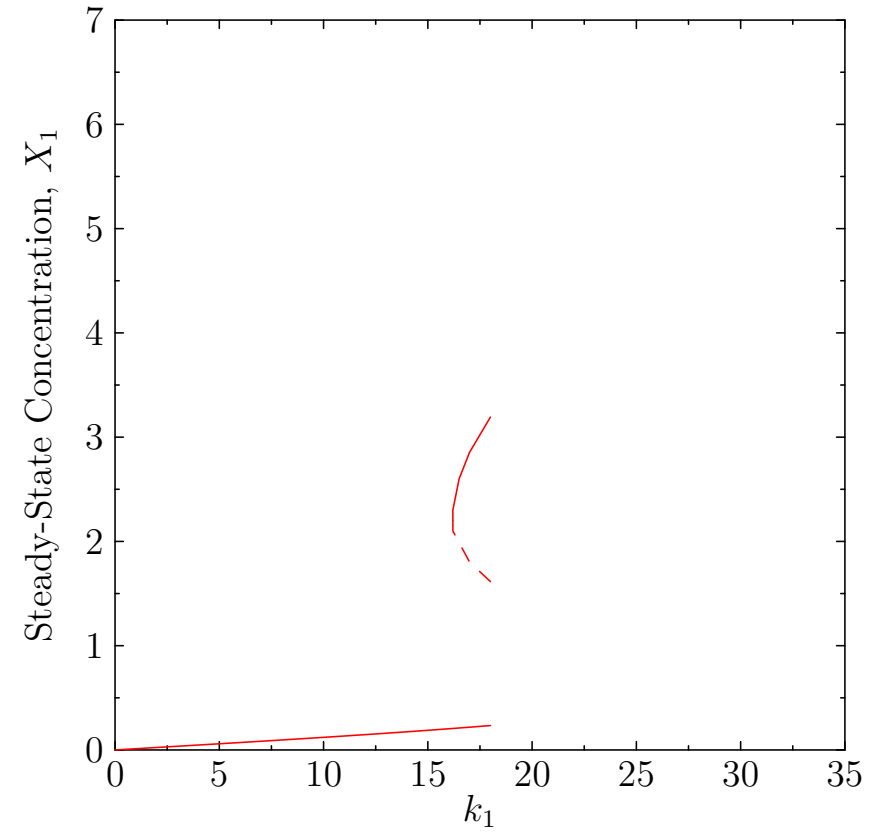
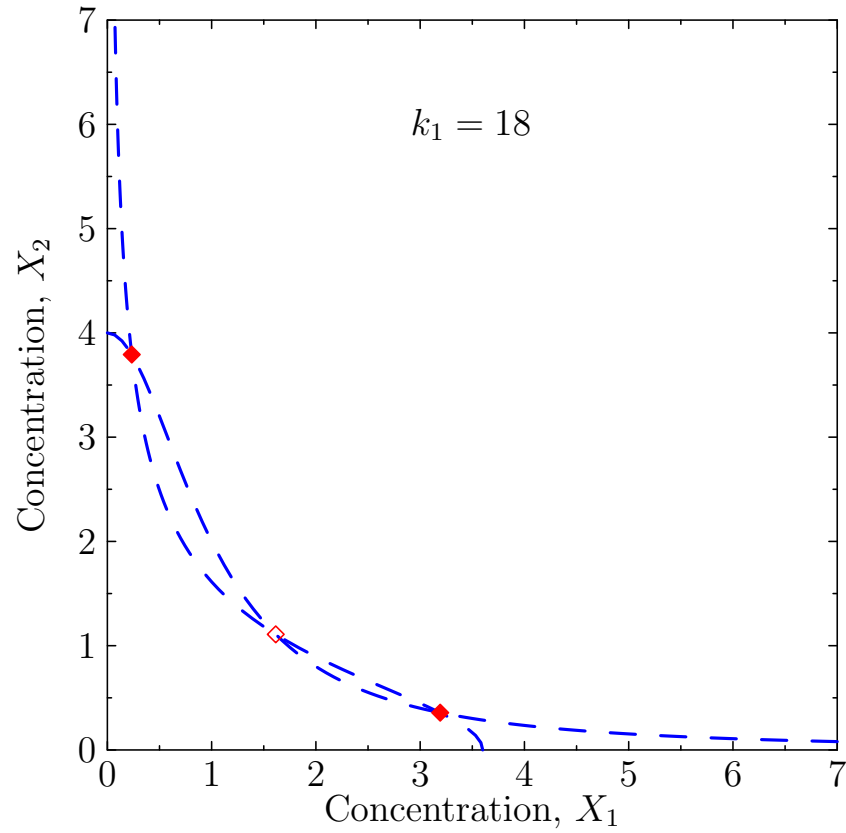
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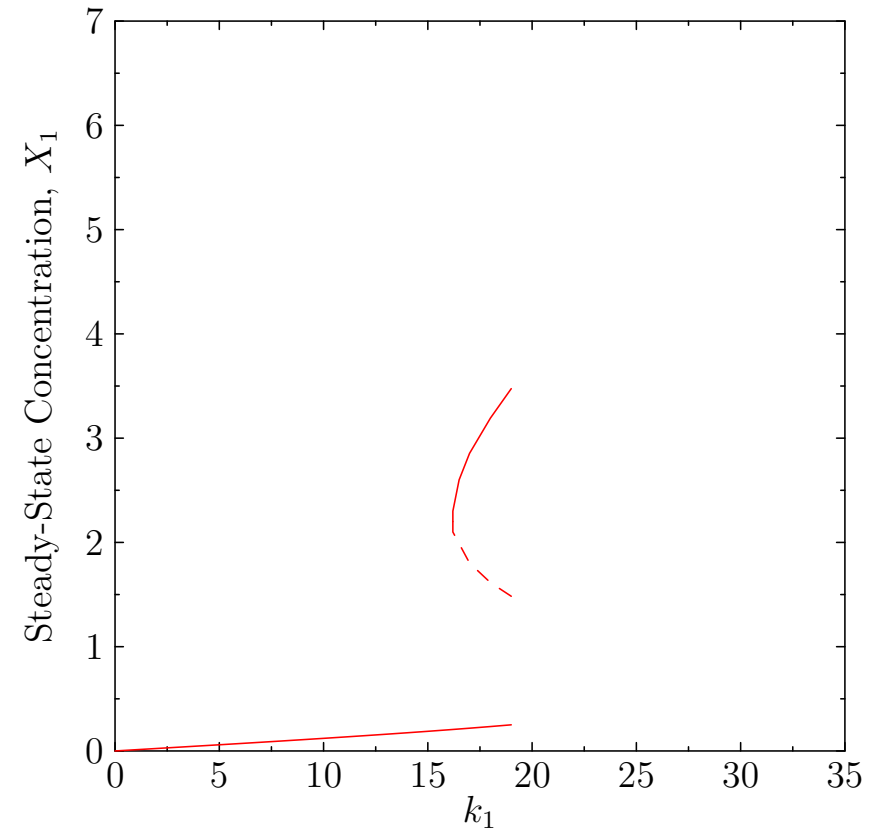
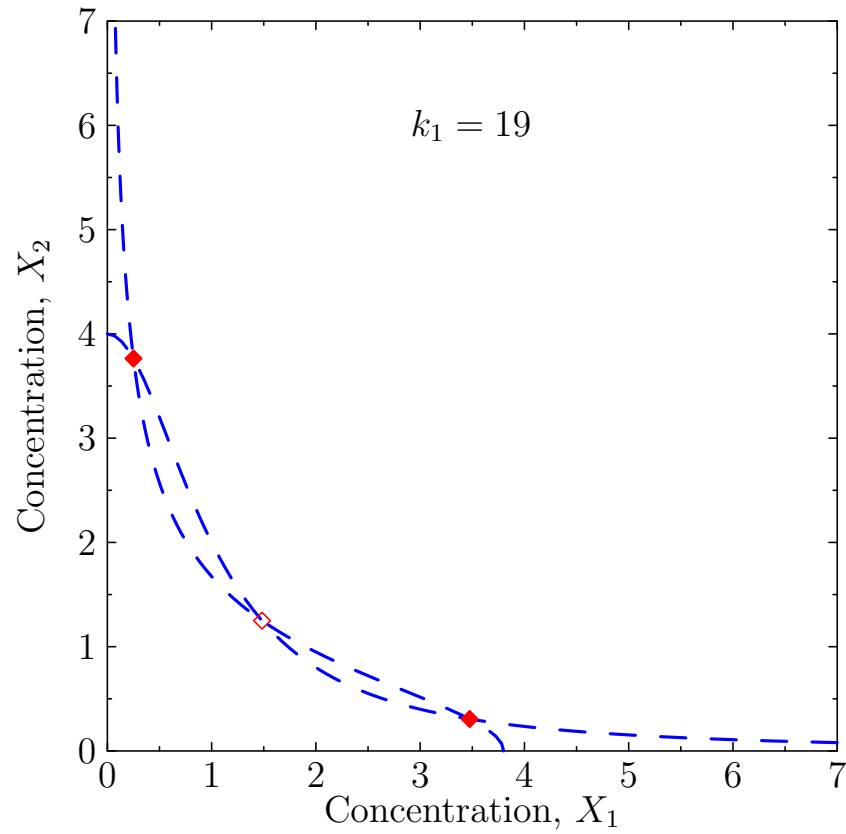
Varying k_1



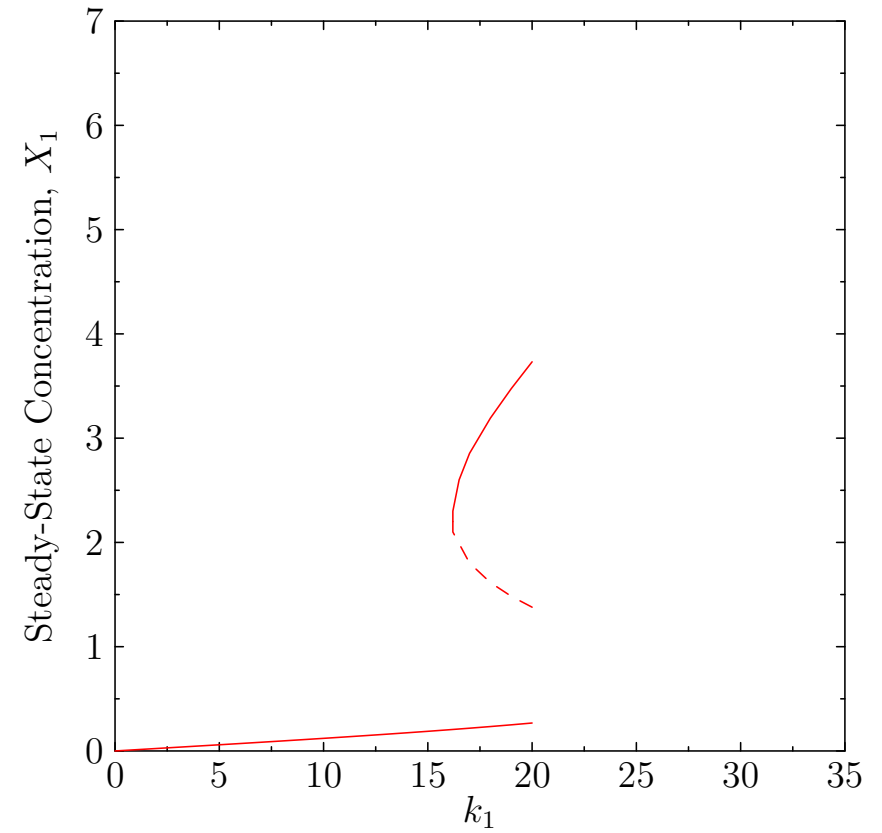
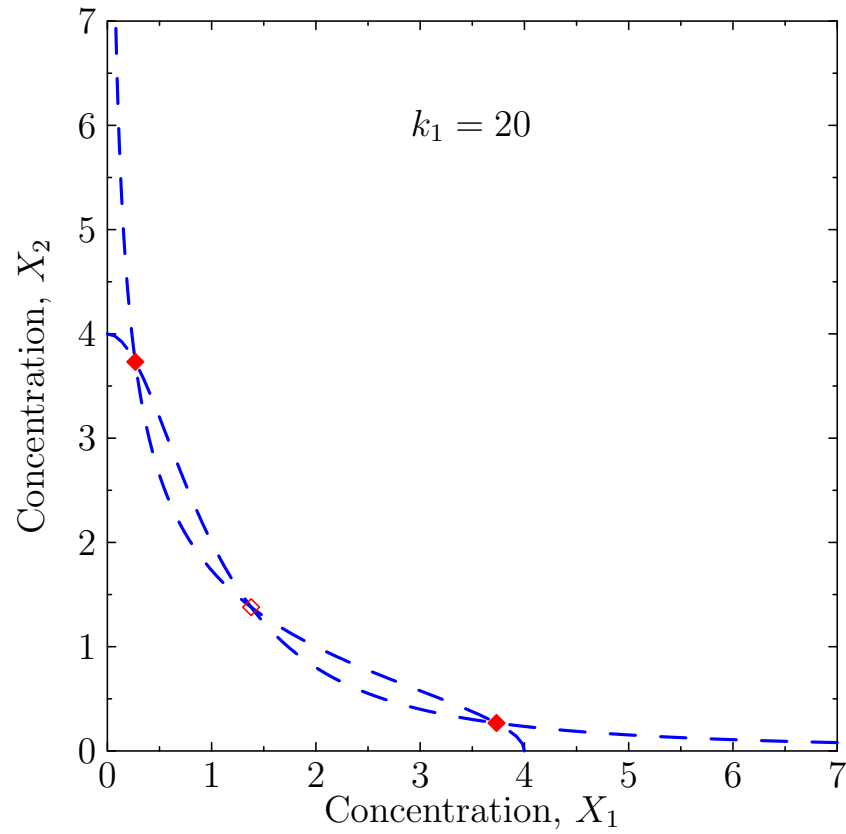
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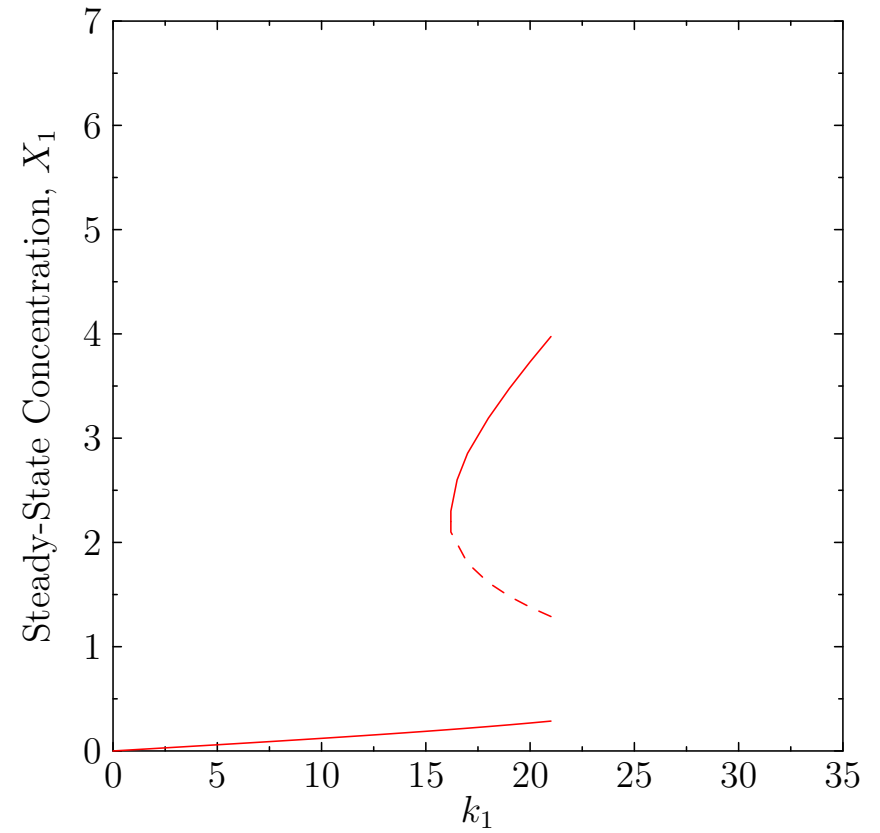
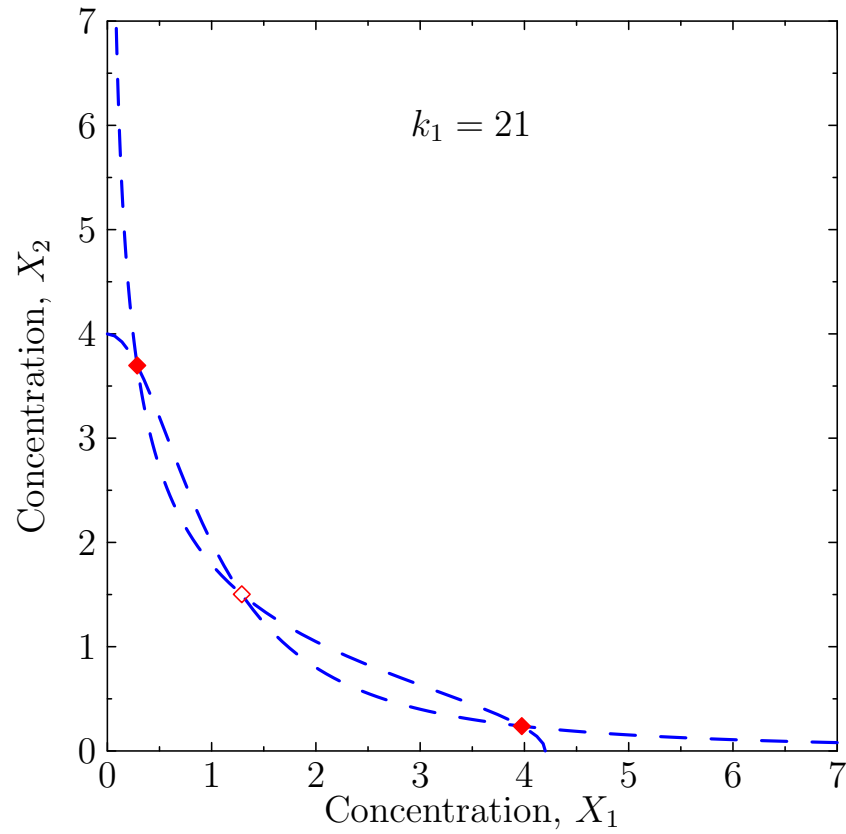
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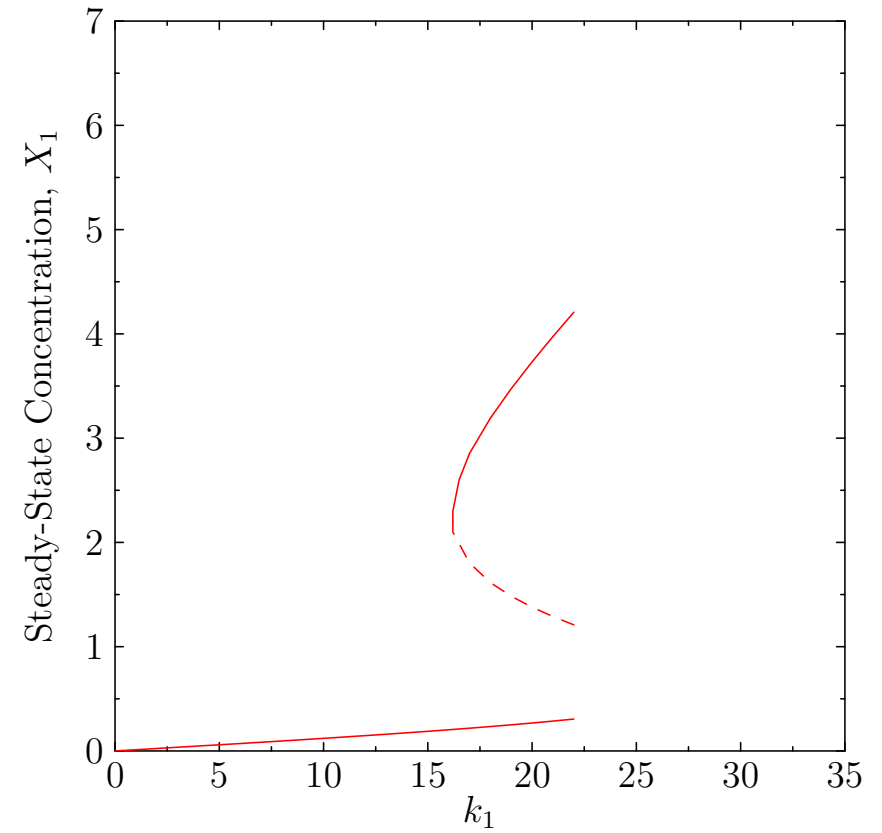
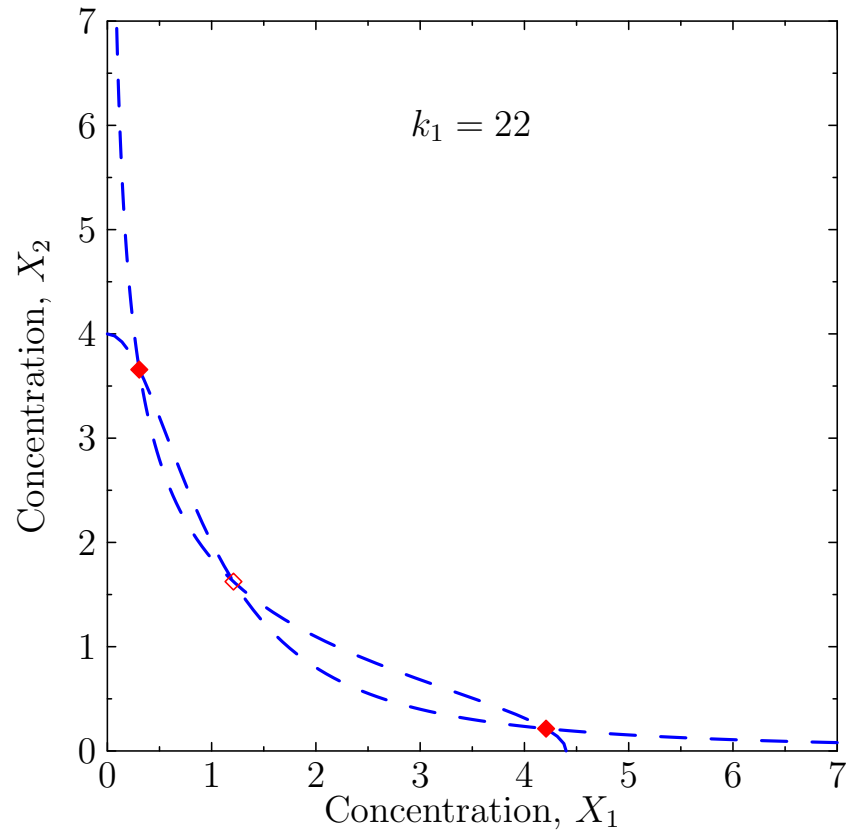
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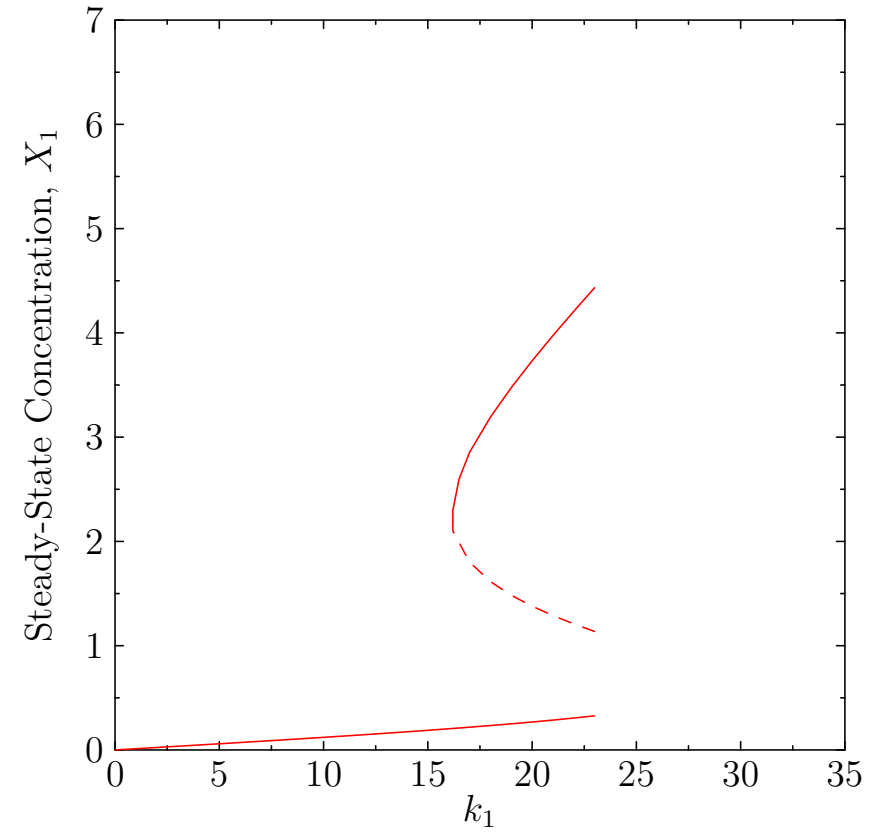
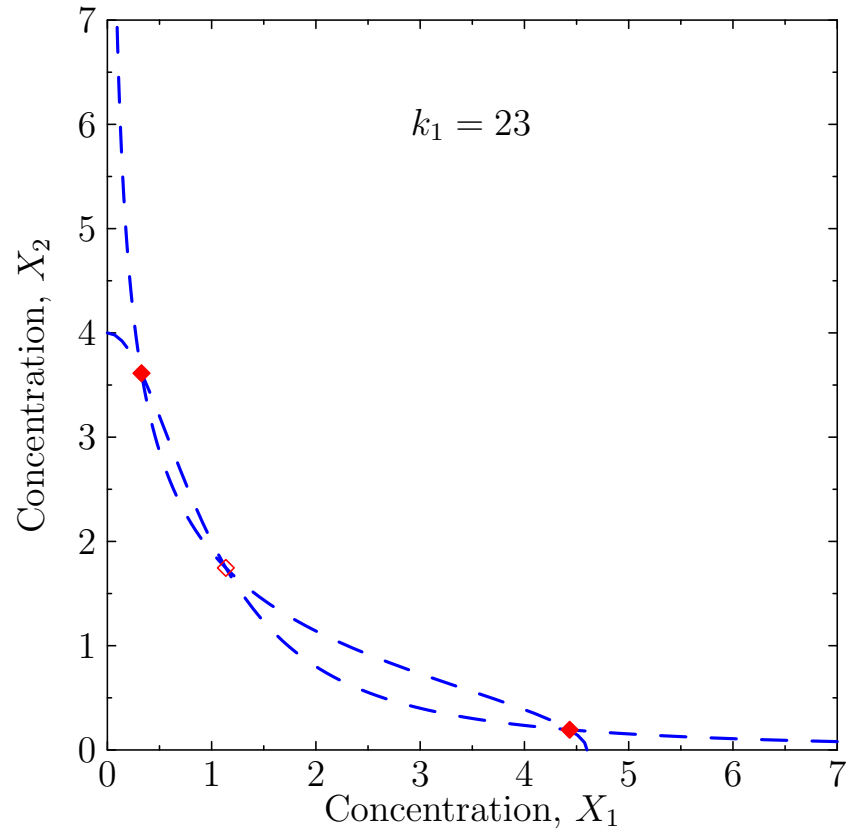
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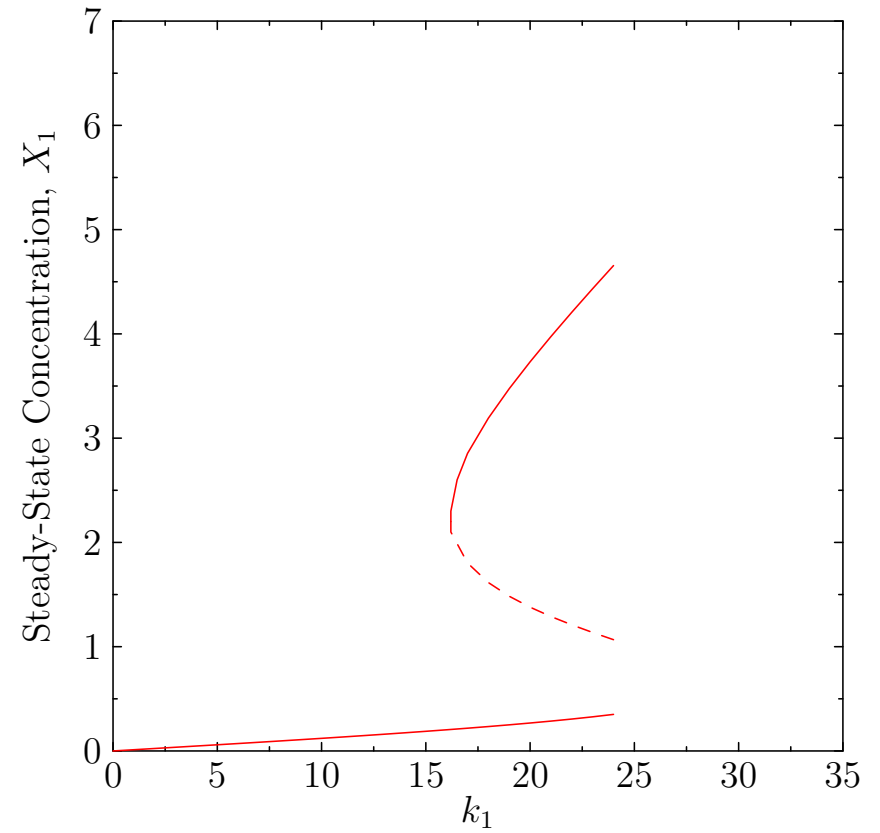
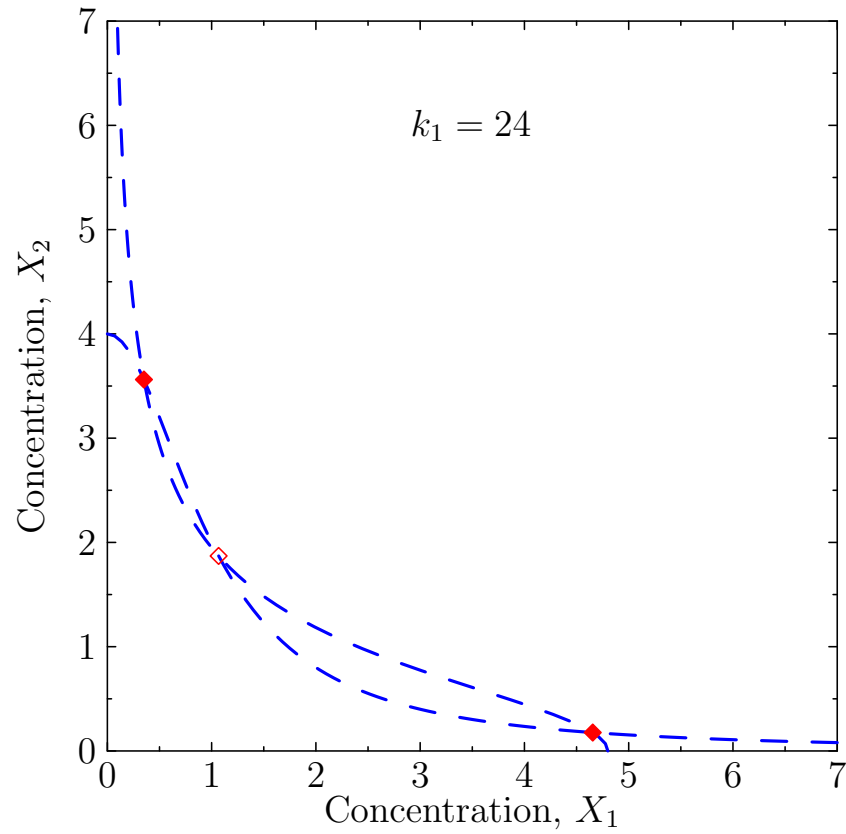
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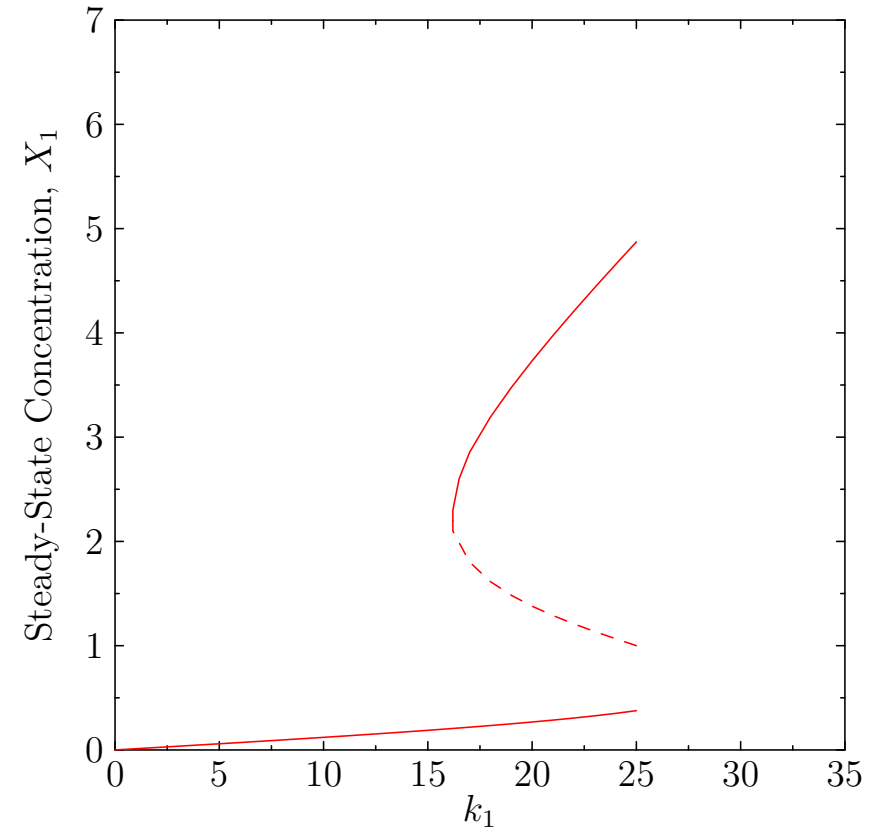
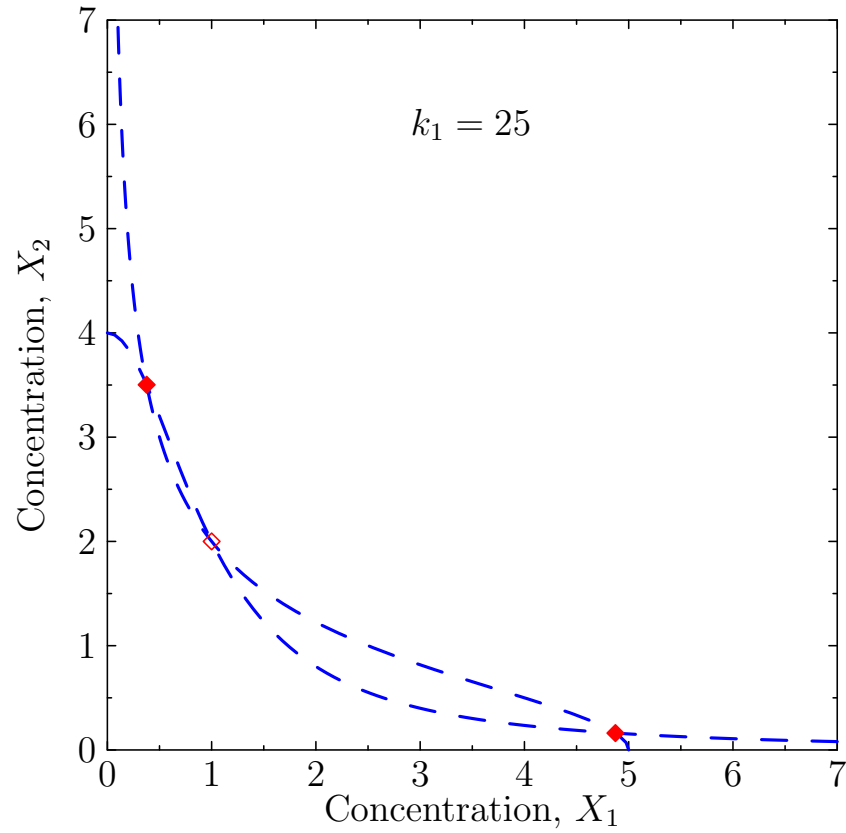
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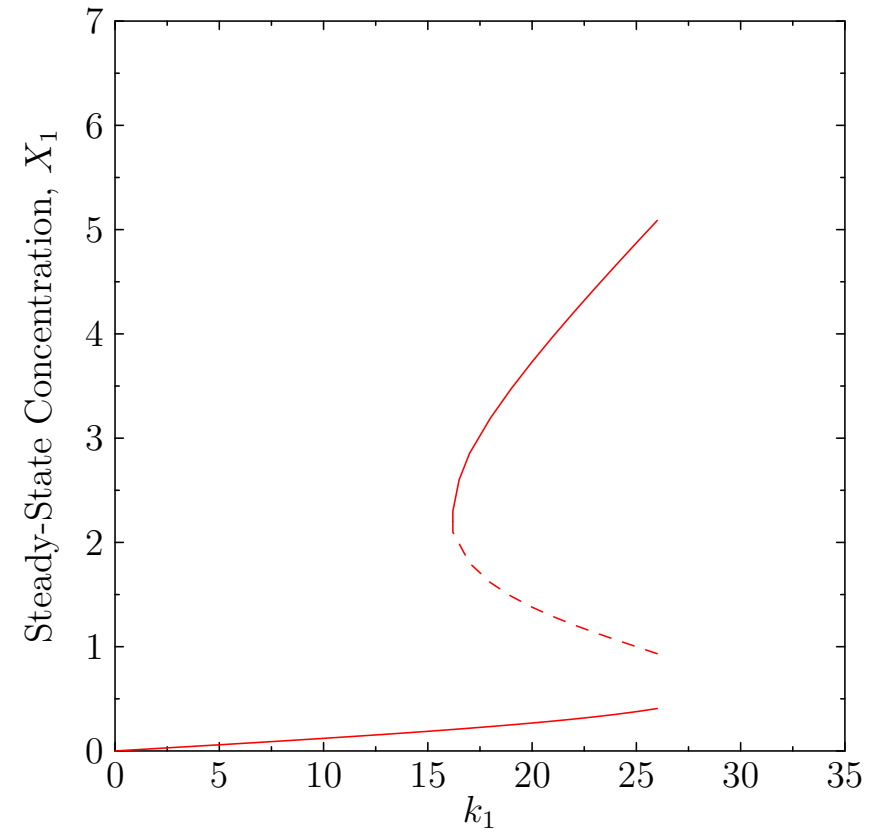
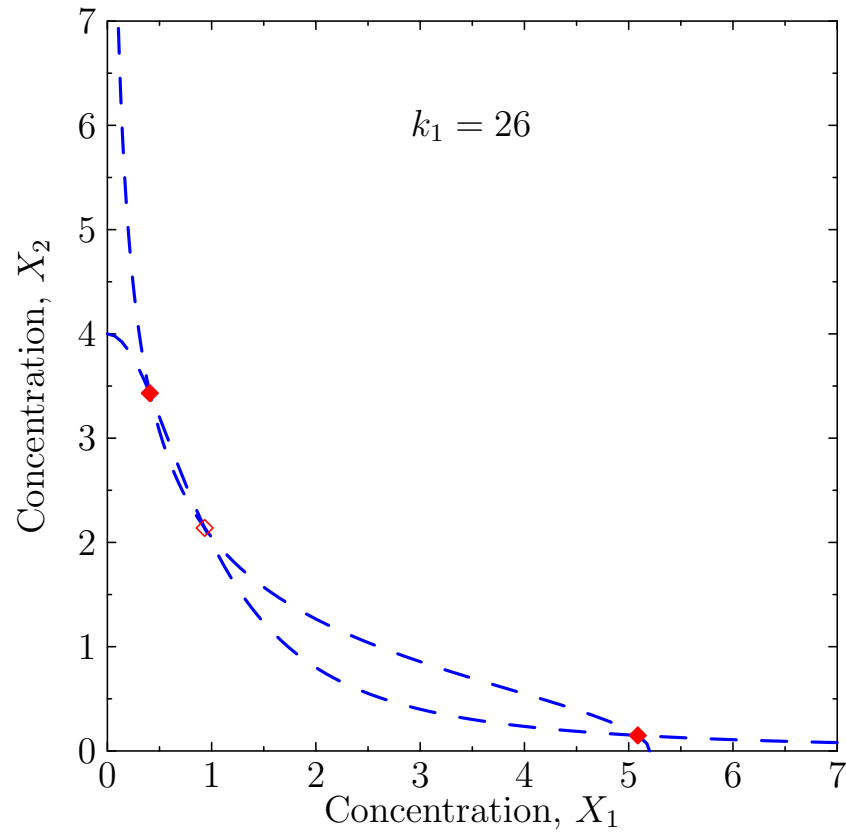
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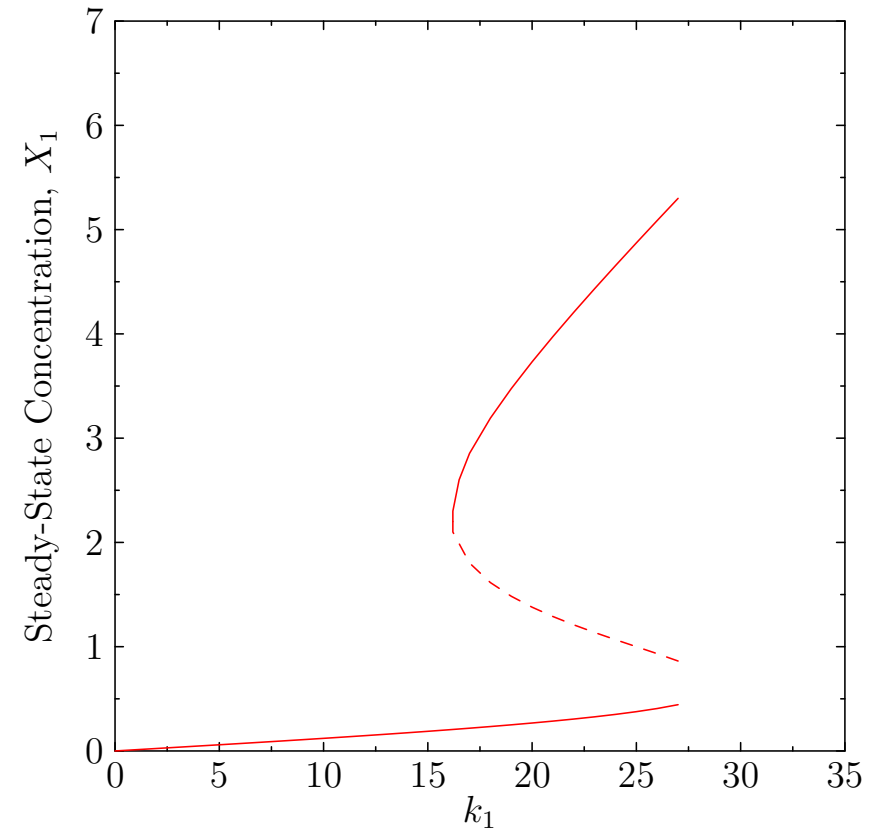
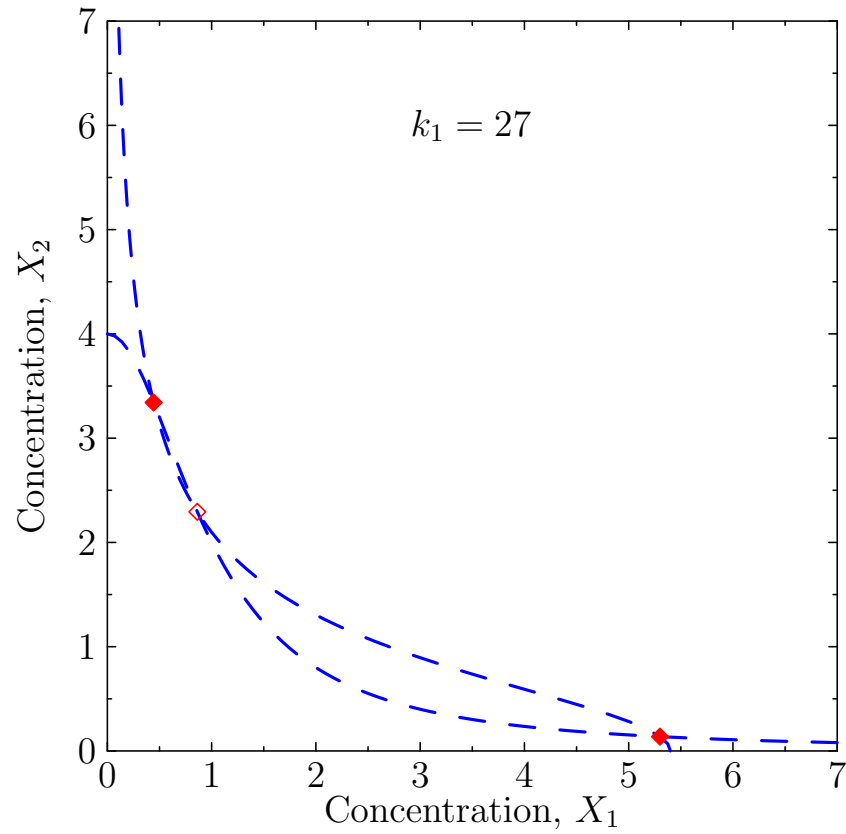
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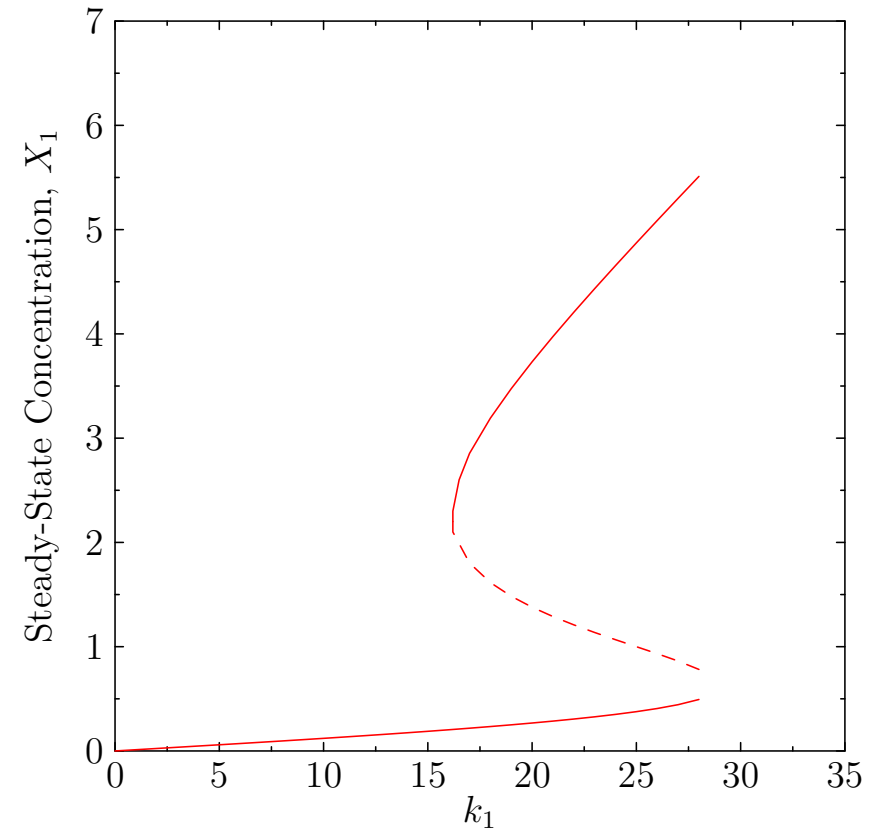
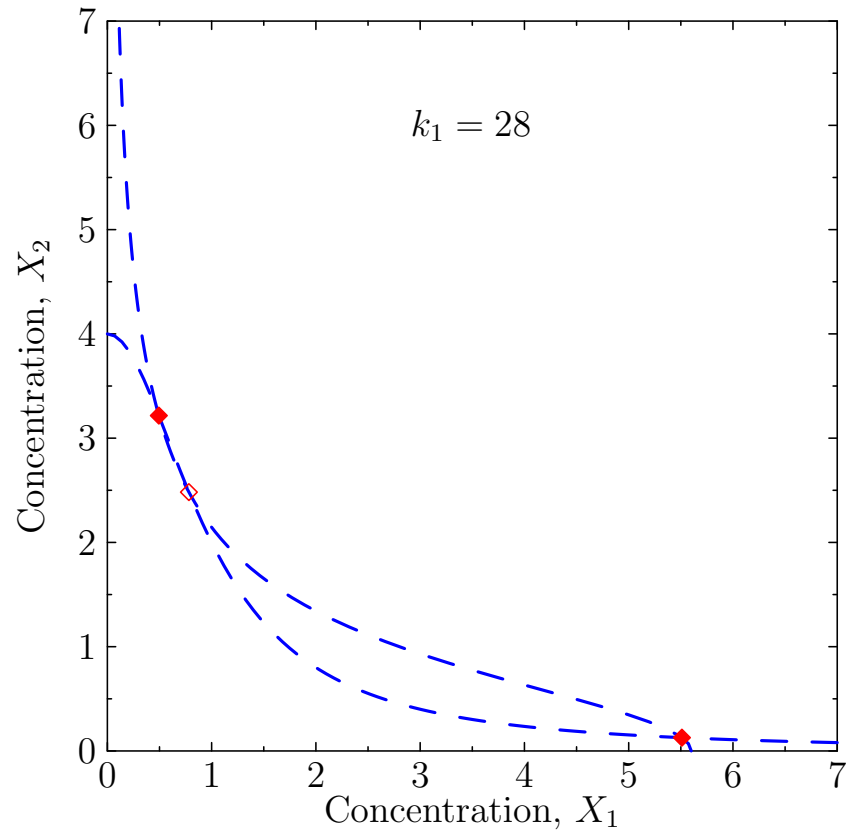
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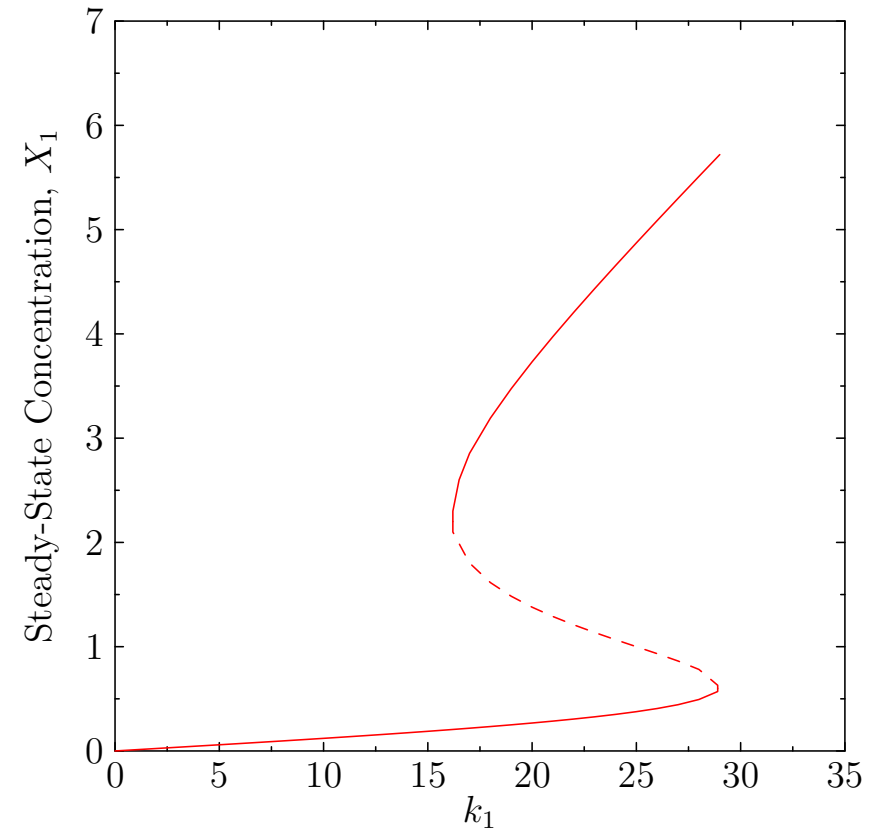
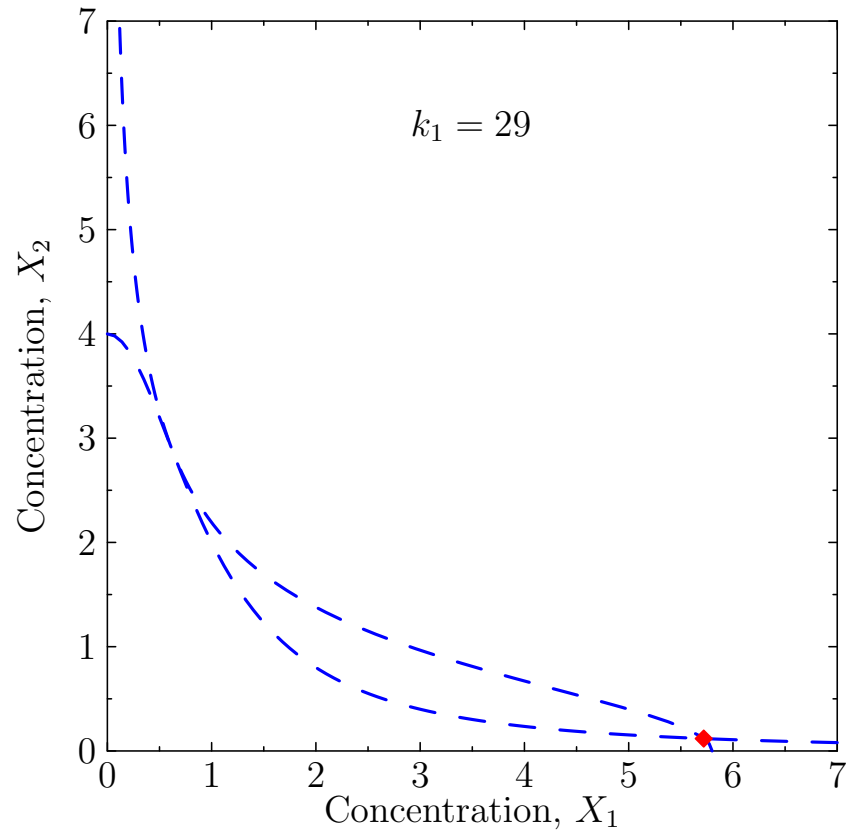
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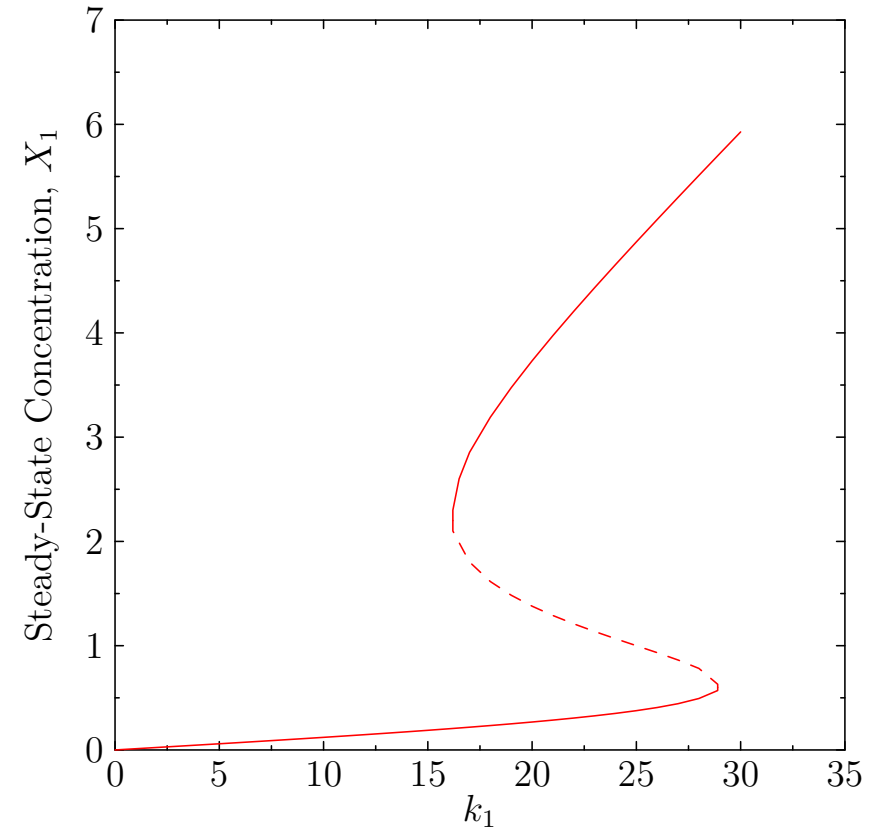
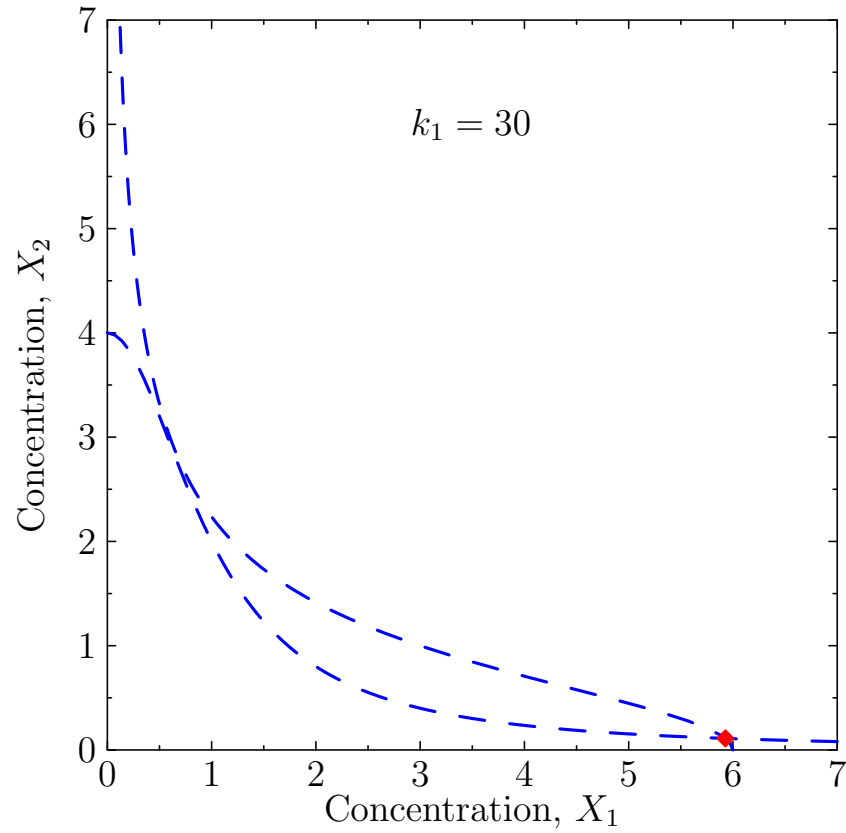
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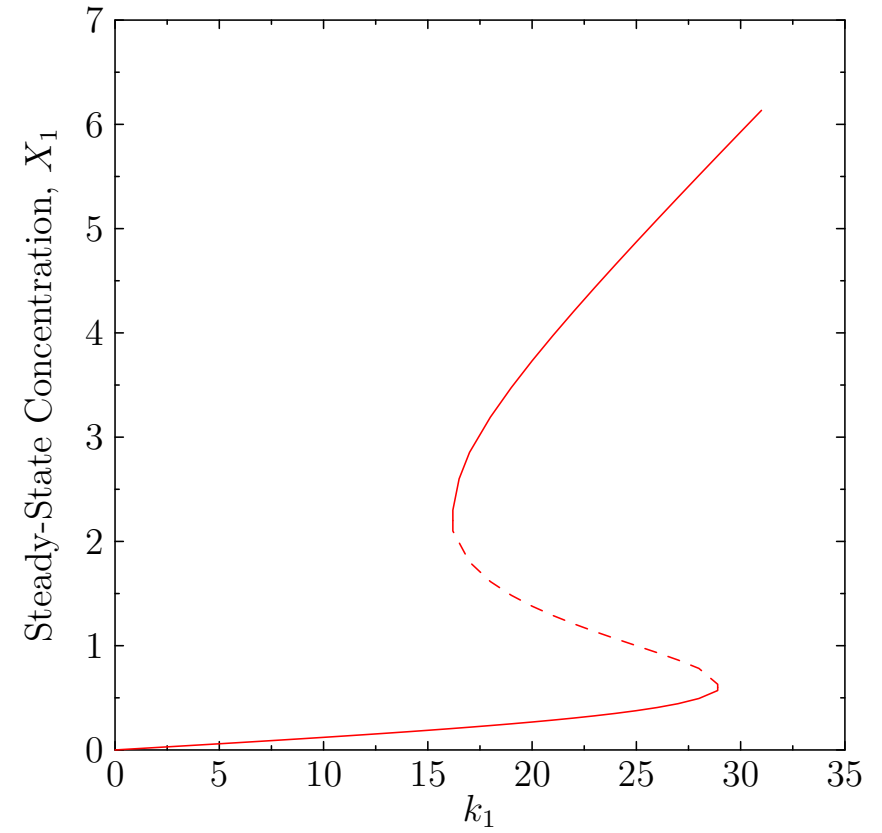
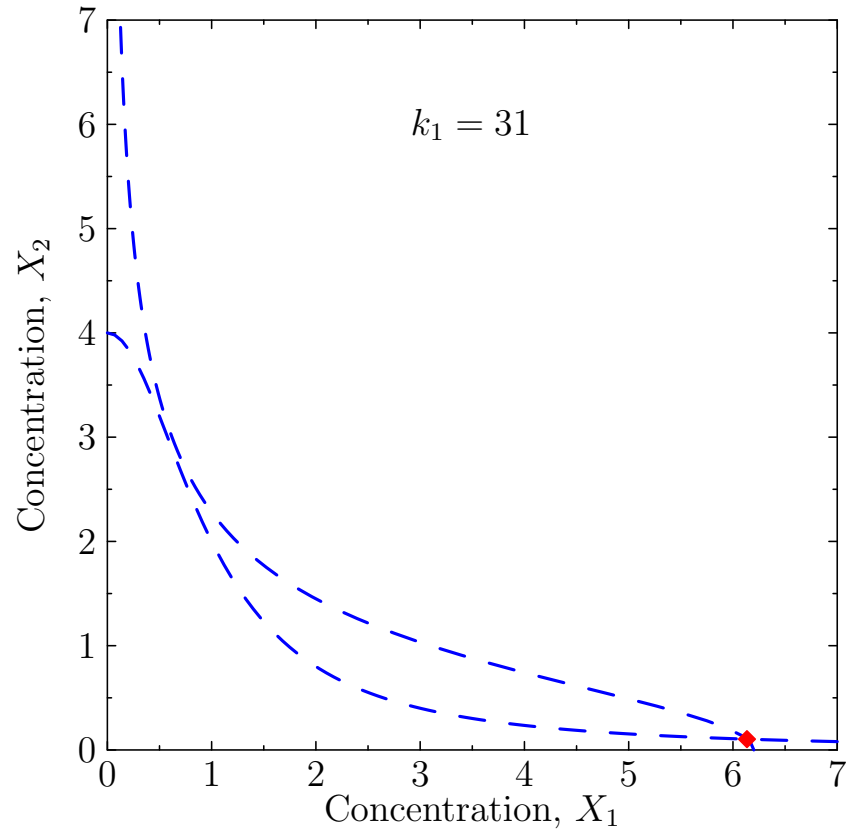
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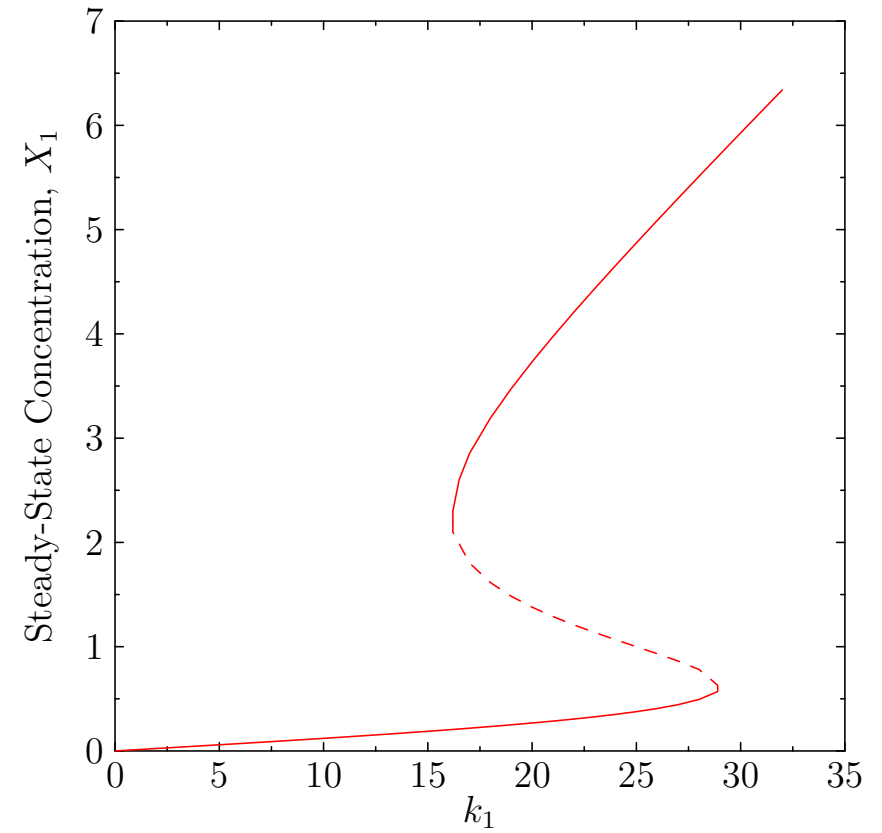
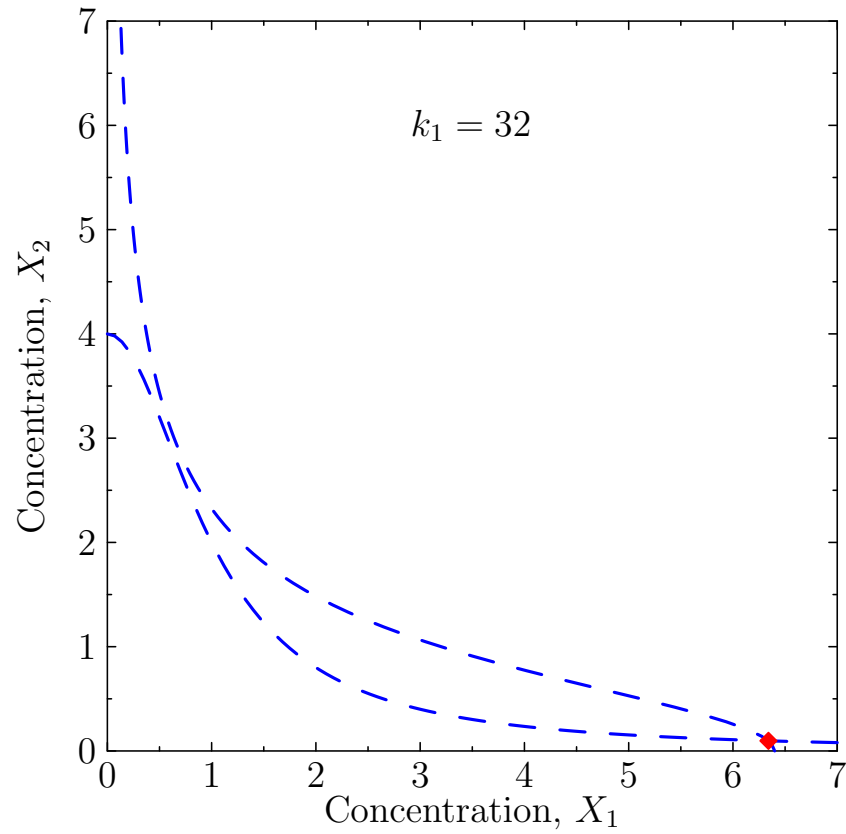
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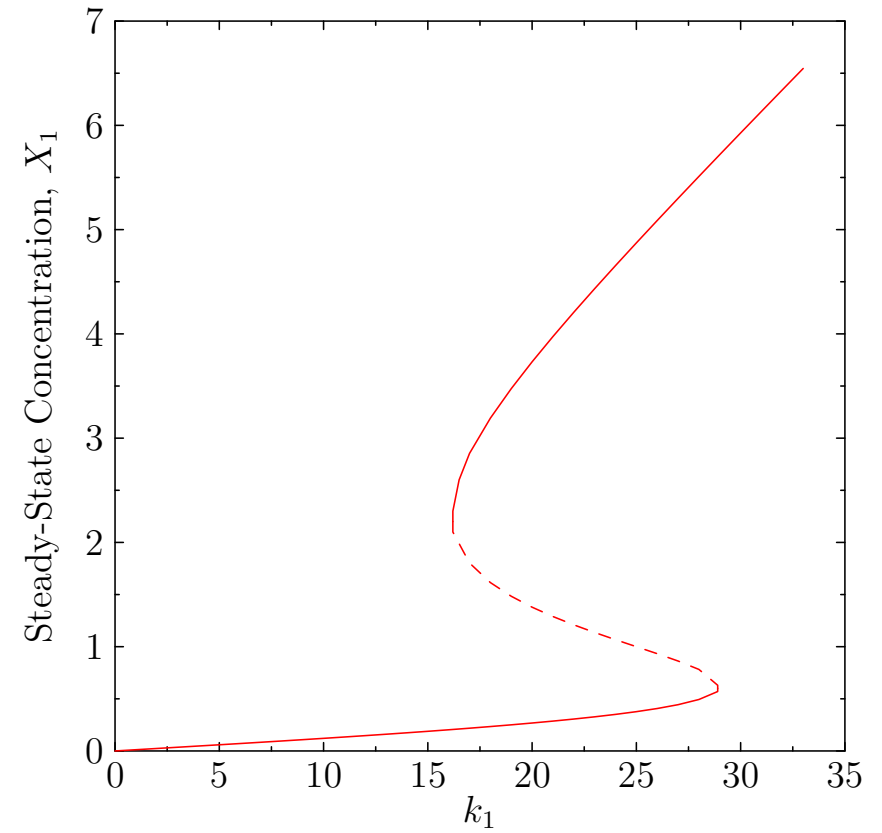
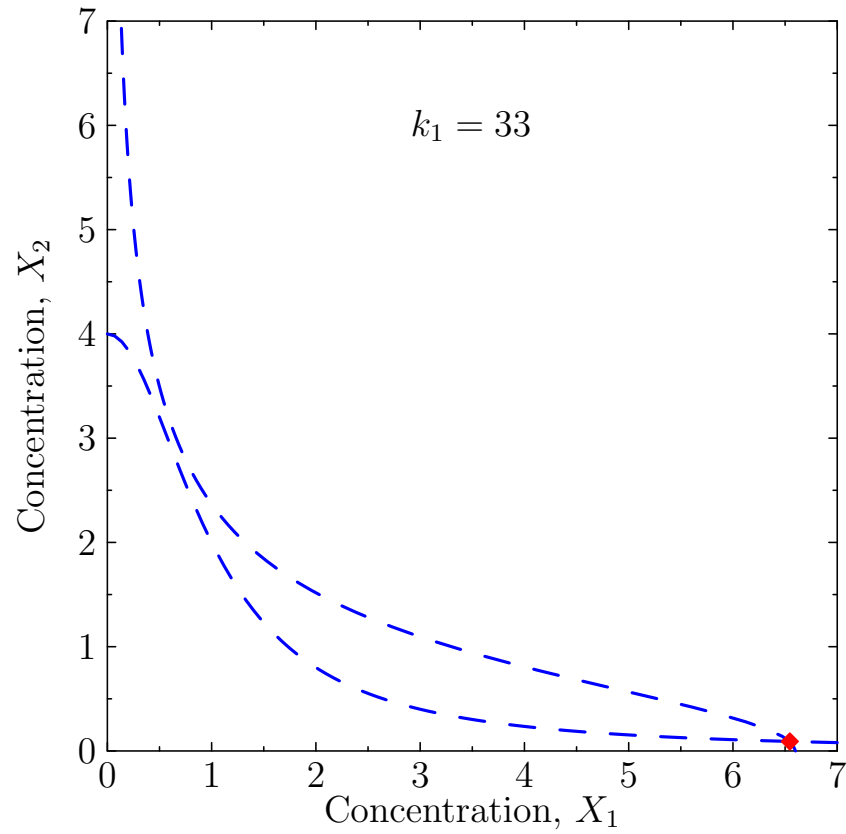
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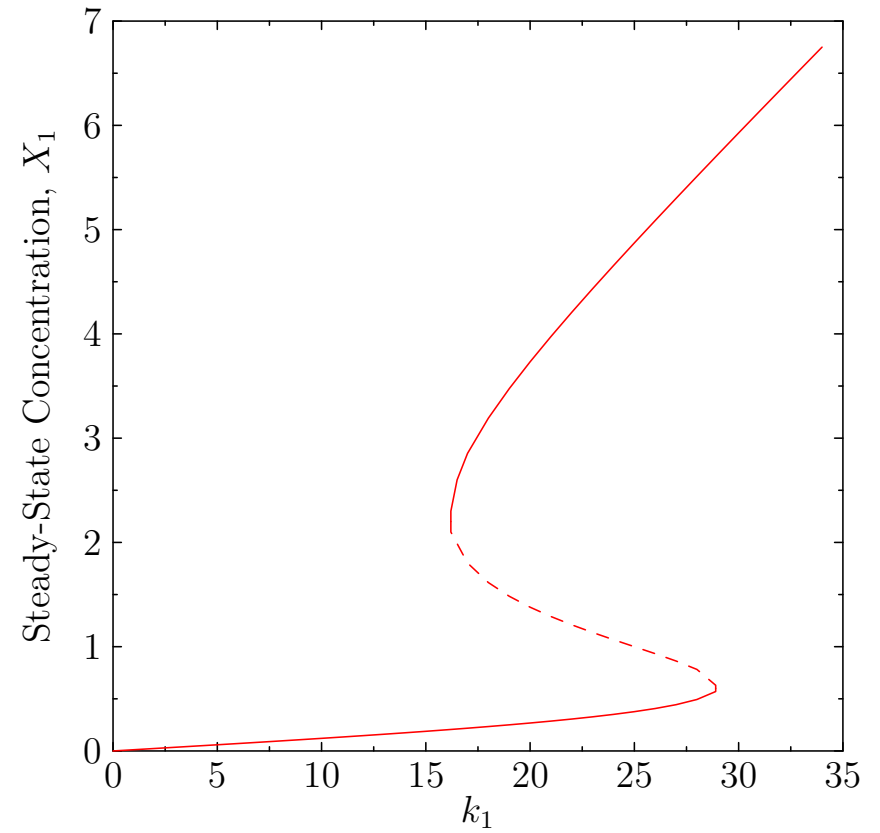
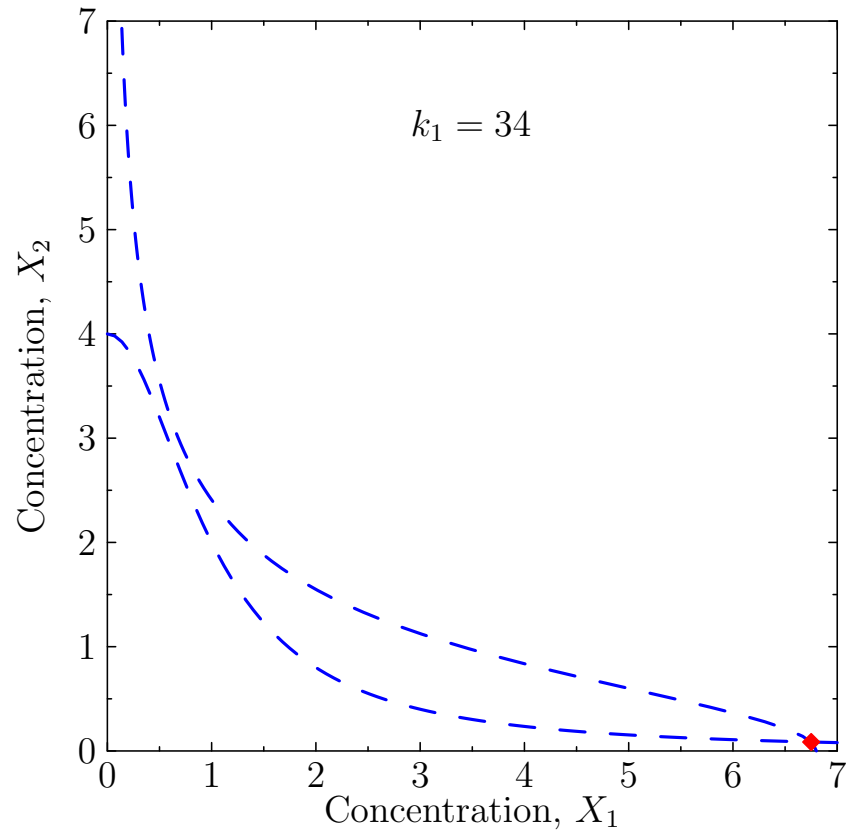
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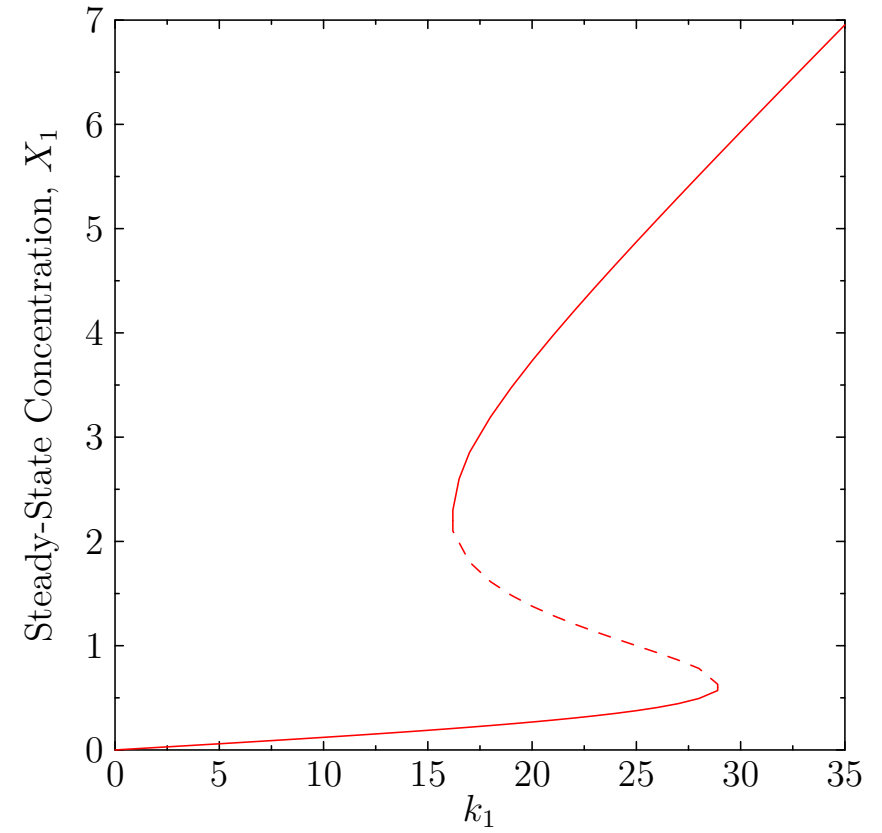
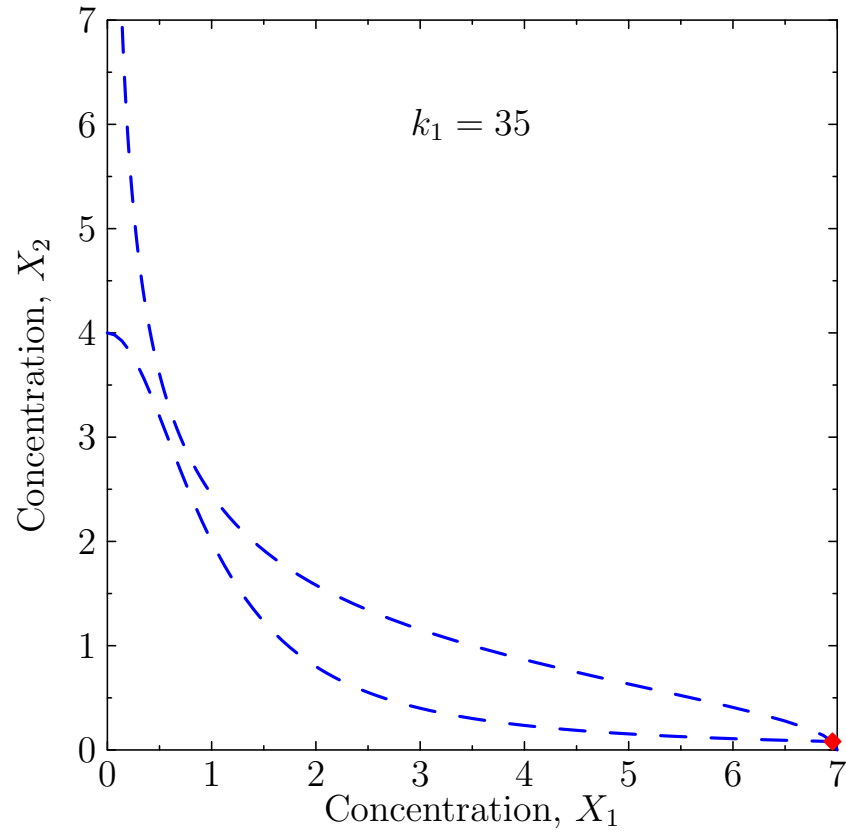
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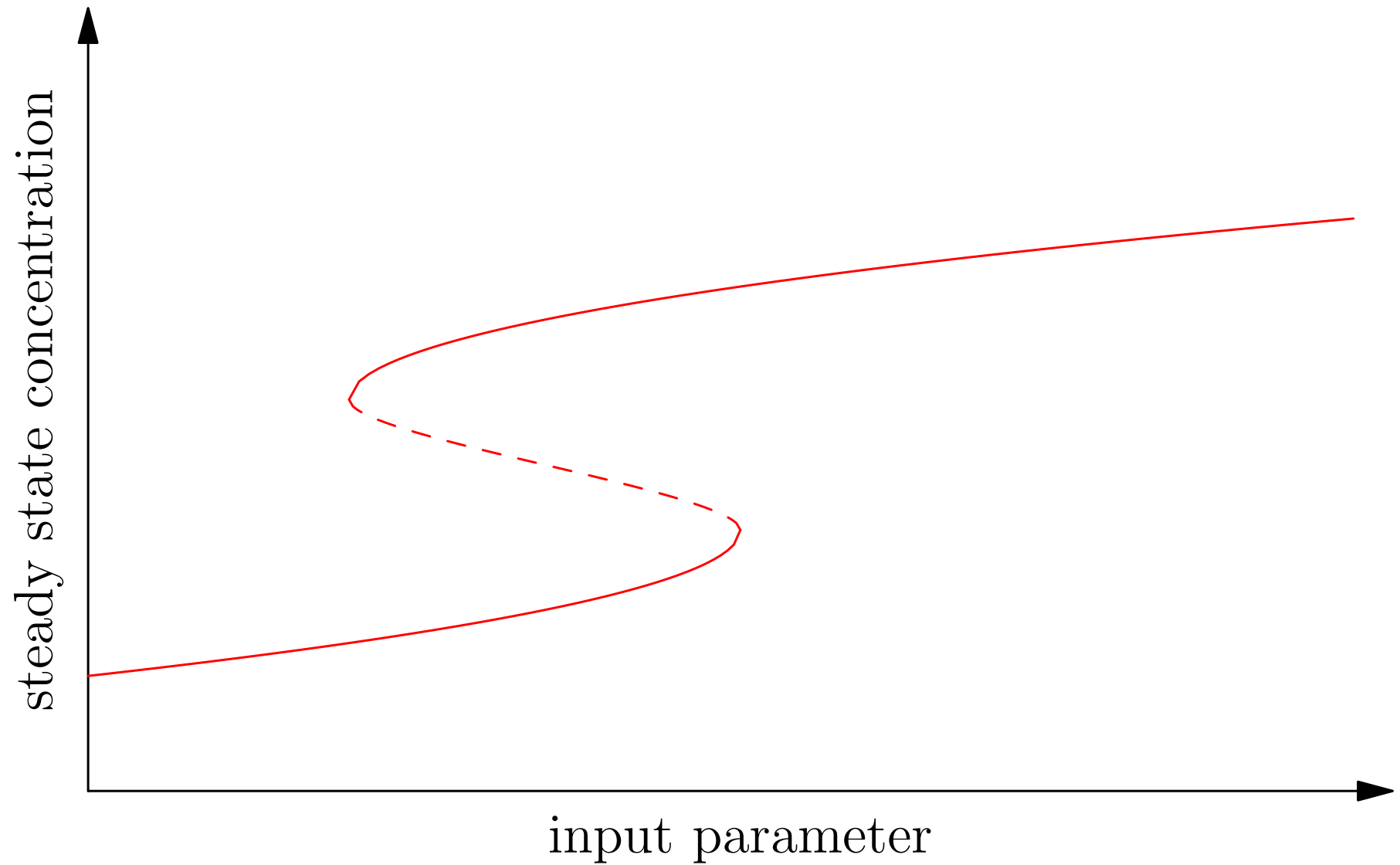
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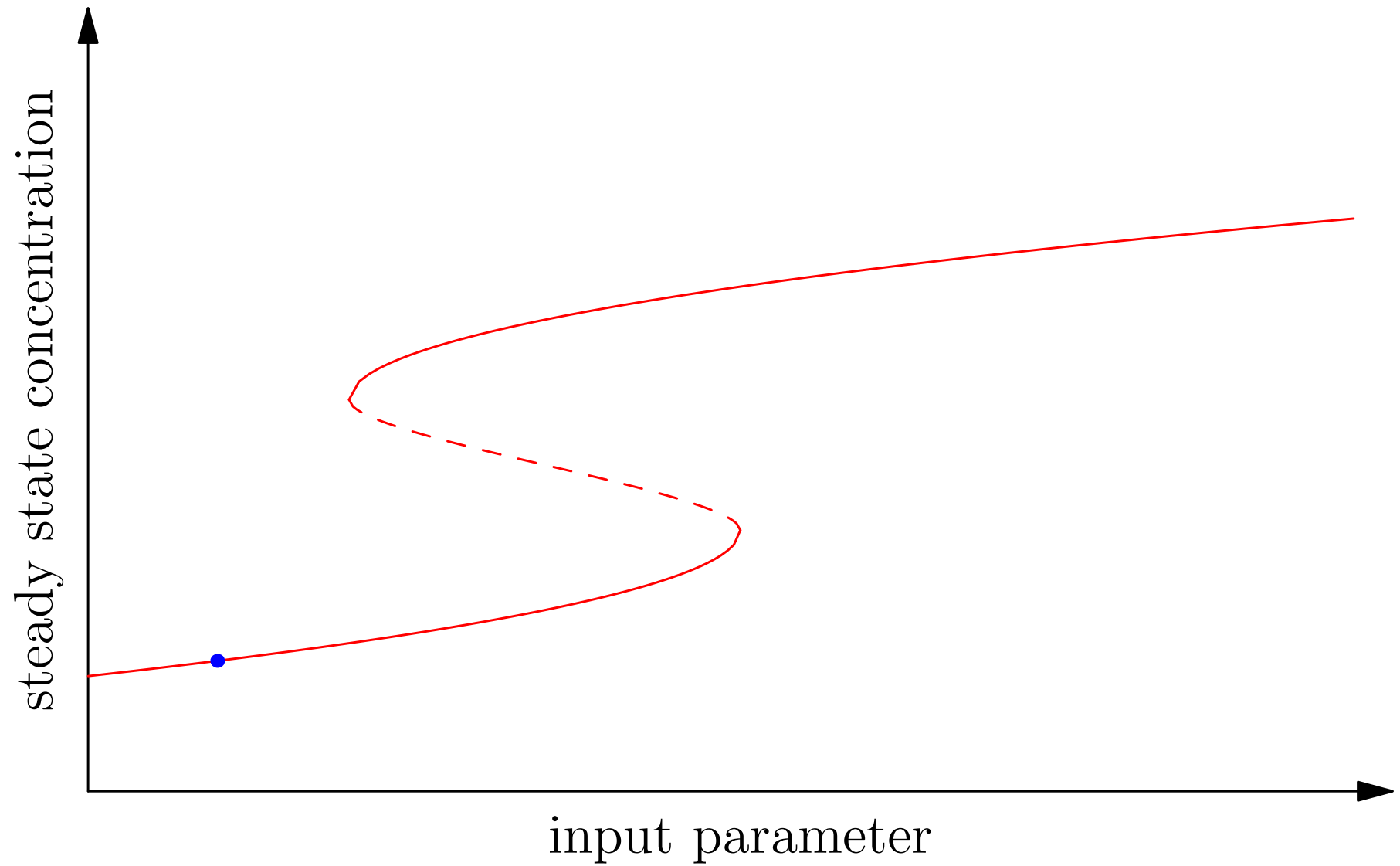
Varying k_1



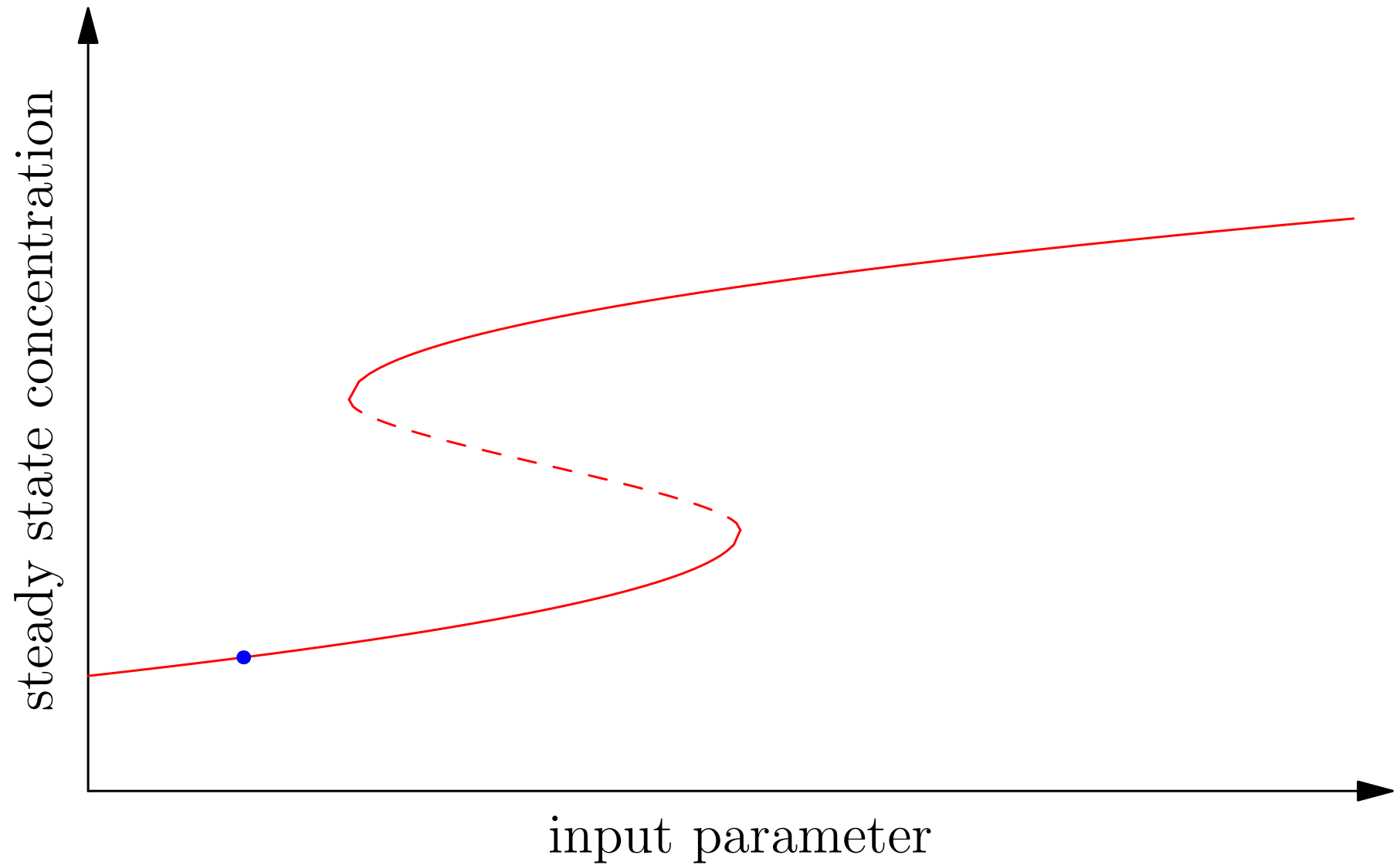
Hysteresis



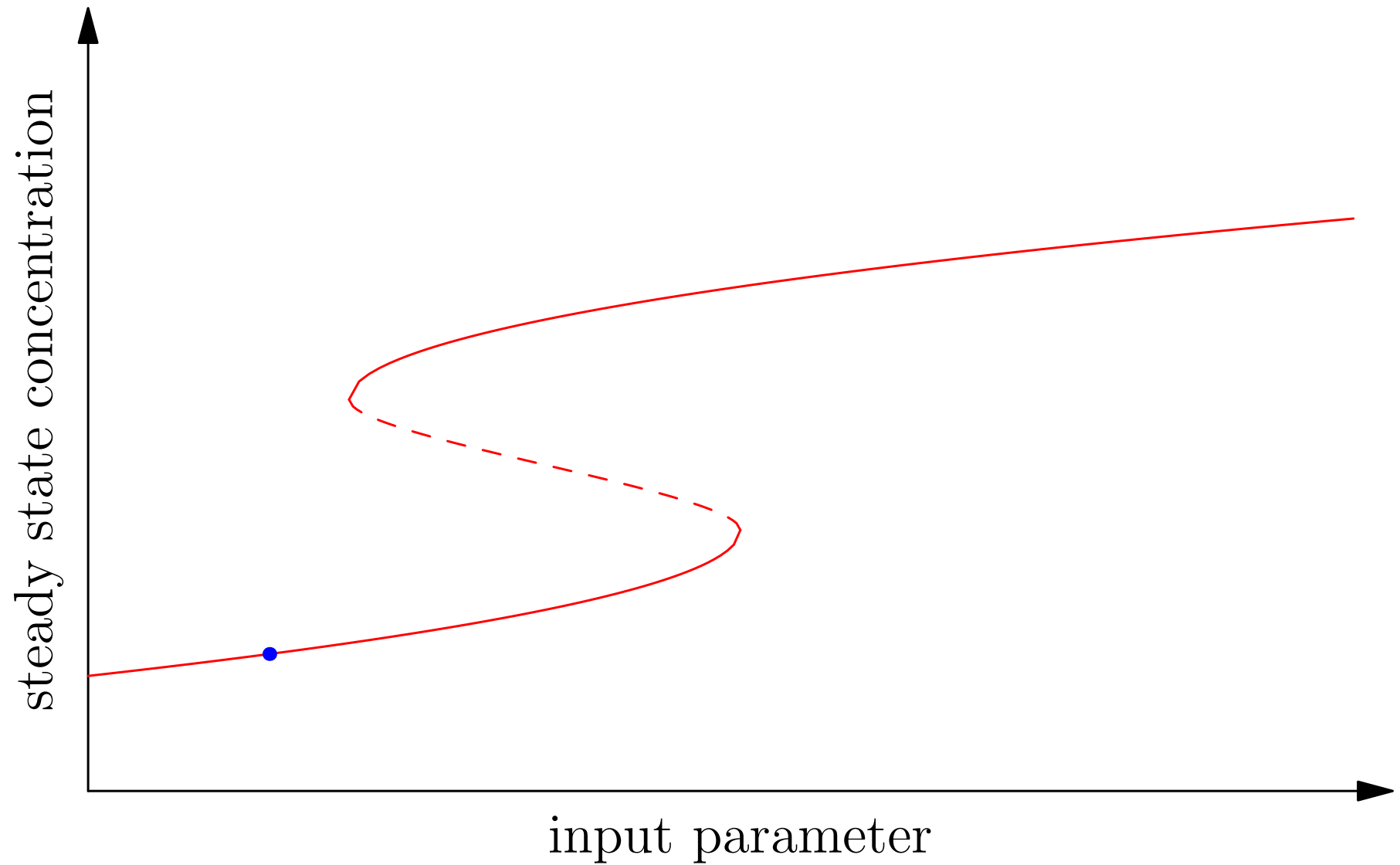
Hysteresis



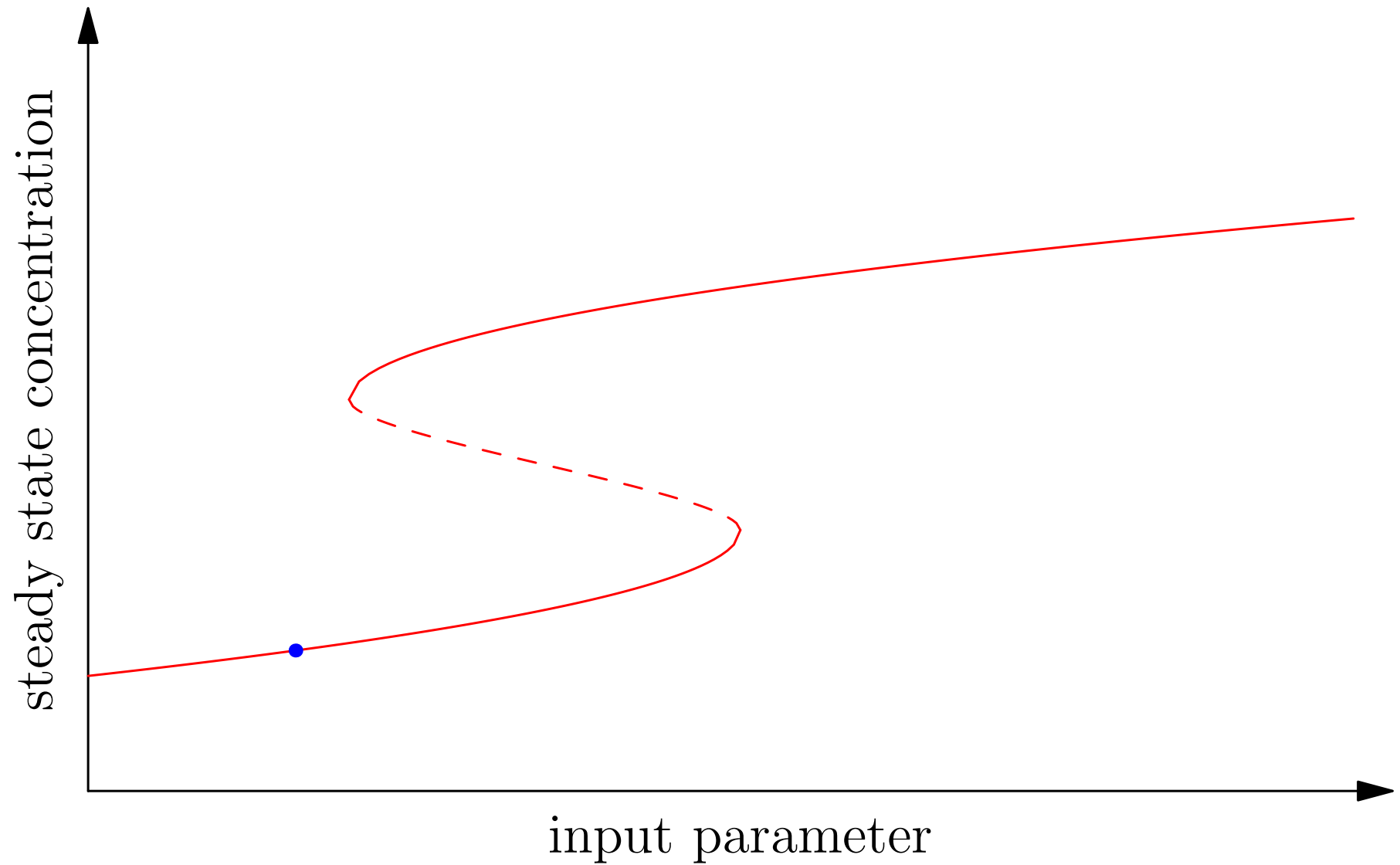
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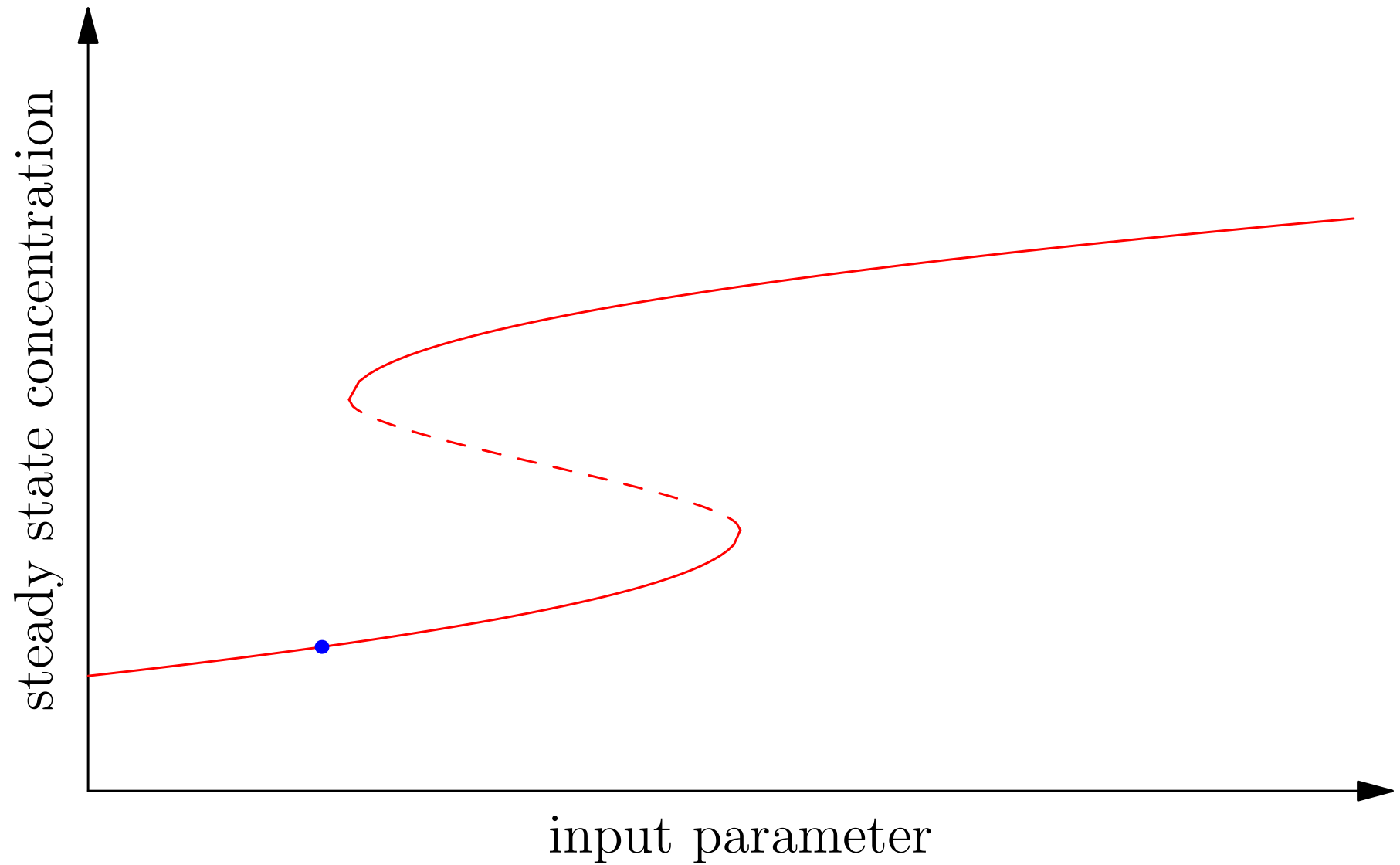
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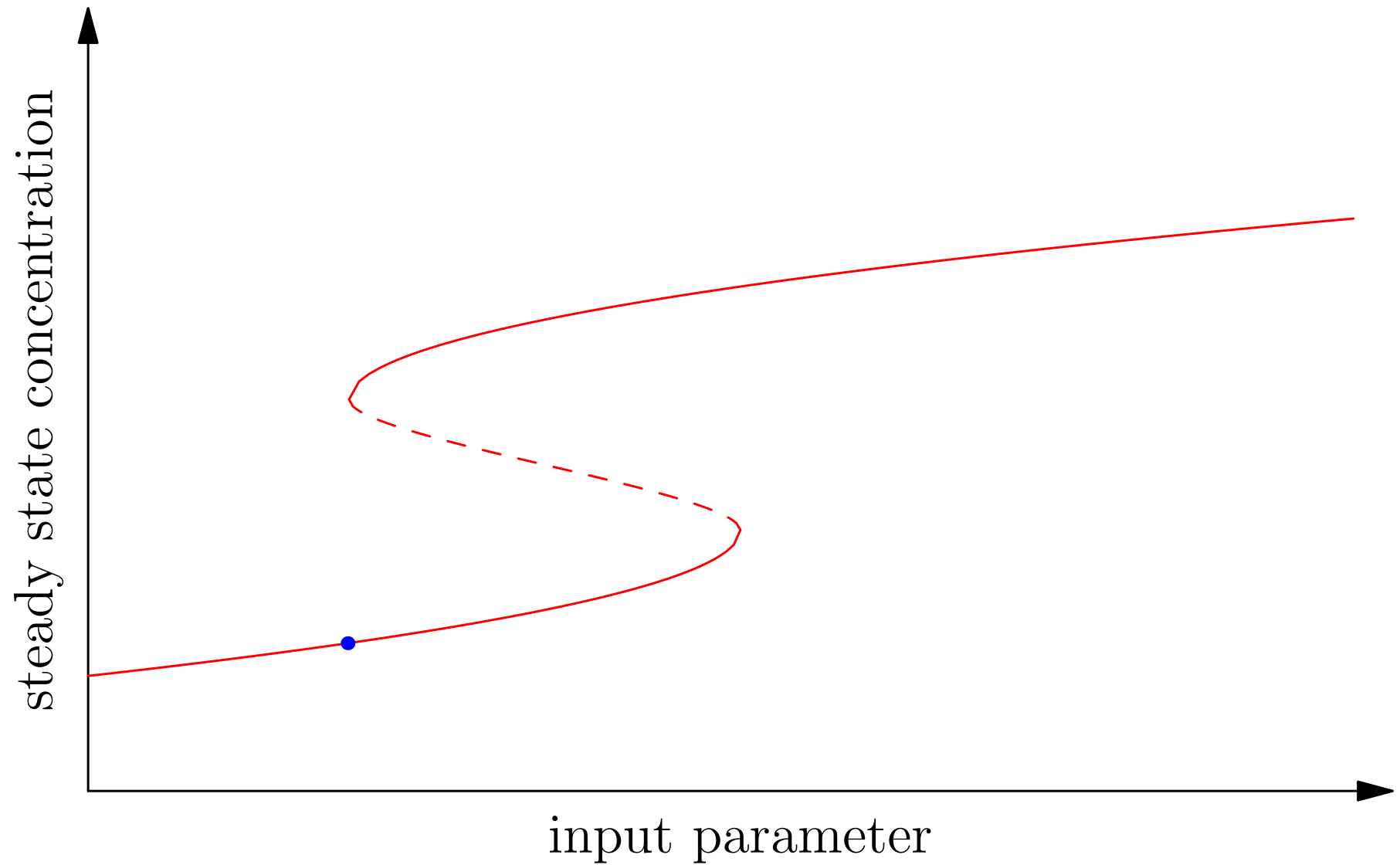
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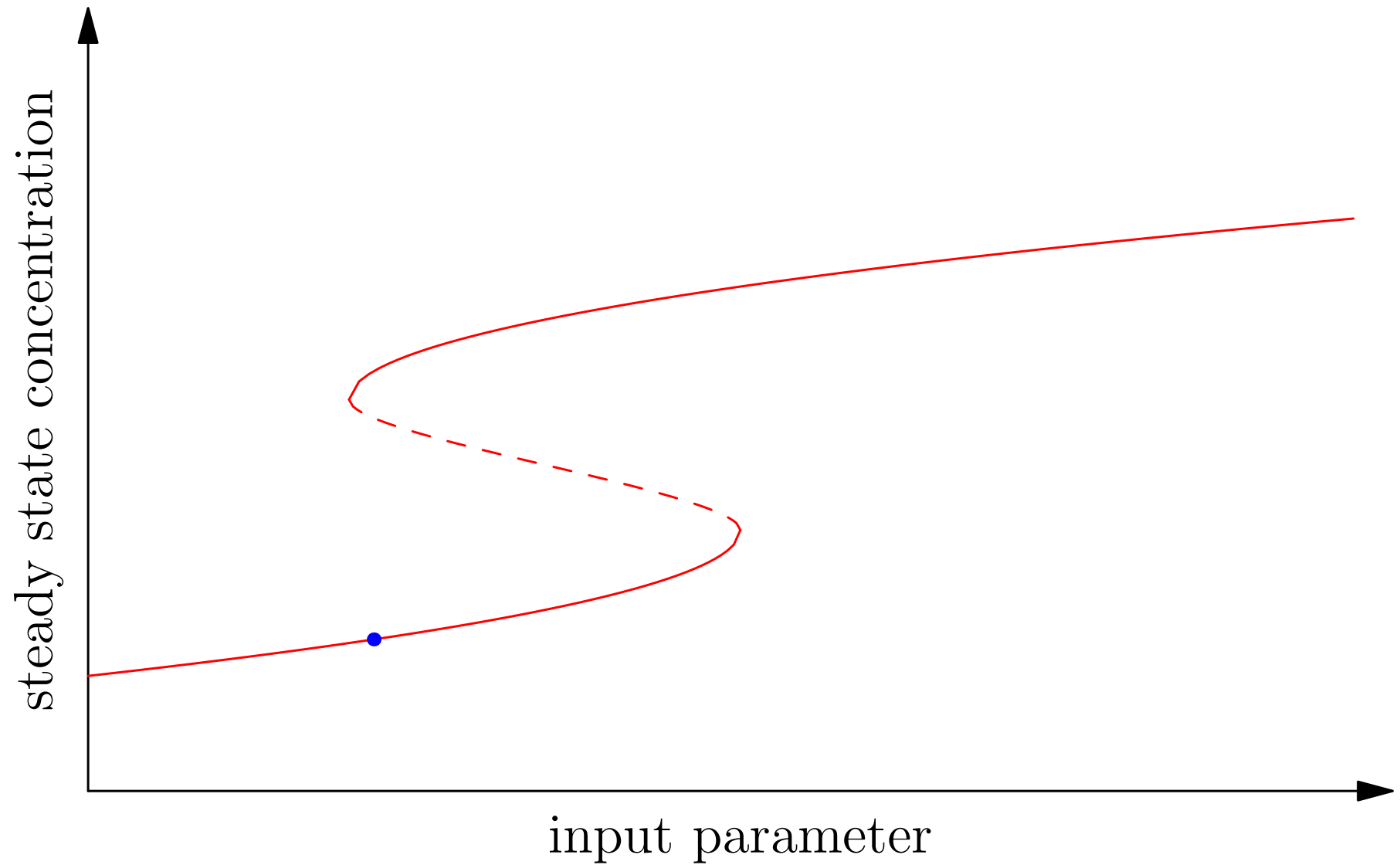
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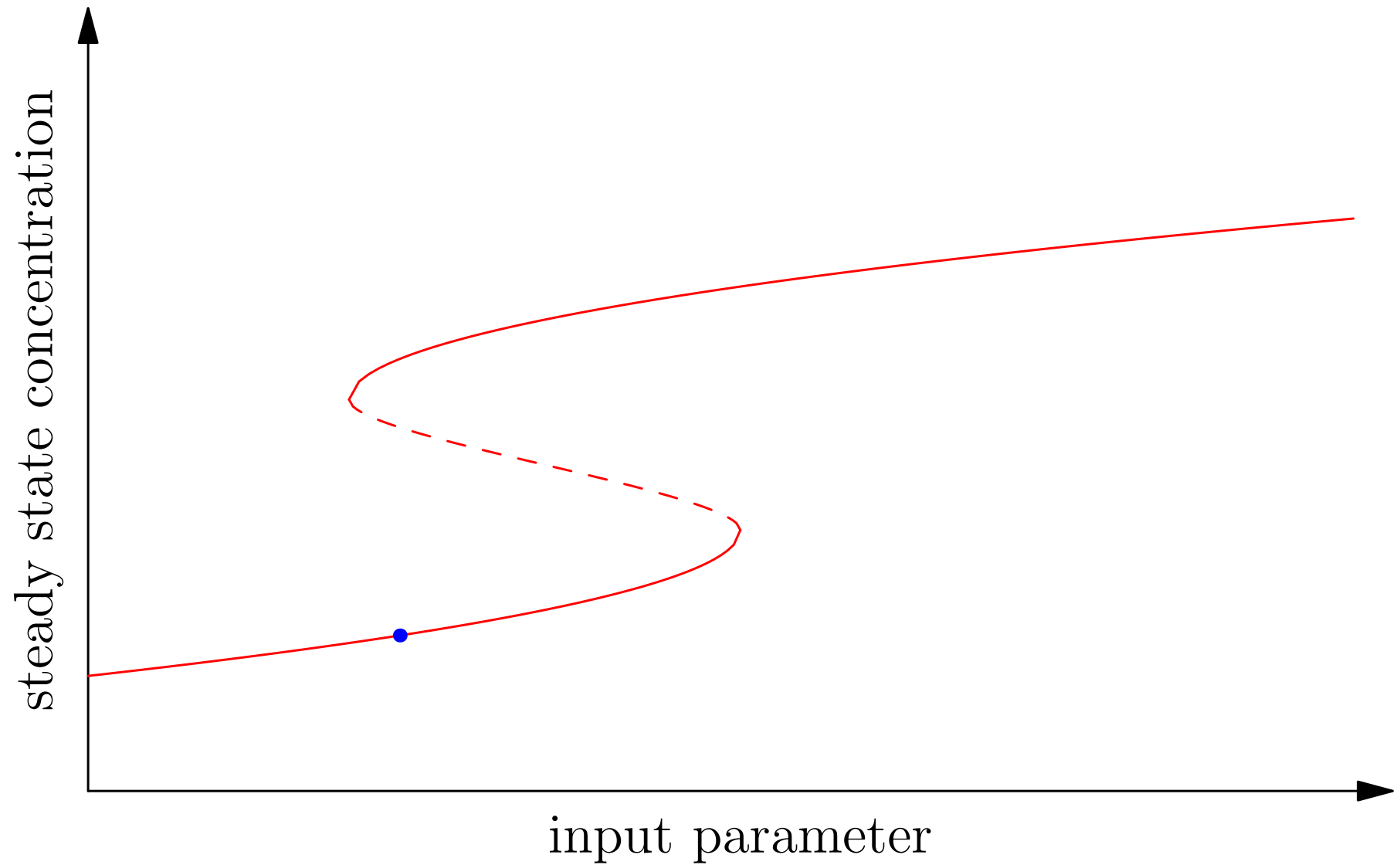
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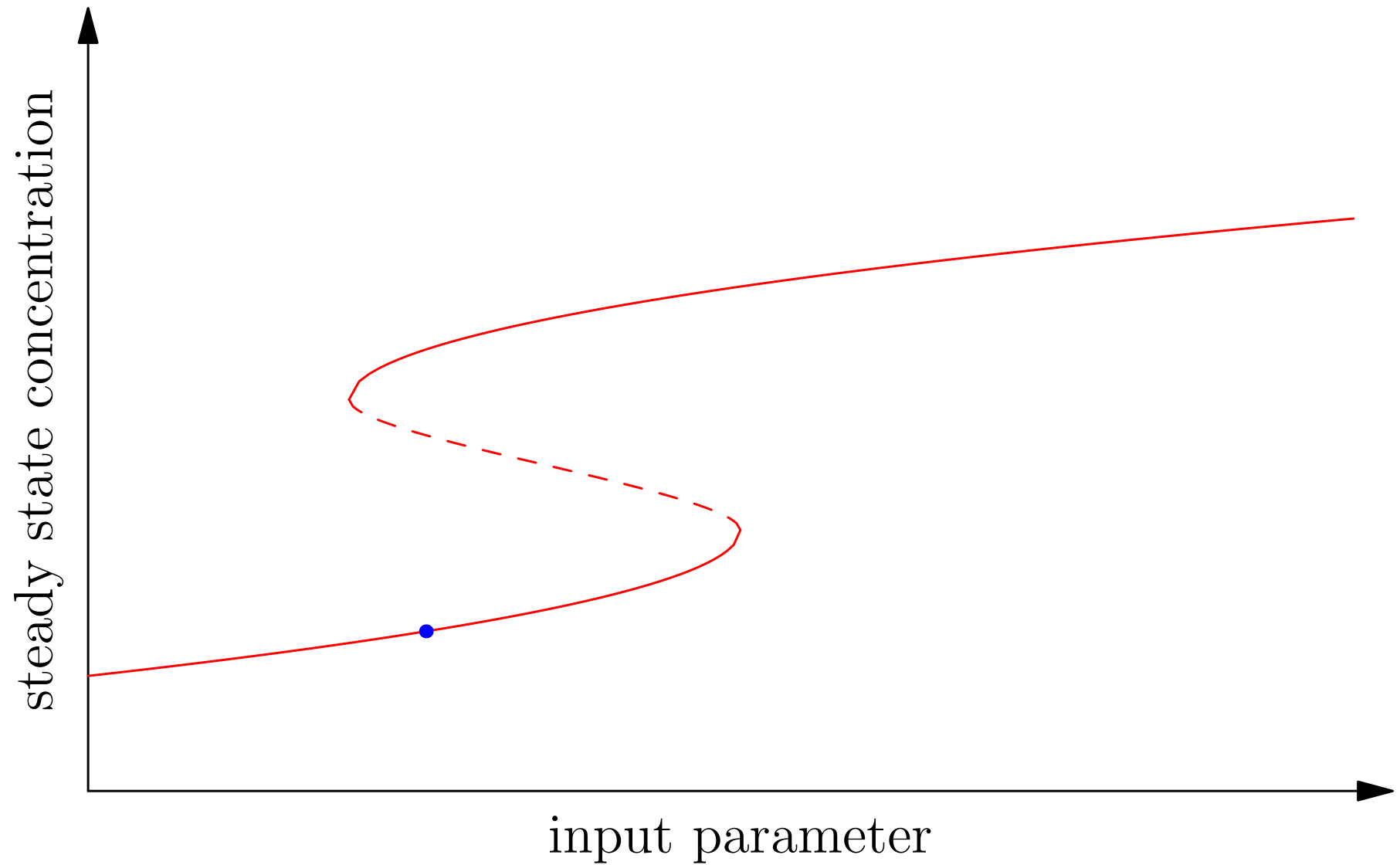
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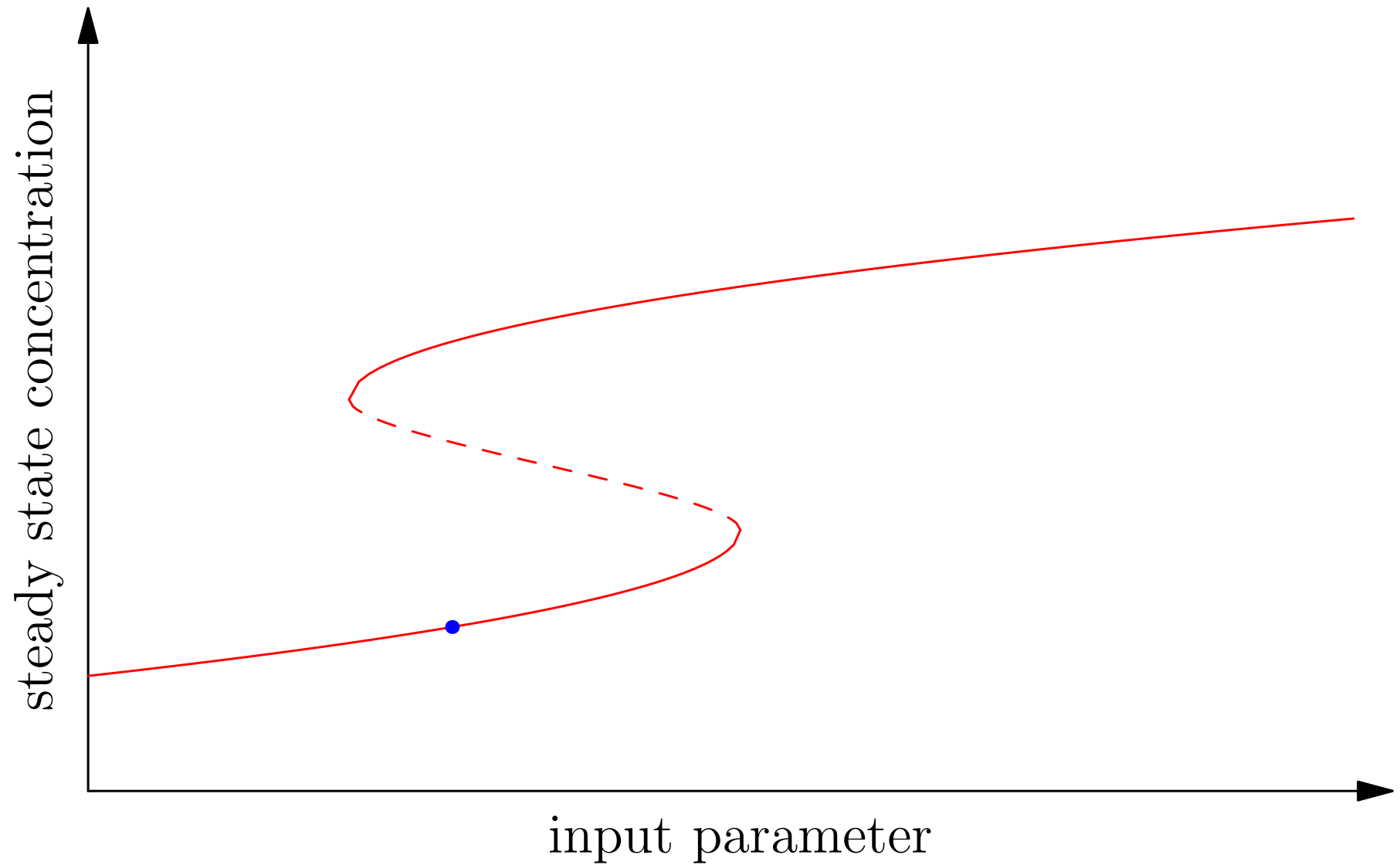
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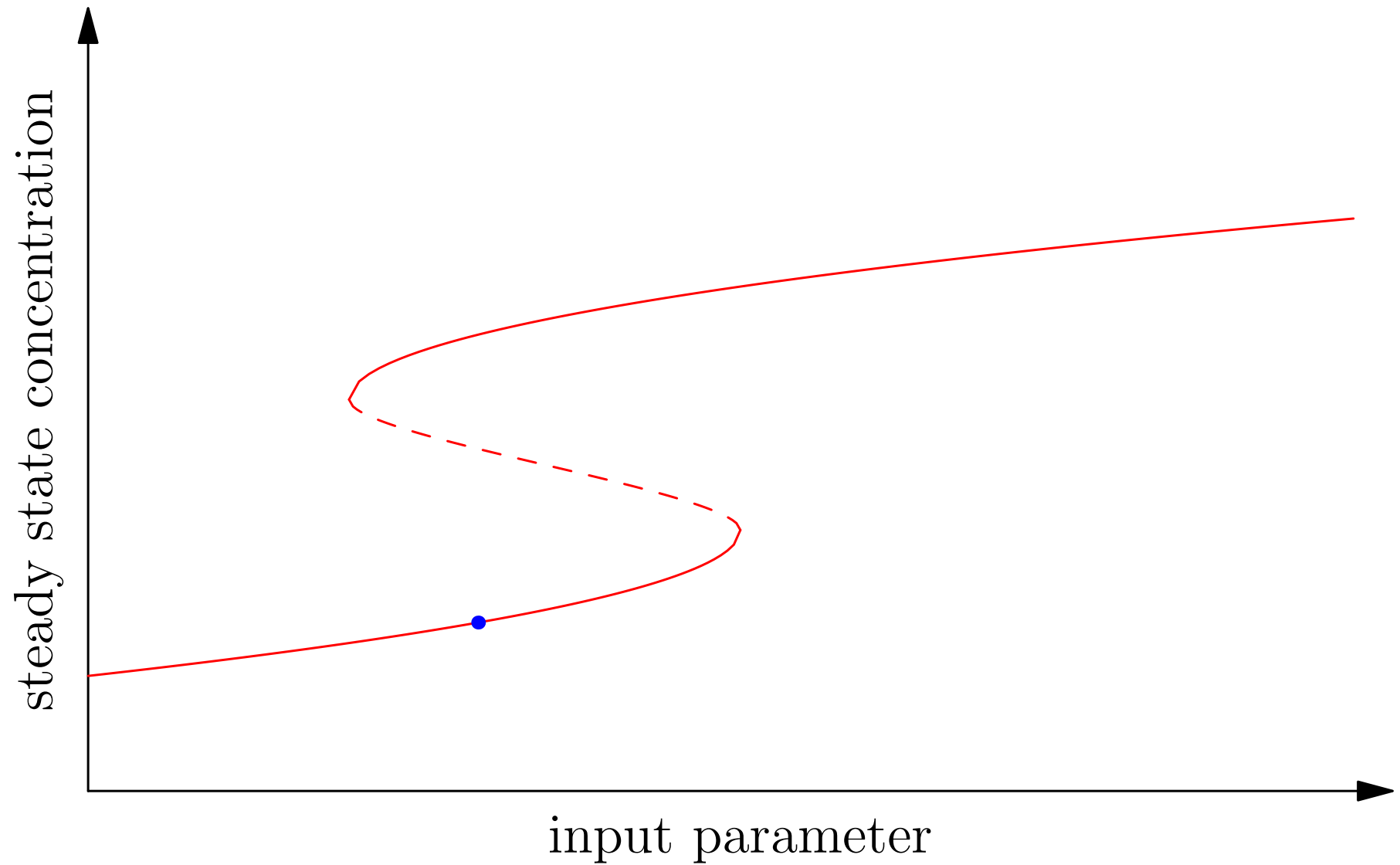
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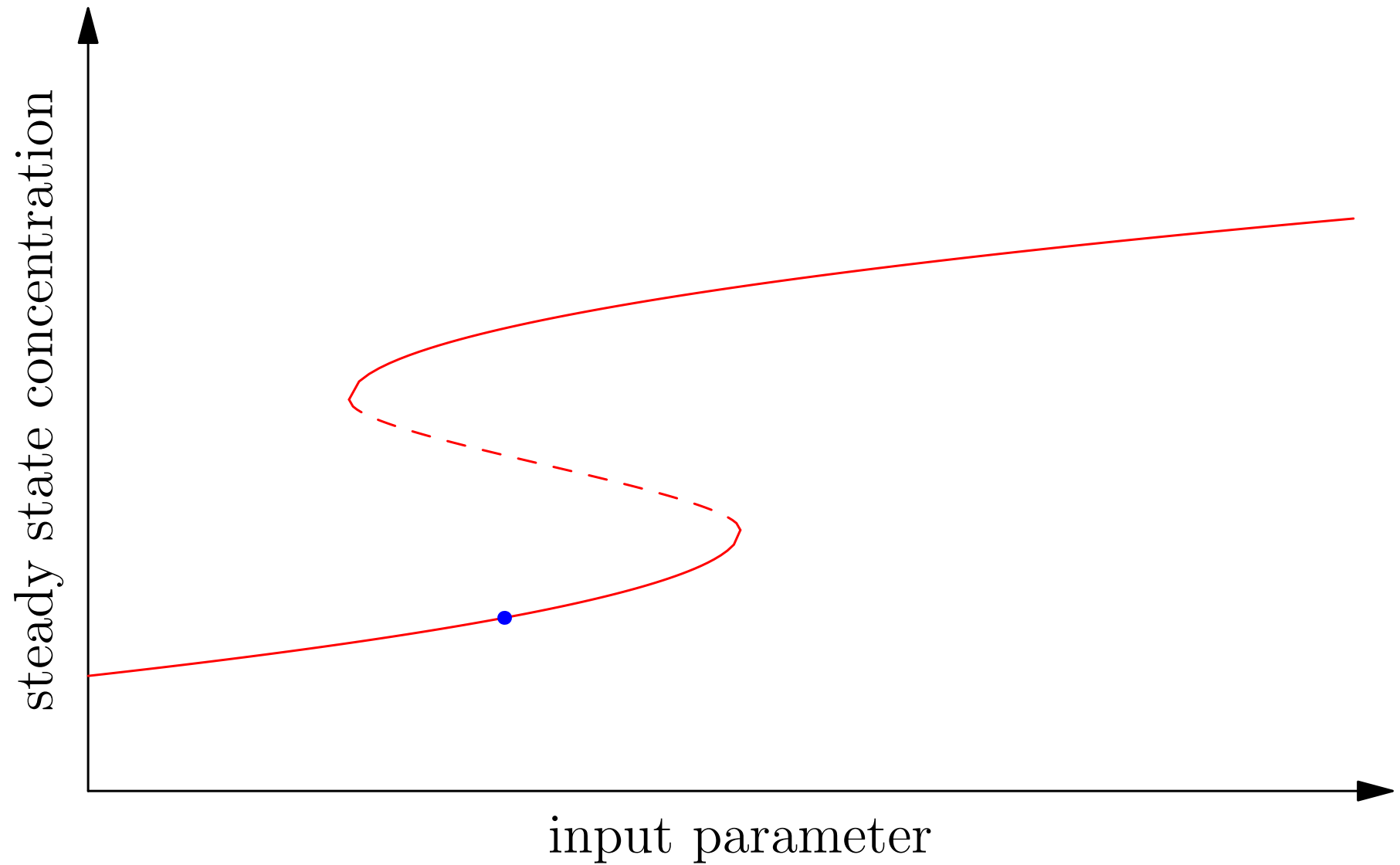
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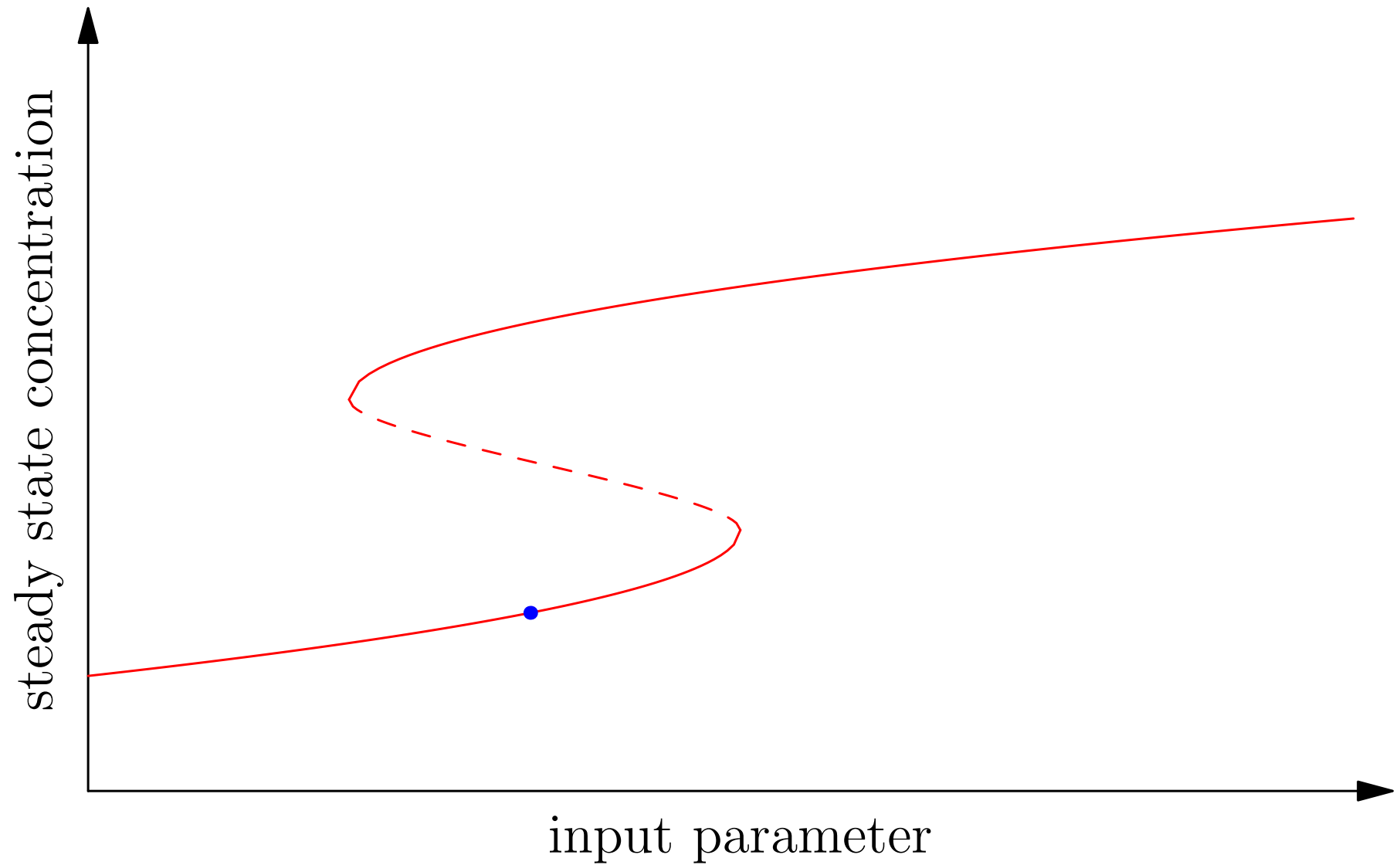
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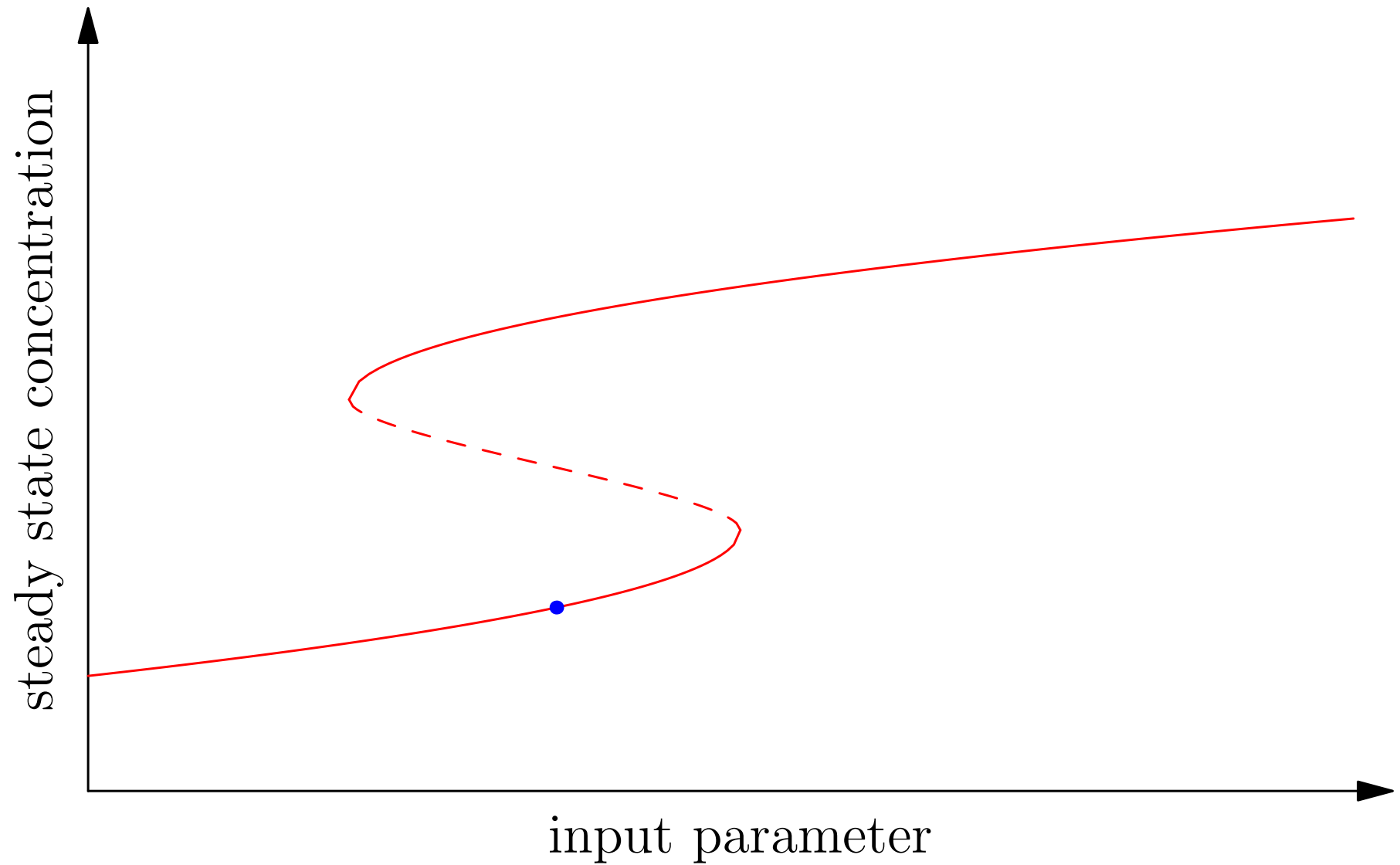
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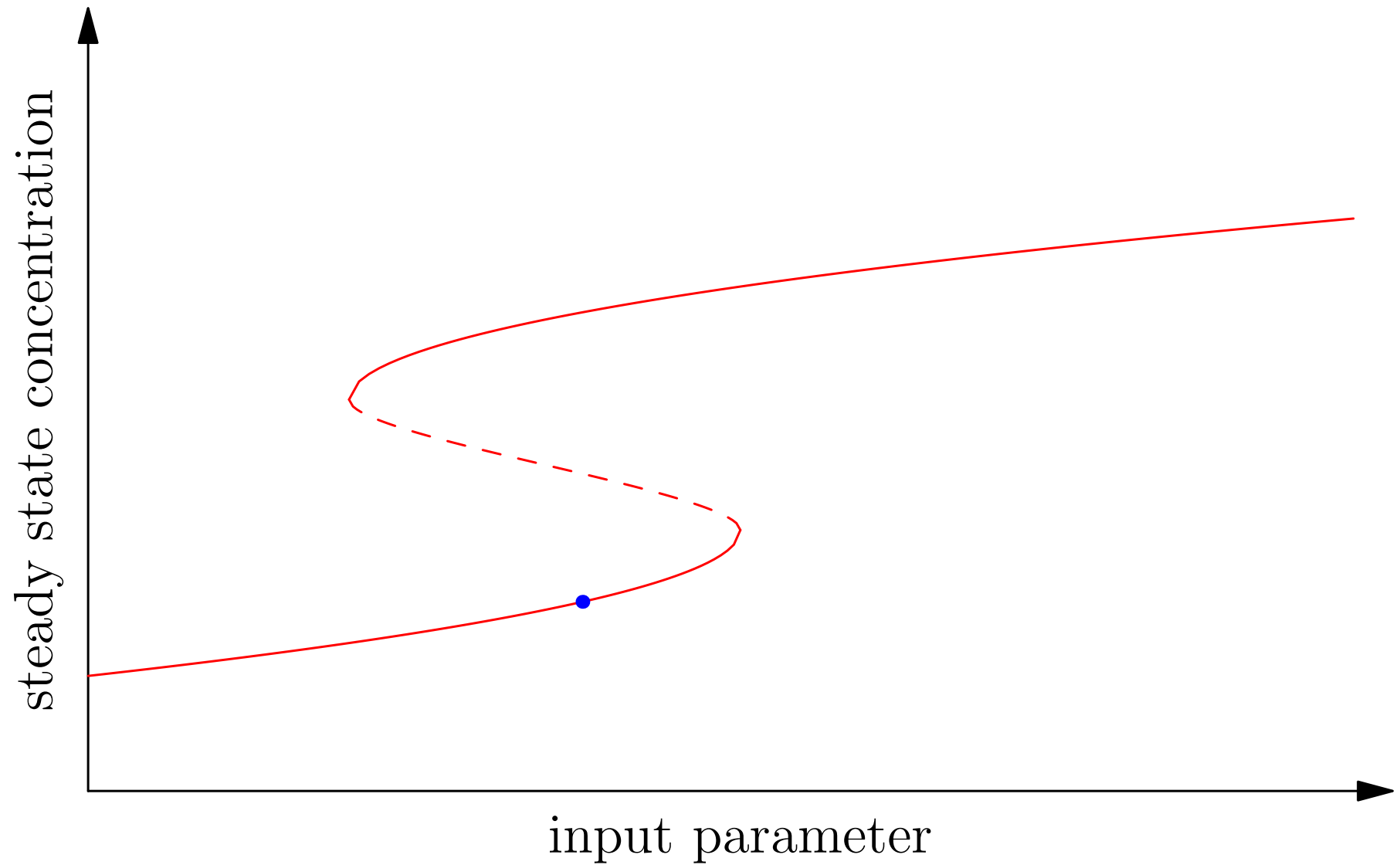
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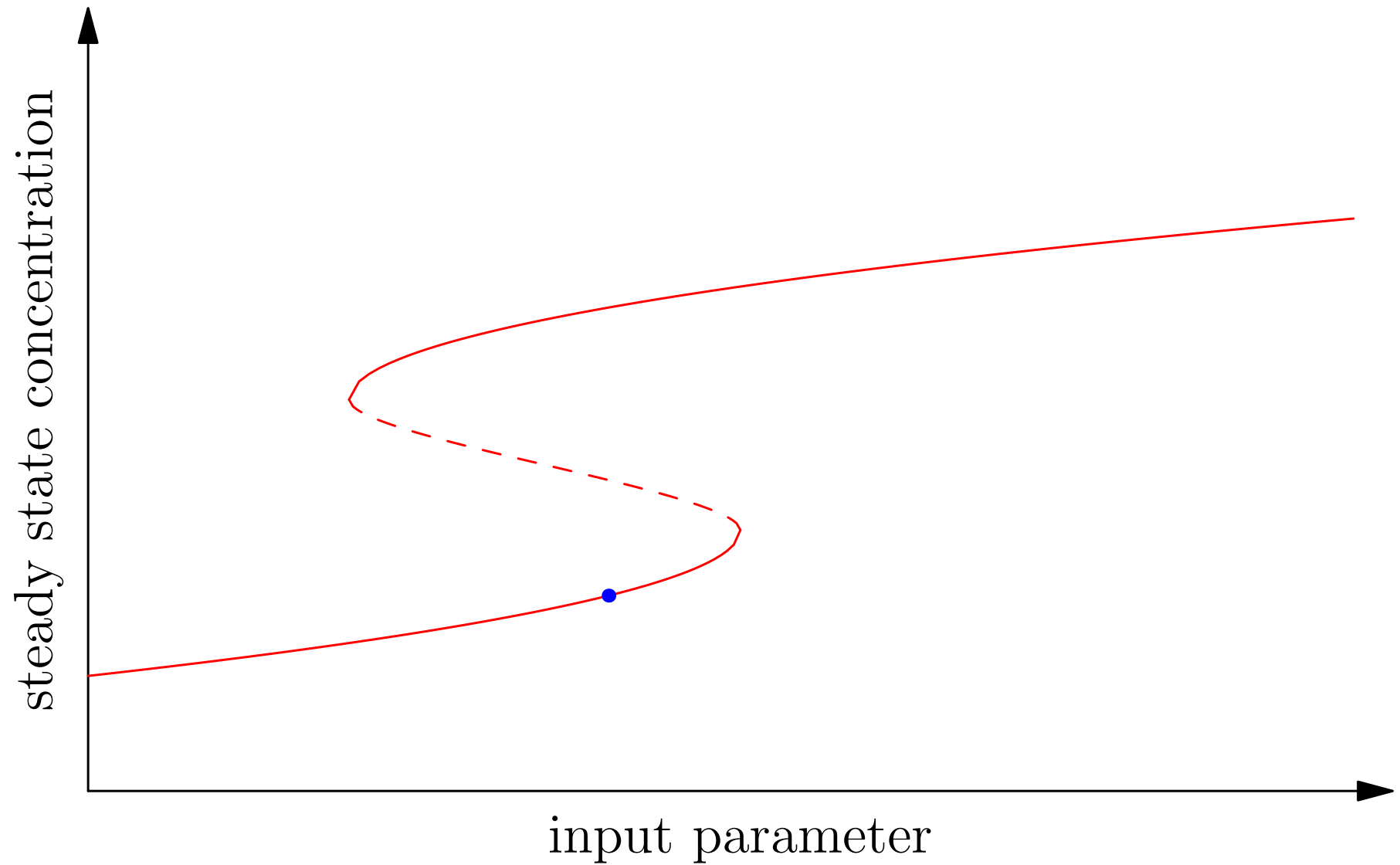
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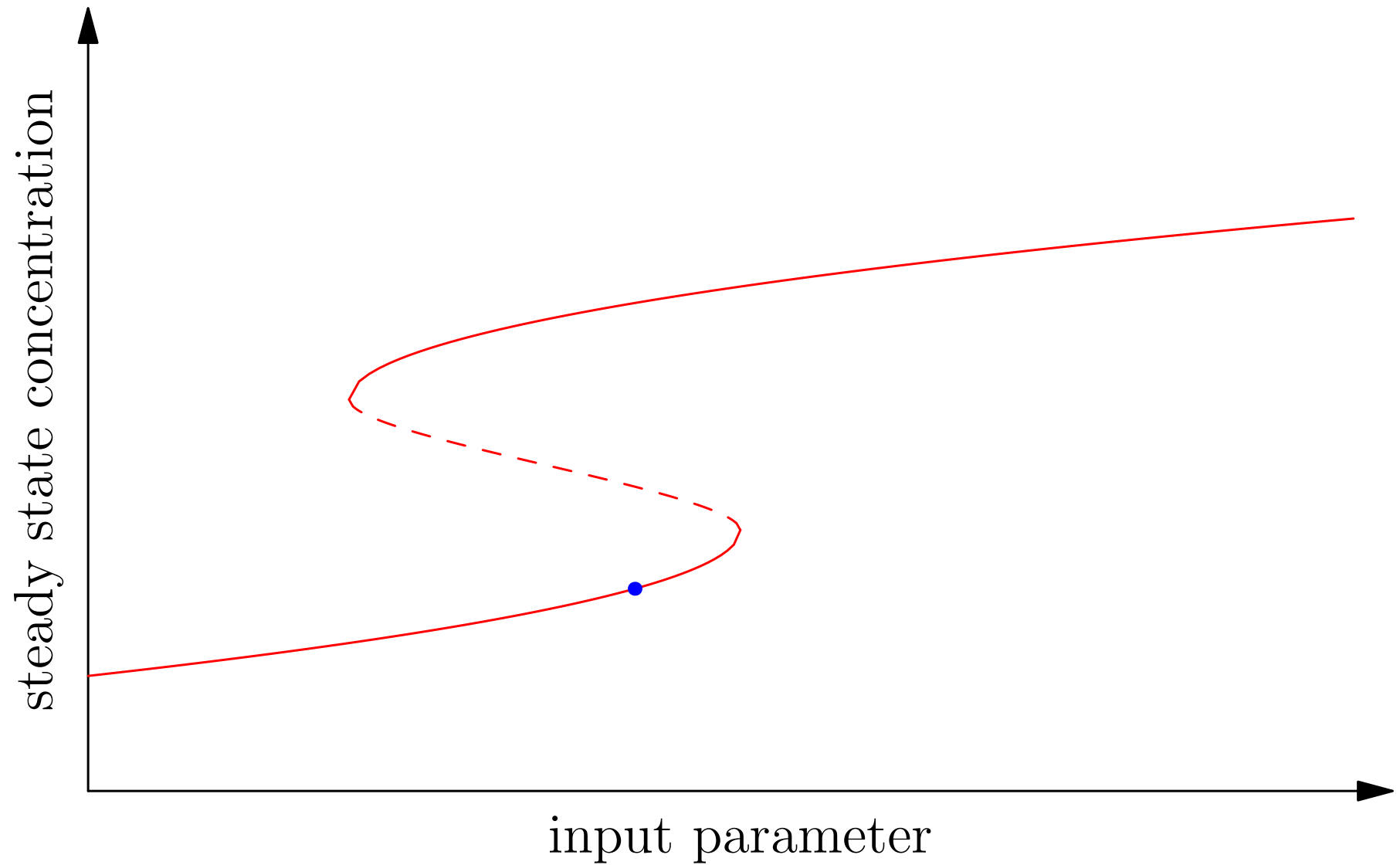
Hysteresis



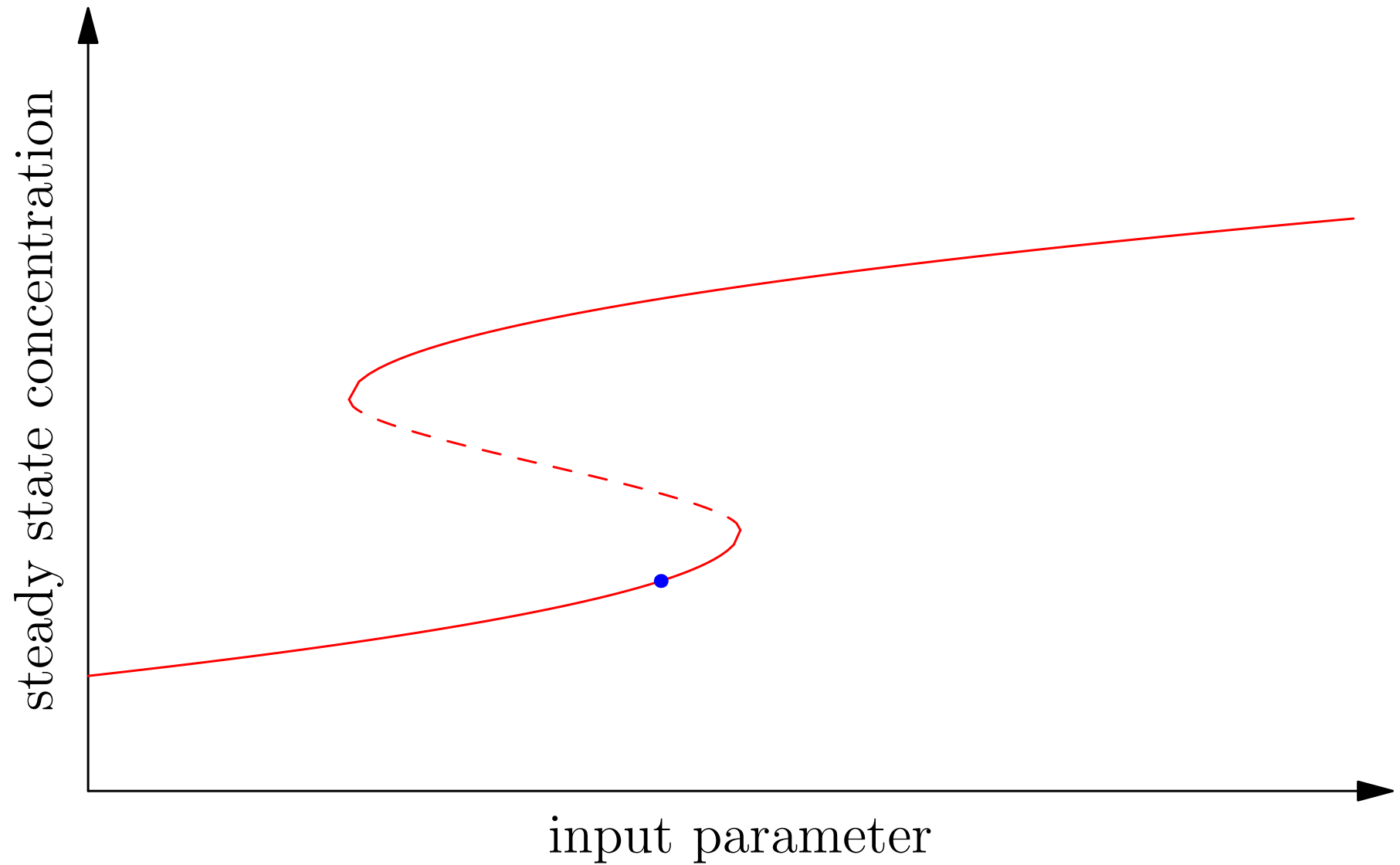
Hysteresis



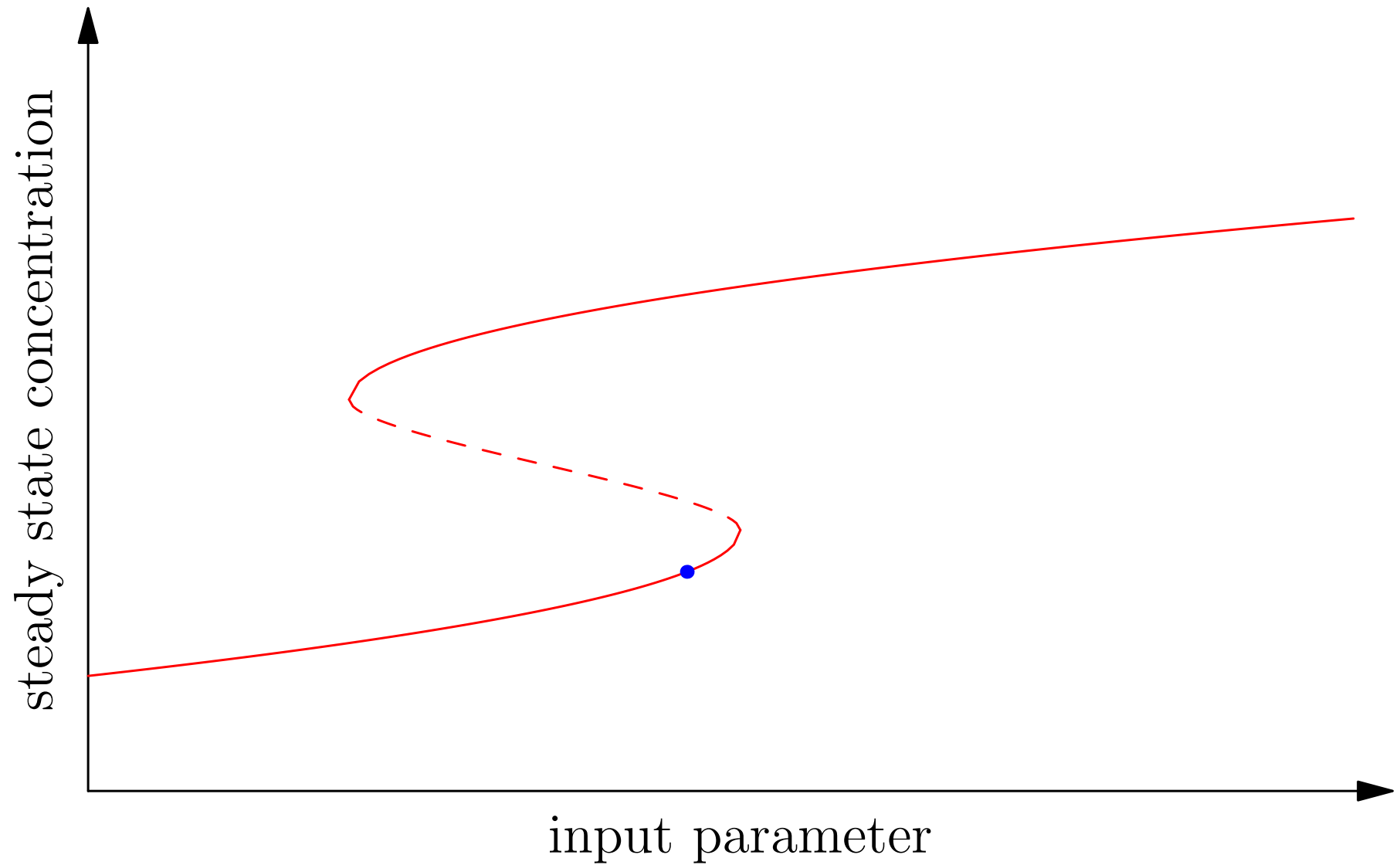
Hysteresis



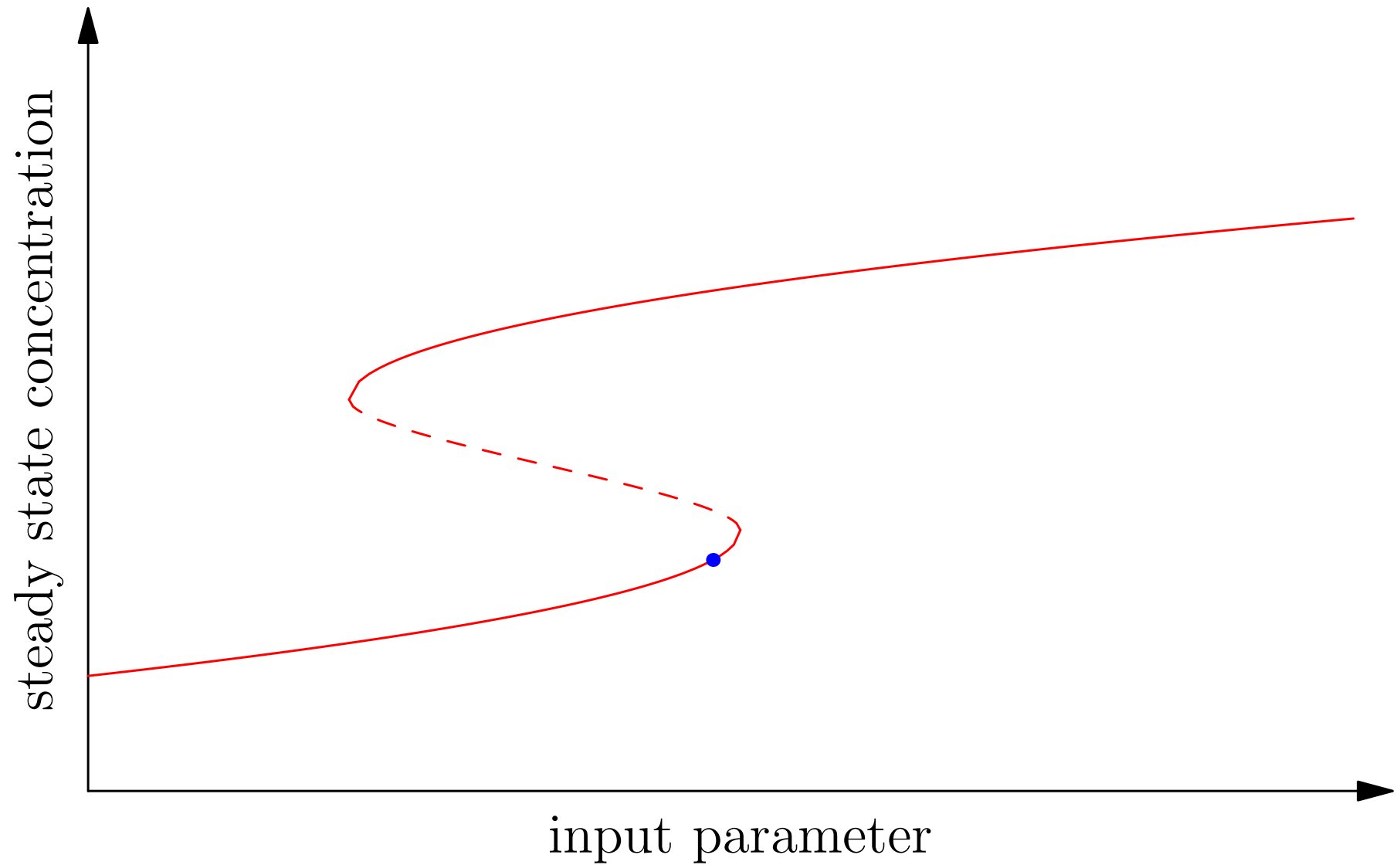
Hysteresis



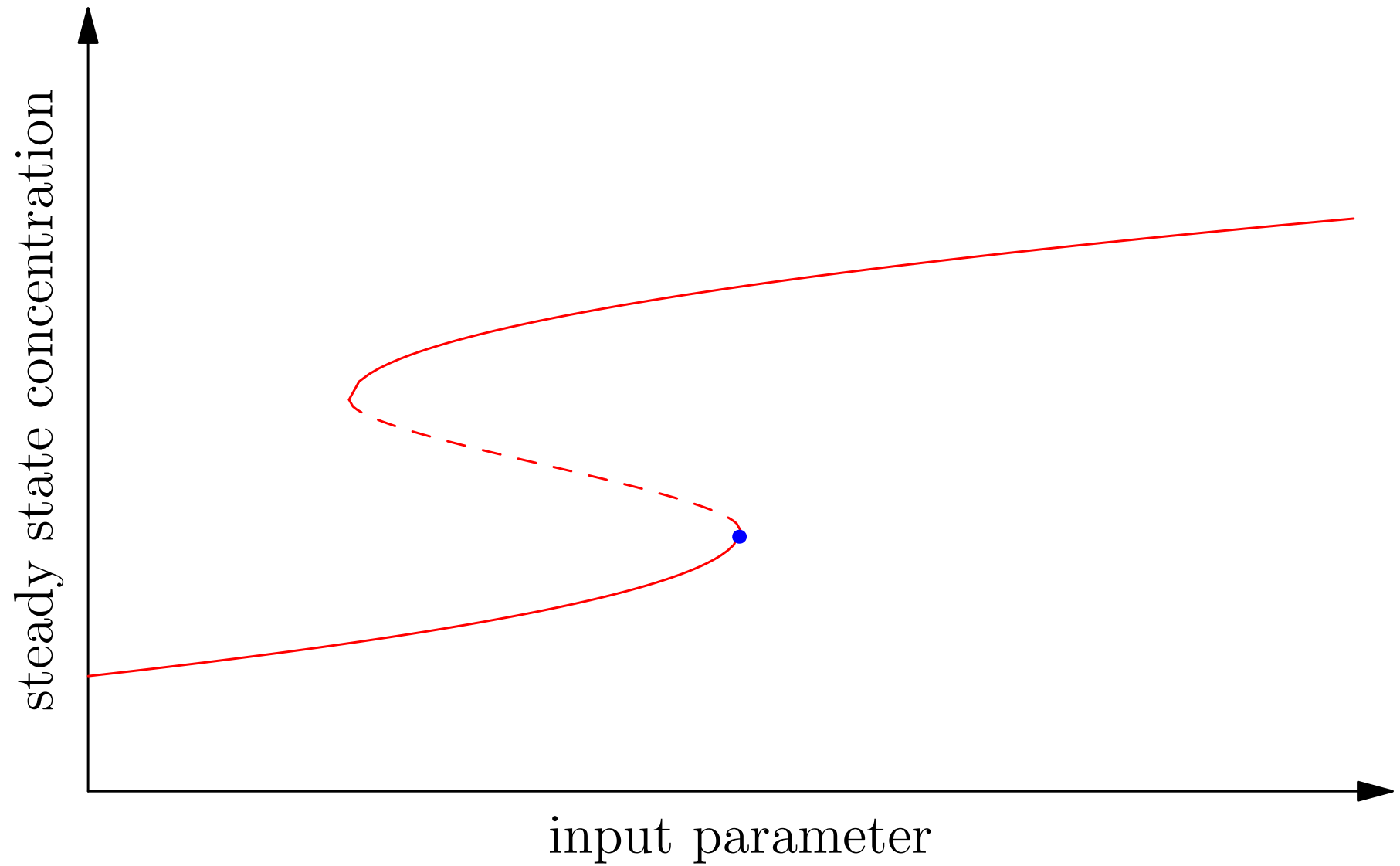
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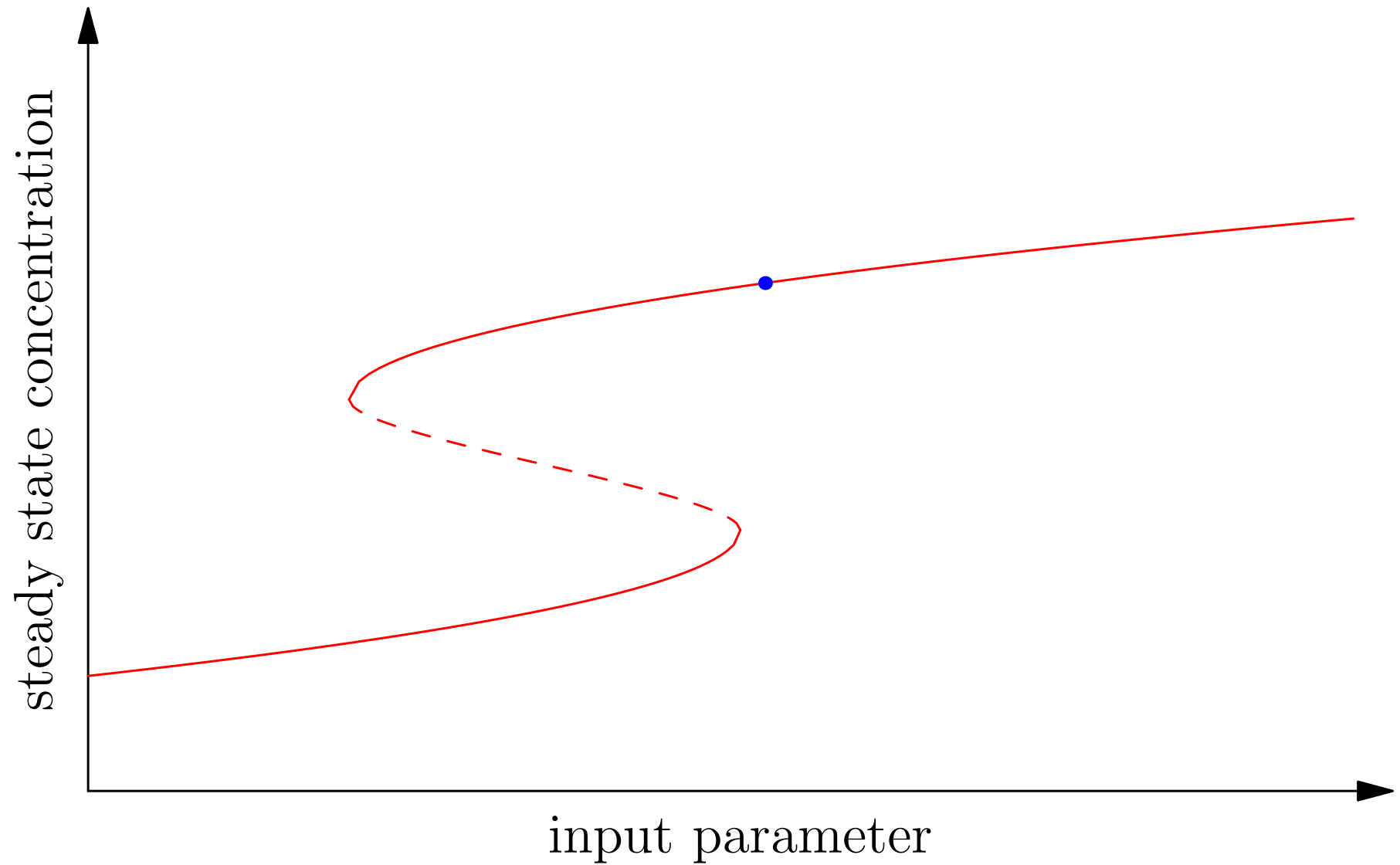
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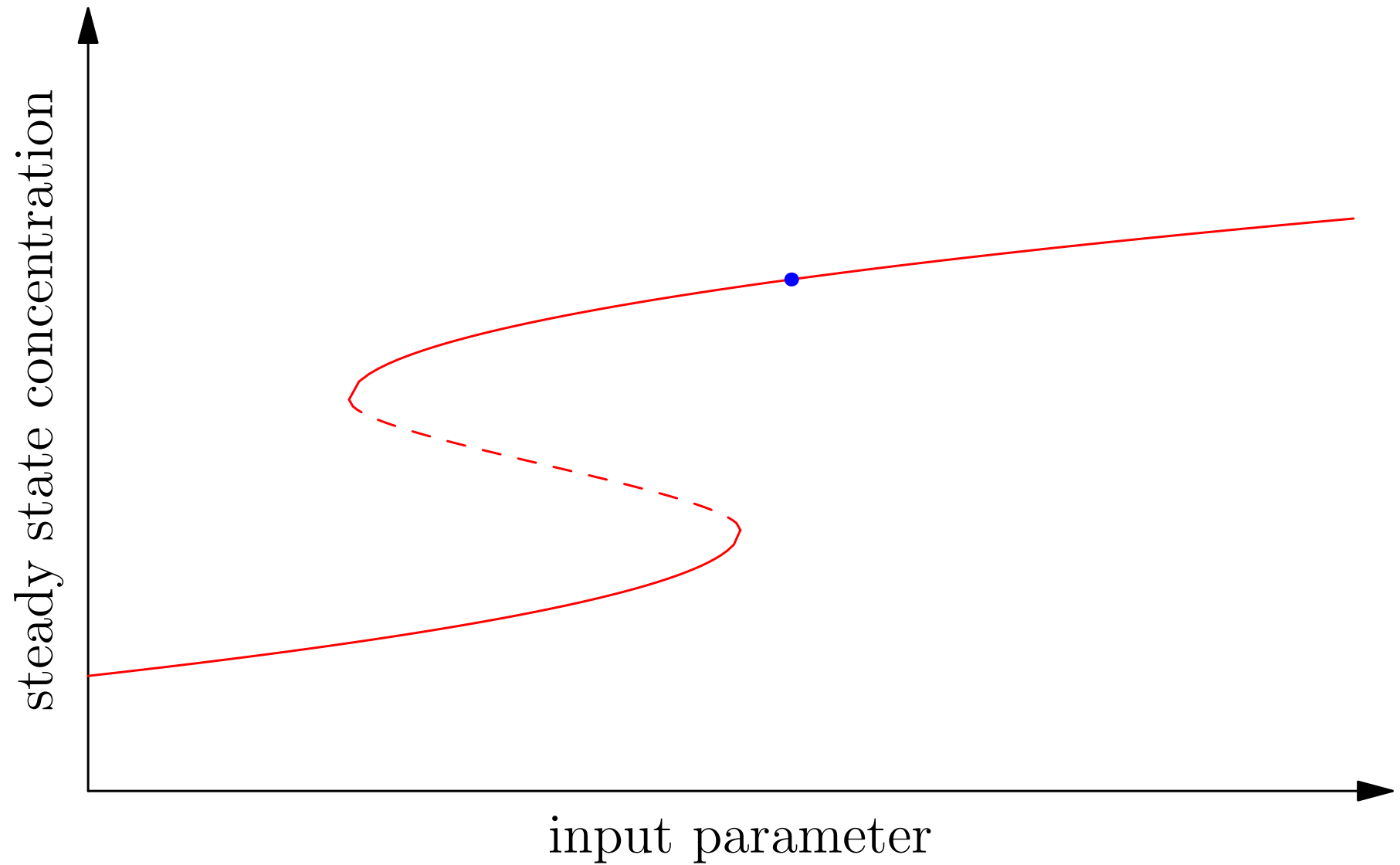
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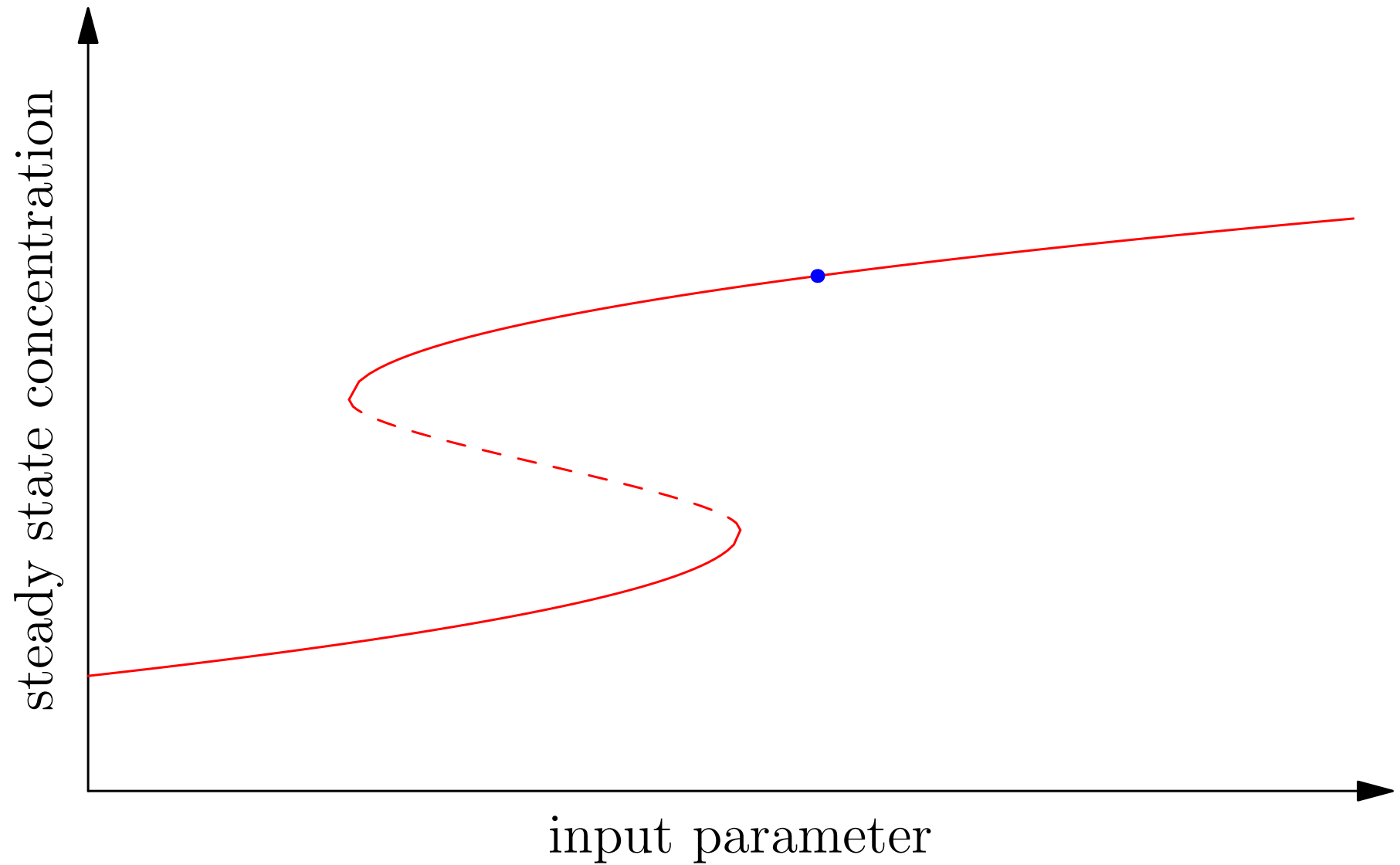
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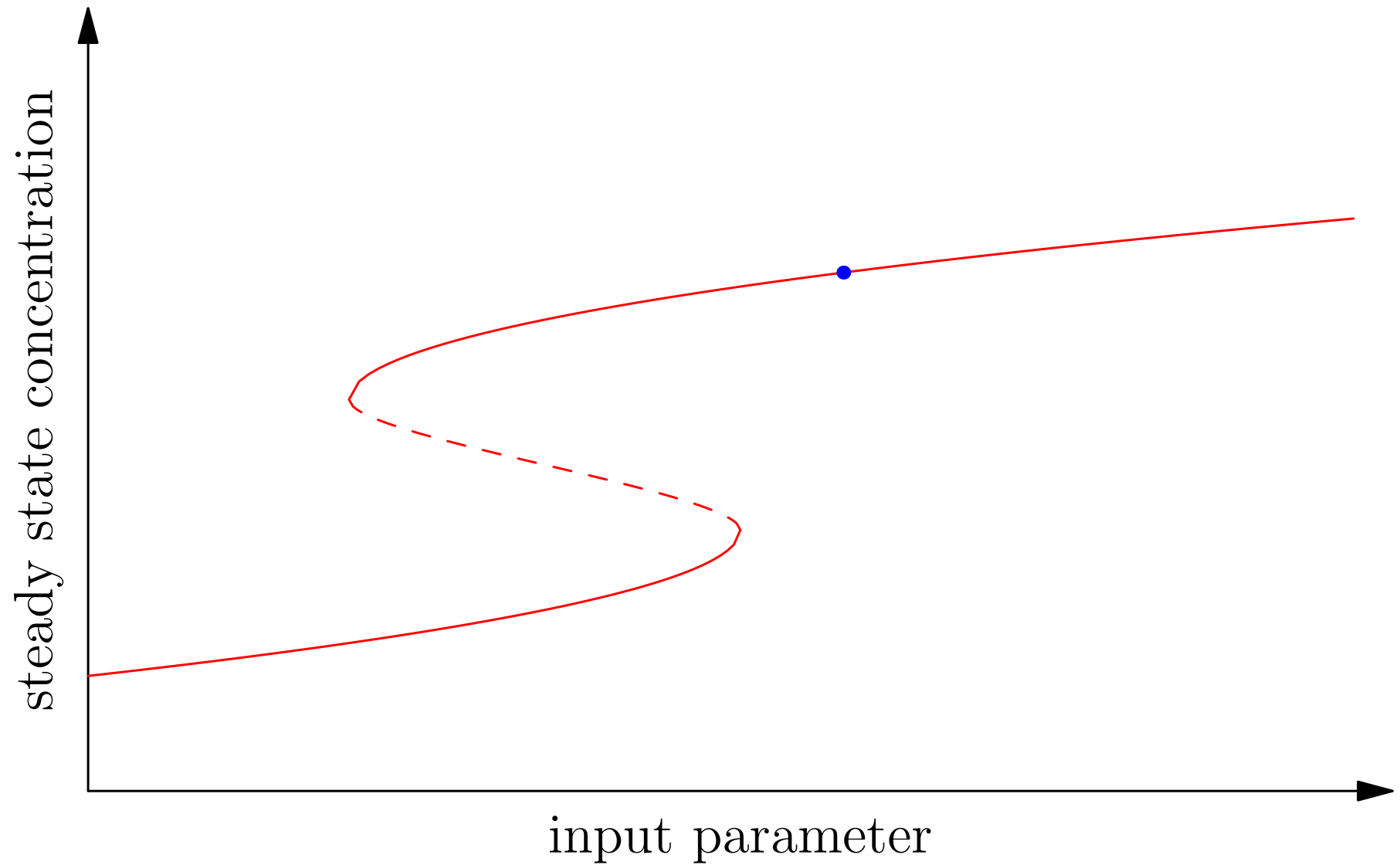
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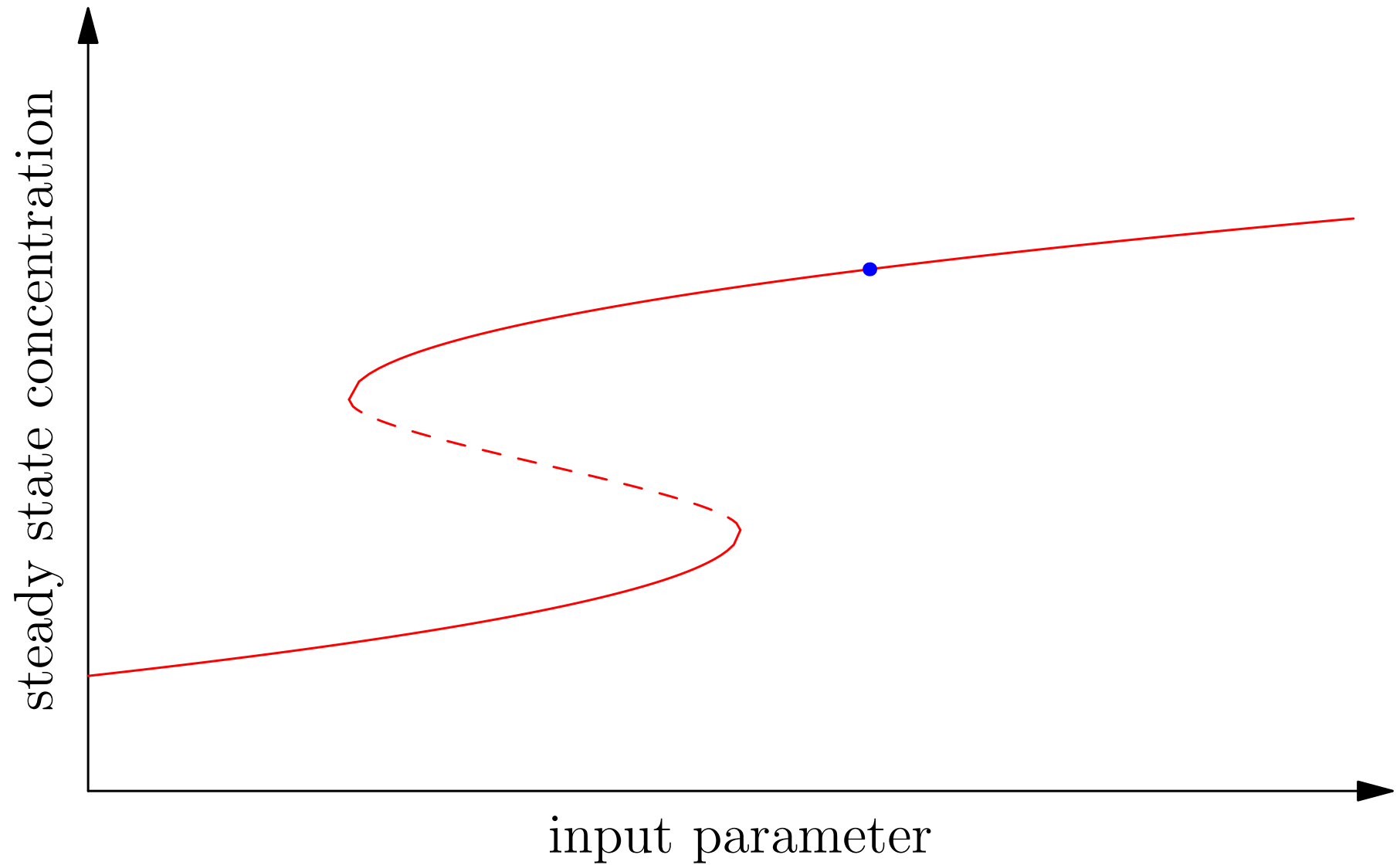
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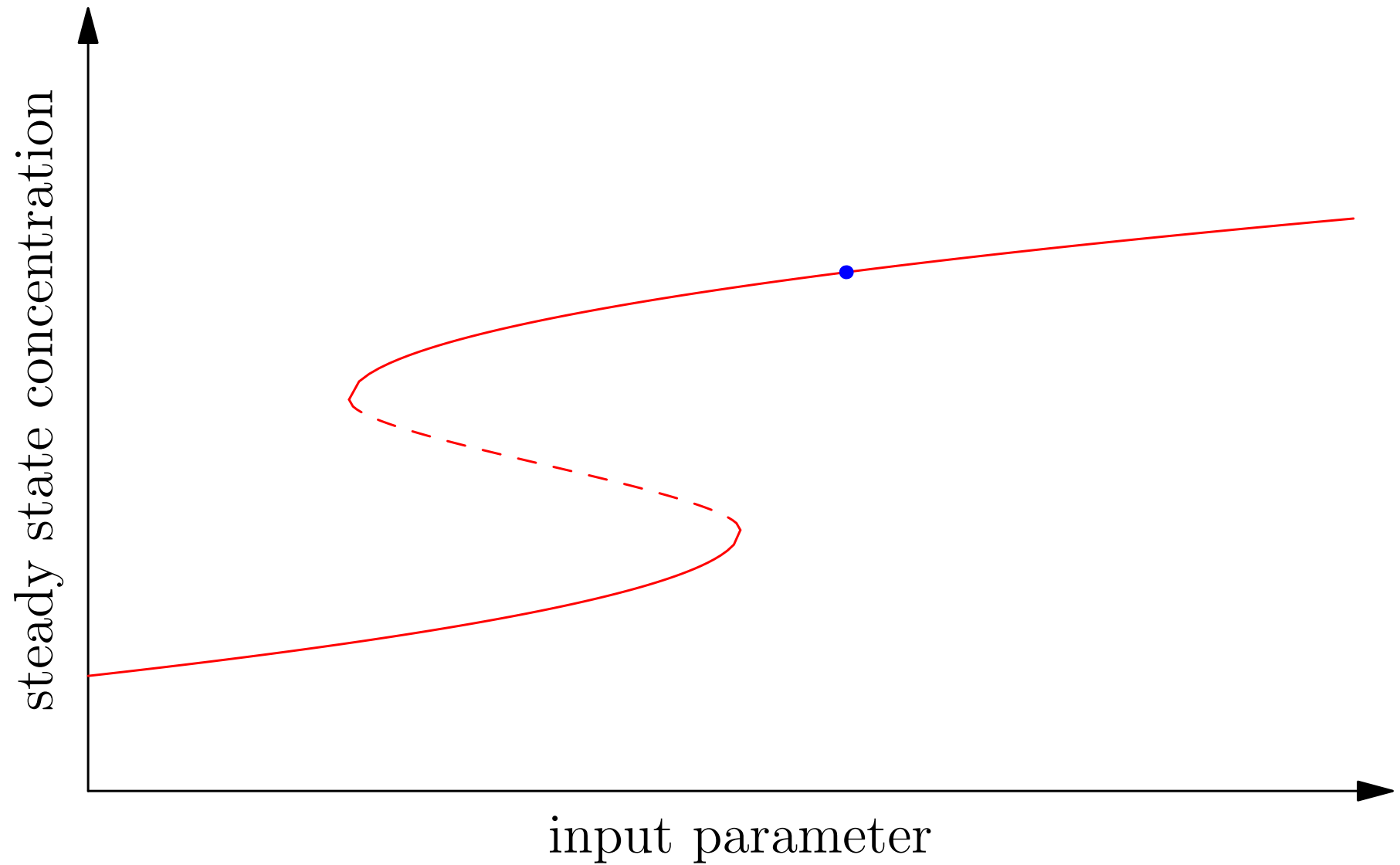
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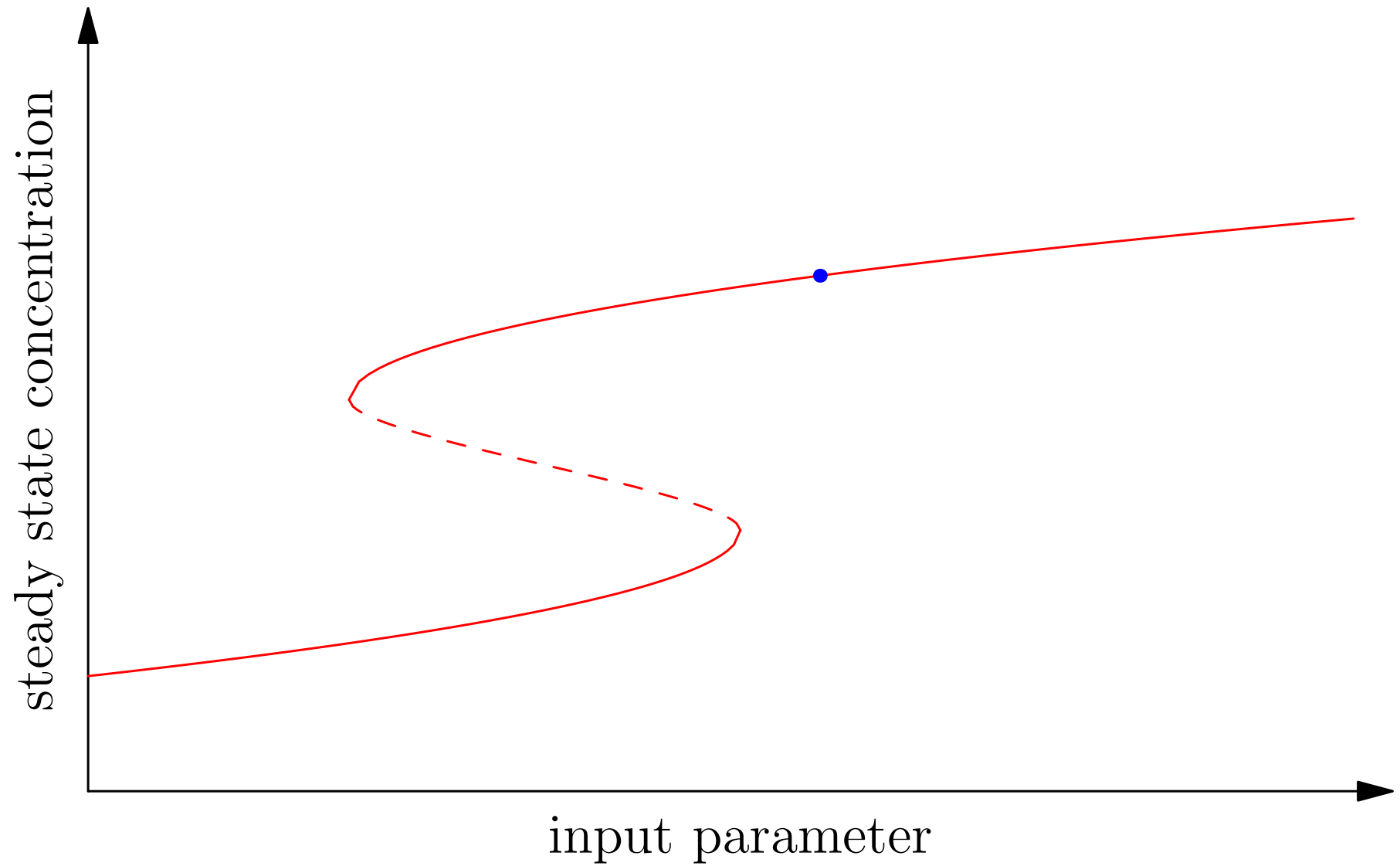
Hysteresis



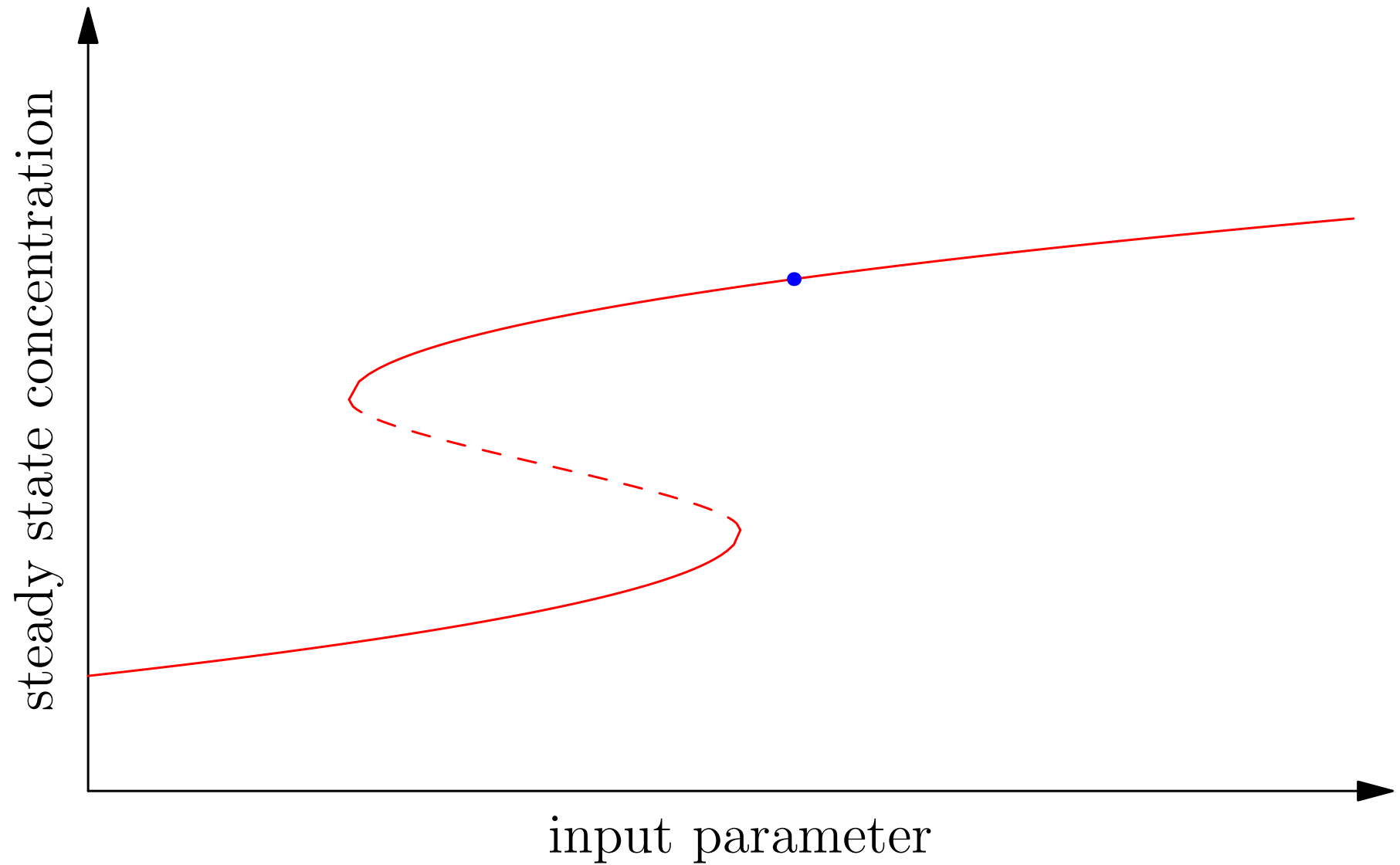
Hysteresis



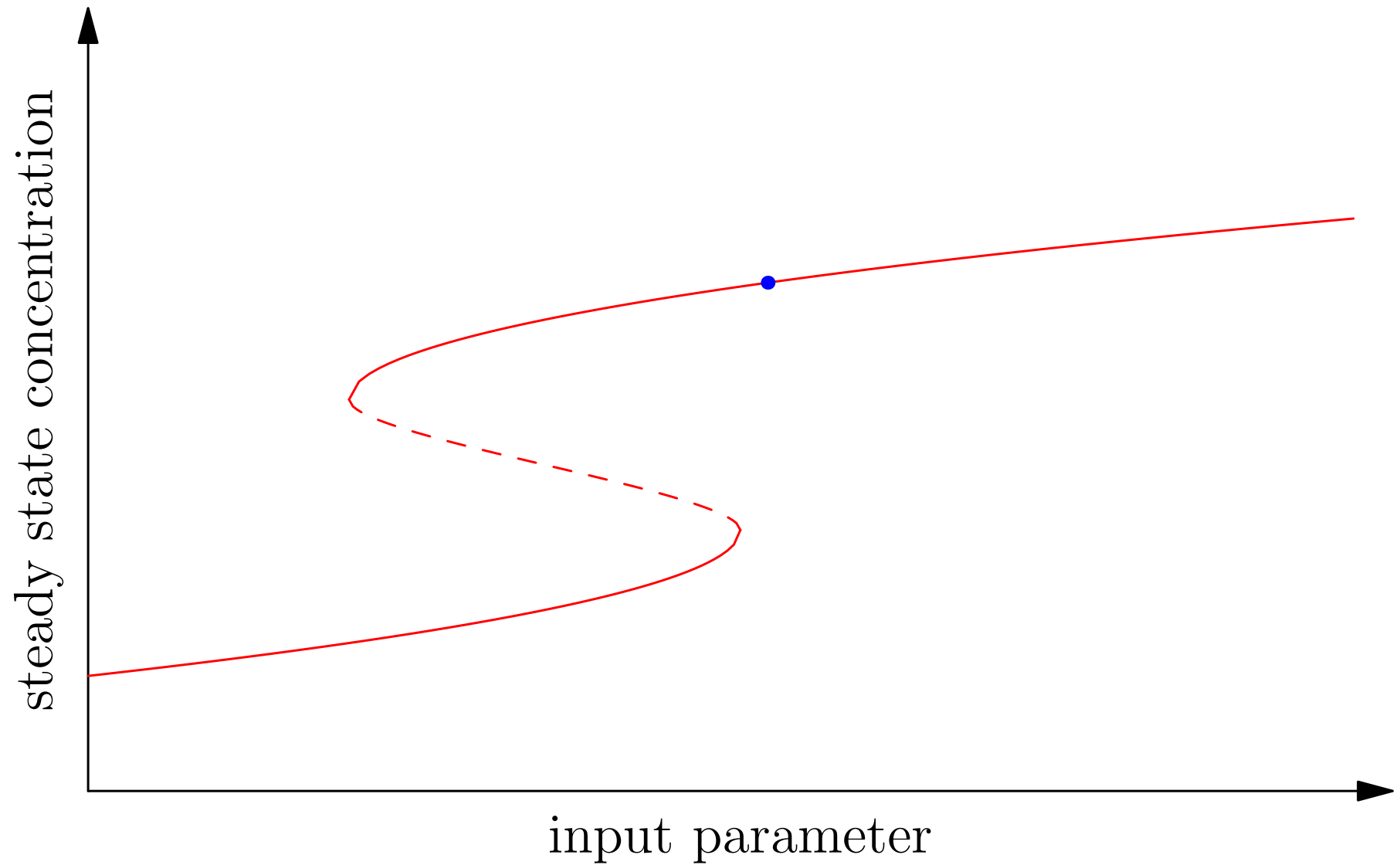
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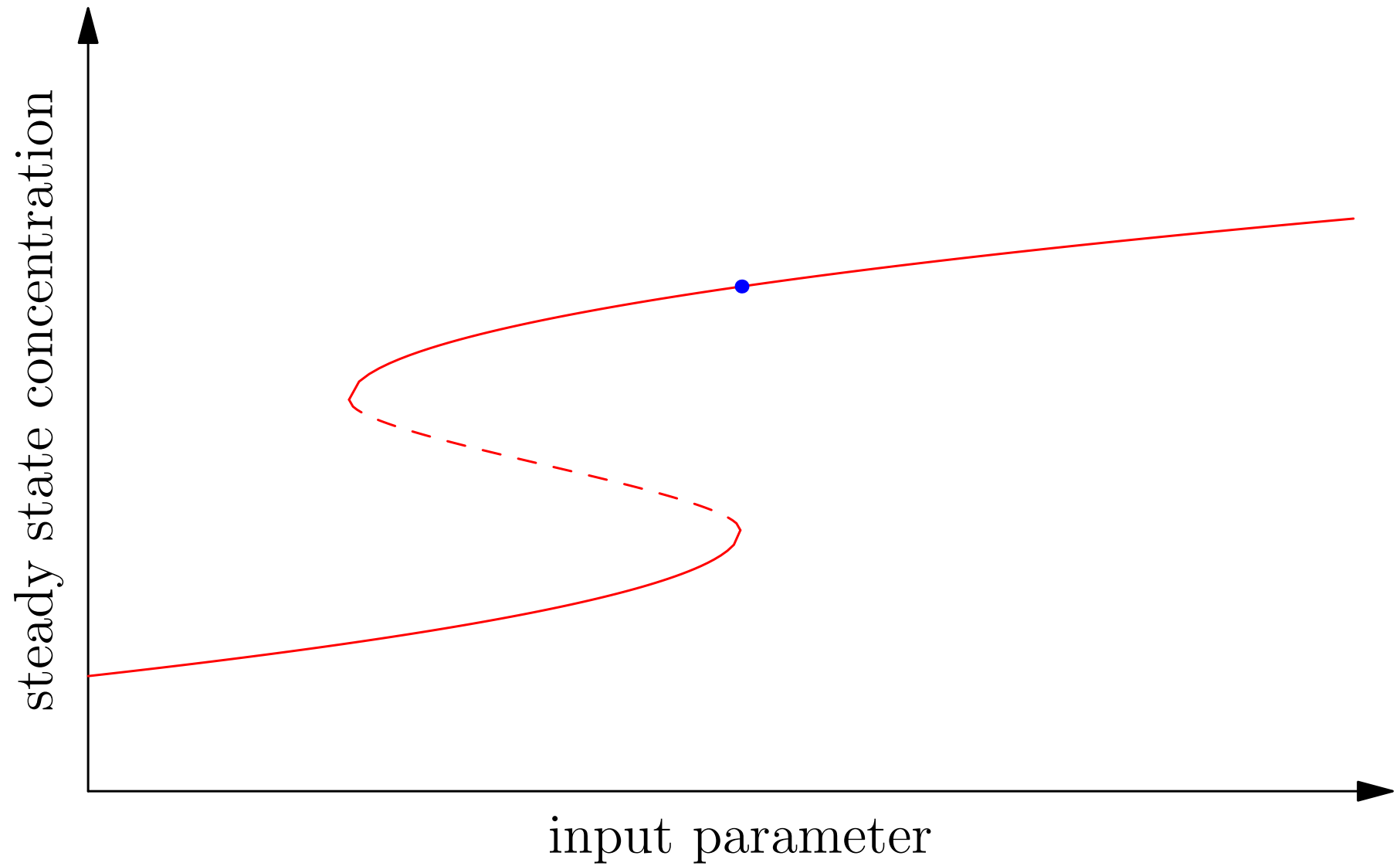
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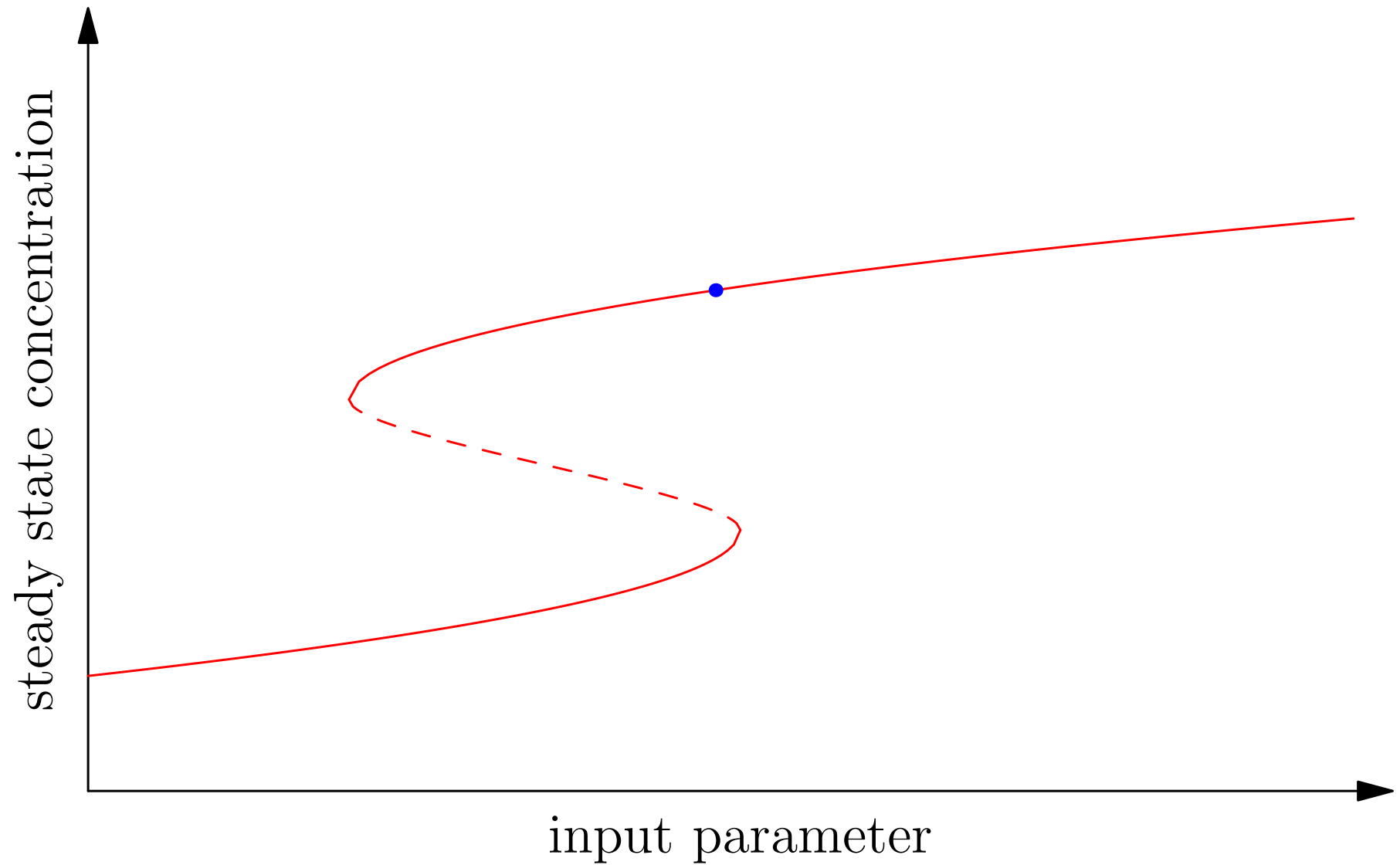
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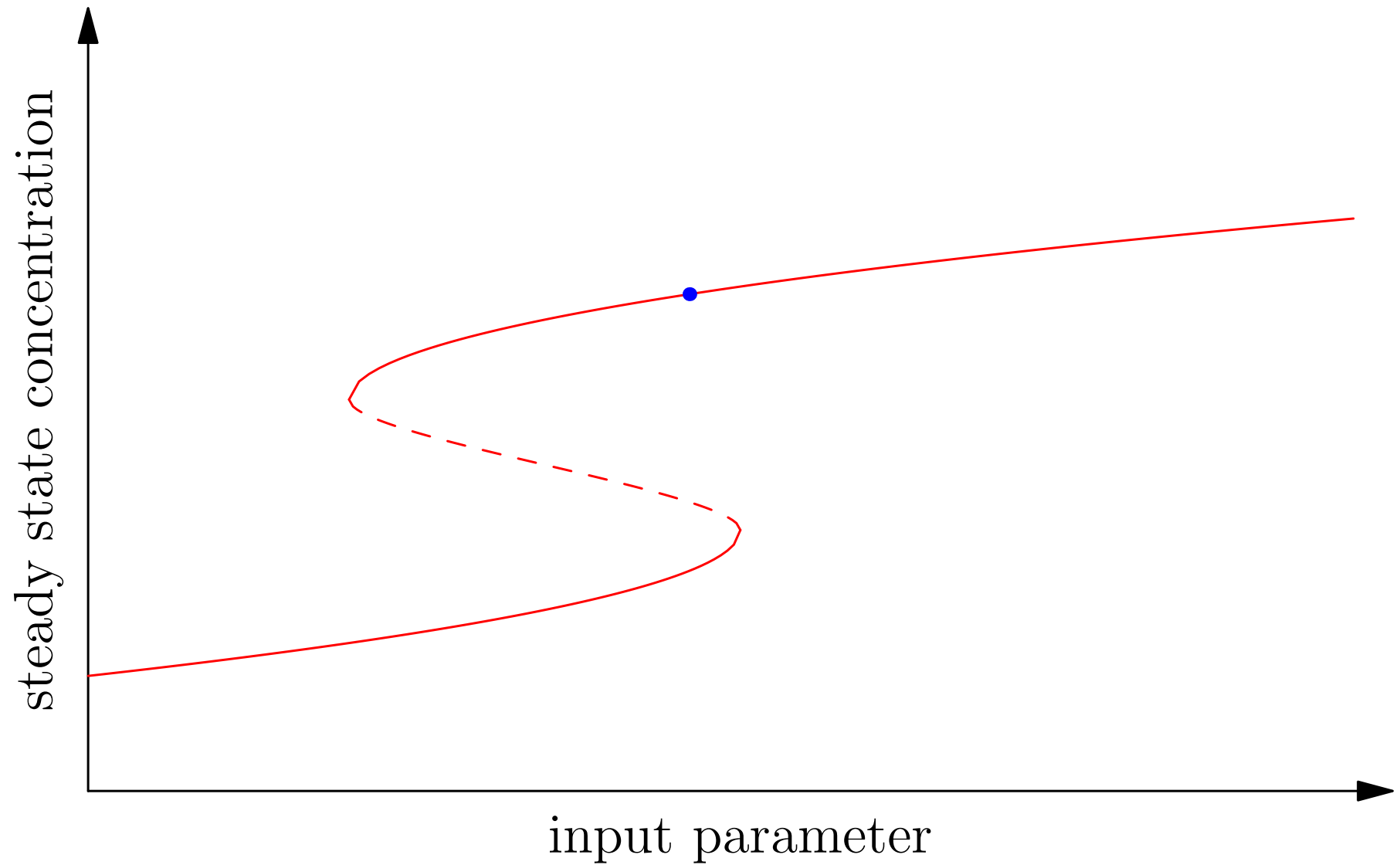
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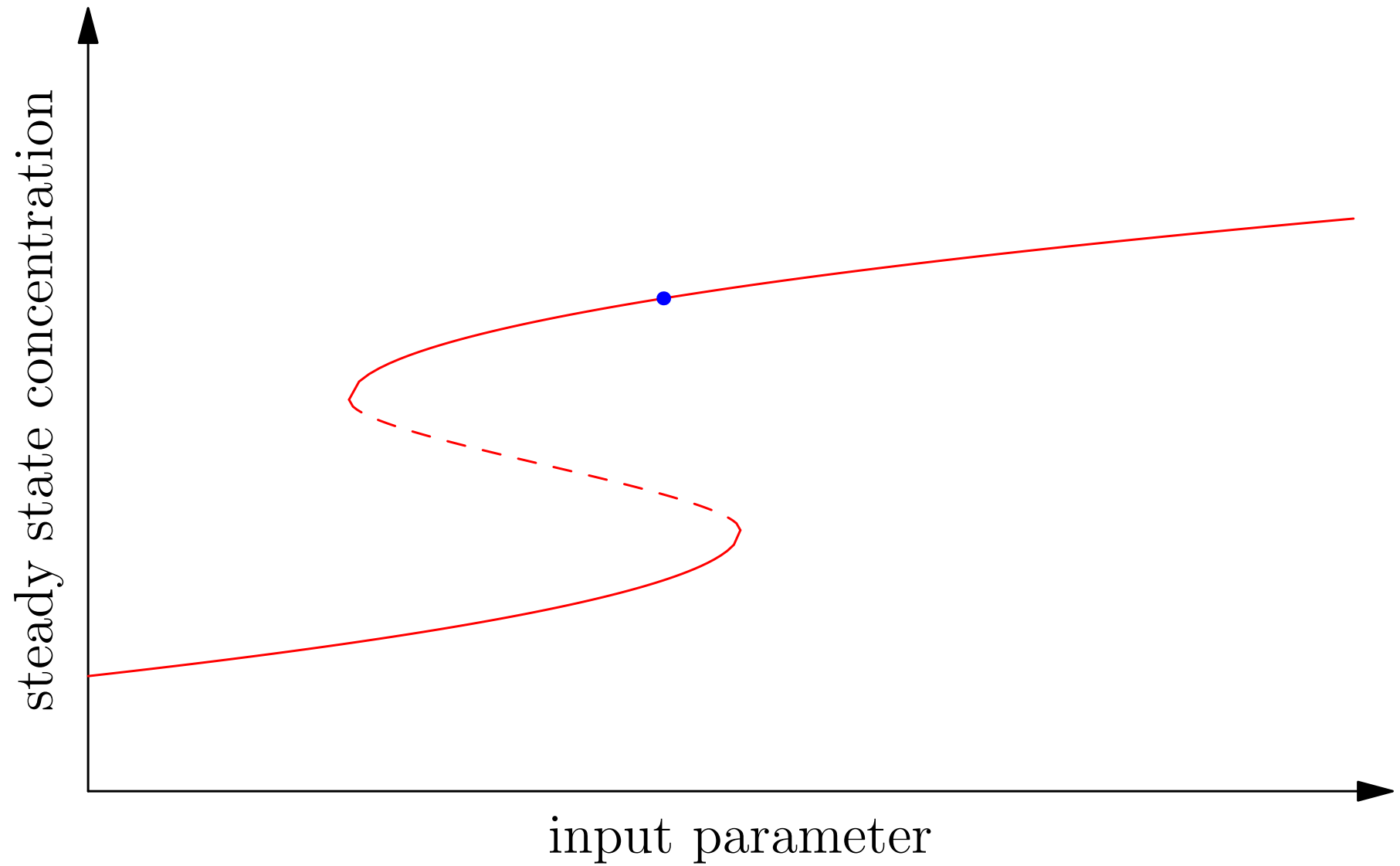
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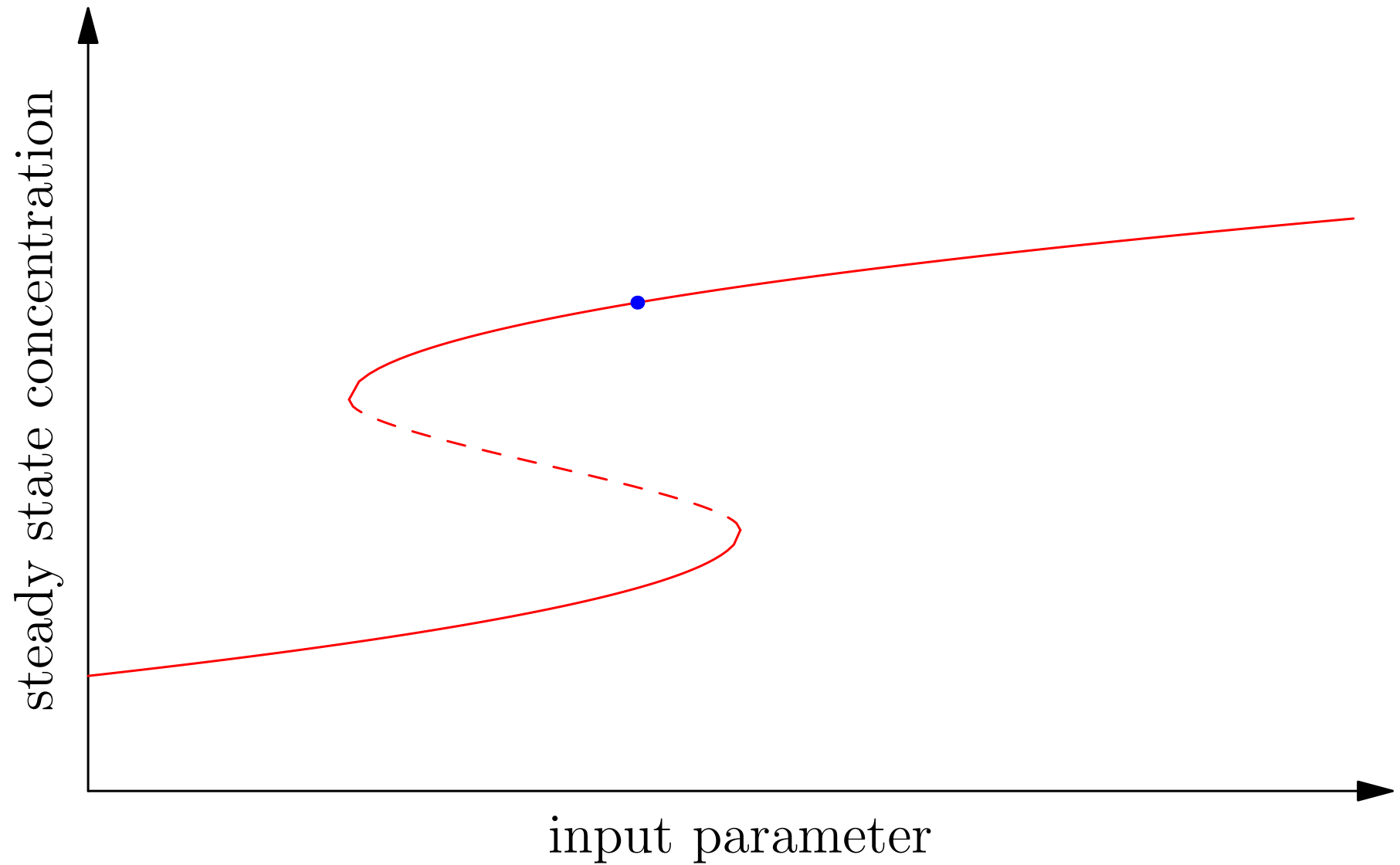
Hysteresis



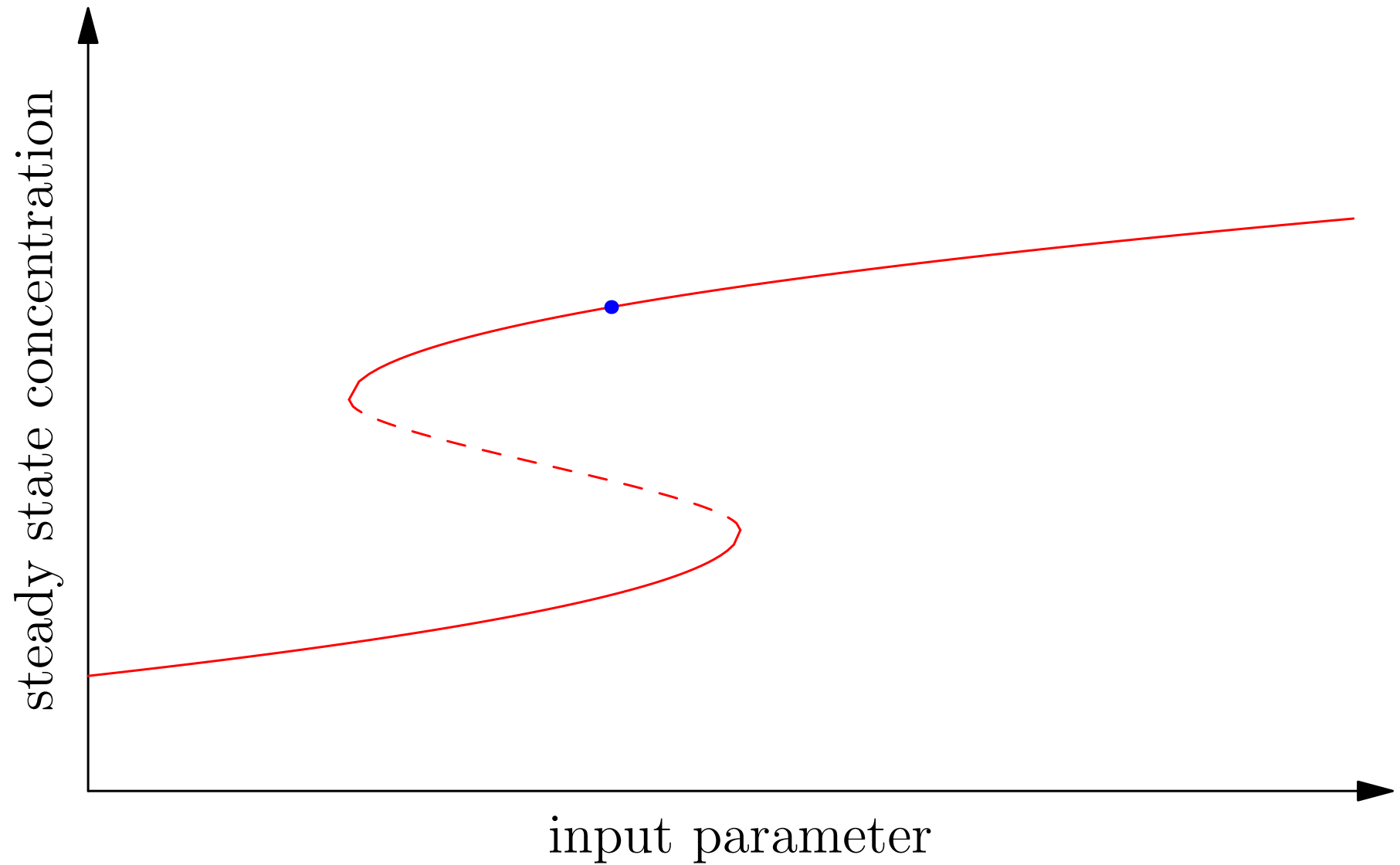
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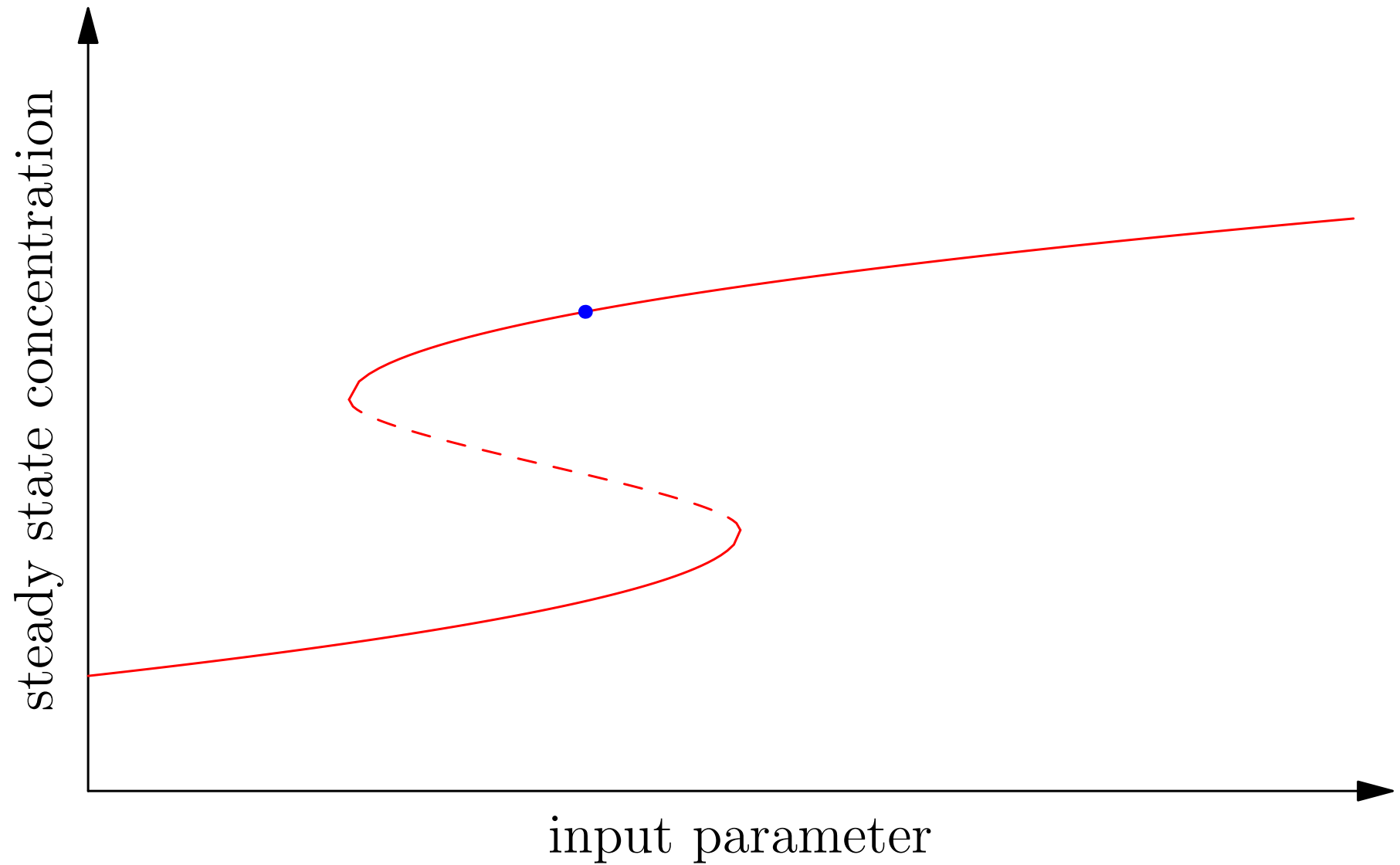
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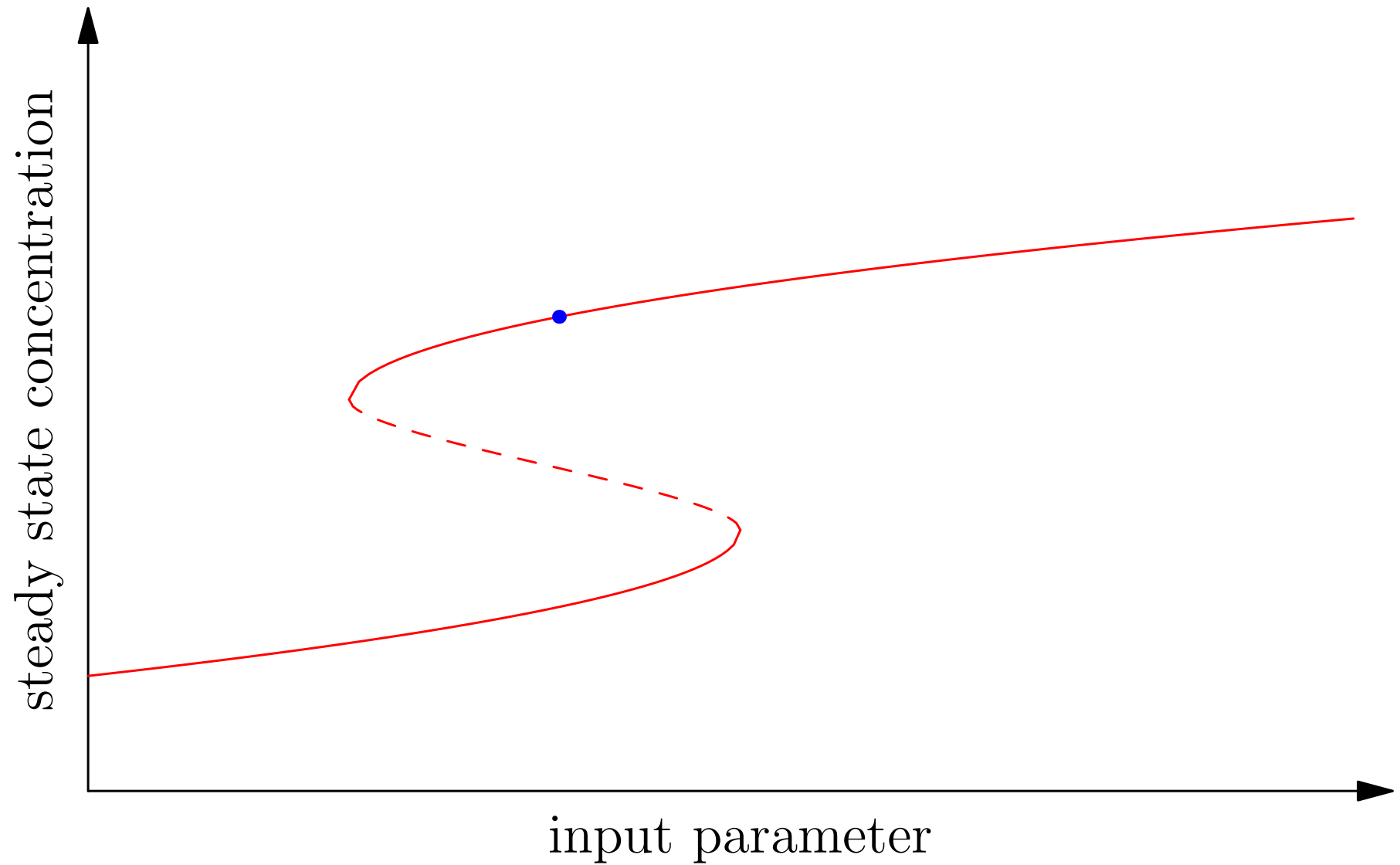
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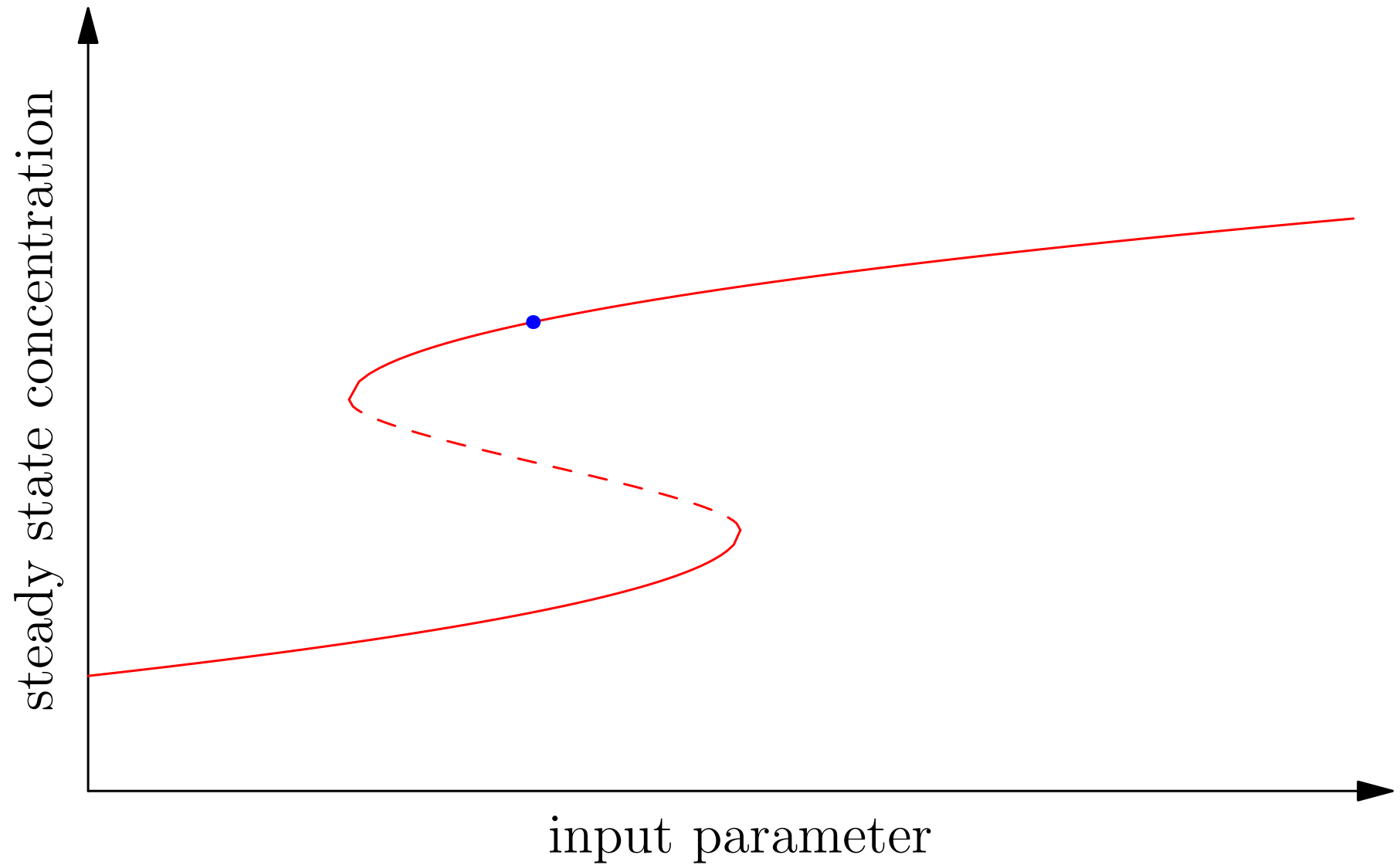
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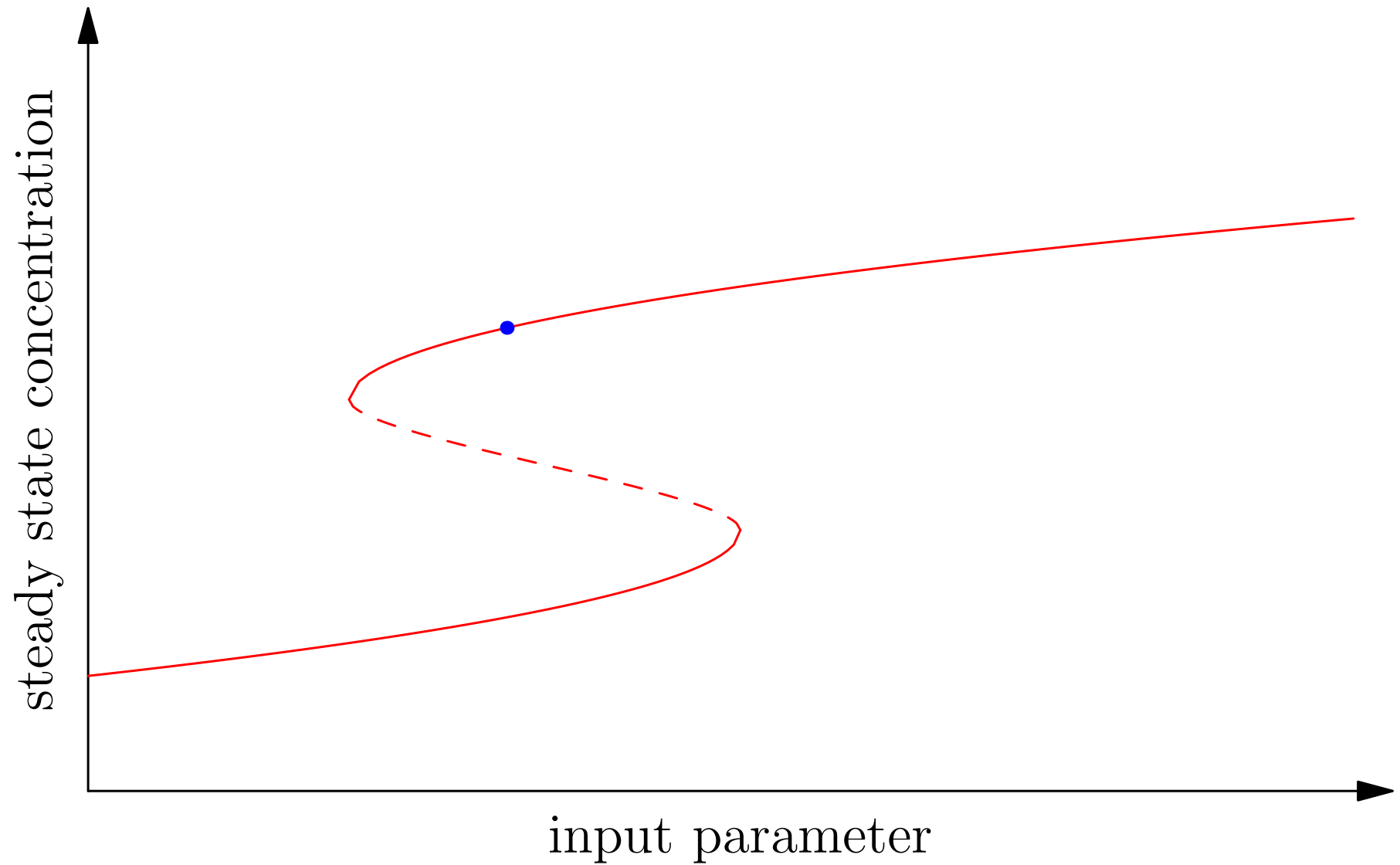
Hysteresis



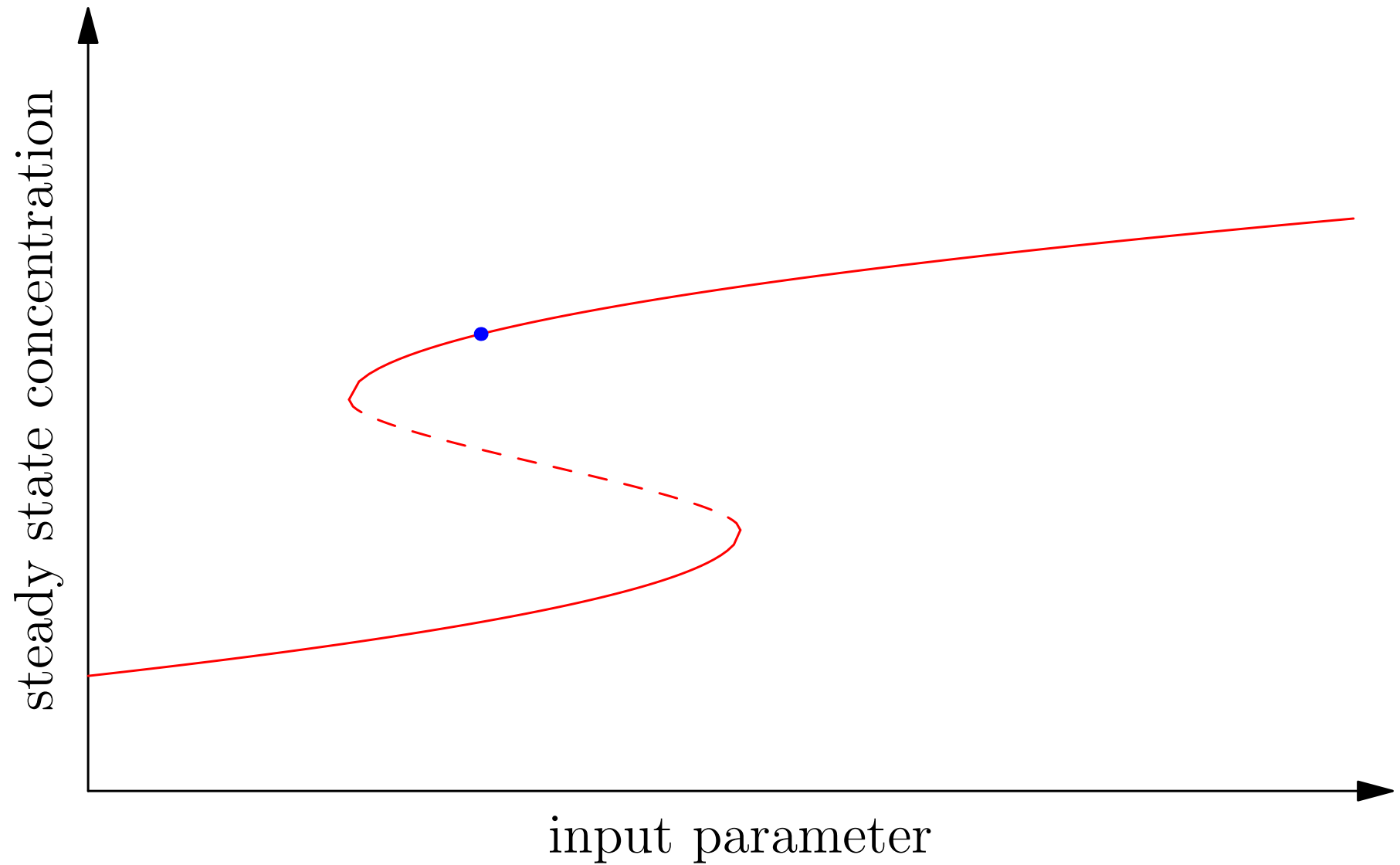
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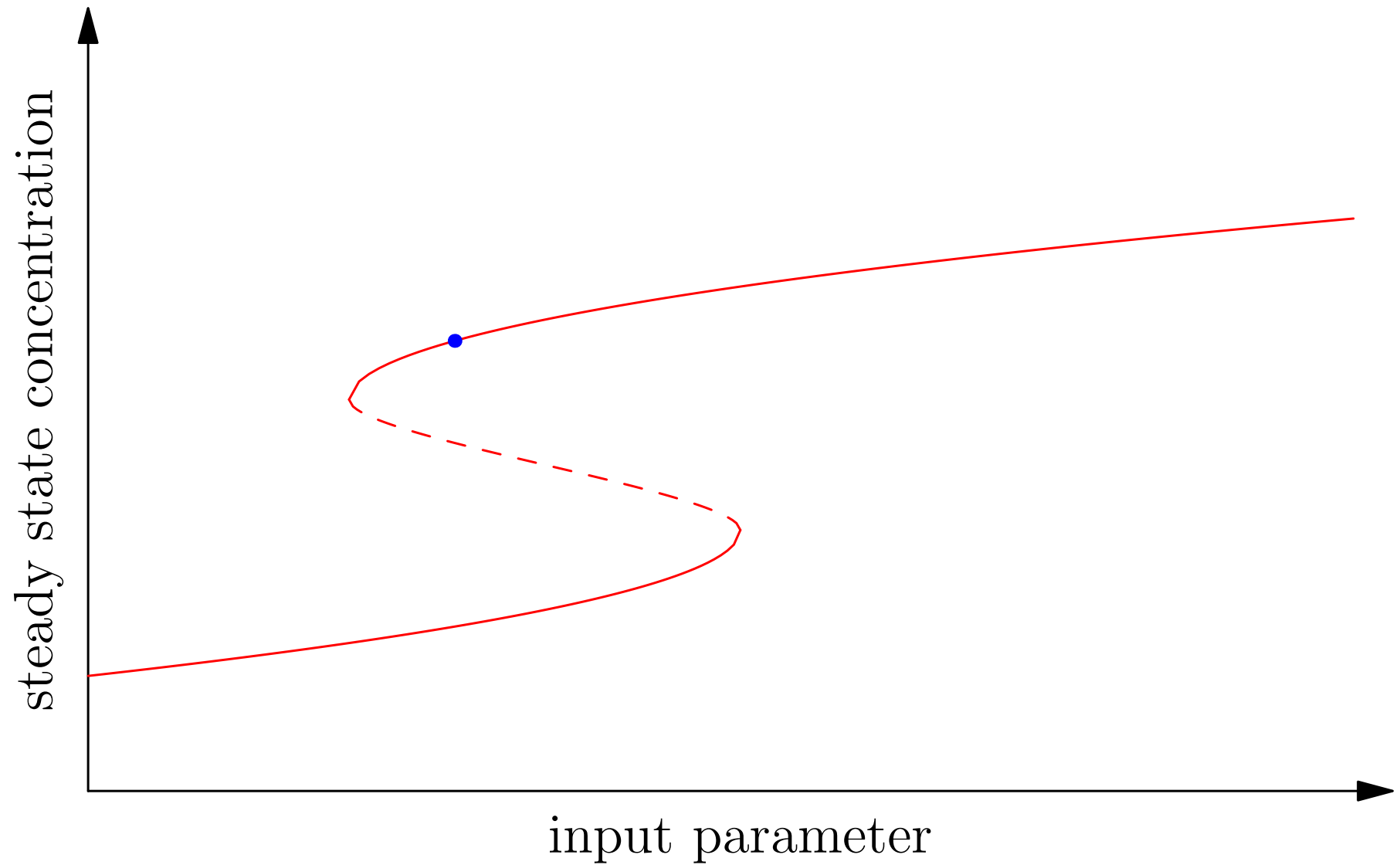
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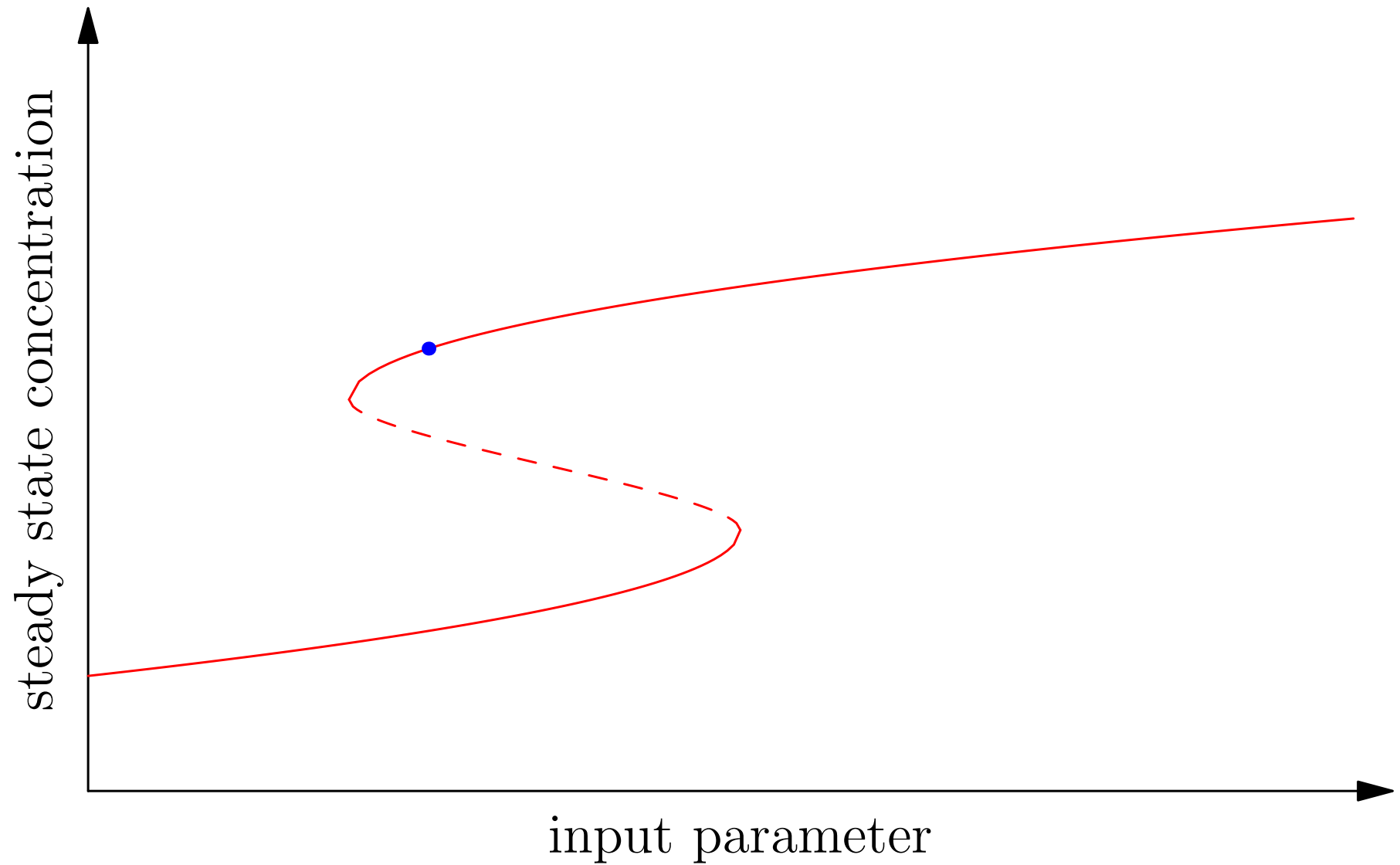
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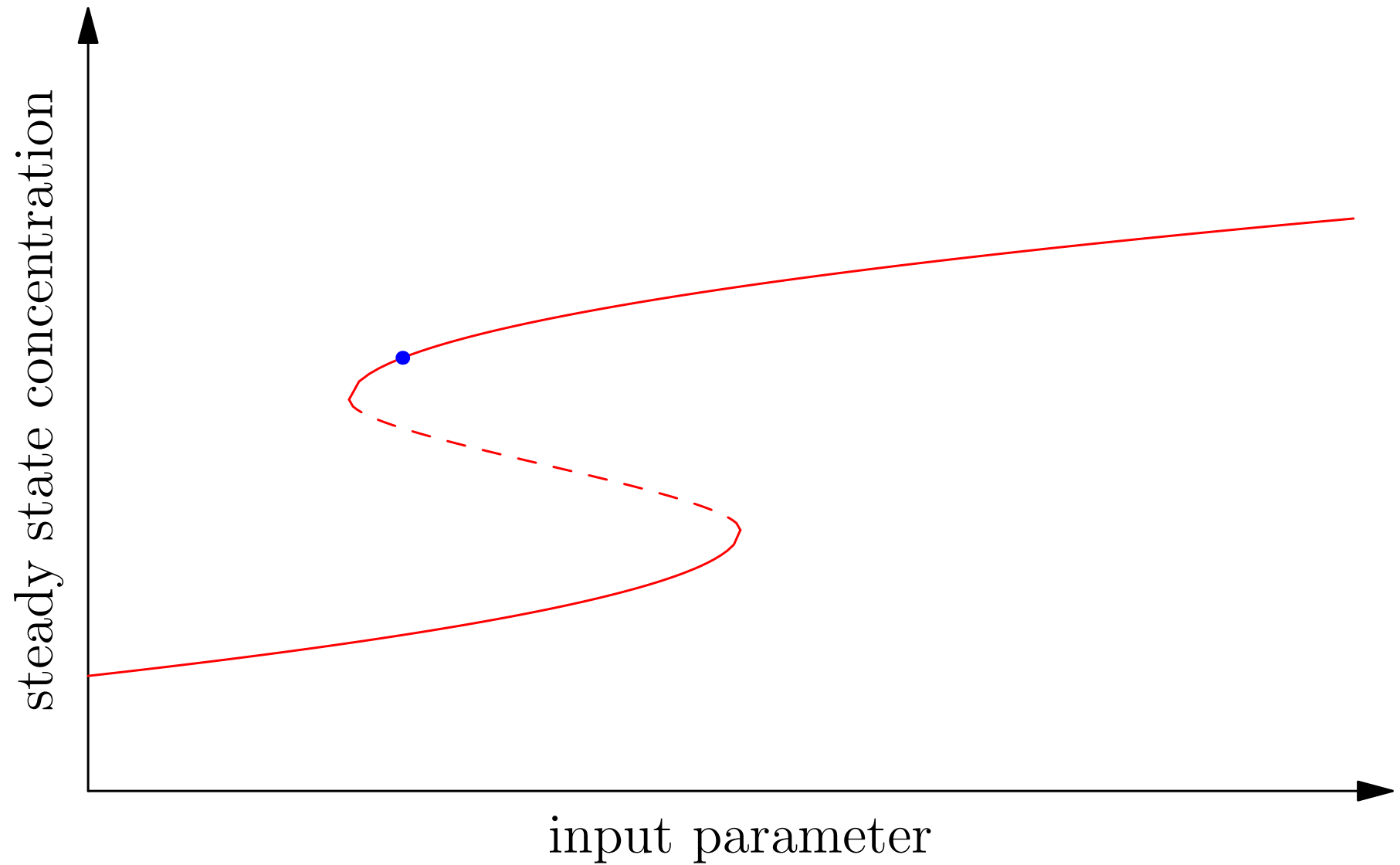
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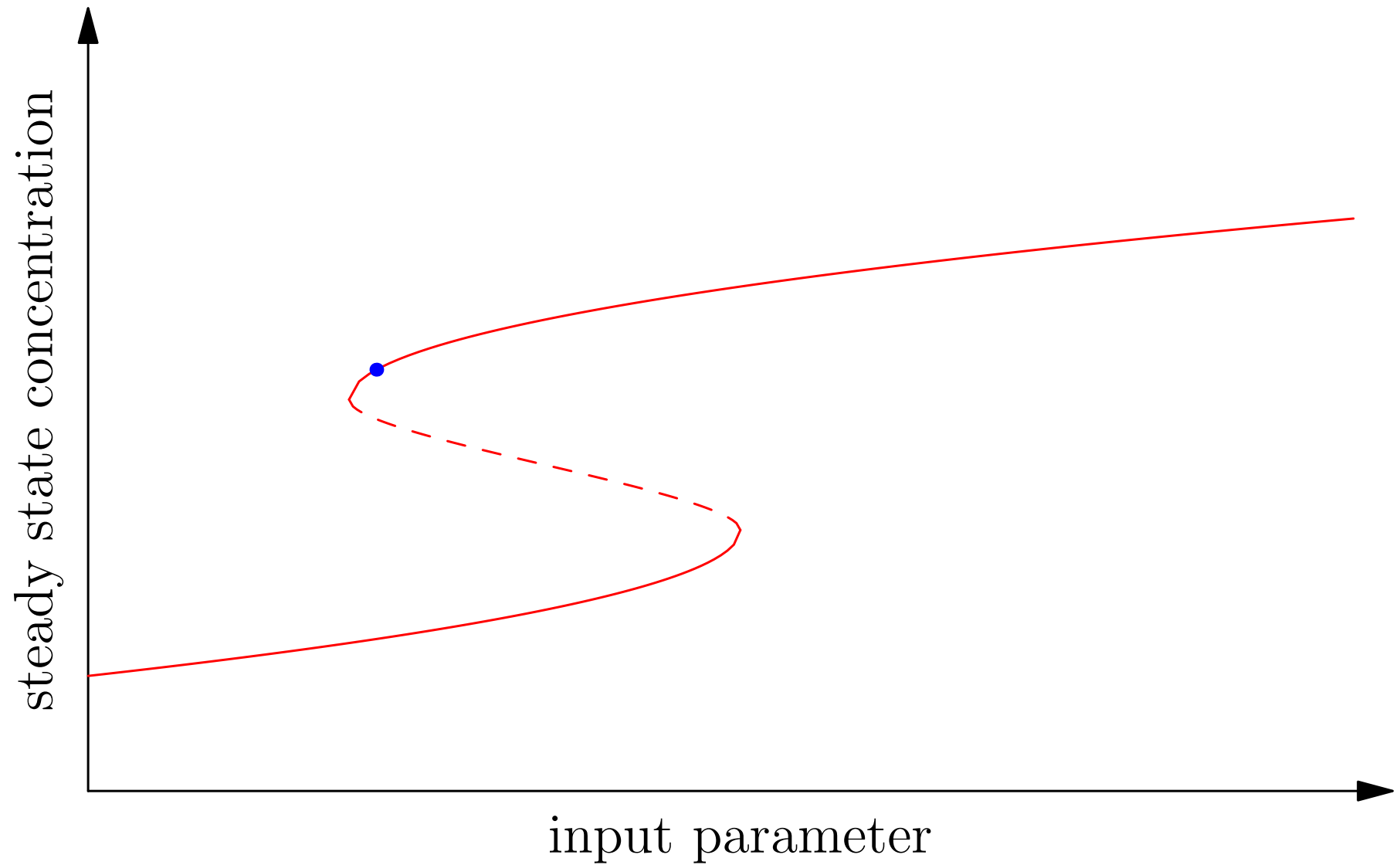
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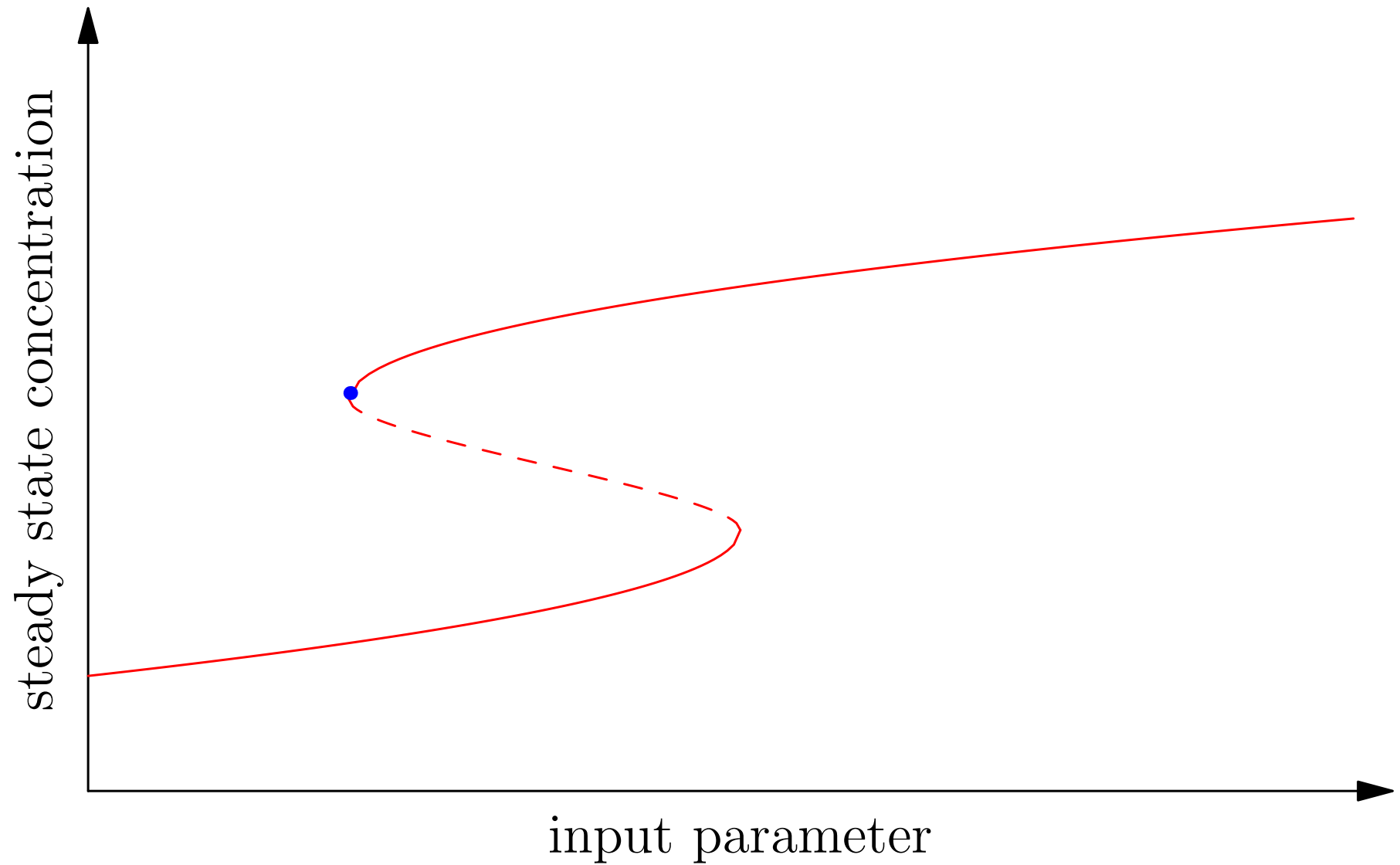
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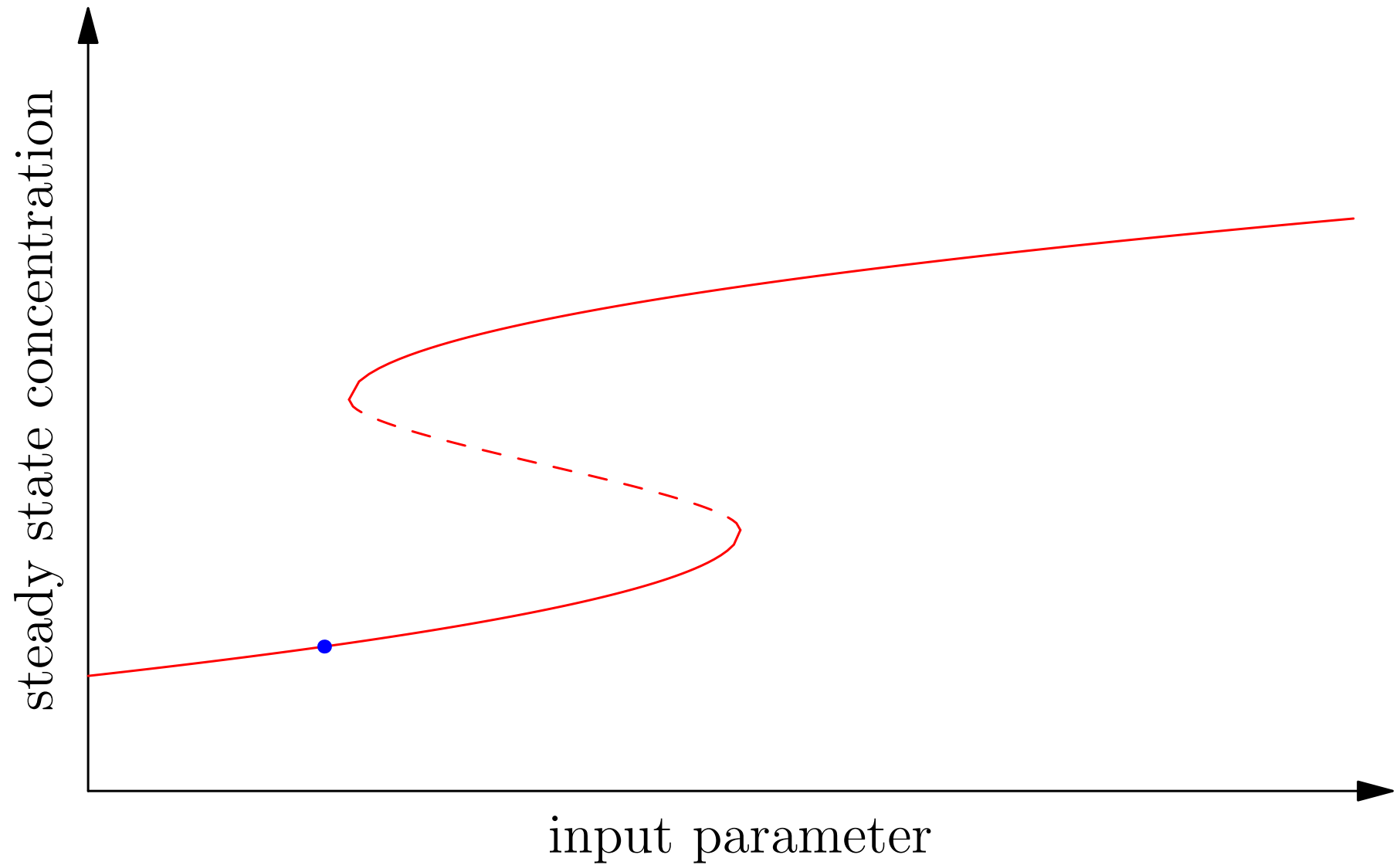
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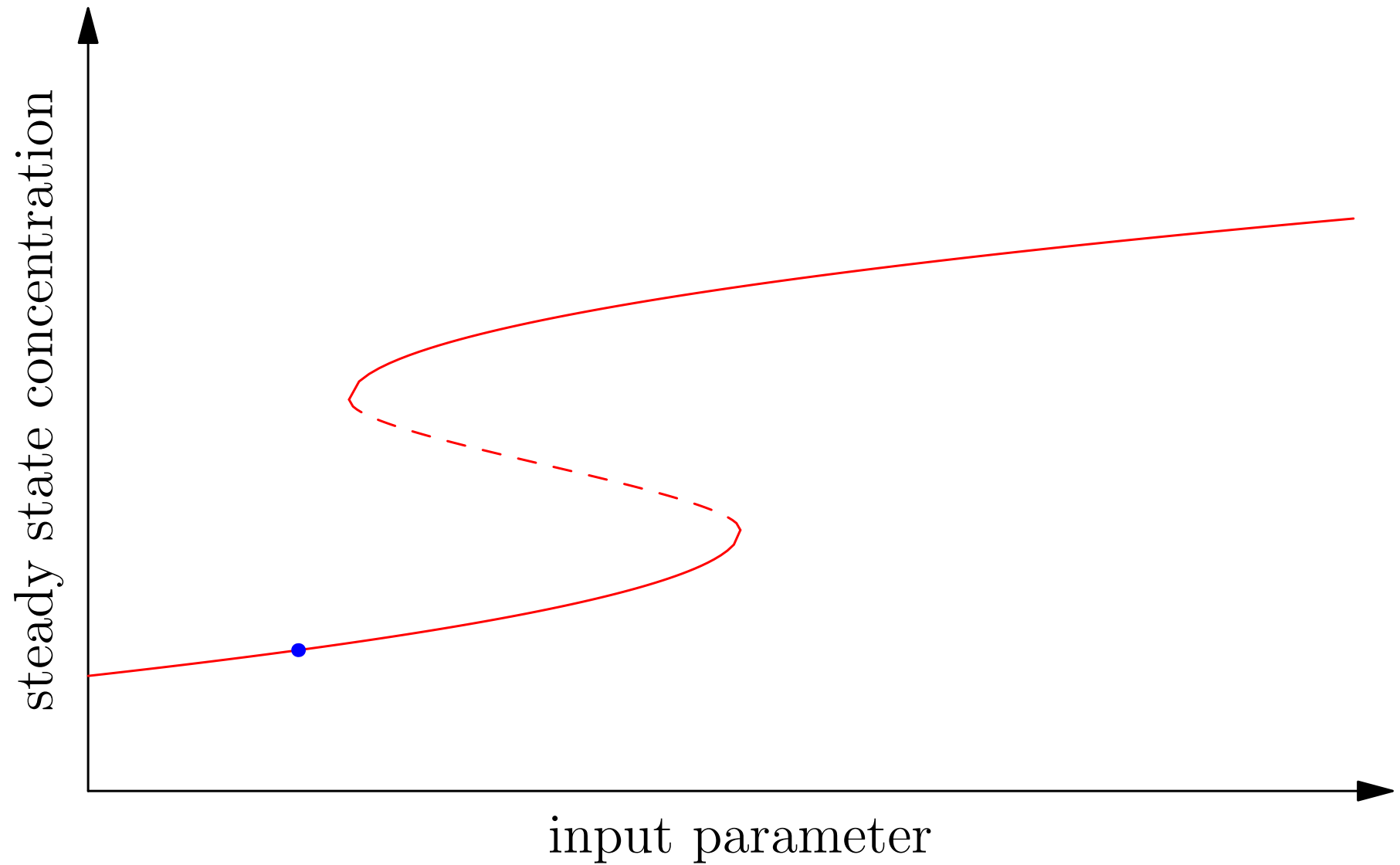
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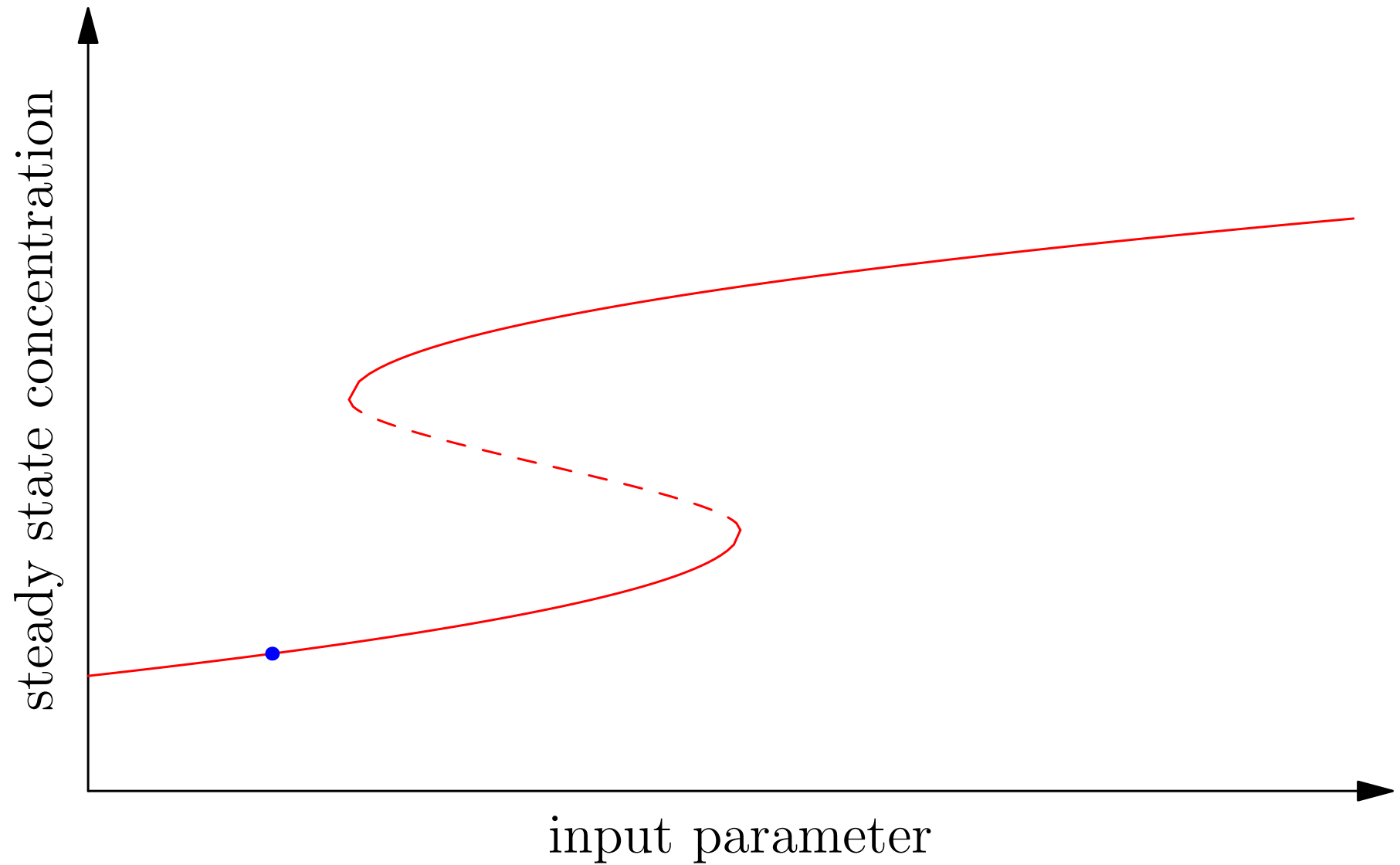
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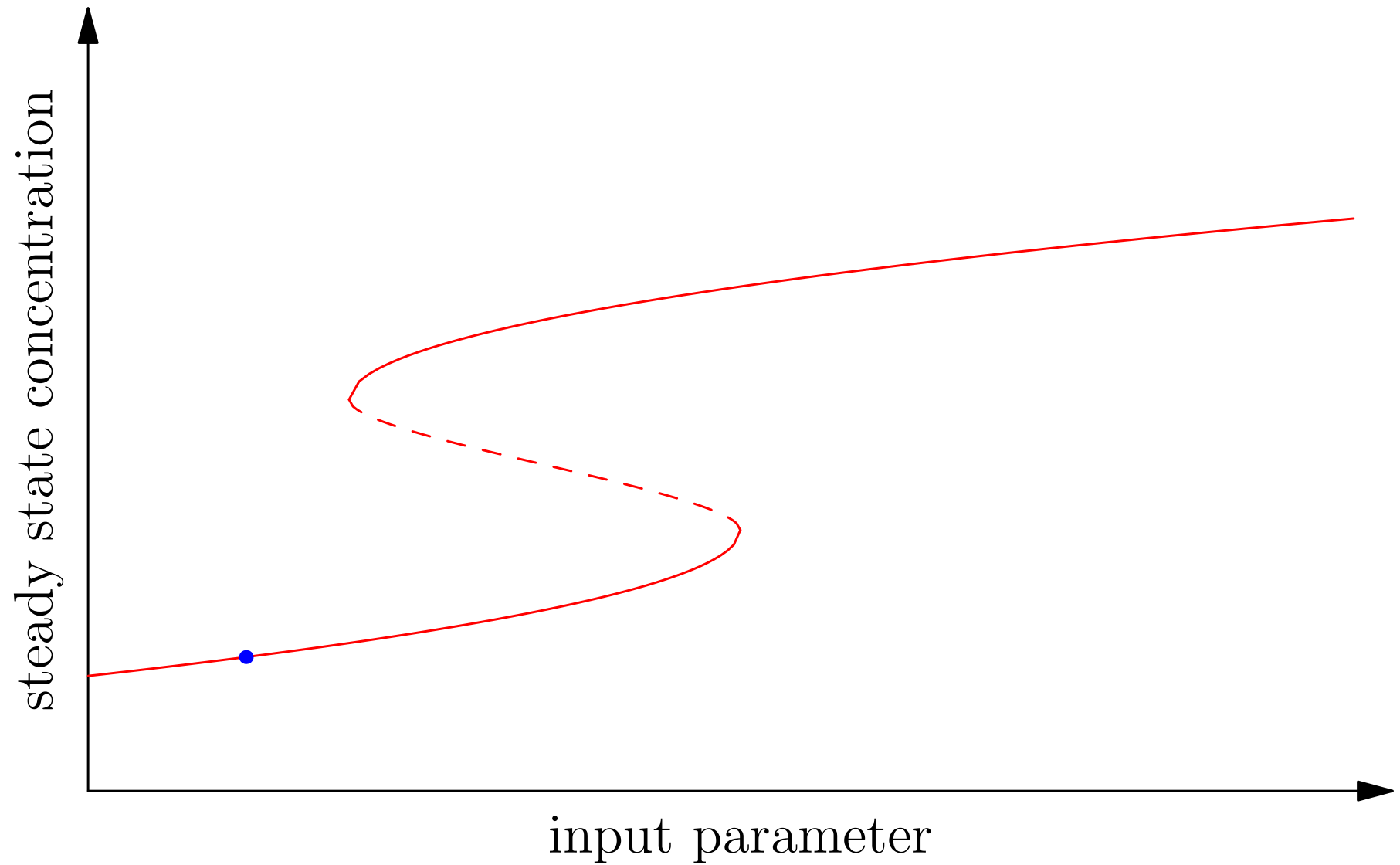
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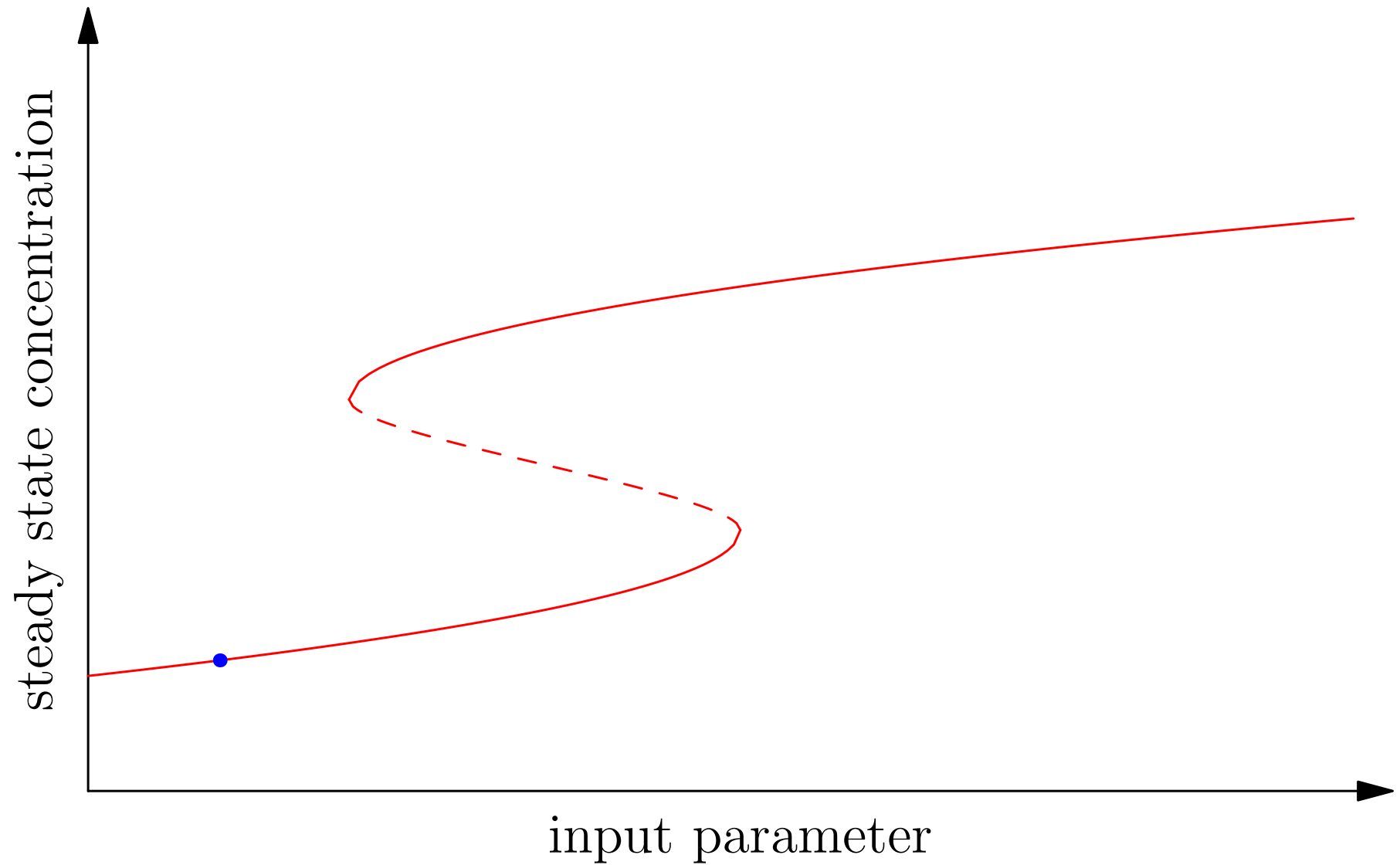
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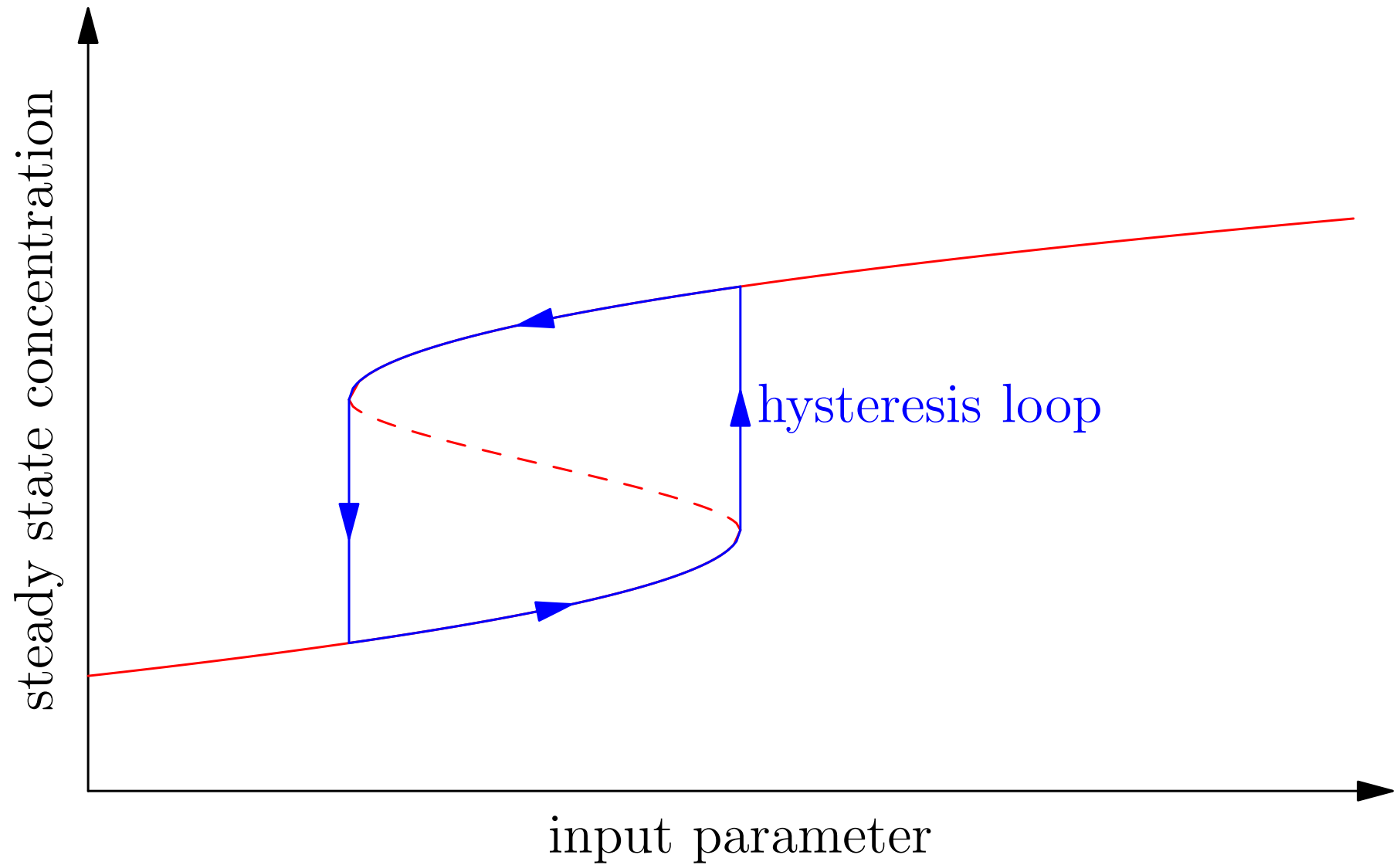
Hysteresis



Hysteresis



Hysteresis



Autocatalysis

- Recall the autocatalytic system

$$\frac{dX_1(t)}{dt} = k_0 - k_1 \left(1 + \left(\frac{X_2(t)}{K} \right)^n \right) X_1(t)$$

$$\frac{dX_2(t)}{dt} = k_1 \left(1 + \left(\frac{X_2(t)}{K} \right)^n \right) X_1(t) - k_2 X_2(t)$$

- With $k_0 = 8$, $k_1 = 1$, $K = 1$, $k_2 = 5$
- We consider what happens when we vary n

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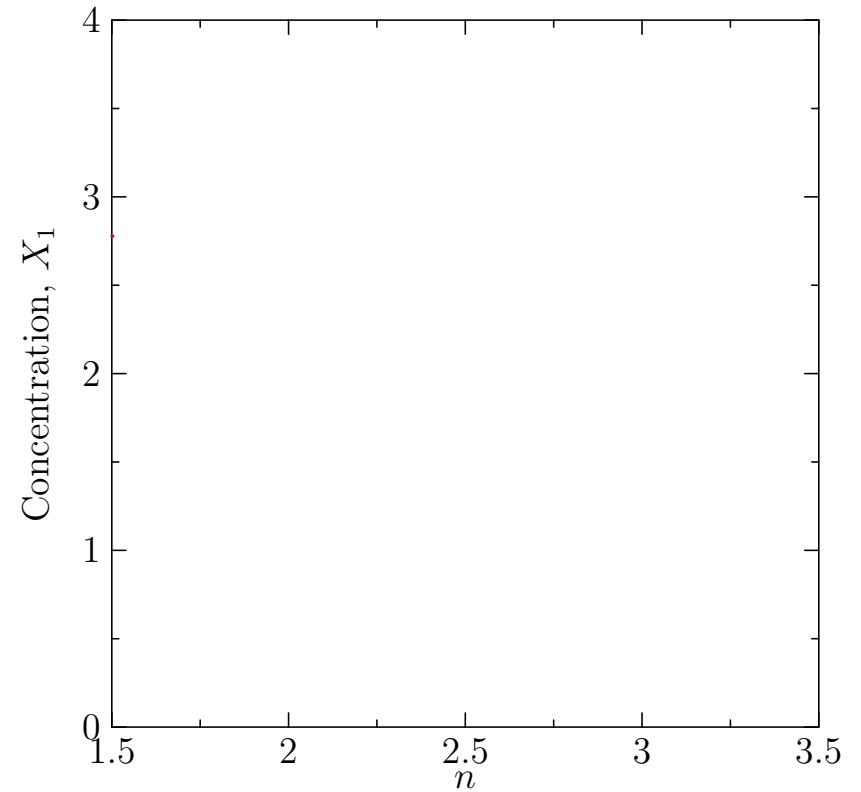
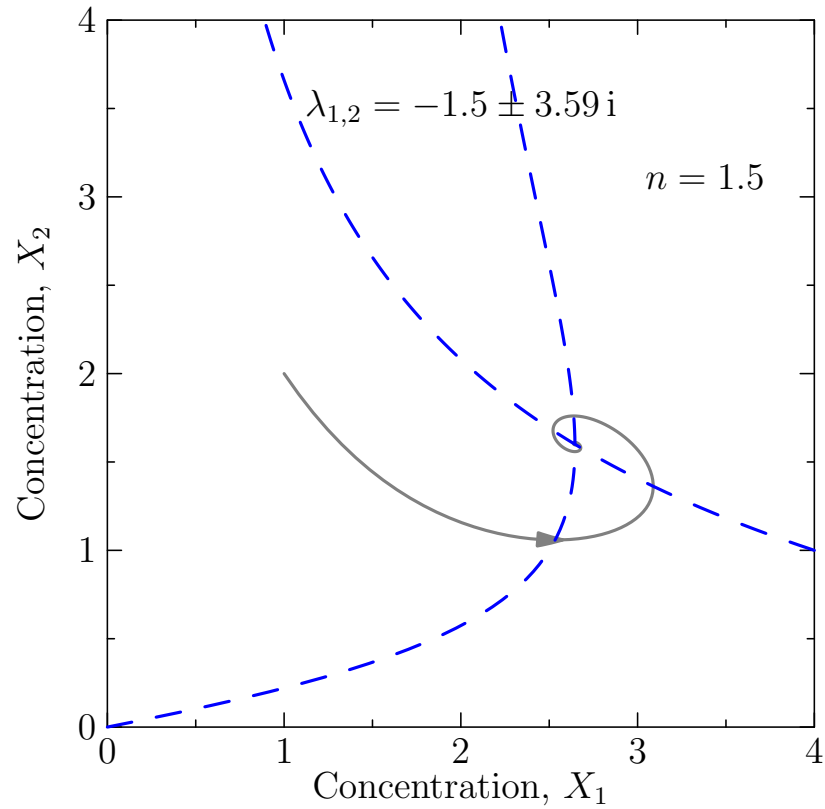
Autocatalysis

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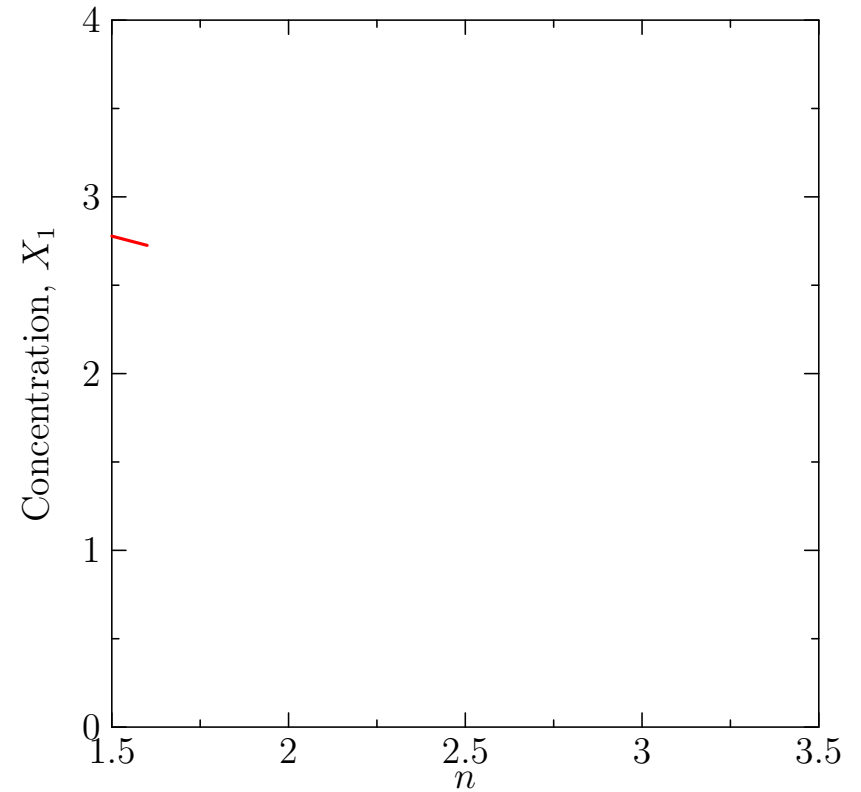
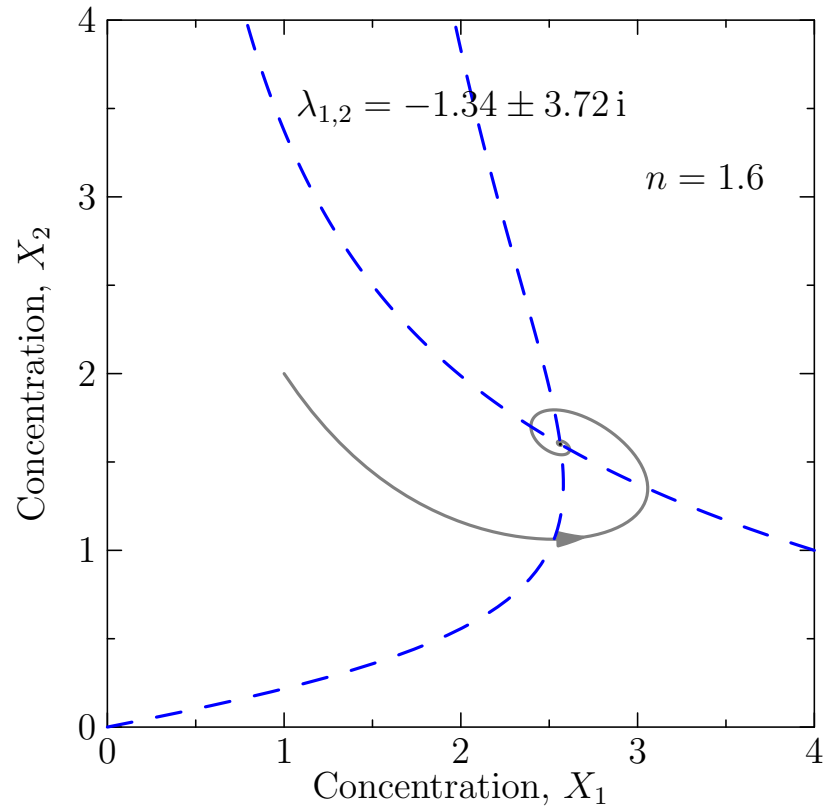
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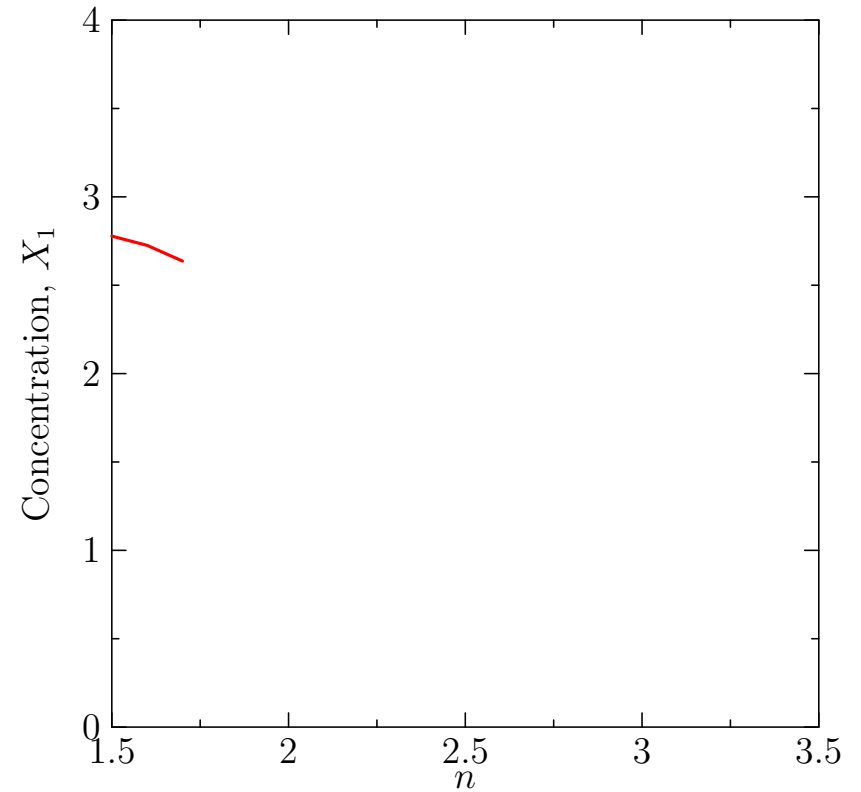
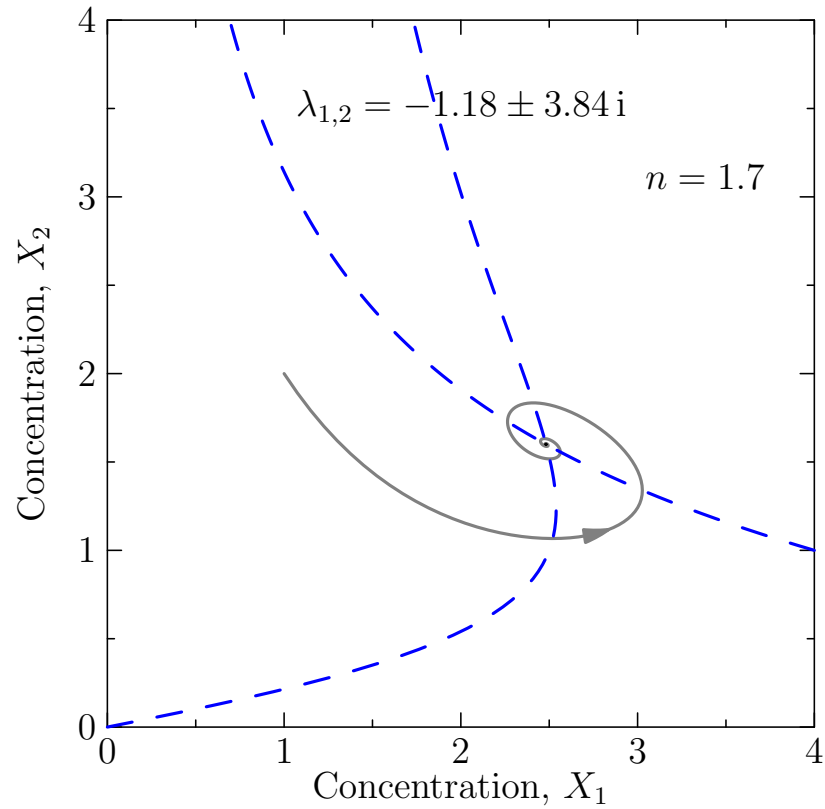
Hopf Bifurcation



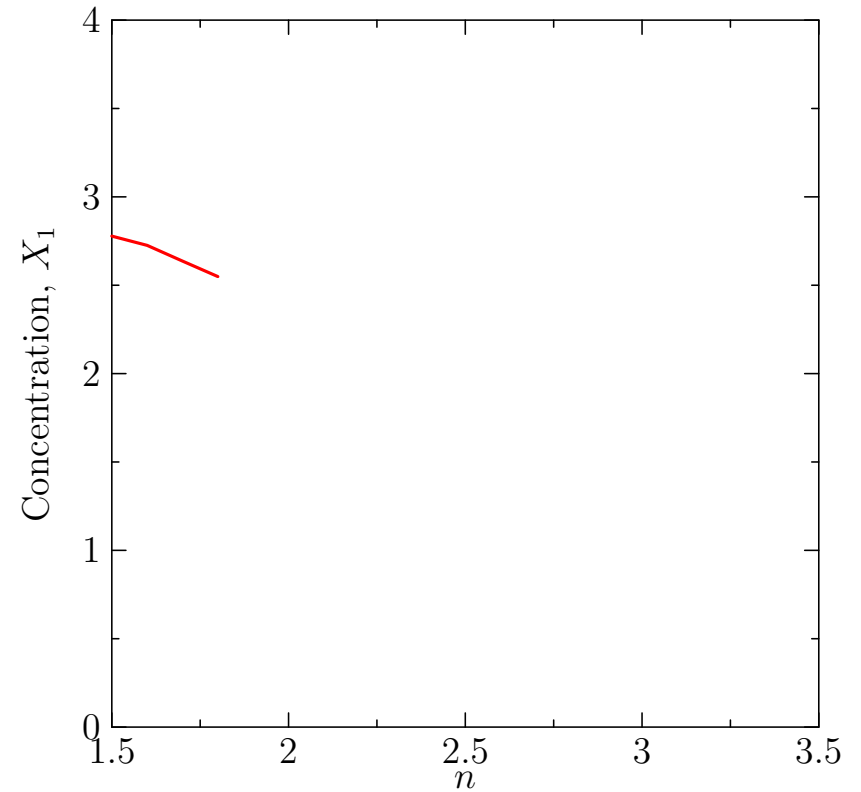
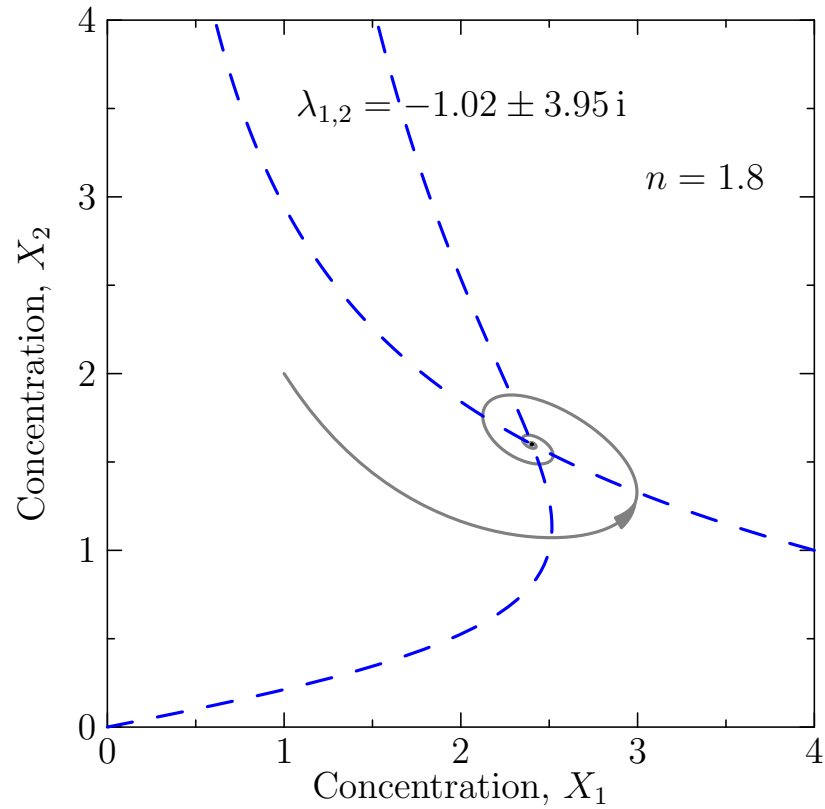
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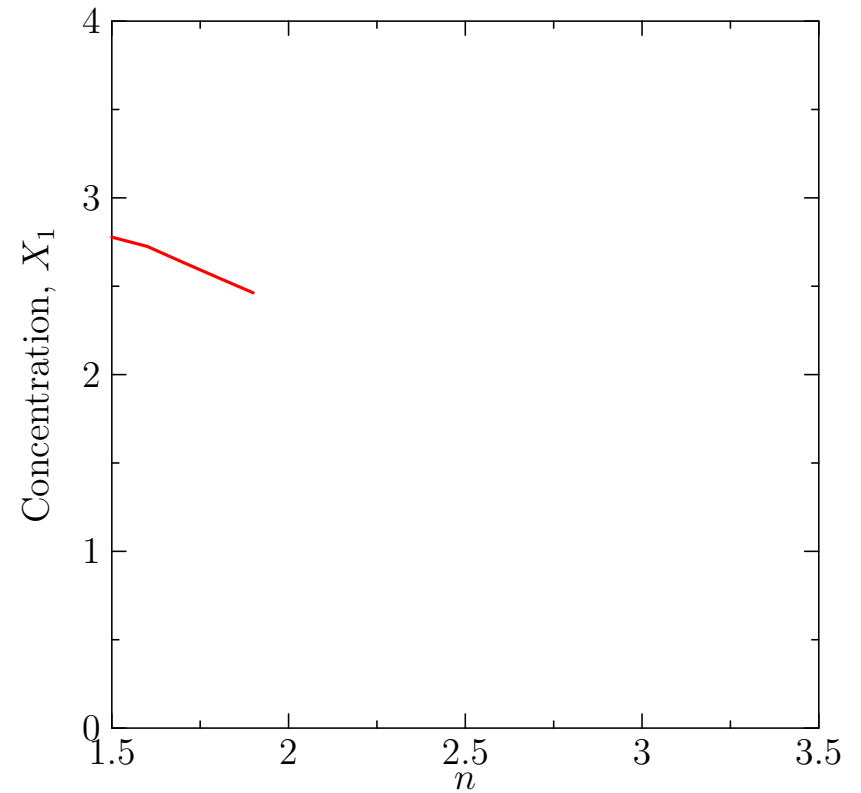
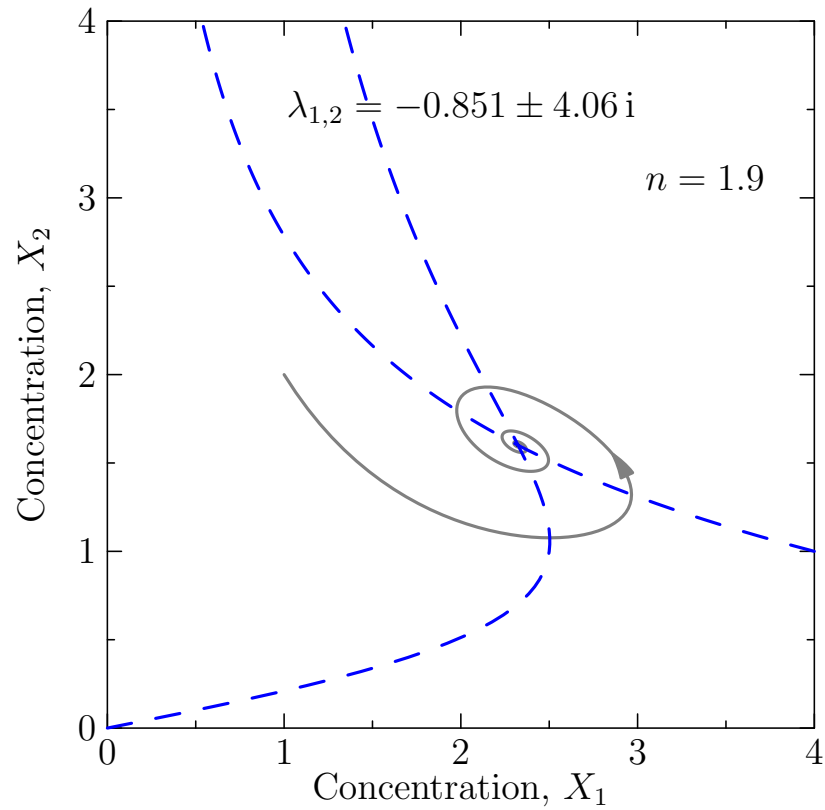
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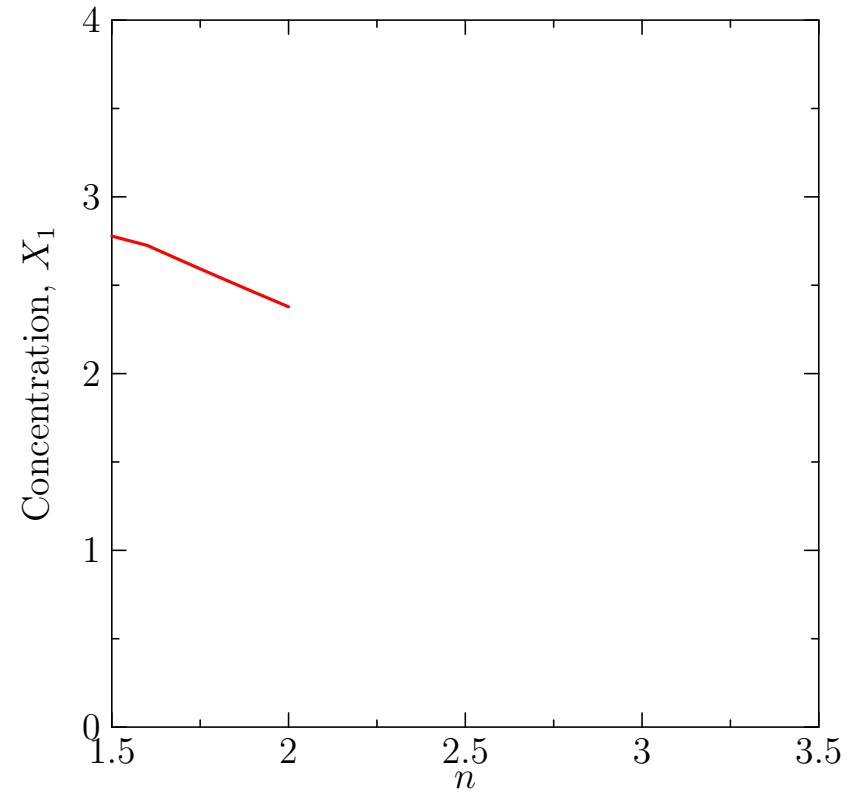
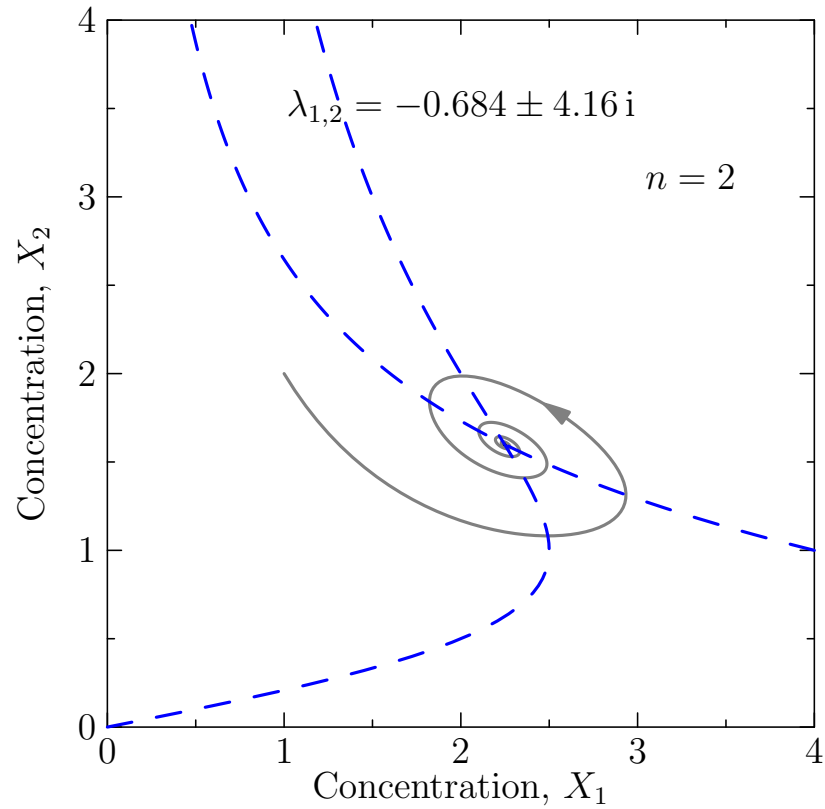
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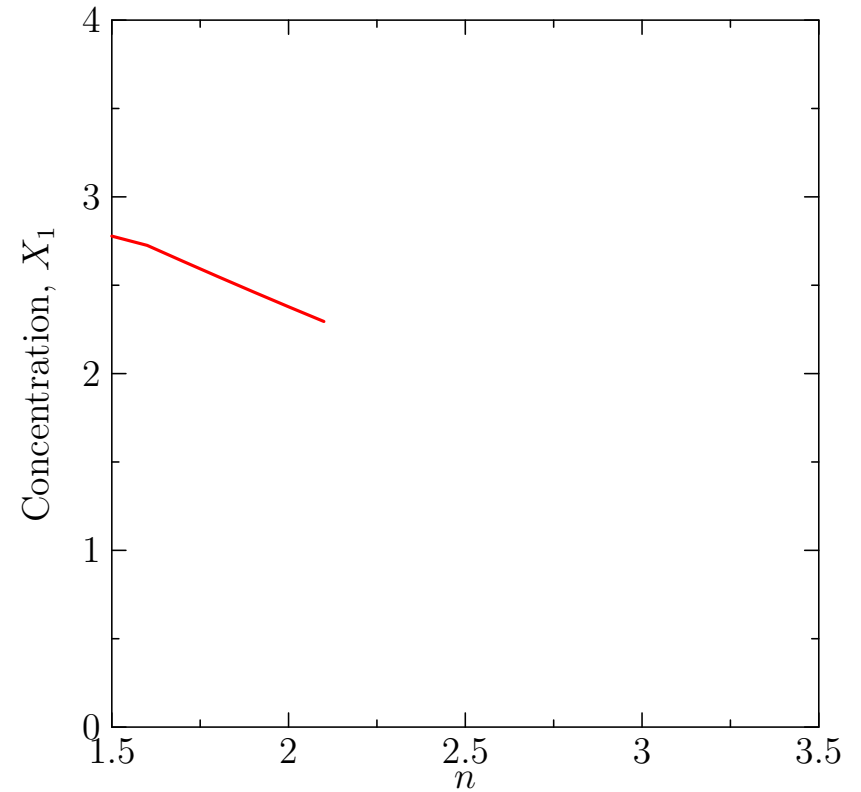
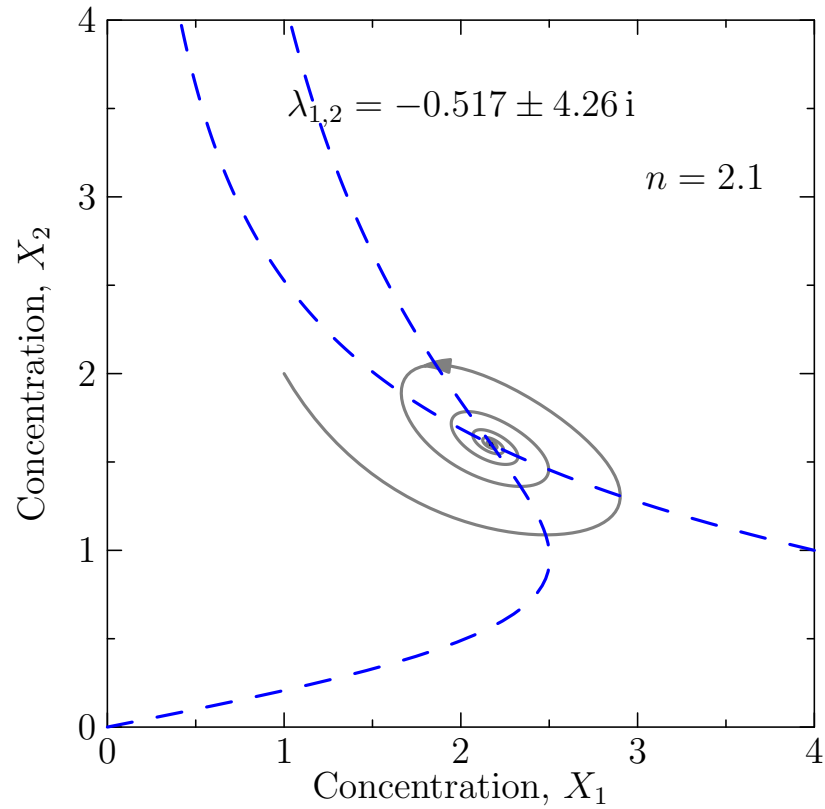
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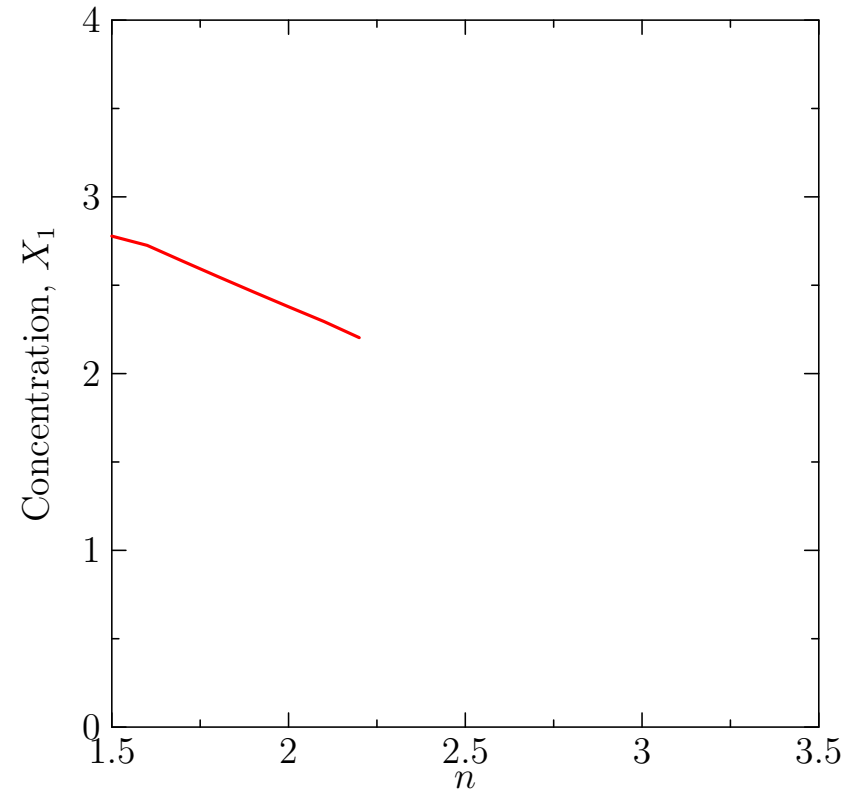
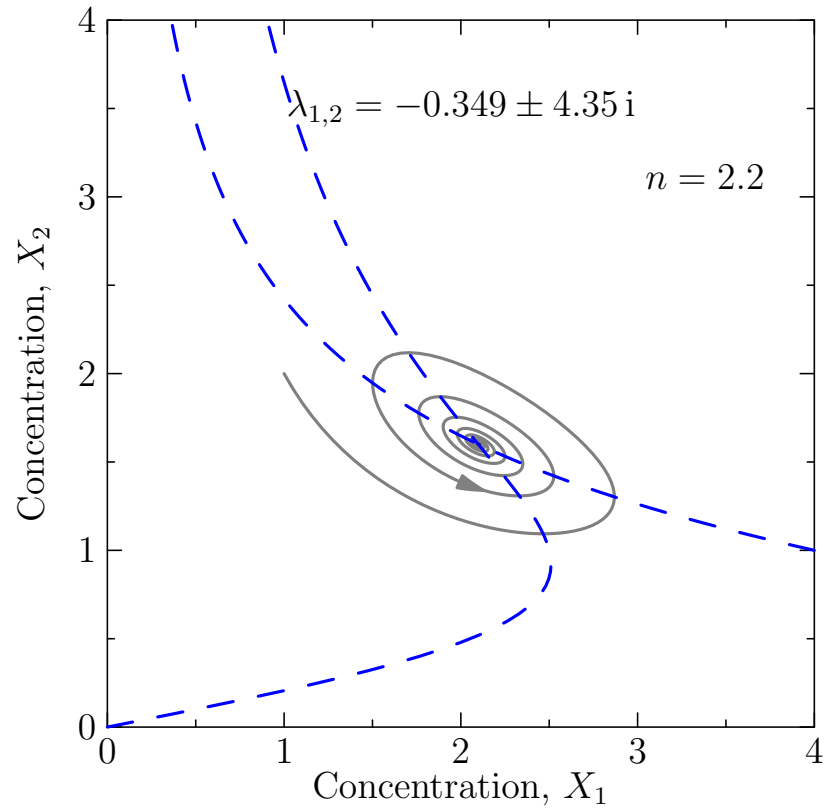
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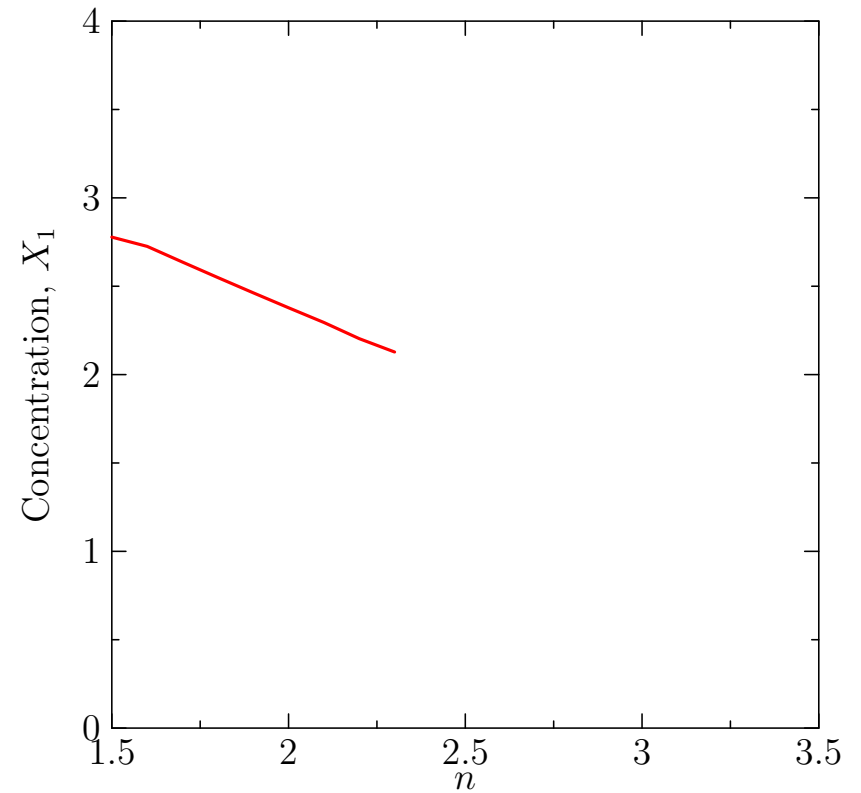
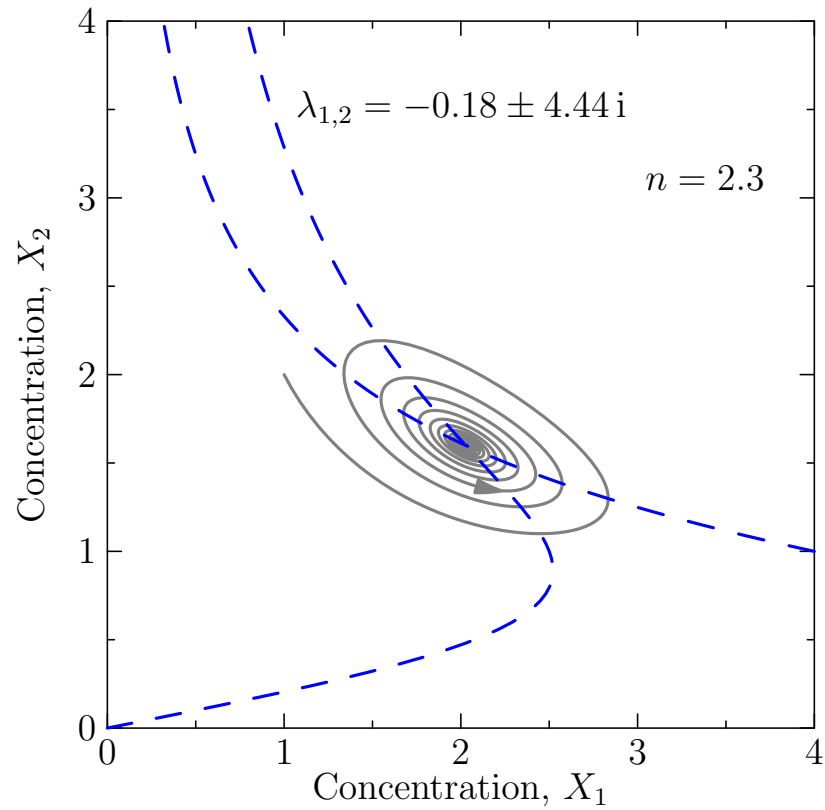
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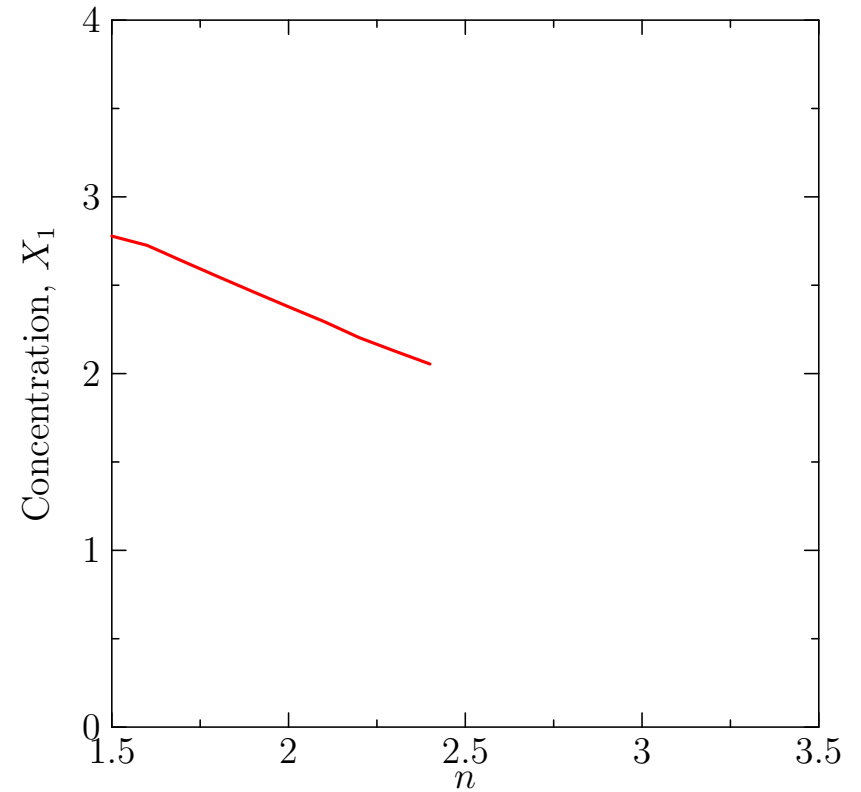
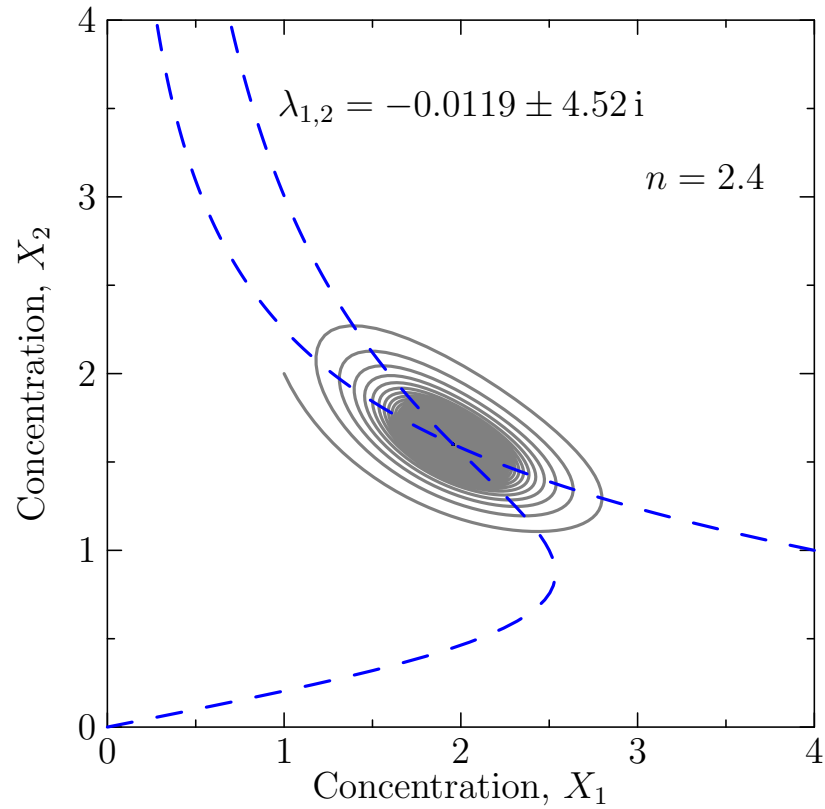
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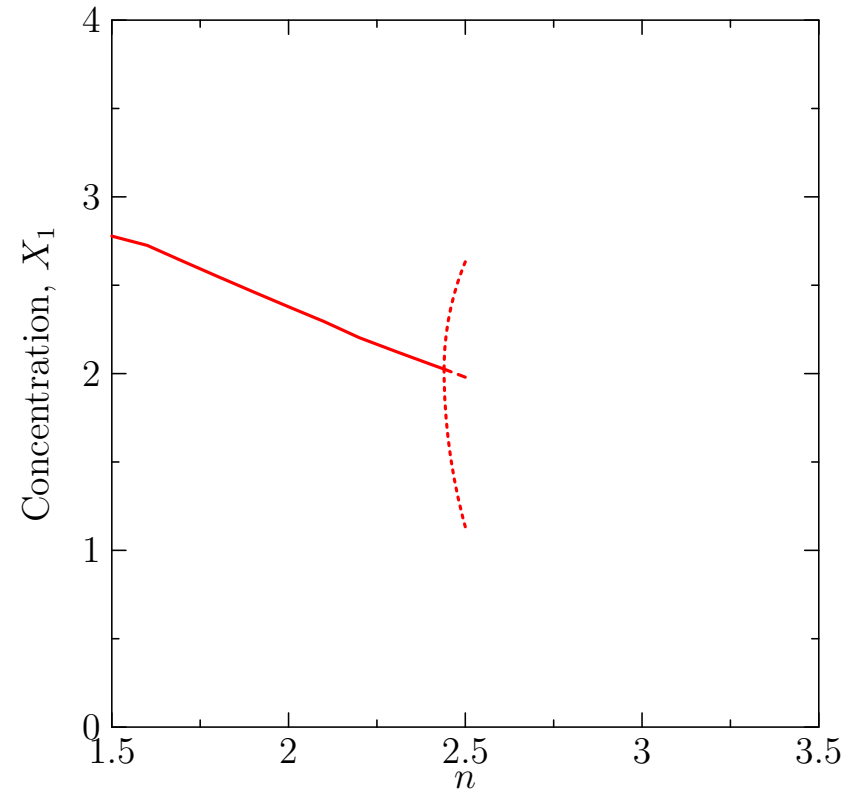
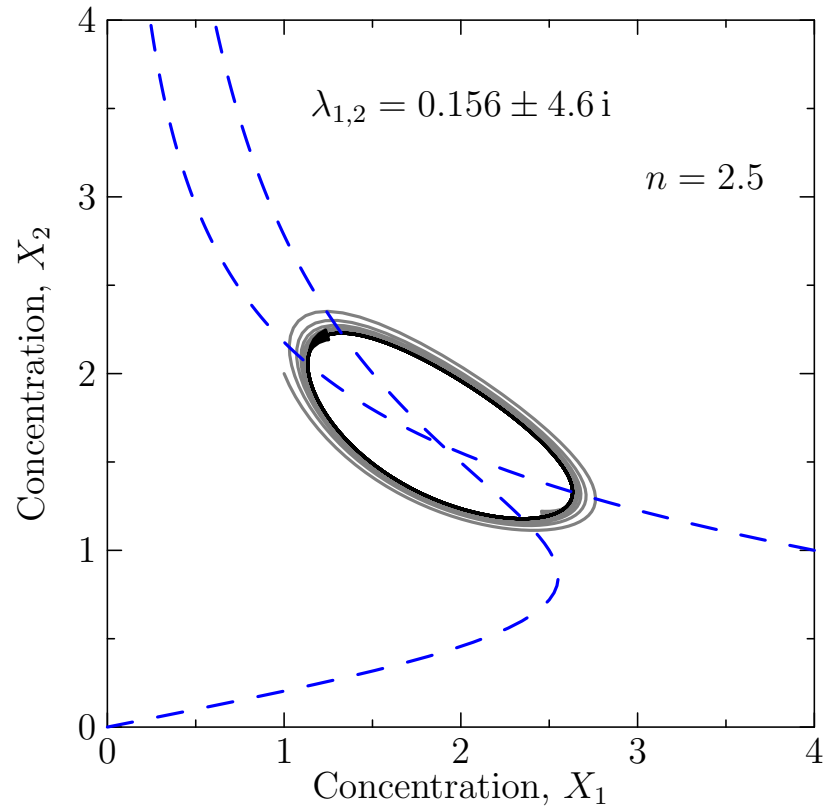
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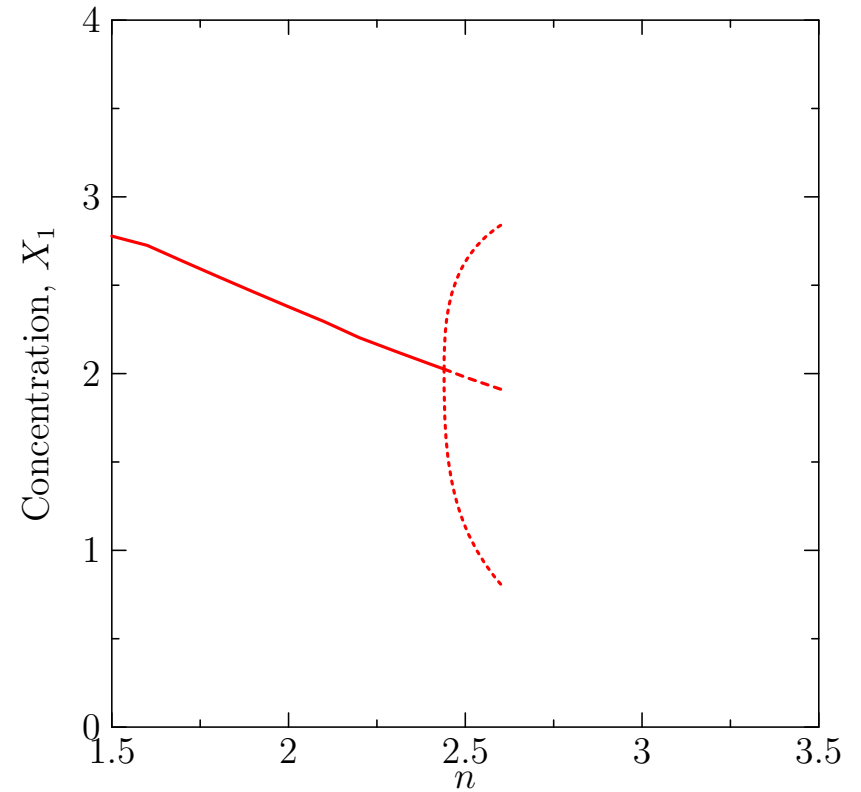
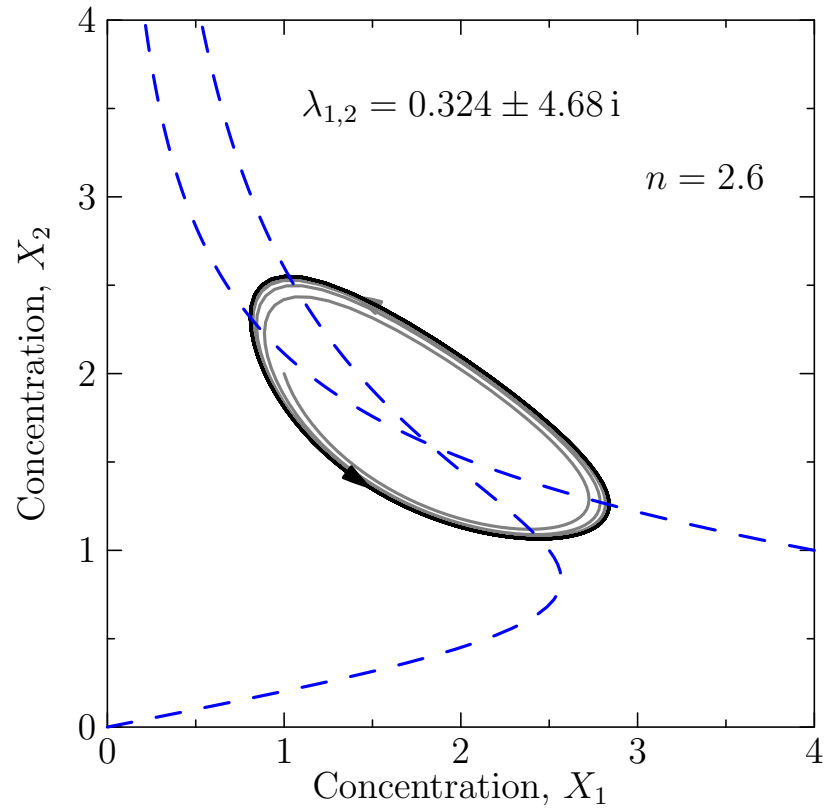
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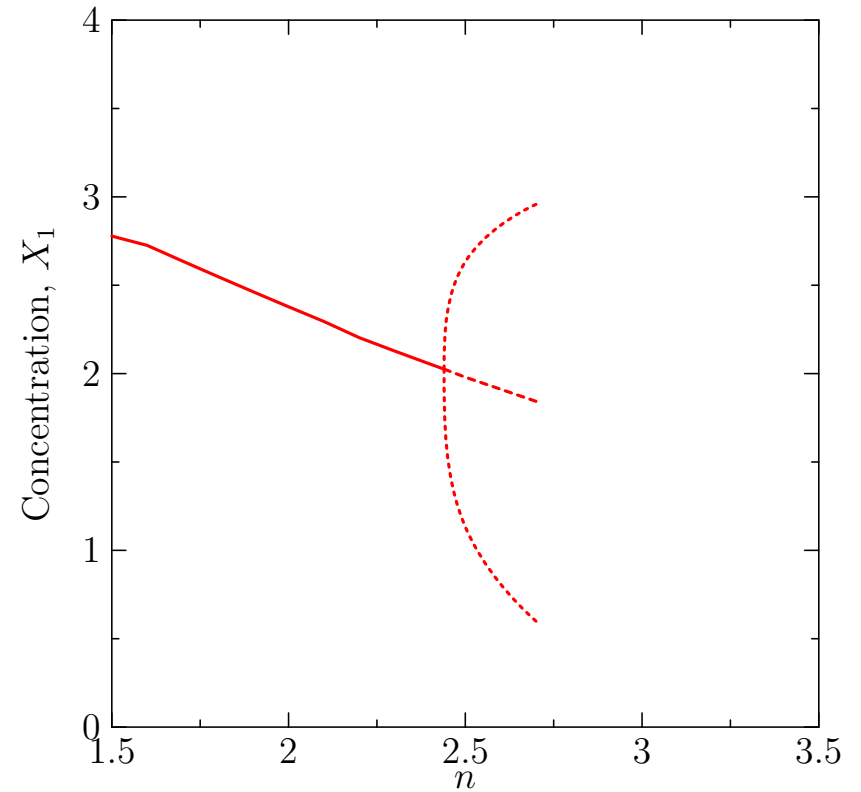
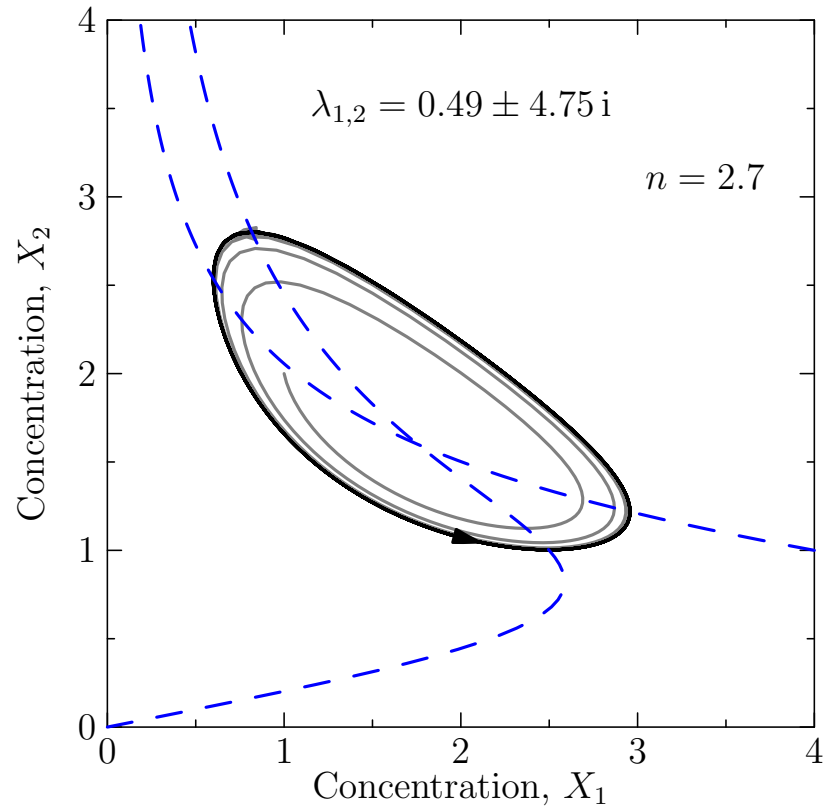
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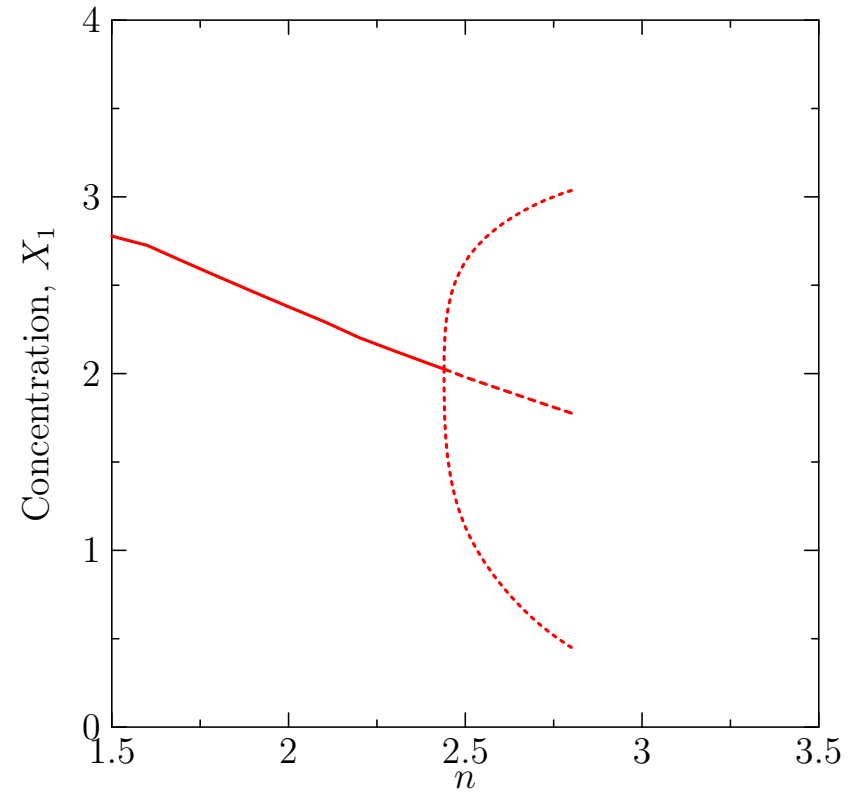
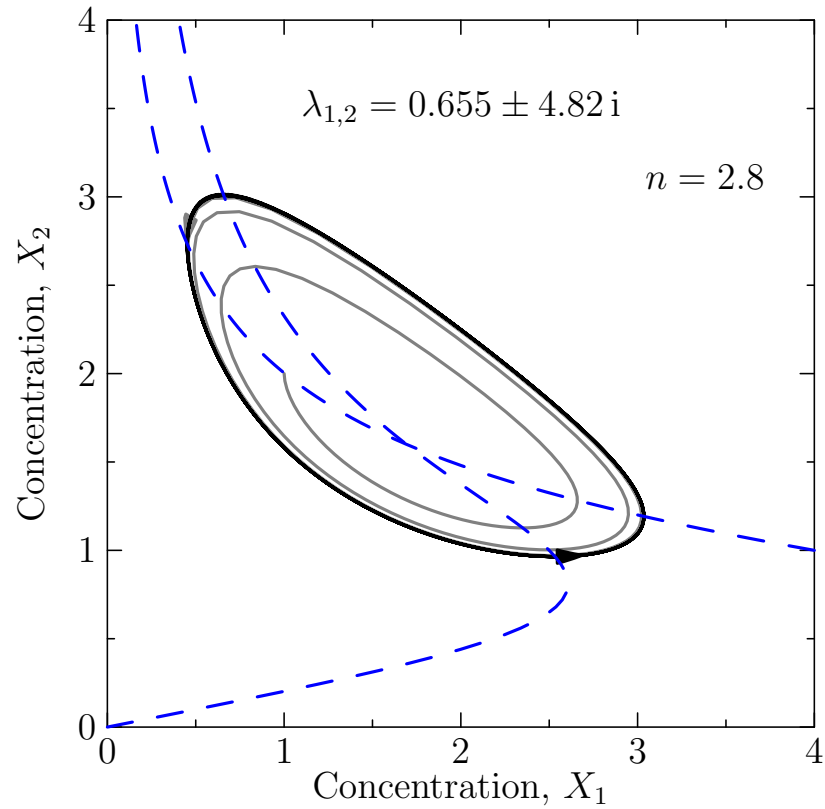
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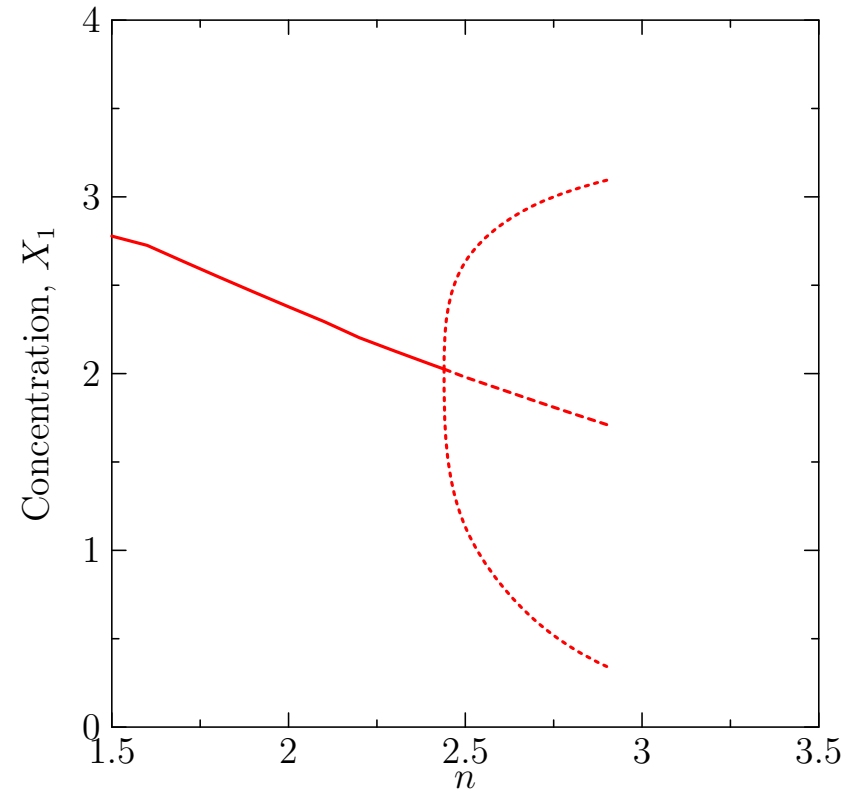
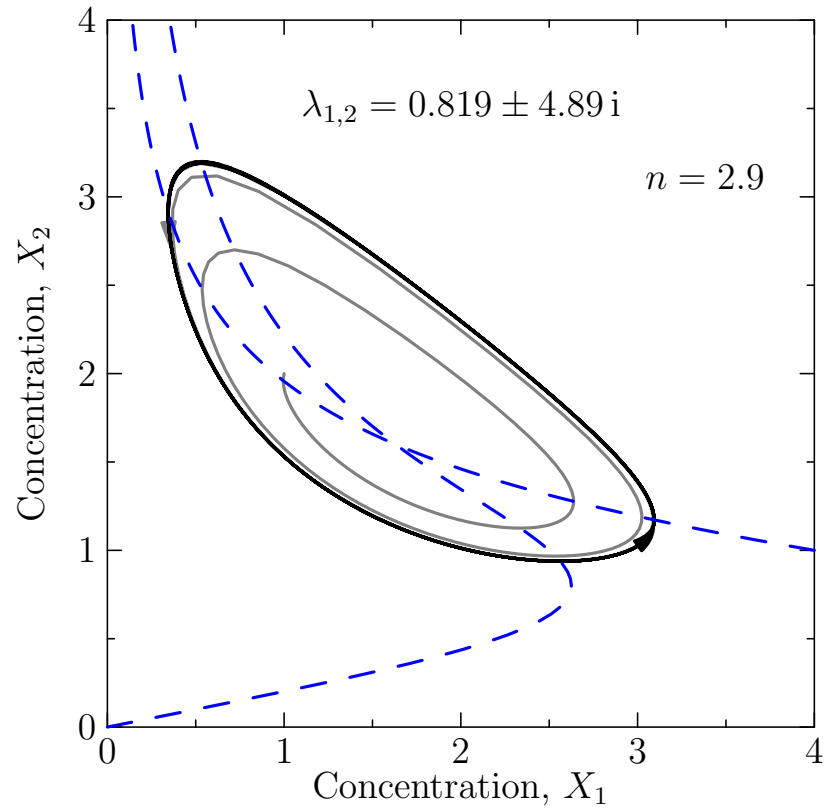
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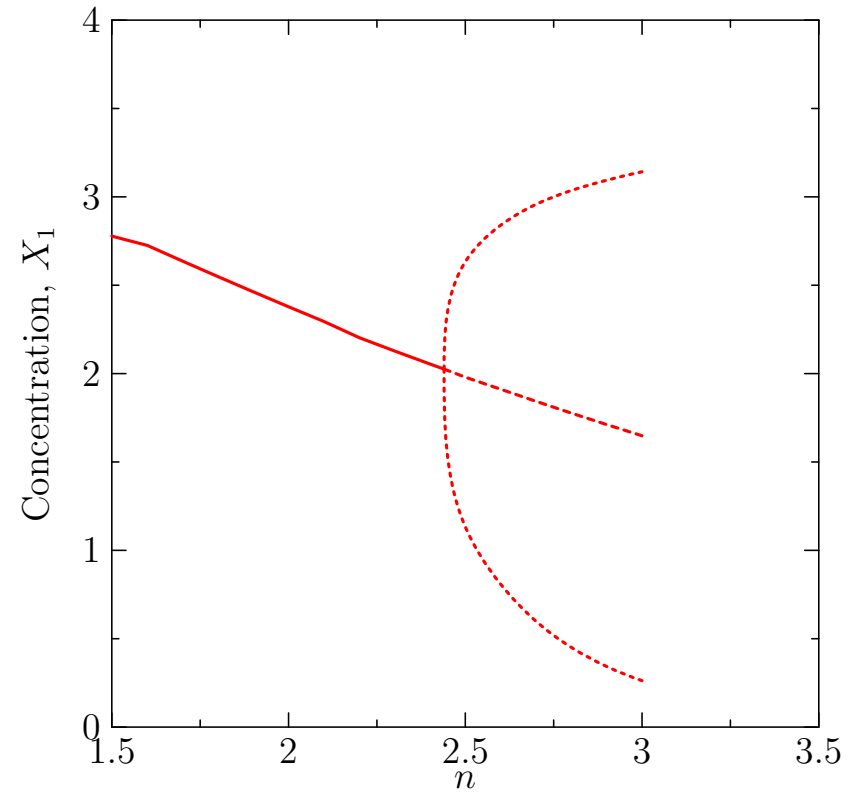
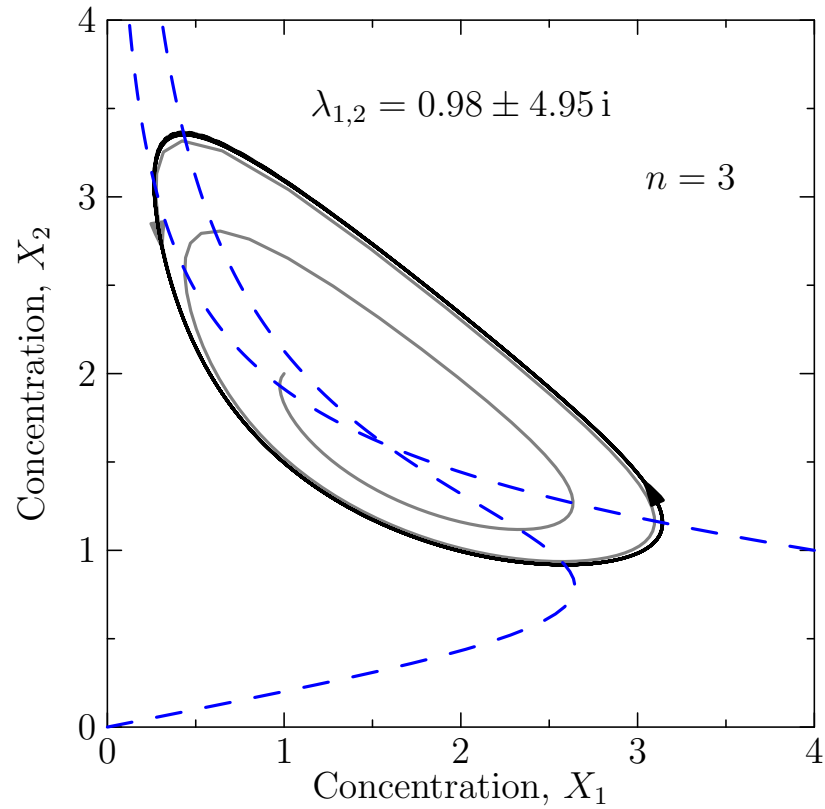
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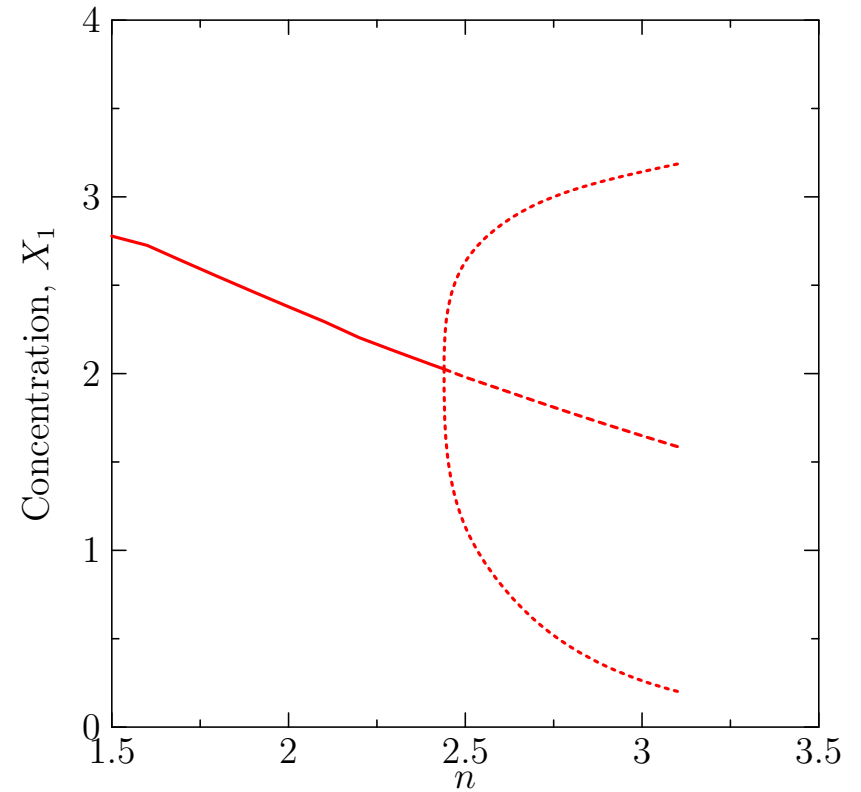
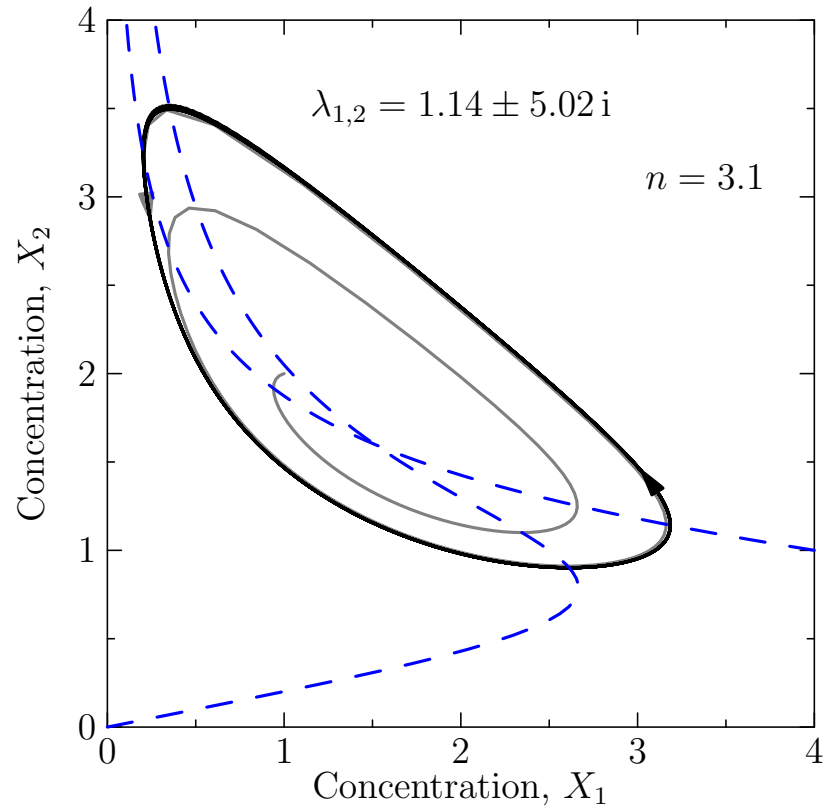
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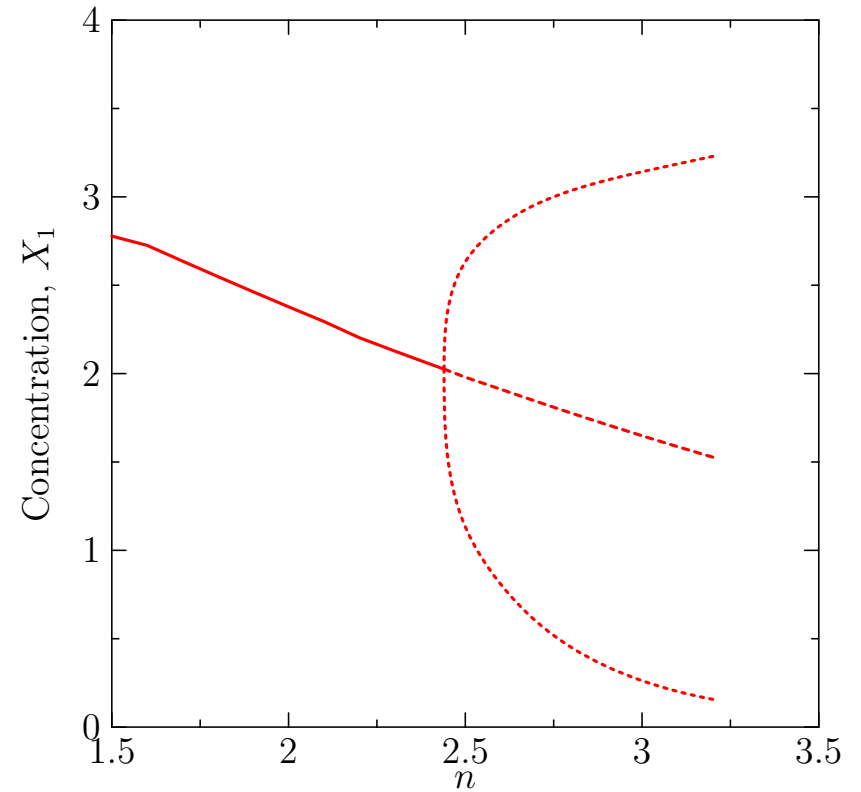
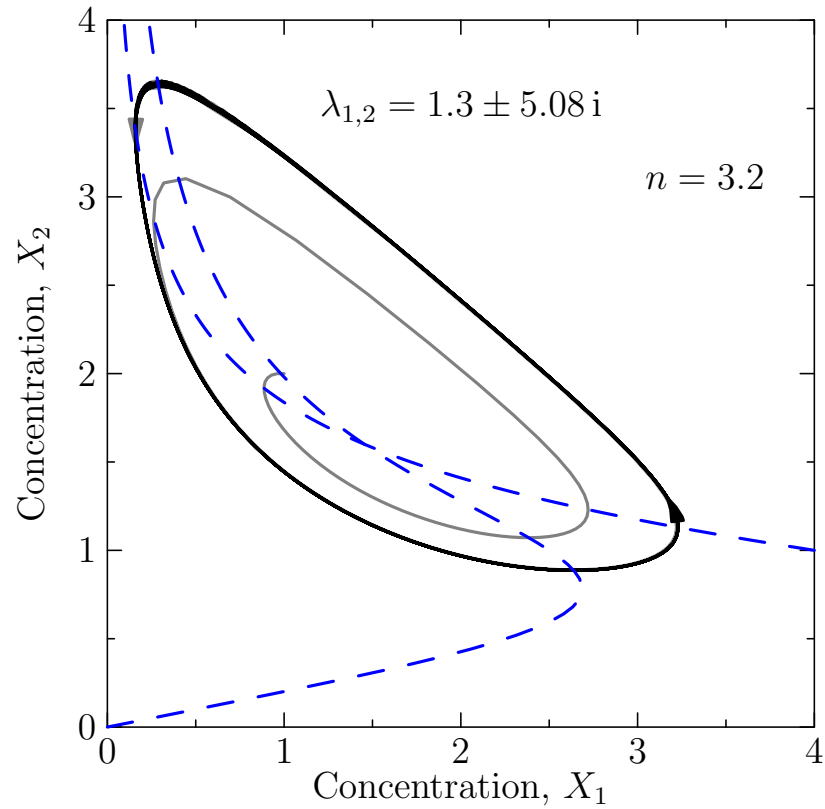
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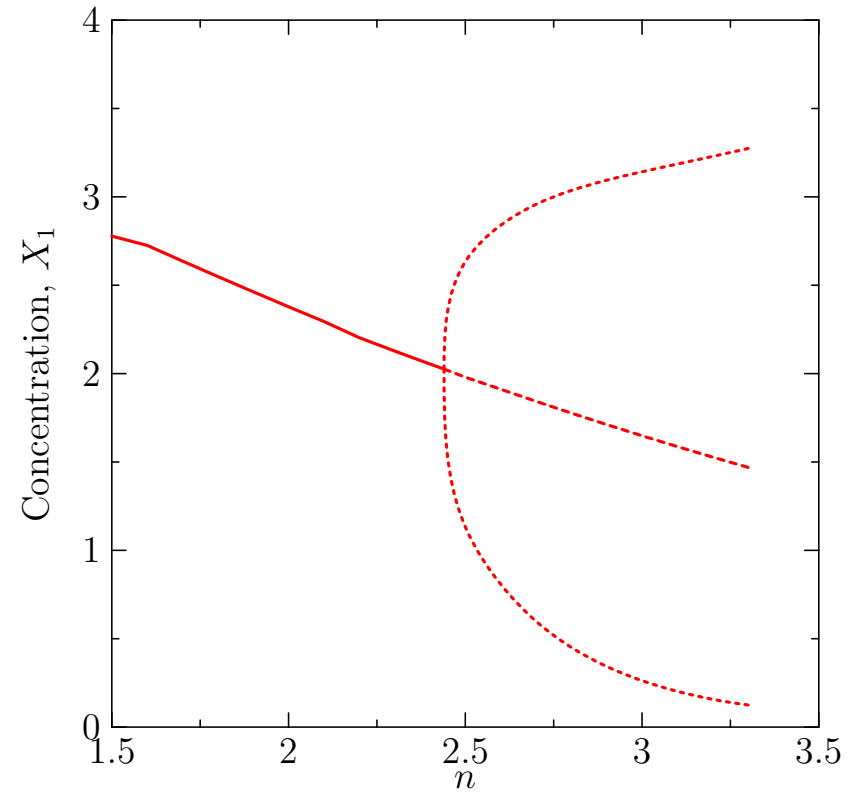
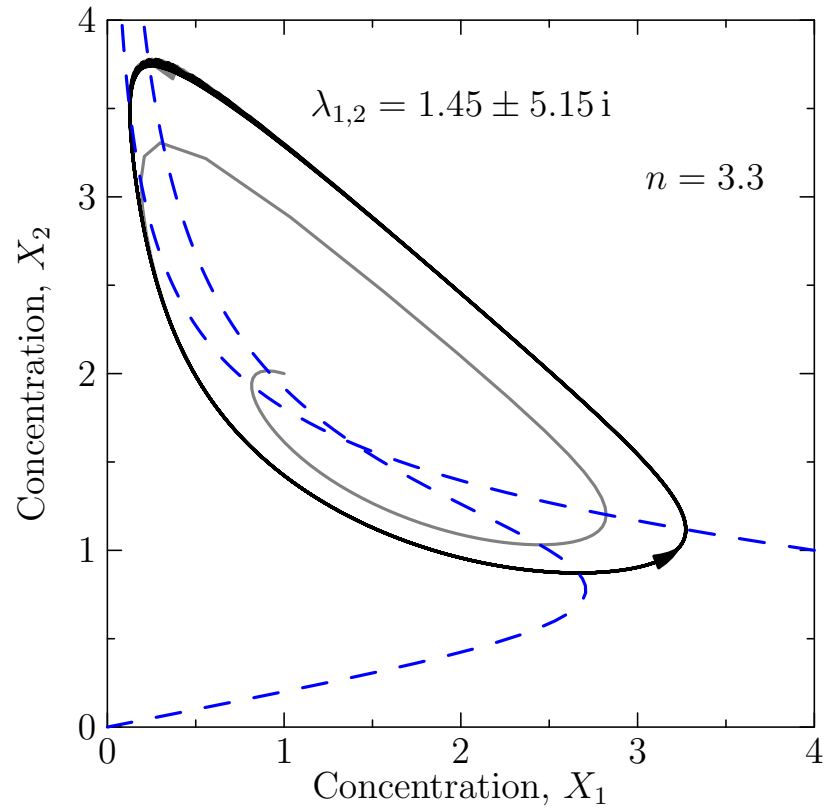
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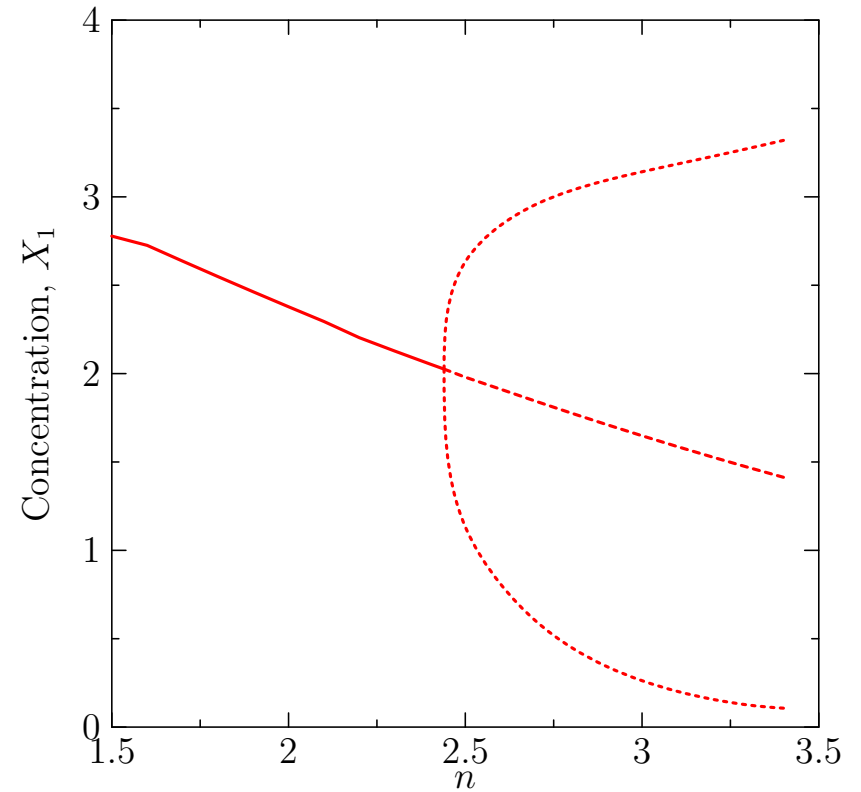
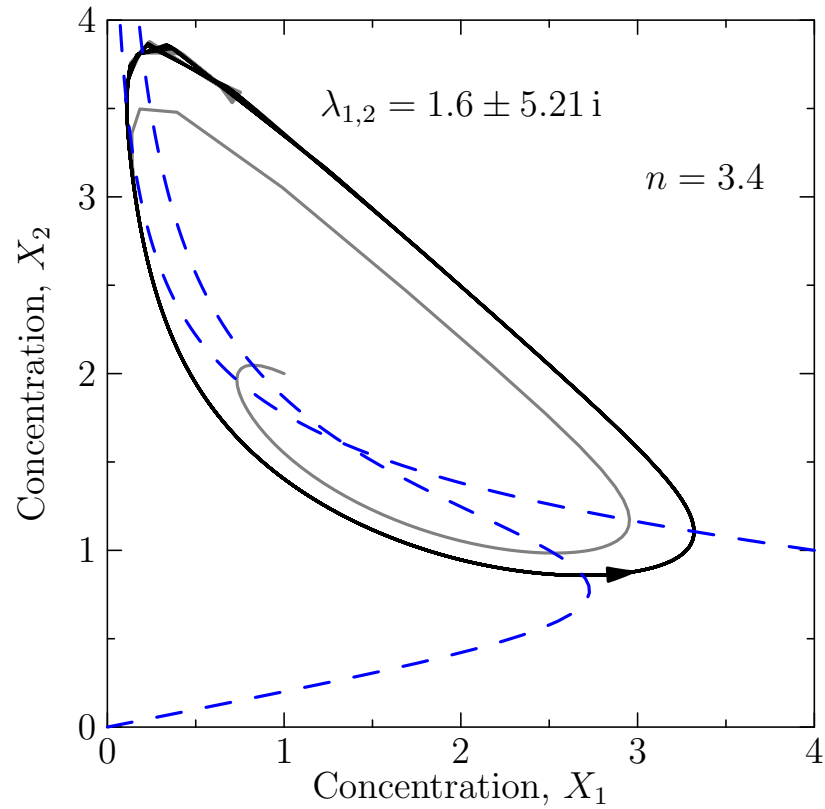
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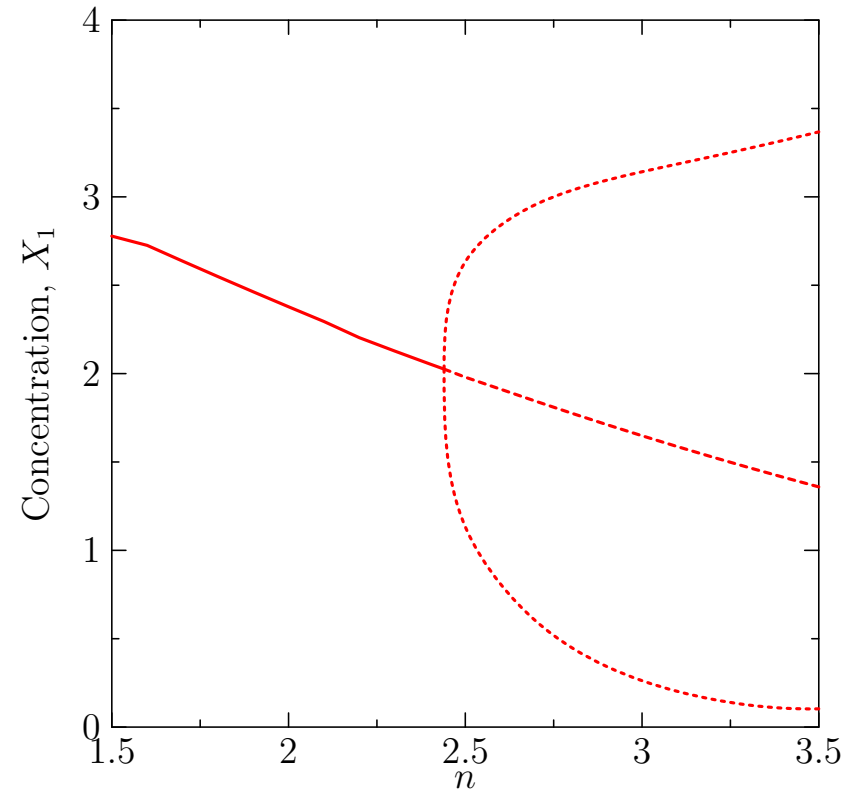
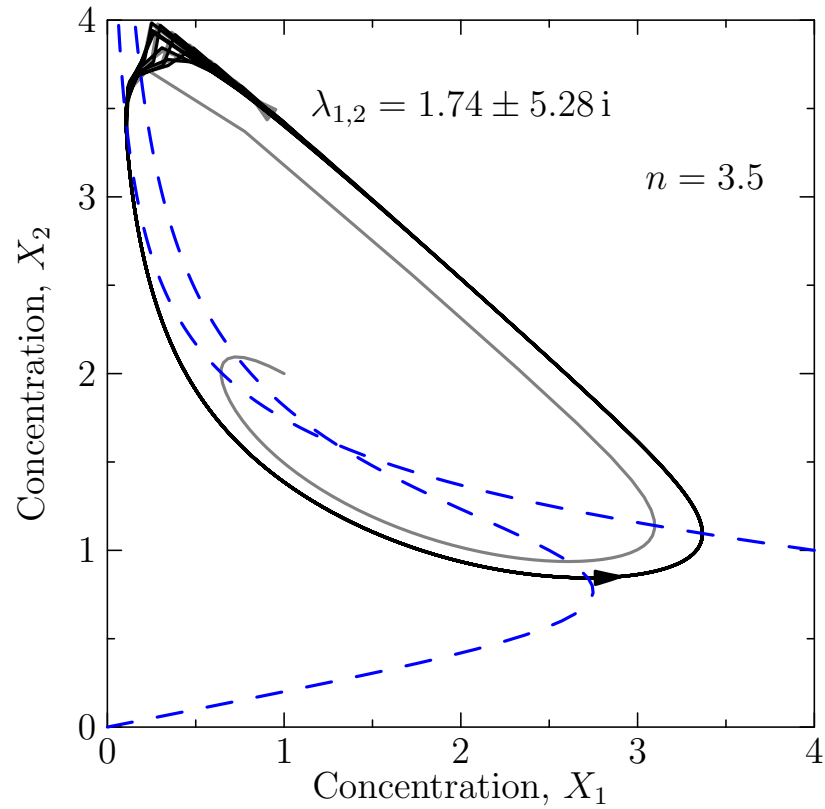
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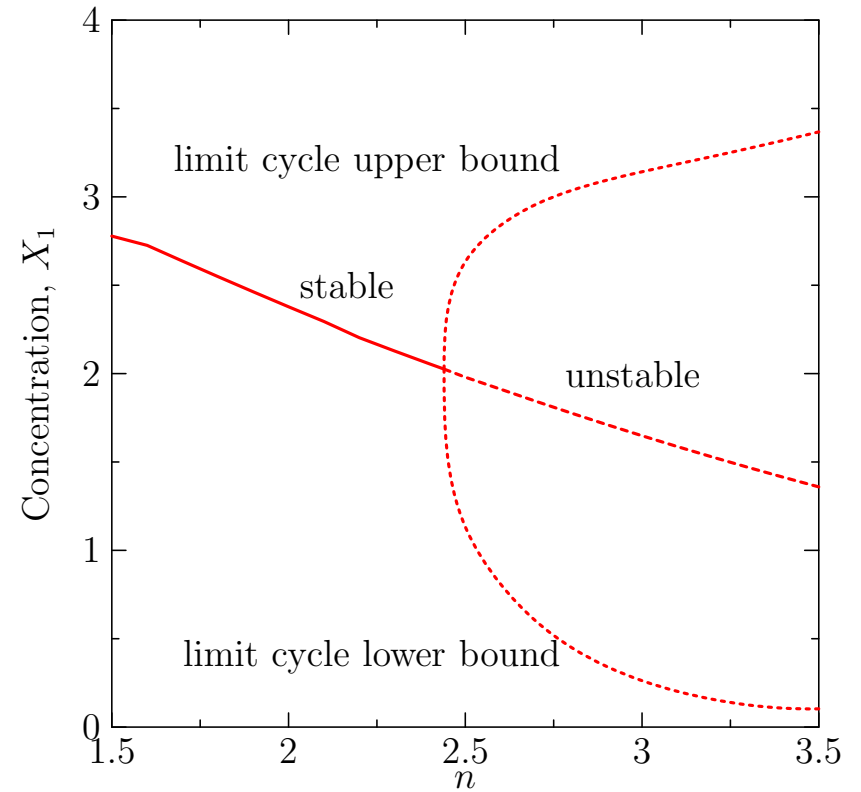
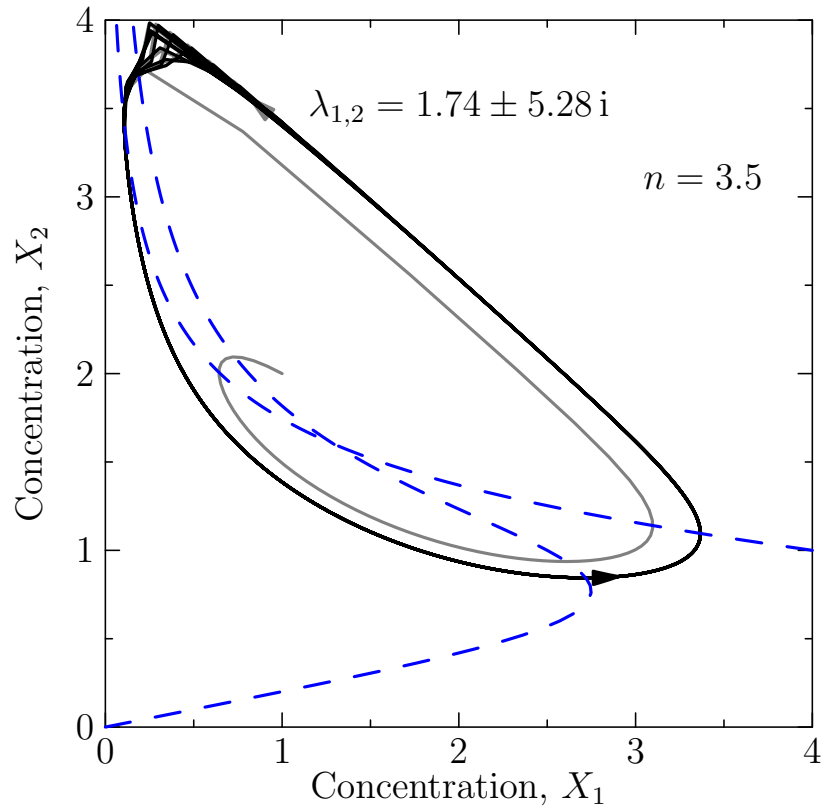
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Summary

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- The behaviour of systems can change radically as new fixed points emerge or fixed points become unstable

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